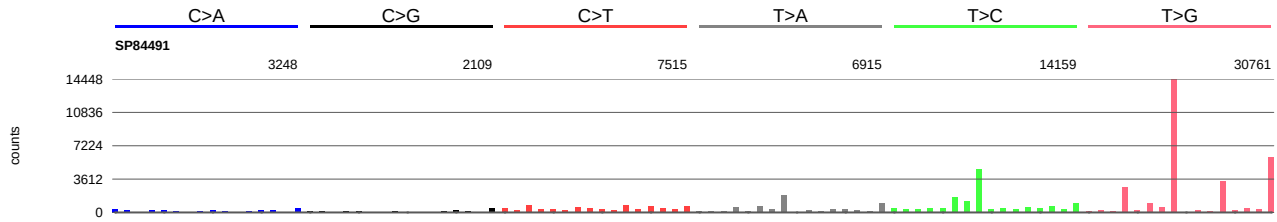
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14448
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3612
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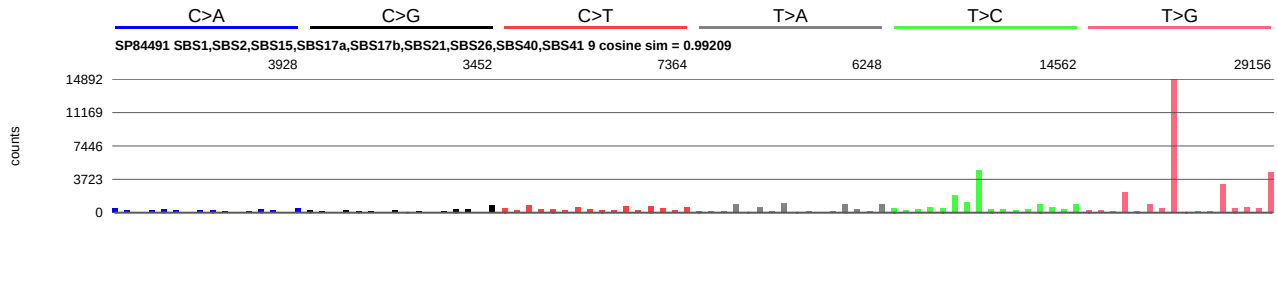
counts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SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.99209
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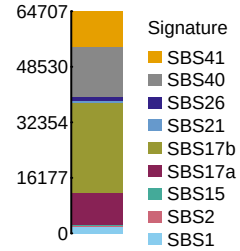
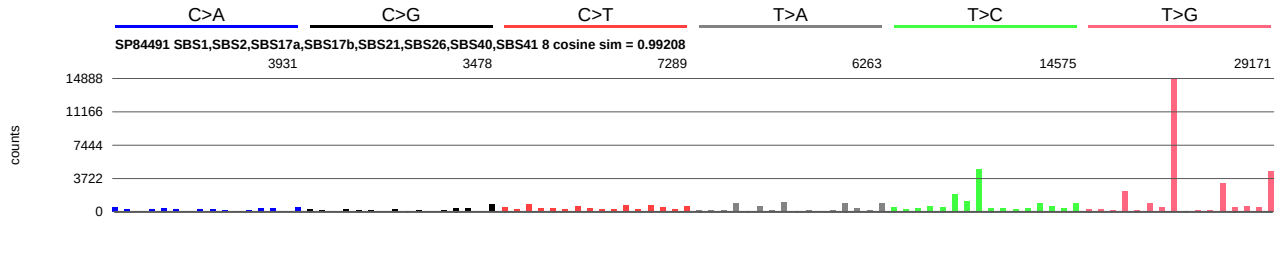
3452
7364
6248
14562
29156
14892
11169
7446
3723
0
counts


64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1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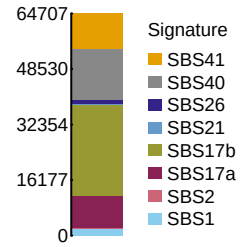
3478
7289
6263
14575
29171
14888
11166
7444
3722
0
counts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48530
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SBS40
SBS26
SBS21
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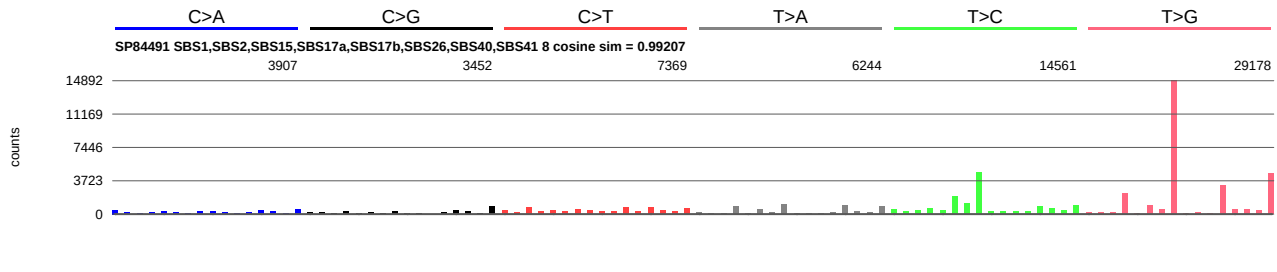
SBS2
SBS1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SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99207
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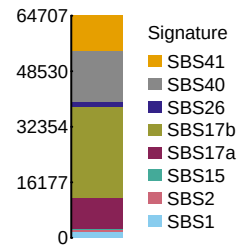
3452
7369
6244
14561
29178
14892
11169
7446
3723
0
counts


64707
48530
32354
16177
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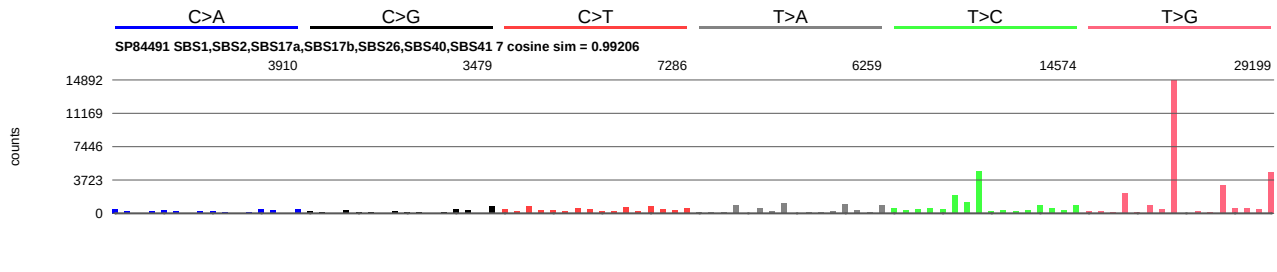
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1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SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99206
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3479
7286
6259
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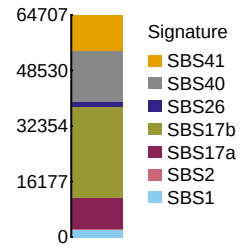
14892
11169
7446
3723
0
counts


64707
48530
32354
16177
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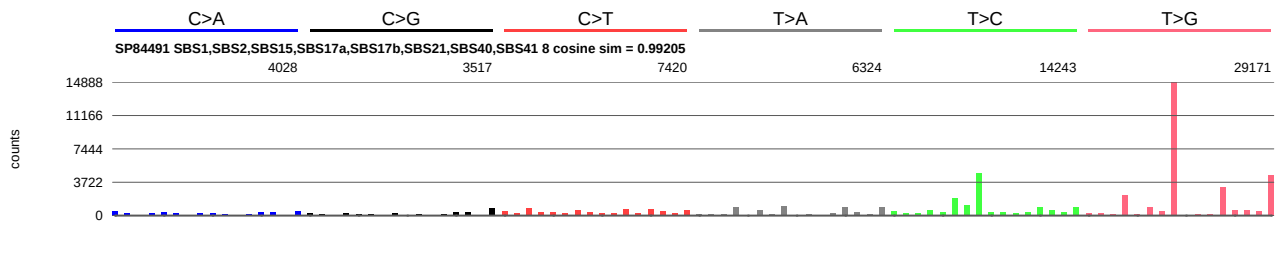
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1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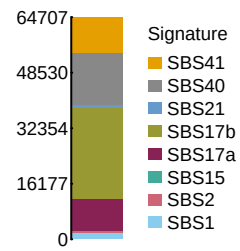
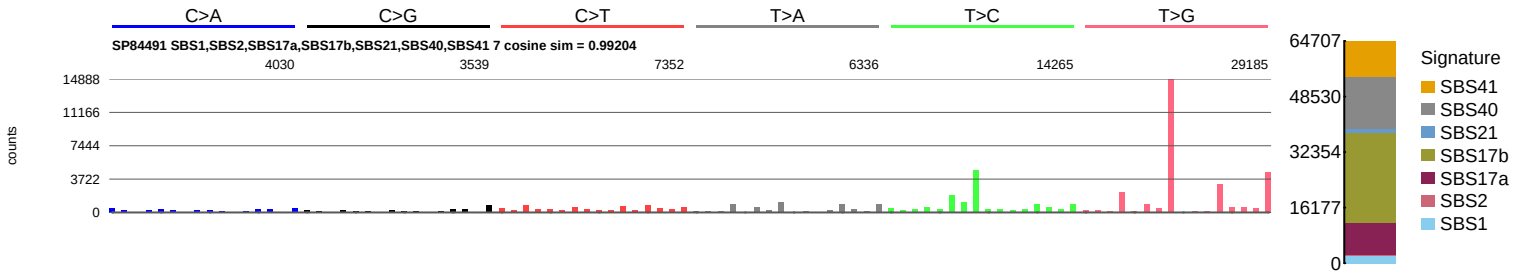
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99205
4028
3517
7420
6324
14243
29171
14888
11166
7444
3722
0
counts


64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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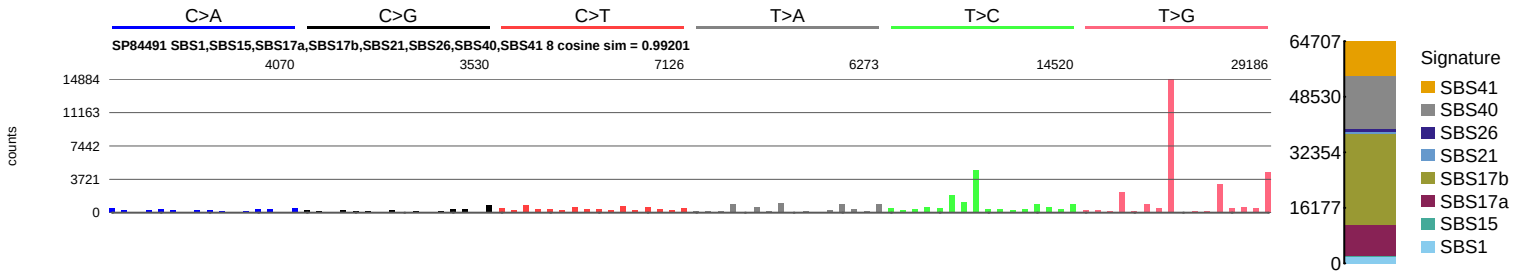
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99204
4030
3539
7352
6336
14265
29185
counts
14888
11166
7444
3722
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1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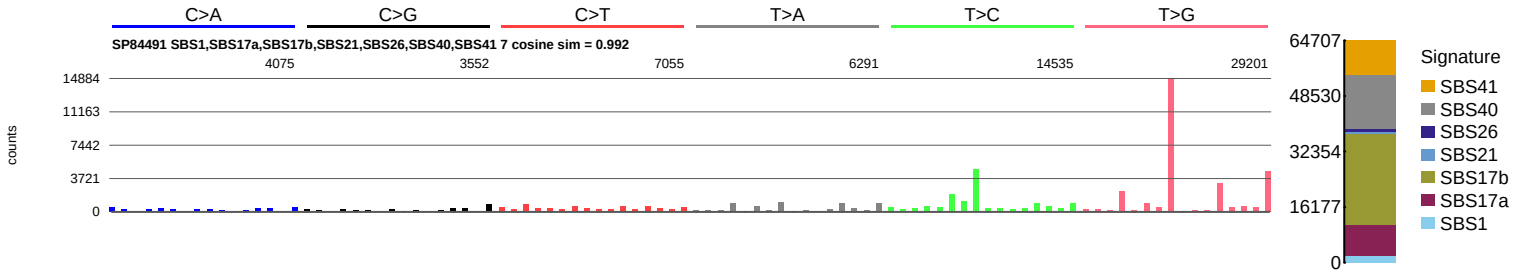
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.99201
4070
3530
7126
6273
14520
29186
counts
14884
11163
7442
3721
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
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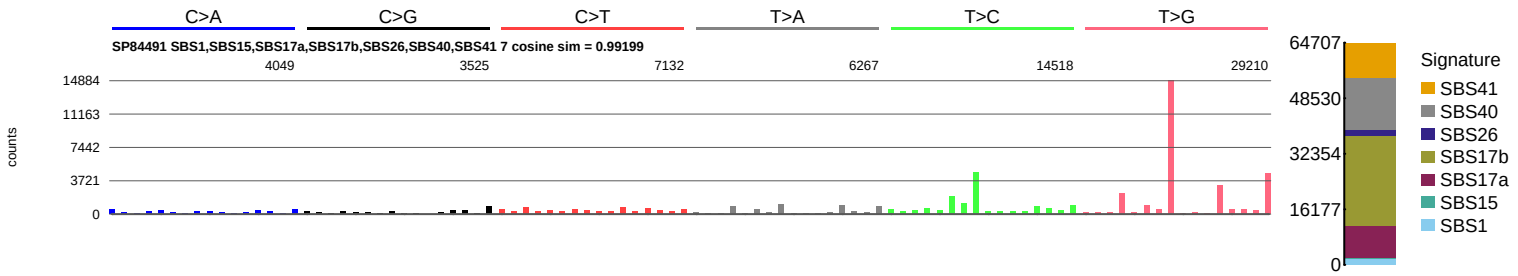
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.992
4075
3552
7055
6291
14535
29201
counts
14884
11163
7442
3721
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1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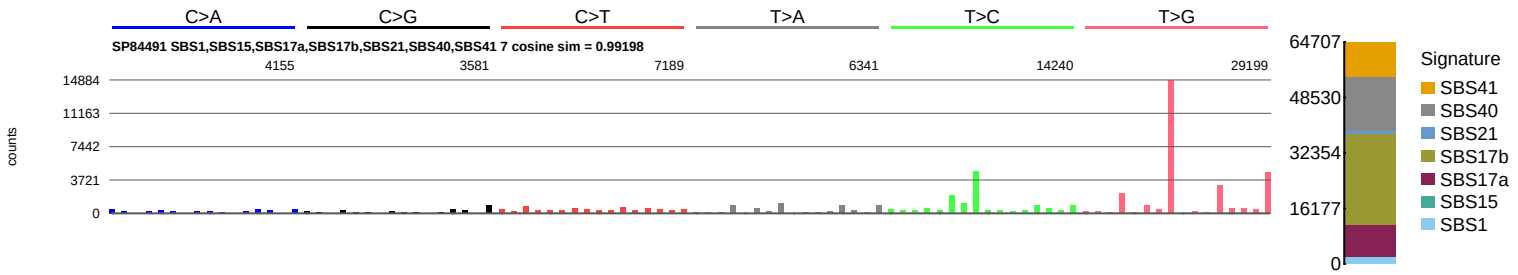
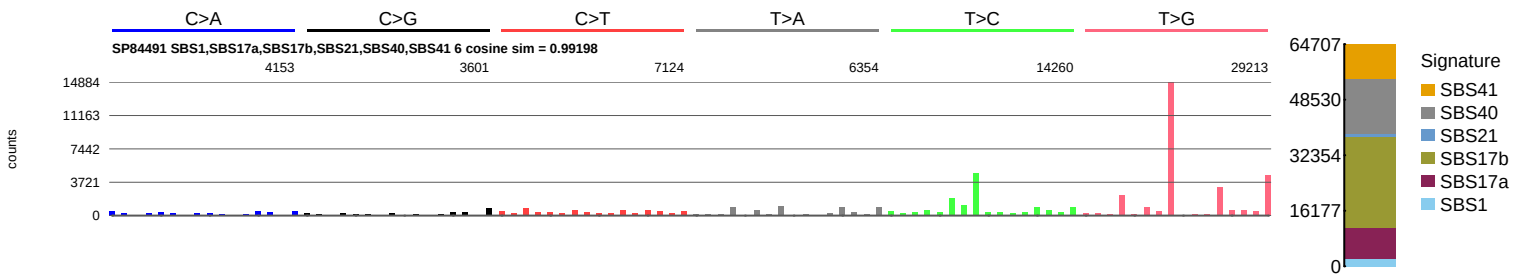
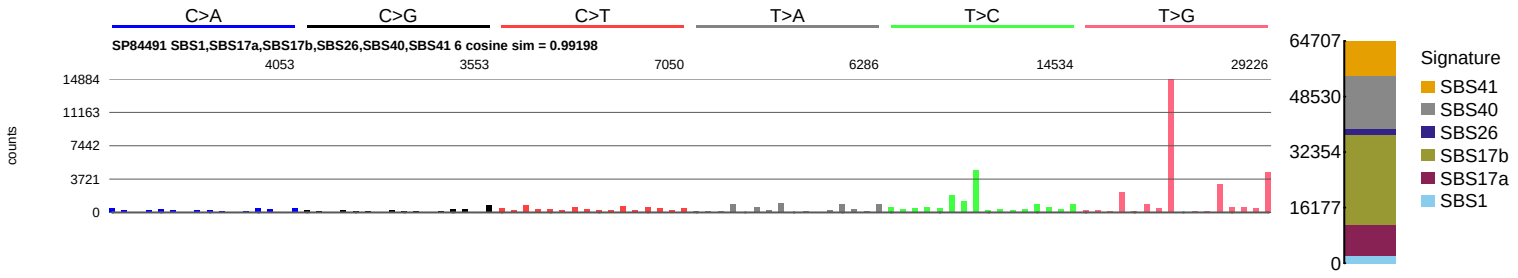
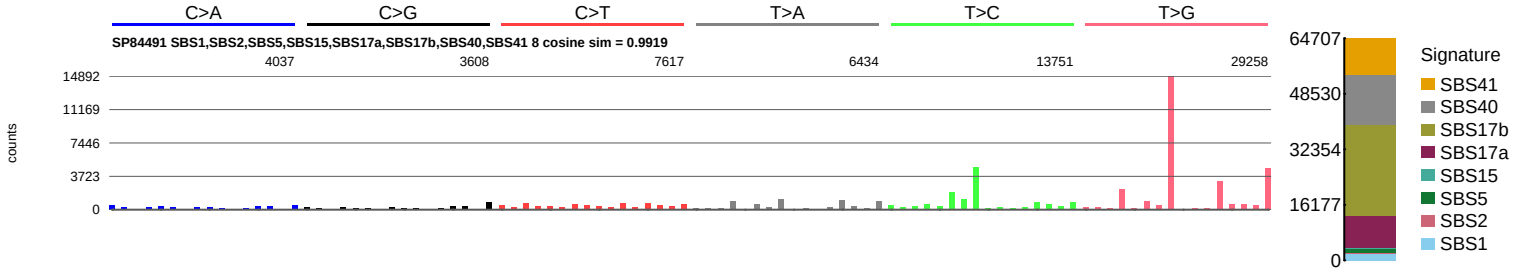
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99199
4049
3525
7132
6267
14518
29210
counts
14884
11163
7442
3721
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99198
4155
3581
7189
6341
14240
29199
counts
14884
11163
7442
3721
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.99198
4153
3601
7124
6354
14260
29213
counts
14884
11163
7442
3721
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.99198
4053
3553
7050
6286
14534
29226
counts
14884
11163
7442
3721
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1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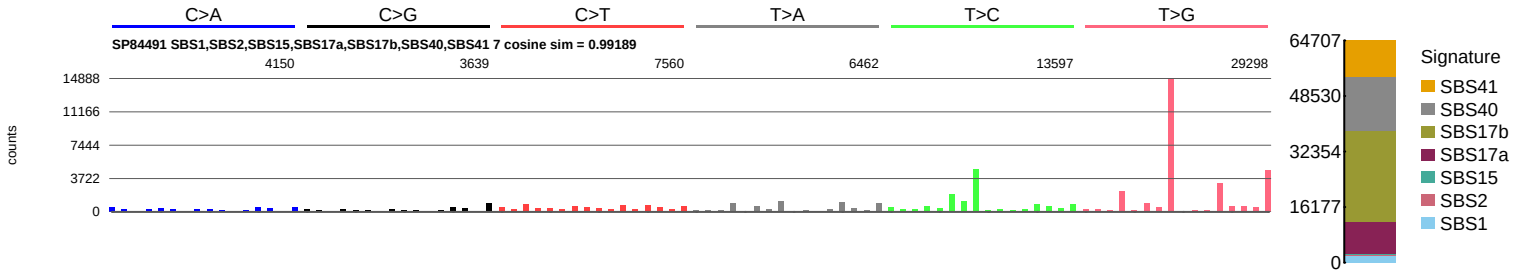
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.9919
4037
3608
7617
6434
13751
29258
counts
14892
11169
7446
3723
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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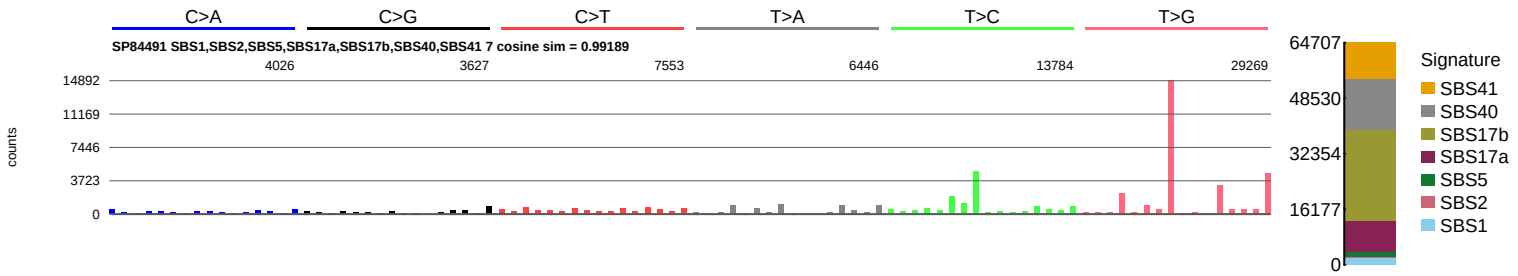
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99189
4150
3639
7560
6462
13597
29298
counts
14888
11166
7444
3722
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99189
4026
3627
7553
6446
13784
29269
counts
14892
11169
7446
3723
0
64707
48530
32354
16177
0
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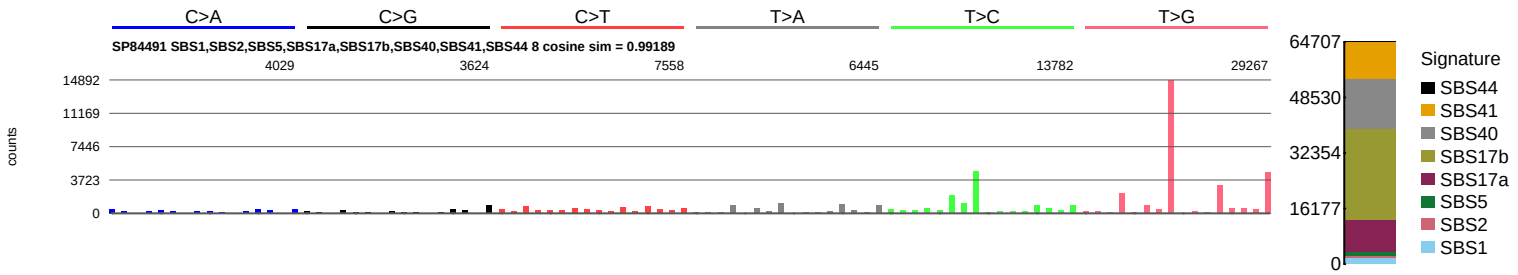
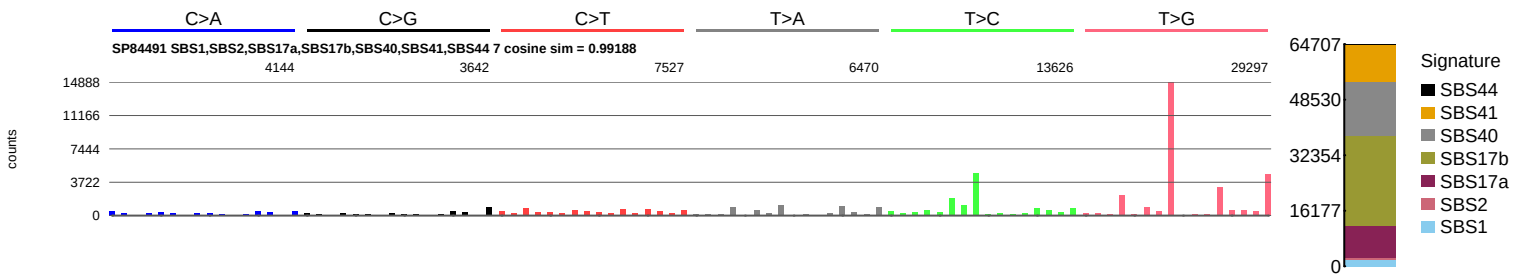
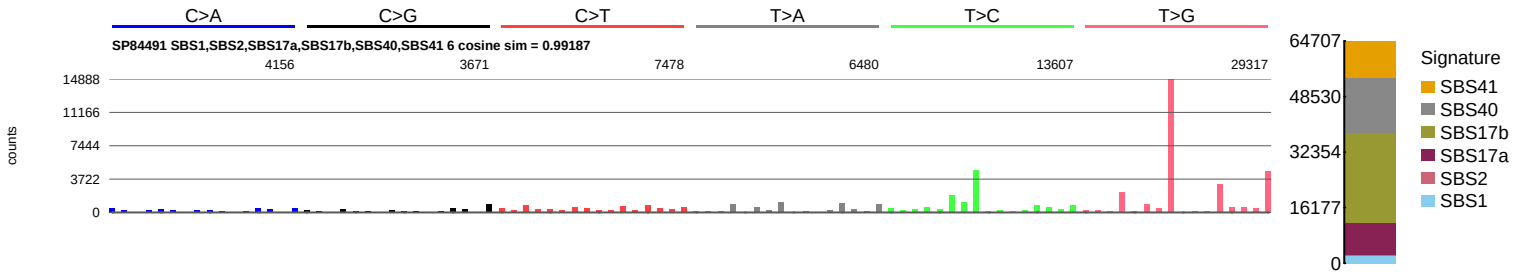
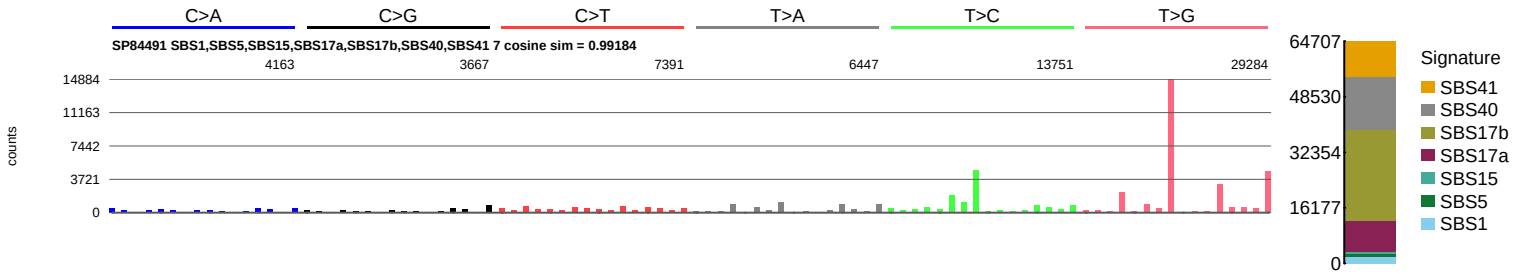
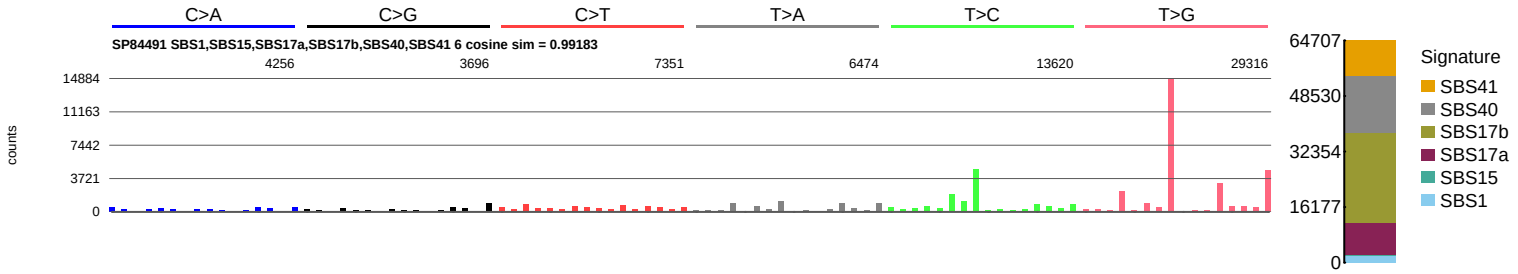
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1

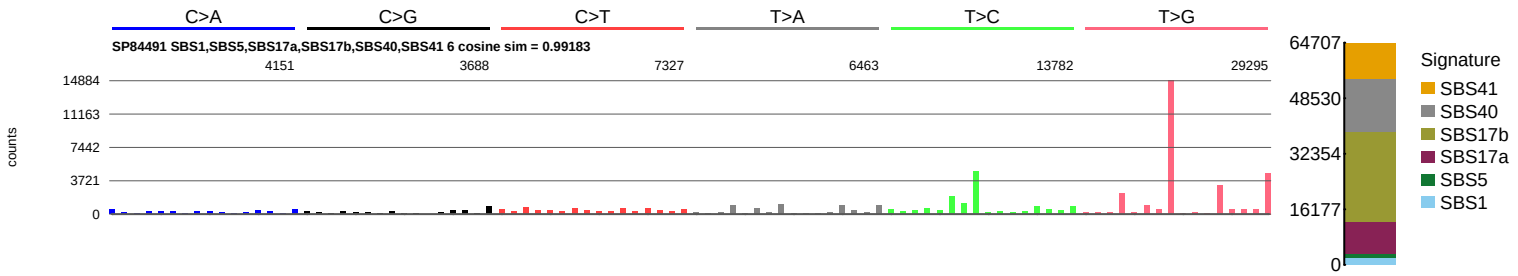

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.99189
4029
3624
7558
6445
13782
29267
counts
14892
11169
7446
3723
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1

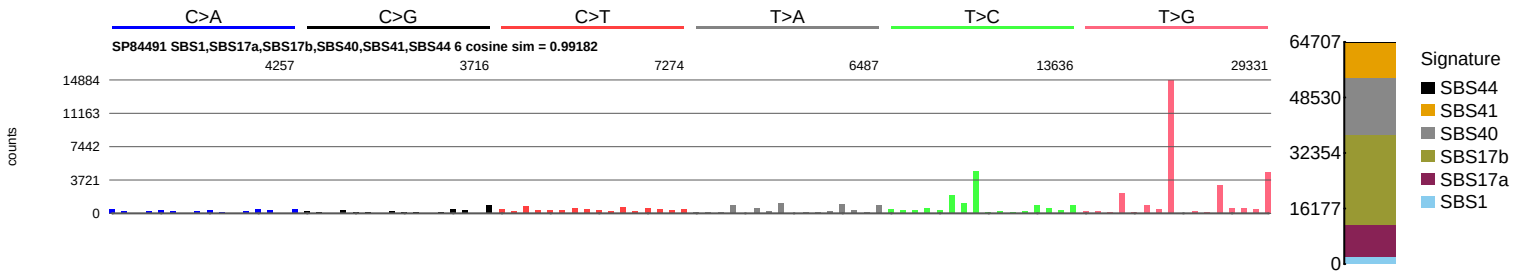

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99188
4144
3642
7527
6470
13626
29297
counts
14888
11166
7444
3722
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1

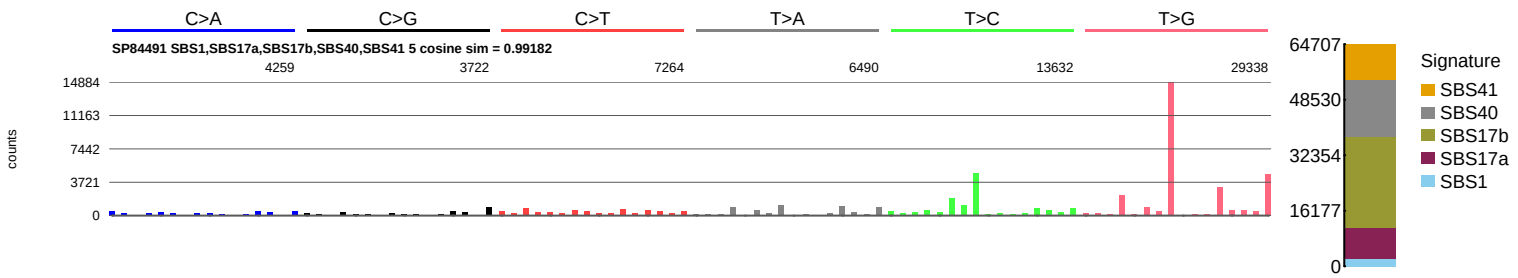

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99187
4156
3671
7478
6480
13607
29317
counts
14888
11166
7444
3722
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99184
4163
3667
7391
6447
13751
29284
counts
14884
11163
7442
3721
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99183
4256
3696
7351
6474
13620
29316
counts
14884
11163
7442
3721
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99183
4151
3688
7327
6463
13782
29295
counts
14884
11163
7442
3721
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.99182
4257
3716
7274
6487
13636
29331
counts
14884
11163
7442
3721
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS1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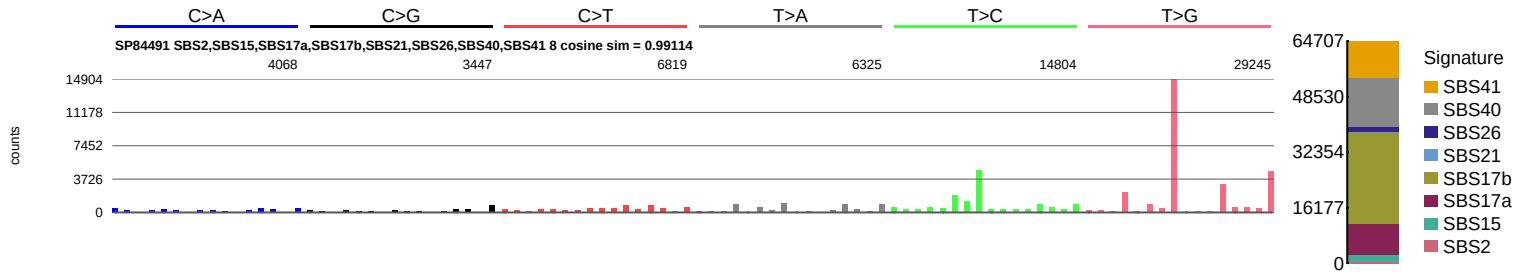
SP84491 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.99182
4259
3722
7264
6490
13632
29338
counts
14884
11163
7442
3721
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1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C>G
C>T
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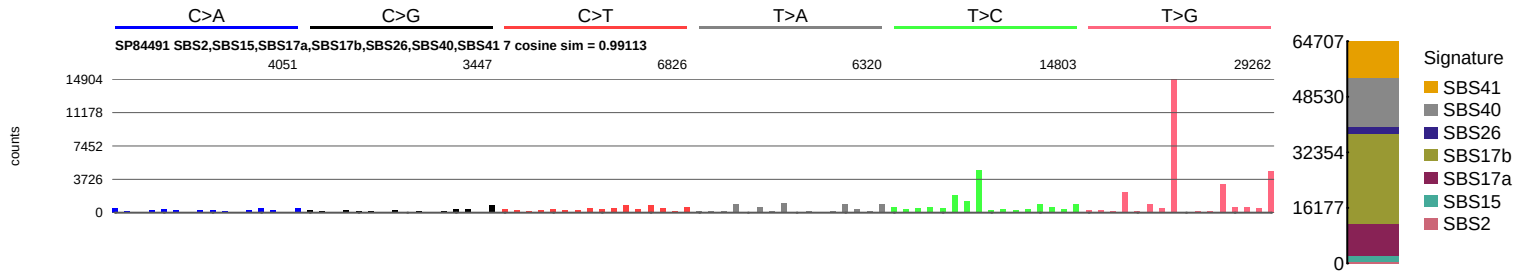
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.99114
4068
3447
6819
6325
14804
29245
counts
14904
11178
7452
3726
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2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C>G
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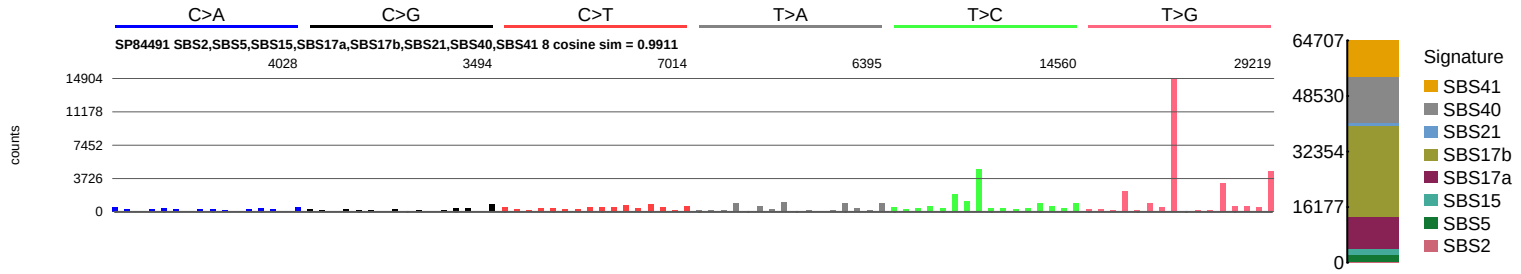
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99113
4051
3447
6826
6320
14803
29262
counts
14904
11178
7452
3726
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2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C>G
C>T
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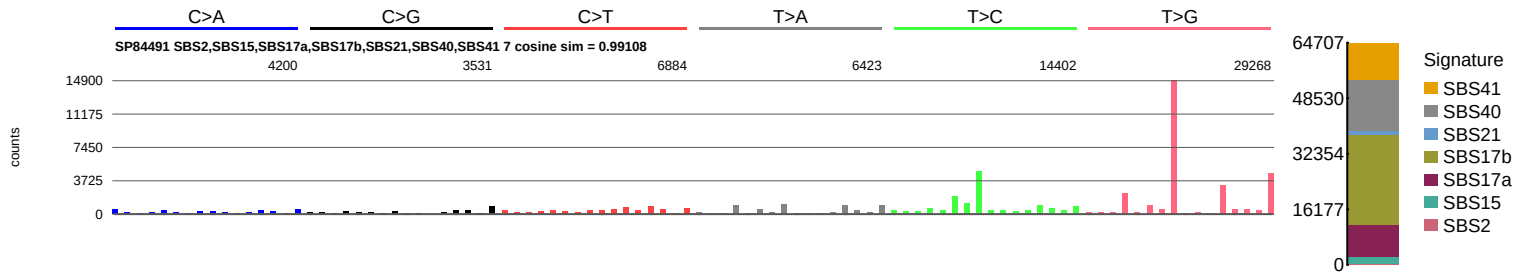
T>C
T>G
SP84491 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.9911
4028
3494
7014
6395
14560
29219
counts
14904
11178
7452
3726
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2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C>G
C>T
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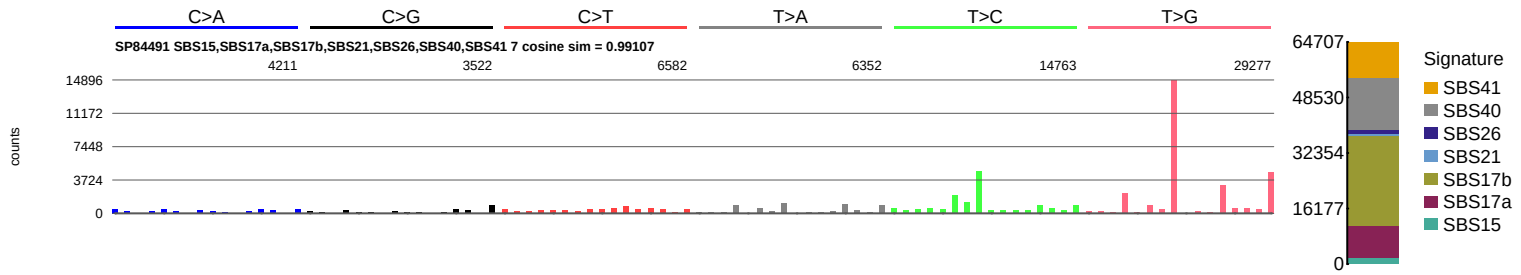
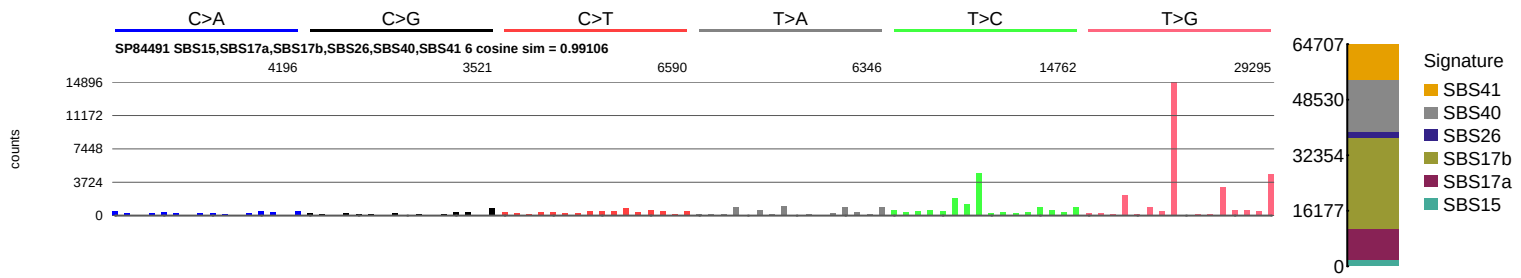
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99108
4200
3531
6884
6423
14402
29268
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.99107
4211
3522
6582
6352
14763
29277
counts
14896
11172
7448
3724
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15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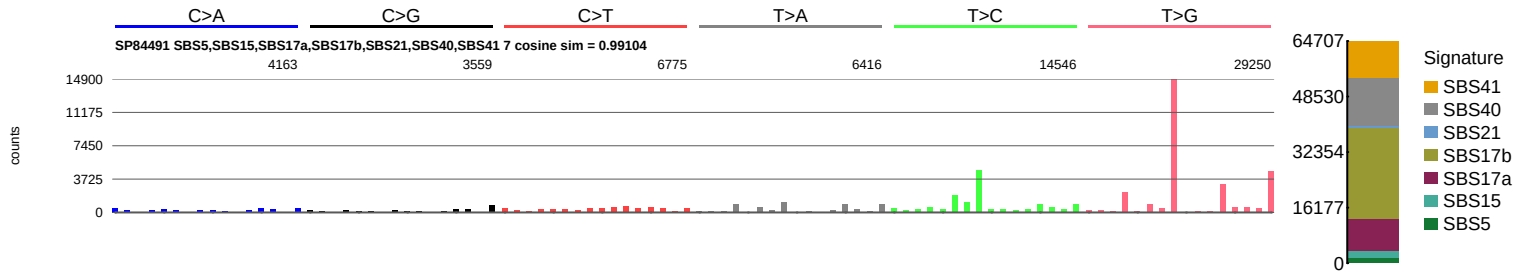
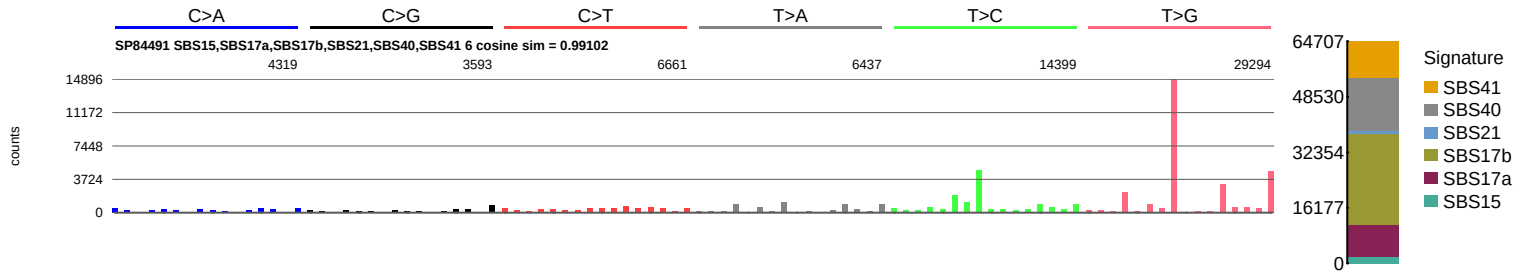
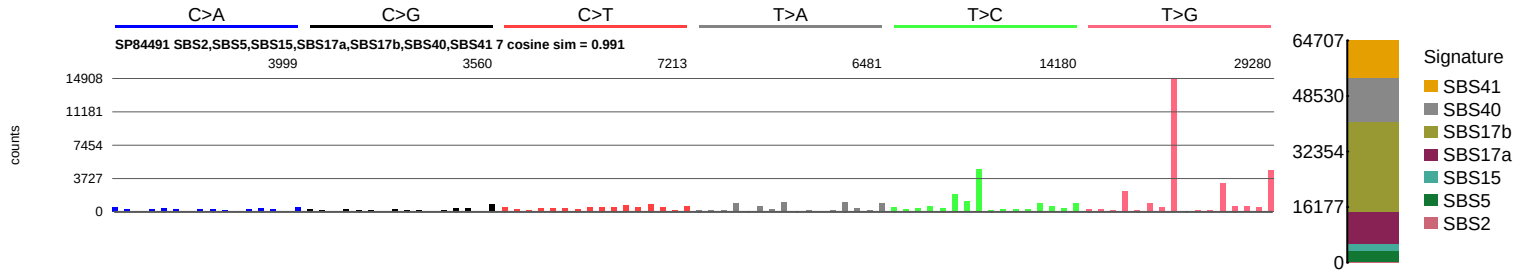
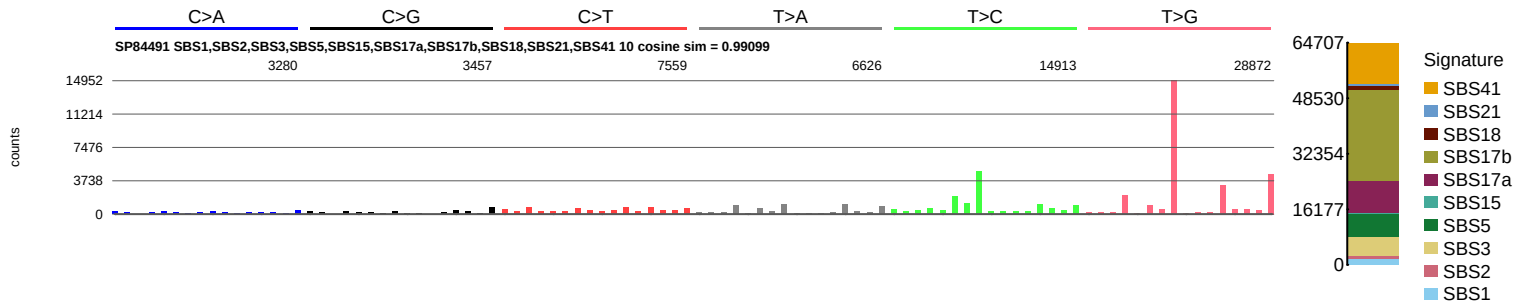
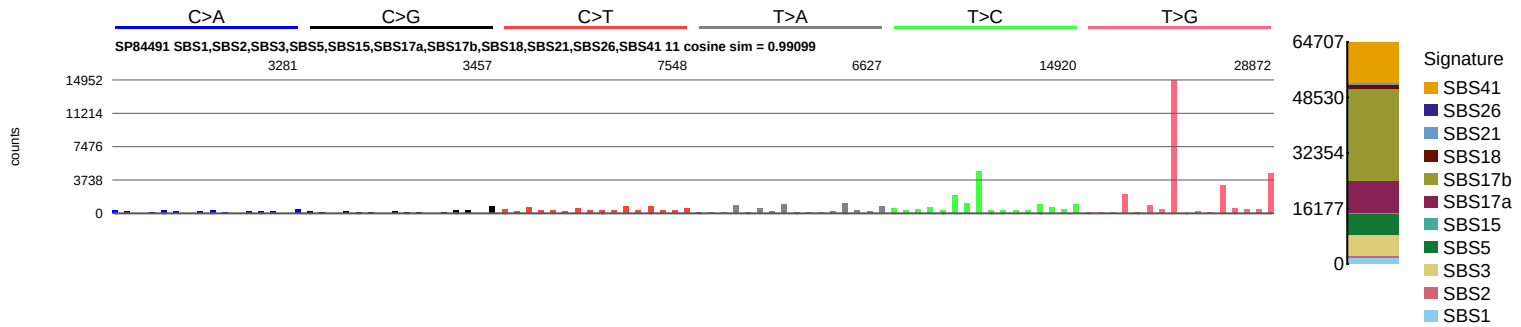
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.99106
4196
3521
6590
6346
14762
29295
counts
14896
11172
7448
3724
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15

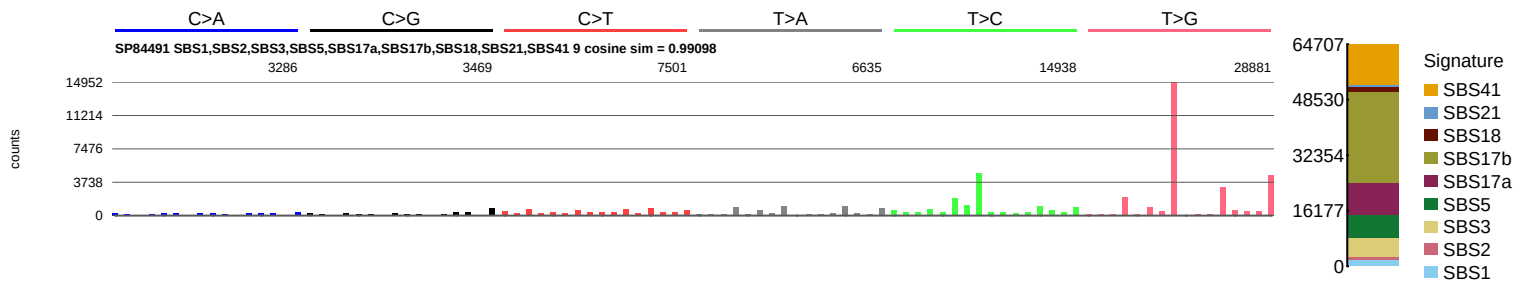

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99104
4163
3559
6775
6416
14546
29250
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5

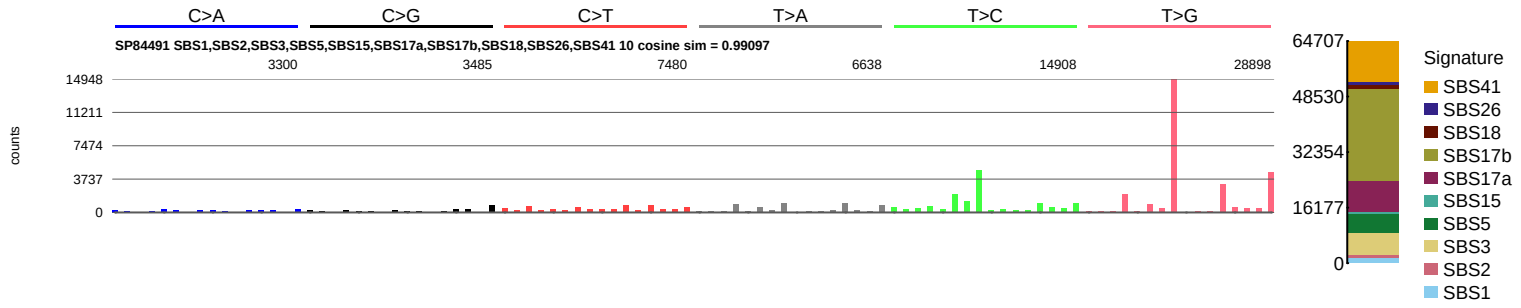

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.99102
4319
3593
6661
6437
14399
29294
counts
14896
11172
7448
3724
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15

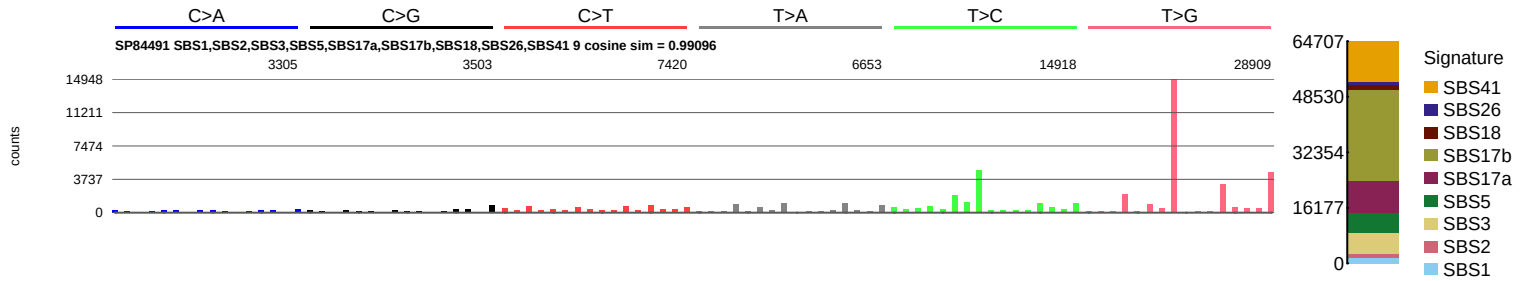

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.991
3999
3560
7213
6481
14180
29280
counts
14908
11181
7454
3727
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99099
3280
3457
7559
6626
14913
28872
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 11 cosine sim = 0.99099
3281
3457
7548
6627
14920
28872
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99098
3286
3469
7501
6635
14938
28881
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.99097
3300
3485
7480
6638
14908
28898
counts
14948
11211
7474
3737
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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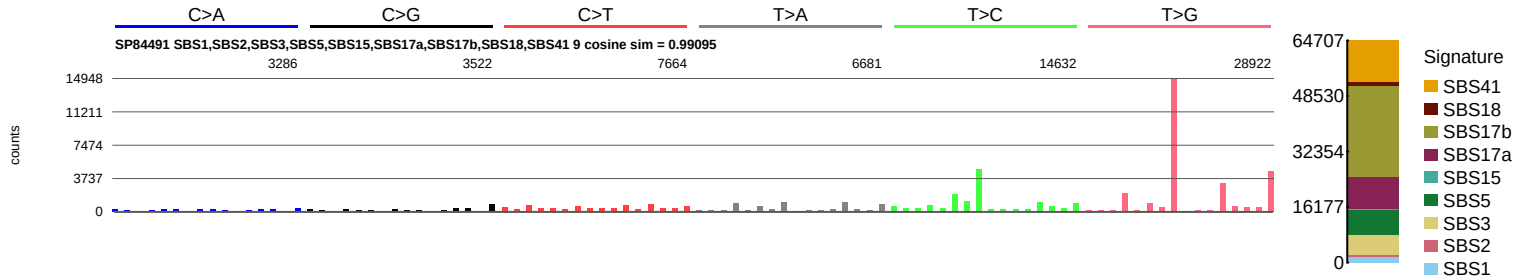
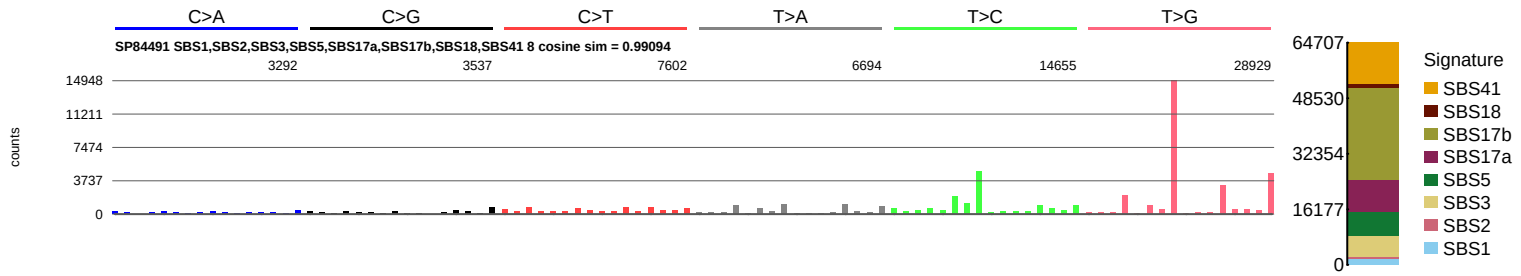
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99096
3305
3503
7420
6653
14918
28909
counts
14948
11211
7474
3737
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99095
3286
3522
7664
6681
14632
28922
counts
14948
11211
7474
3737
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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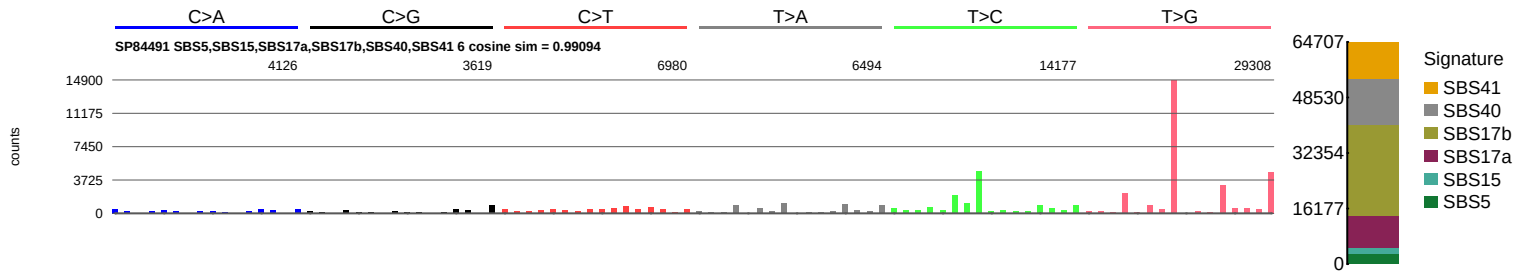
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99094
3292
3537
7602
6694
14655
28929
counts
14948
11211
7474
3737
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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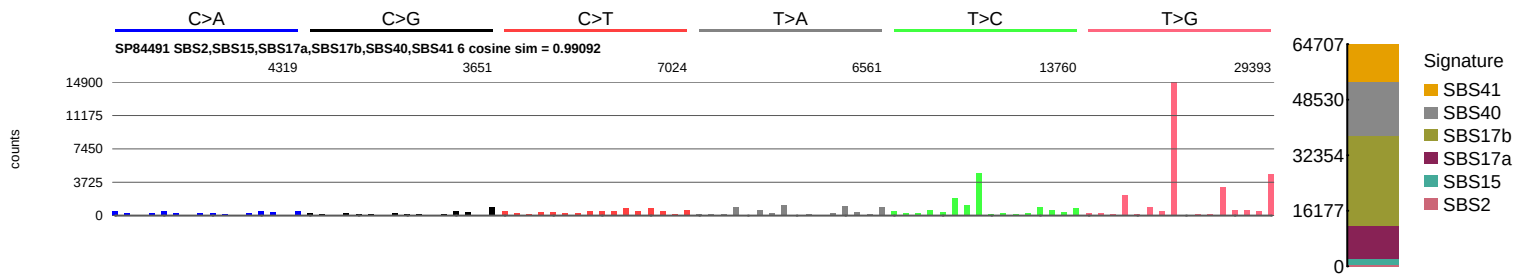
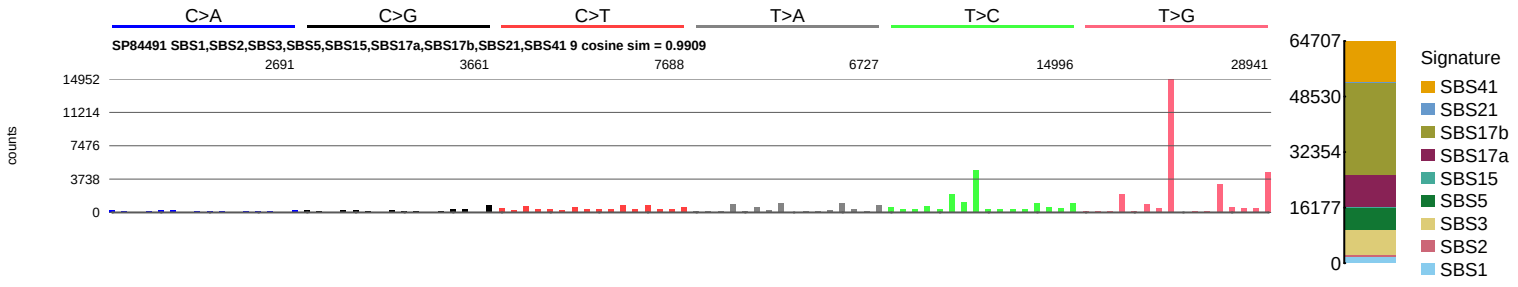
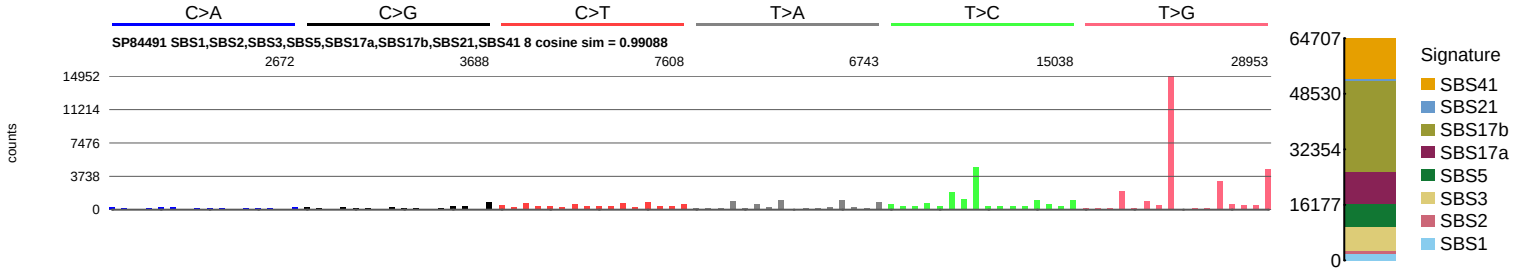
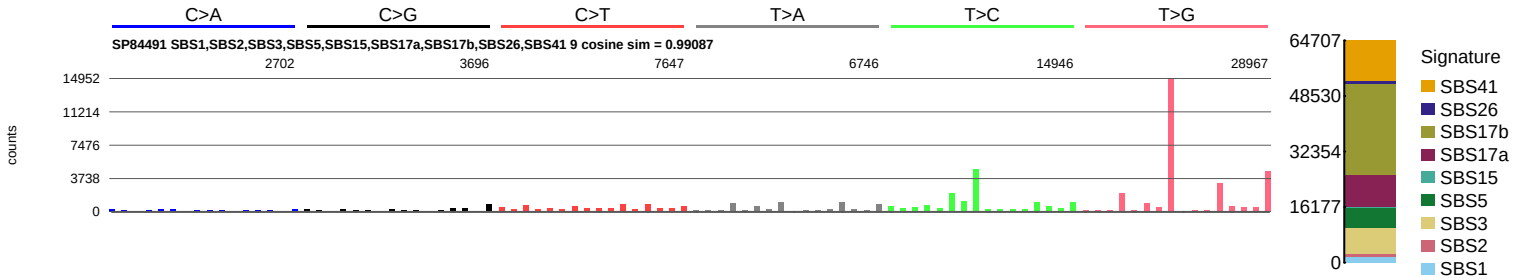
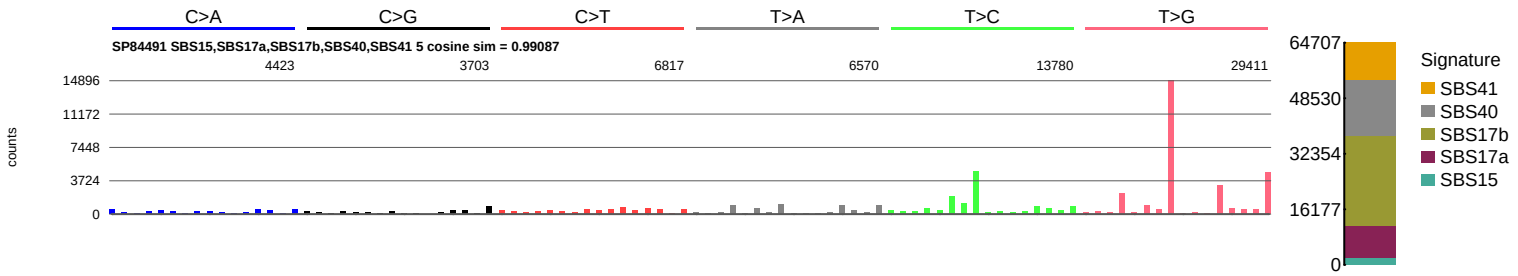
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99094
4126
3619
6980
6494
14177
29308
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5

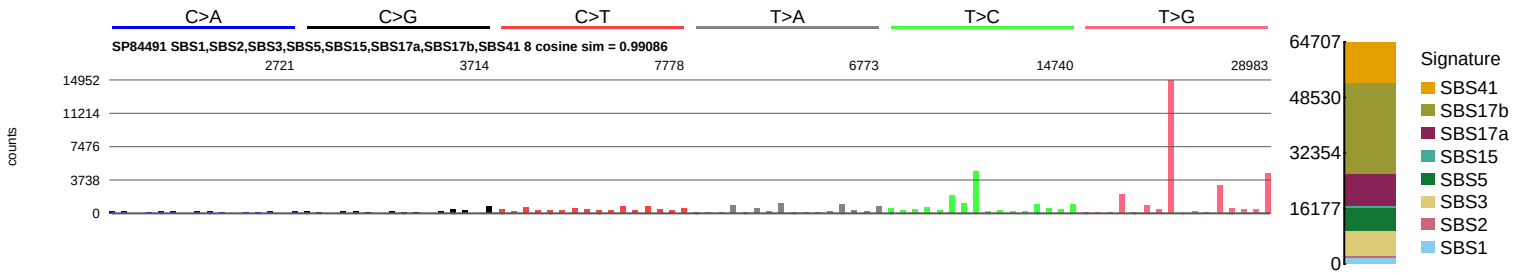

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99092
4319
3651
7024
6561
13760
29393
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2

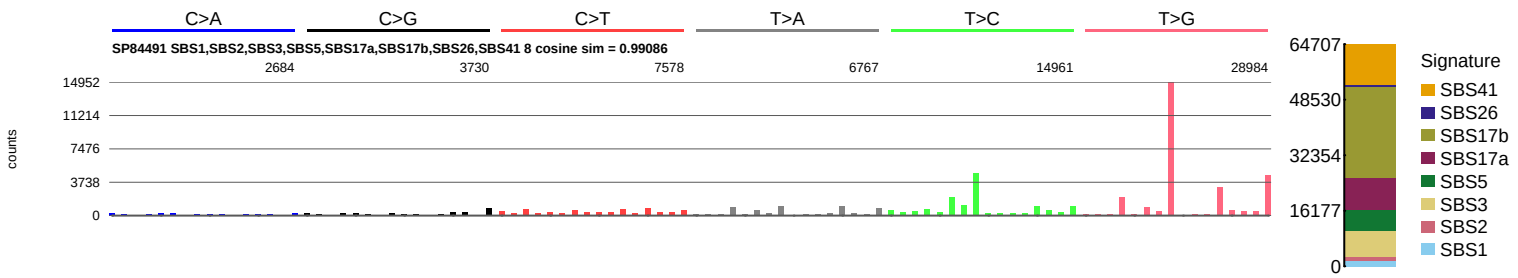

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.9909
2691
3661
7688
6727
14996
28941
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99088
2672
3688
7608
6743
15038
28953
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS41 9 cosine sim = 0.99087
2702
3696
7647
6746
14946
28967
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.99087
4423
3703
6817
6570
13780
29411
counts
14896
11172
7448
3724
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.99086
2721
3714
7778
6773
14740
28983
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99086
2684
3730
7578
6767
14961
28984
counts
14952
11214
7476
3738
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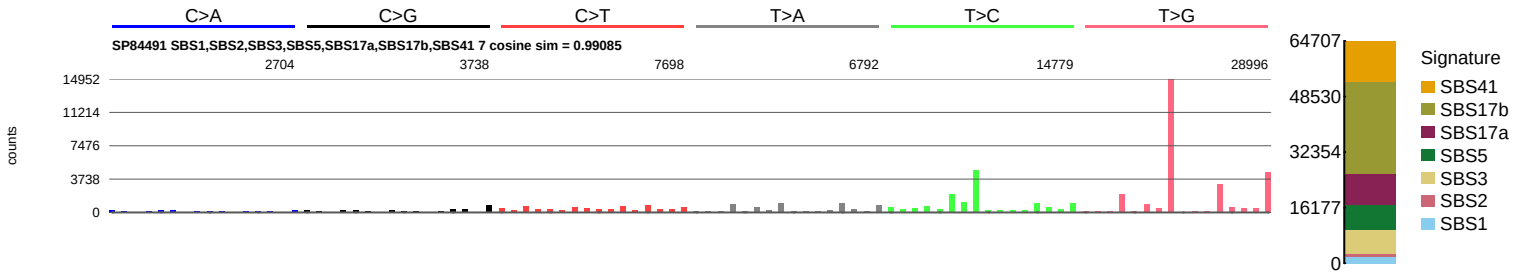
SBS1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SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99085
2704
3738
7698
6792
14779
28996
counts
14952
11214
7476
3738
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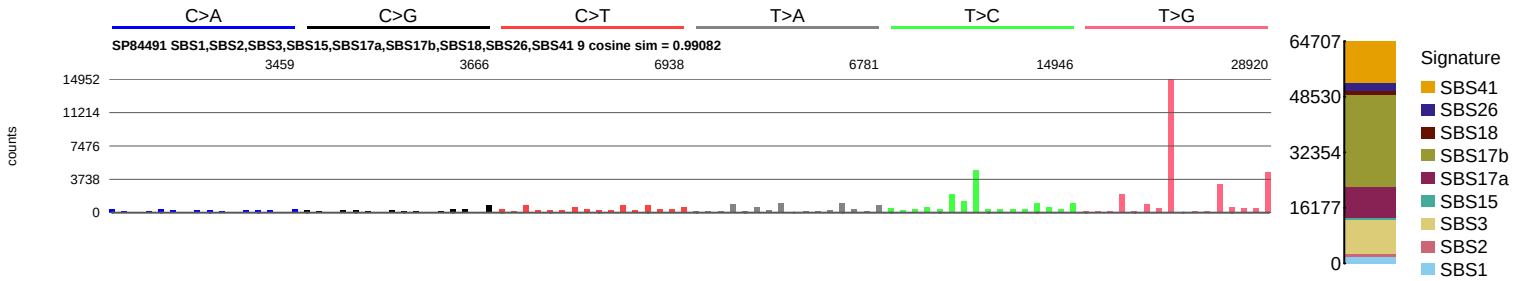
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SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99082
3459
3666
6938
6781
14946
28920
counts
14952
11214
7476
3738
0
64707
48530
32354
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SBS2
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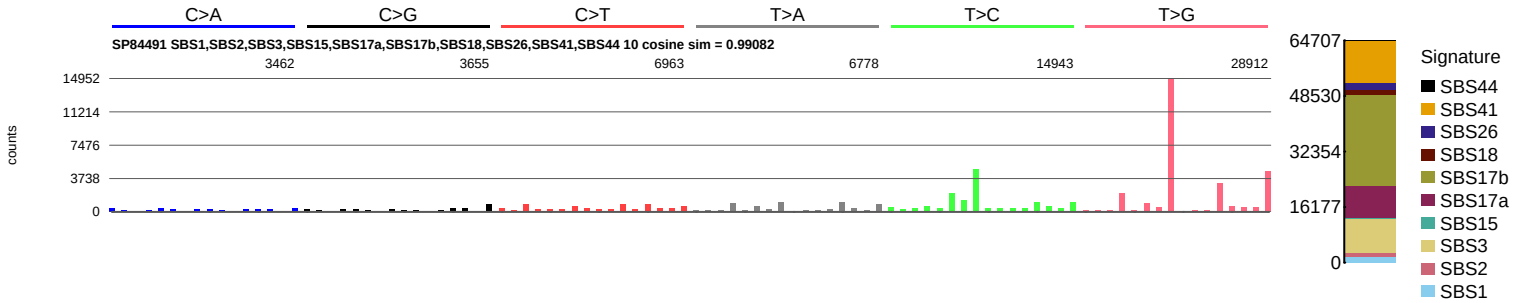
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T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99082
3462
3655
6963
6778
14943
28912
counts
14952
11214
7476
3738
0
64707
48530
32354
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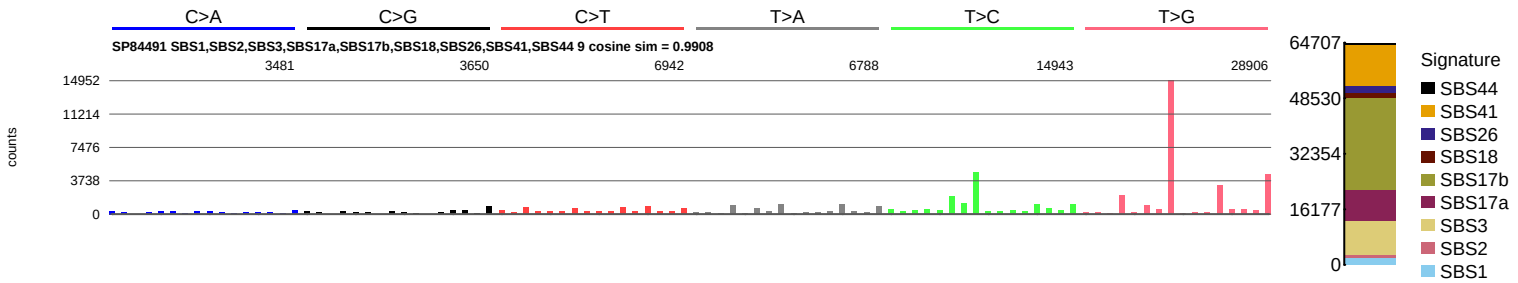
SBS2
SBS1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C>T
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T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.9908
3481
3650
6942
6788
14943
28906
counts
14952
11214
7476
3738
0
64707
48530
32354
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SBS18
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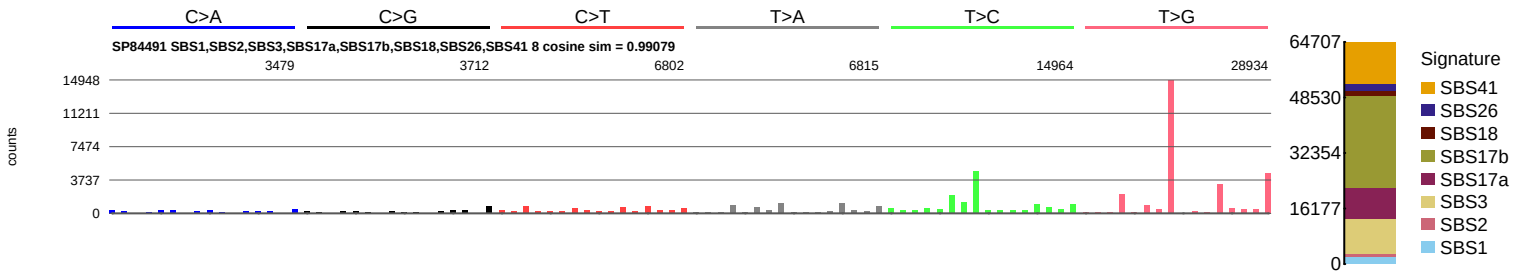
SBS2
SBS1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C>G
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T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99079
3479
3712
6802
6815
14964
28934
counts
14948
11211
7474
3737
0
64707
48530
32354
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SBS41
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SBS17b
SBS17a
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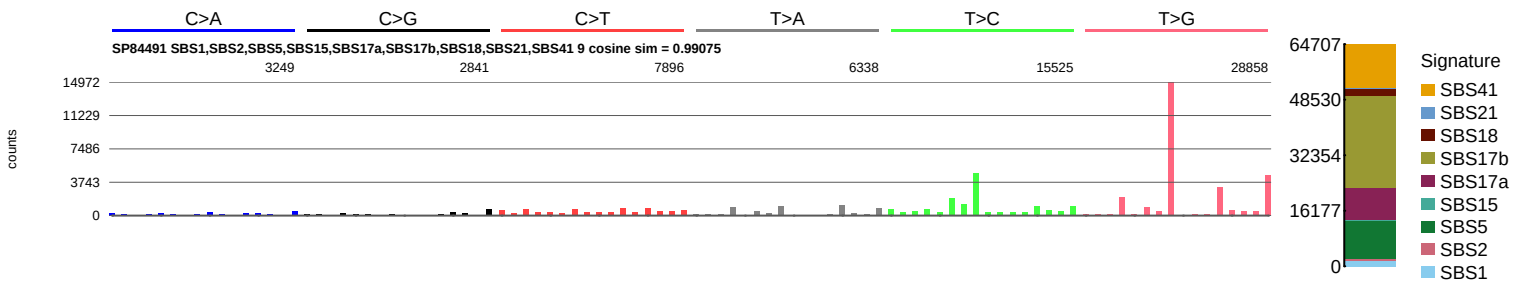
SBS2
SBS1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C>G
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T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99075
3249
2841
7896
6338
15525
28858
counts
14972
11229
7486
3743
0
64707
48530
32354
16177
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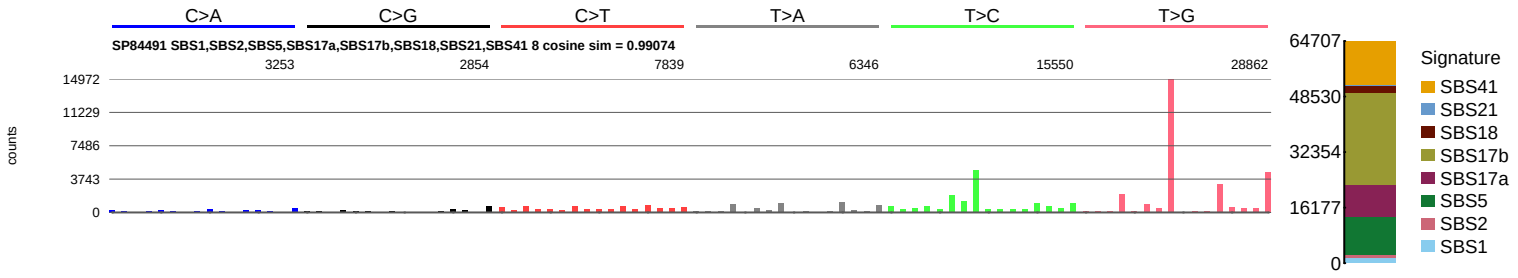
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
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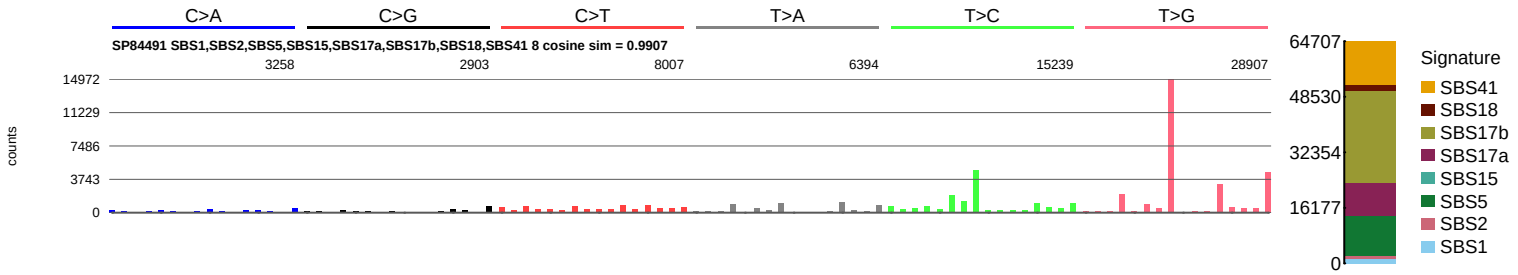
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99074
3253
2854
7839
6346
15550
28862
counts
14972
11229
7486
3743
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9907
3258
2903
8007
6394
15239
28907
counts
14972
11229
7486
3743
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
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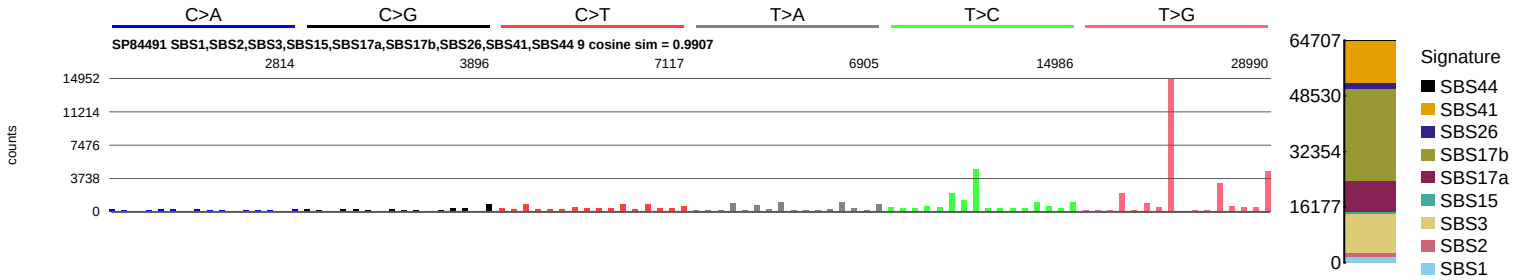
SBS2
SBS1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C>G
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T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.9907
2814
3896
7117
6905
14986
28990
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
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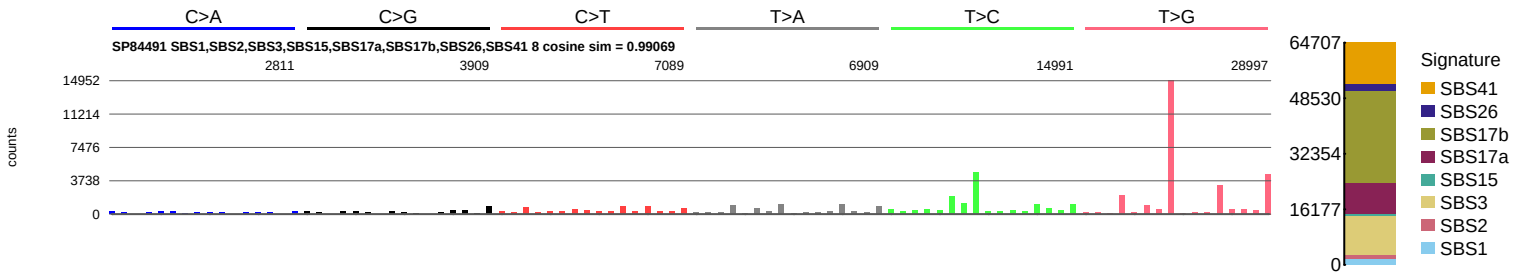
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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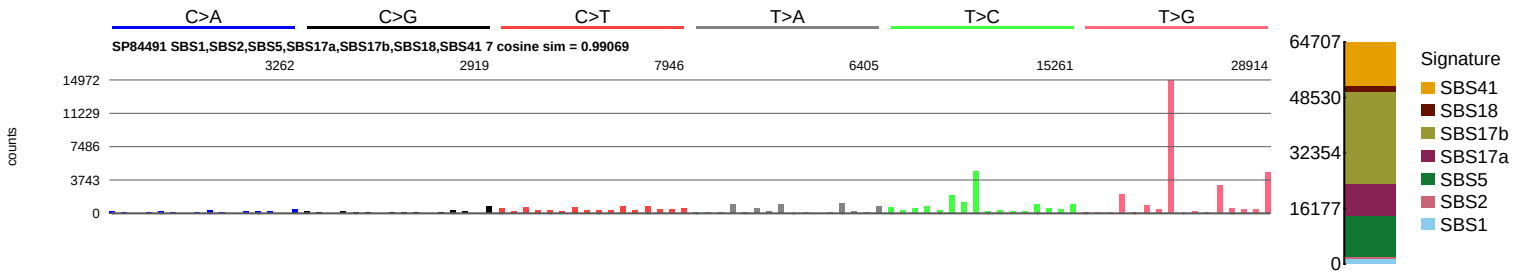
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99069
2811
3909
7089
6909
14991
28997
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99069
3262
2919
7946
6405
15261
28914
counts
14972
11229
7486
3743
0
64707
48530
32354
16177
0
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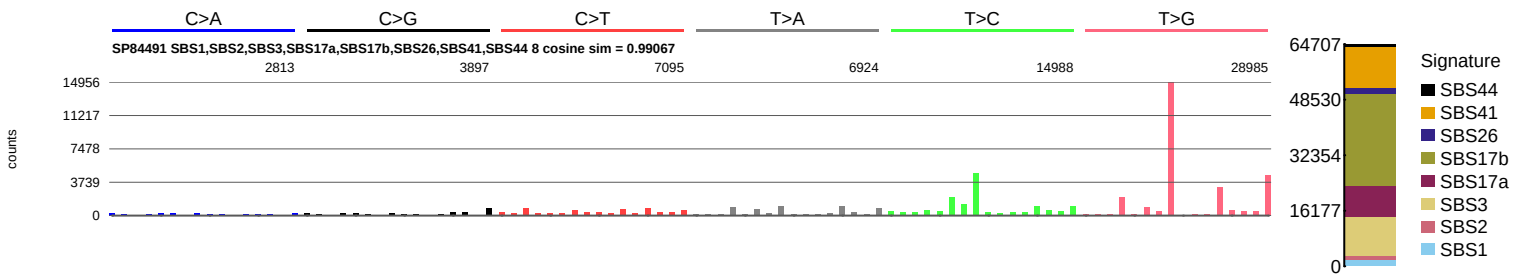
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.99067
2813
3897
7095
6924
14988
28985
counts
14956
11217
7478
3739
0
64707
48530
32354
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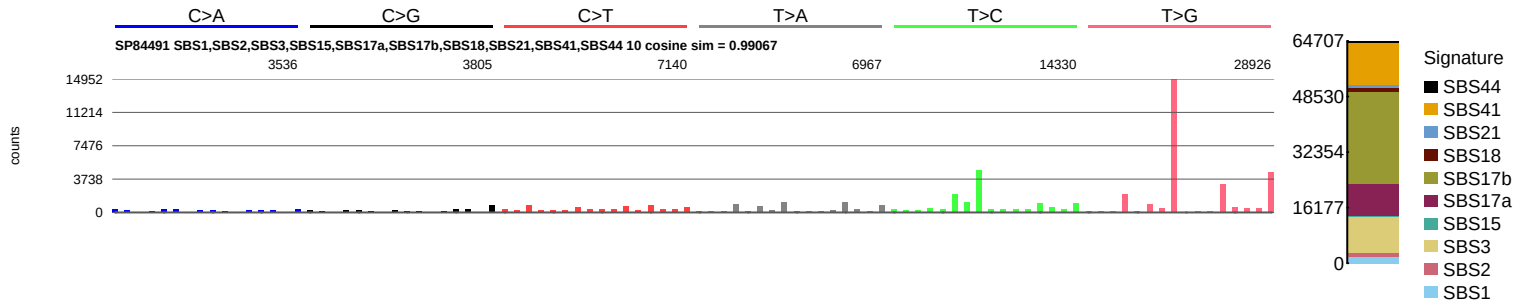
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99067
3536
3805
7140
6967
14330
28926
counts
14952
11214
7476
3738
0
64707
48530
32354
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SBS18
SBS17b
SBS17a
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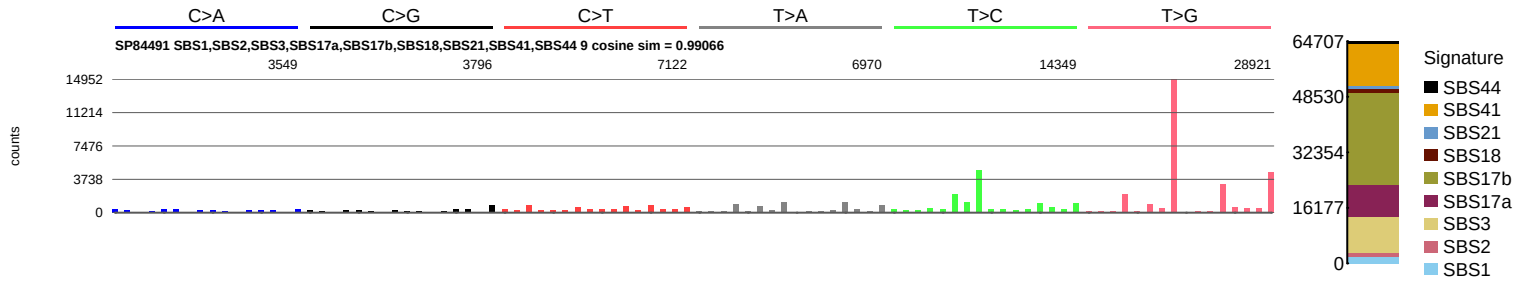
SBS3
SBS2
SBS1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T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99066
3549
3796
7122
6970
14349
28921
counts
14952
11214
7476
3738
0
64707
48530
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SBS18
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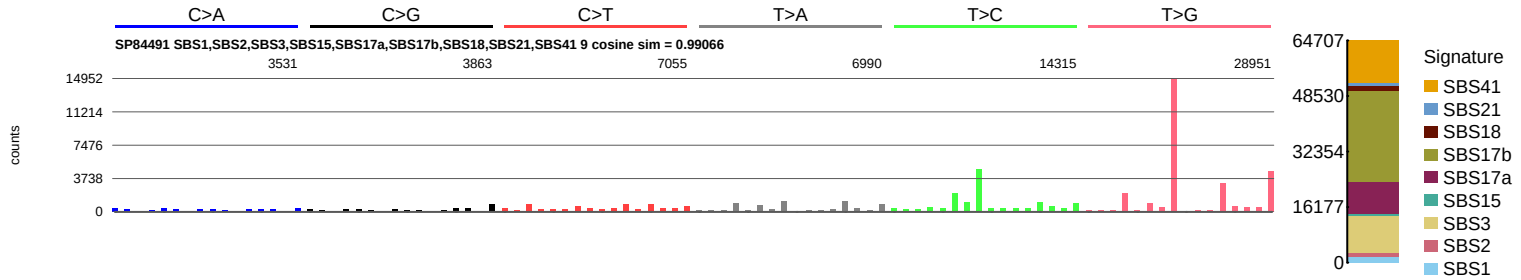
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99066
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3863
7055
6990
14315
28951
counts
14952
11214
7476
3738
0
64707
48530
32354
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SBS17b
SBS17a
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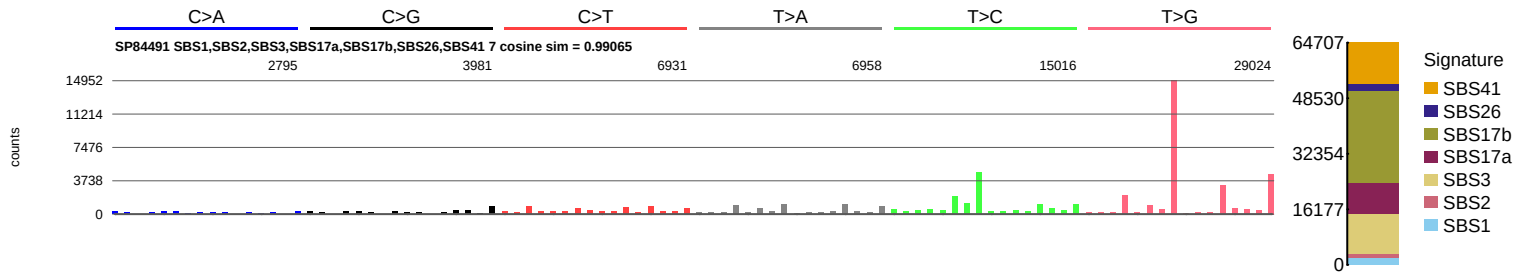
SBS2
SBS1


C>A
C>G
C>T
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T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.99065
2795
3981
6931
6958
15016
29024
counts
14952
11214
7476
3738
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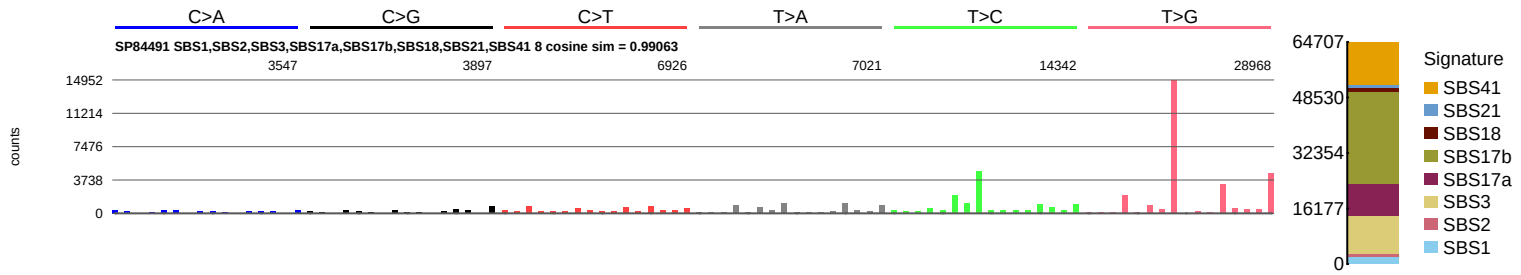
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99063
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3897
6926
7021
14342
28968
counts
14952
11214
7476
3738
0
64707
48530
32354
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SBS18
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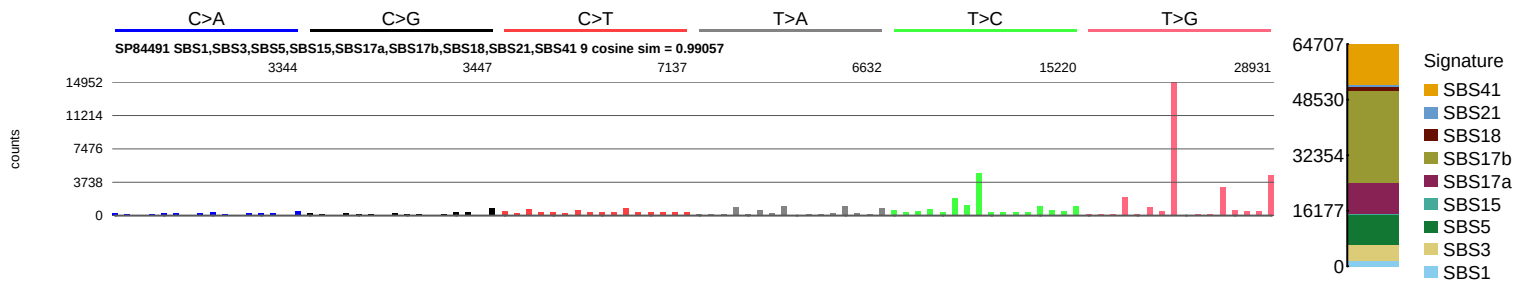
SBS17a
SBS3
SBS2
SBS1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C>G
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T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99057
3344
3447
7137
6632
15220
28931
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
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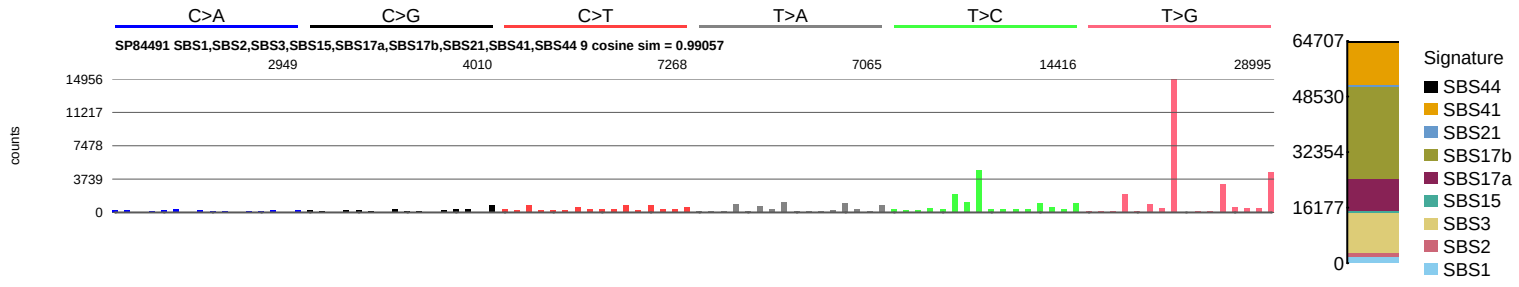
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.99057
2949
4010
7268
7065
14416
28995
counts
14956
11217
7478
3739
0
64707
48530
32354
16177
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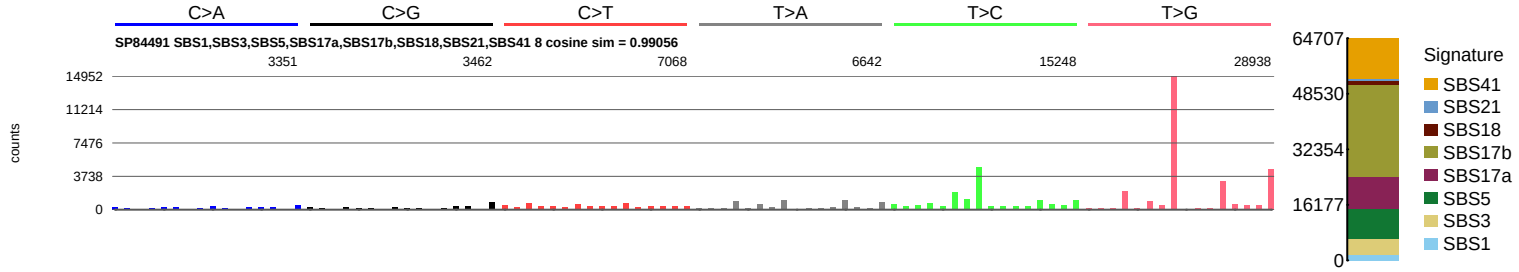
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99056
3351
3462
7068
6642
15248
28938
counts
14952
11214
7476
3738
0
64707
48530
32354
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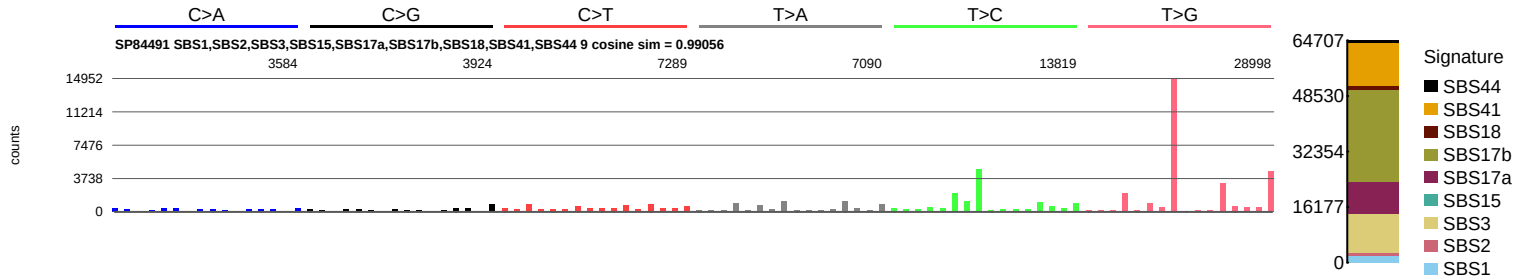
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99056
3584
3924
7289
7090
13819
28998
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
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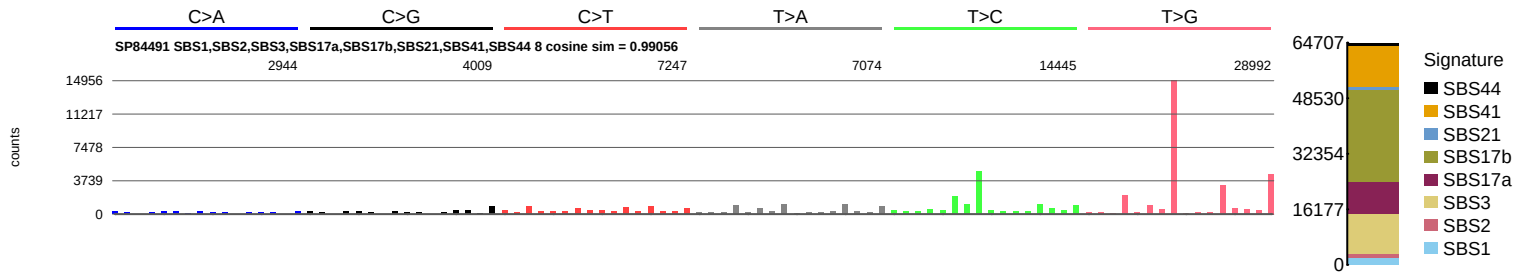
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.99056
2944
4009
7247
7074
14445
28992
counts
14956
11217
7478
3739
0
64707
48530
32354
16177
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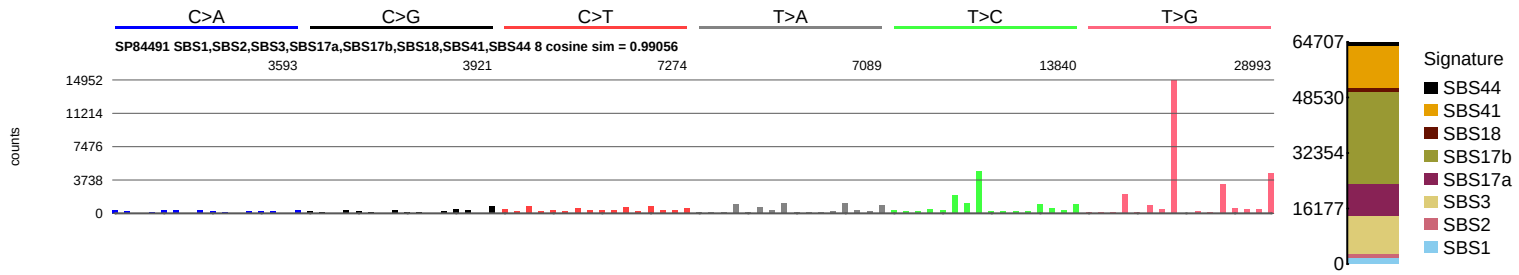
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99056
3593
3921
7274
7089
13840
28993
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
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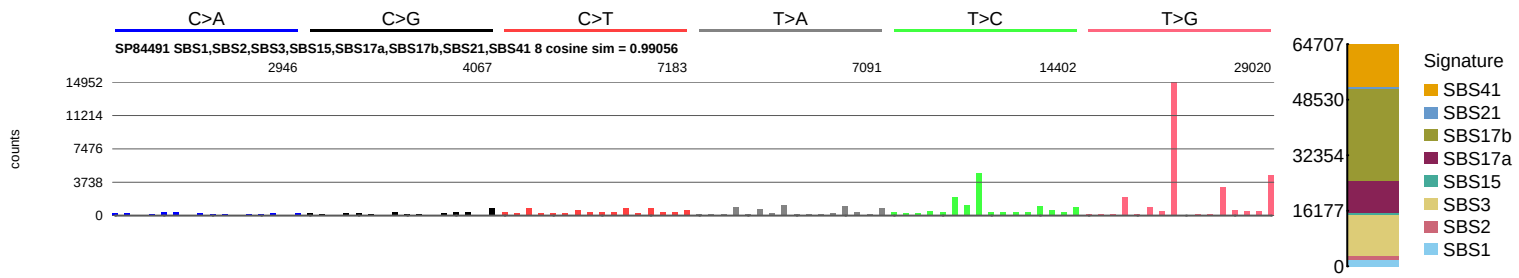
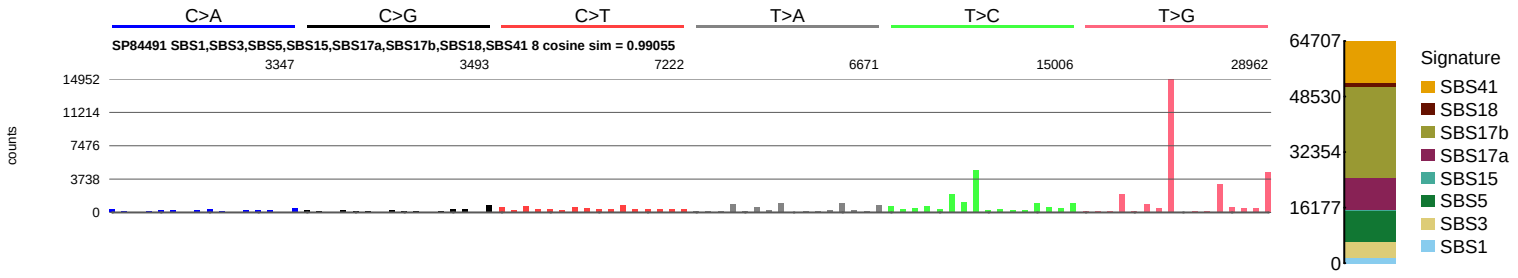
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99056
2946
4067
7183
7091
14402
29020
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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SP84491 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99055
3347
3493
7222
6671
15006
28962
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
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SBS41
SBS18
SBS17b
SBS17a
SBS15
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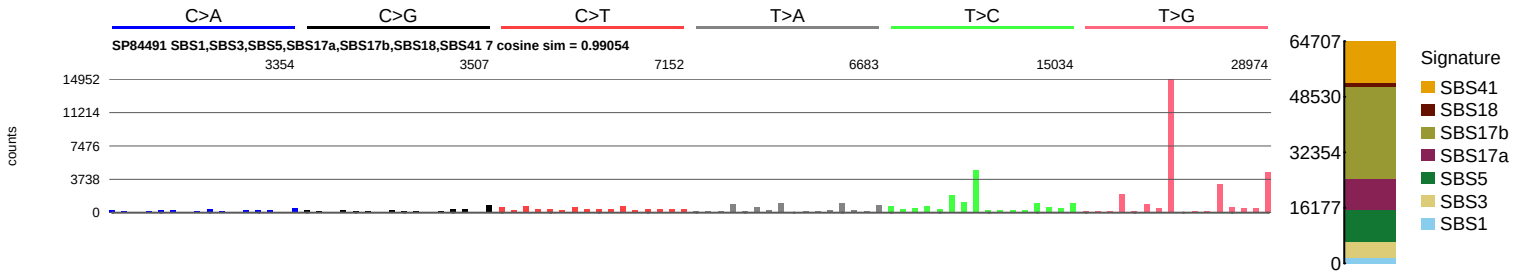
SBS3
SBS1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SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99054
3354
3507
7152
6683
15034
28974
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
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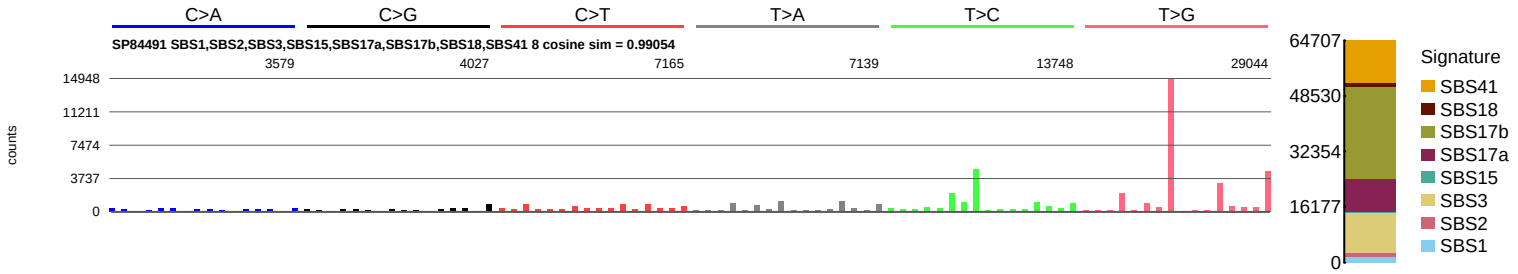
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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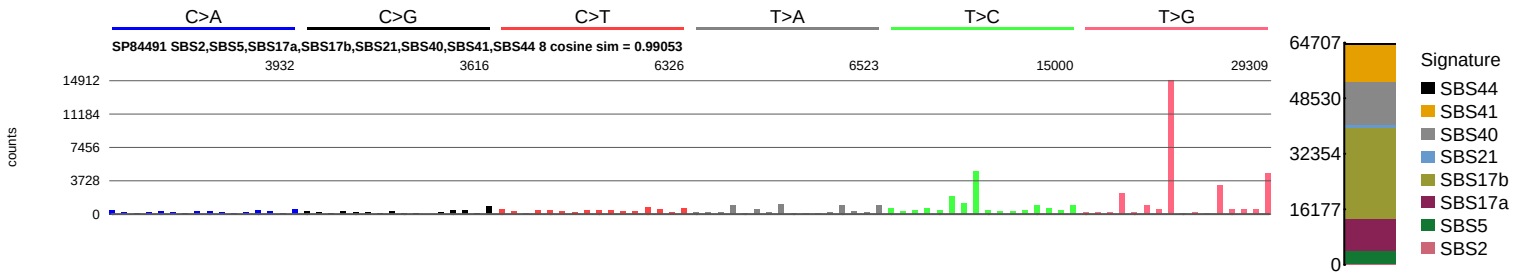
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99054
3579
4027
7165
7139
13748
29044
counts
14948
11211
7474
3737
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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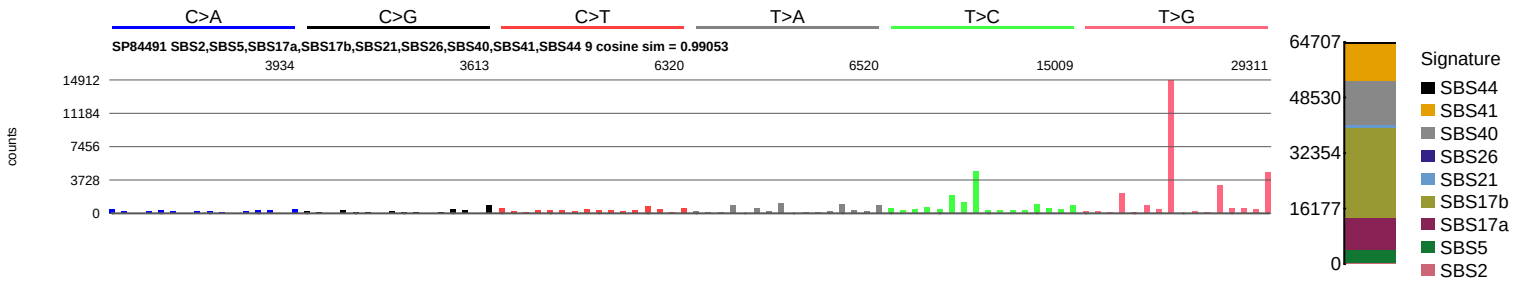
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.99053
3932
3616
6326
6523
15000
29309
counts
14912
11184
7456
3728
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2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C>G
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T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99053
3934
3613
6320
6520
15009
29311
counts
14912
11184
7456
3728
0
64707
48530
32354
16177
0
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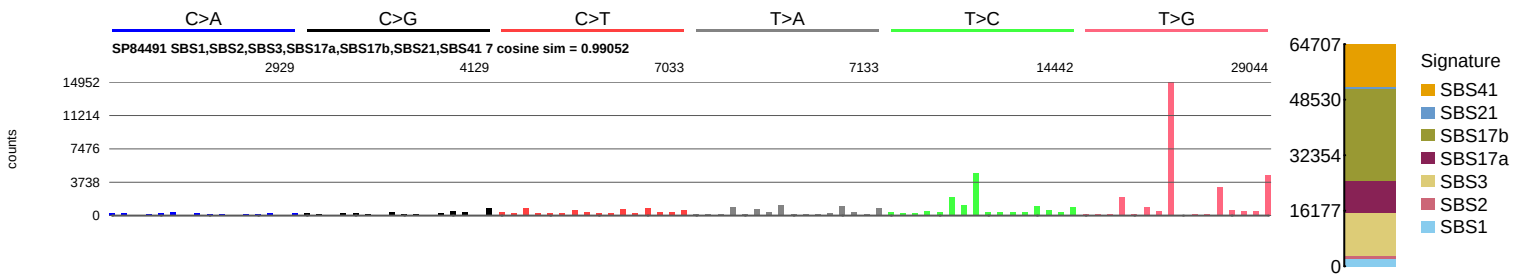
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS5
SBS2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T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99052
2929
4129
7033
7133
14442
29044
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
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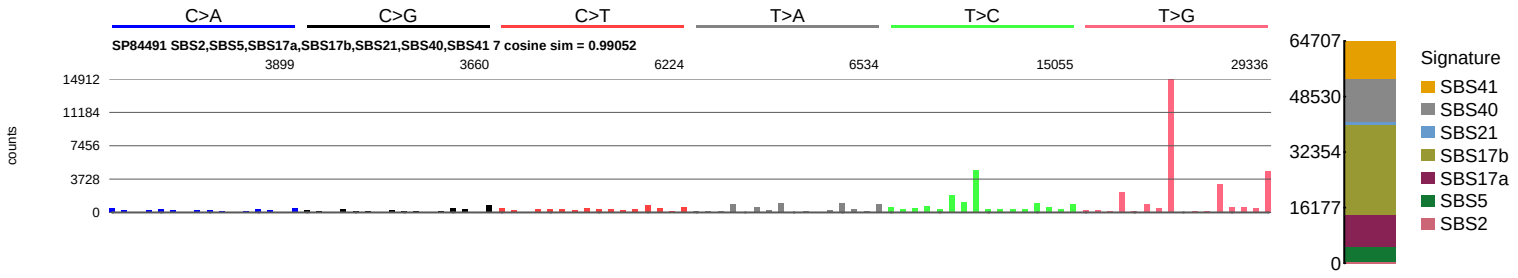
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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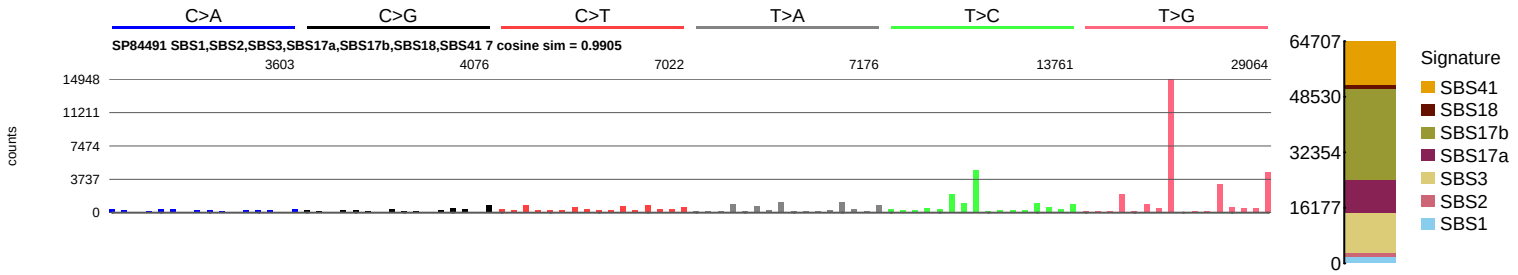
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99052
3899
3660
6224
6534
15055
29336
counts
14912
11184
7456
3728
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2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C>G
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T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9905
3603
4076
7022
7176
13761
29064
counts
14948
11211
7474
3737
0
64707
48530
32354
16177
0
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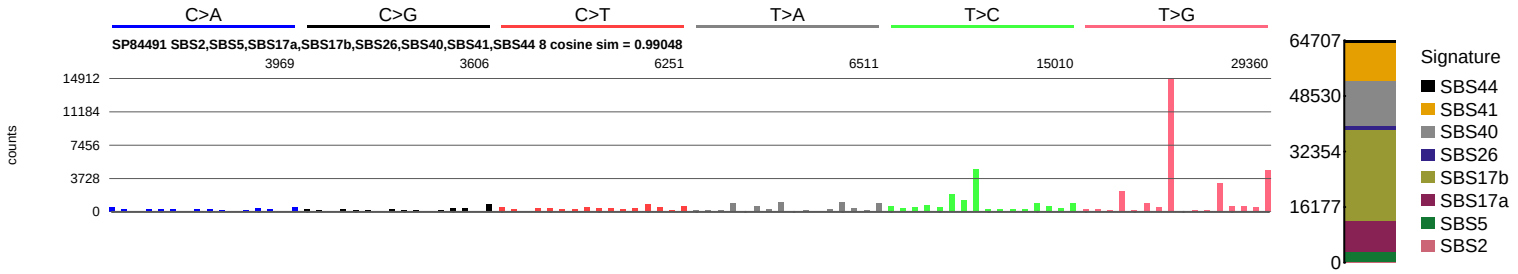
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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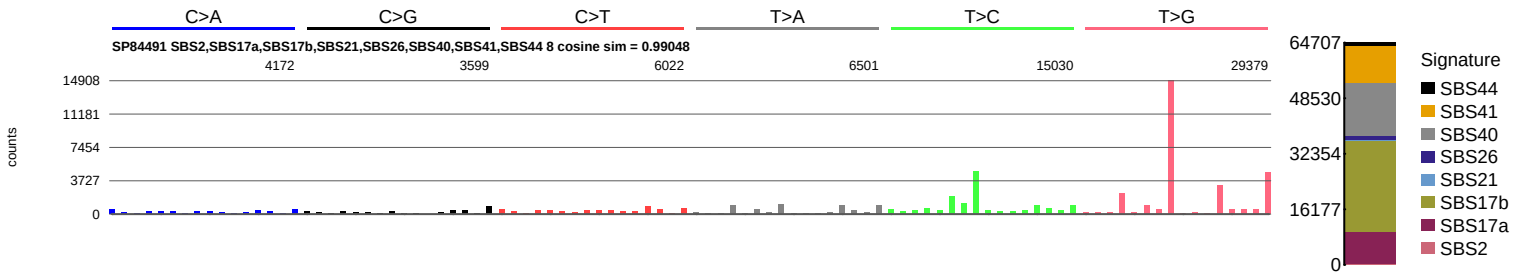
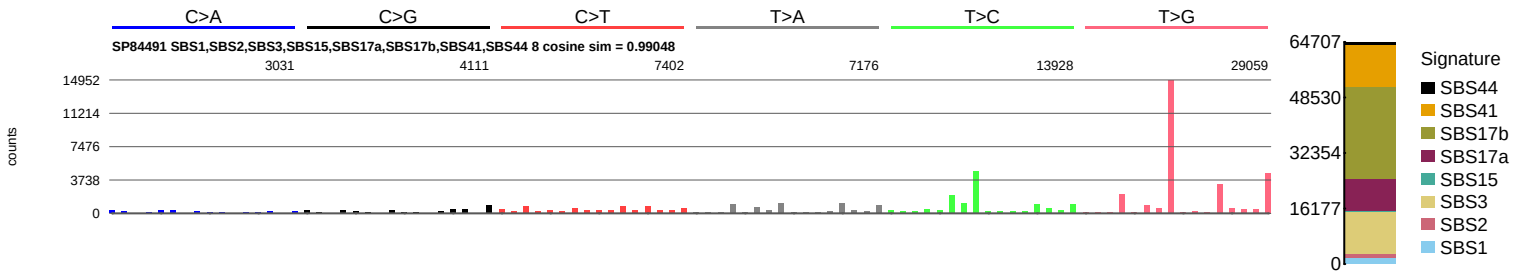
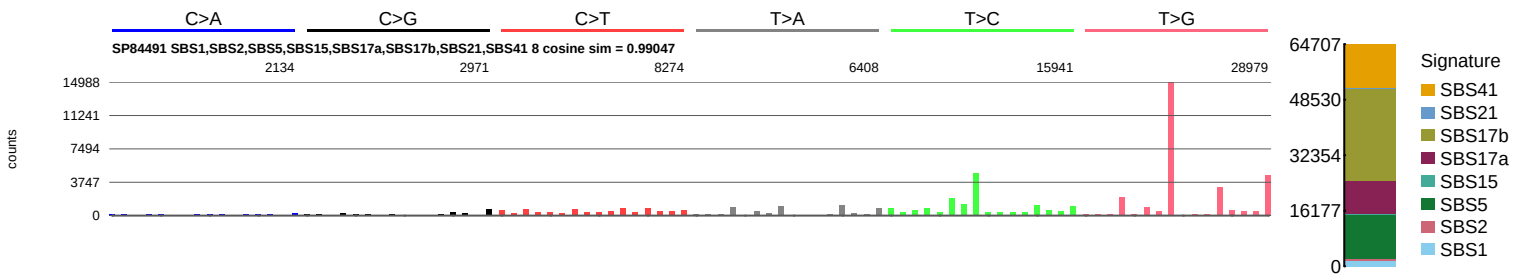
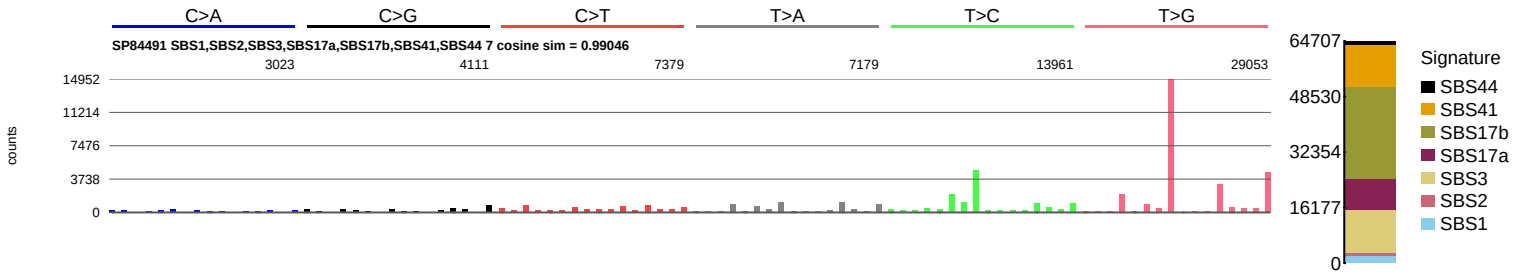
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99048
3969
3606
6251
6511
15010
29360
counts
14912
11184
7456
3728
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99048
4172
3599
6022
6501
15030
29379
counts
14908
11181
7454
3727
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.99048
3031
4111
7402
7176
13928
29059
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99047
2134
2971
8274
6408
15941
28979
counts
14988
11241
7494
3747
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.99046
3023
4111
7379
7179
13961
29053
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
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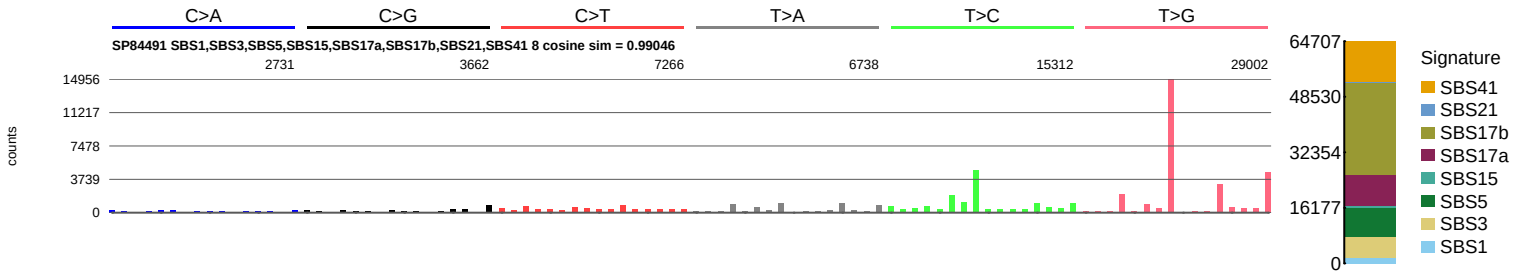
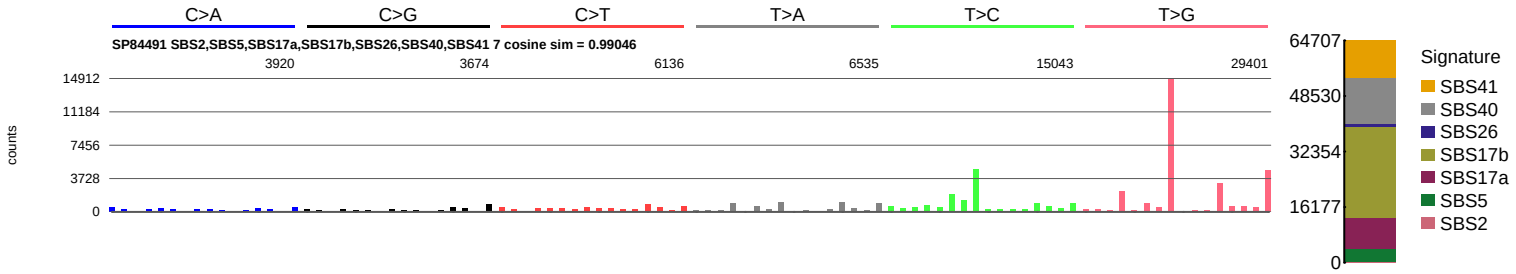
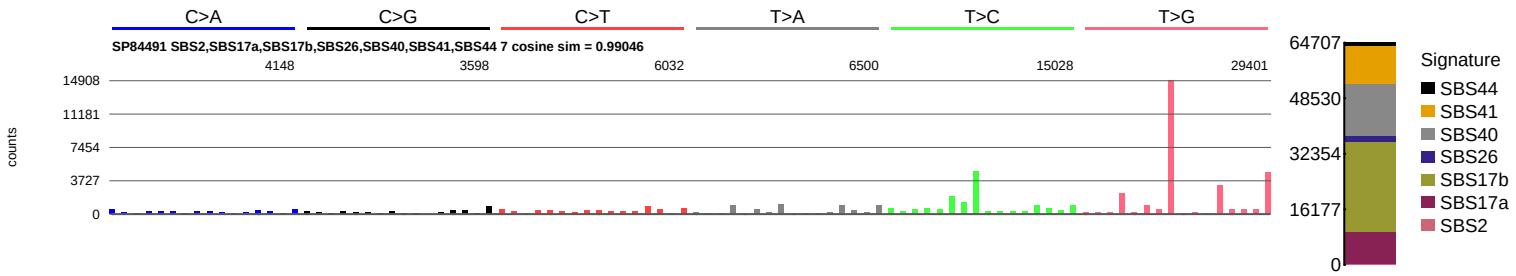
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99046
2731
3662
7266
6738
15312
29002
counts
14956
11217
7478
3739
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99046
3920
3674
6136
6535
15043
29401
counts
14912
11184
7456
3728
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.99046
4148
3598
6032
6500
15028
29401
counts
14908
11181
7454
3727
0
64707
48530
32354
16177
0
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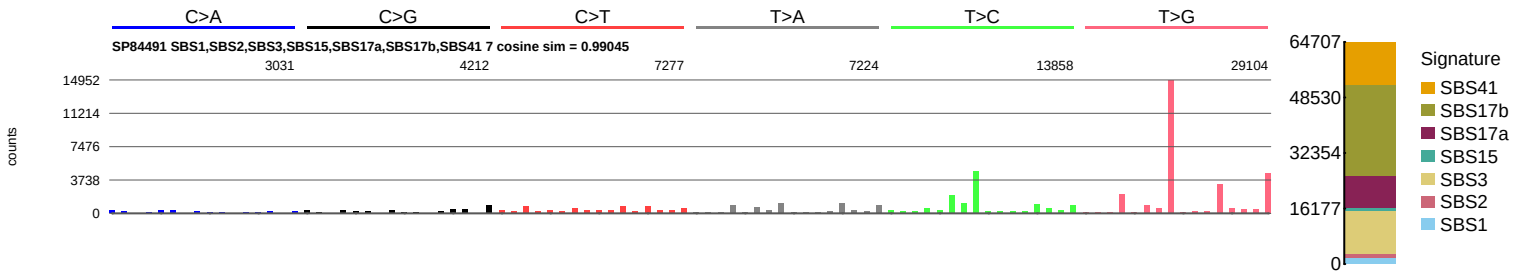
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99045
3031
4212
7277
7224
13858
29104
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
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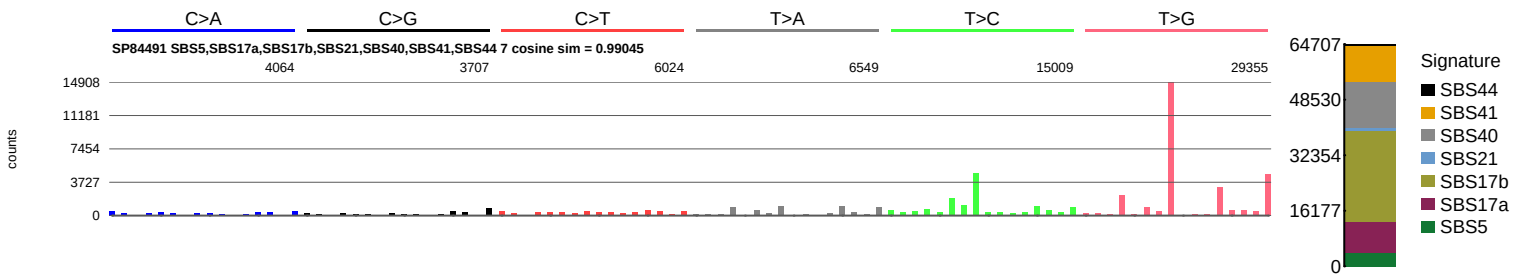
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.99045
4064
3707
6024
6549
15009
29355
counts
14908
11181
7454
3727
0
64707
48530
32354
16177
0
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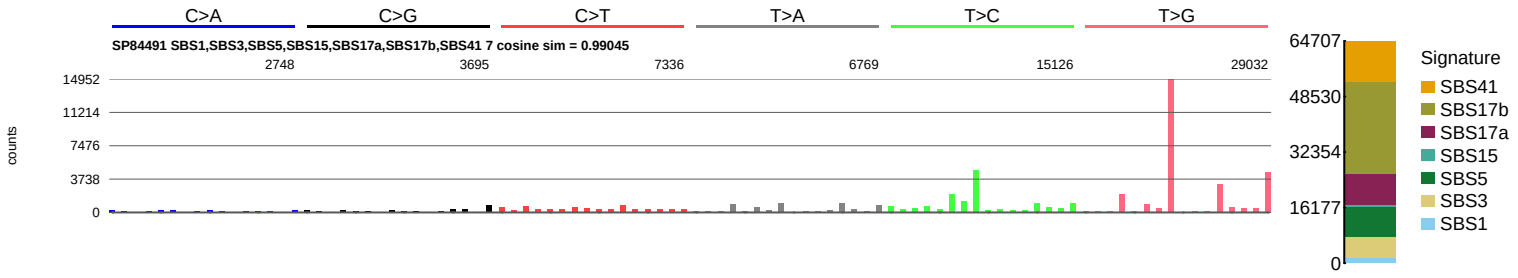
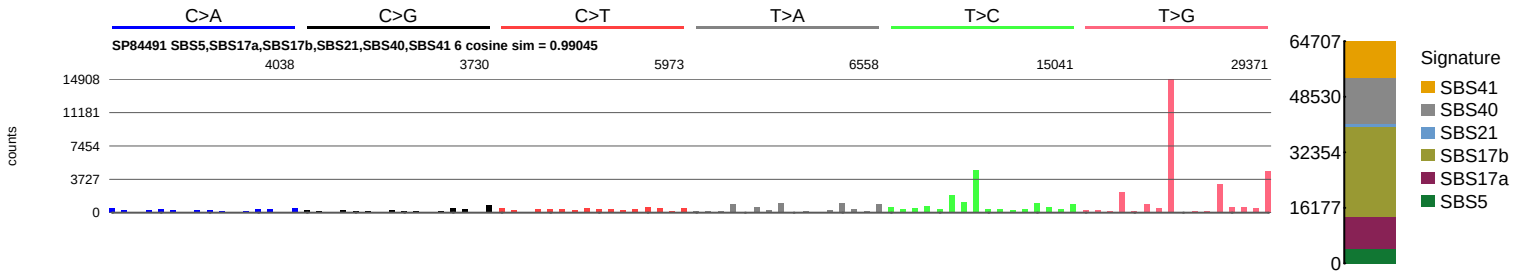
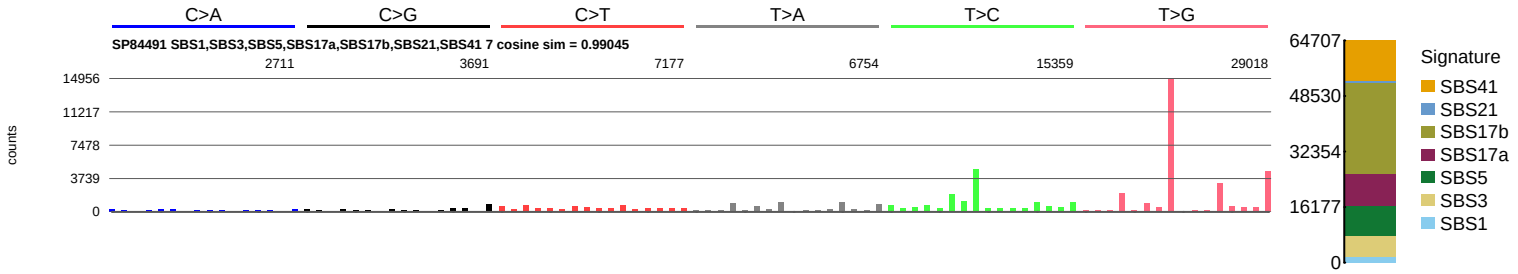
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99045
2748
3695
7336
6769
15126
29032
14952
11214
7476
3738
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.99045
4038
3730
5973
6558
15041
29371
14908
11181
7454
3727
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99045
2711
3691
7177
6754
15359
29018
14956
11217
7478
3739
0
counts
64707
48530
32354
16177
0
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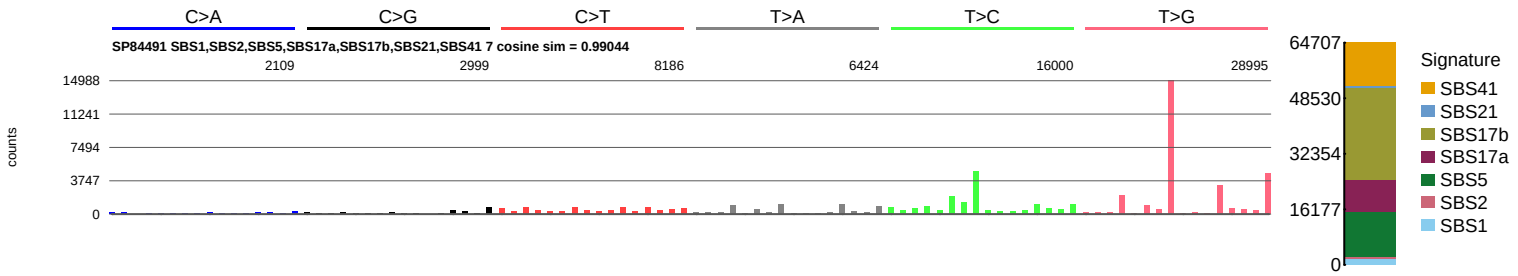
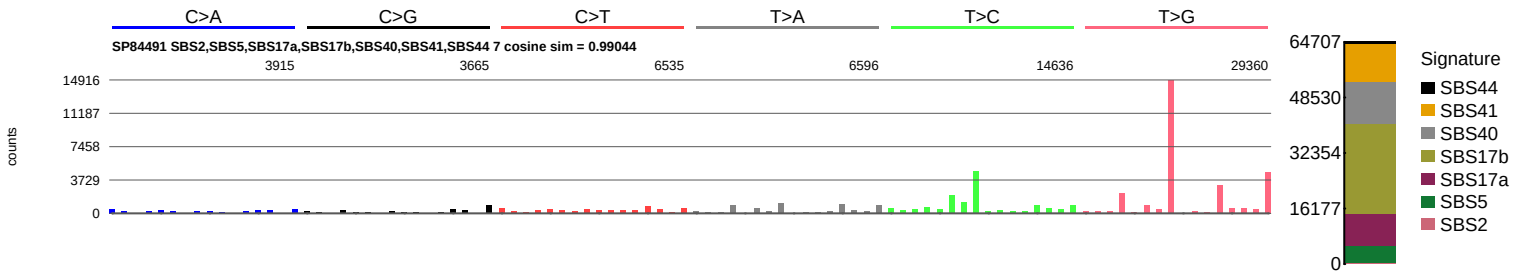
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99044
2109
2999
8186
6424
16000
28995
14988
11241
7494
3747
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99044
3915
3665
6535
6596
14636
29360
14916
11187
7458
3729
0
counts
64707
48530
32354
16177
0
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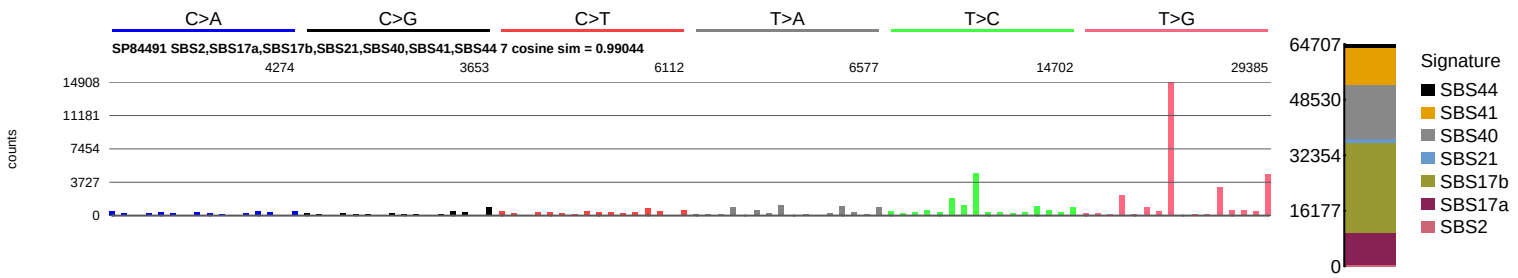
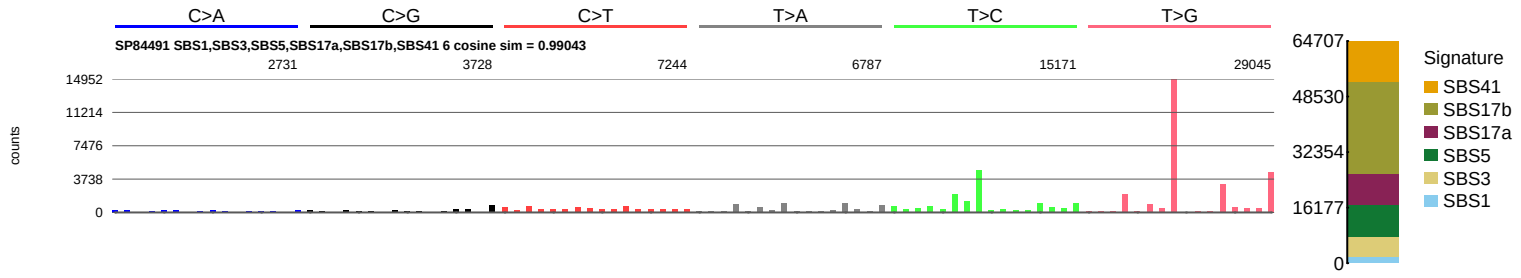
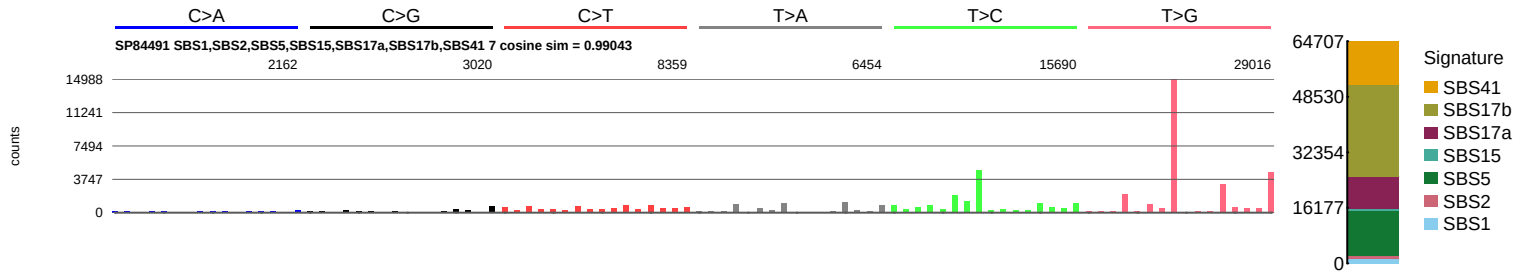
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.99044
4274
3653
6112
6577
14702
29385
14908
11181
7454
3727
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99043
2731
3728
7244
6787
15171
29045
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
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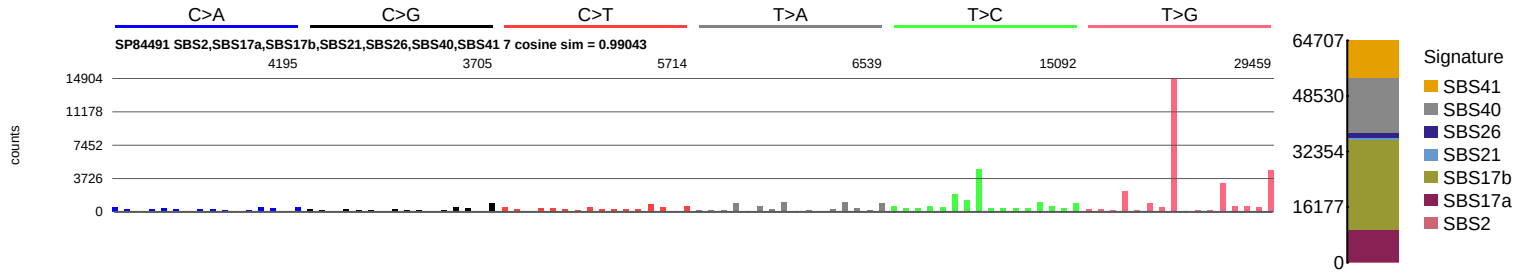
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99043
2162
3020
8359
6454
15690
29016
counts
14988
11241
7494
3747
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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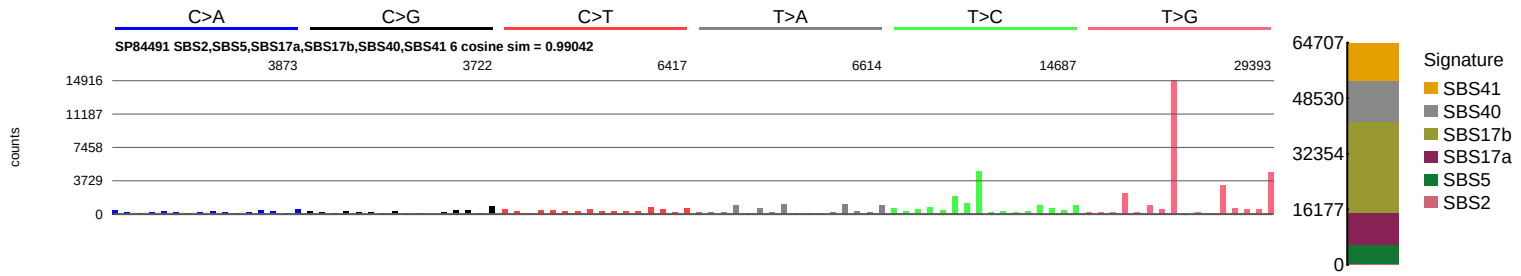
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.99043
4195
3705
5714
6539
15092
29459
counts
14904
11178
7452
3726
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2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C>G
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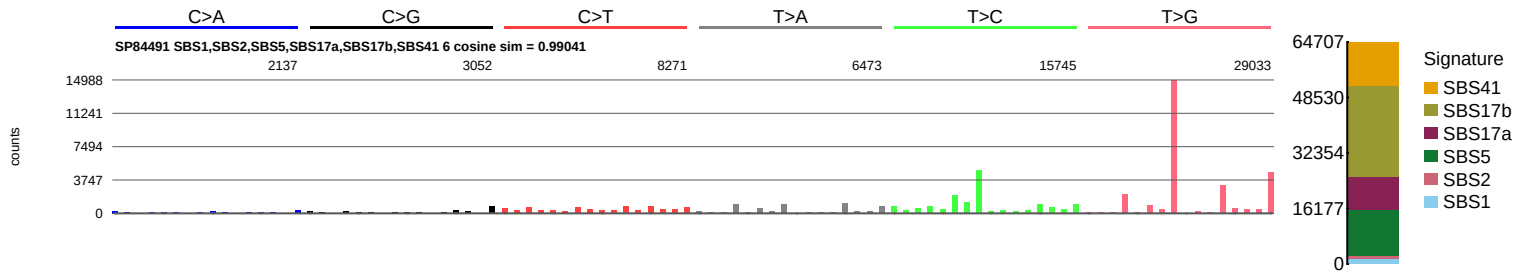
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99042
3873
3722
6417
6614
14687
29393
counts
14916
11187
7458
3729
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2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C>G
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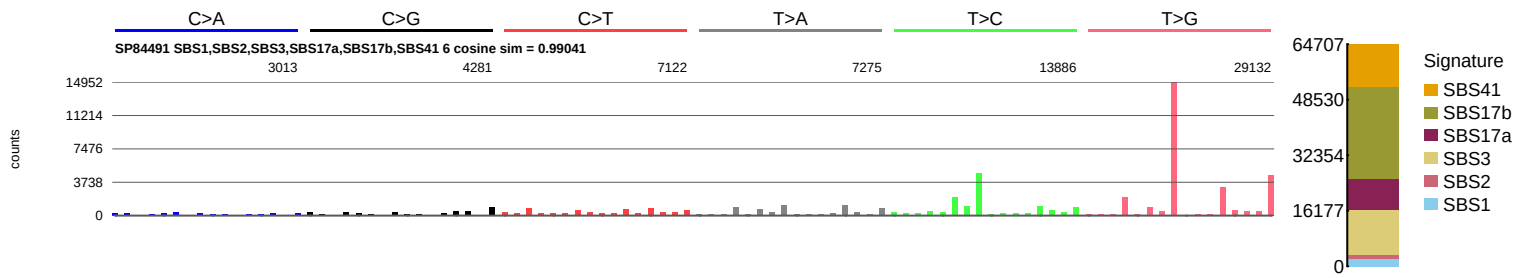
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99041
2137
3052
8271
6473
15745
29033
counts
14988
11241
7494
3747
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS2
SBS1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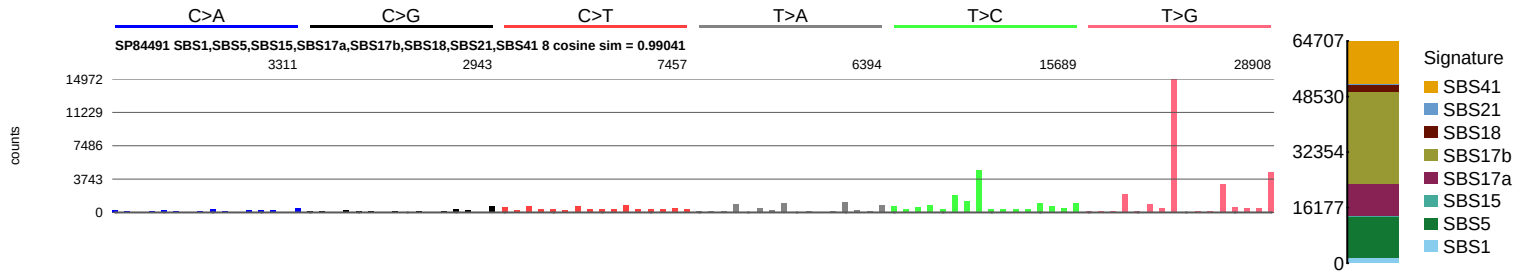
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99041
3013
4281
7122
7275
13886
29132
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1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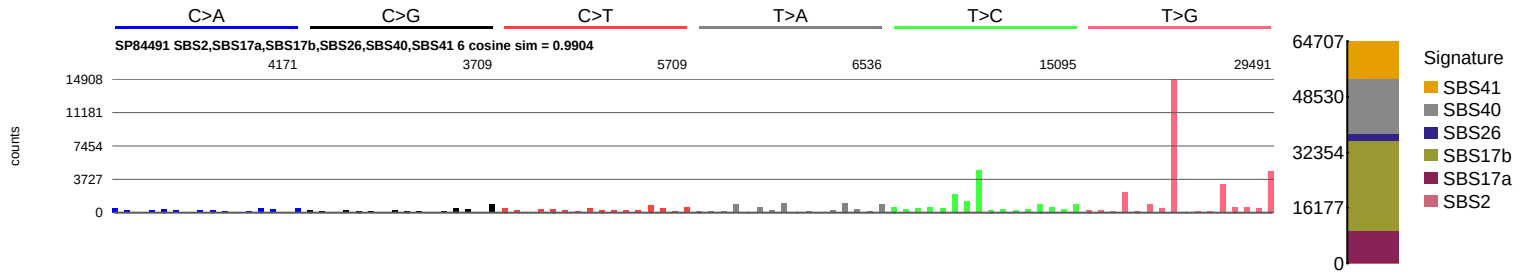
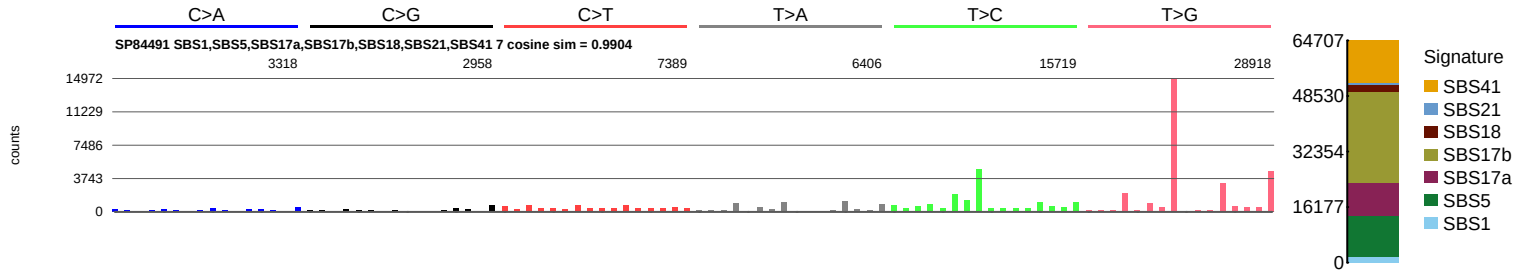
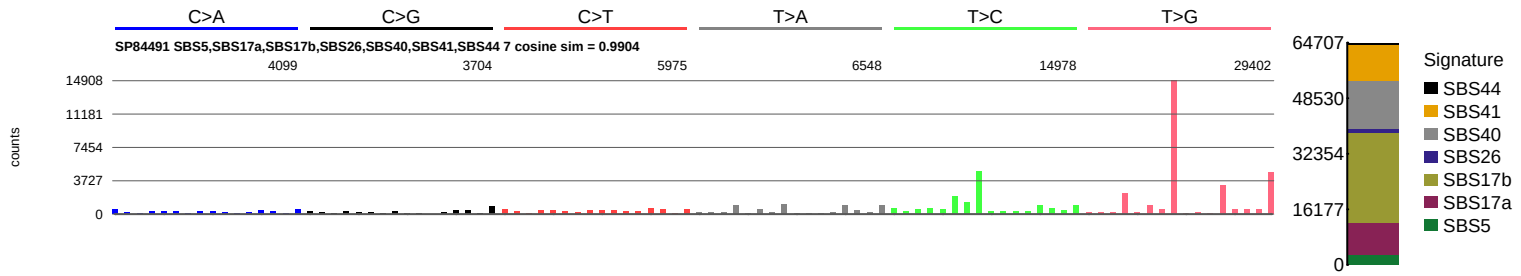
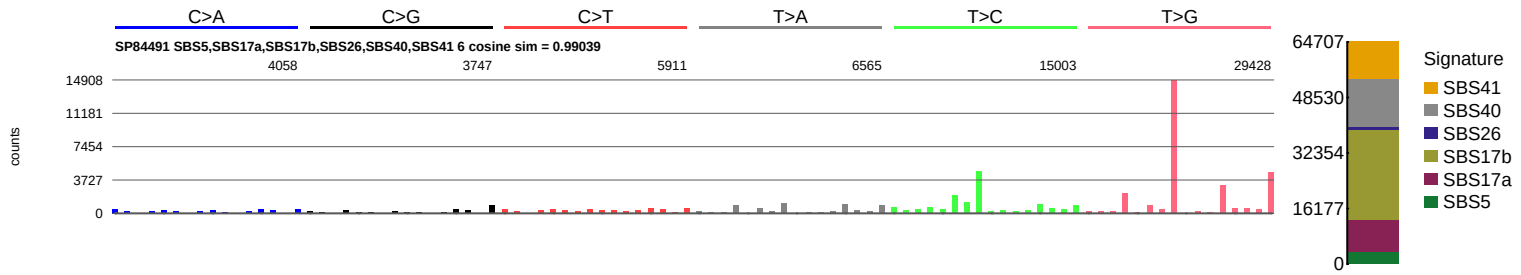
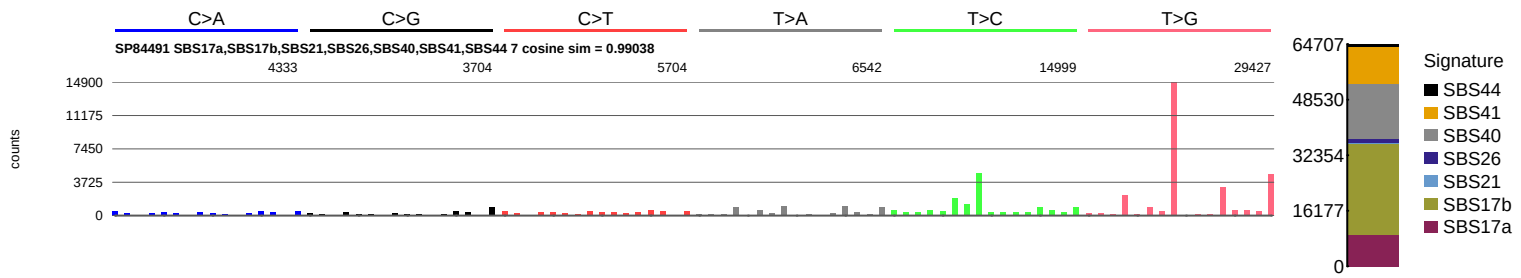
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99041
3311
2943
7457
6394
15689
28908
counts
14972
11229
7486
3743
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

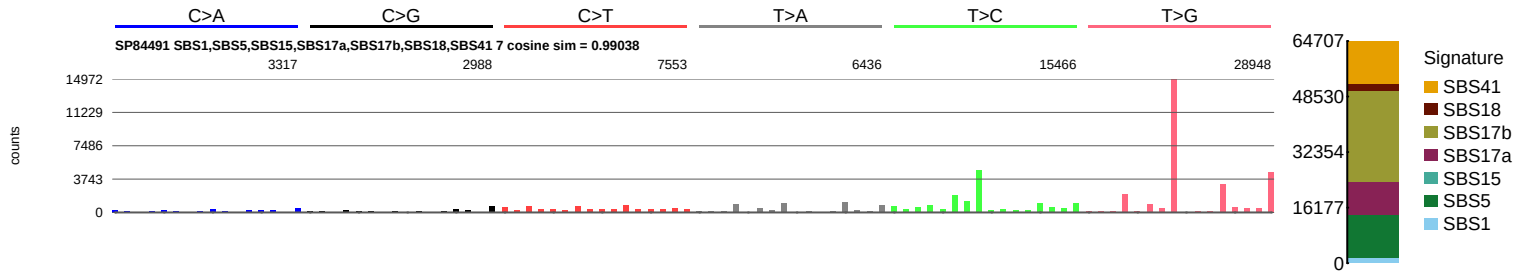

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.9904
4171
3709
5709
6536
15095
29491
counts
14908
11181
7454
3727
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2

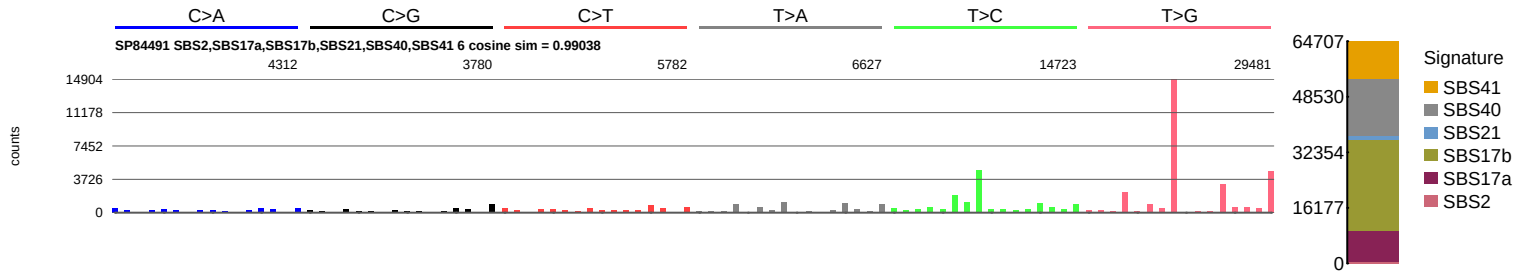

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.9904
3318
2958
7389
6406
15719
28918
counts
14972
11229
7486
3743
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

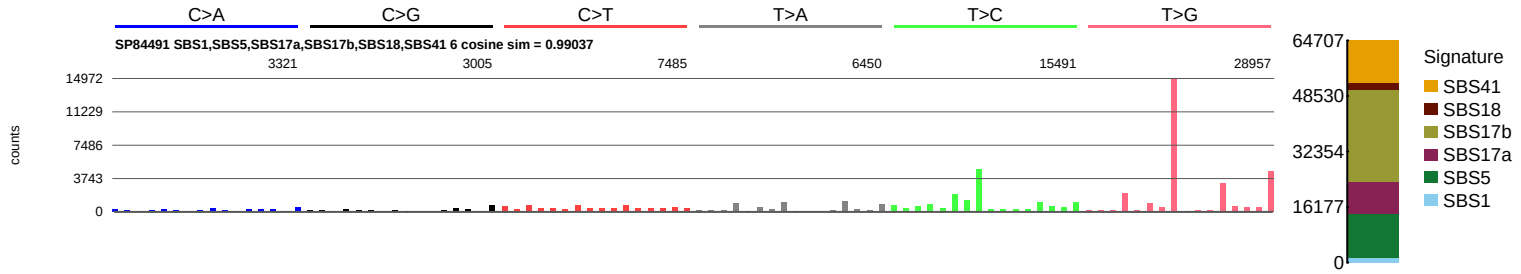

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.9904
4099
3704
5975
6548
14978
29402
counts
14908
11181
7454
3727
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5

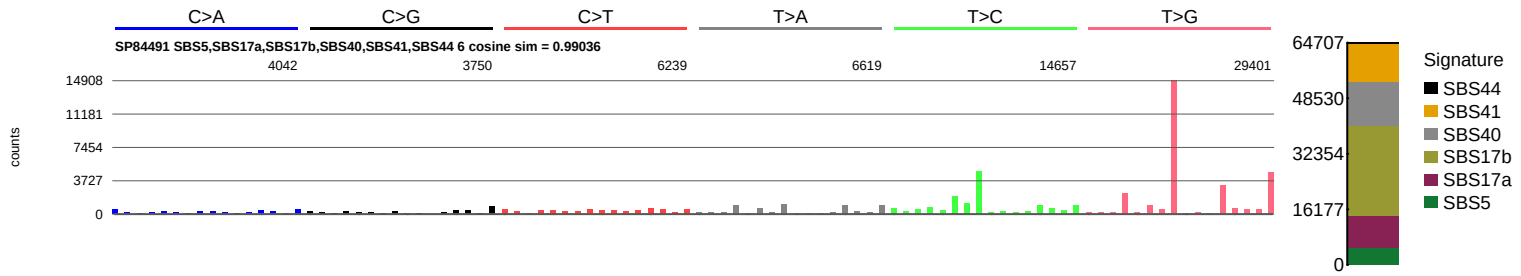

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.99039
4058
3747
5911
6565
15003
29428
counts
14908
11181
7454
3727
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5

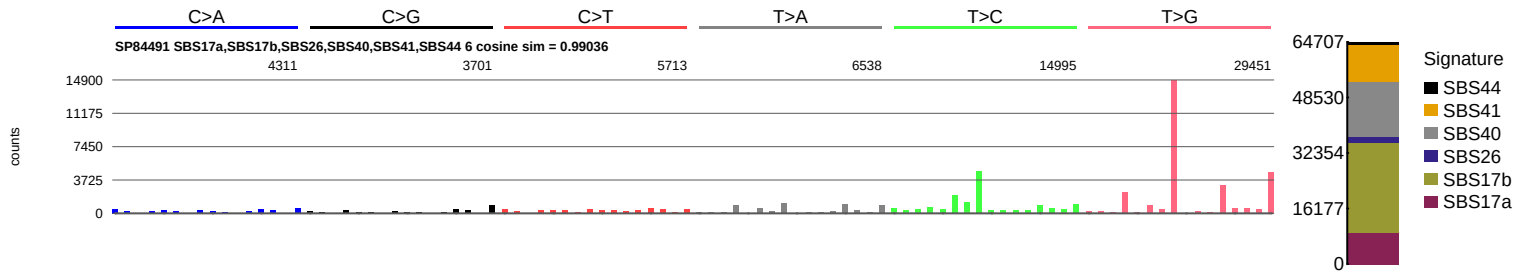

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.99038
4333
3704
5704
6542
14999
29427
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a

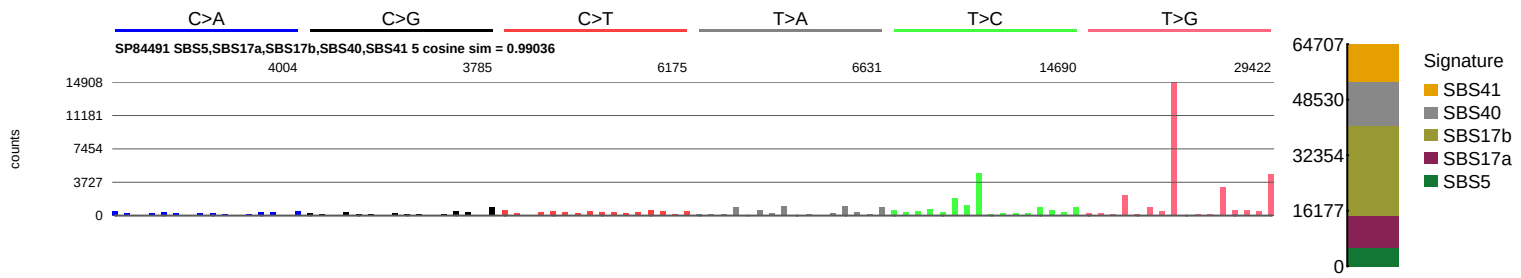

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99038
3317
2988
7553
6436
15466
28948
counts
14972
11229
7486
3743
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

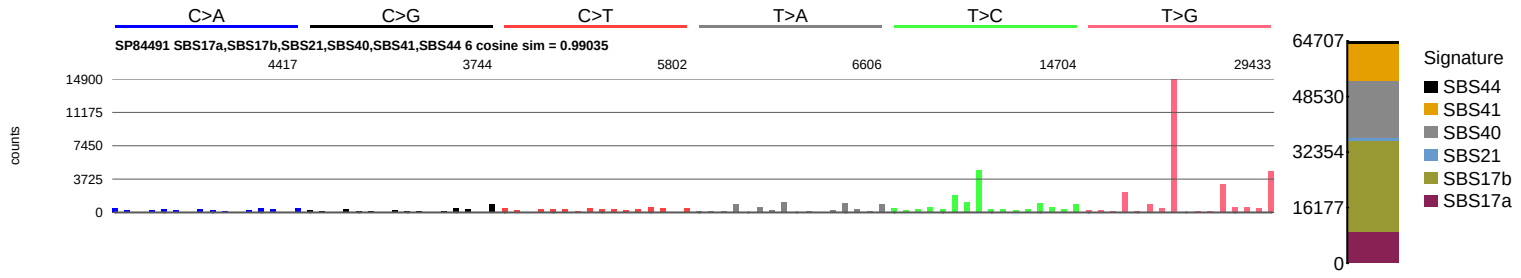

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.99038
4312
3780
5782
6627
14723
29481
counts
14904
11178
7452
3726
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2

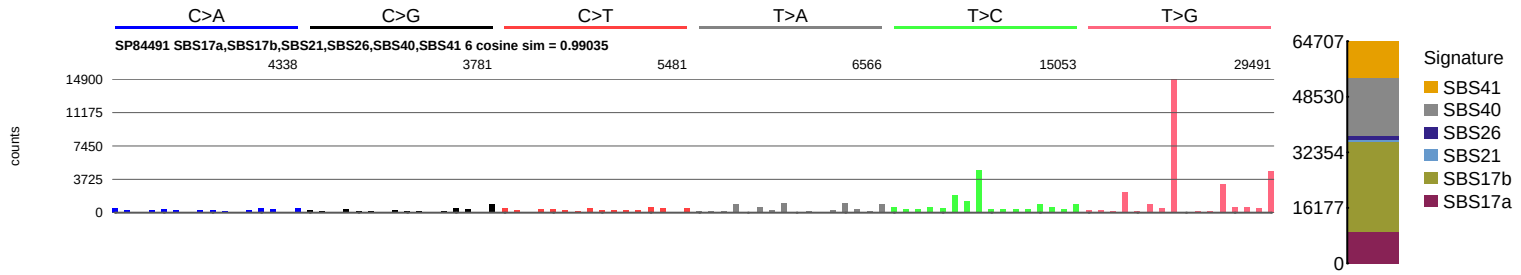

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.99037
3321
3005
7485
6450
15491
28957
counts
14972
11229
7486
3743
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1

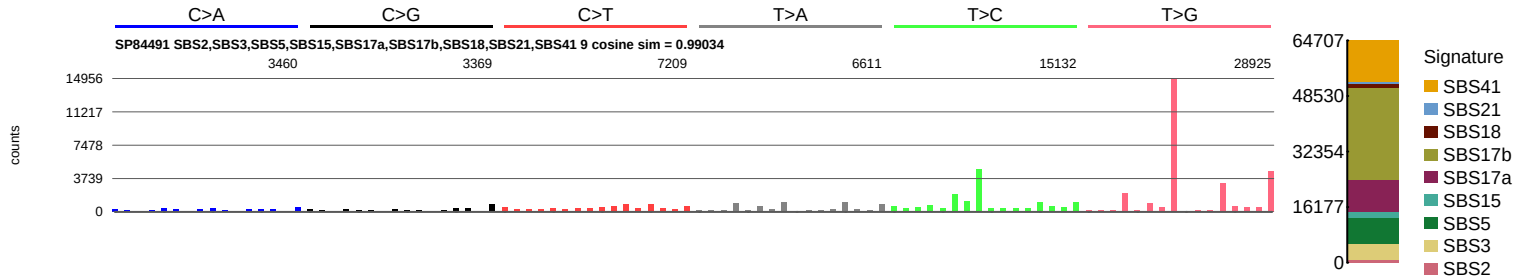

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.99036
4042
3750
6239
6619
14657
29401
counts
14908
11181
7454
3727
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5

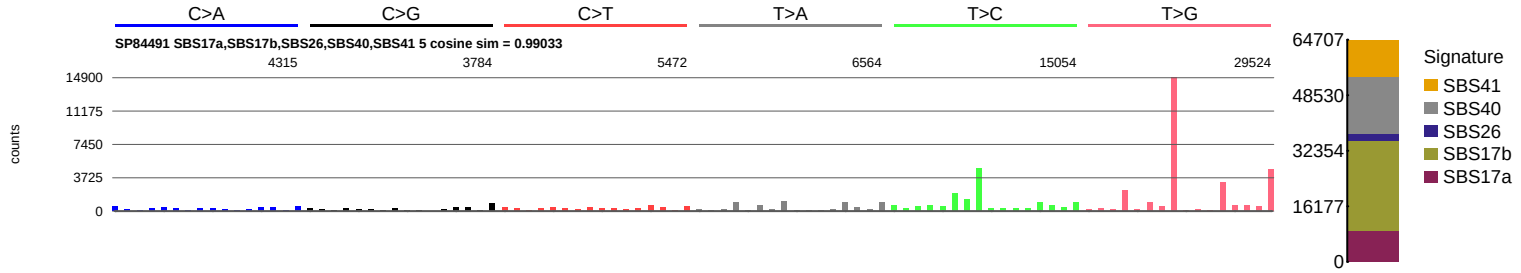

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 6 cosine sim = 0.99036
4311
3701
5713
6538
14995
29451
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a

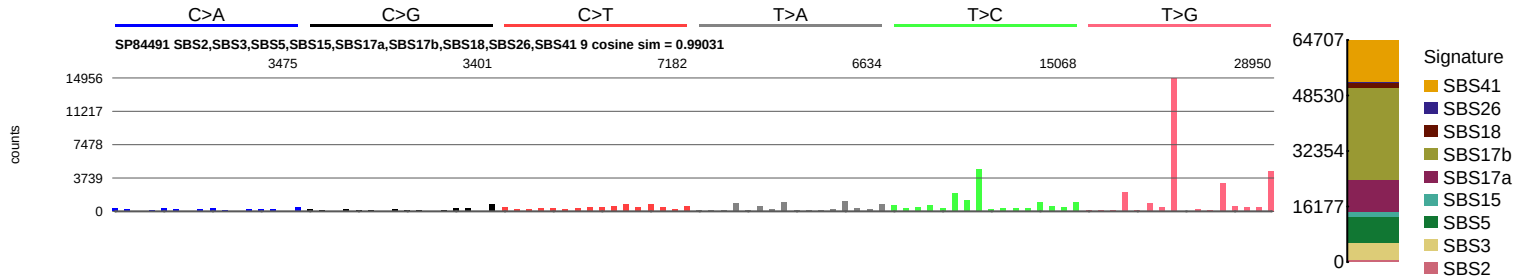

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.99036
4004
3785
6175
6631
14690
29422
counts
14908
11181
7454
3727
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5

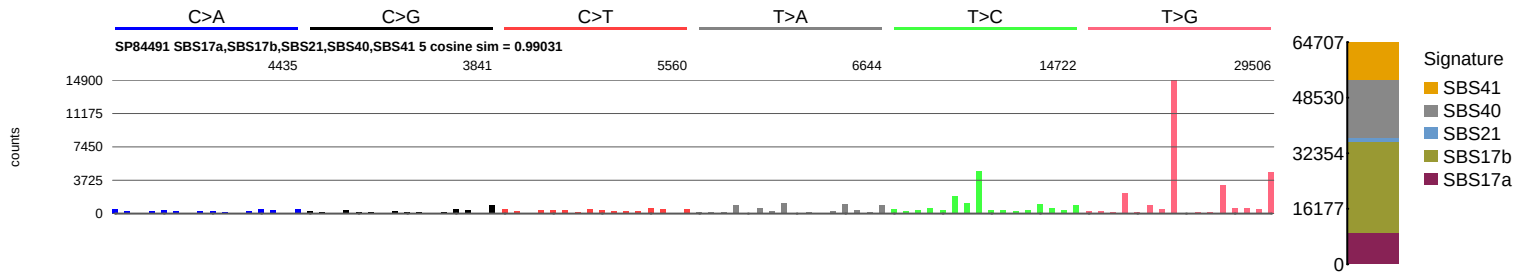

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 6 cosine sim = 0.99035
4417
3744
5802
6606
14704
29433
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a

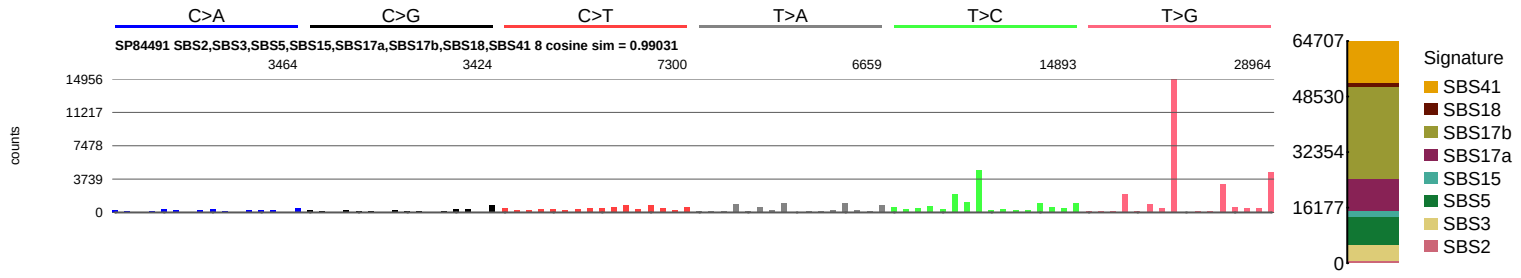

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 6 cosine sim = 0.99035
4338
3781
5481
6566
15053
29491
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a

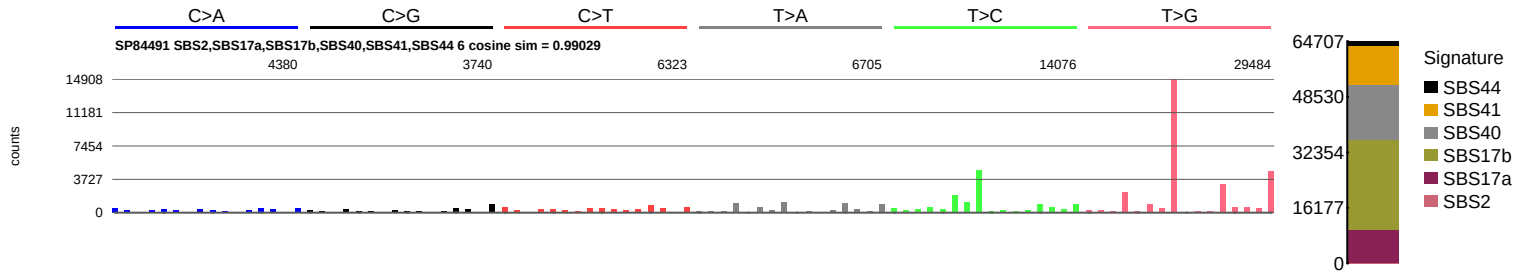

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99034
3460
3369
7209
6611
15132
28925
counts
14956
11217
7478
3739
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

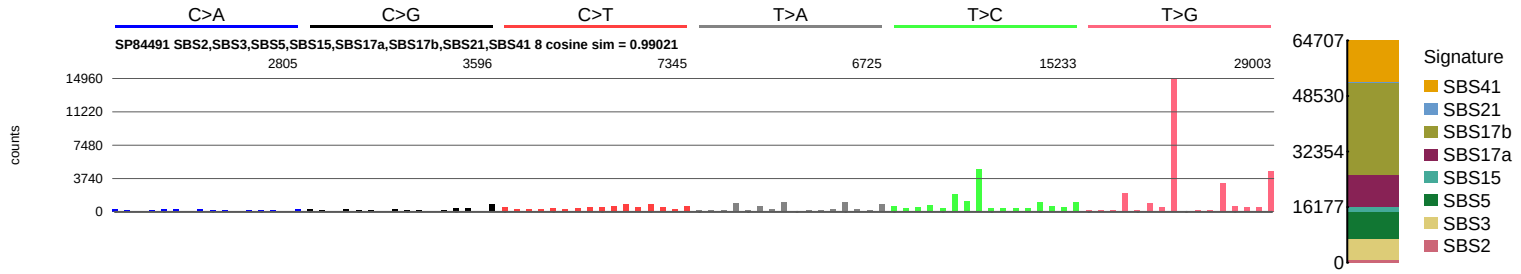

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS26,SBS40,SBS41 5 cosine sim = 0.99033
4315
3784
5472
6564
15054
29524
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a

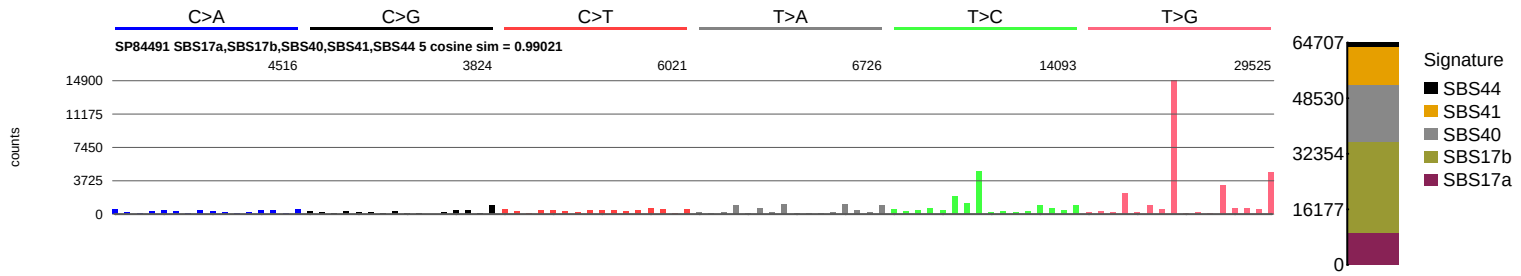

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99031
3475
3401
7182
6634
15068
28950
counts
14956
11217
7478
3739
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

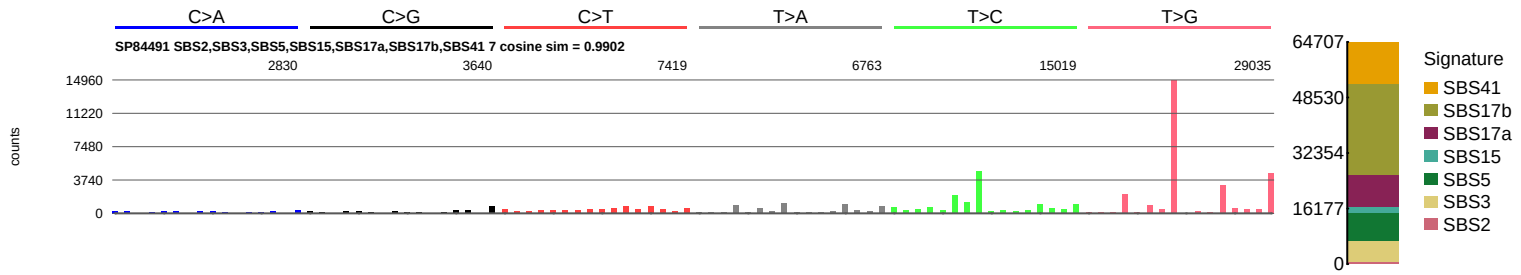

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS40,SBS41 5 cosine sim = 0.99031
4435
3841
5560
6644
14722
29506
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a

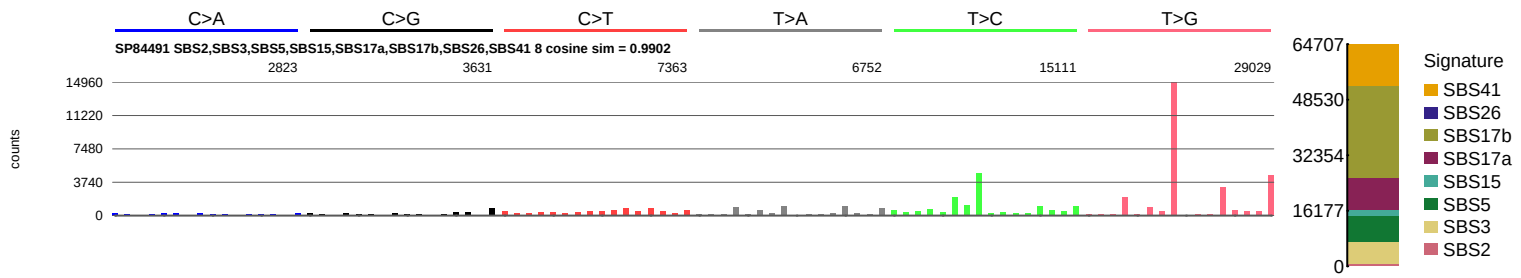

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99031
3464
3424
7300
6659
14893
28964
counts
14956
11217
7478
3739
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

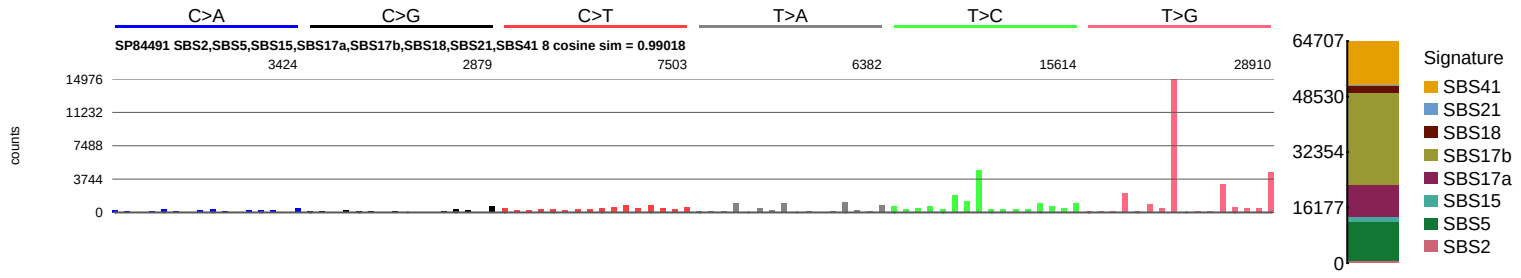

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.99029
4380
3740
6323
6705
14076
29484
counts
14908
11181
7454
3727
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS2

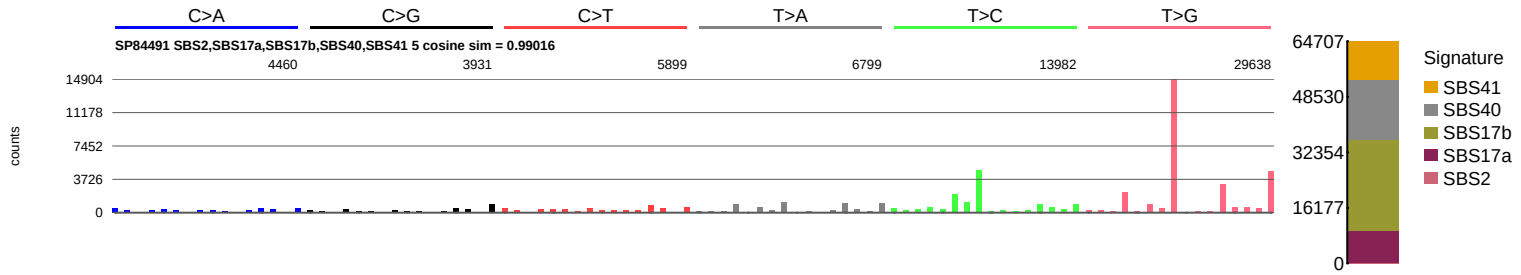

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99021
2805
3596
7345
6725
15233
29003
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

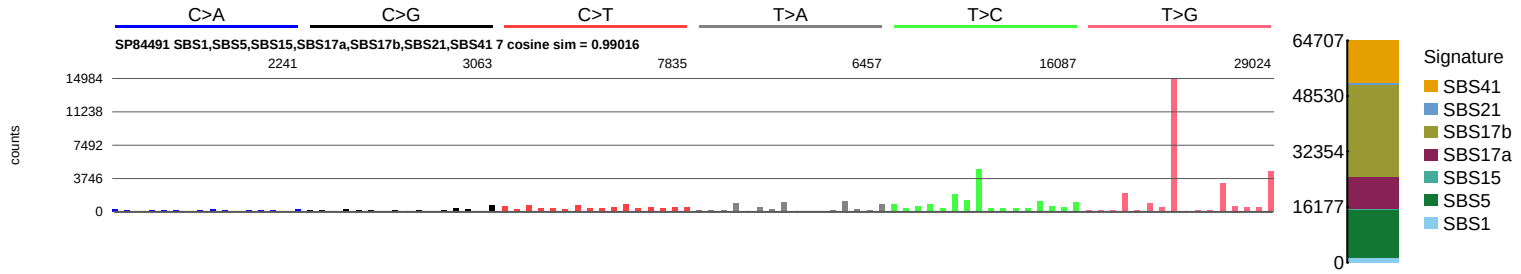

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS40,SBS41,SBS44 5 cosine sim = 0.99021
4516
3824
6021
6726
14093
29525
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a

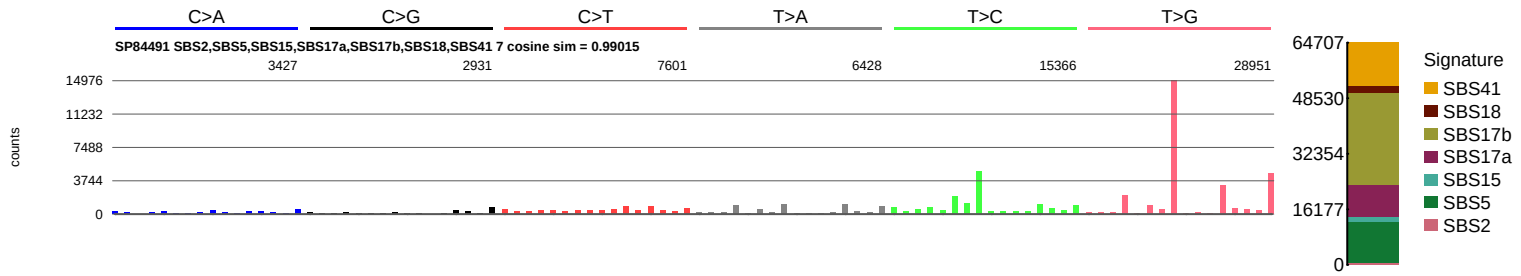

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.9902
2830
3640
7419
6763
15019
29035
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

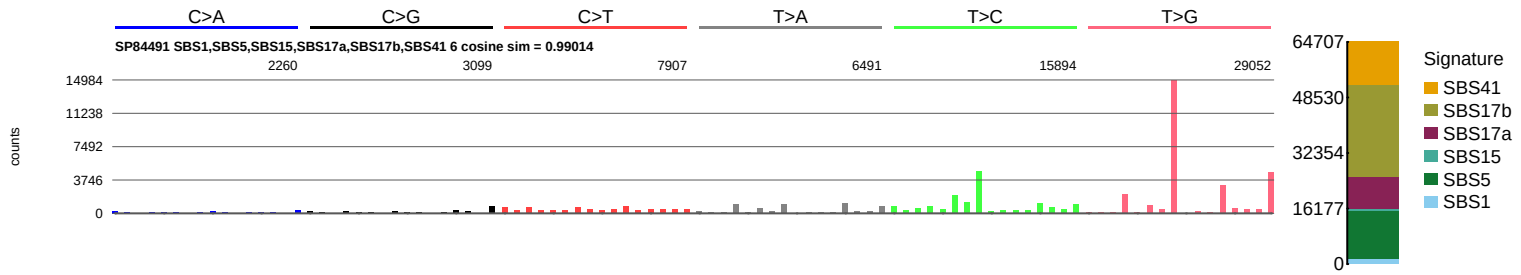

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.9902
2823
3631
7363
6752
15111
29029
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

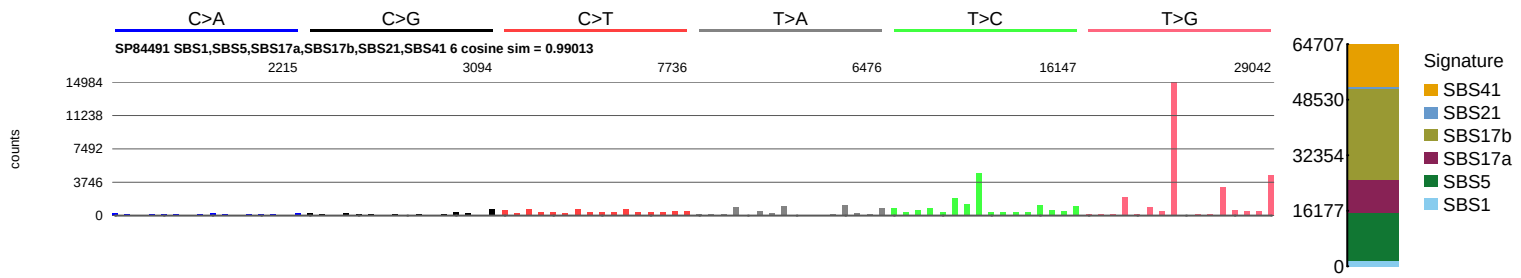

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99018
3424
2879
7503
6382
15614
28910
counts
14976
11232
7488
3744
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

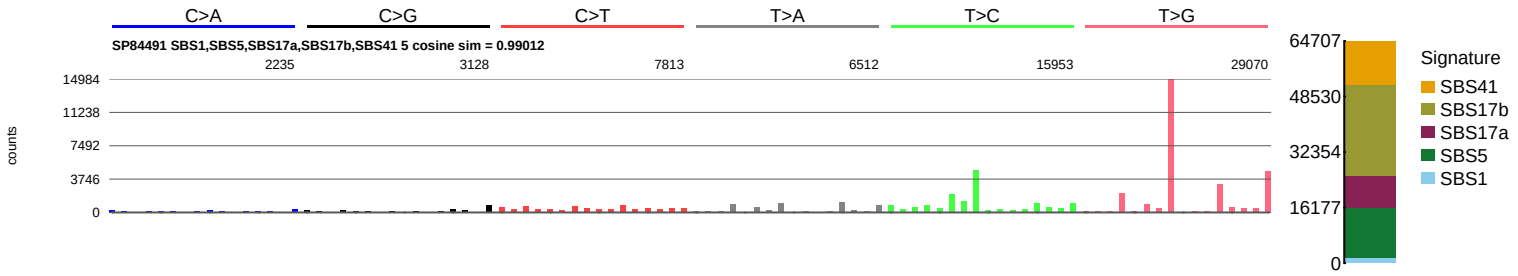

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.99016
4460
3931
5899
6799
13982
29638
counts
14904
11178
7452
3726
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2

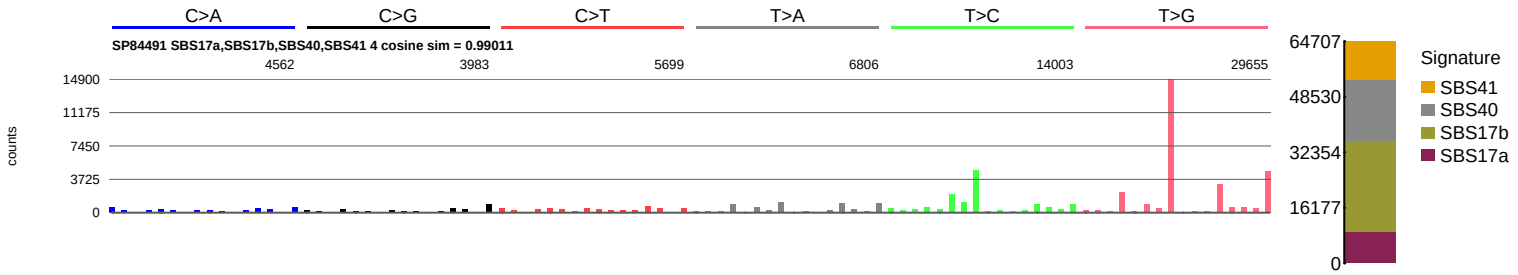

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99016
2241
3063
7835
6457
16087
29024
counts
14984
11238
7492
3746
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

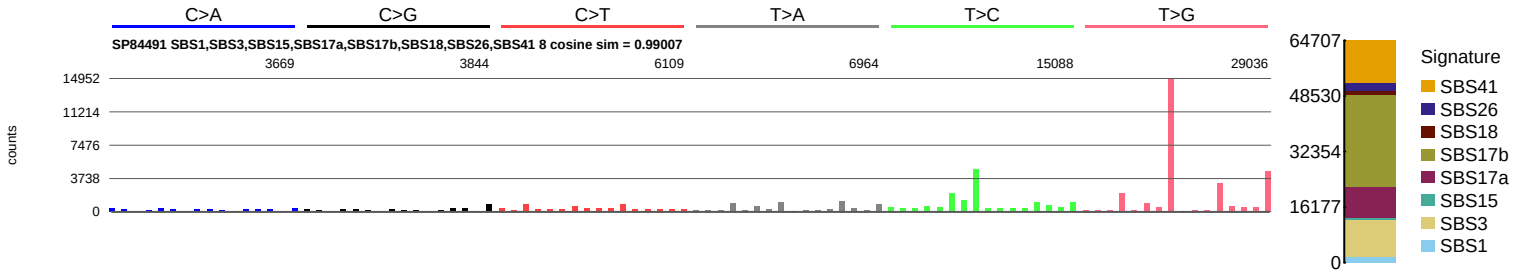

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99015
3427
2931
7601
6428
15366
28951
counts
14976
11232
7488
3744
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

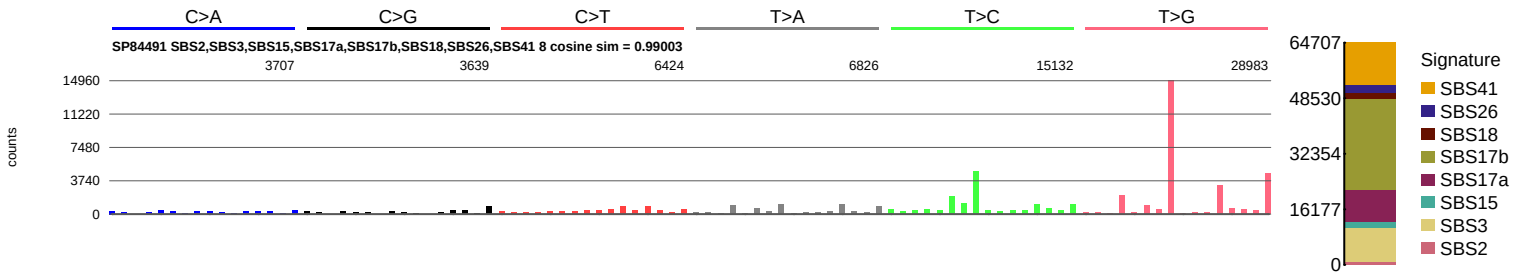

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99014
2260
3099
7907
6491
15894
29052
counts
14984
11238
7492
3746
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS1

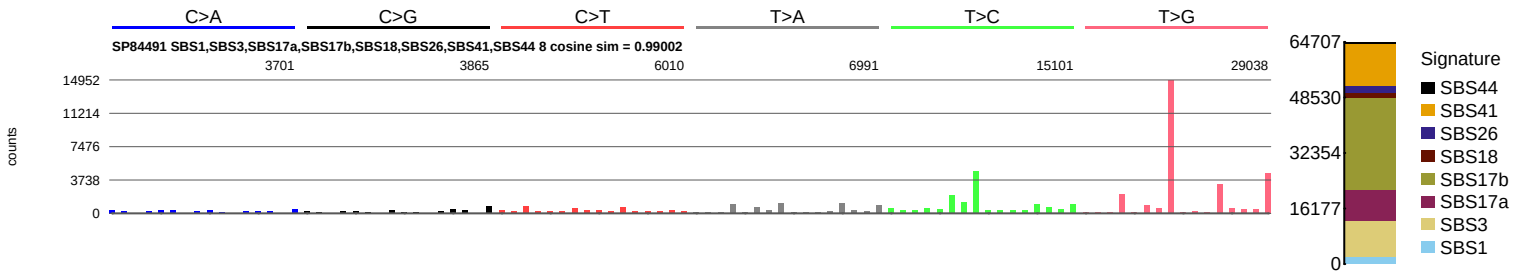

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.99013
2215
3094
7736
6476
16147
29042
counts
14984
11238
7492
3746
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS1

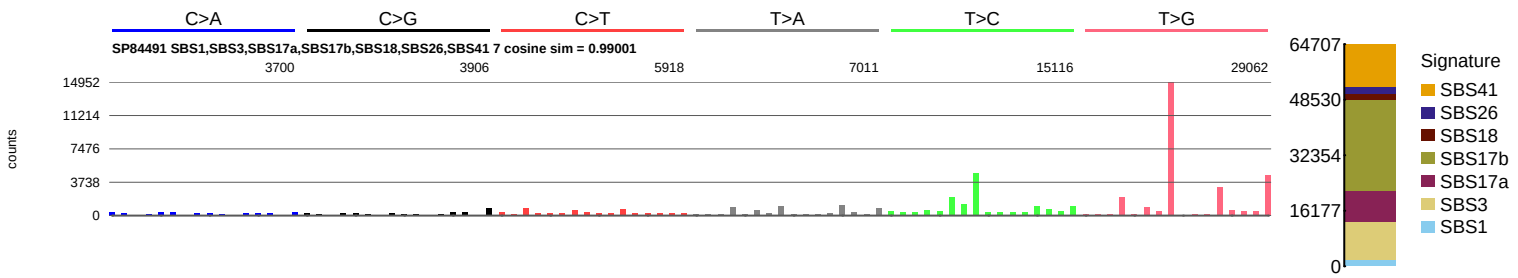

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.99012
2235
3128
7813
6512
15953
29070
counts
14984
11238
7492
3746
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS1

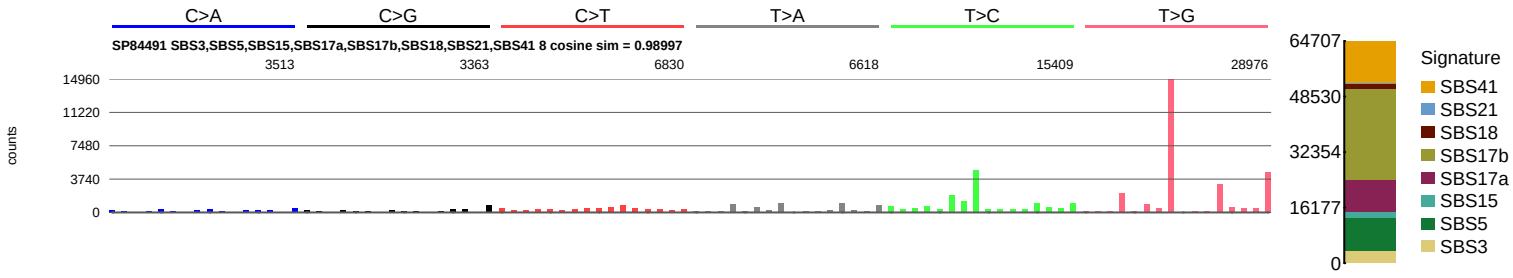

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS40,SBS41 4 cosine sim = 0.99011
4562
3983
5699
6806
14003
29655
counts
14900
11175
7450
3725
0
64707
48530
32354
16177
0
Signature
SBS41
SBS40
SBS17b
SBS17a

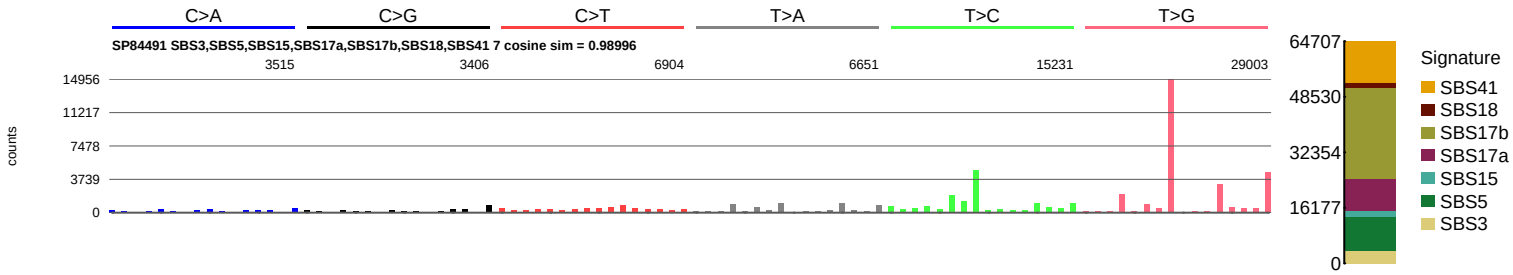

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99007
3669
3844
6109
6964
15088
29036
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

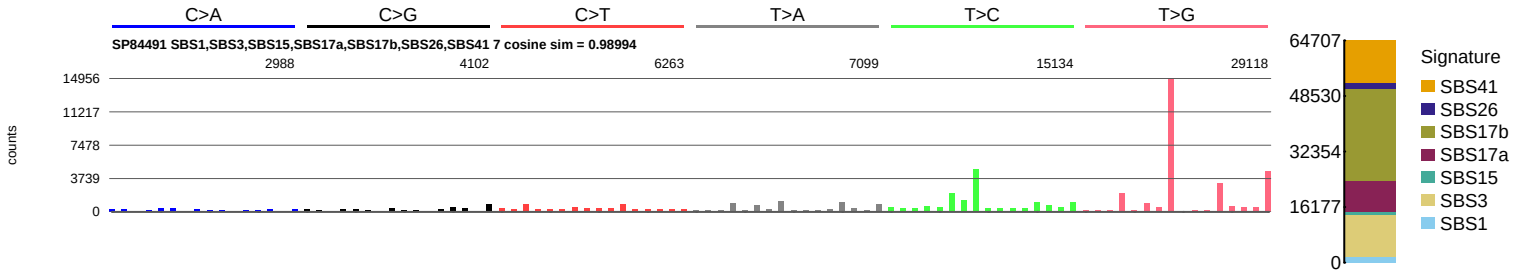

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99003
3707
3639
6424
6826
15132
28983
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

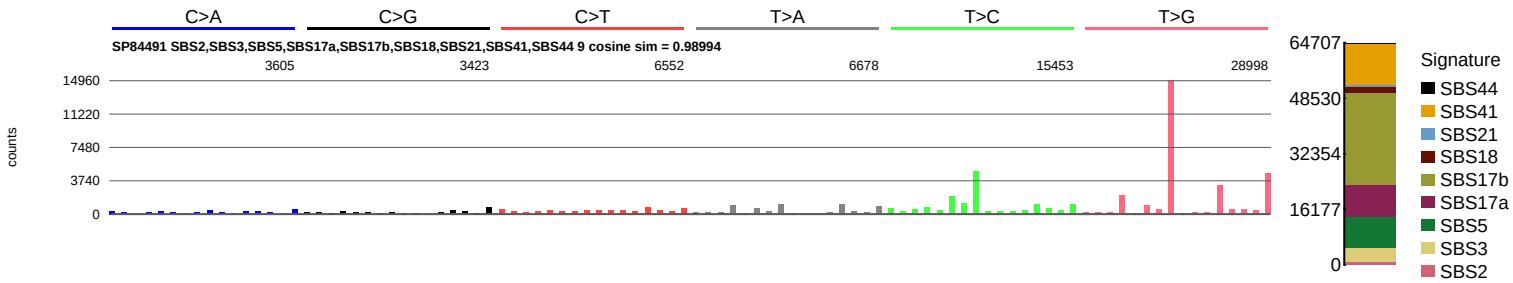

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.99002
3701
3865
6010
6991
15101
29038
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

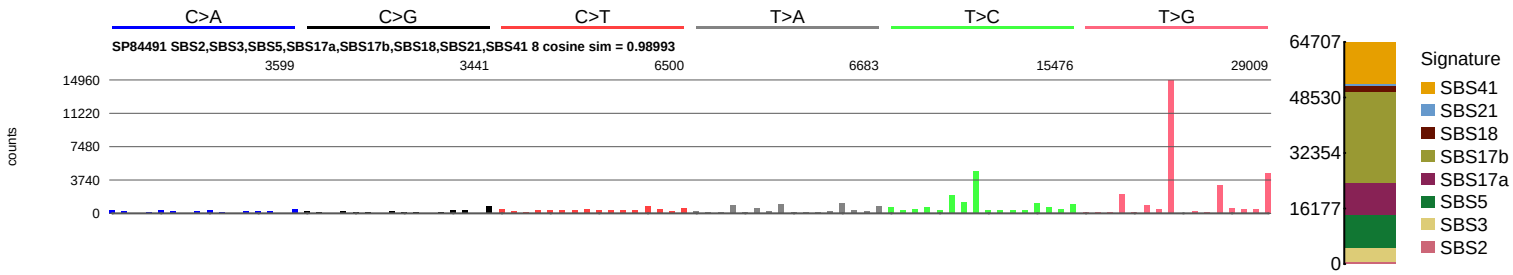

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.99001
3700
3906
5918
7011
15116
29062
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

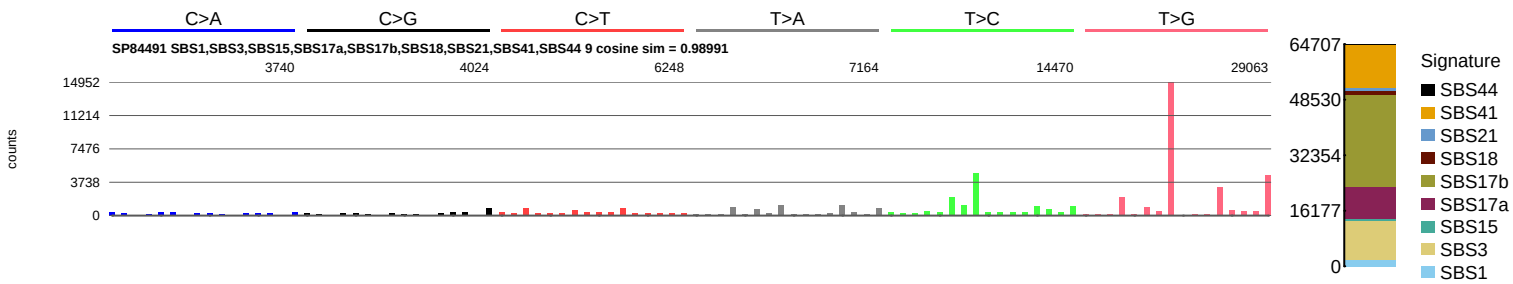

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98997
3513
3363
6830
6618
15409
28976
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

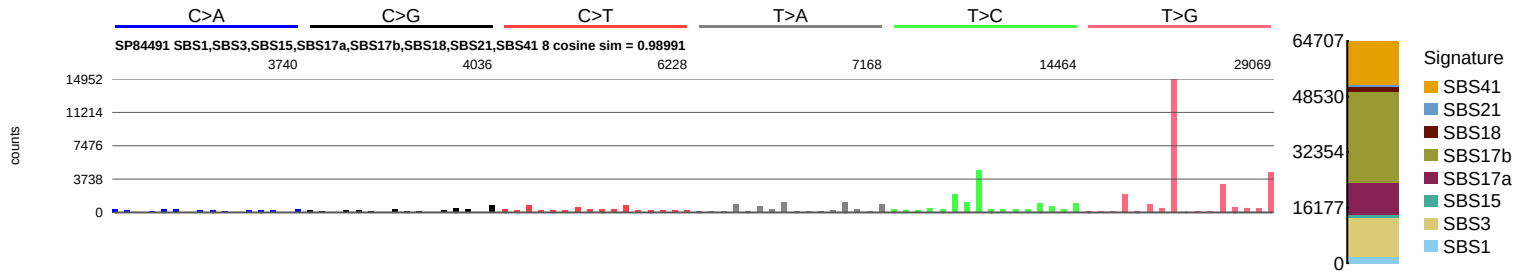

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98996
3515
3406
6904
6651
15231
29003
counts
14956
11217
7478
3739
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

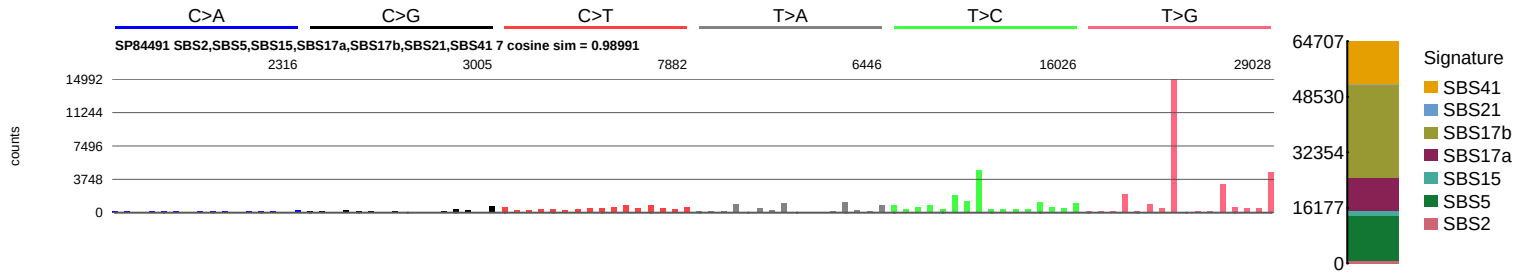

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98994
2988
4102
6263
7099
15134
29118
counts
14956
11217
7478
3739
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1

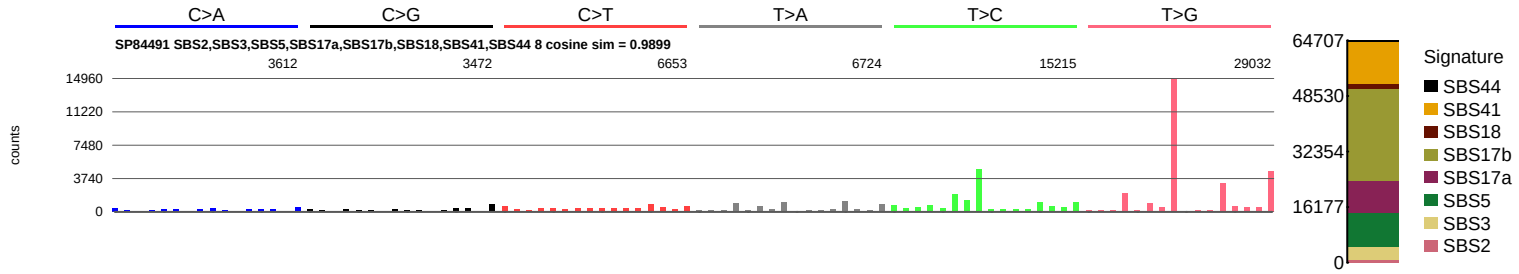

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98994
3605
3423
6552
6678
15453
28998
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

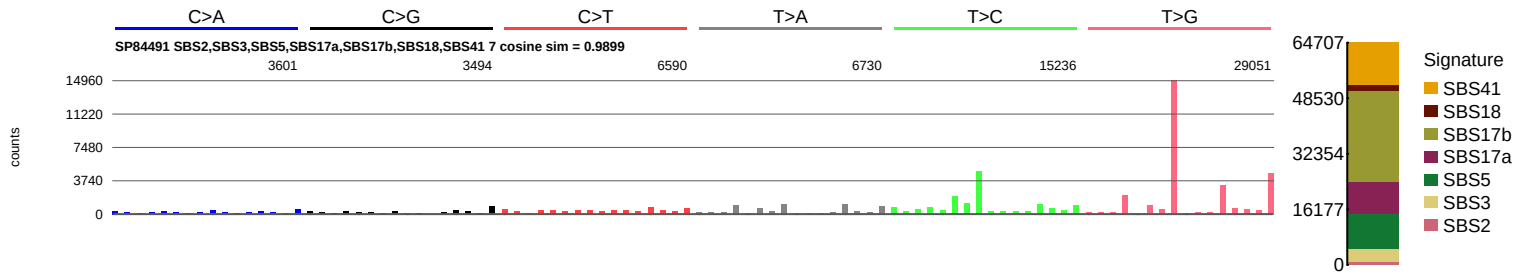

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98993
3599
3441
6500
6683
15476
29009
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98991
3740
4024
6248
7164
14470
29063
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98991
3740
4036
6228
7168
14464
29069
14952
11214
7476
3738
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98991
2316
3005
7882
6446
16026
29028
14992
11244
7496
3748
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.9899
3612
3472
6653
6724
15215
29032
14960
11220
7480
3740
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9899
3601
3494
6590
6730
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29051
14960
11220
7480
3740
0
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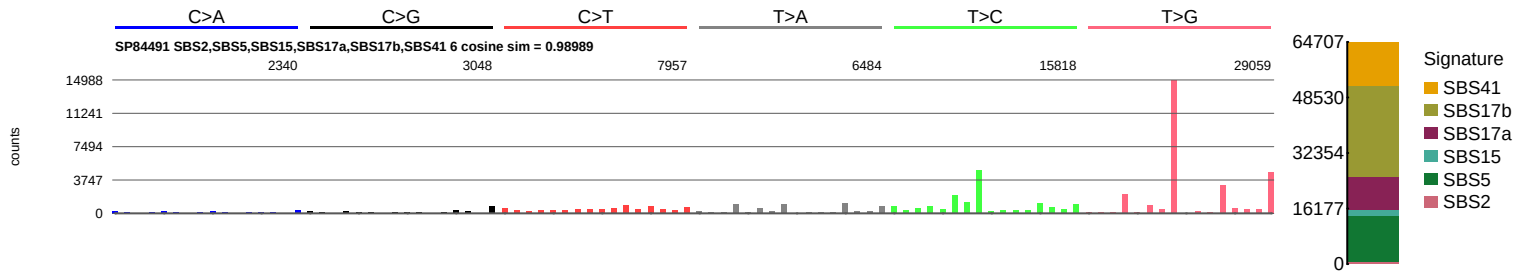
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SP84491 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98989
2340
3048
7957
6484
15818
29059
14988
11241
7494
3747
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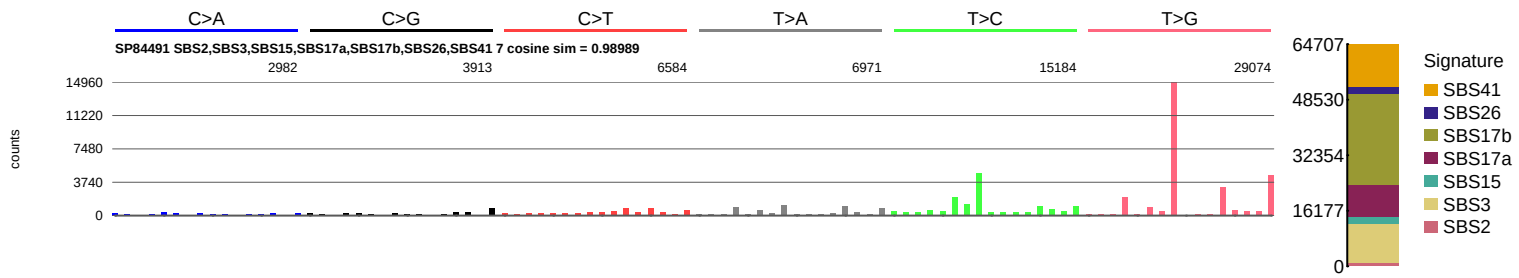
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SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98989
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3913
6584
6971
15184
29074
14960
11220
7480
3740
0
counts
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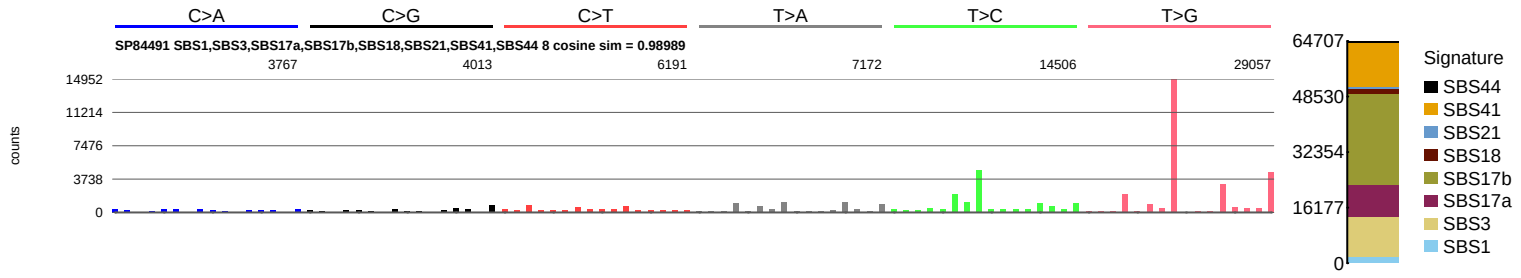
SBS17a
SBS15
SBS3
SBS2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4013
6191
7172
14506
29057
14952
11214
7476
3738
0
counts
64707
48530
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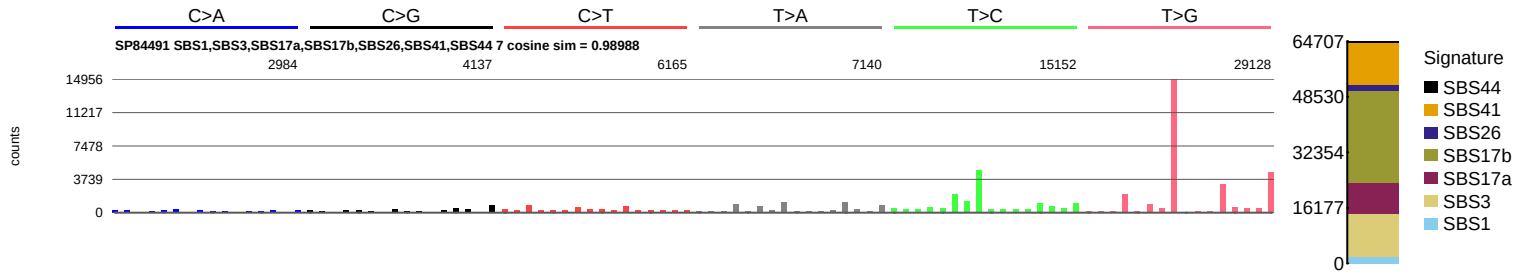
SBS3
SBS1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4137
6165
7140
15152
29128
14956
11217
7478
3739
0
counts
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SBS41
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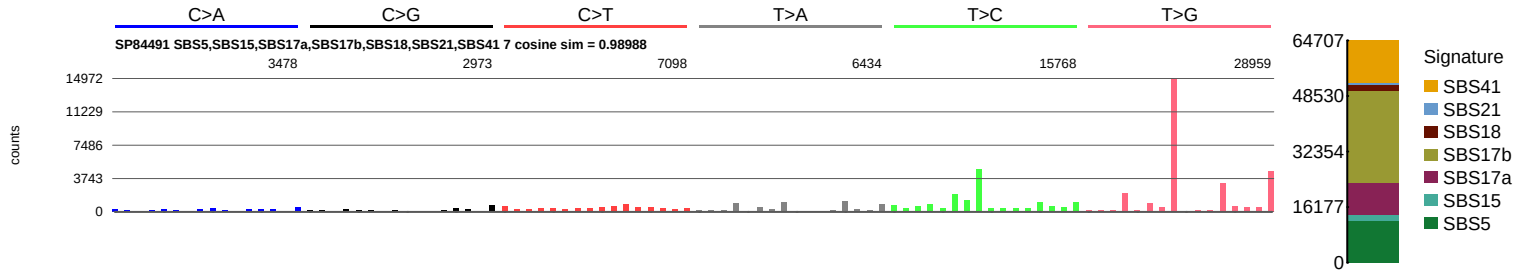
SBS17b
SBS17a
SBS3
SBS1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SP84491 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98988
3478
2973
7098
6434
15768
28959
14972
11229
7486
3743
0
counts
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16177
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SBS21
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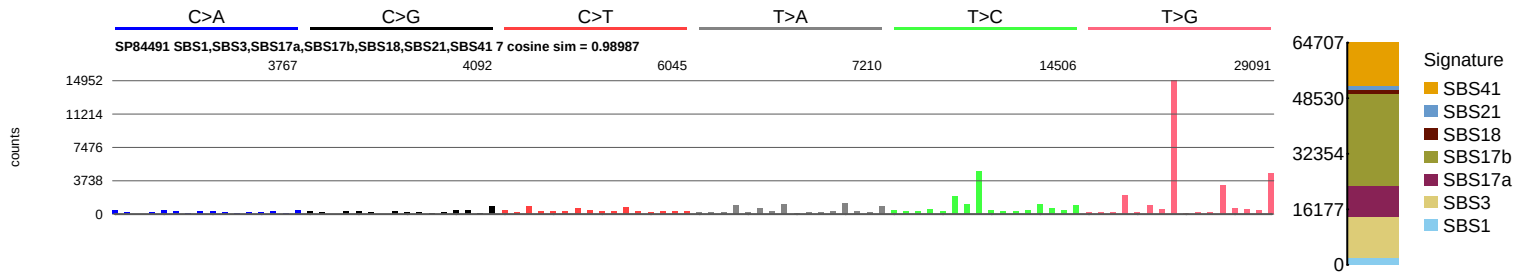
SBS17b
SBS17a
SBS15
SBS5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3767
4092
6045
7210
14506
29091
14952
11214
7476
3738
0
counts
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48530
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16177
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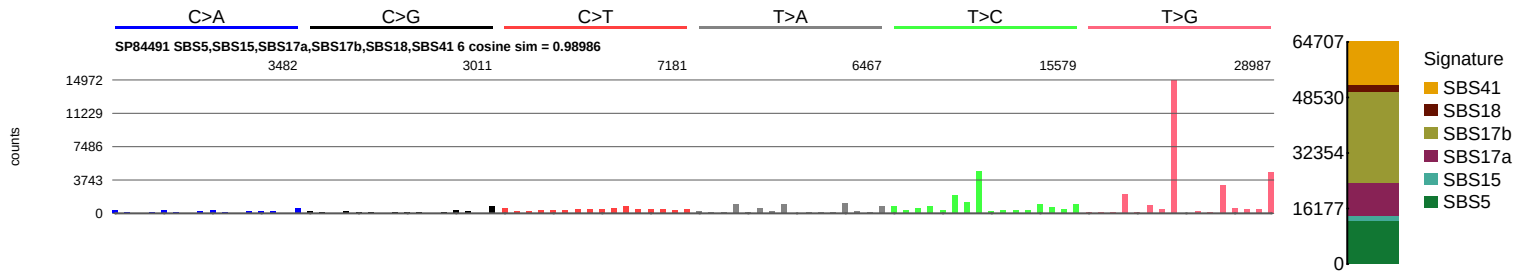
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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SP84491 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98986
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3011
7181
6467
15579
28987
14972
11229
7486
3743
0
counts
64707
48530
32354
16177
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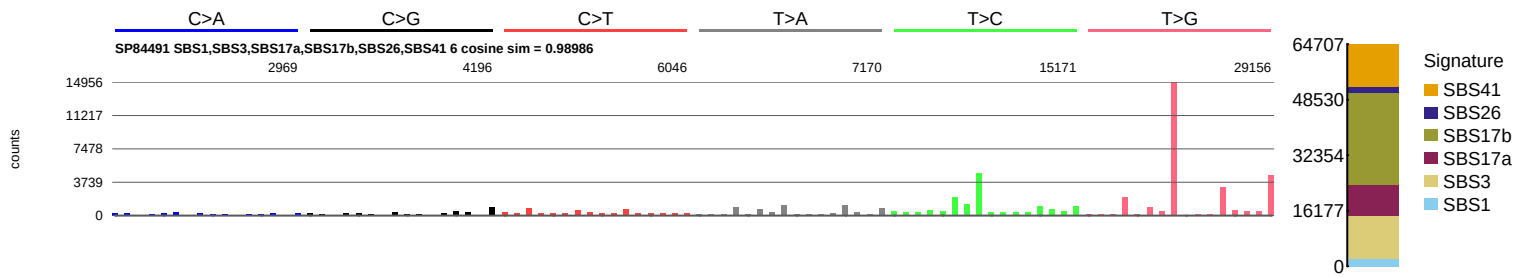
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5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SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98986
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4196
6046
7170
15171
29156
14956
11217
7478
3739
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48530
32354
16177
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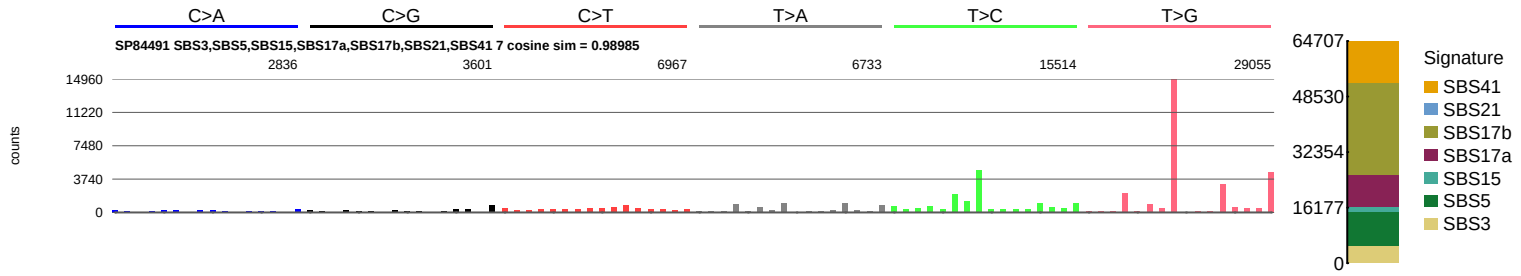
SBS17b
SBS17a
SBS3
SBS1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SP84491 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98985
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3601
6967
6733
15514
29055
counts
14960
11220
7480
3740
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16177
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SBS17a
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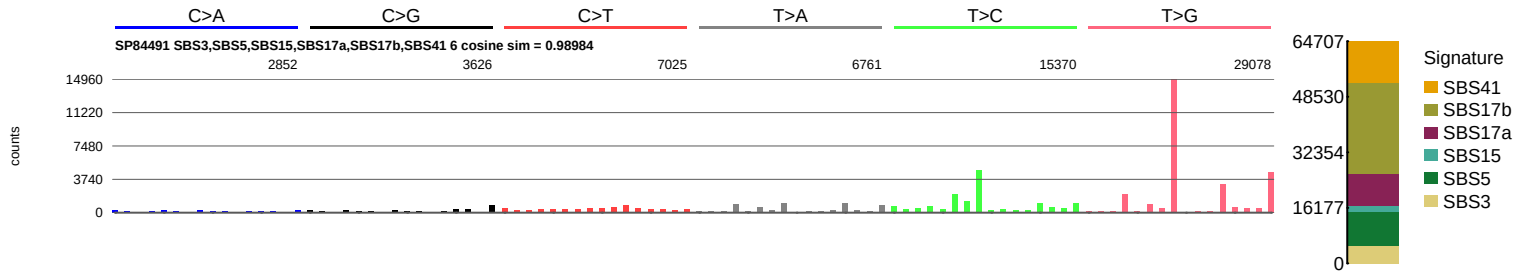
SBS5
SBS3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C>G
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T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98984
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3626
7025
6761
15370
29078
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
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SBS41
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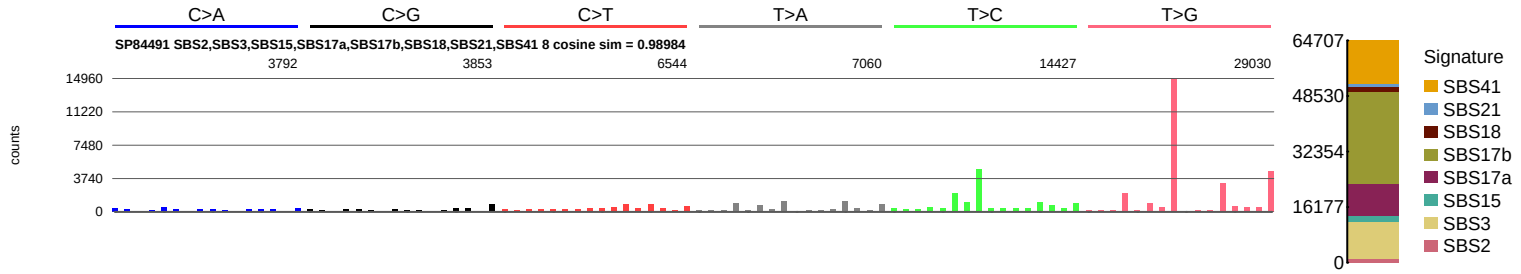
SBS17a
SBS15
SBS5
SBS3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C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98984
3792
3853
6544
7060
14427
29030
counts
14960
11220
7480
3740
0
64707
48530
32354
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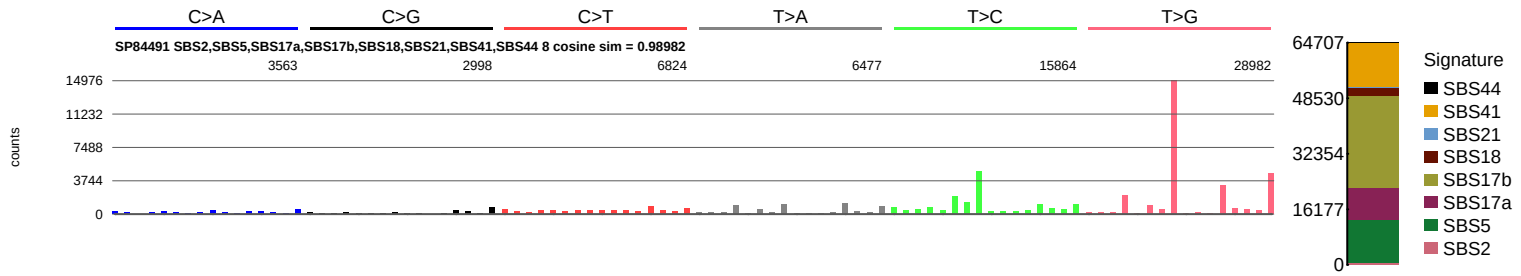
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98982
3563
2998
6824
6477
15864
28982
counts
14976
11232
7488
3744
0
64707
48530
32354
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0
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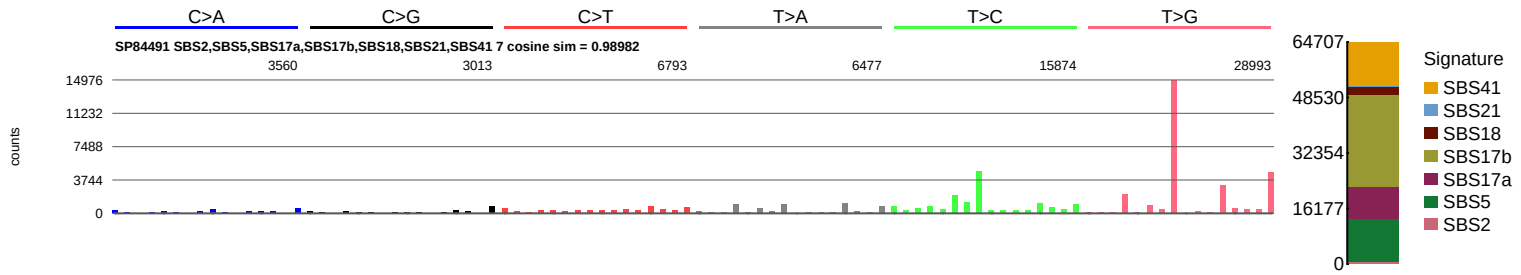
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98982
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3013
6793
6477
15874
28993
counts
14976
11232
7488
3744
0
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32354
16177
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SBS21
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SBS17b
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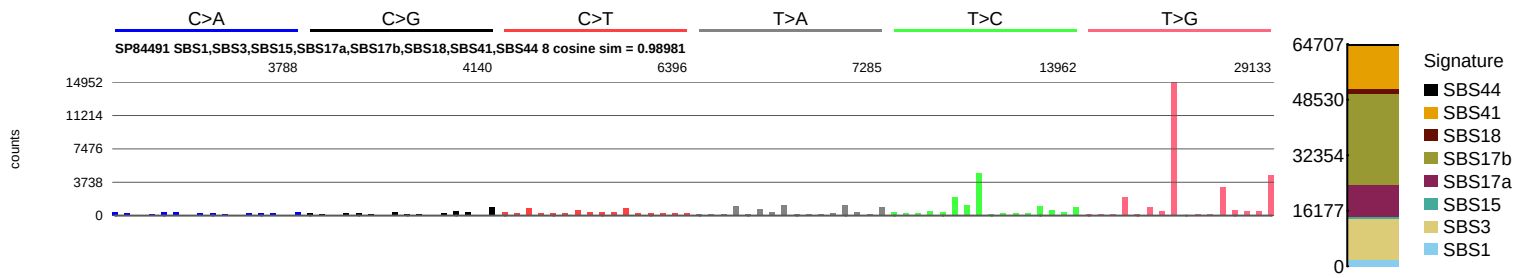
SBS5
SBS2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C>T
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T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98981
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4140
6396
7285
13962
29133
counts
14952
11214
7476
3738
0
64707
48530
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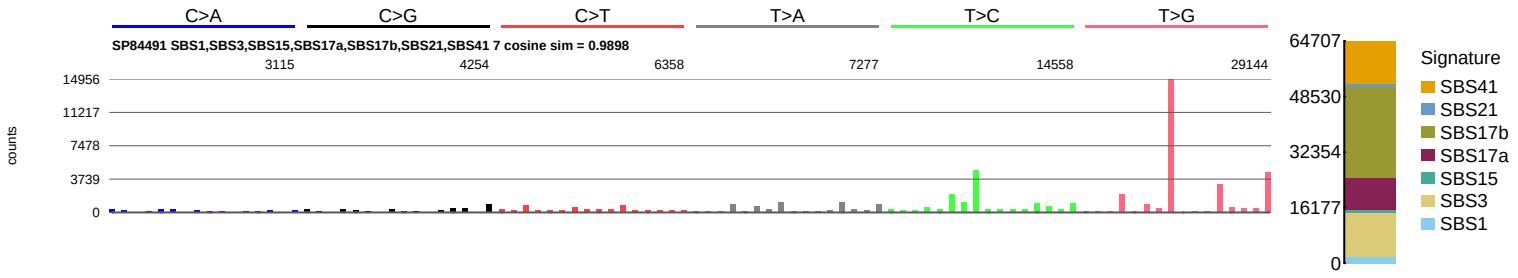
SBS3
SBS1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C>G
C>T
T>A
T>C
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SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.9898
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4254
6358
7277
14558
29144
counts
14956
11217
7478
3739
0
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16177
0
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SBS21
SBS17b
SBS17a
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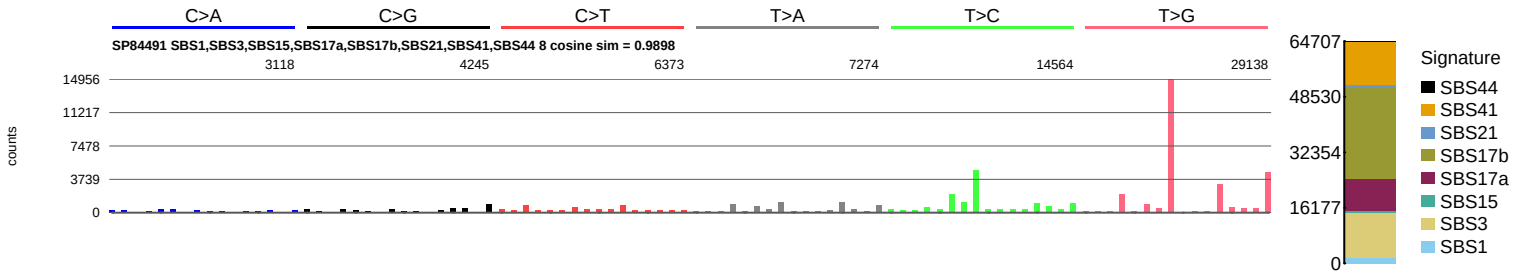
SBS3
SBS1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C>T
T>A
T>C
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SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.9898
3118
4245
6373
7274
14564
29138
counts
14956
11217
7478
3739
0
64707
48530
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SBS17b
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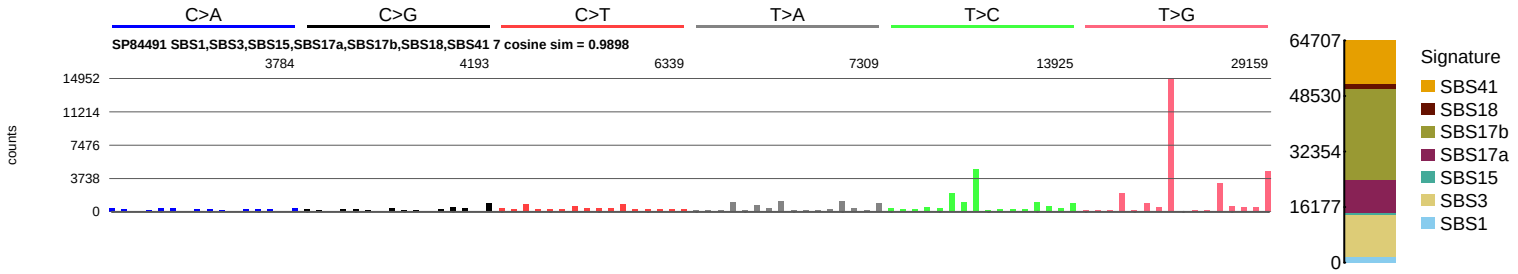
SBS3
SBS1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T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9898
3784
4193
6339
7309
13925
29159
counts
14952
11214
7476
3738
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64707
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16177
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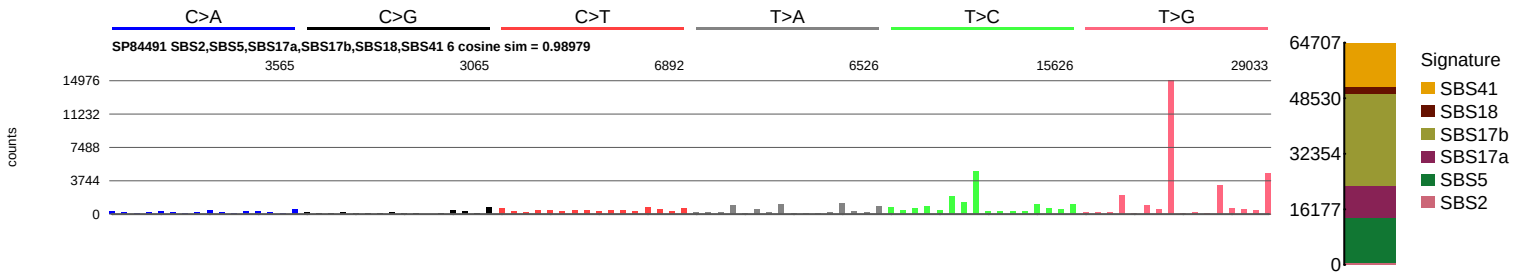
SBS3
SBS1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T>C
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SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98979
3565
3065
6892
6526
15626
29033
counts
14976
11232
7488
3744
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32354
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SBS18
SBS17b
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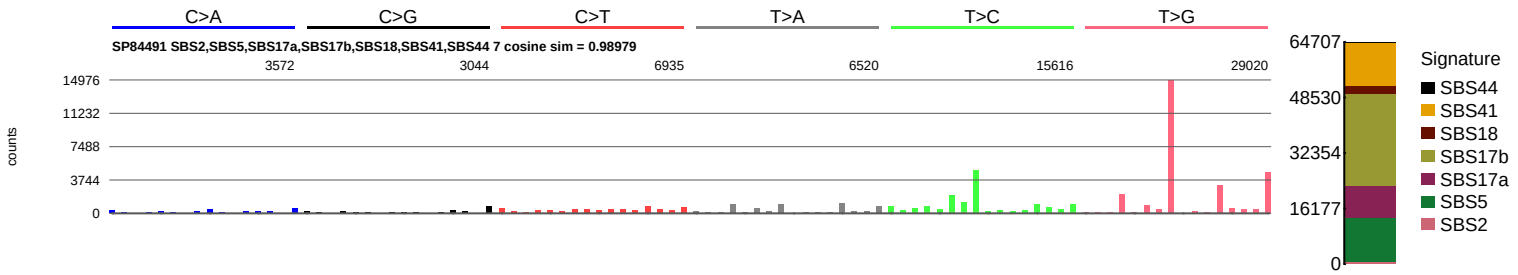
SBS5
SBS2


C>A
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T>C
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SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98979
3572
3044
6935
6520
15616
29020
counts
14976
11232
7488
3744
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64707
48530
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16177
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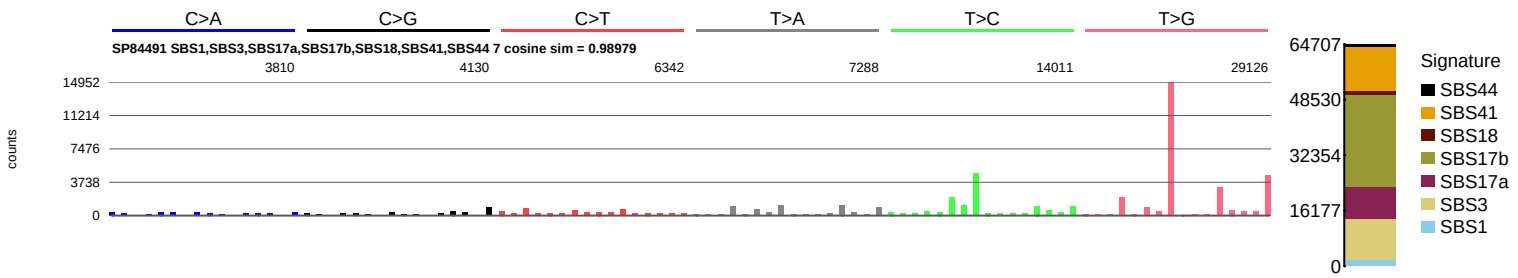
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
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T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98979
3810
4130
6342
7288
14011
29126
counts
14952
11214
7476
3738
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64707
48530
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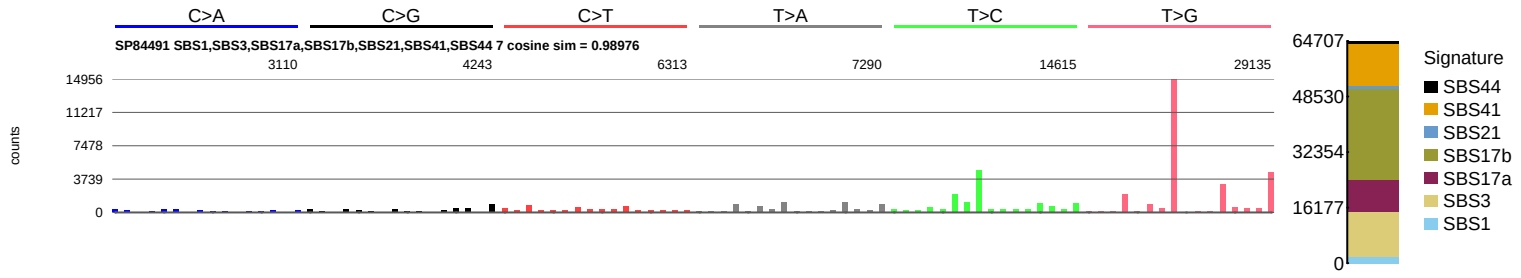
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98976
3110
4243
6313
7290
14615
29135
counts
14956
11217
7478
3739
0
64707
48530
32354
16177
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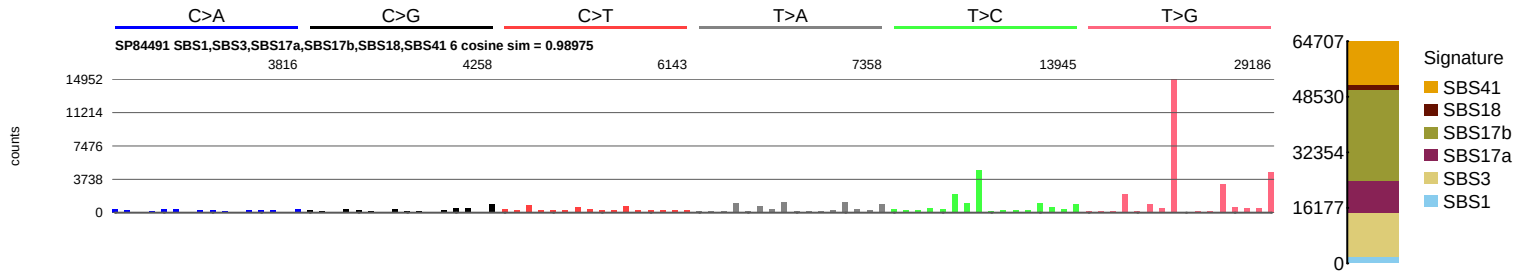
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1


C>A
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C>T
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T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98975
3816
4258
6143
7358
13945
29186
counts
14952
11214
7476
3738
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64707
48530
32354
16177
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SBS17b
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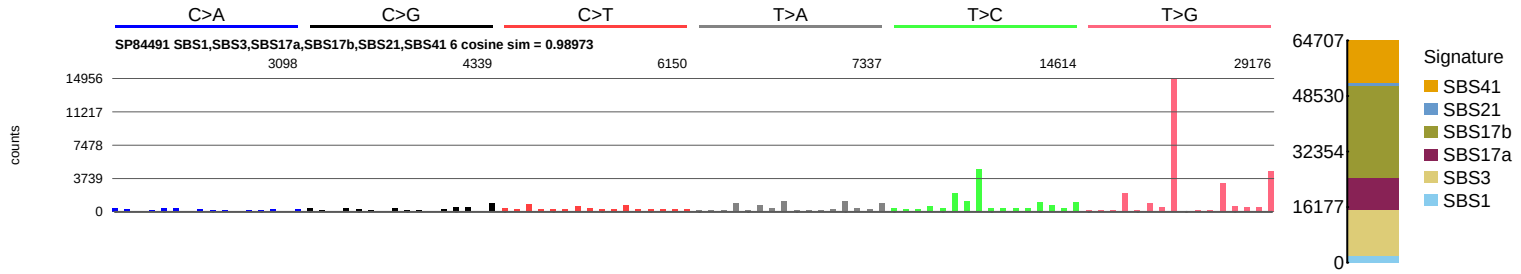
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98973
3098
4339
6150
7337
14614
29176
counts
14956
11217
7478
3739
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64707
48530
32354
16177
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SBS21
SBS17b
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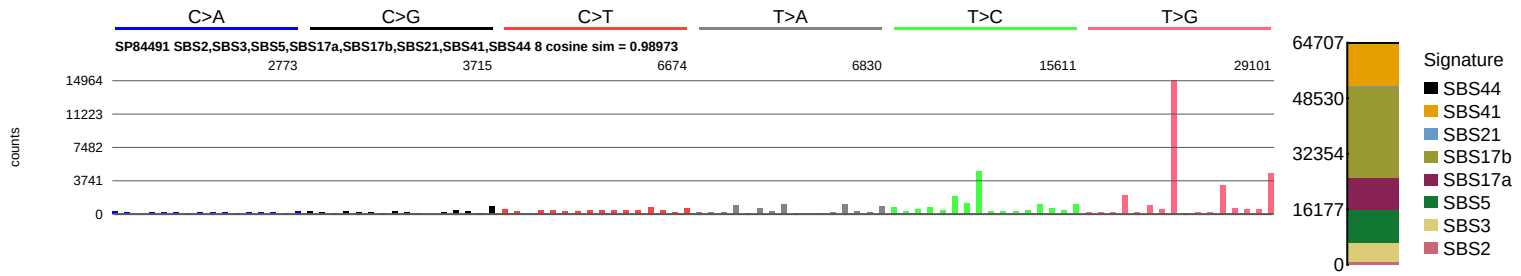
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98973
2773
3715
6674
6830
15611
29101
counts
14964
11223
7482
3741
0
64707
48530
32354
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SBS21
SBS17b
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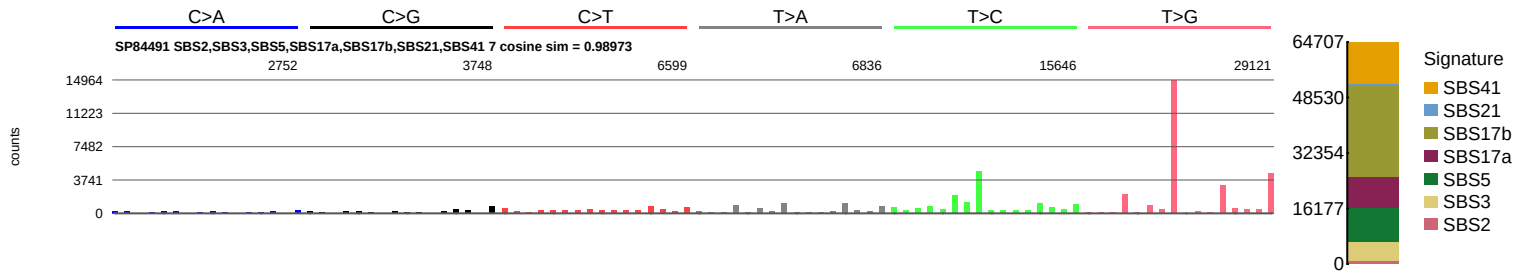
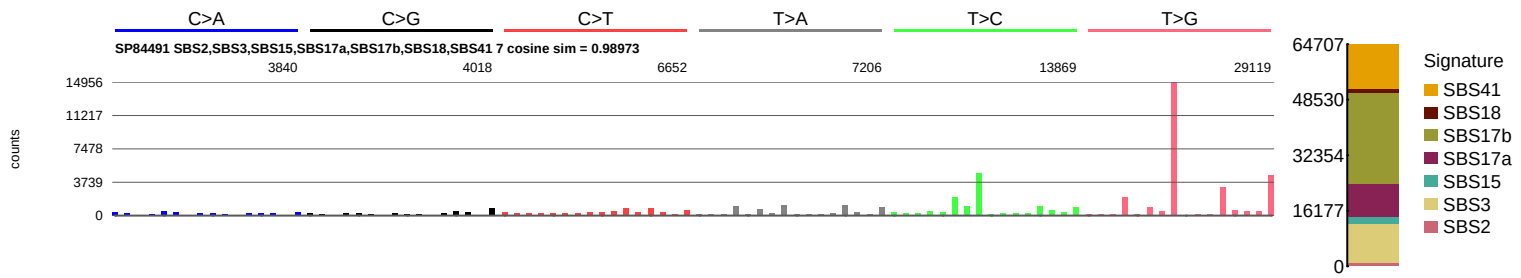
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98973
2752
3748
6599
6836
15646
29121
counts
14964
11223
7482
3741
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2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SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98973
3840
4018
6652
7206
13869
29119
counts
14956
11217
7478
3739
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
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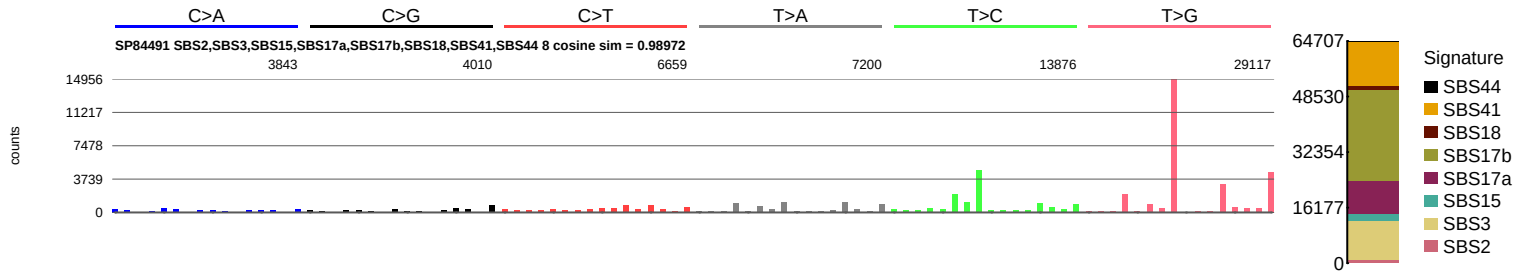
SBS3
SBS2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C>T
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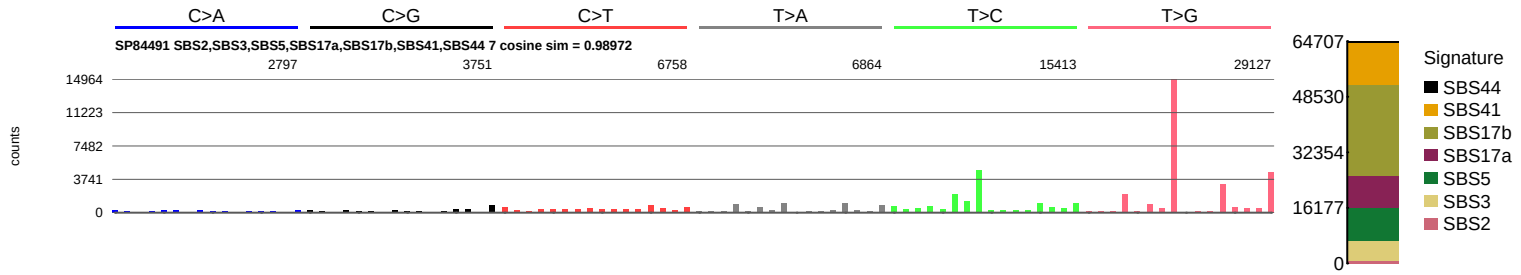
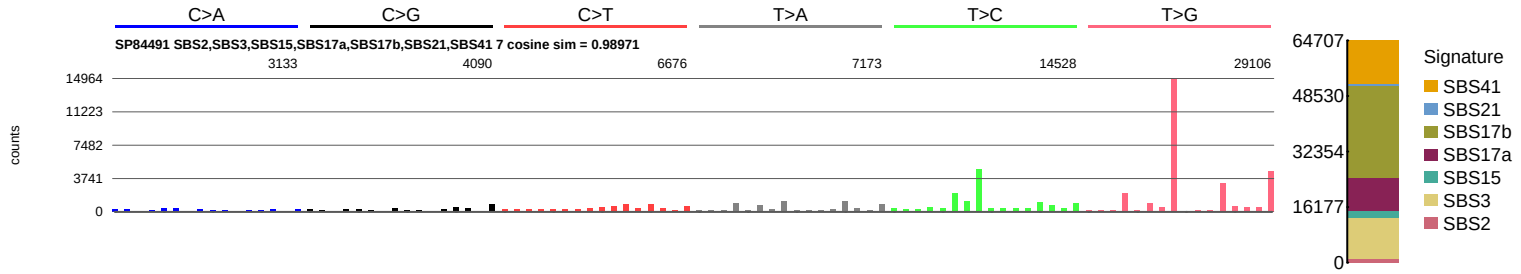
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98972
3843
4010
6659
7200
13876
29117
14956
11217
7478
3739
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98972
2797
3751
6758
6864
15413
29127
14964
11223
7482
3741
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2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C>T
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SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98971
3133
4090
6676
7173
14528
29106
14964
11223
7482
3741
0
counts
64707
48530
32354
16177
0
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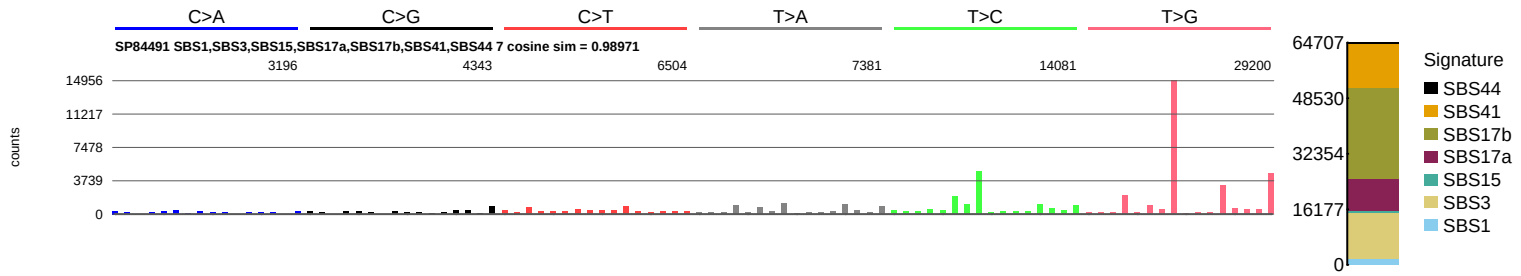
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2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C>G
C>T
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T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98971
3196
4343
6504
7381
14081
29200
14956
11217
7478
3739
0
counts
64707
48530
32354
16177
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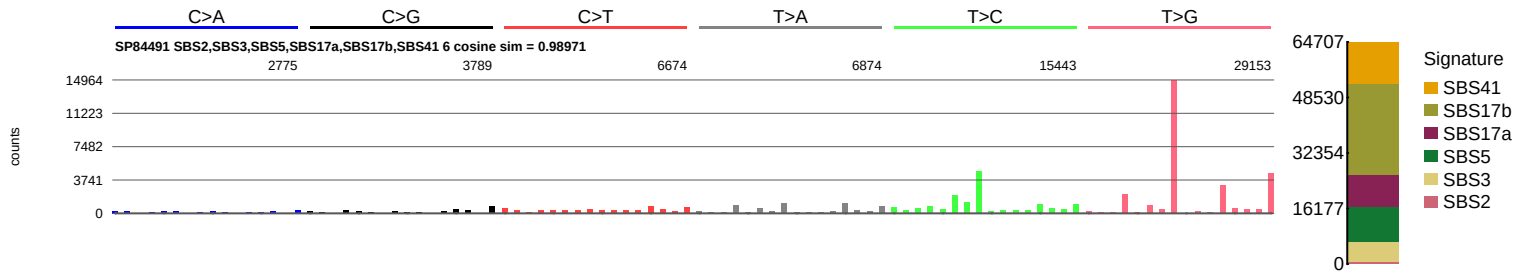
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98971
2775
3789
6674
6874
15443
29153
14964
11223
7482
3741
0
counts
64707
48530
32354
16177
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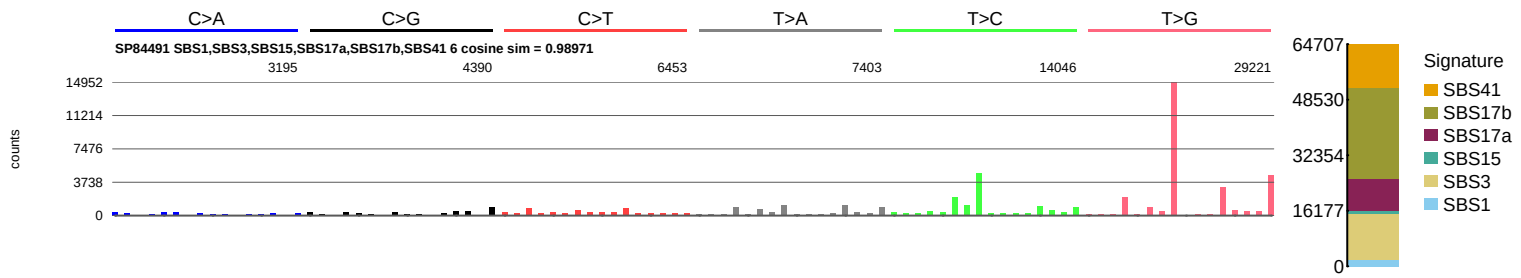
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2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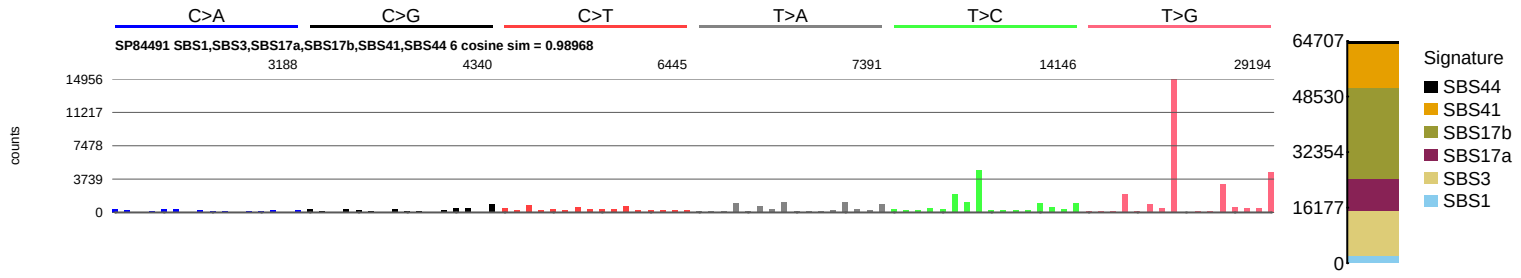
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98971
3195
4390
6453
7403
14046
29221
14952
11214
7476
3738
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS1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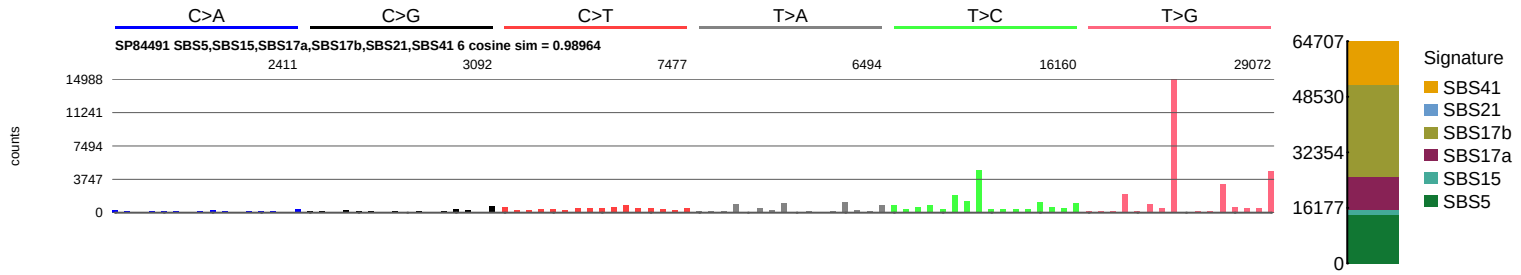
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98968
3188
4340
6445
7391
14146
29194
counts
14956
11217
7478
3739
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS1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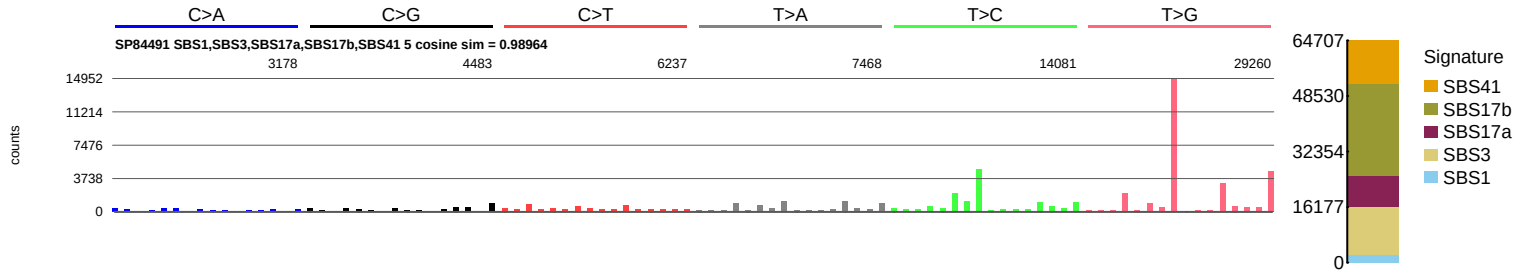
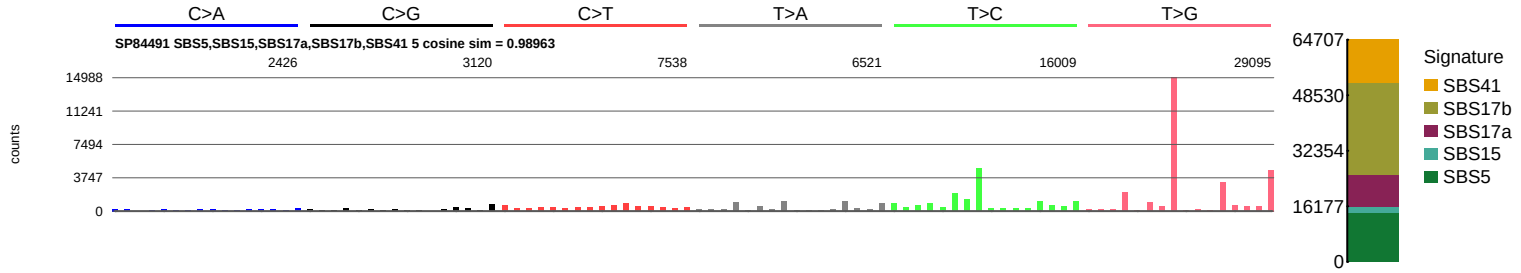
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98964
2411
3092
7477
6494
16160
29072
counts
14988
11241
7494
3747
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98964
3178
4483
6237
7468
14081
29260
counts
14952
11214
7476
3738
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS1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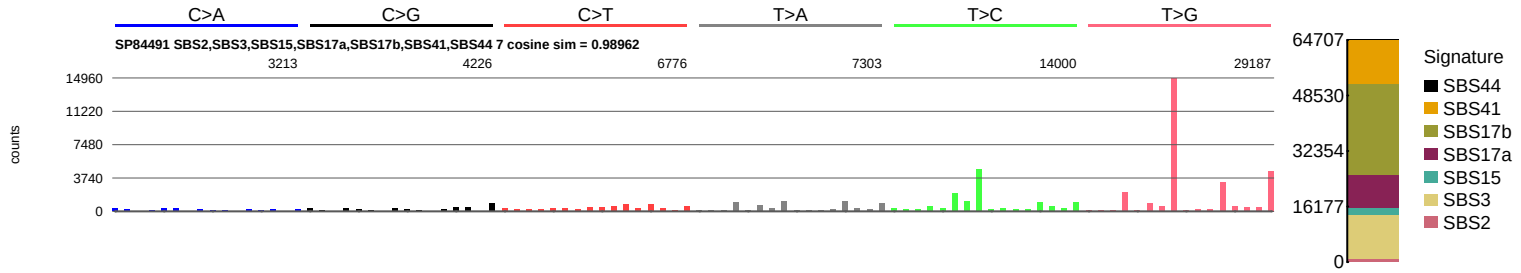
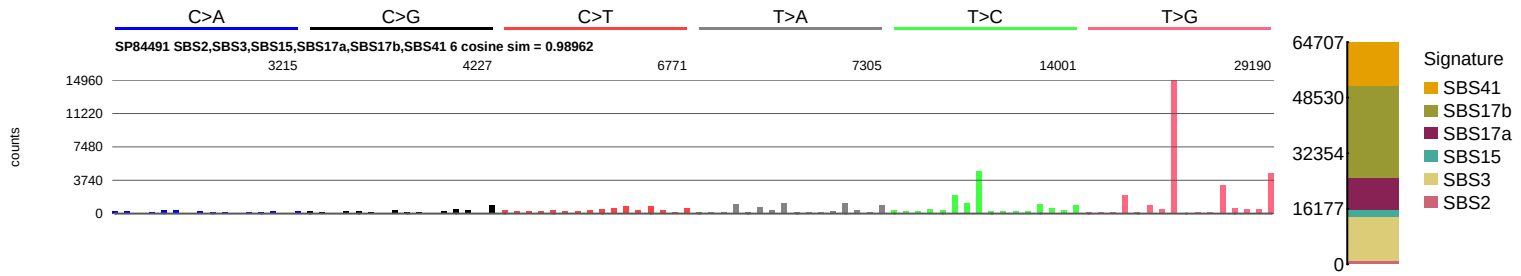
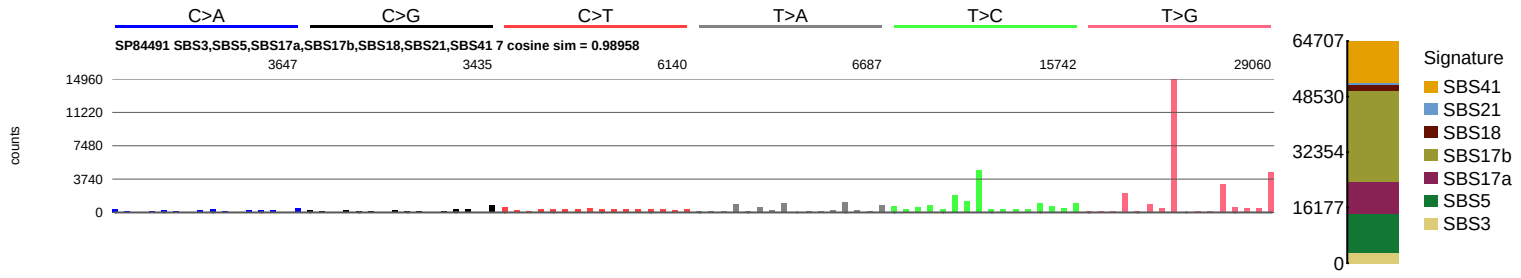
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98963
2426
3120
7538
6521
16009
29095
counts
14988
11241
7494
3747
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98962
3213
4226
6776
7303
14000
29187
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98962
3215
4227
6771
7305
14001
29190
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
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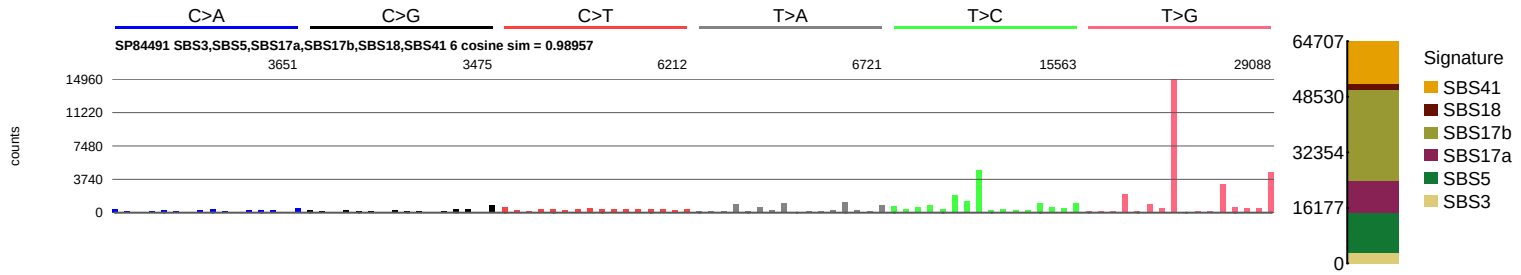
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98958
3647
3435
6140
6687
15742
29060
14960
11220
7480
3740
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98957
3651
3475
6212
6721
15563
29088
14960
11220
7480
3740
0
counts
64707
48530
32354
16177
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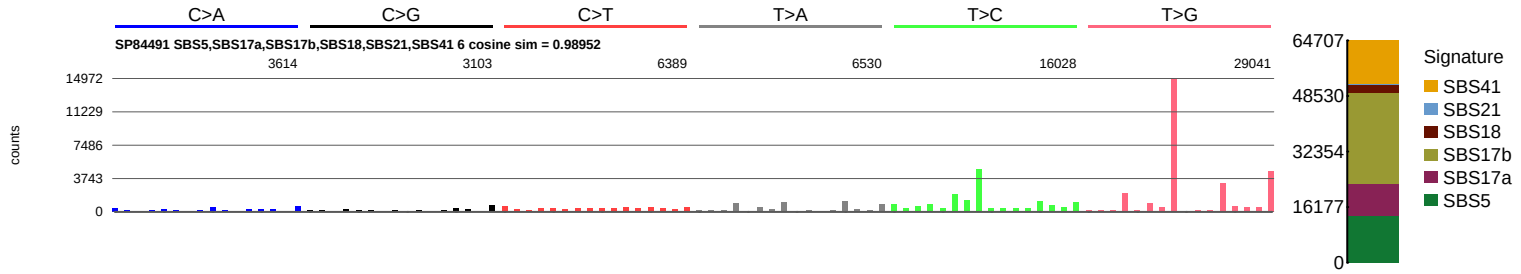
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98952
3614
3103
6389
6530
16028
29041
14972
11229
7486
3743
0
counts
64707
48530
32354
16177
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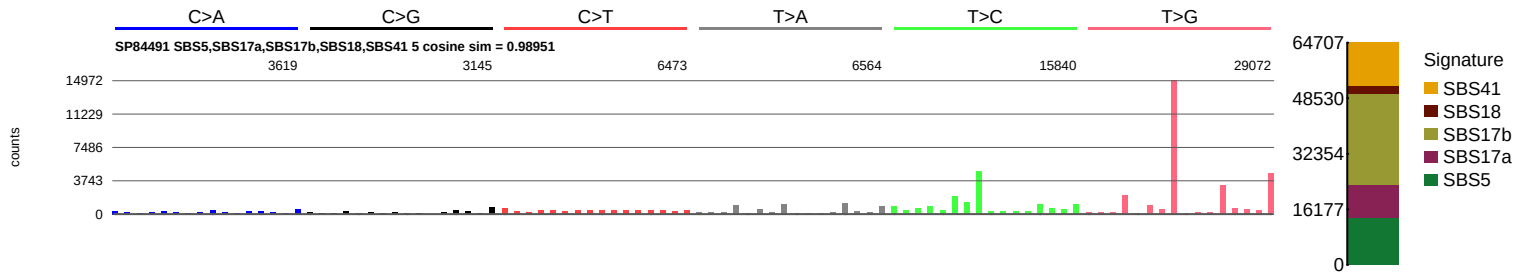
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5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C>G
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T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98951
3619
3145
6473
6564
15840
29072
14972
11229
7486
3743
0
counts
64707
48530
32354
16177
0
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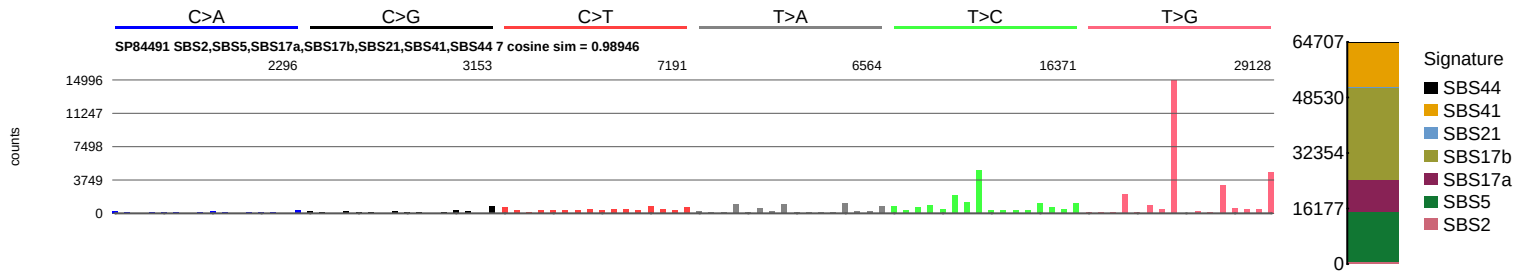
SBS41
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98946
2296
3153
7191
6564
16371
29128
14996
11247
7498
3749
0
counts
64707
48530
32354
16177
0
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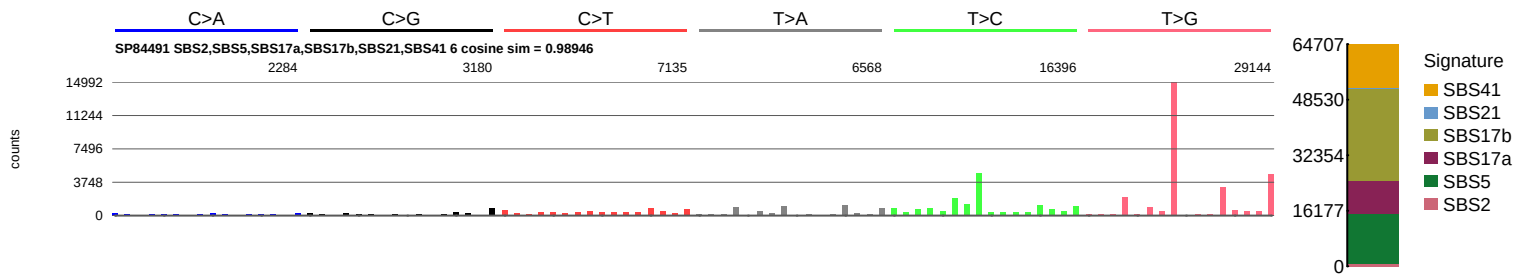
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98946
2284
3180
7135
6568
16396
29144
14992
11244
7496
3748
0
counts
64707
48530
32354
16177
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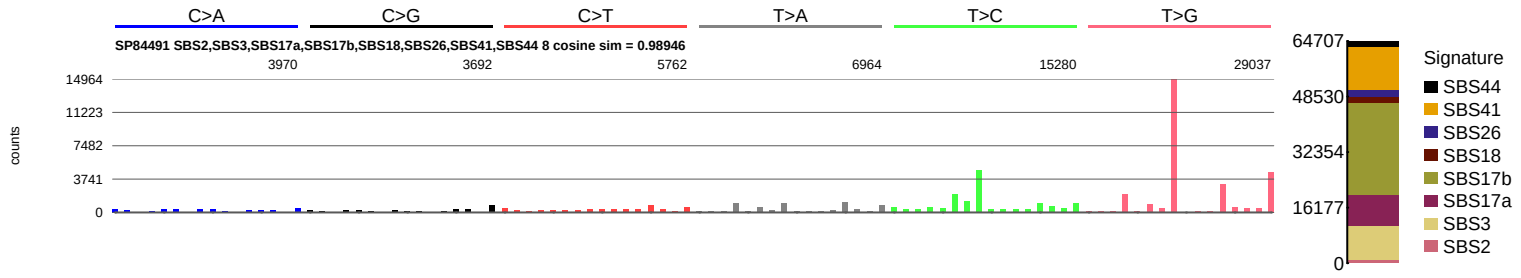
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2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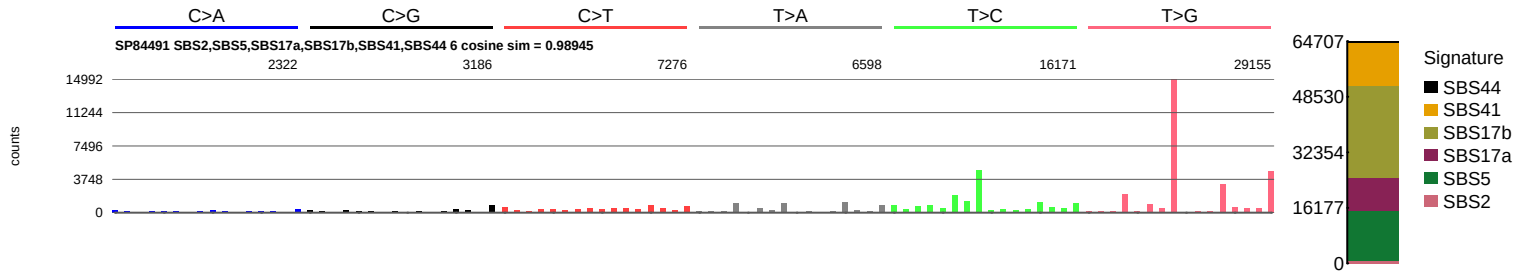
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98946
3970
3692
5762
6964
15280
29037
counts
14964
11223
7482
3741
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98945
2322
3186
7276
6598
16171
29155
counts
14992
11244
7496
3748
0
64707
48530
32354
16177
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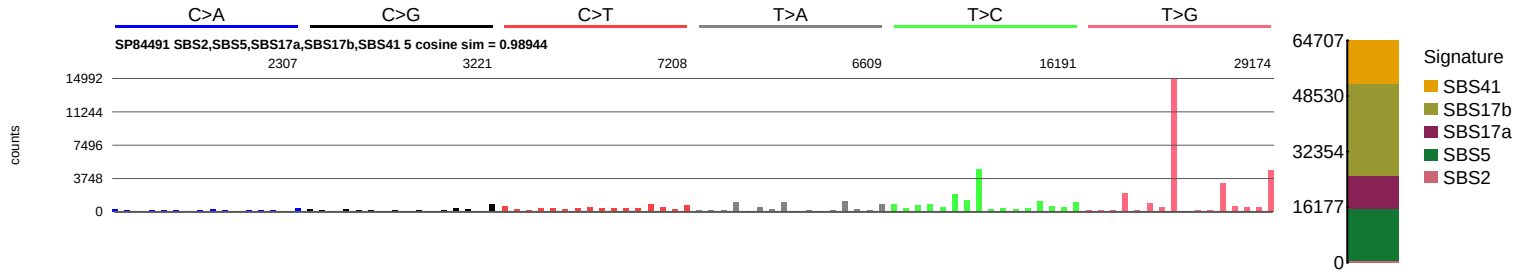
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98944
2307
3221
7208
6609
16191
29174
counts
14992
11244
7496
3748
0
64707
48530
32354
16177
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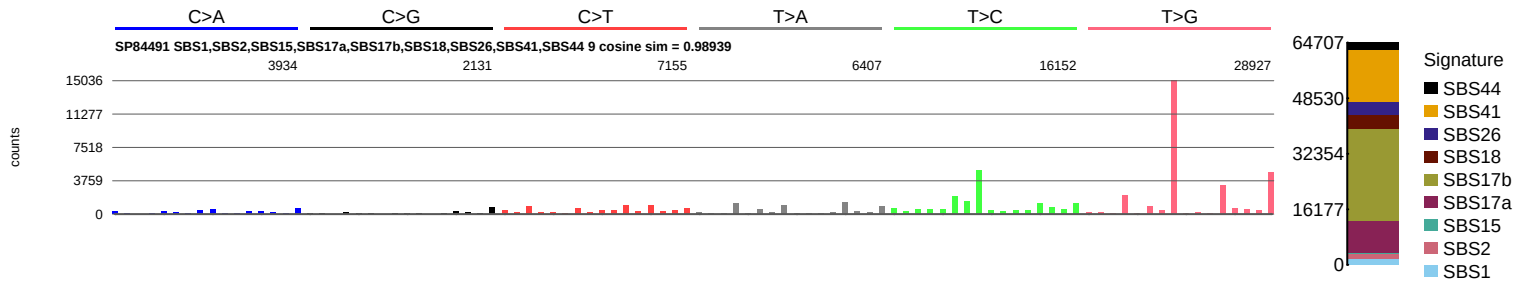
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98939
3934
2131
7155
6407
16152
28927
counts
15036
11277
7518
3759
0
64707
48530
32354
16177
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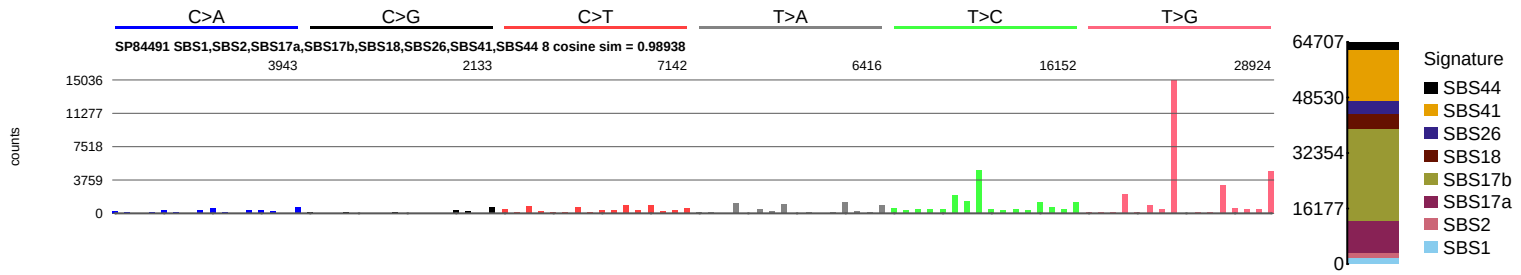
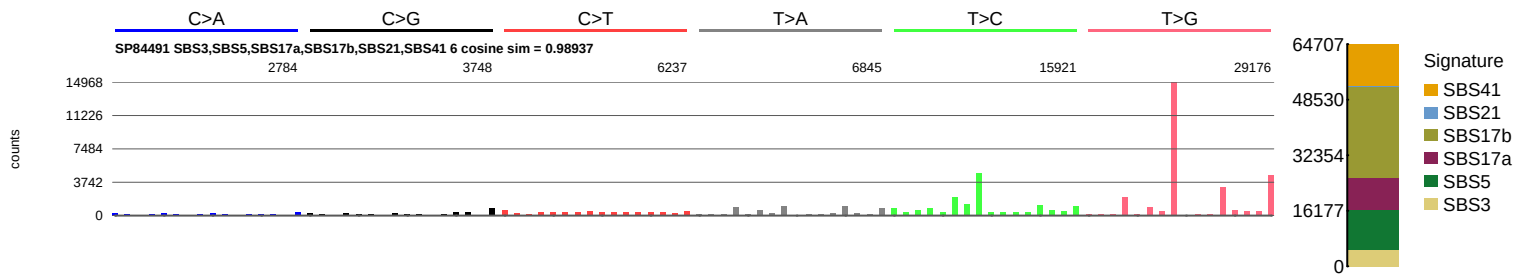
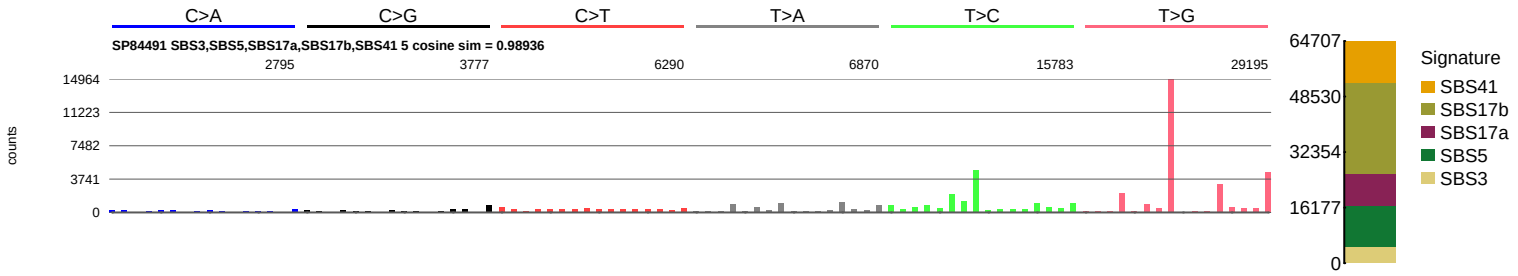
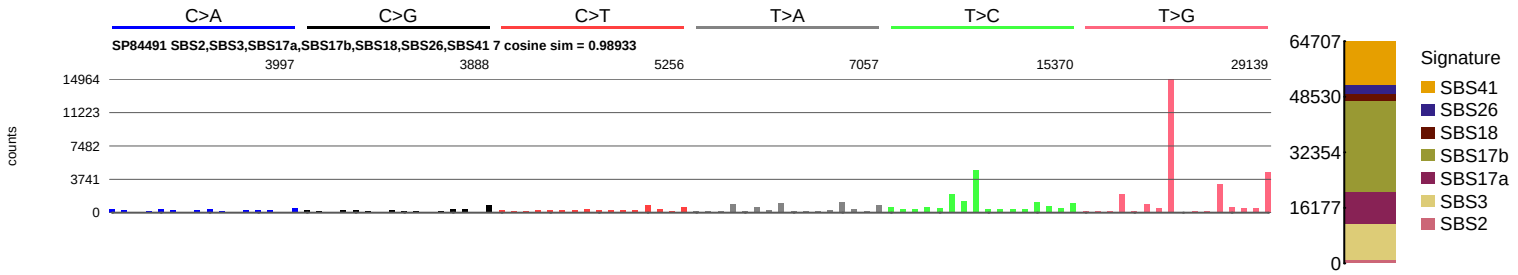
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98938
3943
2133
7142
6416
16152
28924
counts
15036
11277
7518
3759
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98937
2784
3748
6237
6845
15921
29176
counts
14968
11226
7484
3742
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98936
2795
3777
6290
6870
15783
29195
14964
11223
7482
3741
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
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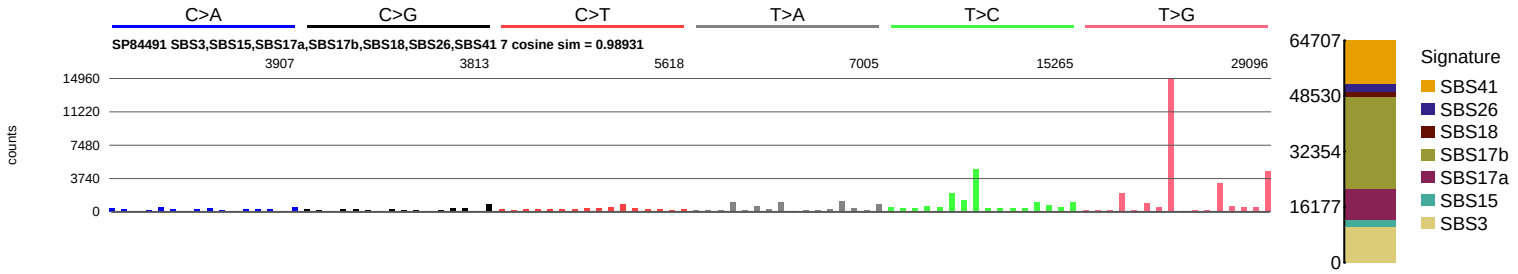
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98933
3997
3888
5256
7057
15370
29139
14964
11223
7482
3741
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2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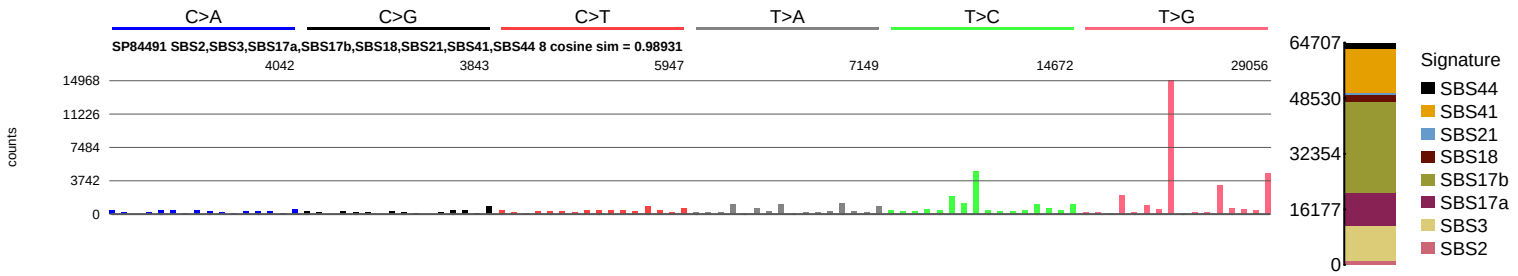
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98931
3907
3813
5618
7005
15265
29096
14960
11220
7480
3740
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3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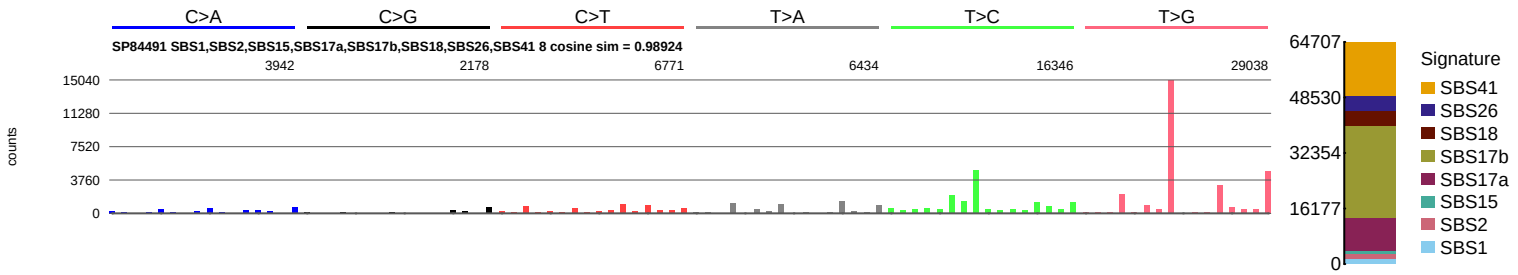
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98931
4042
3843
5947
7149
14672
29056
14968
11226
7484
3742
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98924
3942
2178
6771
6434
16346
29038
15040
11280
7520
3760
0
counts
64707
48530
32354
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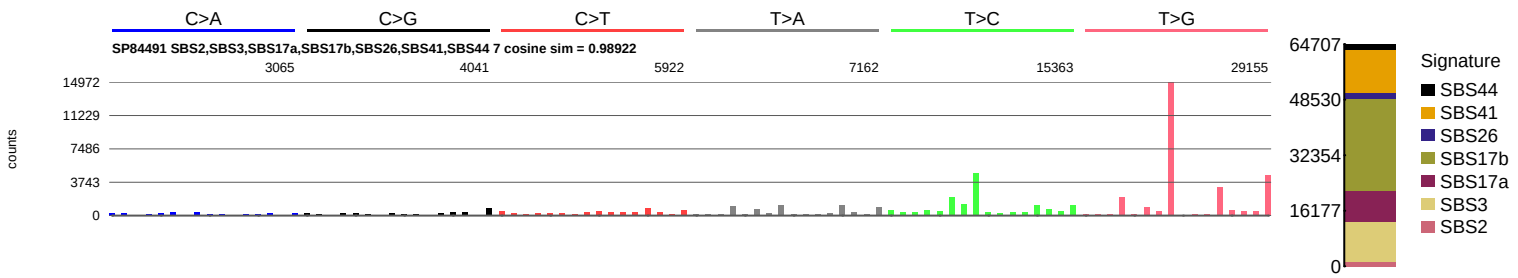
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98922
3065
4041
5922
7162
15363
29155
14972
11229
7486
3743
0
counts
64707
48530
32354
16177
0
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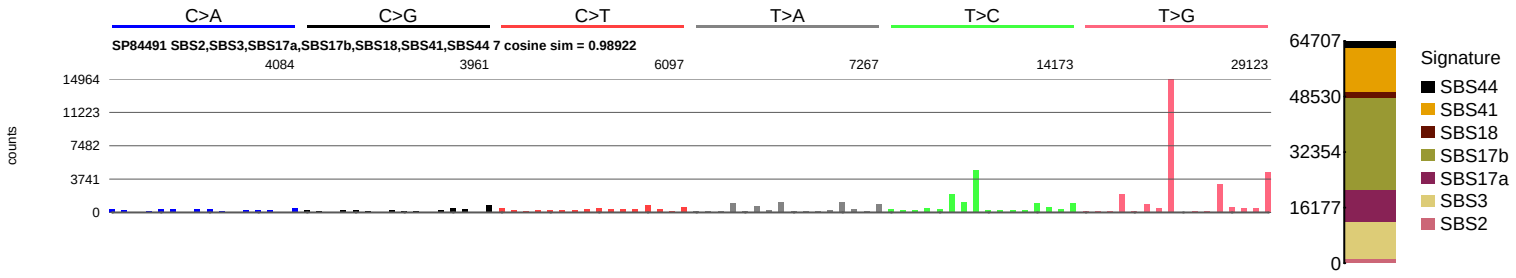
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98922
4084
3961
6097
7267
14173
29123
counts
14964
11223
7482
3741
0
64707
48530
32354
16177
0
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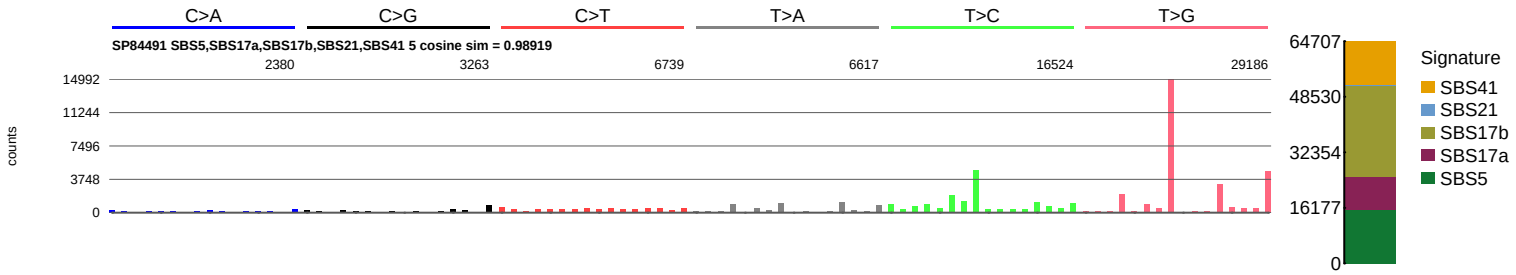
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2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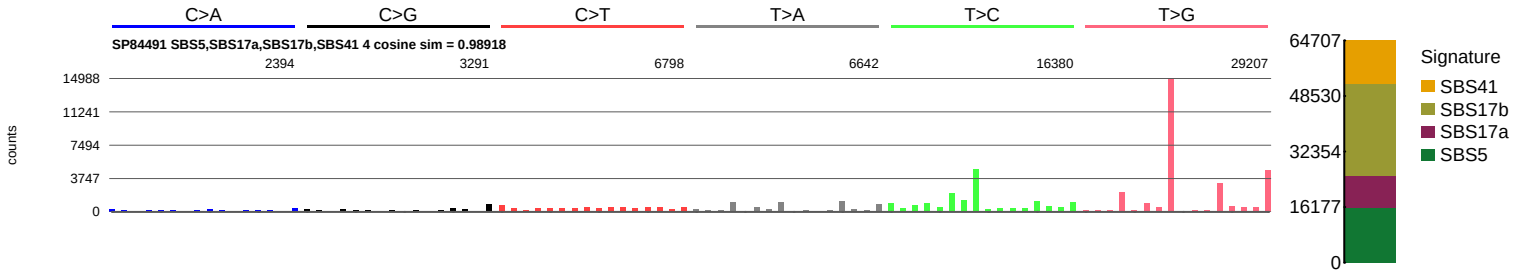
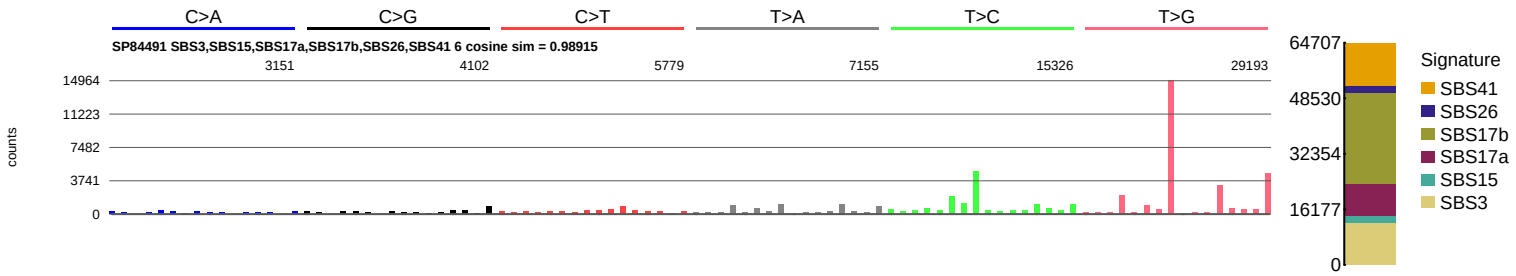
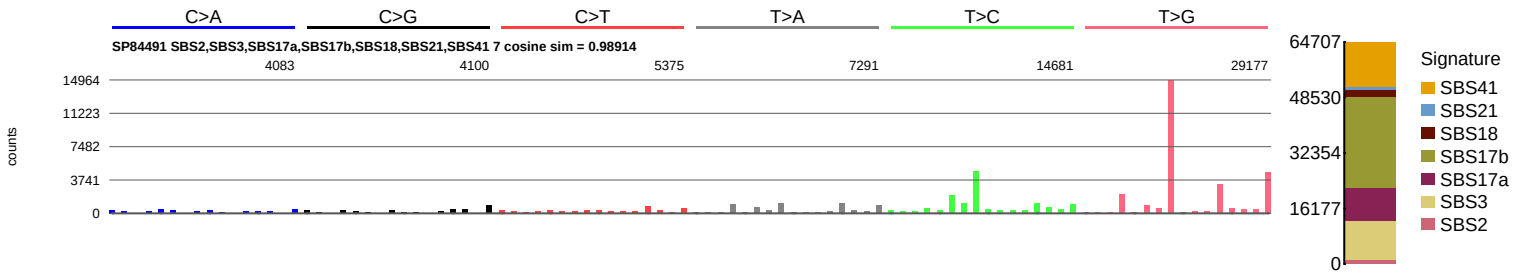
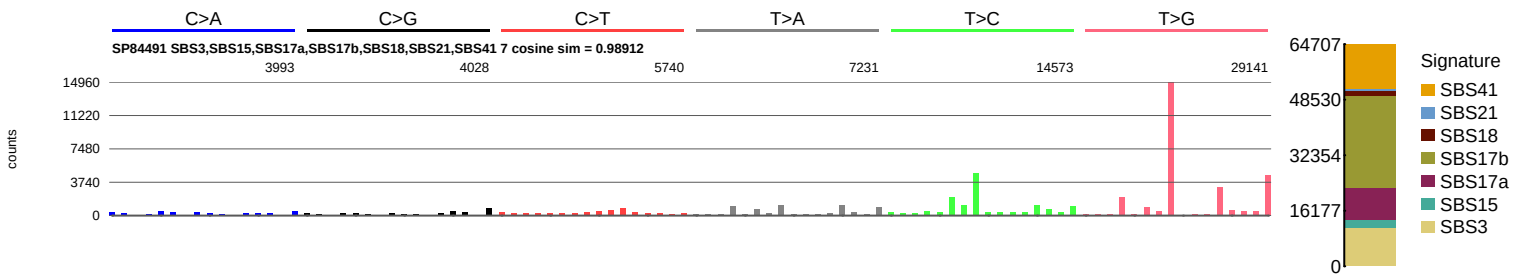
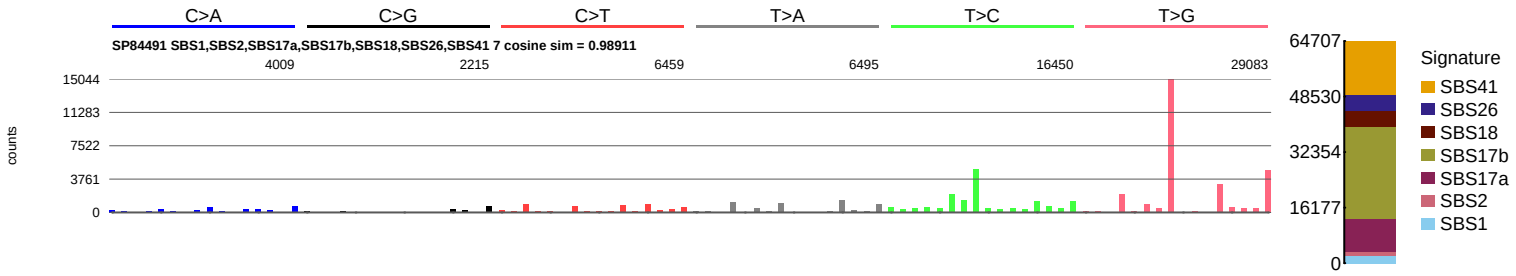
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98919
2380
3263
6739
6617
16524
29186
counts
14992
11244
7496
3748
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5

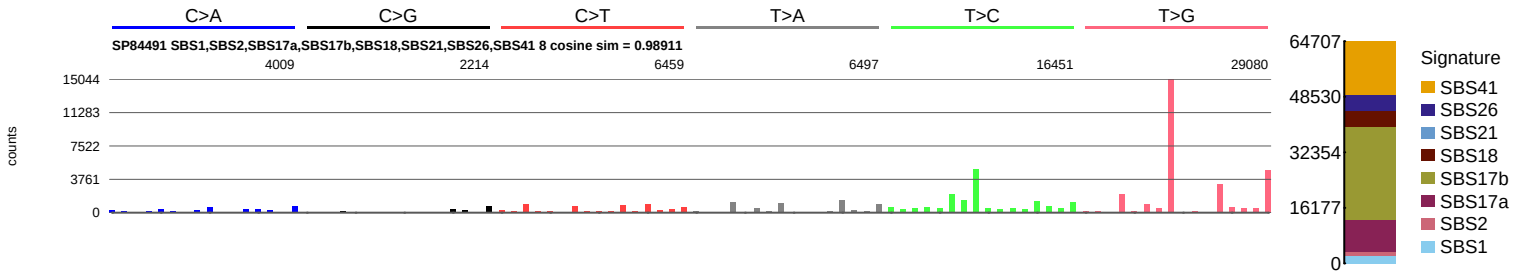

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS41 4 cosine sim = 0.98918
2394
3291
6798
6642
16380
29207
counts
14988
11241
7494
3747
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS5

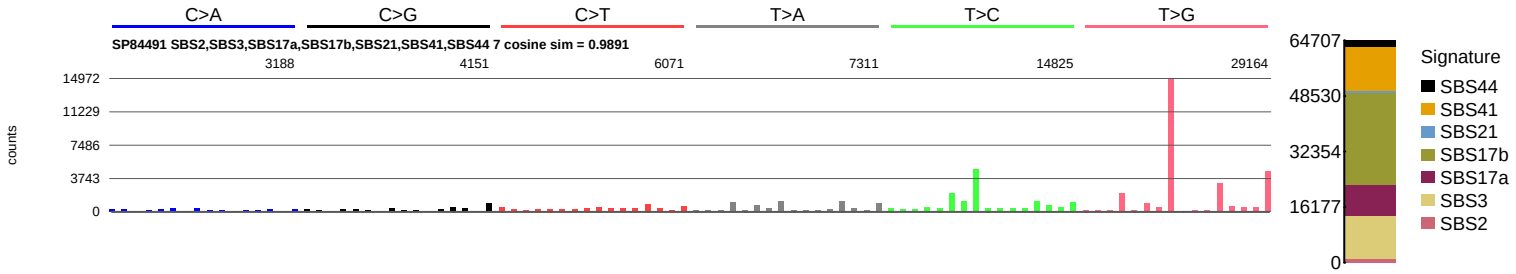

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98915
3151
4102
5779
7155
15326
29193
counts
14964
11223
7482
3741
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3

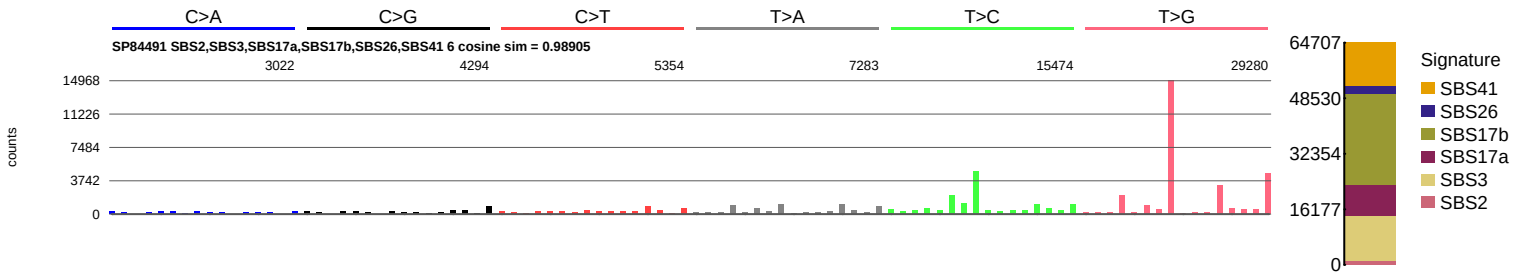

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98914
4083
4100
5375
7291
14681
29177
counts
14964
11223
7482
3741
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98912
3993
4028
5740
7231
14573
29141
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98911
4009
2215
6459
6495
16450
29083
counts
15044
11283
7522
3761
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98911
4009
2214
6459
6497
16451
29080
counts
15044
11283
7522
3761
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.9891
3188
4151
6071
7311
14825
29164
counts
14972
11229
7486
3743
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98905
3022
4294
5354
7283
15474
29280
counts
14968
11226
7484
3742
0
64707
48530
32354
16177
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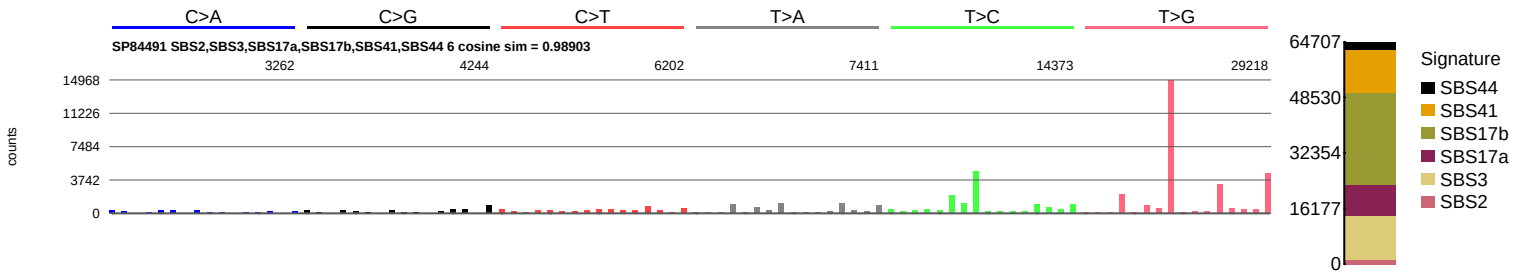
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98903
3262
4244
6202
7411
14373
29218
counts
14968
11226
7484
3742
0
64707
48530
32354
16177
0
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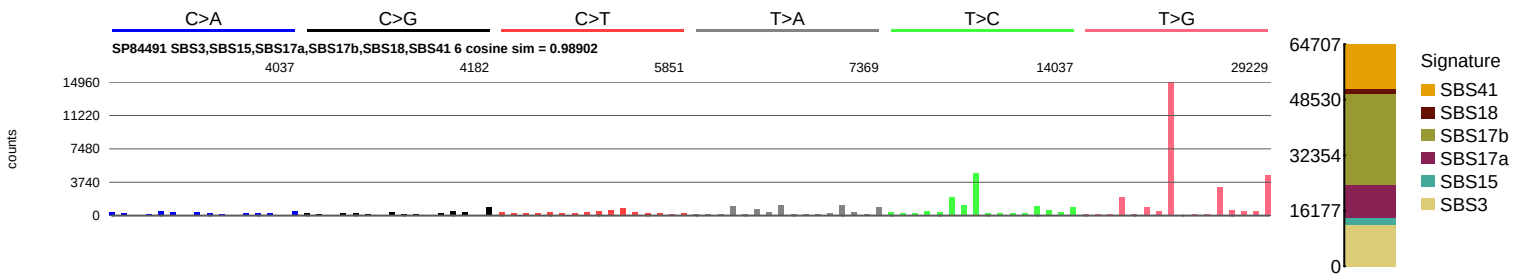
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98902
4037
4182
5851
7369
14037
29229
counts
14960
11220
7480
3740
0
64707
48530
32354
16177
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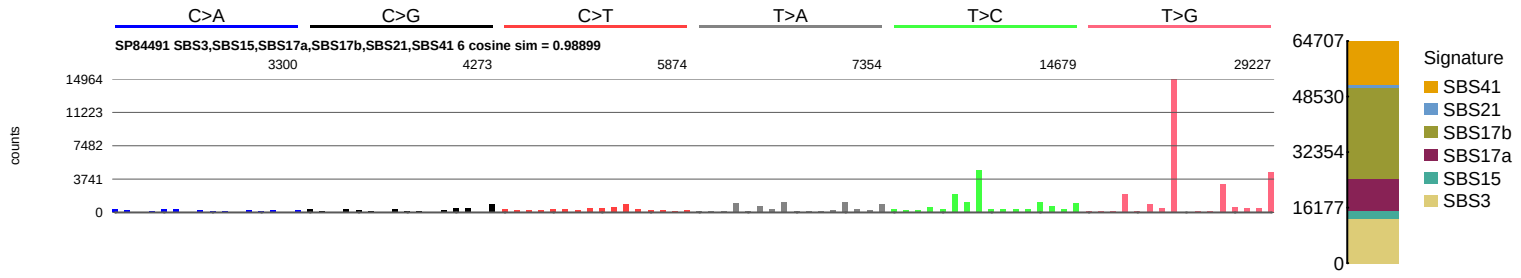
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98899
3300
4273
5874
7354
14679
29227
14964
11223
7482
3741
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counts
64707
48530
32354
16177
0
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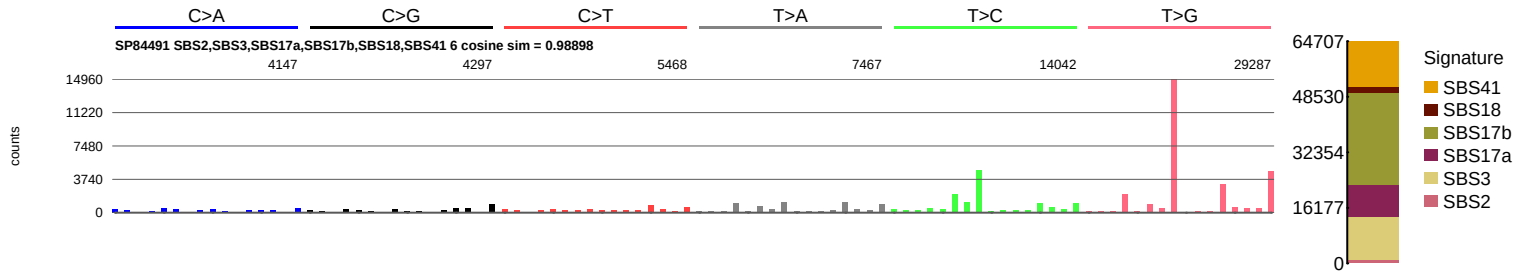
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98898
4147
4297
5468
7467
14042
29287
14960
11220
7480
3740
0
counts
64707
48530
32354
16177
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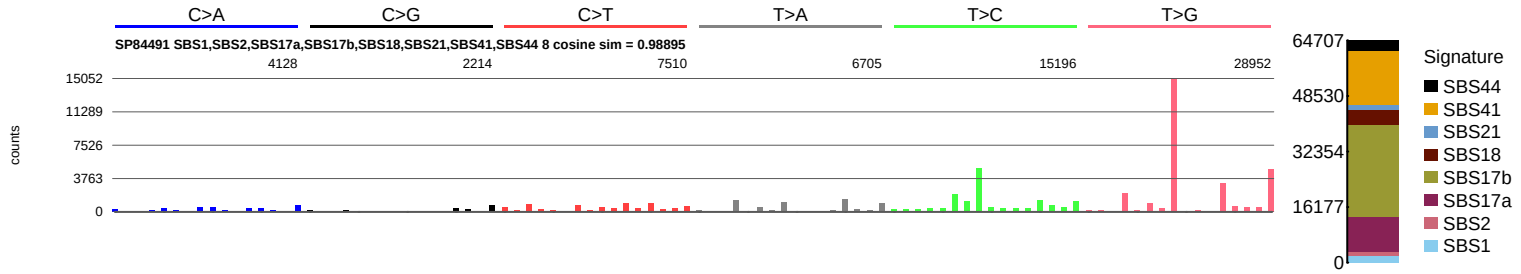
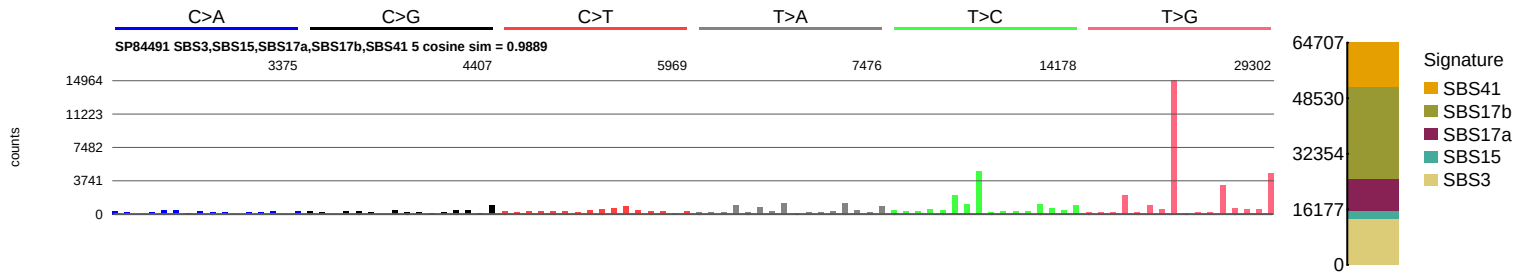
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98895
4128
2214
7510
6705
15196
28952
15052
11289
7526
3763
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.9889
3375
4407
5969
7476
14178
29302
14964
11223
7482
3741
0
counts
64707
48530
32354
16177
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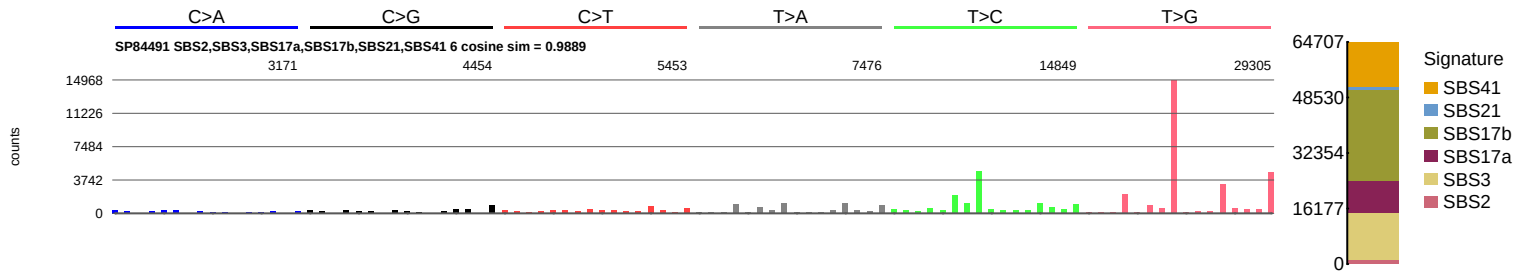
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.9889
3171
4454
5453
7476
14849
29305
14968
11226
7484
3742
0
counts
64707
48530
32354
16177
0
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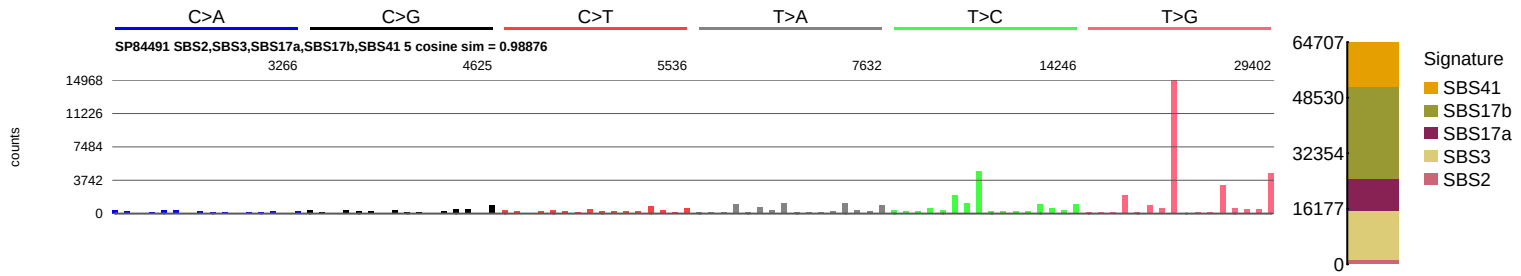
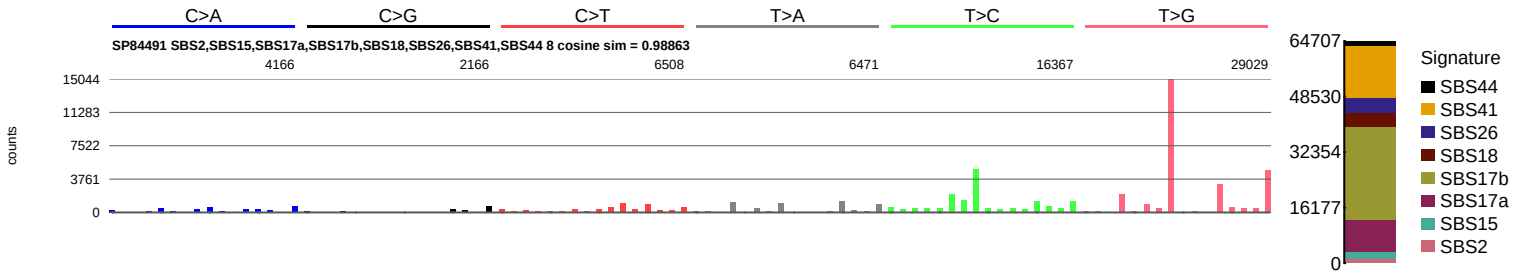
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98876
3266
4625
5536
7632
14246
29402
14968
11226
7484
3742
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98863
4166
2166
6508
6471
16367
29029
counts
15044
11283
7522
3761
0
64707
48530
32354
16177
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Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
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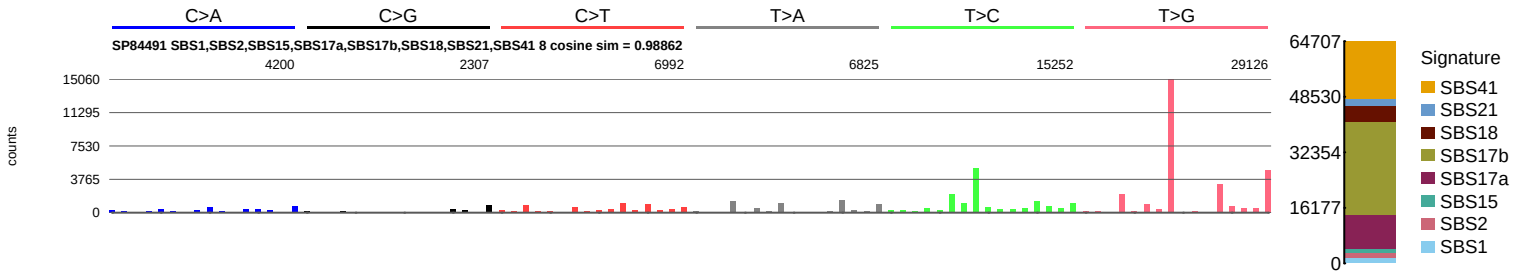
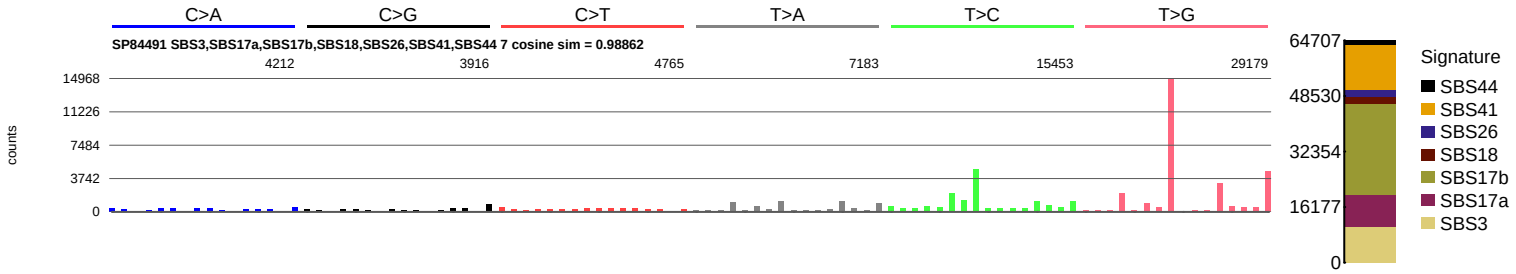
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98862
4200
2307
6992
6825
15252
29126
counts
15060
11295
7530
3765
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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4212
3916
4765
7183
15453
29179
counts
14968
11226
7484
3742
0
64707
48530
32354
16177
0
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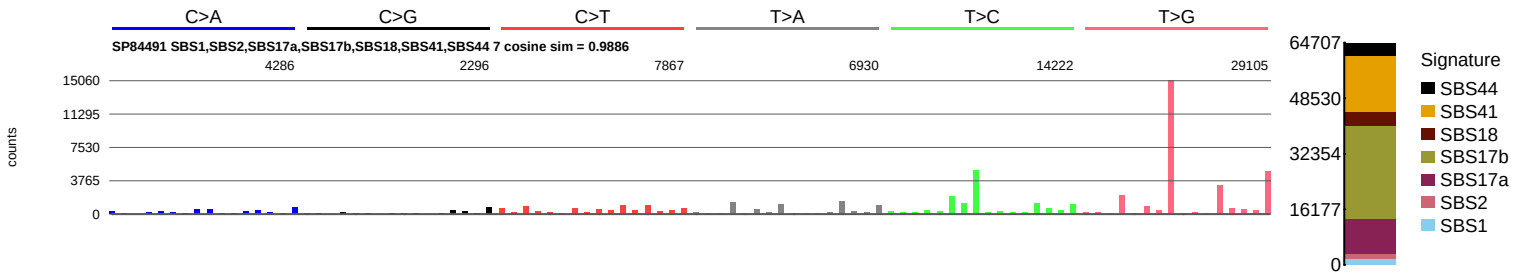
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3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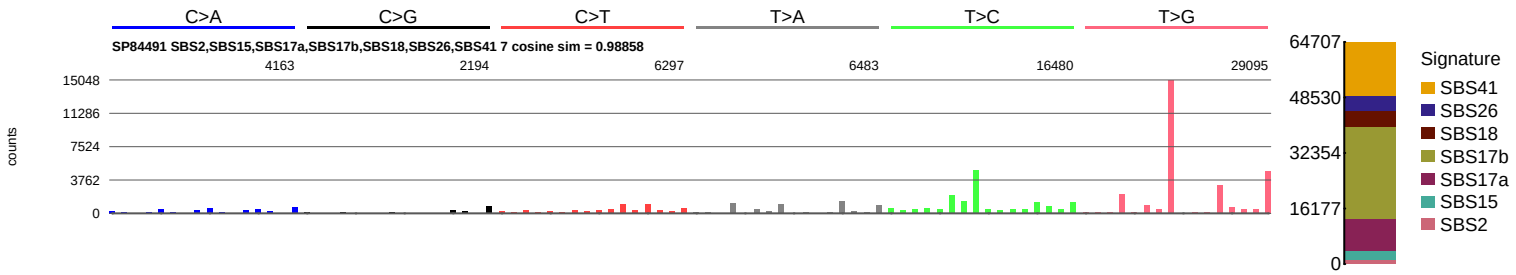
4286
2296
7867
6930
14222
29105
counts
15060
11295
7530
3765
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1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4163
2194
6297
6483
16480
29095
counts
15048
11286
7524
3762
0
64707
48530
32354
16177
0
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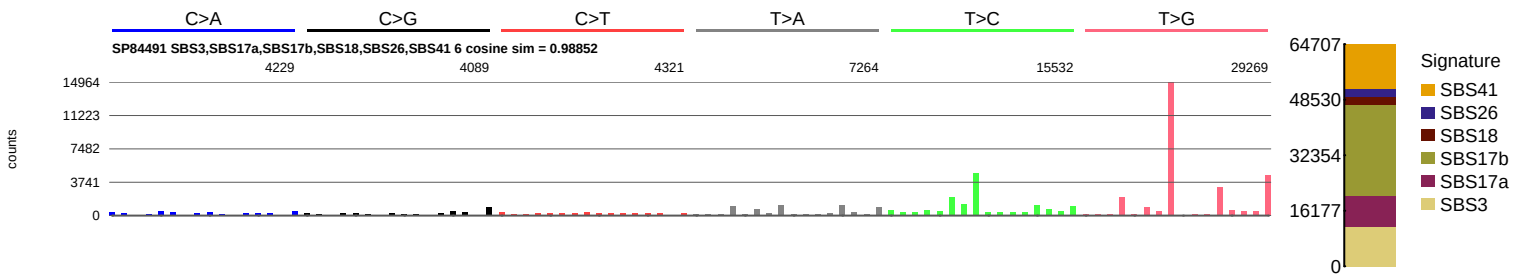
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2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SP84491 SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98852
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4089
4321
7264
15532
29269
counts
14964
11223
7482
3741
0
64707
48530
32354
16177
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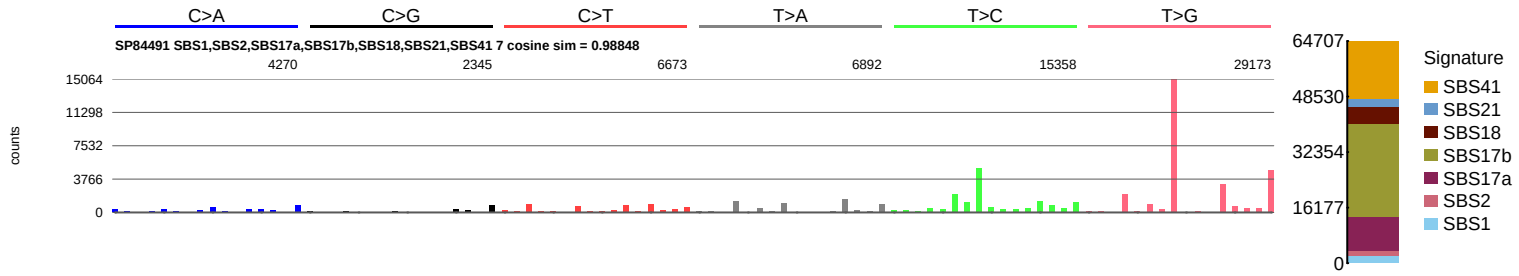
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3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SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98848
4270
2345
6673
6892
15358
29173
15064
11298
7532
3766
0
counts
64707
48530
32354
16177
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SBS21
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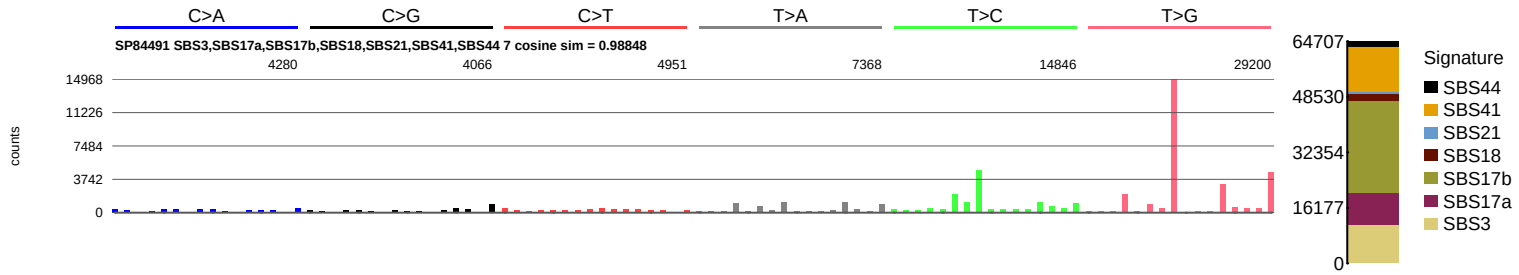
SBS17b
SBS17a
SBS2
SBS1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SP84491 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98848
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4066
4951
7368
14846
29200
14968
11226
7484
3742
0
counts
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48530
32354
16177
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SBS41
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SBS18
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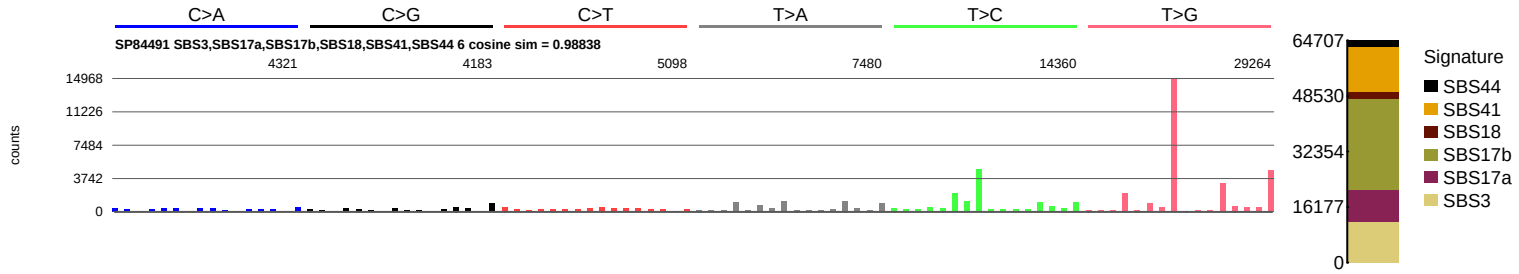
SBS17a
SBS3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SP84491 SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98838
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4183
5098
7480
14360
29264
14968
11226
7484
3742
0
counts
64707
48530
32354
16177
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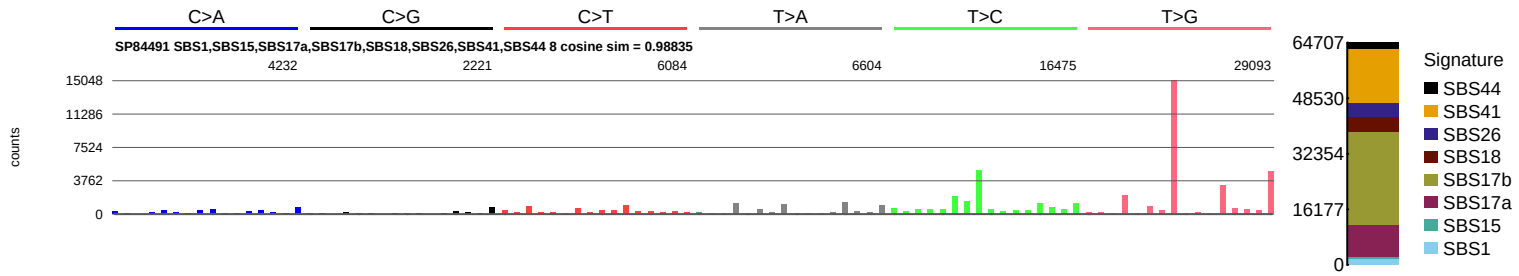
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3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SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98835
4232
2221
6084
6604
16475
29093
15048
11286
7524
3762
0
counts
64707
48530
32354
16177
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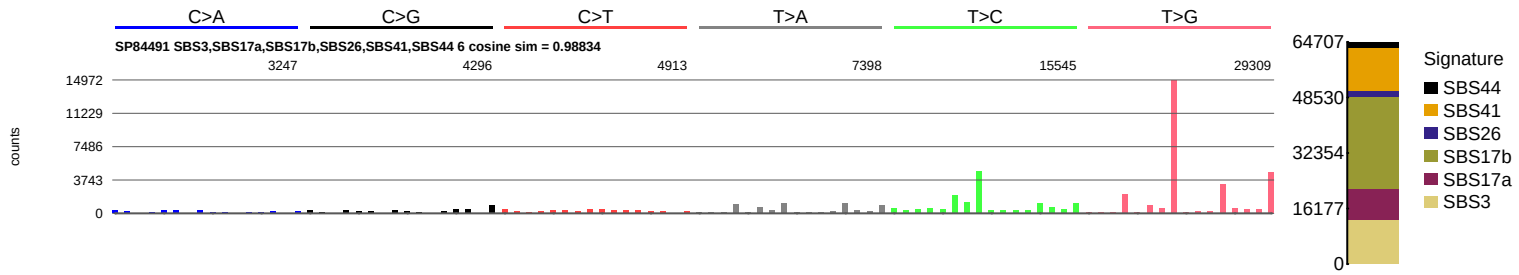
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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SP84491 SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98834
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4296
4913
7398
15545
29309
14972
11229
7486
3743
0
counts
64707
48530
32354
16177
0
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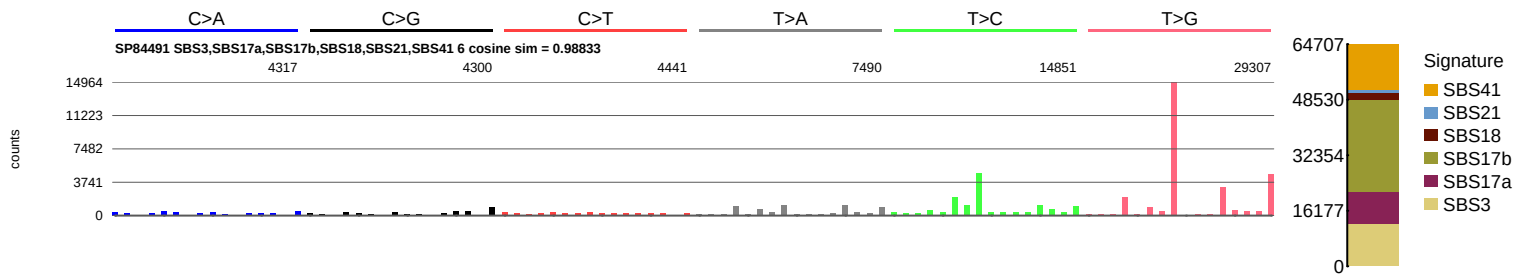
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3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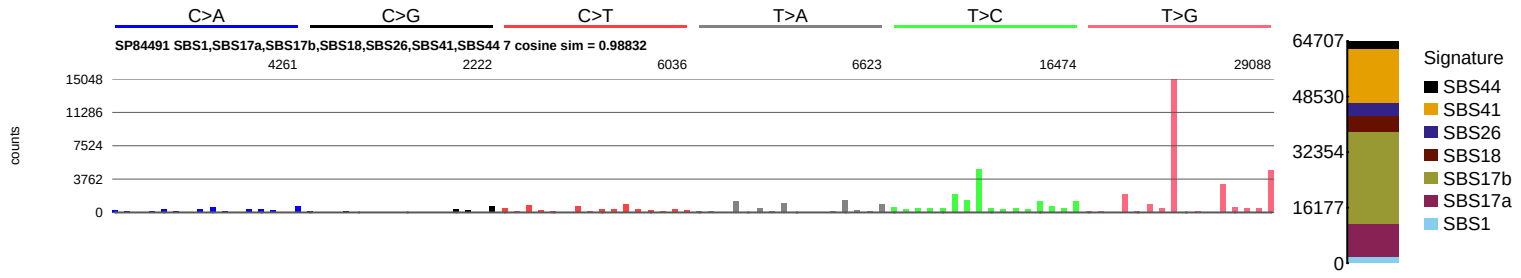
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98833
4317
4300
4441
7490
14851
29307
14964
11223
7482
3741
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98832
4261
2222
6036
6623
16474
29088
15048
11286
7524
3762
0
counts
64707
48530
32354
16177
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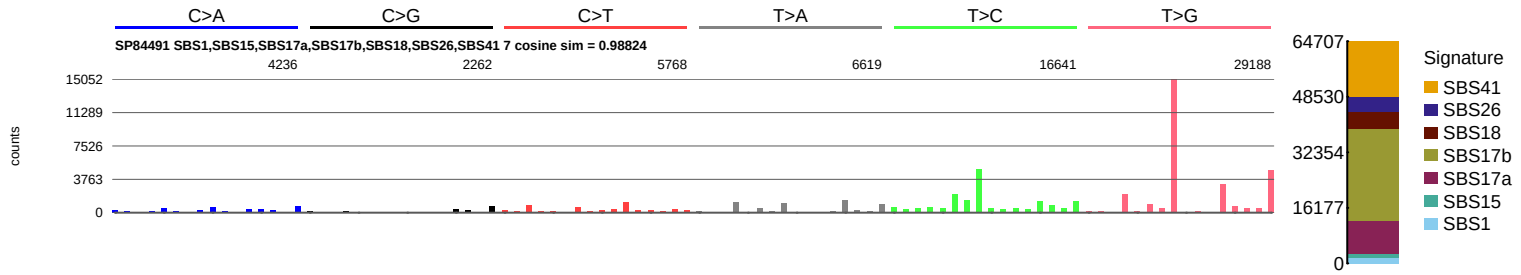
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1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C>G
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T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98824
4236
2262
5768
6619
16641
29188
15052
11289
7526
3763
0
counts
64707
48530
32354
16177
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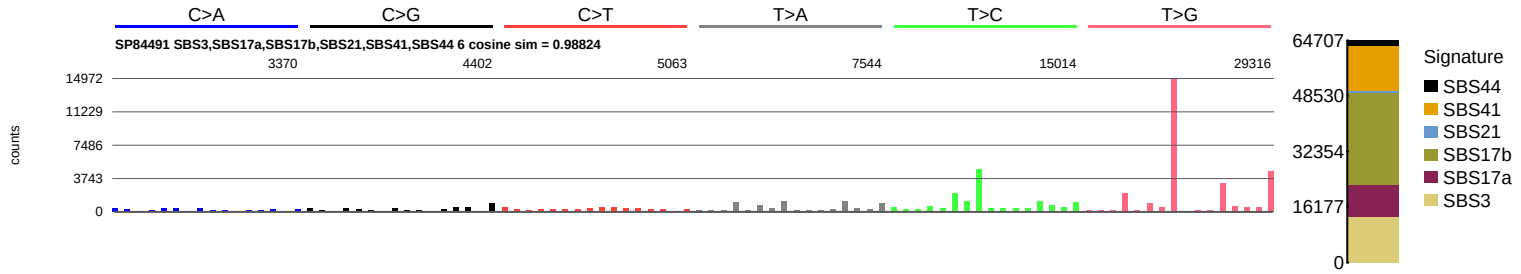
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.98824
3370
4402
5063
7544
15014
29316
14972
11229
7486
3743
0
counts
64707
48530
32354
16177
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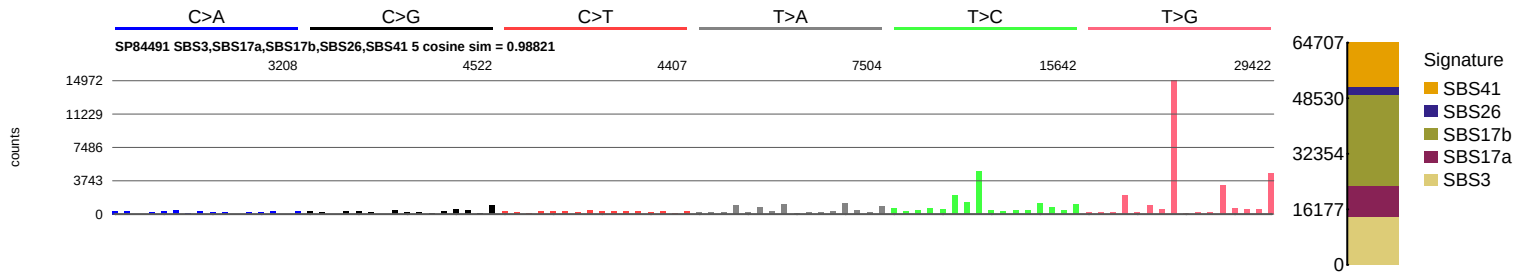
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3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C>G
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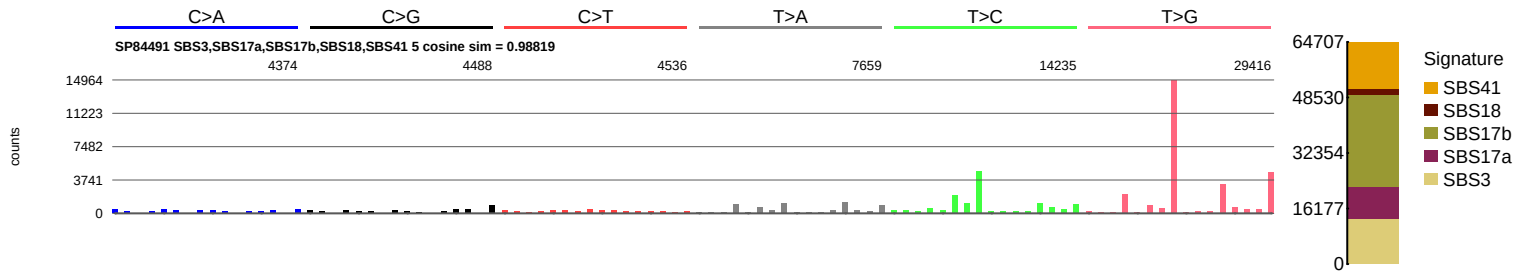
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS26,SBS41 5 cosine sim = 0.98821
3208
4522
4407
7504
15642
29422
14972
11229
7486
3743
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3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T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98819
4374
4488
4536
7659
14235
29416
14964
11223
7482
3741
0
counts
64707
48530
32354
16177
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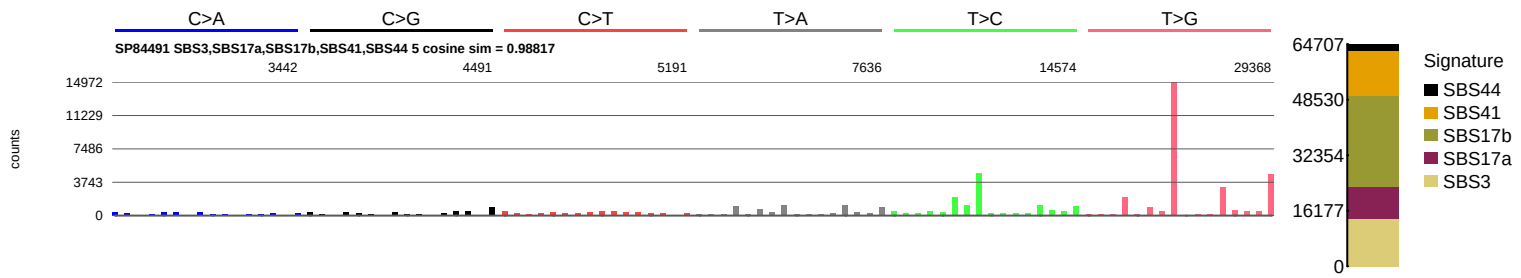
SBS41
SBS18
SBS17b
SBS17a
SBS3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SP84491 SBS3,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.98817
3442
4491
5191
7636
14574
29368
14972
11229
7486
3743
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48530
32354
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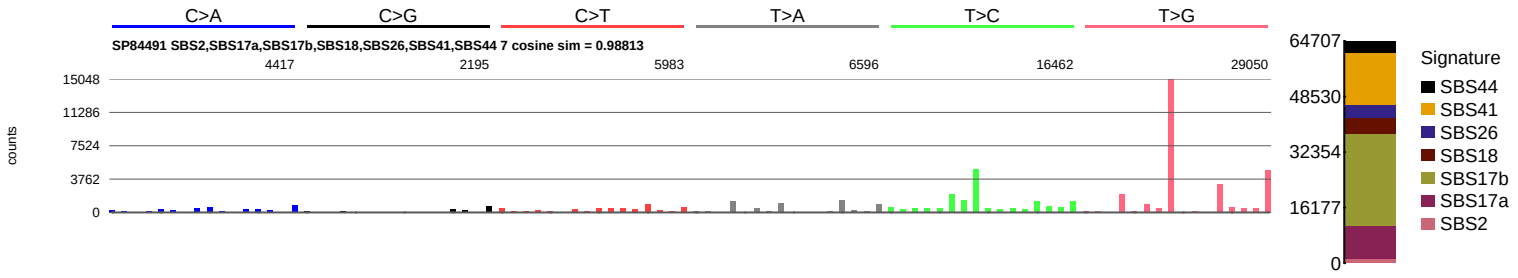
SBS41
SBS17b
SBS17a
SBS3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T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98813
4417
2195
5983
6596
16462
29050
15048
11286
7524
3762
0
counts
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48530
32354
16177
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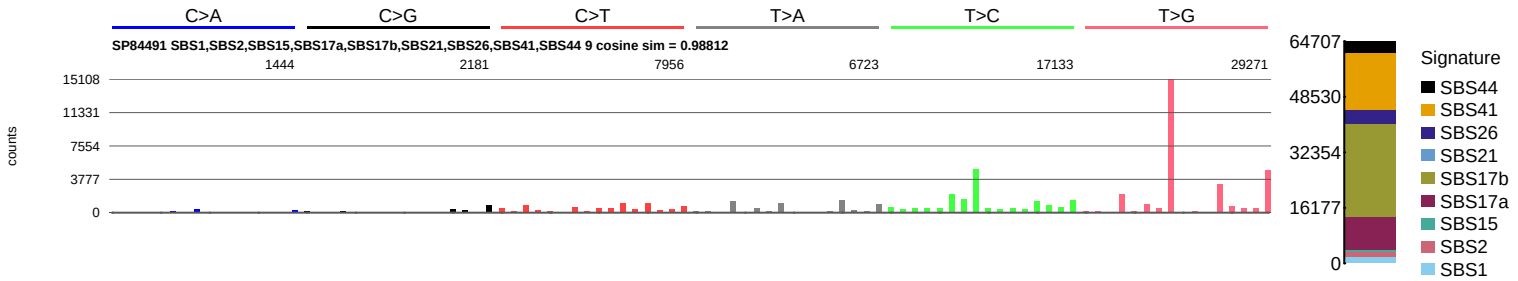
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2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T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98812
1444
2181
7956
6723
17133
29271
15108
11331
7554
3777
0
counts
64707
48530
32354
16177
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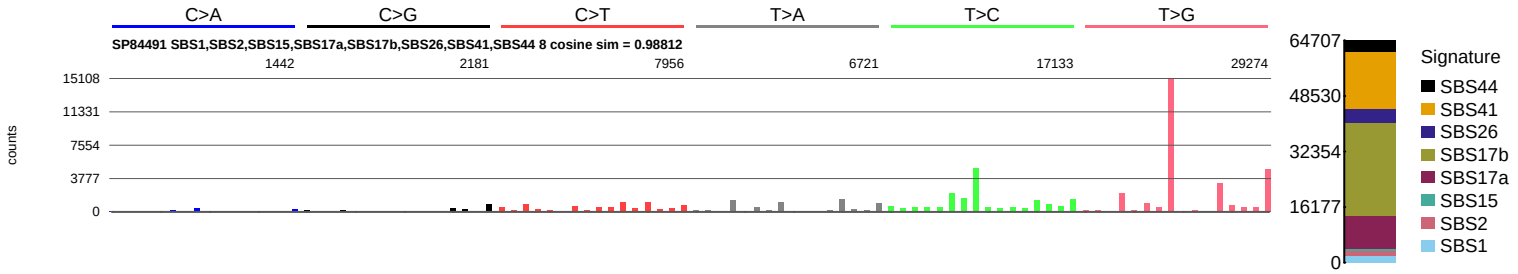
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
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T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98812
1442
2181
7956
6721
17133
29274
15108
11331
7554
3777
0
counts
64707
48530
32354
16177
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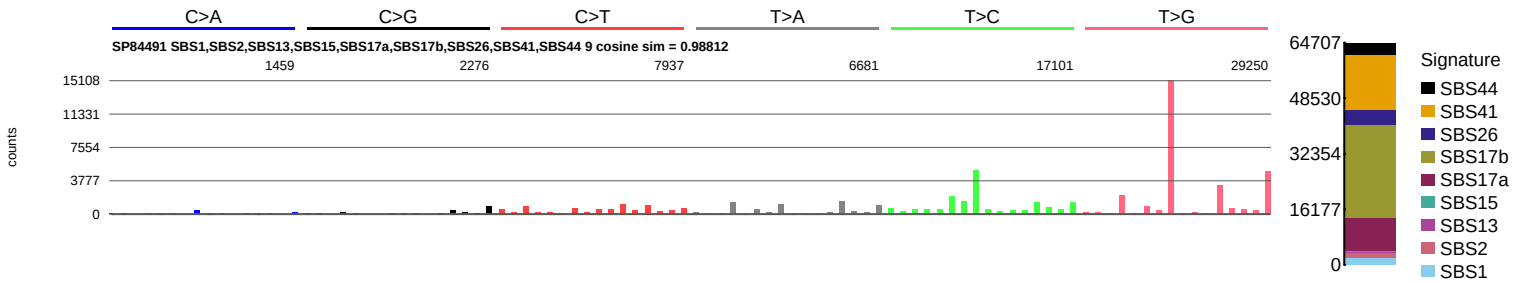
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98812
1459
2276
7937
6681
17101
29250
15108
11331
7554
3777
0
counts
64707
48530
32354
16177
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SBS44
SBS41
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SBS17b
SBS17a
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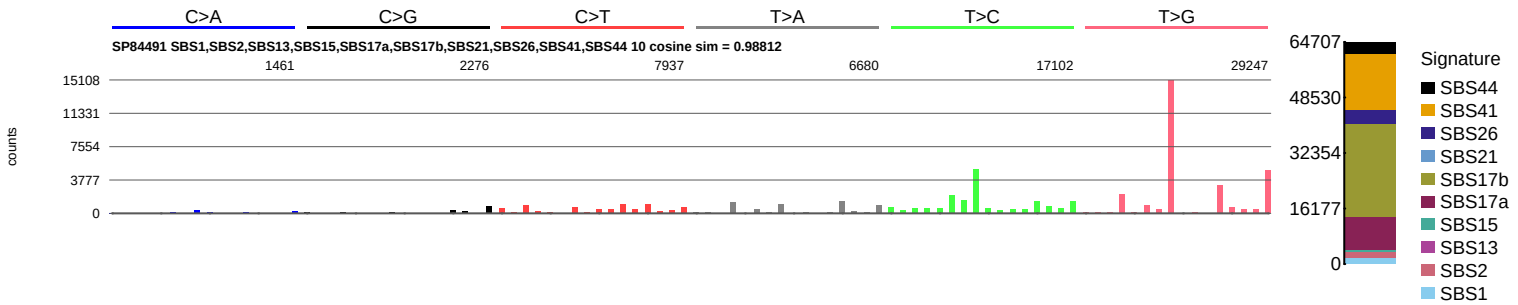
SBS13
SBS2
SBS1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C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98812
1461
2276
7937
6680
17102
29247
15108
11331
7554
3777
0
counts
64707
48530
32354
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SBS26
SBS21
SBS17b
SBS17a
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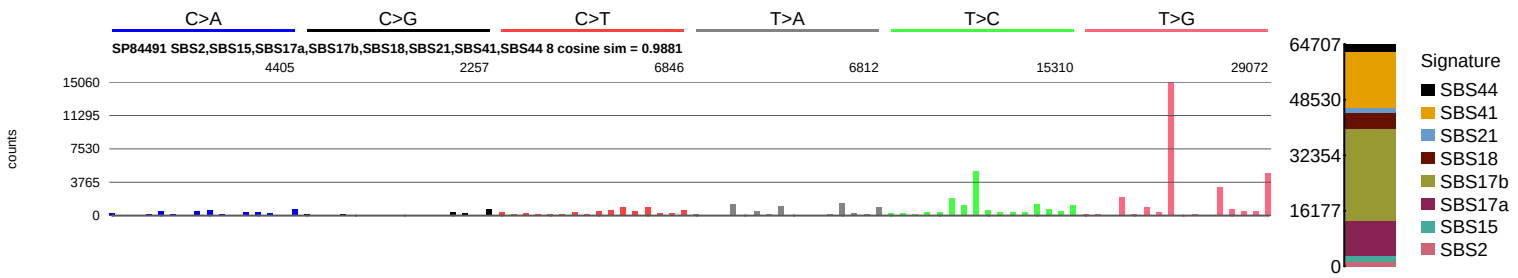
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.9881
4405
2257
6846
6812
15310
29072
15060
11295
7530
3765
0
counts
64707
48530
32354
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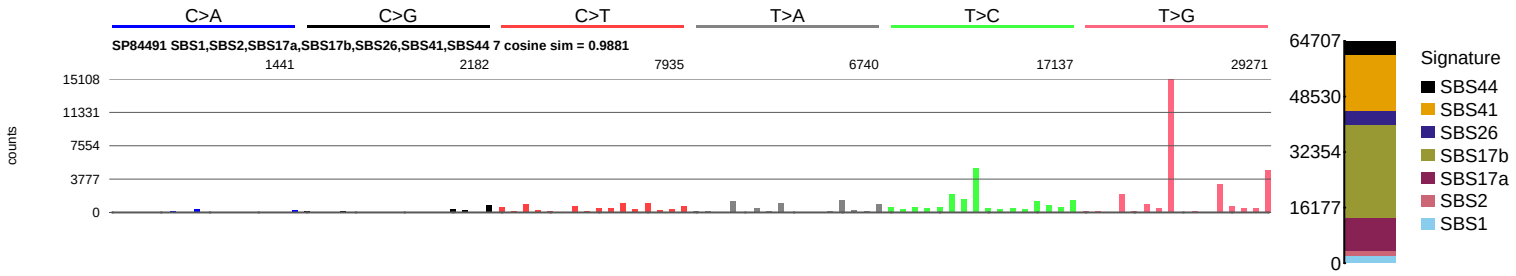
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.9881
1441
2182
7935
6740
17137
29271
counts
15108
11331
7554
3777
0
64707
48530
32354
16177
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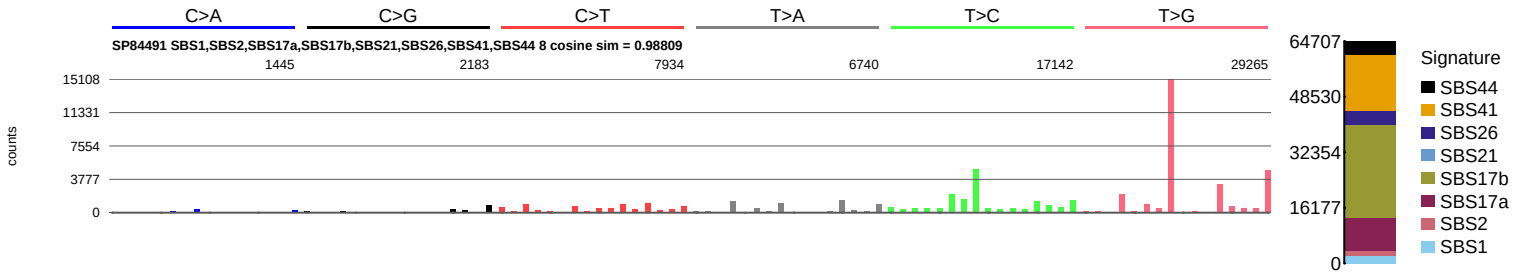
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98809
1445
2183
7934
6740
17142
29265
counts
15108
11331
7554
3777
0
64707
48530
32354
16177
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Signature
SBS44
SBS41
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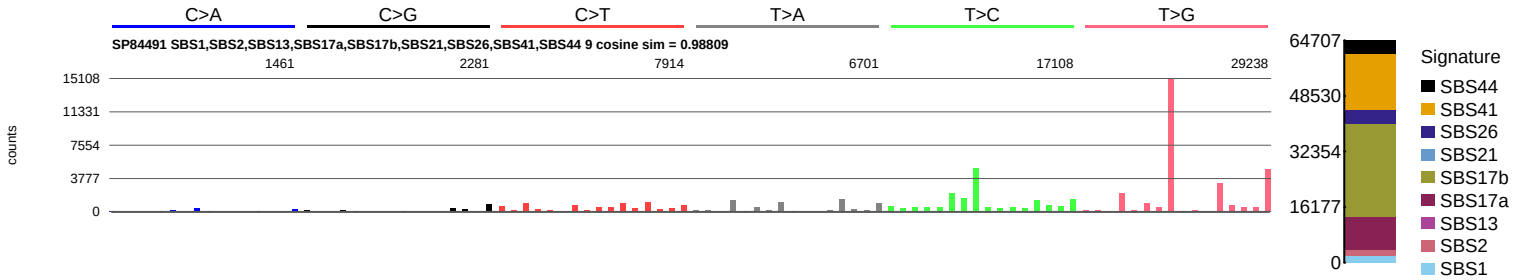
SBS2
SBS1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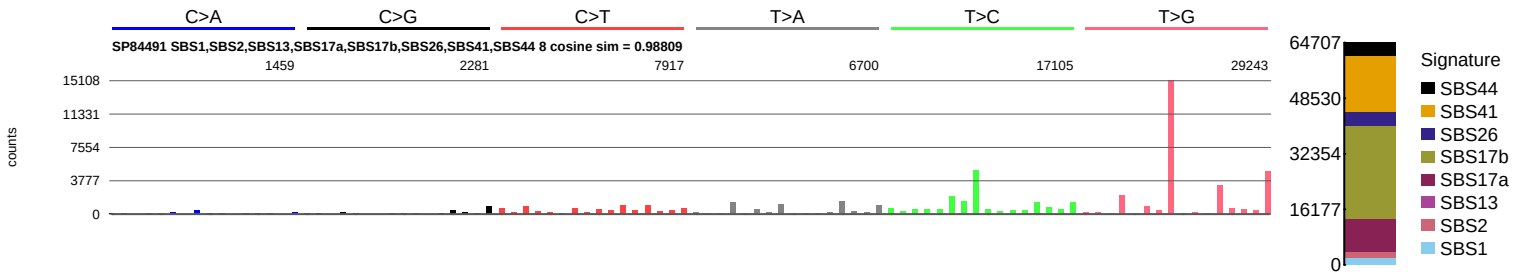
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98809
1461
2281
7914
6701
17108
29238
counts
15108
11331
7554
3777
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
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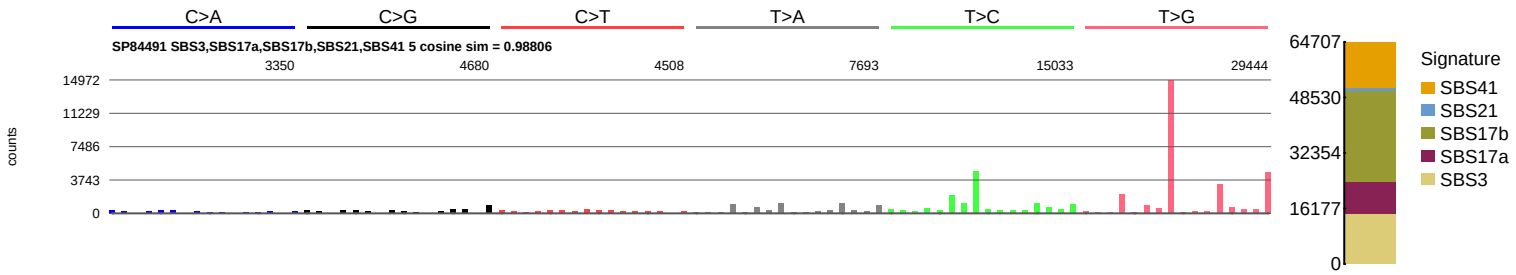
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98809
1459
2281
7917
6700
17105
29243
counts
15108
11331
7554
3777
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
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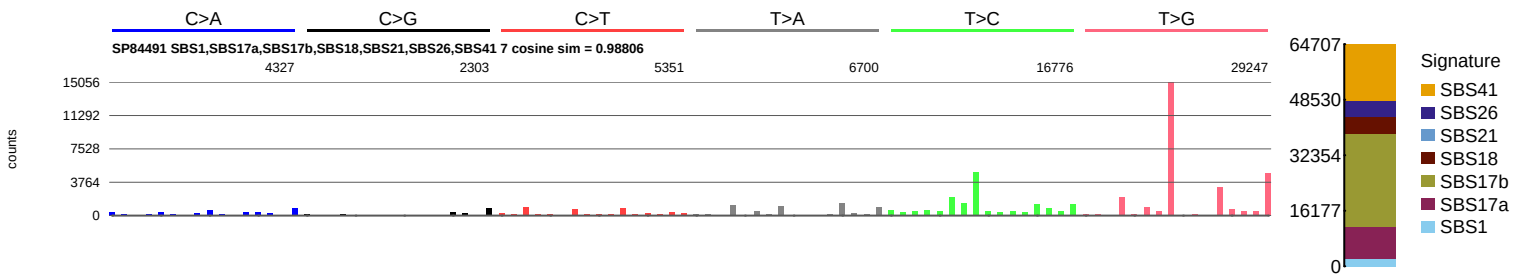
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98806
3350
4680
4508
7693
15033
29444
counts
14972
11229
7486
3743
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3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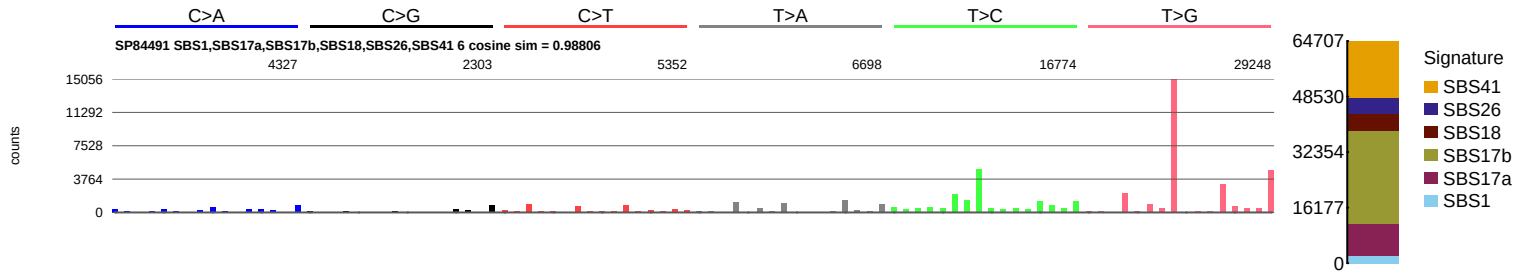
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.98806
4327
2303
5351
6700
16776
29247
counts
15056
11292
7528
3764
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1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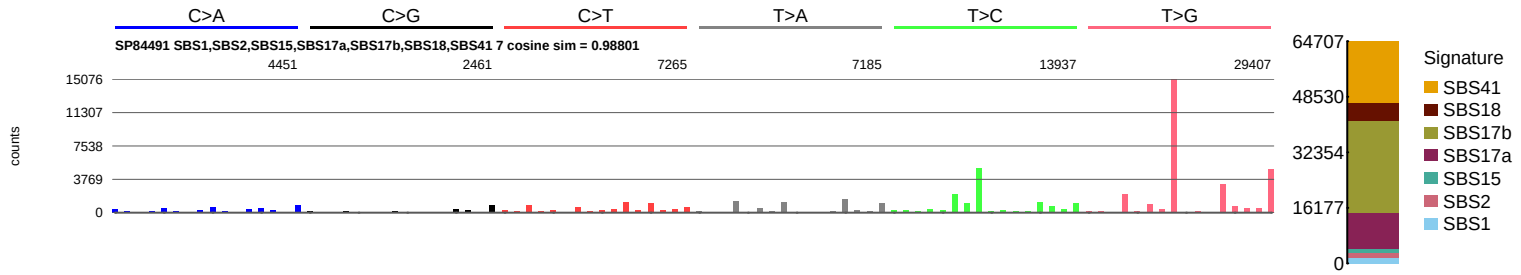
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98806
4327
2303
5352
6698
16774
29248
15056
11292
7528
3764
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1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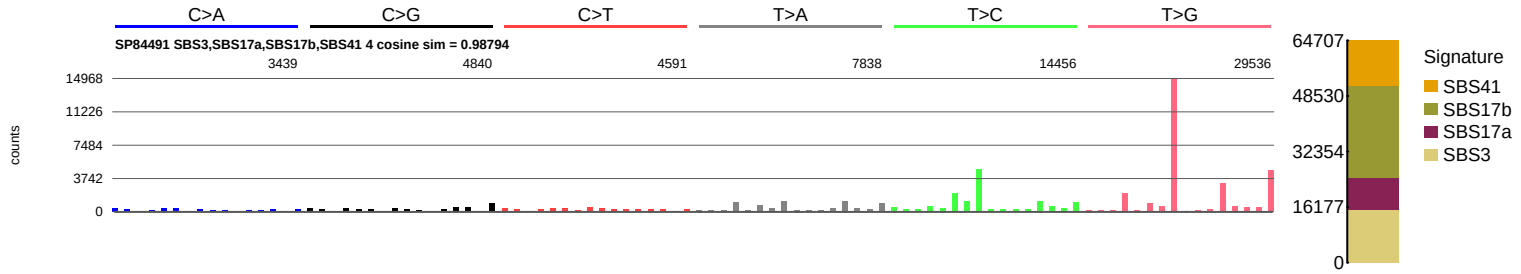
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98801
4451
2461
7265
7185
13937
29407
15076
11307
7538
3769
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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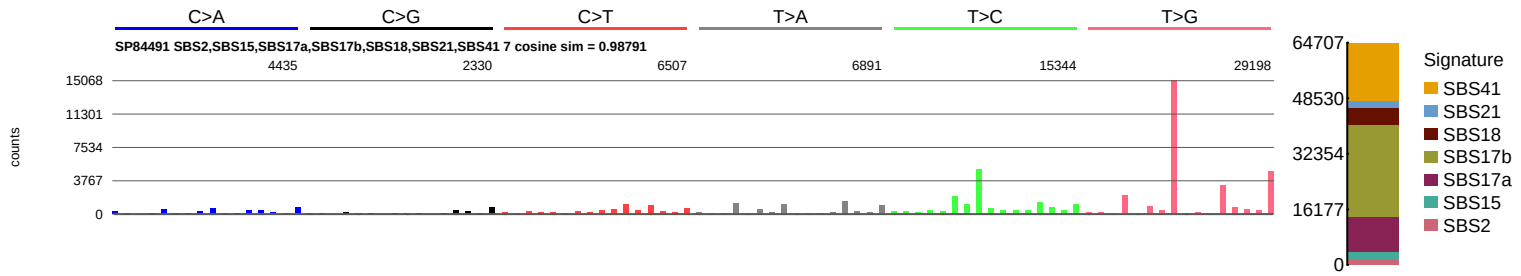
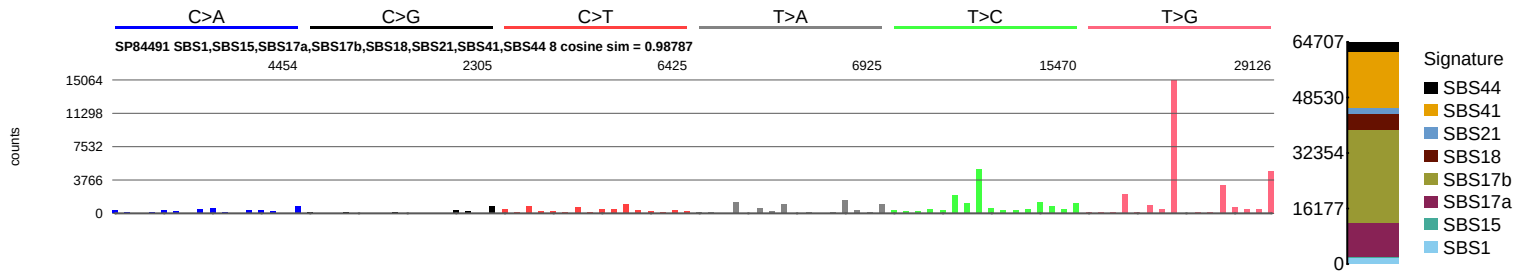
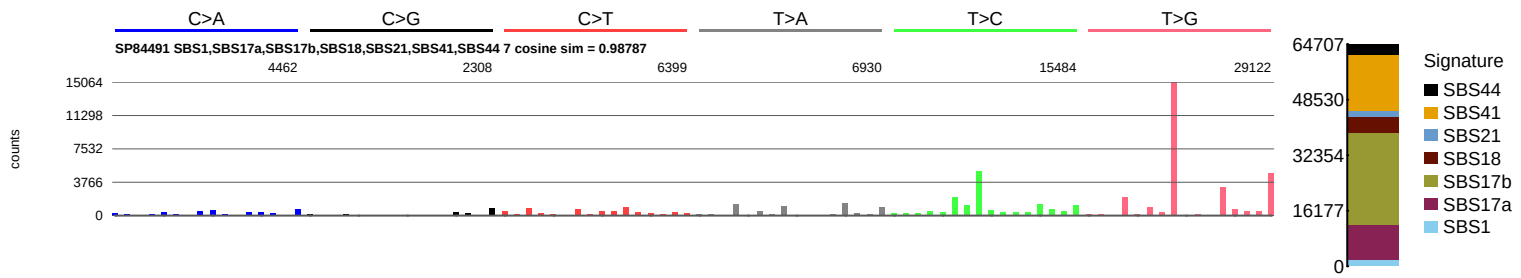
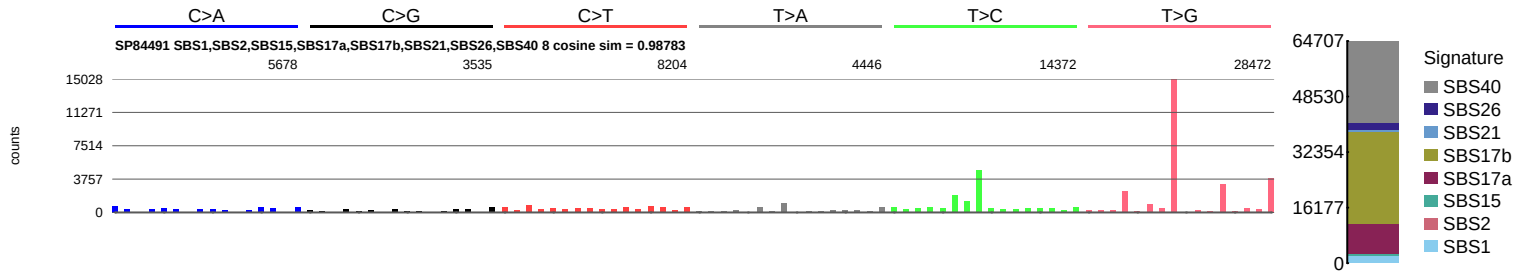
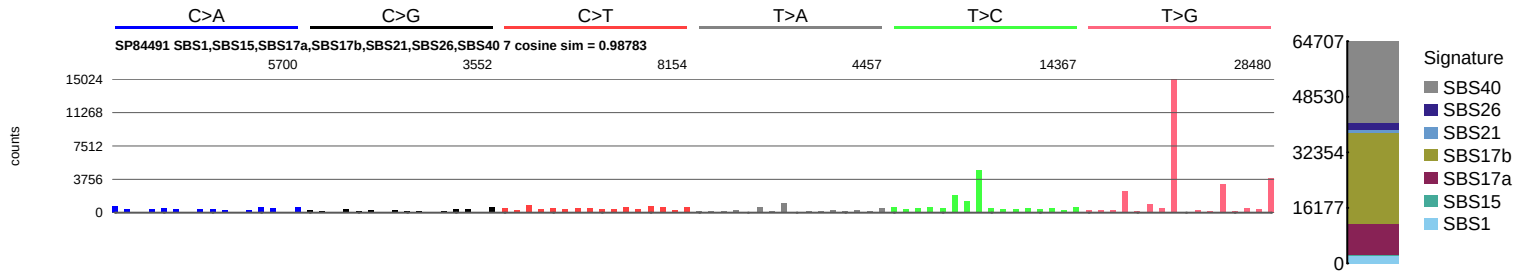
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS41 4 cosine sim = 0.98794
3439
4840
4591
7838
14456
29536
14968
11226
7484
3742
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS3

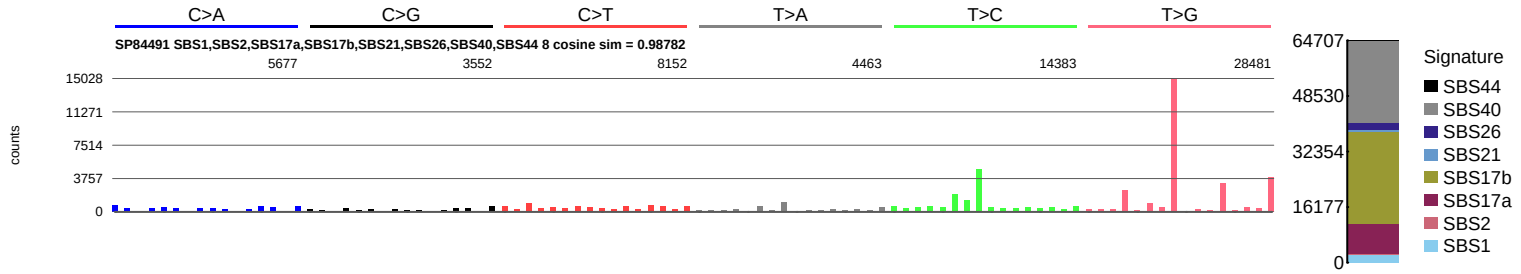

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98791
4435
2330
6507
6891
15344
29198
15068
11301
7534
3767
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2

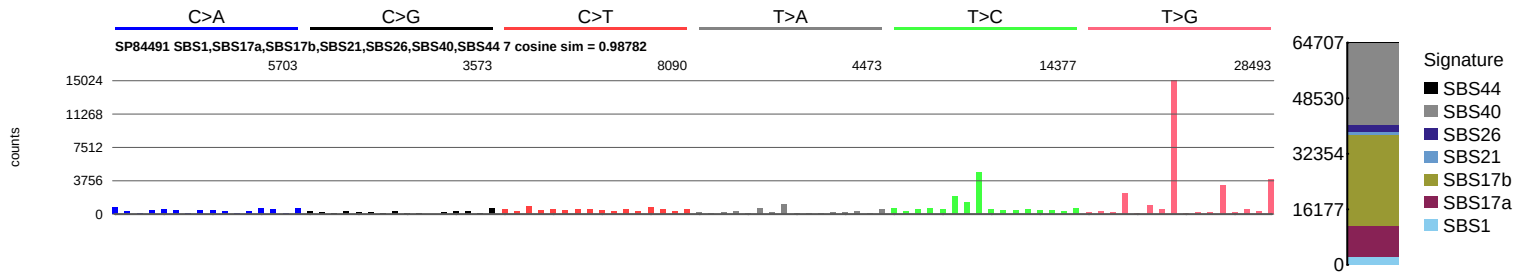

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98787
4454
2305
6425
6925
15470
29126
15064
11298
7532
3766
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

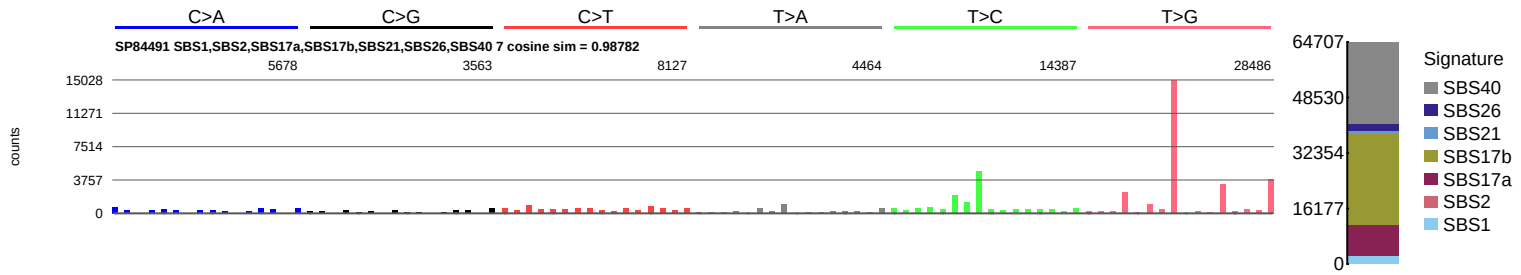

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98787
4462
2308
6399
6930
15484
29122
15064
11298
7532
3766
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1

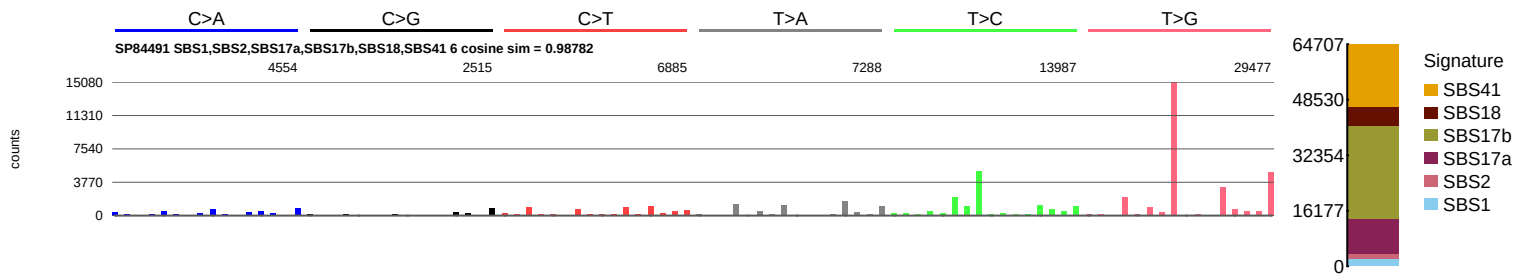

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98783
5678
3535
8204
4446
14372
28472
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

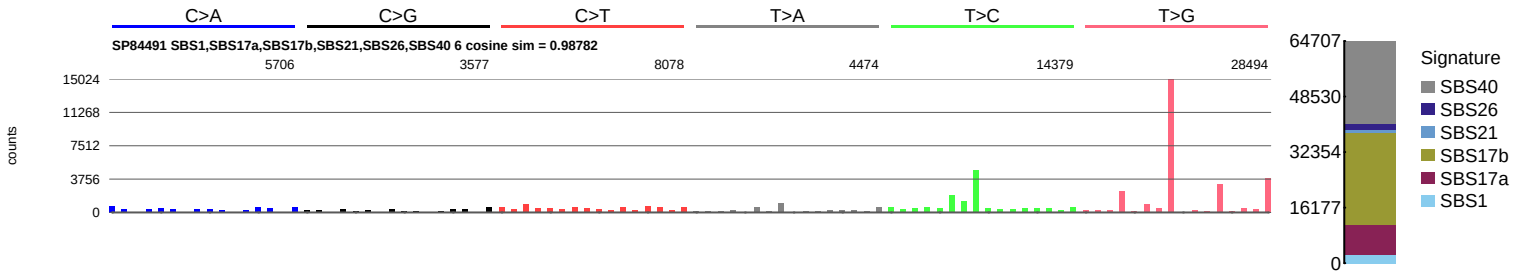

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98783
5700
3552
8154
4457
14367
28480
counts
15024
11268
7512
3756
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1

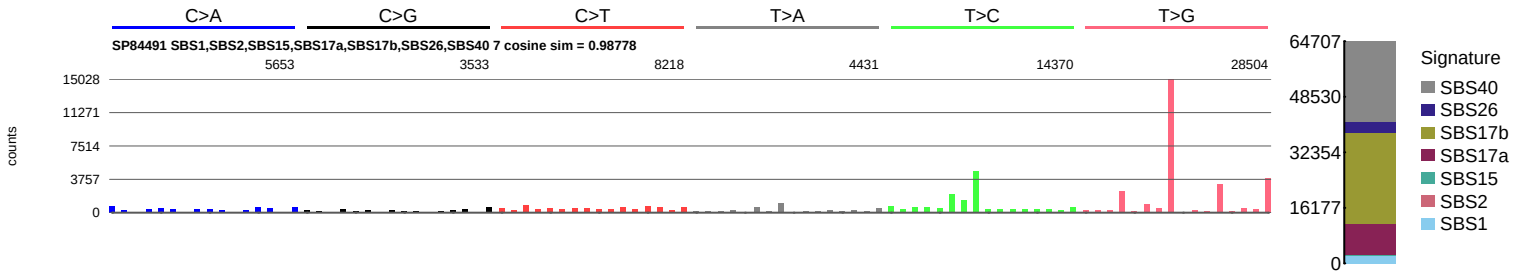

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98782
5677
3552
8152
4463
14383
28481
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1

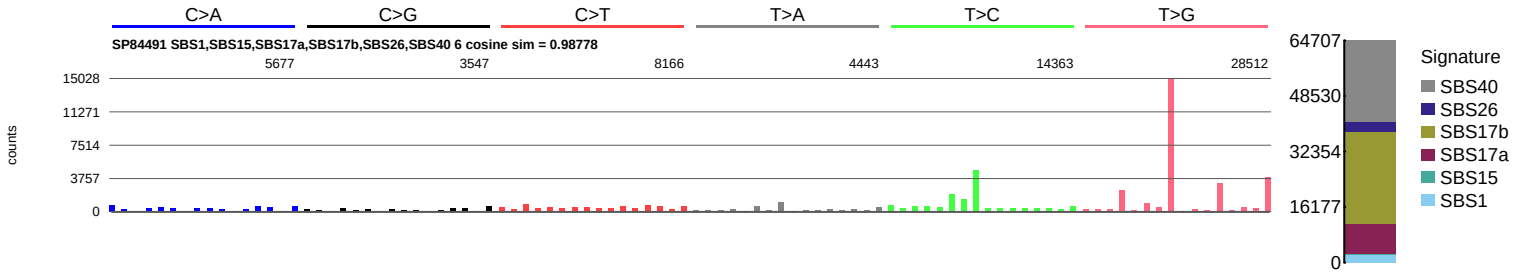

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98782
5703
3573
8090
4473
14377
28493
counts
15024
11268
7512
3756
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1

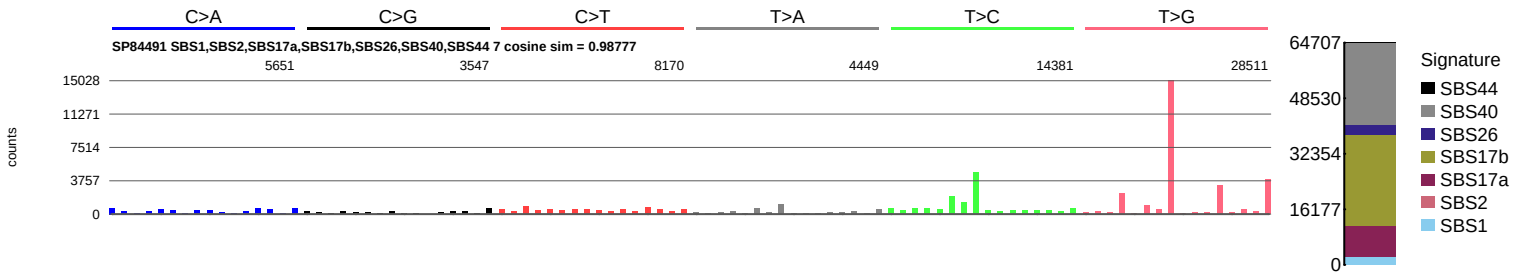

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98782
5678
3563
8127
4464
14387
28486
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98782
4554
2515
6885
7288
13987
29477
counts
15080
11310
7540
3770
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.98782
5706
3577
8078
4474
14379
28494
counts
15024
11268
7512
3756
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98778
5653
3533
8218
4431
14370
28504
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98778
5677
3547
8166
4443
14363
28512
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98777
5651
3547
8170
4449
14381
28511
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
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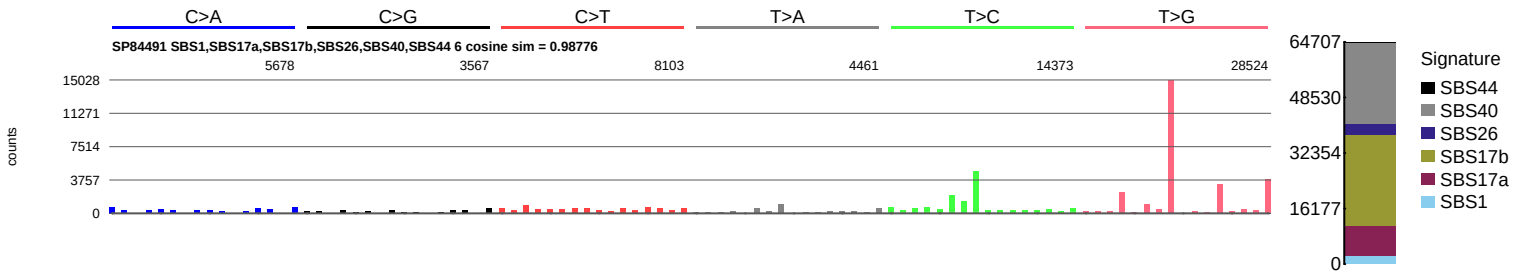
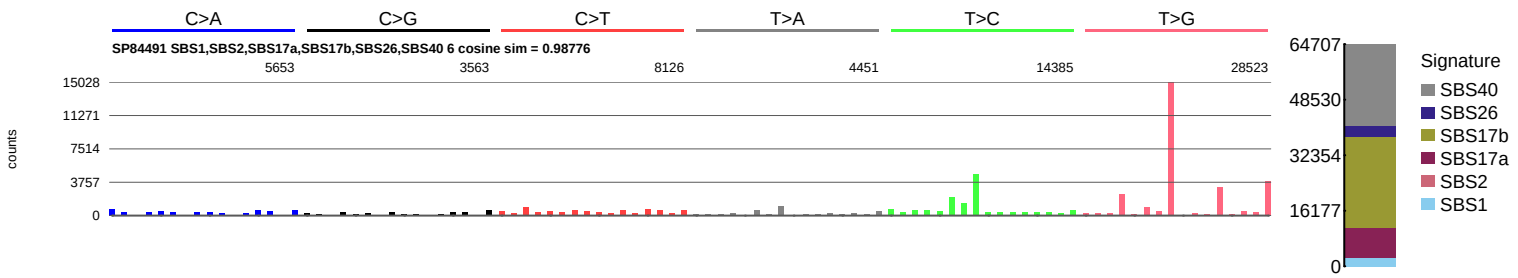
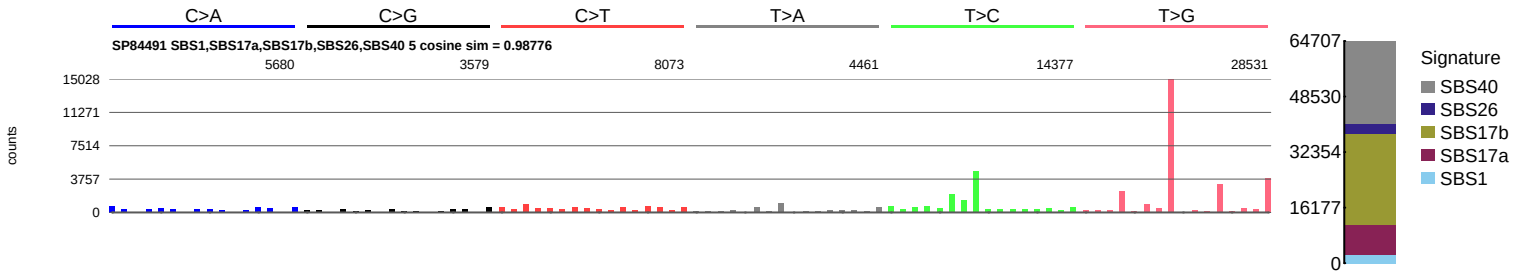
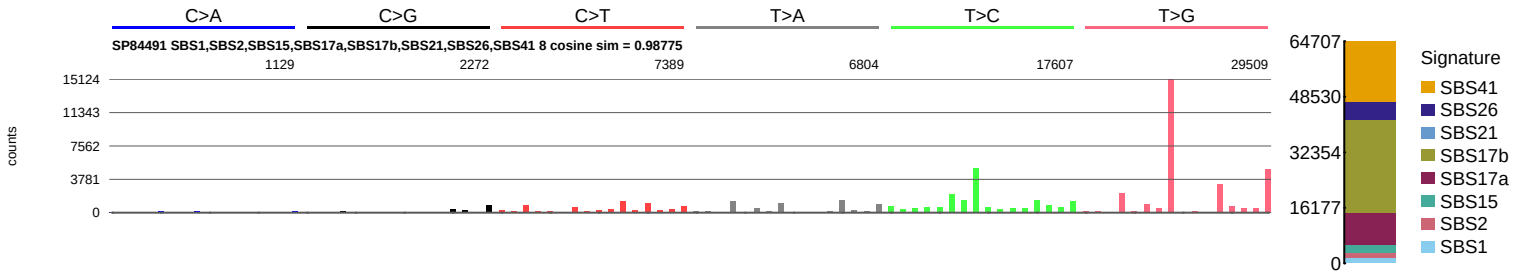
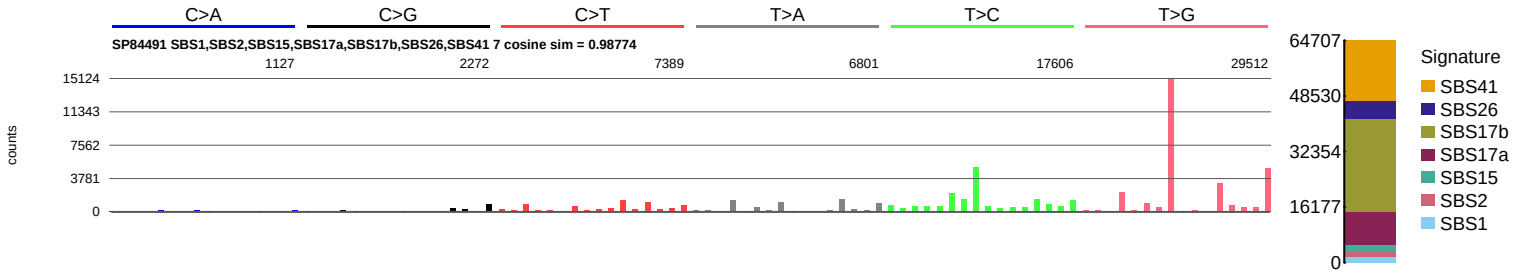
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1

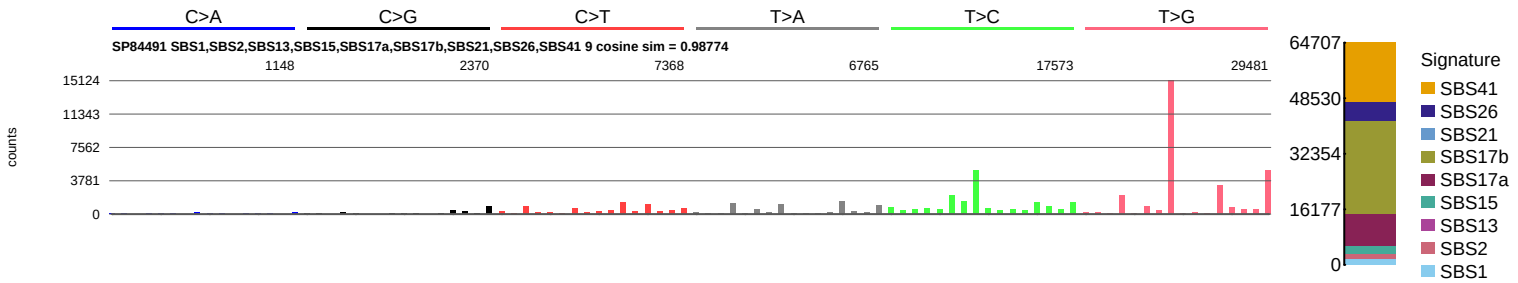

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98776
5678
3567
8103
4461
14373
28524
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS1

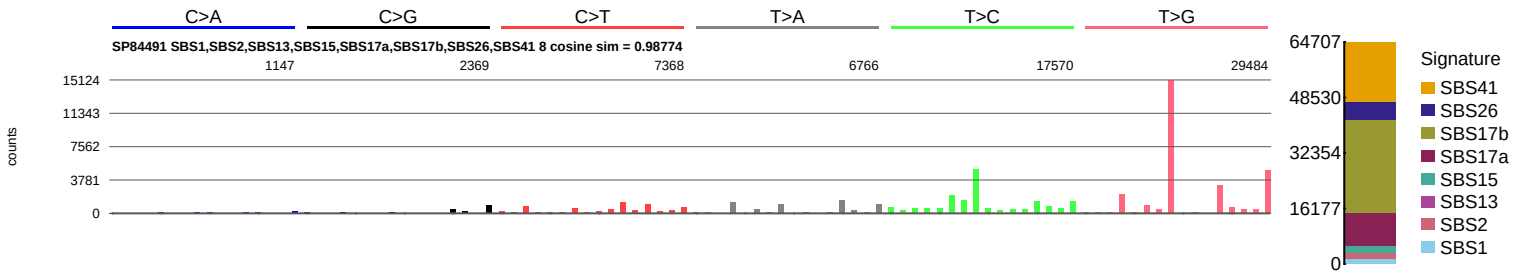

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98776
5653
3563
8126
4451
14385
28523
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1

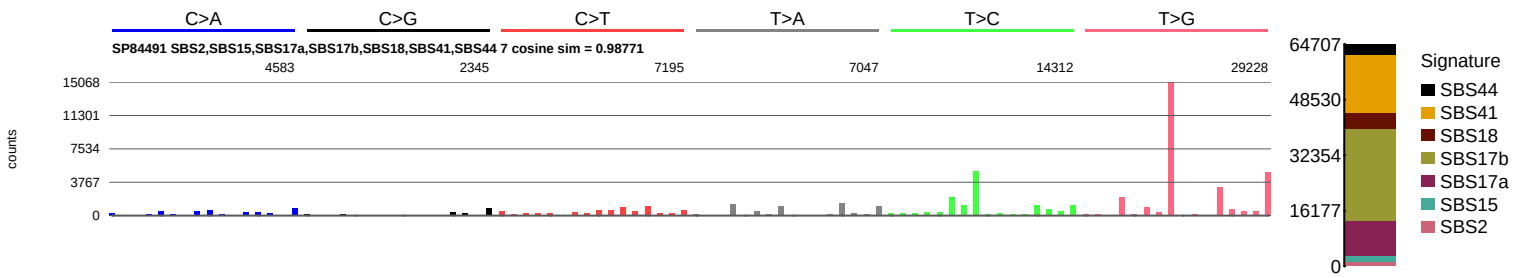

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98776
5680
3579
8073
4461
14377
28531
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS1

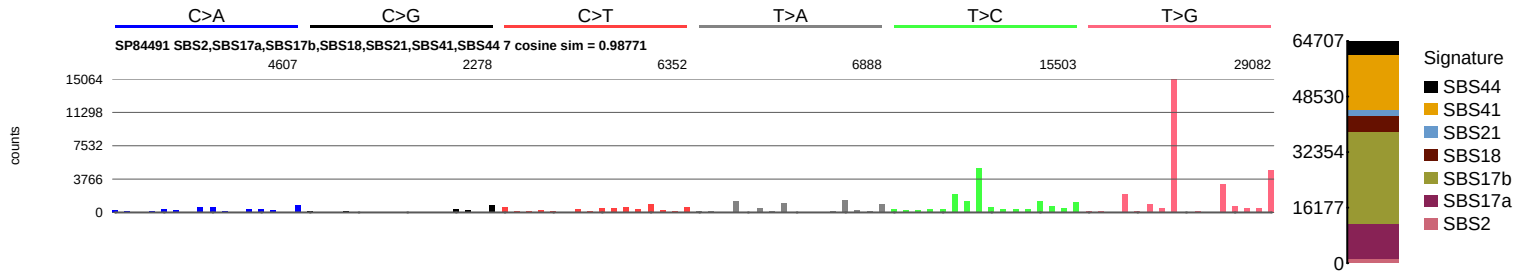

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98775
1129
2272
7389
6804
17607
29509
counts
15124
11343
7562
3781
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

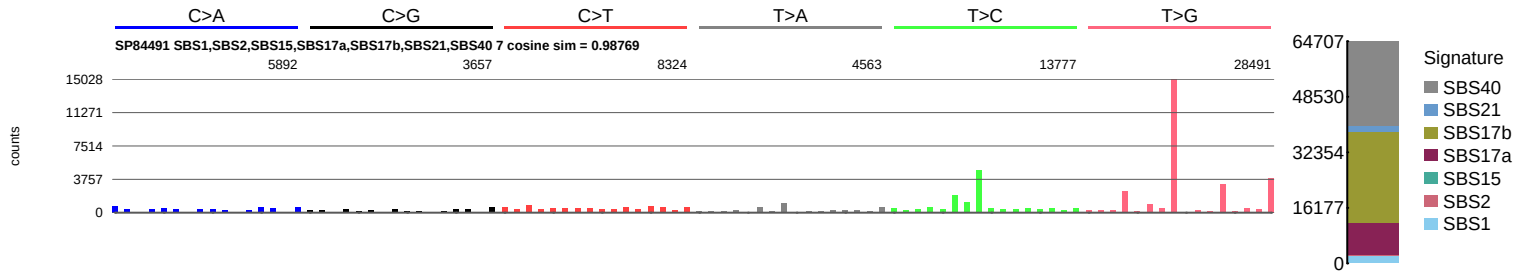

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98774
1127
2272
7389
6801
17606
29512
counts
15124
11343
7562
3781
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.98774
1148
2370
7368
6765
17573
29481
counts
15124
11343
7562
3781
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98774
1147
2369
7368
6766
17570
29484
counts
15124
11343
7562
3781
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98771
4583
2345
7195
7047
14312
29228
counts
15068
11301
7534
3767
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98771
4607
2278
6352
6888
15503
29082
counts
15064
11298
7532
3766
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2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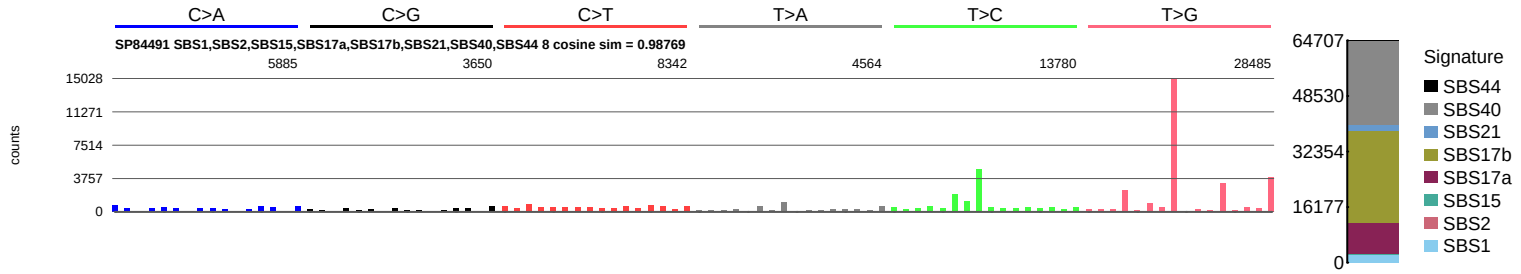
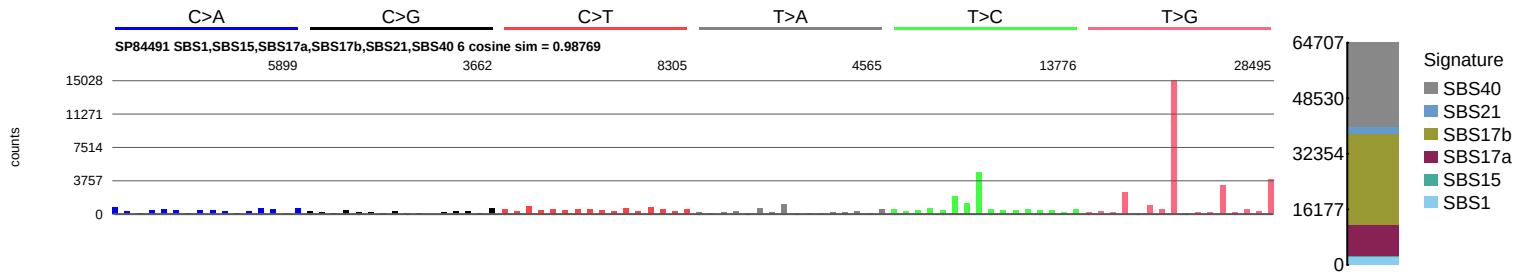
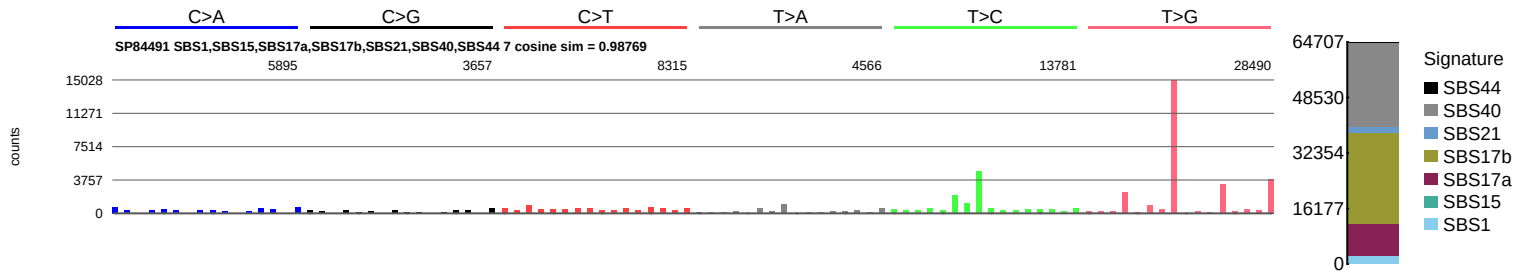
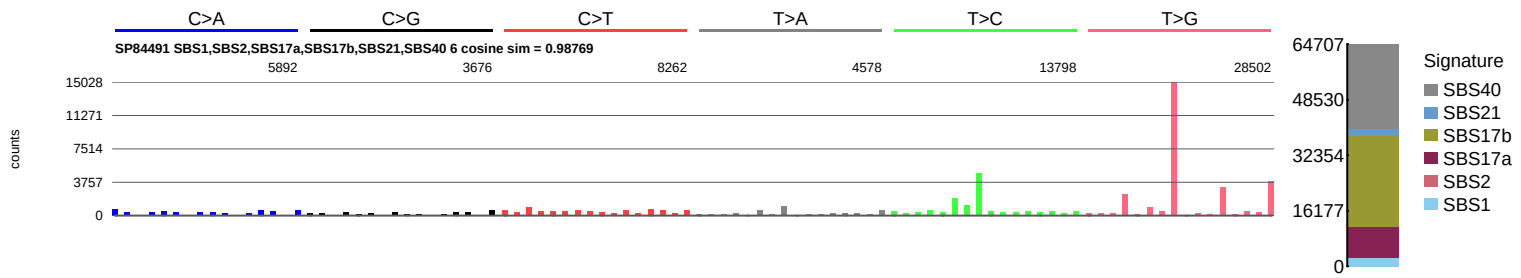
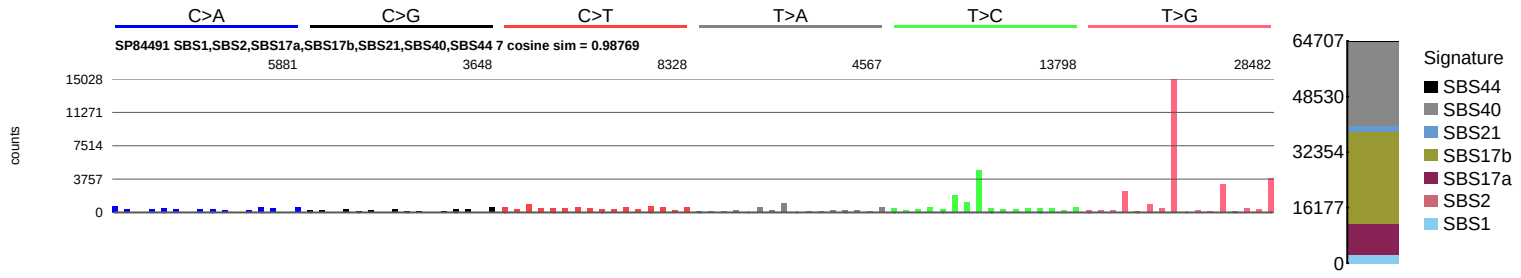
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98769
5892
3657
8324
4563
13777
28491
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

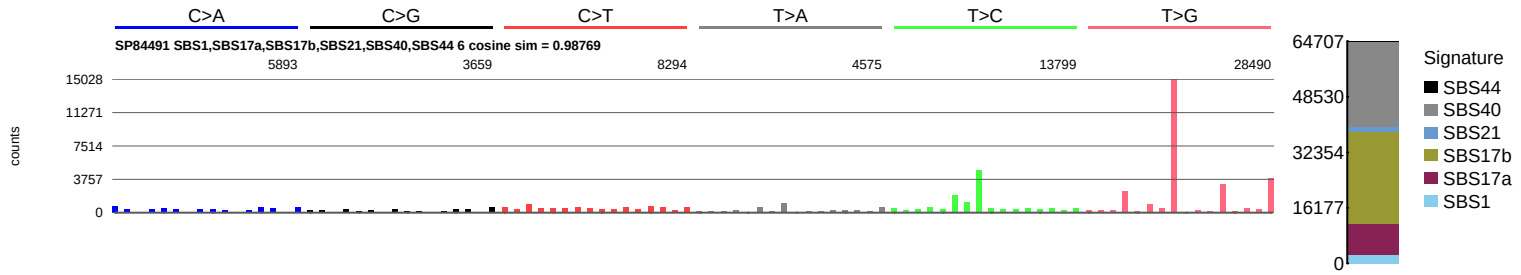

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98769
5885
3650
8342
4564
13780
28485
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

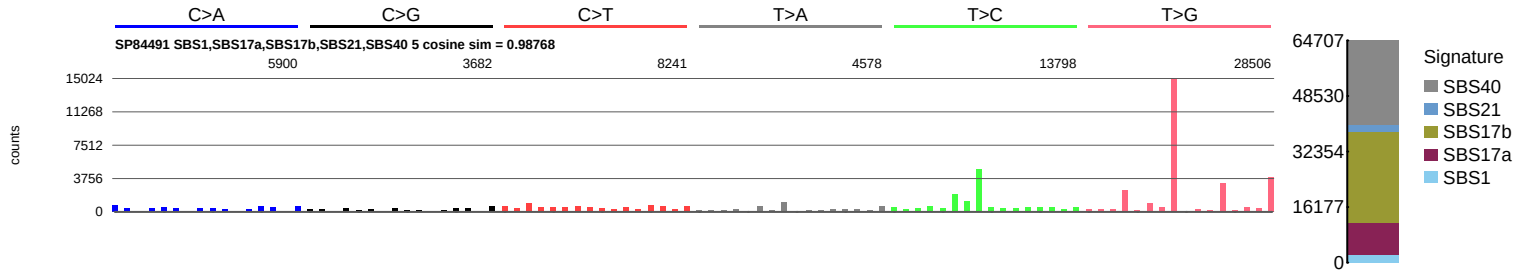

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98769
5899
3662
8305
4565
13776
28495
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

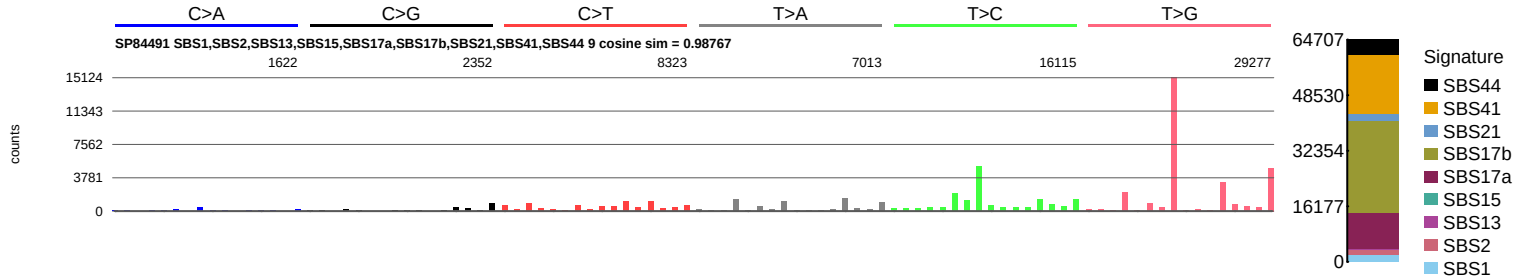

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98769
5895
3657
8315
4566
13781
28490
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

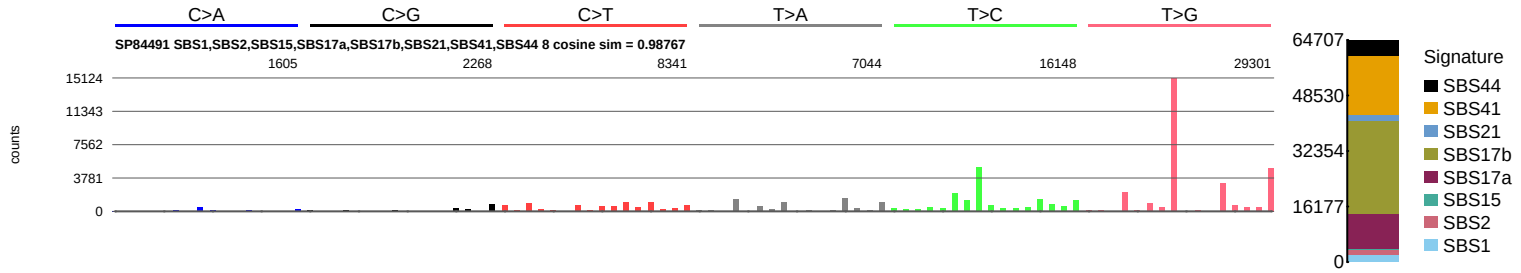

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98769
5892
3676
8262
4578
13798
28502
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1

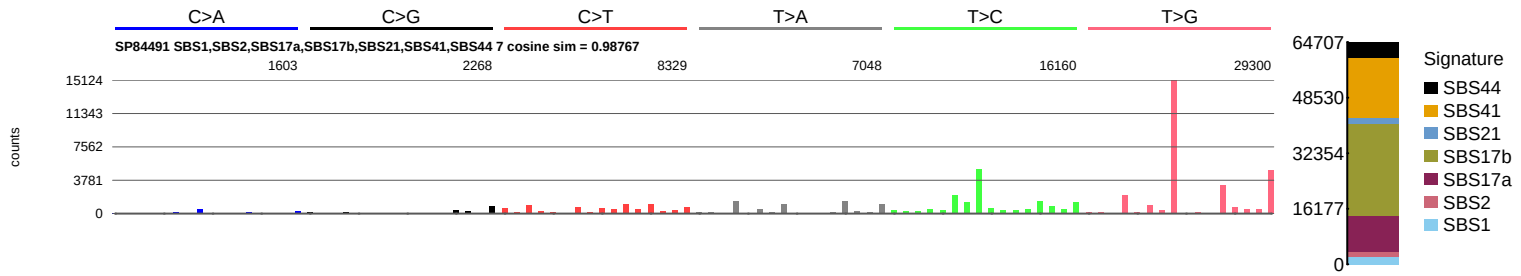

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98769
5881
3648
8328
4567
13798
28482
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1

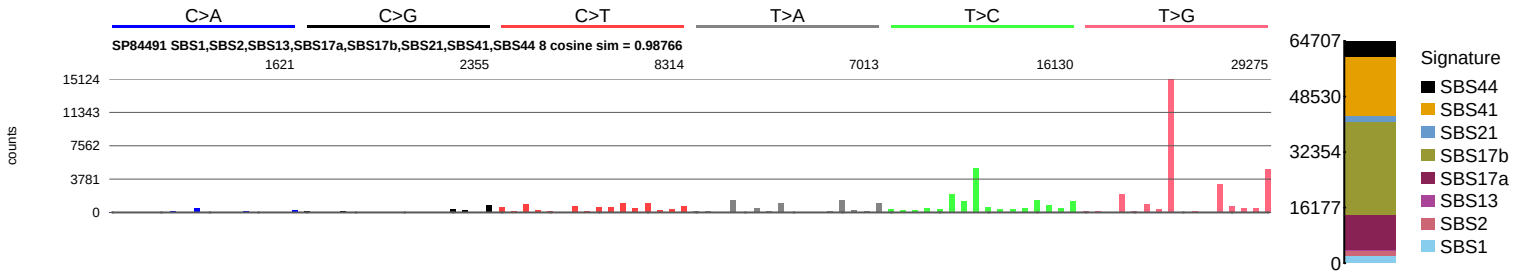

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98769
5893
3659
8294
4575
13799
28490
counts
15028
11271
7514
3757
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS1

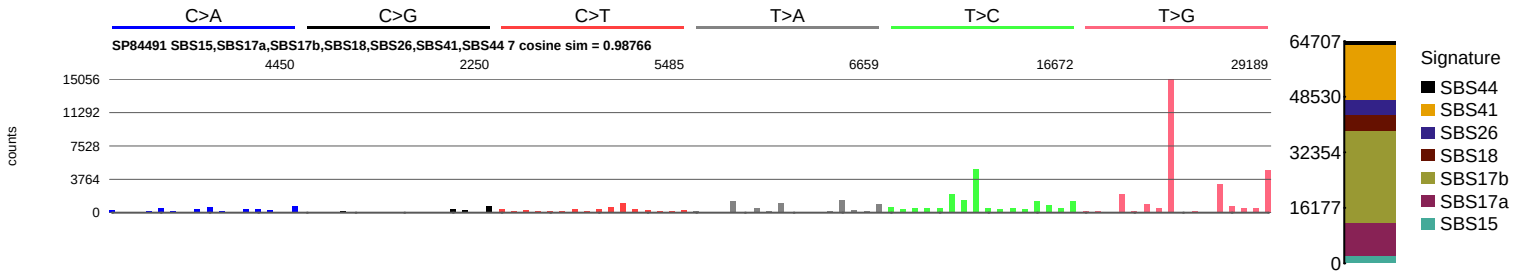

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98768
5900
3682
8241
4578
13798
28506
counts
15024
11268
7512
3756
0
64707
48530
32354
16177
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS1

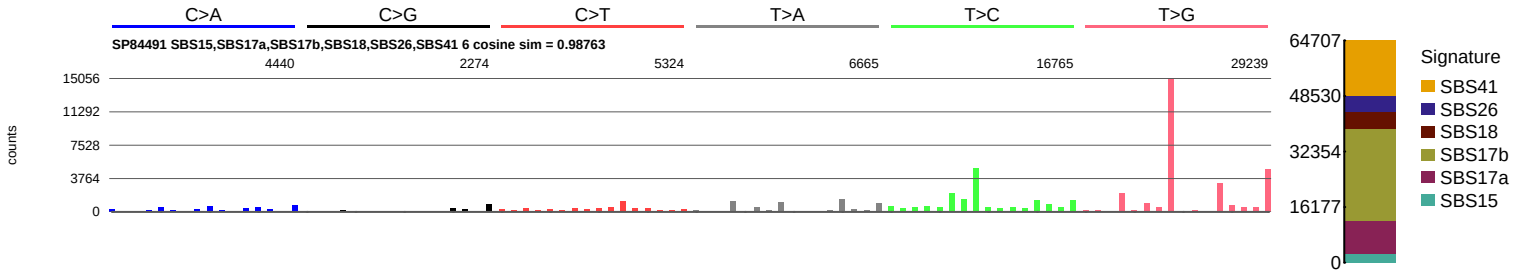

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98767
1622
2352
8323
7013
16115
29277
counts
15124
11343
7562
3781
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

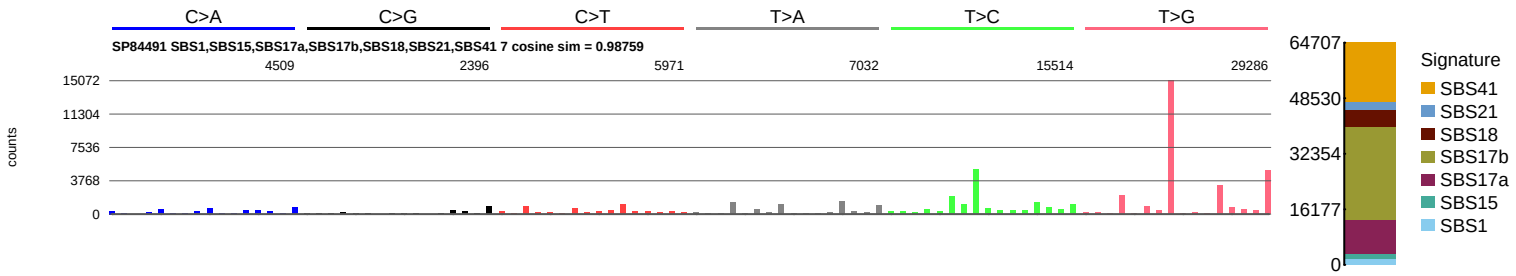

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98767
1605
2268
8341
7044
16148
29301
counts
15124
11343
7562
3781
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

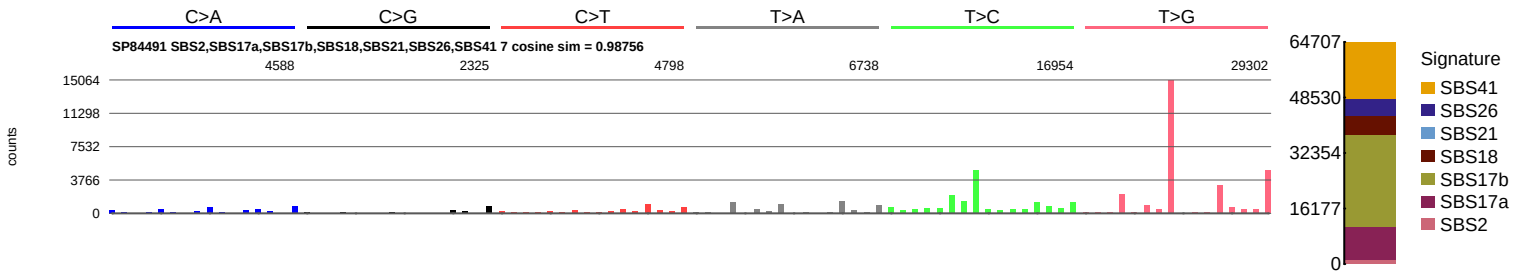

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98767
1603
2268
8329
7048
16160
29300
counts
15124
11343
7562
3781
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS2
SBS1

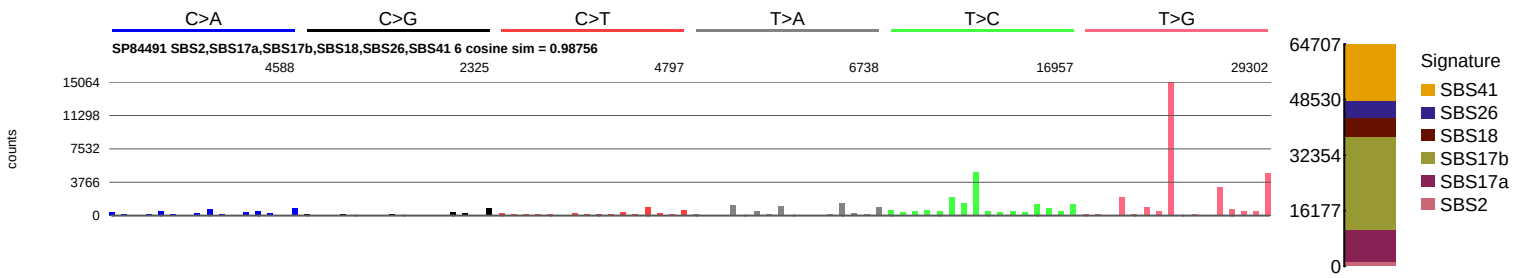

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98766
1621
2355
8314
7013
16130
29275
15124
11343
7562
3781
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

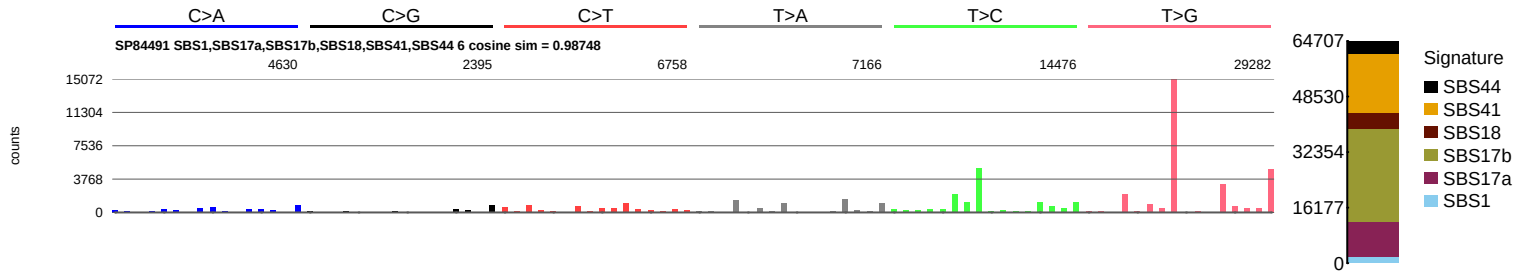

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98766
4450
2250
5485
6659
16672
29189
15056
11292
7528
3764
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15

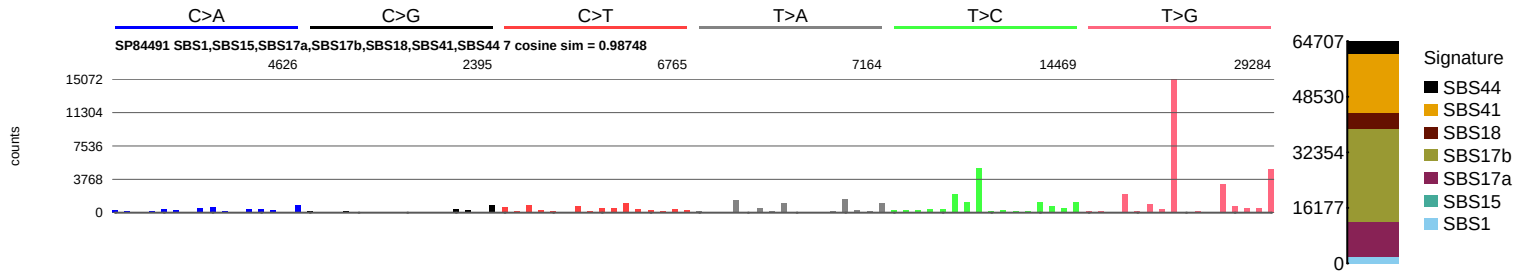

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98763
4440
2274
5324
6665
16765
29239
15056
11292
7528
3764
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15

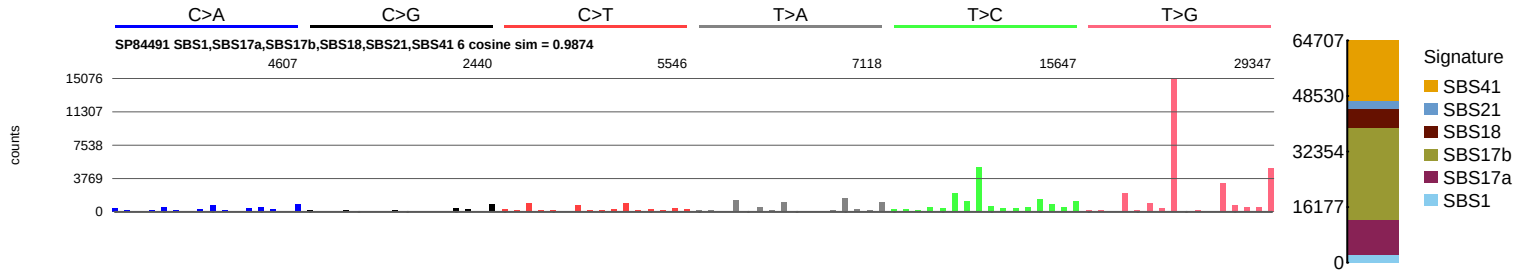

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98759
4509
2396
5971
7032
15514
29286
15072
11304
7536
3768
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

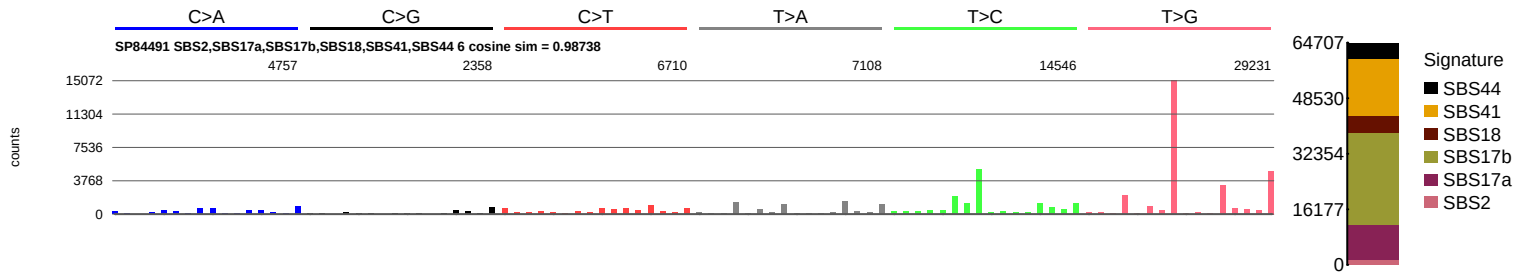

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.98756
4588
2325
4798
6738
16954
29302
15064
11298
7532
3766
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2

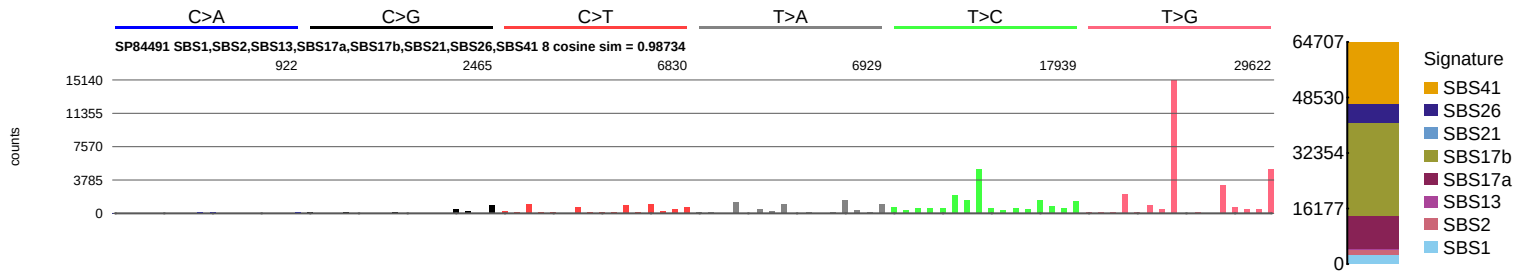

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98756
4588
2325
4797
6738
16957
29302
15064
11298
7532
3766
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2

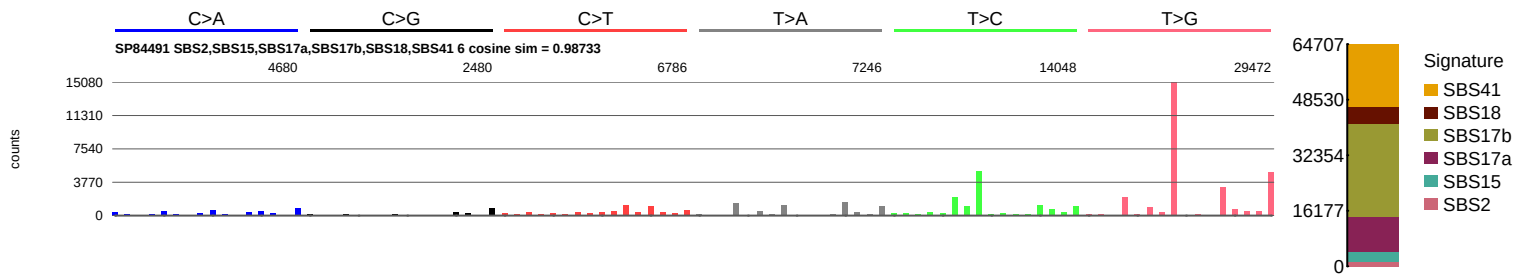

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98748
4630
2395
6758
7166
14476
29282
counts
15072
11304
7536
3768
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98748
4626
2395
6765
7164
14469
29284
counts
15072
11304
7536
3768
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.9874
4607
2440
5546
7118
15647
29347
counts
15076
11307
7538
3769
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98738
4757
2358
6710
7108
14546
29231
counts
15072
11304
7536
3768
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98734
922
2465
6830
6929
17939
29622
counts
15140
11355
7570
3785
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98733
4680
2480
6786
7246
14048
29472
counts
15080
11310
7540
3770
0
64707
48530
32354
16177
0
Signature
SBS41
SBS18
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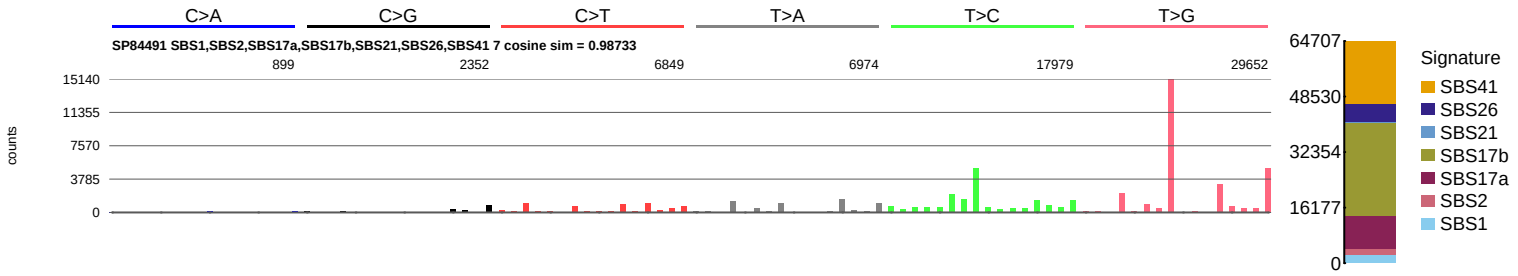
SBS17a
SBS15
SBS2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SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98733
899
2352
6849
6974
17979
29652
counts
15140
11355
7570
3785
0
64707
48530
32354
16177
0
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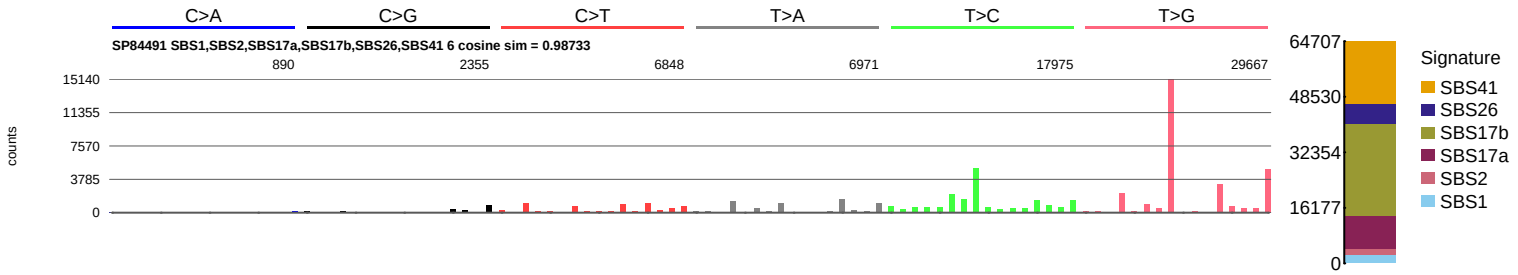
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1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SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98733
890
2355
6848
6971
17975
29667
counts
15140
11355
7570
3785
0
64707
48530
32354
16177
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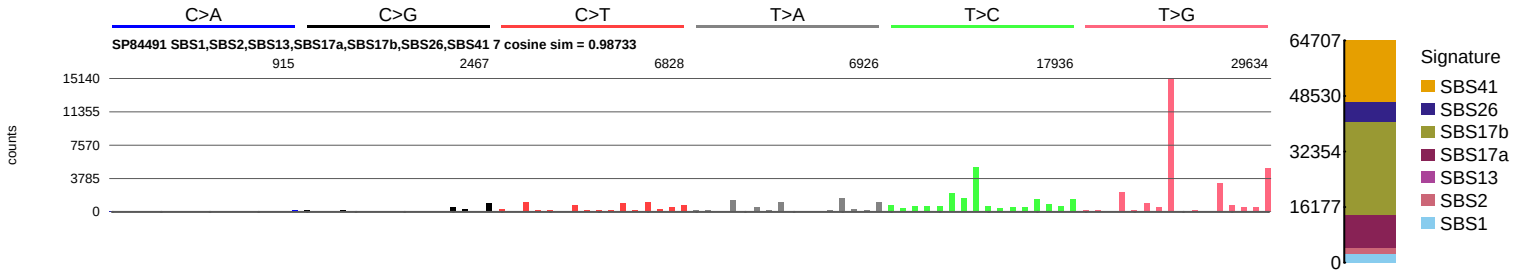
SBS17a
SBS2
SBS1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SP84491 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98733
915
2467
6828
6926
17936
29634
counts
15140
11355
7570
3785
0
64707
48530
32354
16177
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SBS26
SBS17b
SBS17a
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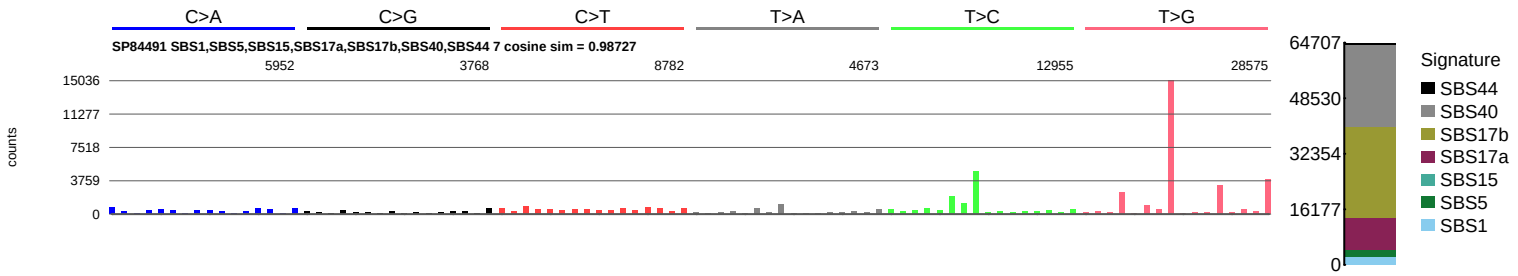
SBS2
SBS1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5952
3768
8782
4673
12955
28575
counts
15036
11277
7518
3759
0
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48530
32354
16177
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SBS44
SBS40
SBS17b
SBS17a
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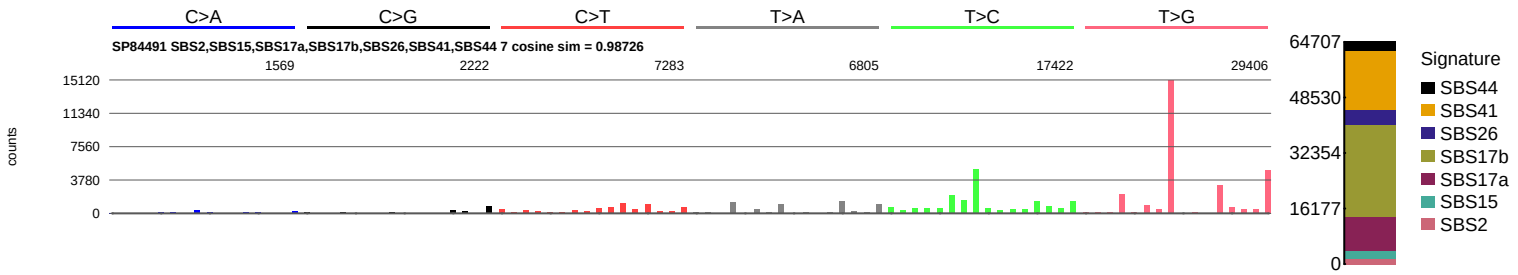
SBS5
SBS1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SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98726
1569
2222
7283
6805
17422
29406
counts
15120
11340
7560
3780
0
64707
48530
32354
16177
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SBS44
SBS41
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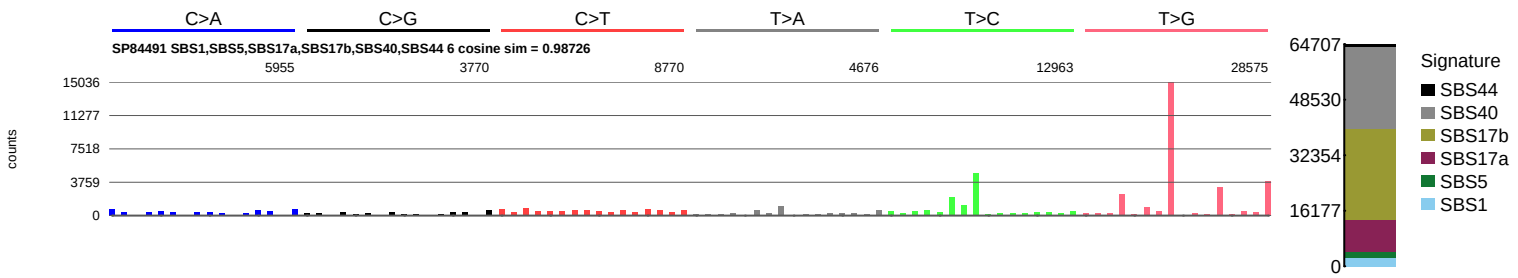
SBS17b
SBS17a
SBS15
SBS2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SP84491 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98726
5955
3770
8770
4676
12963
28575
counts
15036
11277
7518
3759
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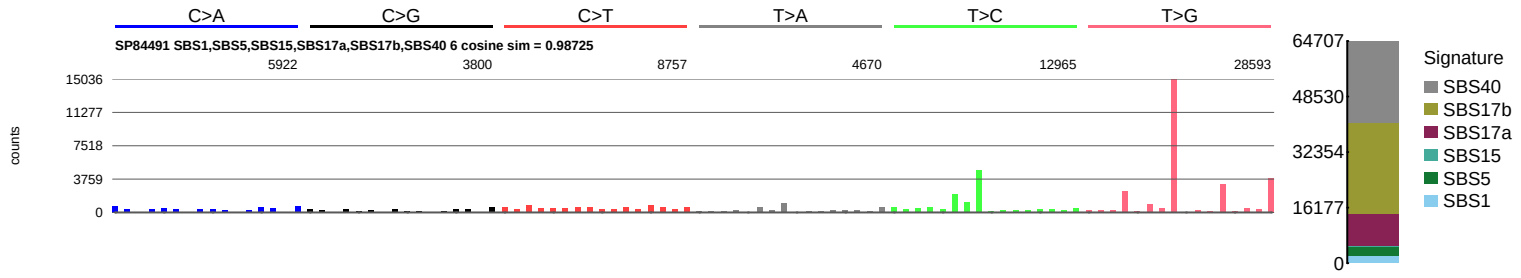
SBS5
SBS1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SP84491 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98725
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3800
8757
4670
12965
28593
15036
11277
7518
3759
0
counts
64707
48530
32354
16177
0
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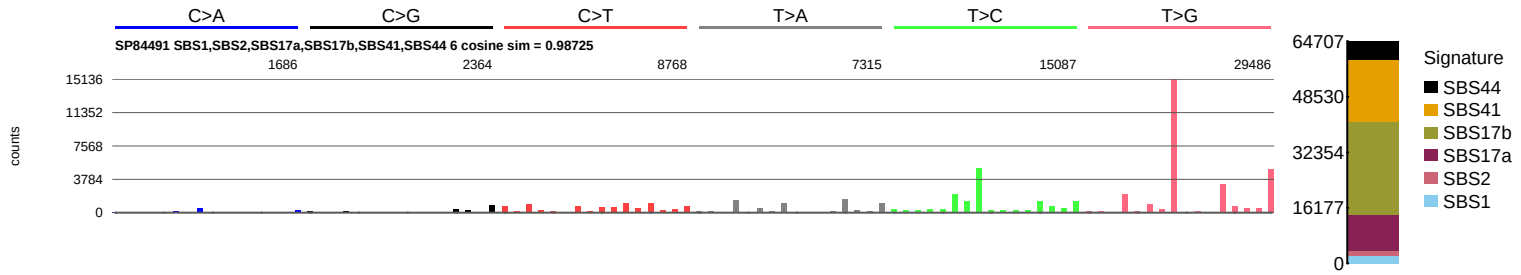
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1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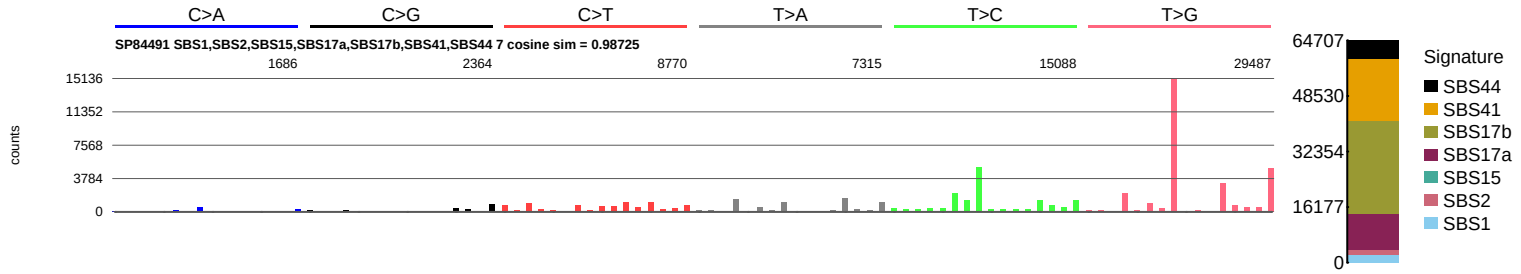
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98725
1686
2364
8768
7315
15087
29486
15136
11352
7568
3784
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS2
SBS1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C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98725
1686
2364
8770
7315
15088
29487
15136
11352
7568
3784
0
counts
64707
48530
32354
16177
0
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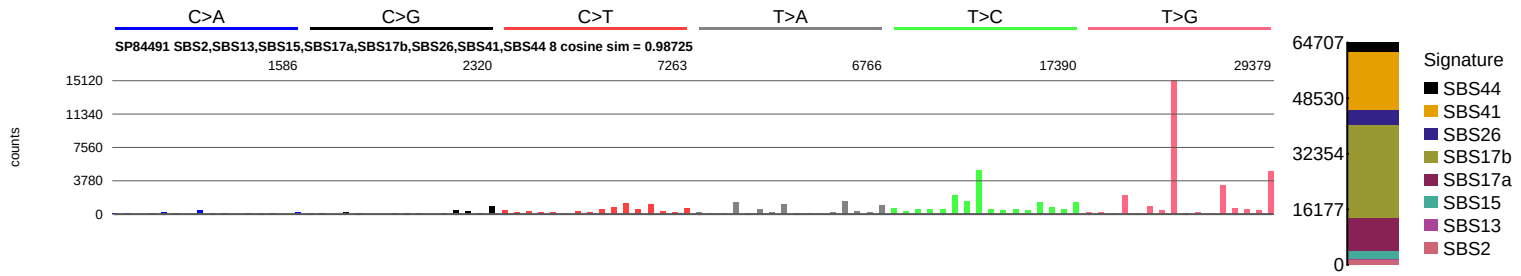
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>T
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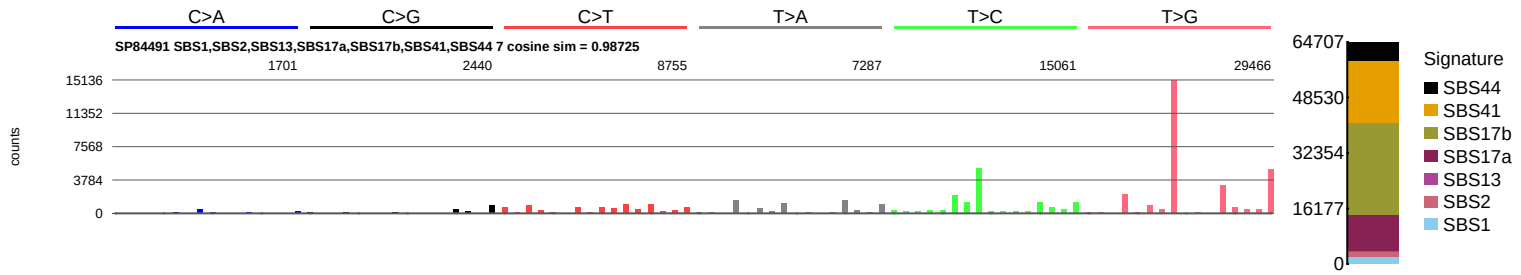
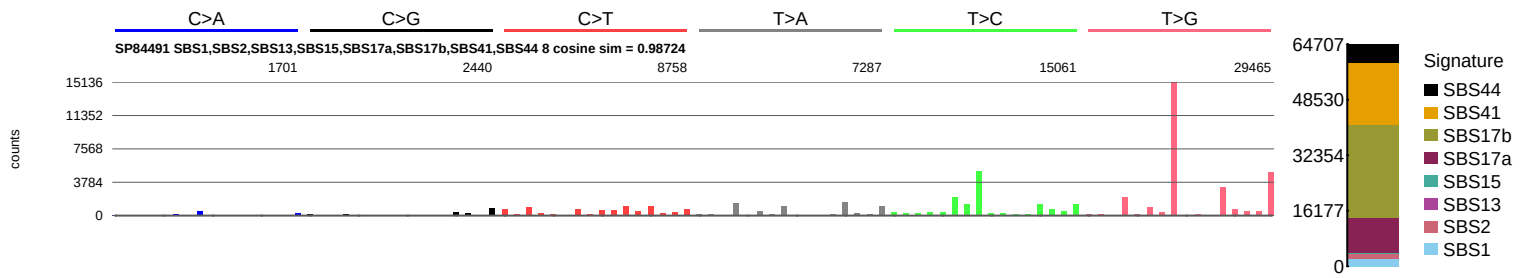
T>C
T>G
SP84491 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98725
1586
2320
7263
6766
17390
29379
15120
11340
7560
3780
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98725
1701
2440
8755
7287
15061
29466
15136
11352
7568
3784
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
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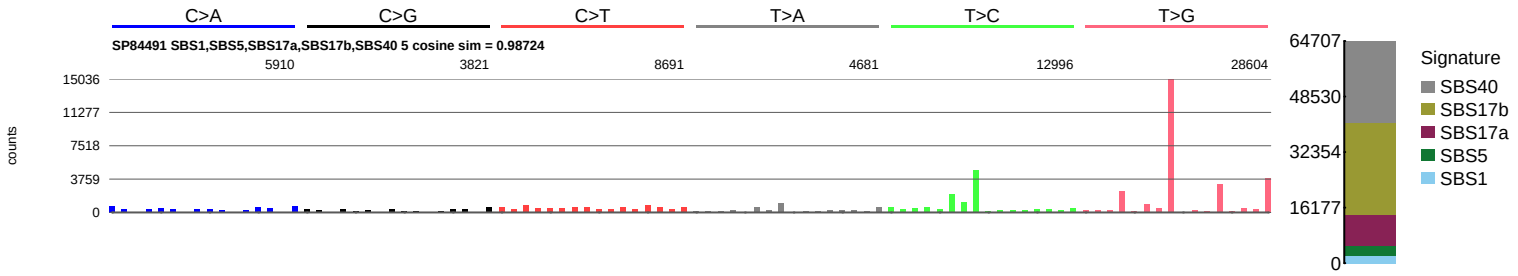
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98724
1701
2440
8758
7287
15061
29465
15136
11352
7568
3784
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
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SP84491 SBS1,SBS5,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98724
5910
3821
8691
4681
12996
28604
counts
15036
11277
7518
3759
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64707
48530
32354
16177
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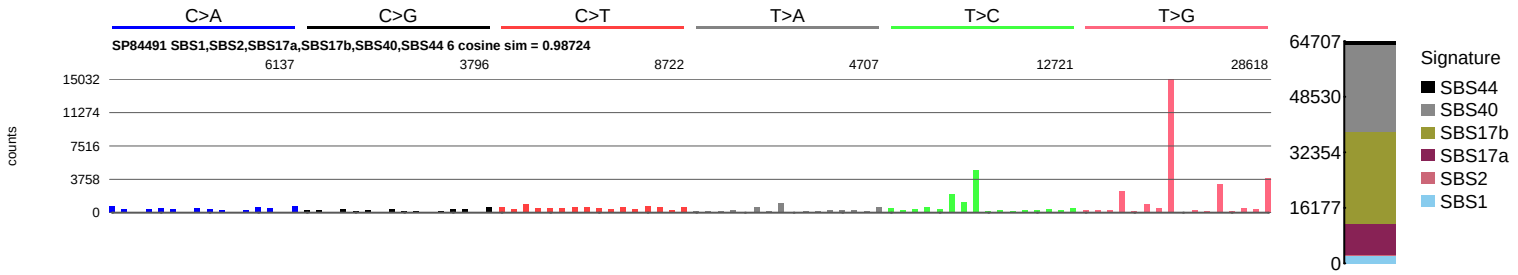
SBS40
SBS17b
SBS17a
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98724
6137
3796
8722
4707
12721
28618
counts
15032
11274
7516
3758
0
64707
48530
32354
16177
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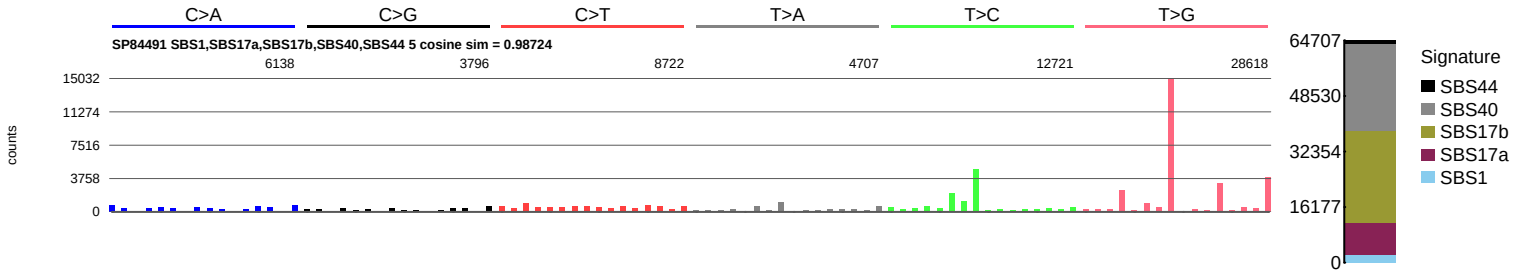
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS2
SBS1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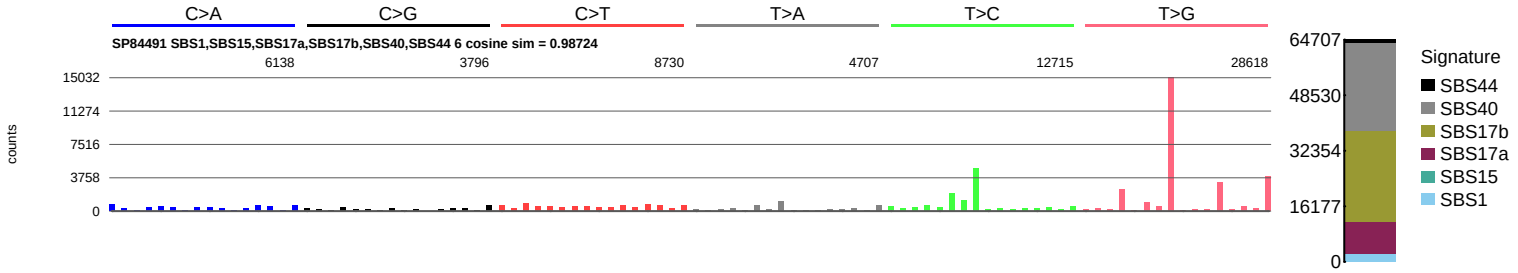
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98724
6138
3796
8722
4707
12721
28618
counts
15032
11274
7516
3758
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98724
6138
3796
8730
4707
12715
28618
counts
15032
11274
7516
3758
0
64707
48530
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16177
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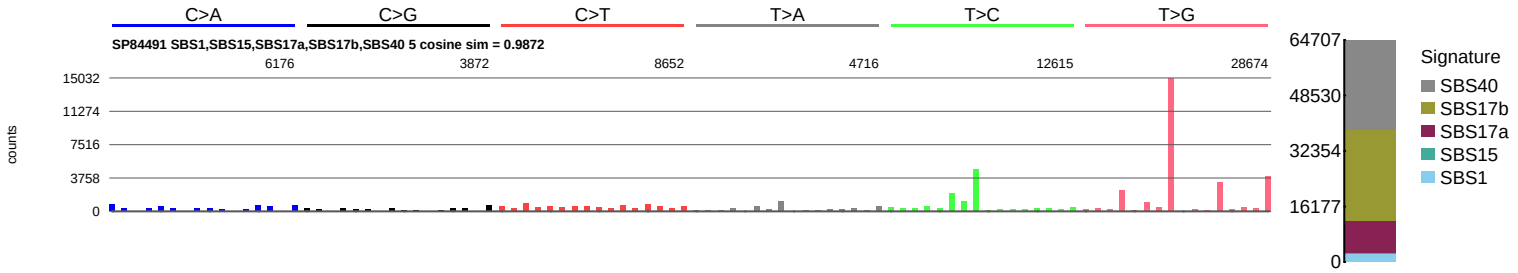
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS1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3872
8652
4716
12615
28674
counts
15032
11274
7516
3758
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64707
48530
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16177
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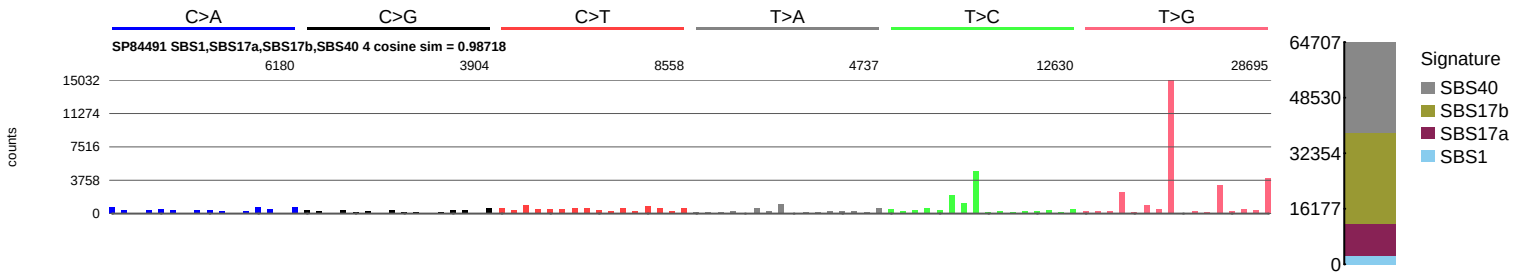
SBS40
SBS17b
SBS17a
SBS15
SBS1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T>C
T>G
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3904
8558
4737
12630
28695
counts
15032
11274
7516
3758
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64707
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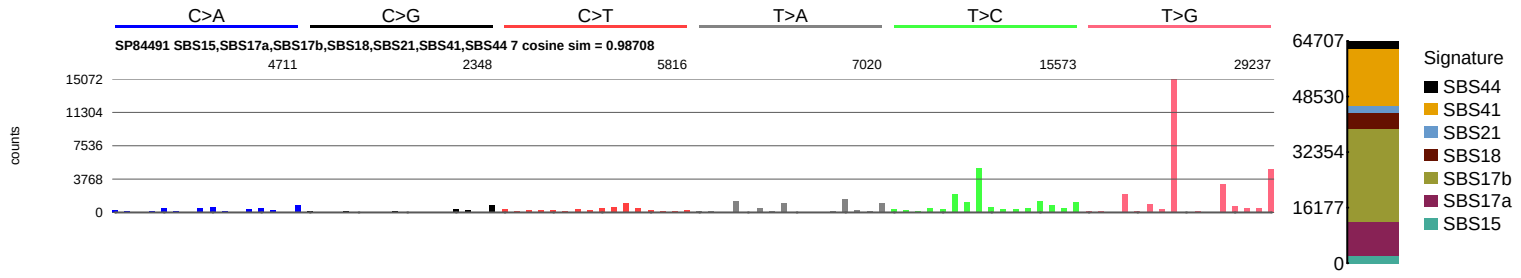
SBS17b
SBS17a
SBS1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T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98708
4711
2348
5816
7020
15573
29237
15072
11304
7536
3768
0
counts
64707
48530
32354
16177
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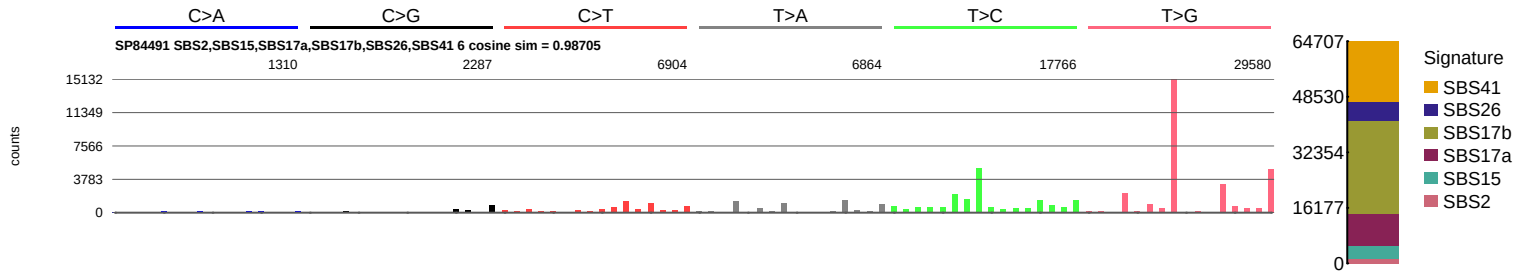
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15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T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98705
1310
2287
6904
6864
17766
29580
15132
11349
7566
3783
0
counts
64707
48530
32354
16177
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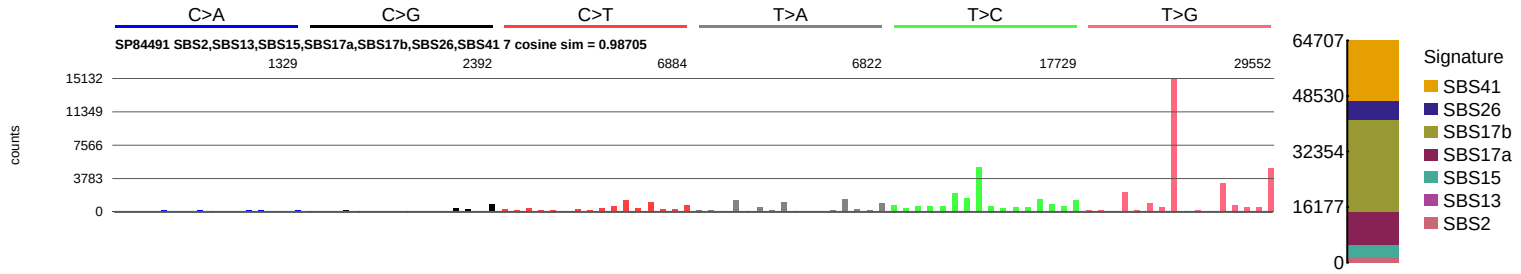
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2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T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98705
1329
2392
6884
6822
17729
29552
15132
11349
7566
3783
0
counts
64707
48530
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16177
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SBS41
SBS26
SBS17b
SBS17a
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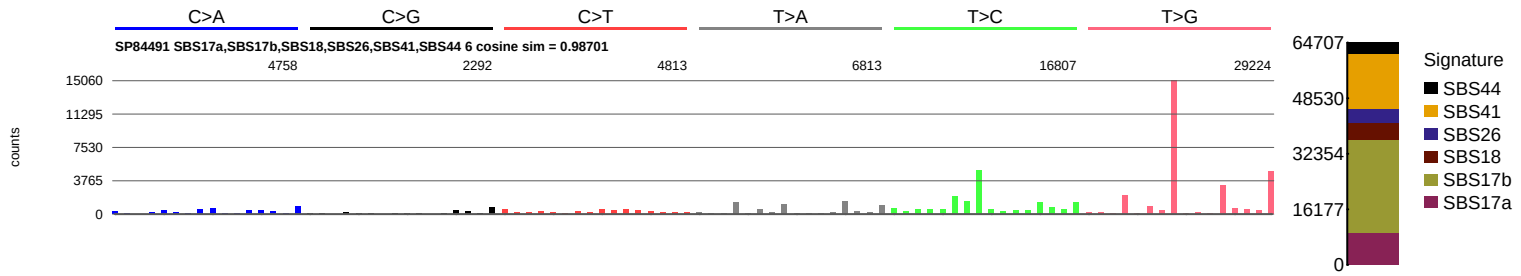
SBS13
SBS2


C>A
C>G
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T>C
T>G
SP84491 SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 6 cosine sim = 0.98701
4758
2292
4813
6813
16807
29224
15060
11295
7530
3765
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counts
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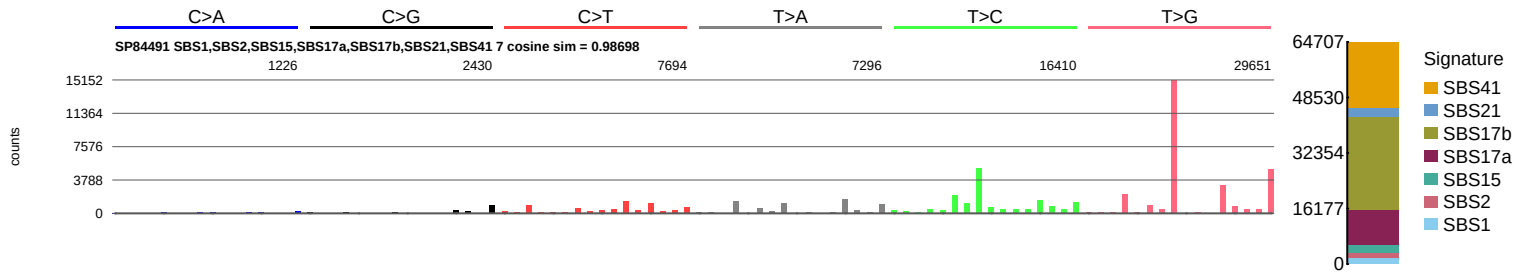
SBS41
SBS26
SBS18
SBS17b
SBS17a


C>A
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T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98698
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2430
7694
7296
16410
29651
15152
11364
7576
3788
0
counts
64707
48530
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16177
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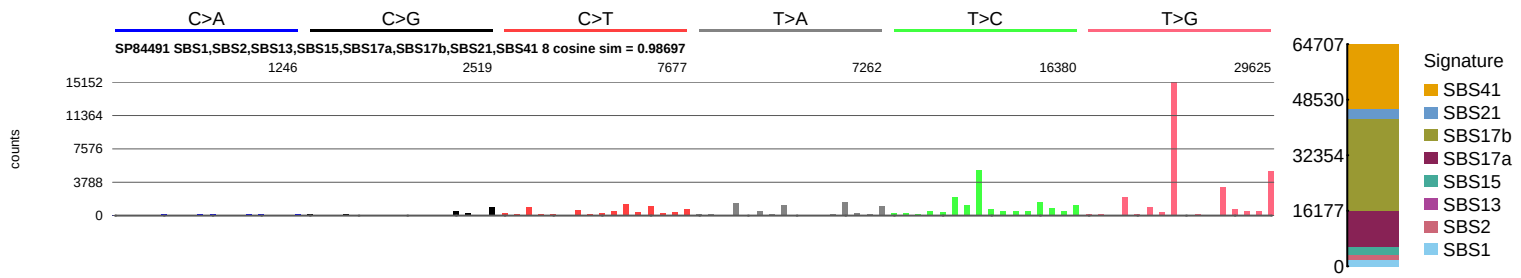
SBS2
SBS1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C>G
C>T
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T>C
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SP84491 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98697
1246
2519
7677
7262
16380
29625
15152
11364
7576
3788
0
counts
64707
48530
32354
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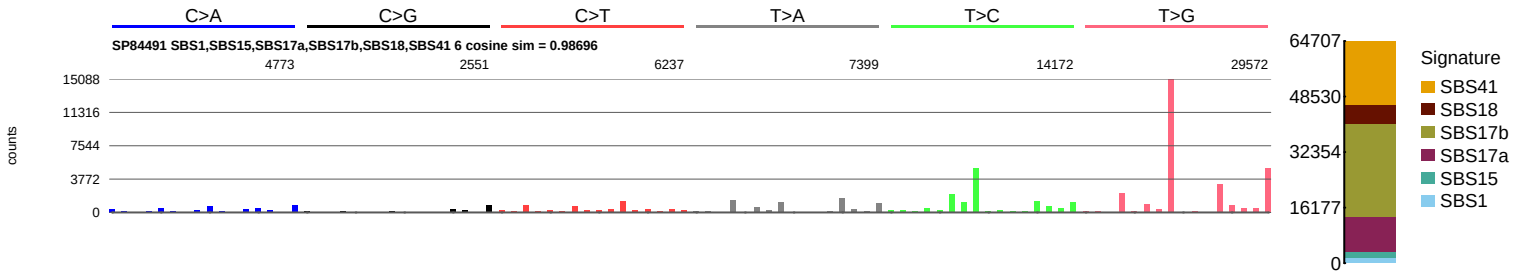
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98696
4773
2551
6237
7399
14172
29572
15088
11316
7544
3772
0
counts
64707
48530
32354
16177
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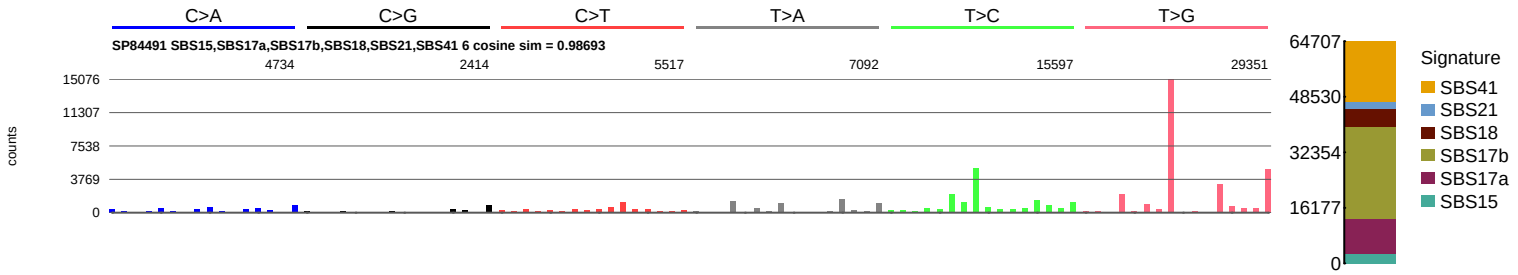
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS1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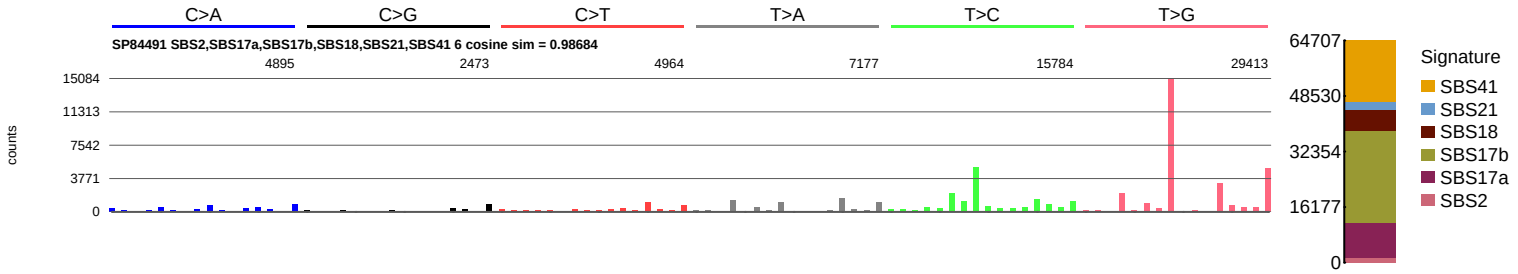
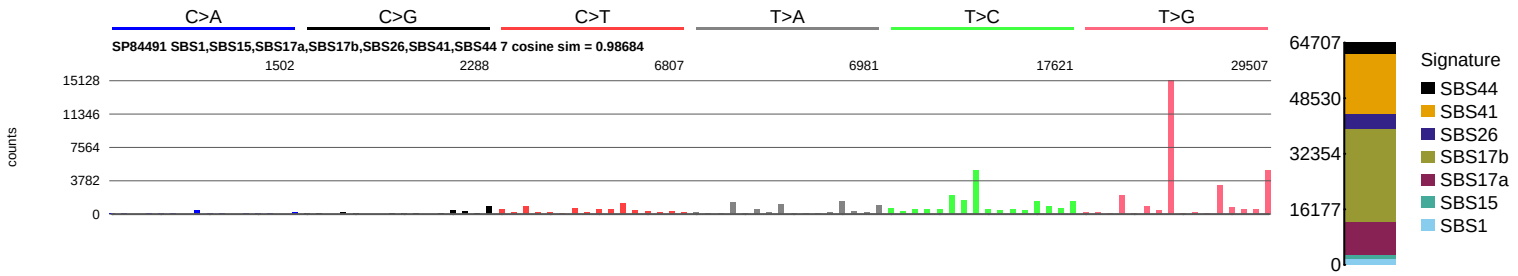
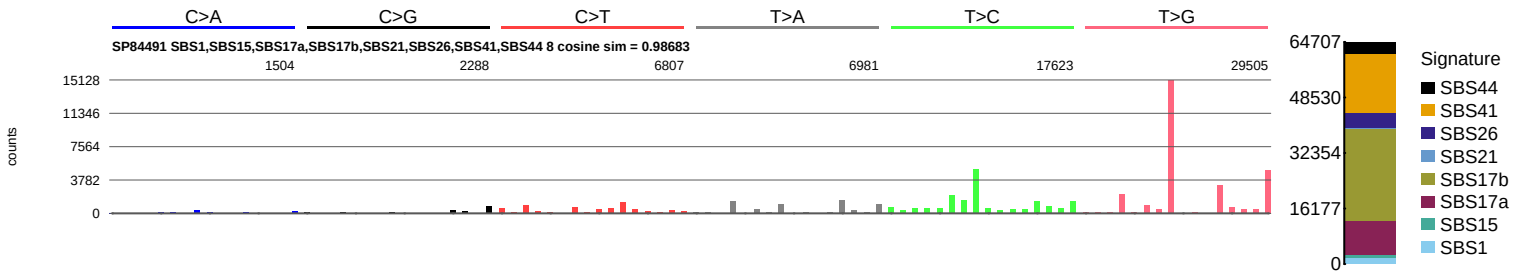
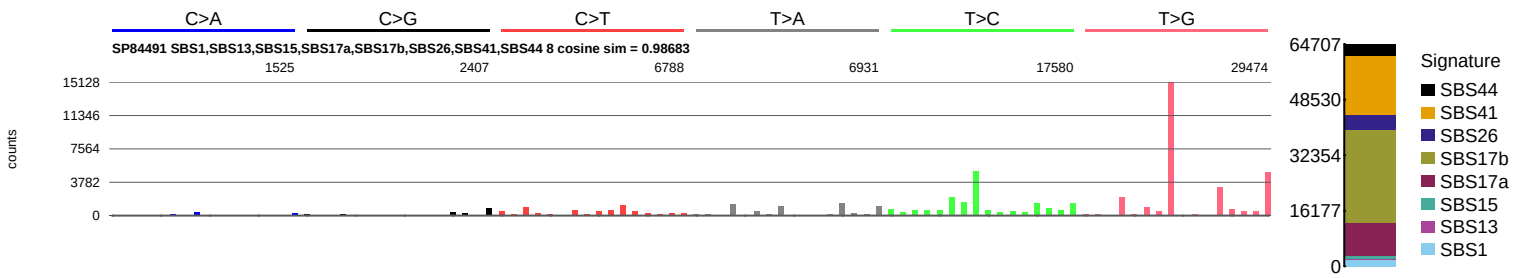
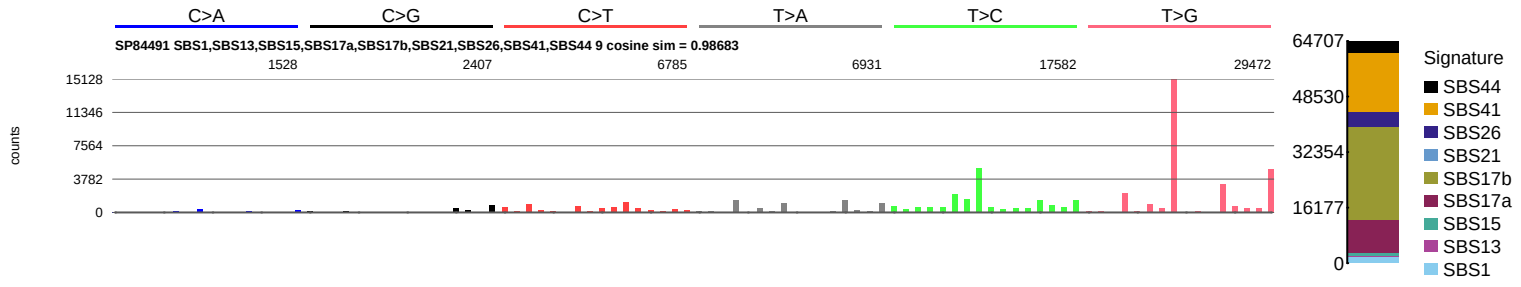
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98693
4734
2414
5517
7092
15597
29351
15076
11307
7538
3769
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15

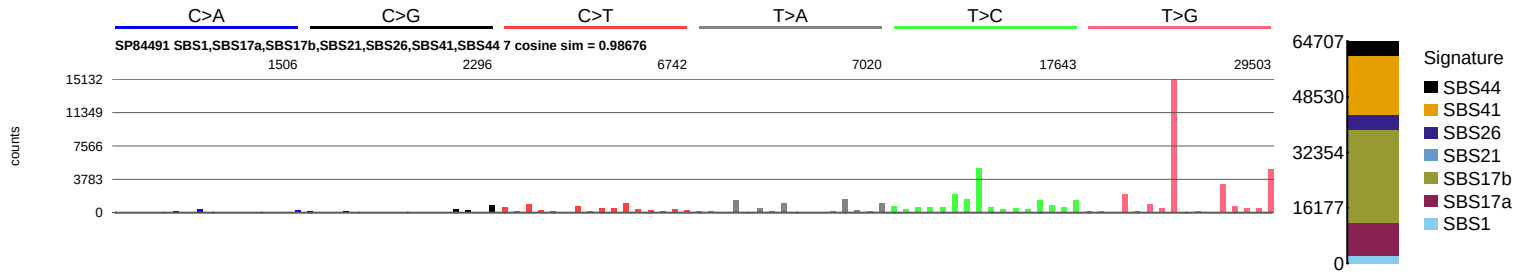

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98684
4895
2473
4964
7177
15784
29413
15084
11313
7542
3771
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2

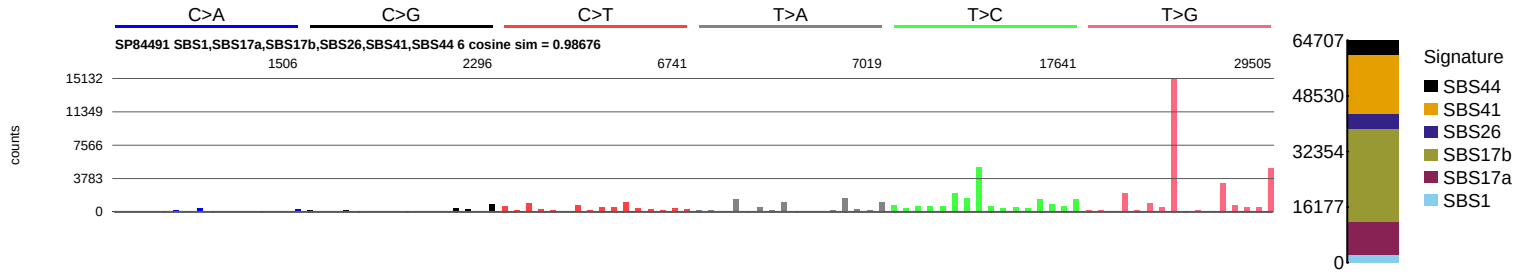

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98684
1502
2288
6807
6981
17621
29507
15128
11346
7564
3782
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS1

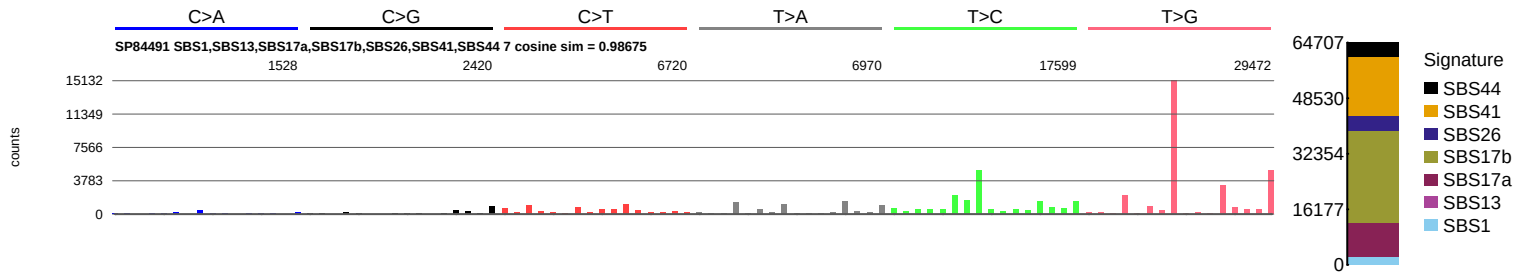

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98683
1504
2288
6807
6981
17623
29505
15128
11346
7564
3782
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1

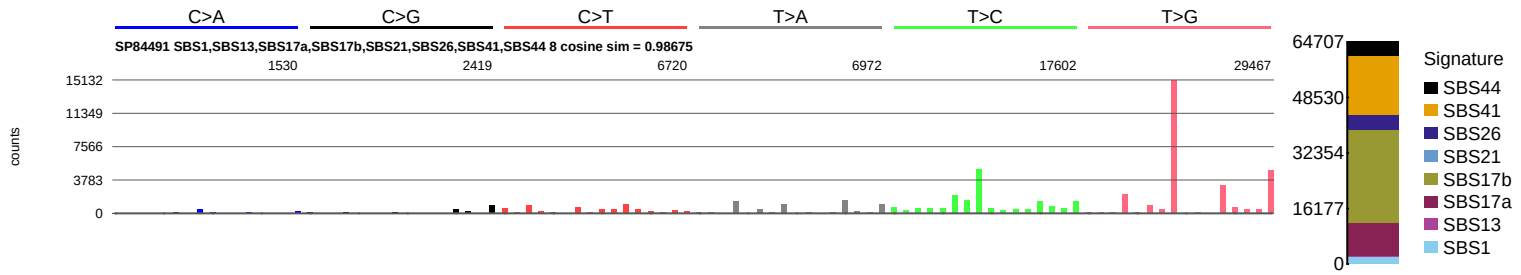

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98683
1525
2407
6788
6931
17580
29474
15128
11346
7564
3782
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1

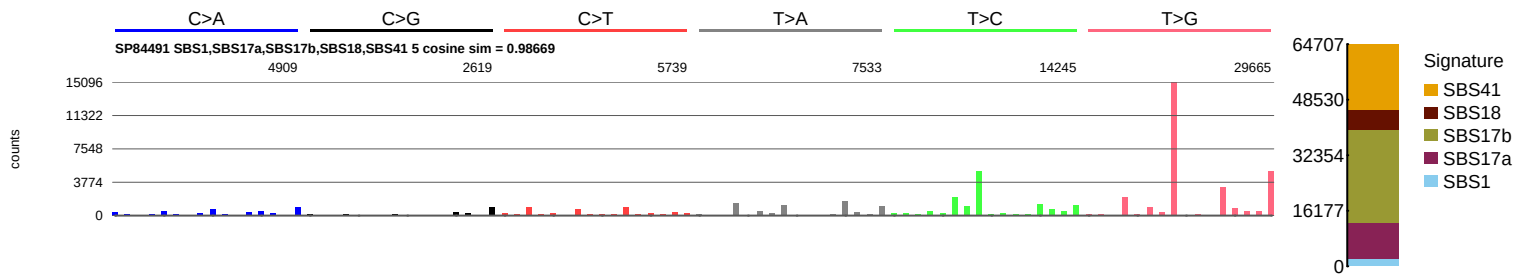

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98683
1528
2407
6785
6931
17582
29472
15128
11346
7564
3782
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

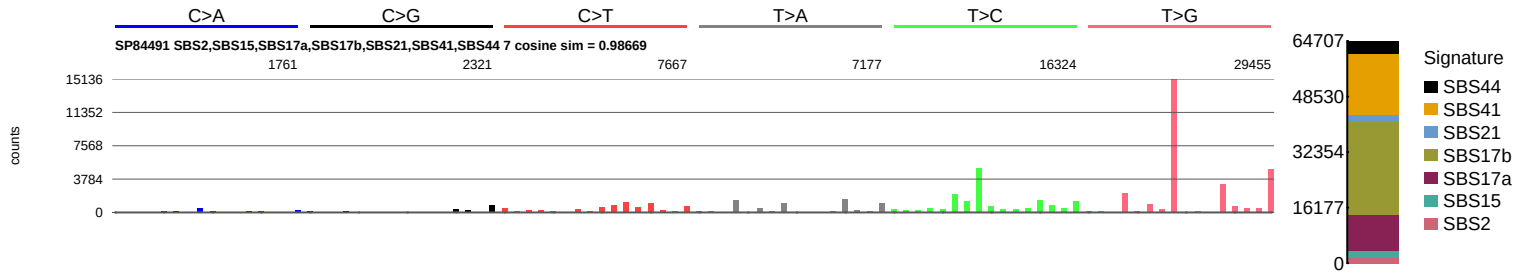

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 7 cosine sim = 0.98676
1506
2296
6742
7020
17643
29503
15132
11349
7566
3783
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS1

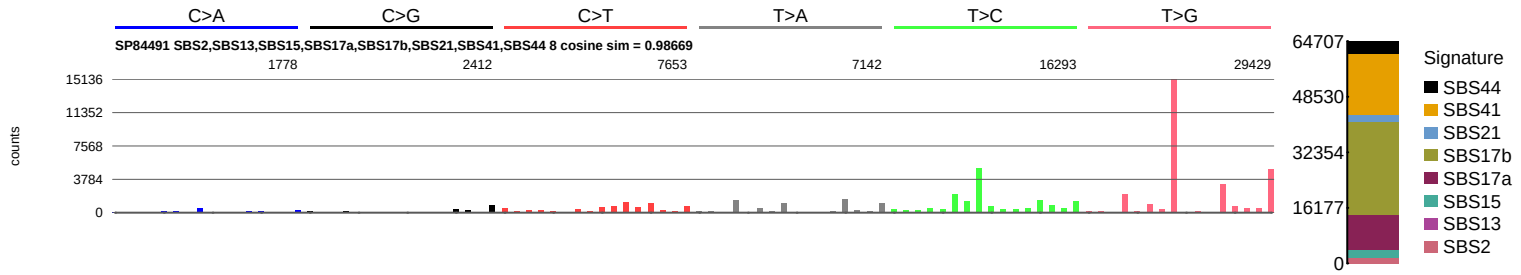

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98676
1506
2296
6741
7019
17641
29505
15132
11349
7566
3783
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS1

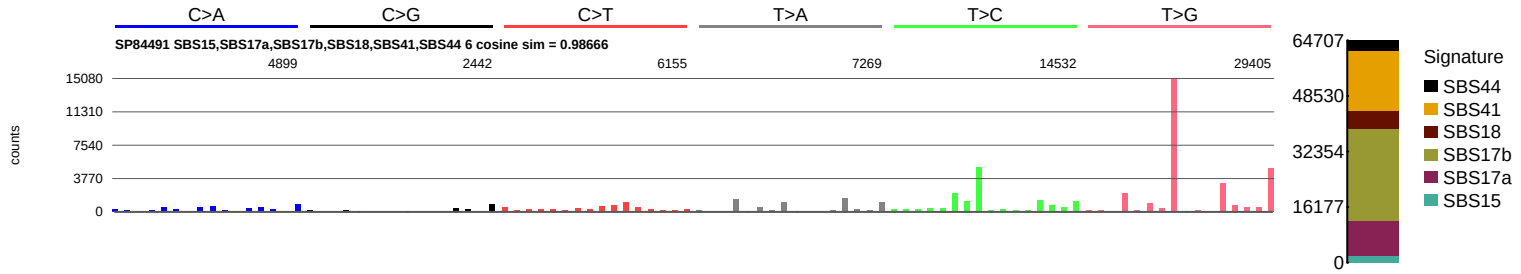

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98675
1528
2420
6720
6970
17599
29472
15132
11349
7566
3783
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS1

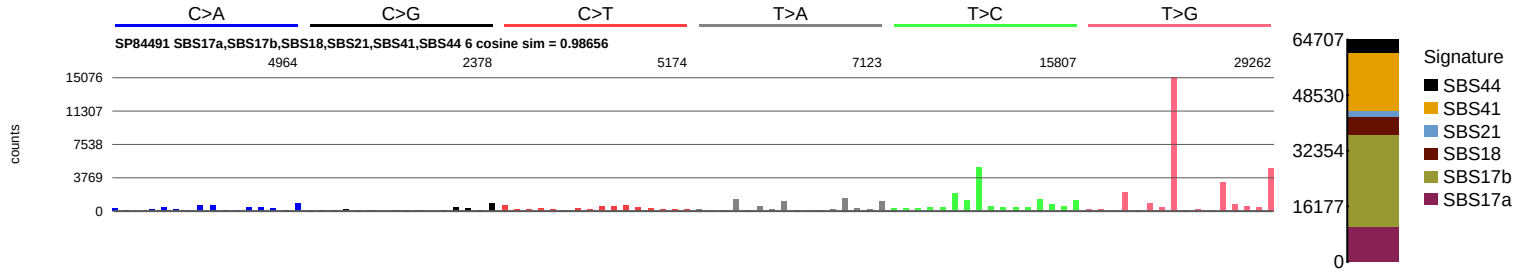

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98675
1530
2419
6720
6972
17602
29467
15132
11349
7566
3783
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1

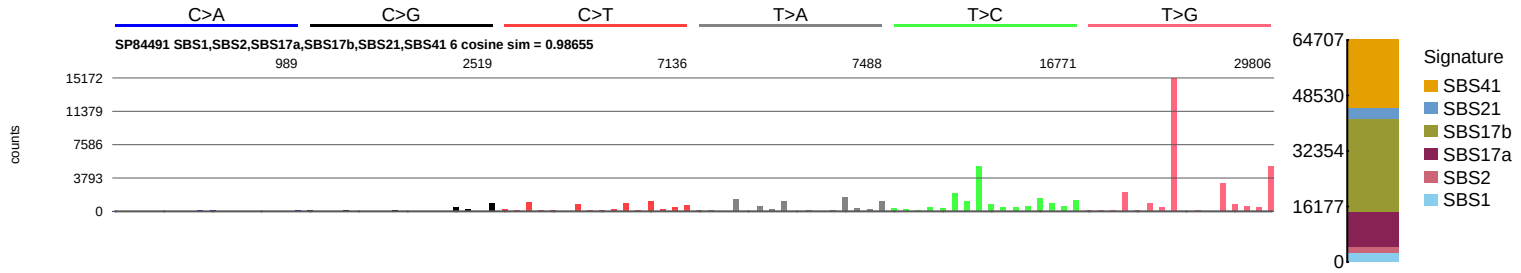

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98669
4909
2619
5739
7533
14245
29665
15096
11322
7548
3774
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS1

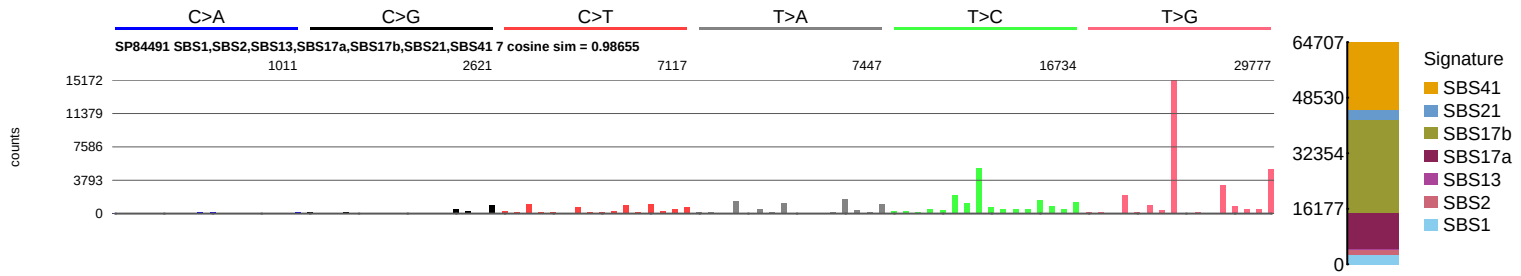

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98669
1761
2321
7667
7177
16324
29455
15136
11352
7568
3784
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2

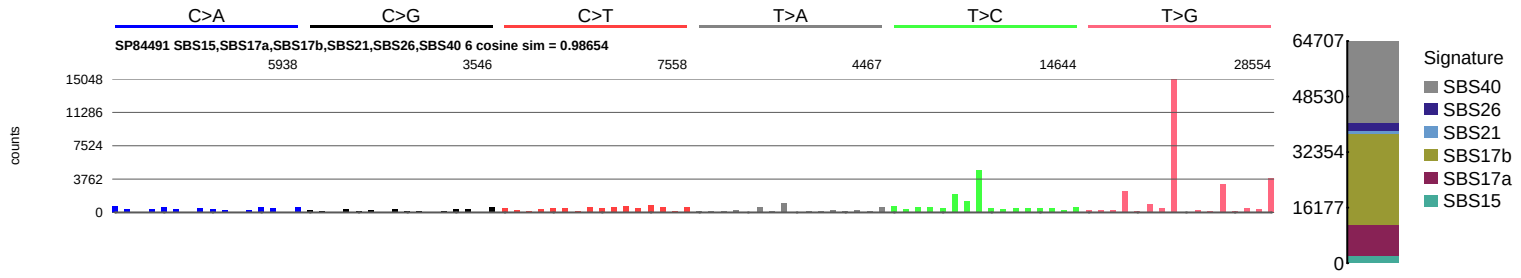

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98669
1778
2412
7653
7142
16293
29429
15136
11352
7568
3784
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2

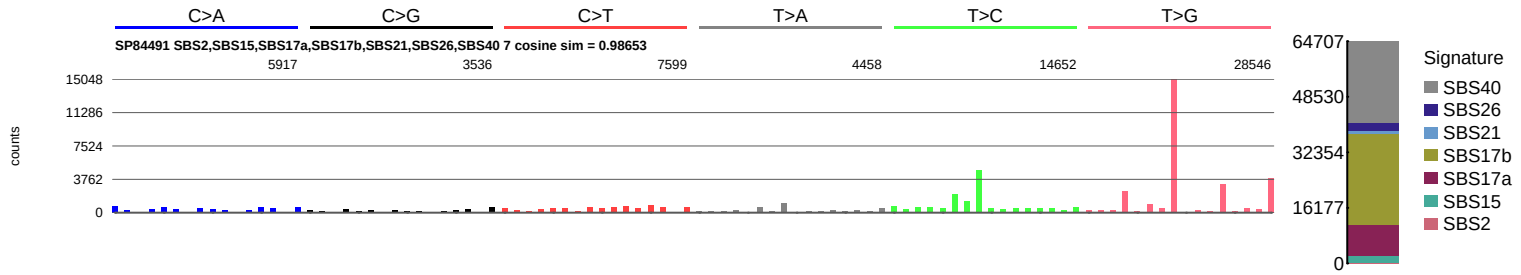

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98666
4899
2442
6155
7269
14532
29405
15080
11310
7540
3770
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15

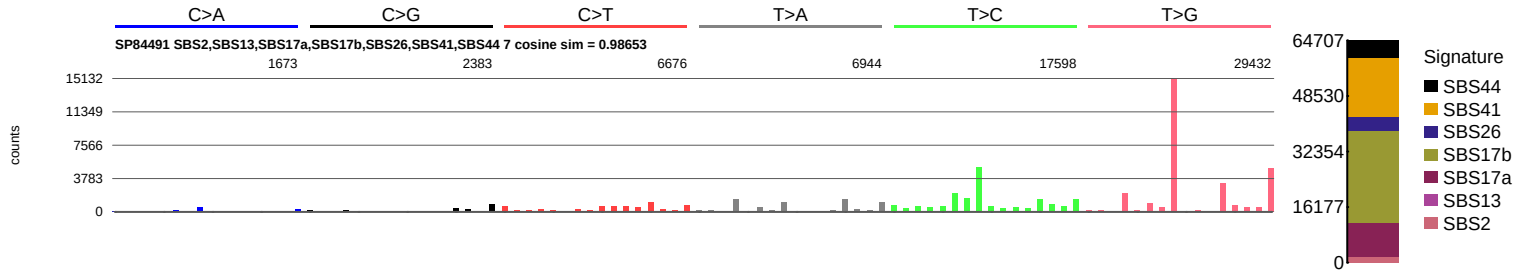

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 6 cosine sim = 0.98656
4964
2378
5174
7123
15807
29262
15076
11307
7538
3769
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a

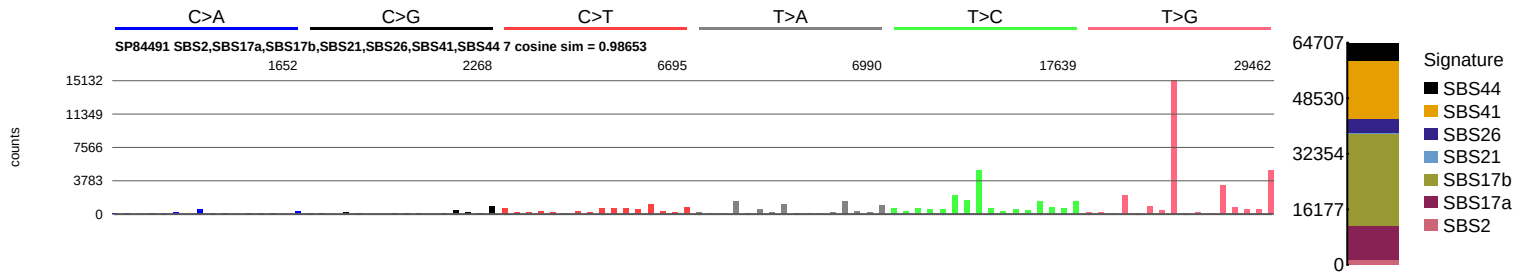

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98655
989
2519
7136
7488
16771
29806
15172
11379
7586
3793
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS2
SBS1

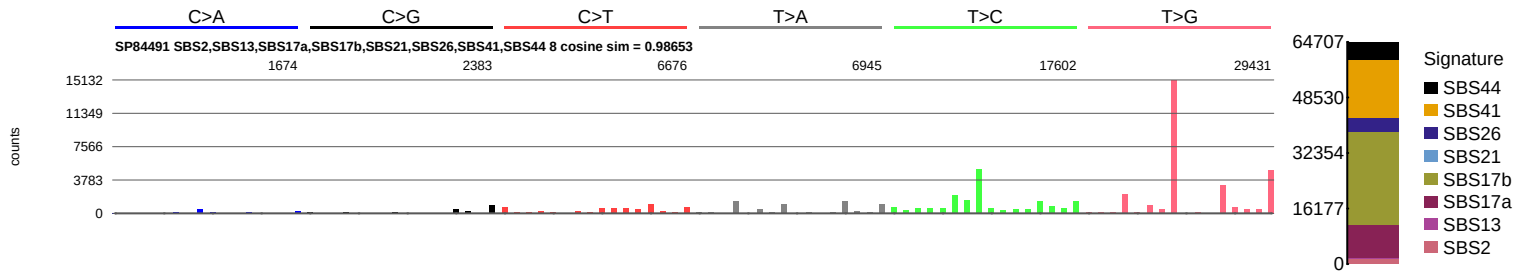

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98655
1011
2621
7117
7447
16734
29777
15172
11379
7586
3793
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

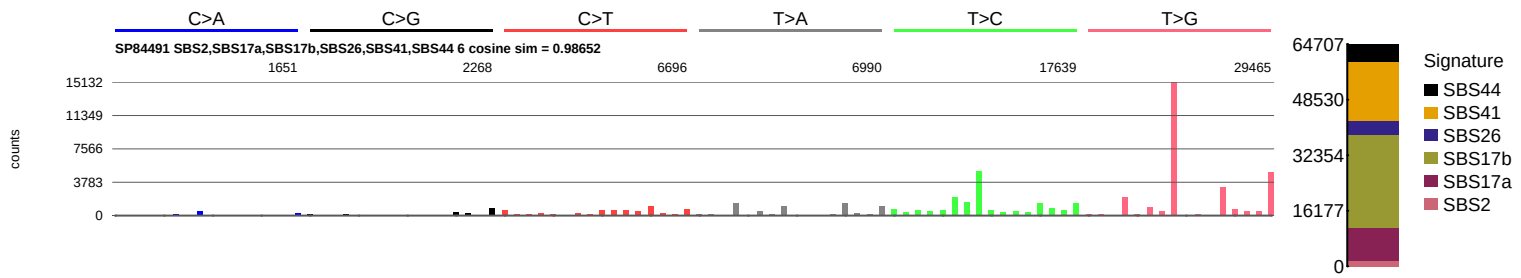

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.98654
5938
3546
7558
4467
14644
28554
counts
15048
11286
7524
3762
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15

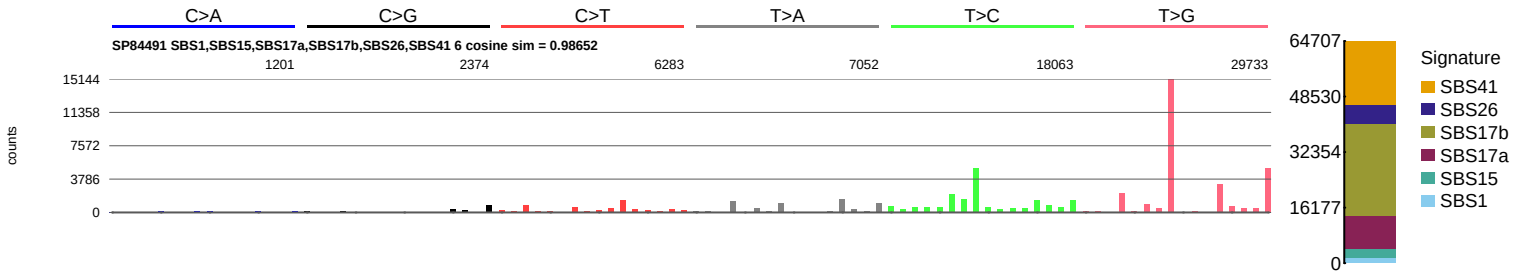

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98653
5917
3536
7599
4458
14652
28546
counts
15048
11286
7524
3762
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2

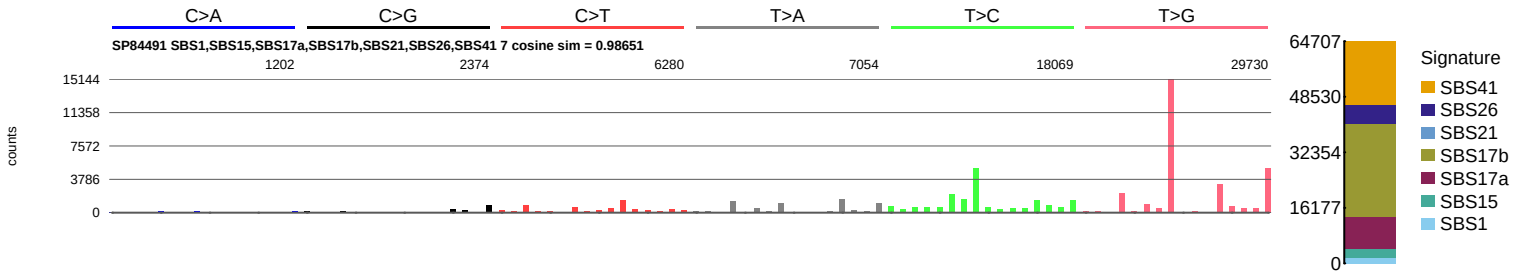

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98653
1673
2383
6676
6944
17598
29432
counts
15132
11349
7566
3783
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2

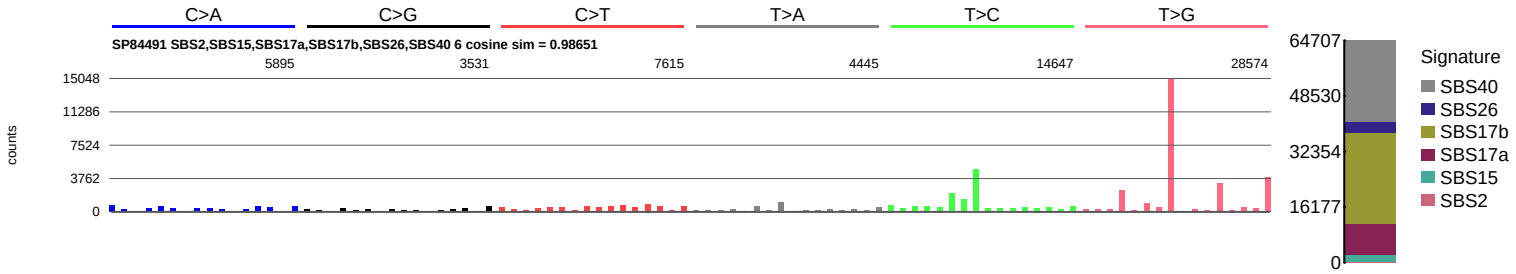

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 7 cosine sim = 0.98653
1652
2268
6695
6990
17639
29462
counts
15132
11349
7566
3783
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS2

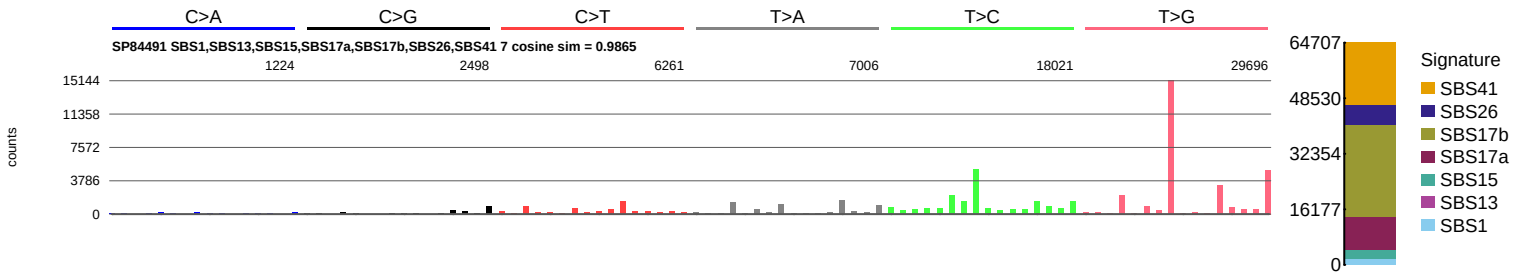

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98653
1674
2383
6676
6945
17602
29431
counts
15132
11349
7566
3783
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98652
1651
2268
6696
6990
17639
29465
counts
15132
11349
7566
3783
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98652
1201
2374
6283
7052
18063
29733
counts
15144
11358
7572
3786
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98651
1202
2374
6280
7054
18069
29730
counts
15144
11358
7572
3786
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98651
5895
3531
7615
4445
14647
28574
counts
15048
11286
7524
3762
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2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C>G
C>T
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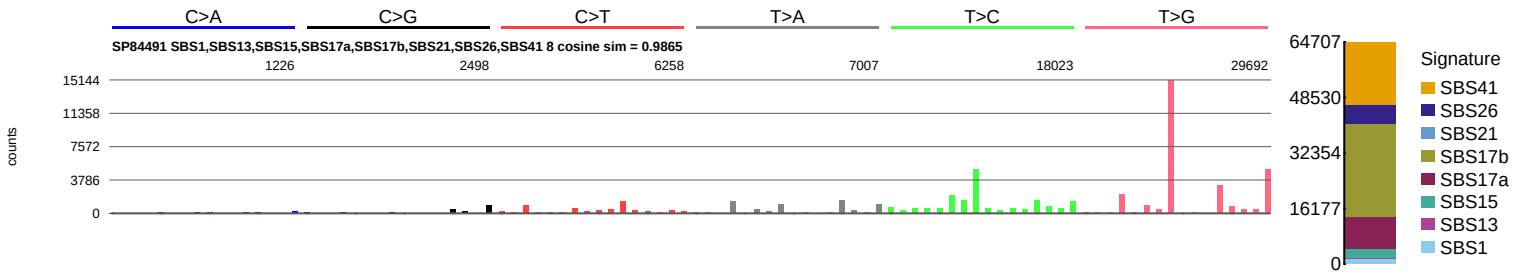
T>C
T>G
SP84491 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.9865
1224
2498
6261
7006
18021
29696
counts
15144
11358
7572
3786
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1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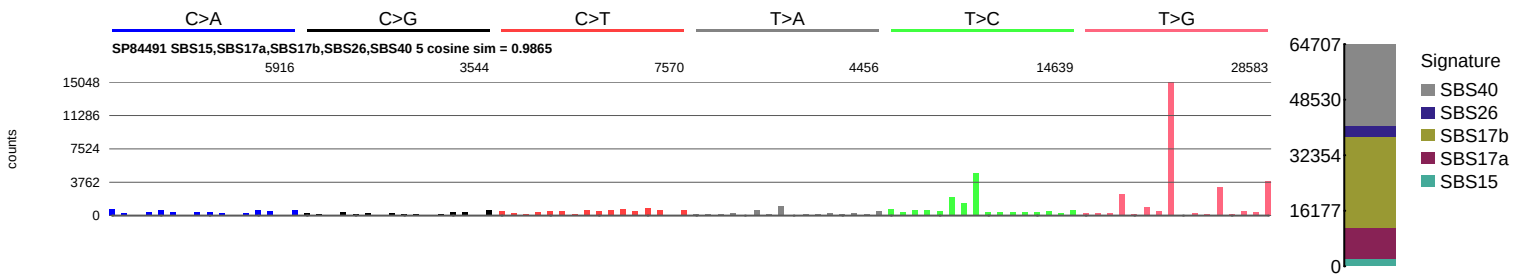
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.9865
1226
2498
6258
7007
18023
29692
counts
15144
11358
7572
3786
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
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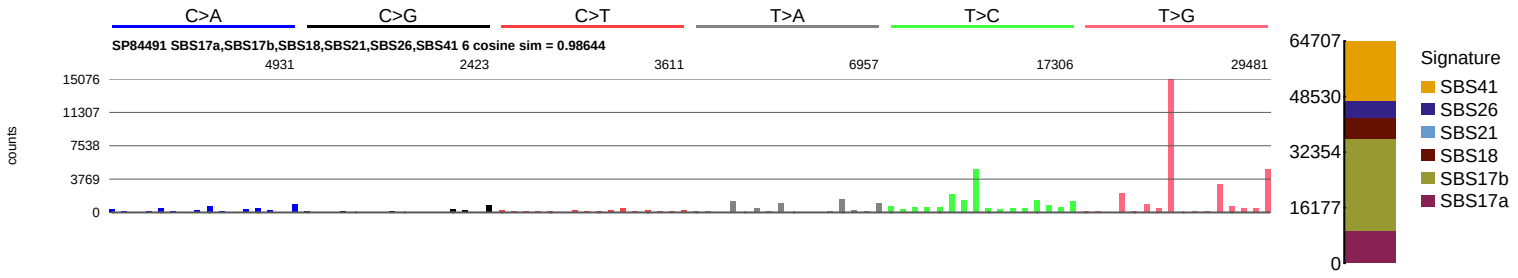
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.9865
5916
3544
7570
4456
14639
28583
counts
15048
11286
7524
3762
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15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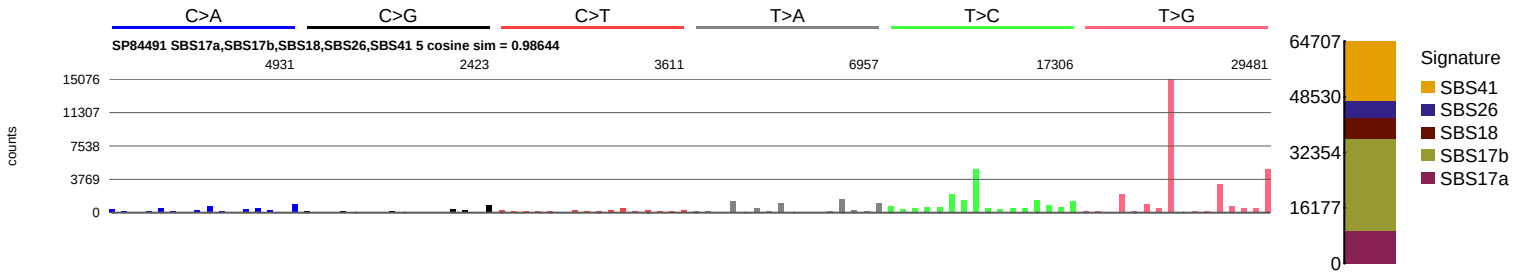
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 6 cosine sim = 0.98644
4931
2423
3611
6957
17306
29481
counts
15076
11307
7538
3769
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
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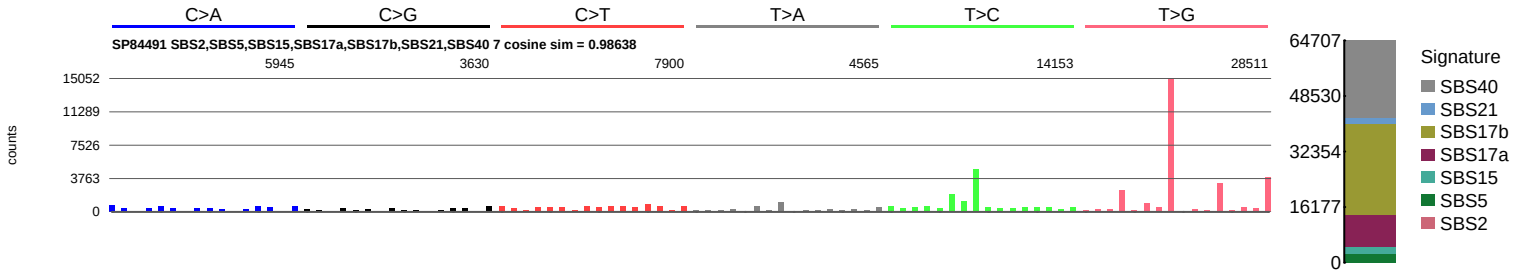
T>C
T>G
SP84491 SBS17a,SBS17b,SBS18,SBS26,SBS41 5 cosine sim = 0.98644
4931
2423
3611
6957
17306
29481
counts
15076
11307
7538
3769
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a


C>A
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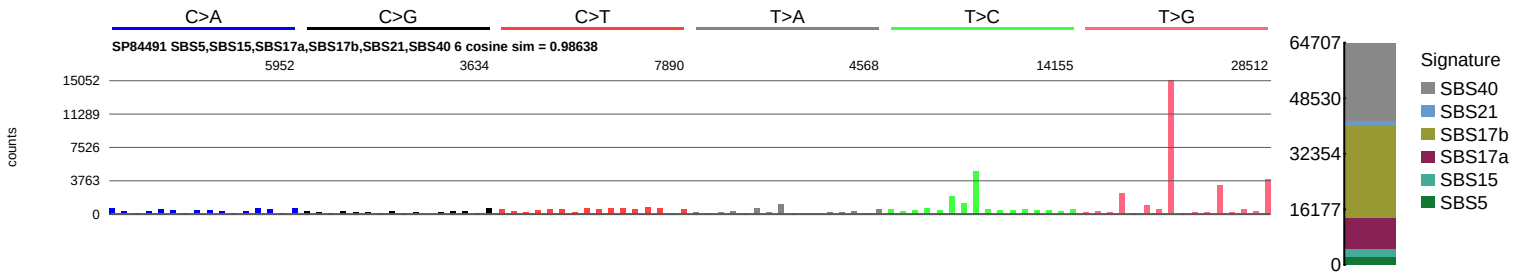
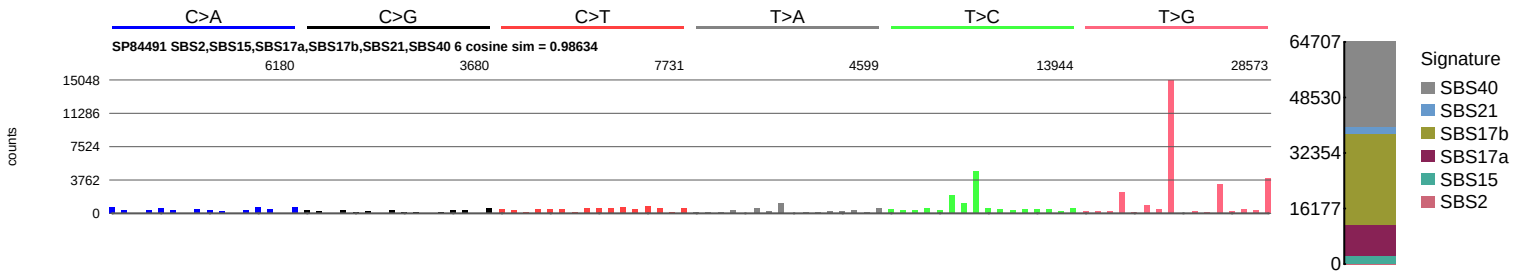
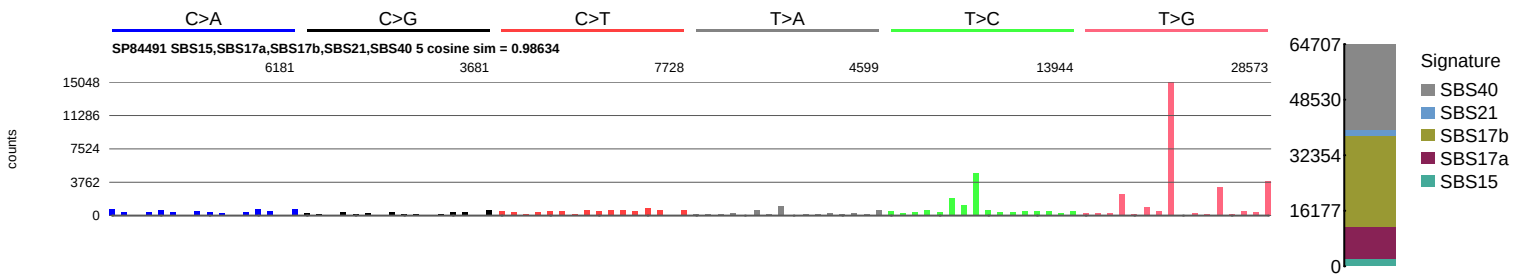
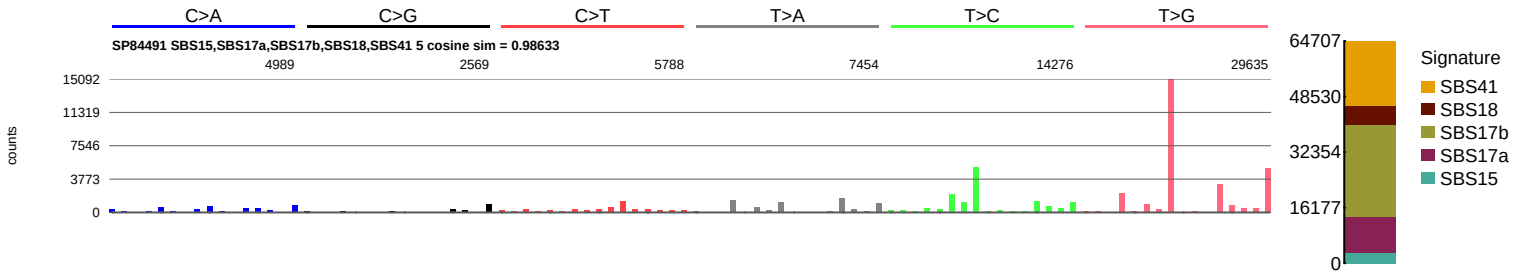
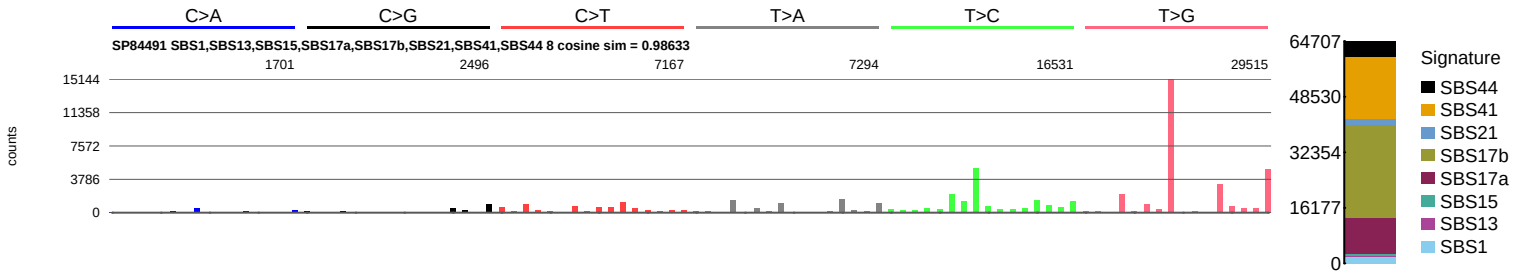
T>C
T>G
SP84491 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98638
5945
3630
7900
4565
14153
28511
counts
15052
11289
7526
3763
0
64707
48530
32354
16177
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2

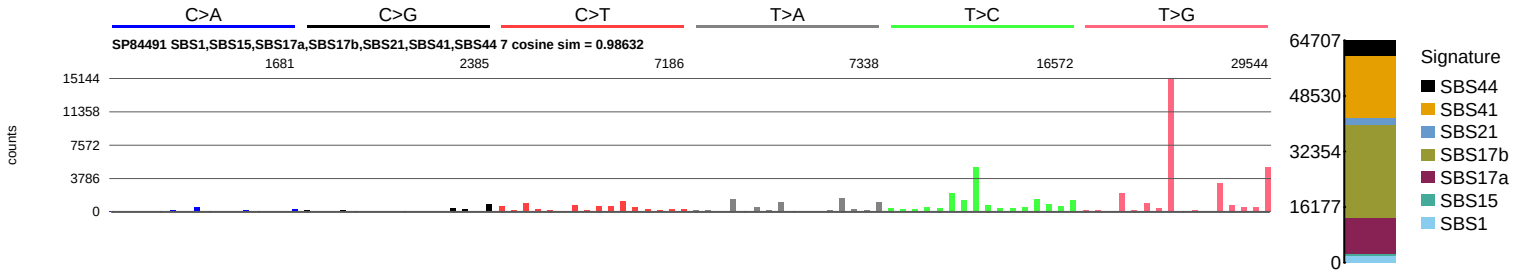

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98638
5952
3634
7890
4568
14155
28512
counts
15052
11289
7526
3763
0
64707
48530
32354
16177
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5

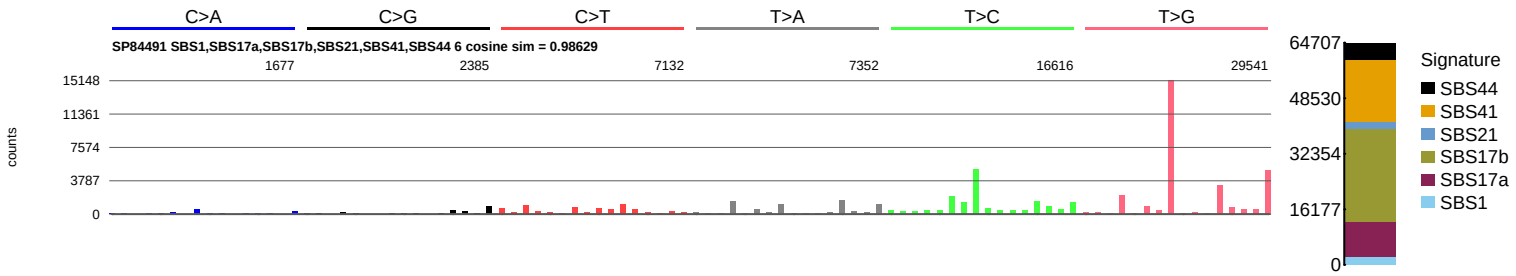

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98634
6180
3680
7731
4599
13944
28573
counts
15048
11286
7524
3762
0
64707
48530
32354
16177
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2

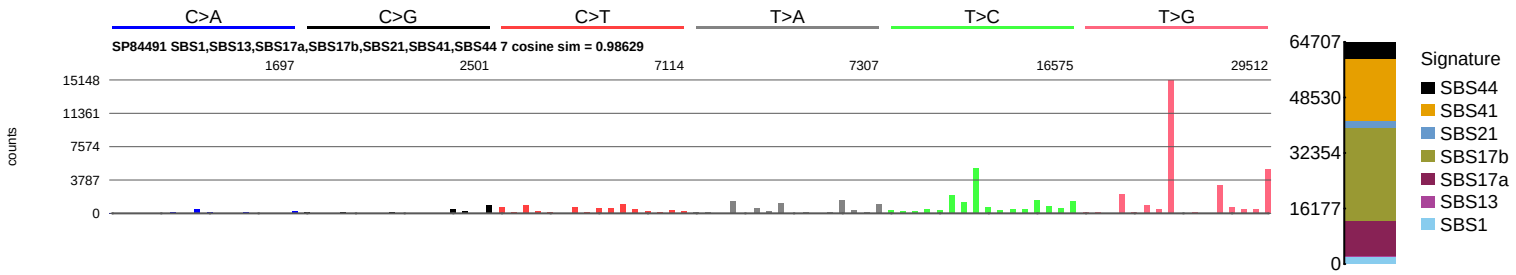

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98634
6181
3681
7728
4599
13944
28573
counts
15048
11286
7524
3762
0
64707
48530
32354
16177
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15

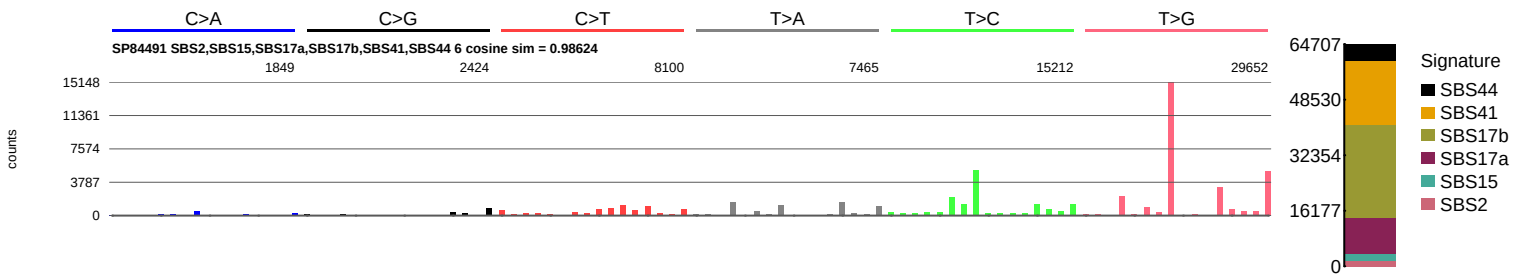

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98633
4989
2569
5788
7454
14276
29635
15092
11319
7546
3773
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15

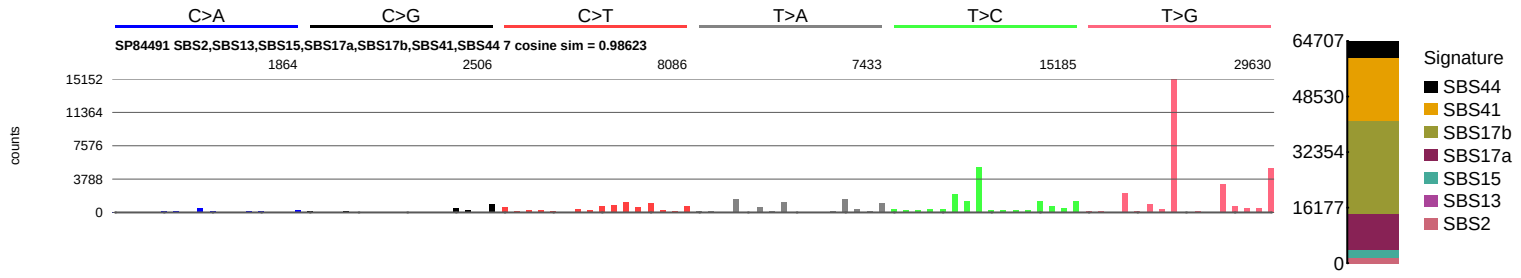

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98633
1701
2496
7167
7294
16531
29515
15144
11358
7572
3786
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1

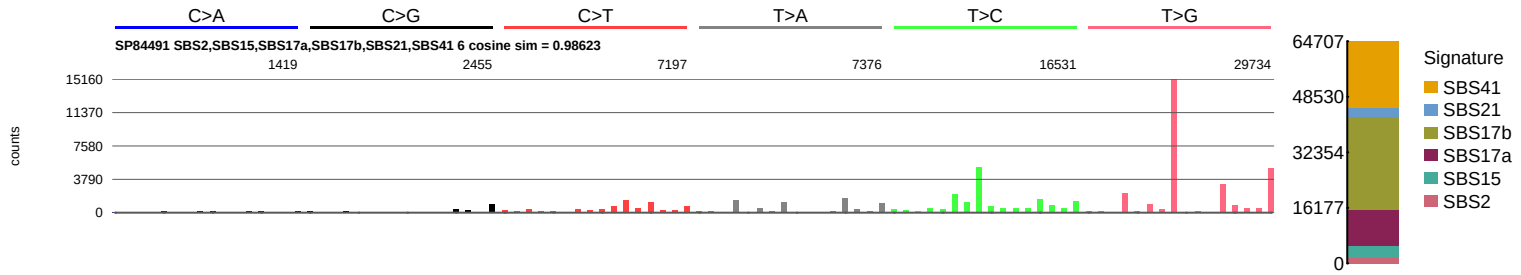

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98632
1681
2385
7186
7338
16572
29544
15144
11358
7572
3786
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS1

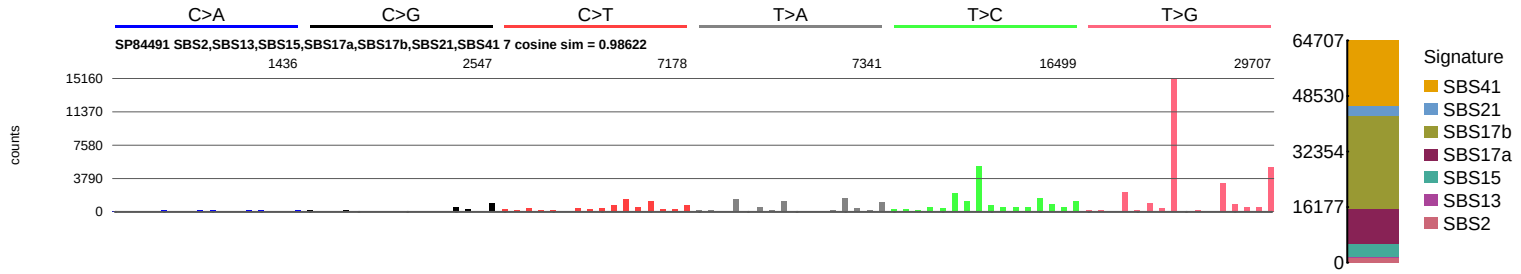

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.98629
1677
2385
7132
7352
16616
29541
15148
11361
7574
3787
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS1

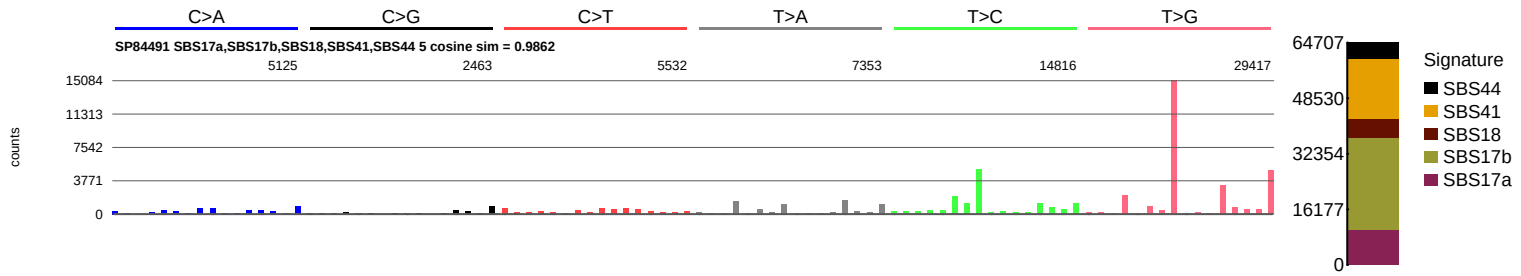

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98629
1697
2501
7114
7307
16575
29512
15148
11361
7574
3787
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS1

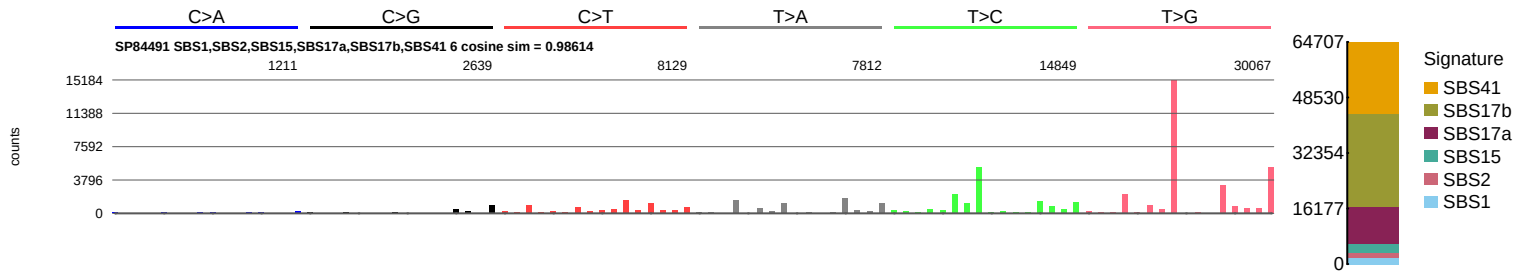

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98624
1849
2424
8100
7465
15212
29652
15148
11361
7574
3787
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS2

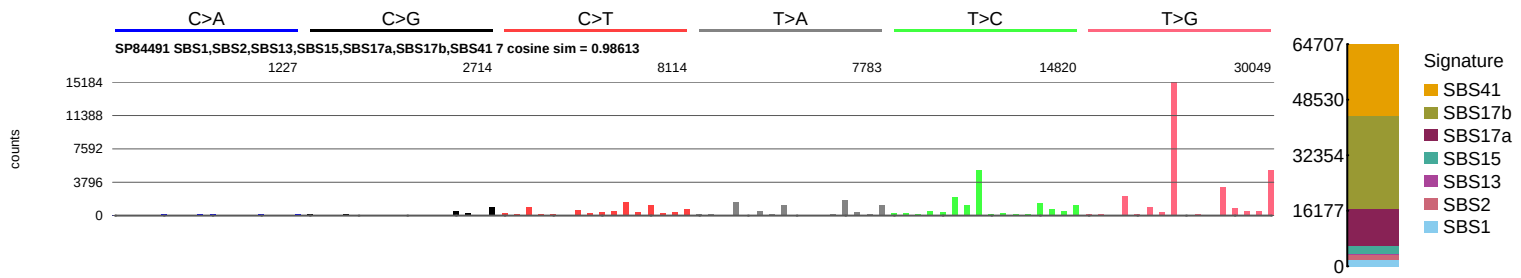

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98623
1864
2506
8086
7433
15185
29630
15152
11364
7576
3788
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2

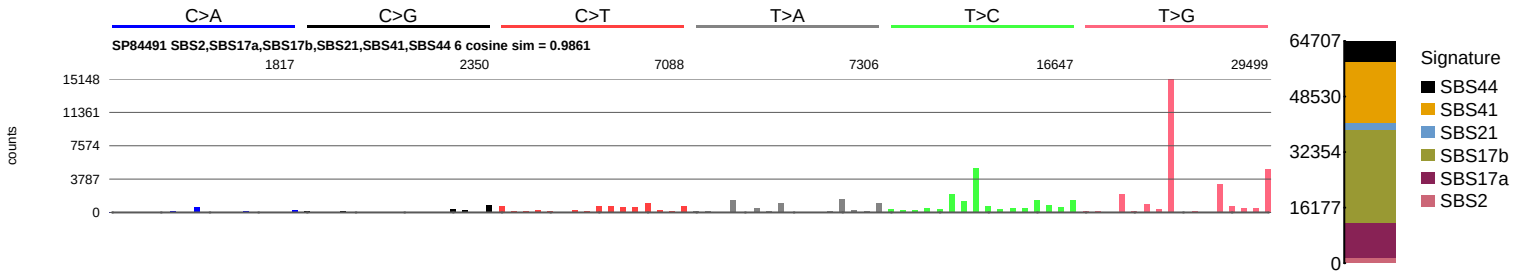

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98623
1419
2455
7197
7376
16531
29734
15160
11370
7580
3790
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98622
1436
2547
7178
7341
16499
29707
15160
11370
7580
3790
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS18,SBS41,SBS44 5 cosine sim = 0.9862
5125
2463
5532
7353
14816
29417
15084
11313
7542
3771
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98614
1211
2639
8129
7812
14849
30067
15184
11388
7592
3796
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98613
1227
2714
8114
7783
14820
30049
15184
11388
7592
3796
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.9861
1817
2350
7088
7306
16647
29499
15148
11361
7574
3787
0
counts
64707
48530
32354
16177
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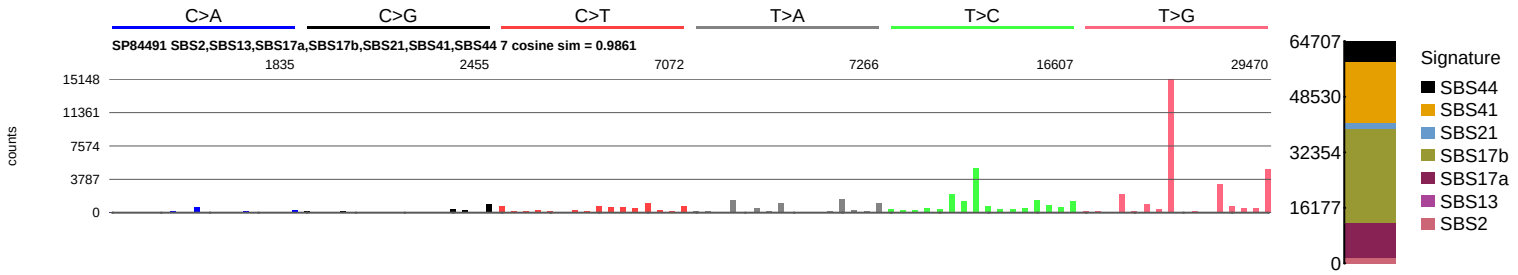
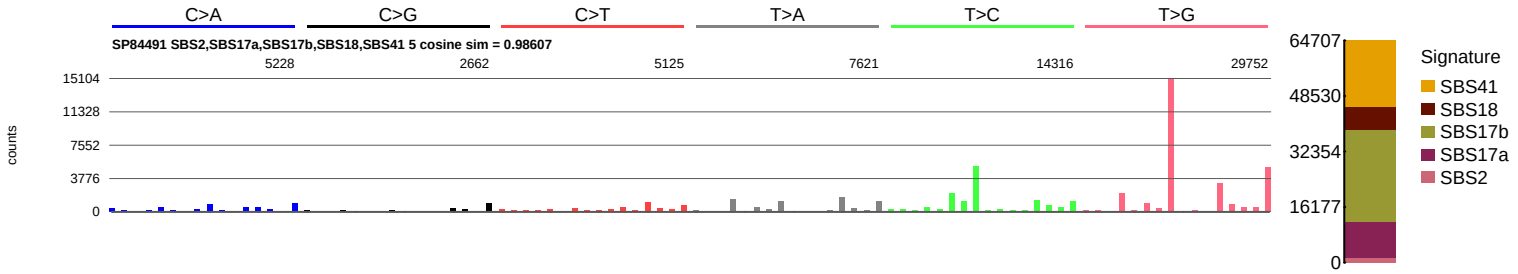
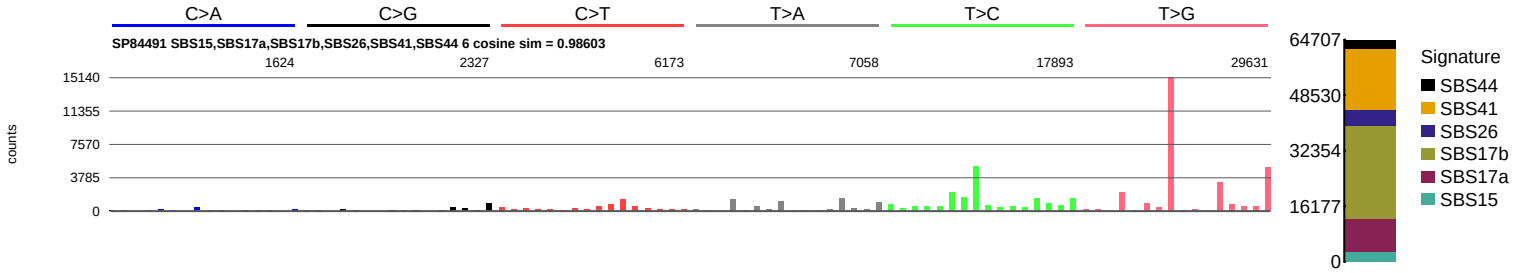
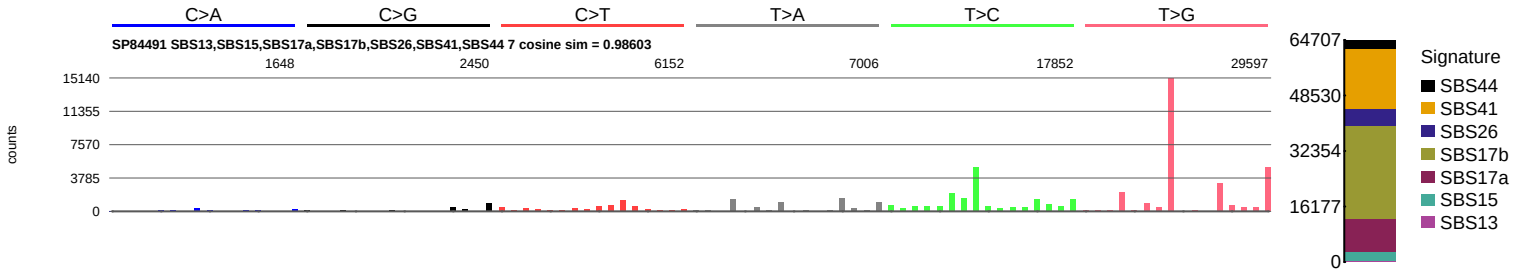
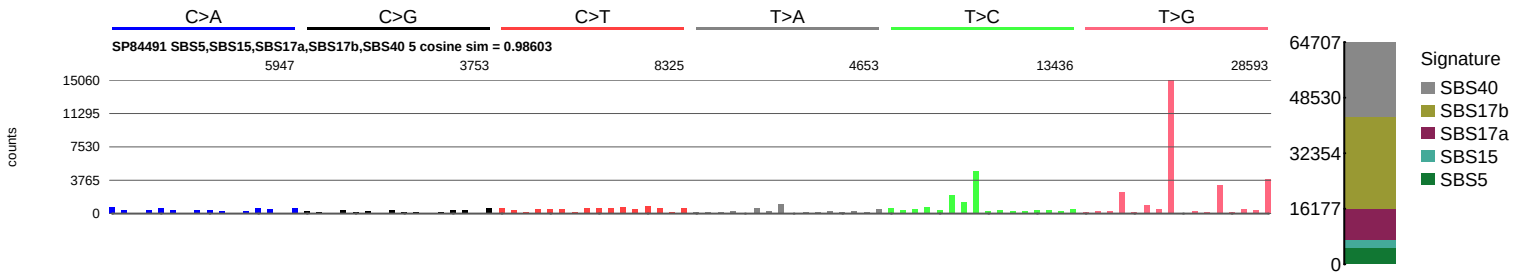
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS2

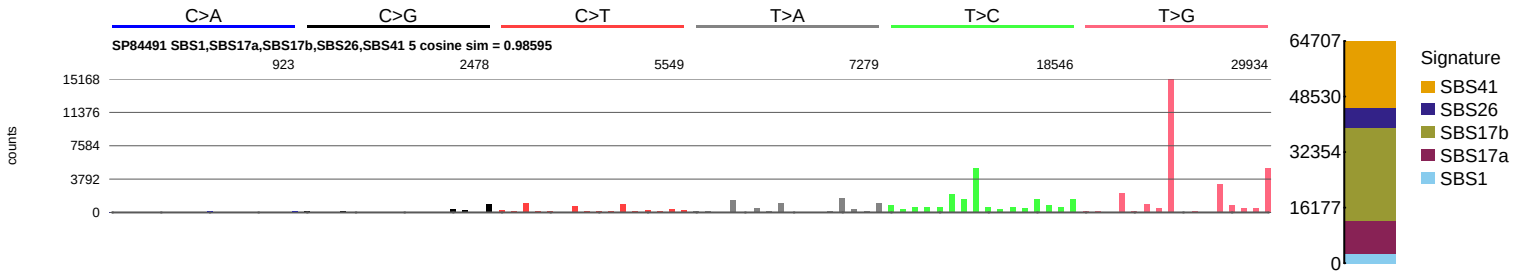

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.9861
1835
2455
7072
7266
16607
29470
15148
11361
7574
3787
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS2

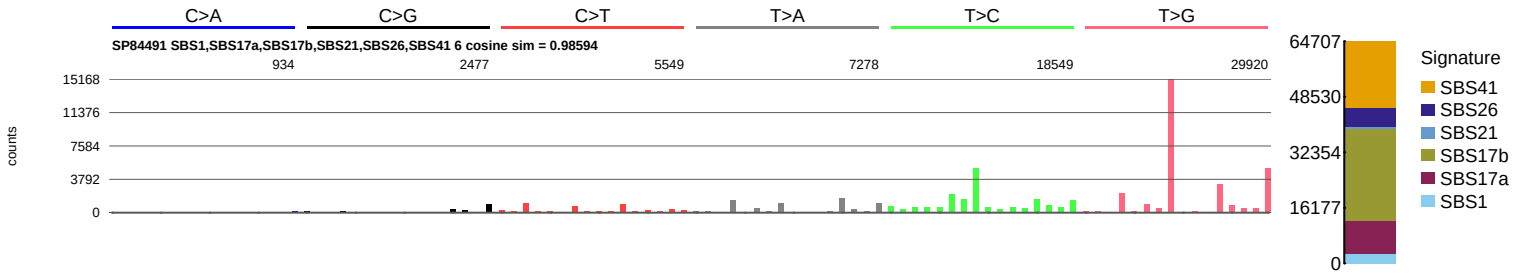

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98607
5228
2662
5125
7621
14316
29752
15104
11328
7552
3776
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS2

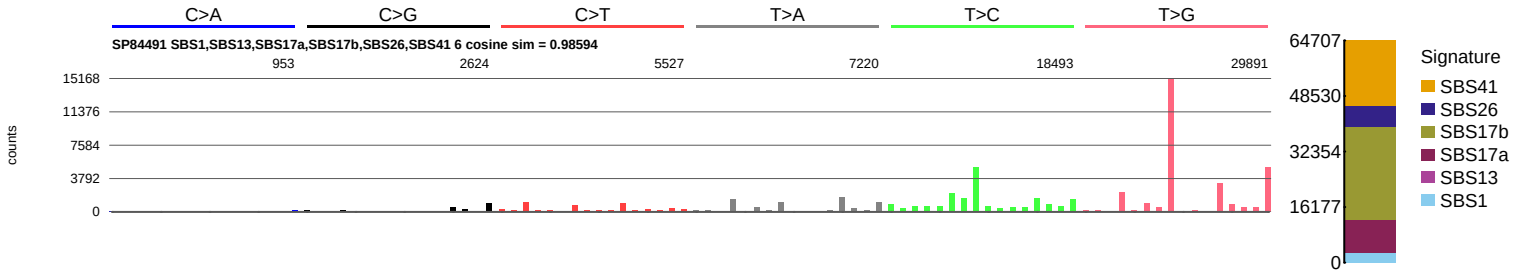

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98603
1624
2327
6173
7058
17893
29631
15140
11355
7570
3785
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15

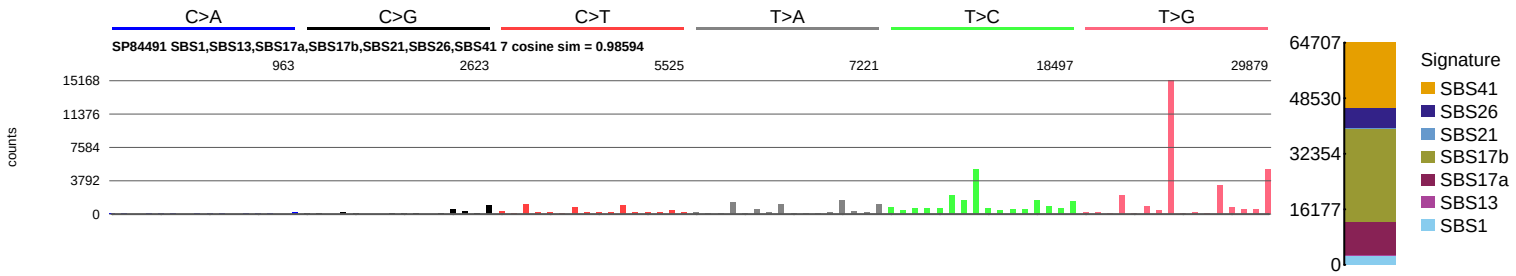

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98603
1648
2450
6152
7006
17852
29597
15140
11355
7570
3785
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13

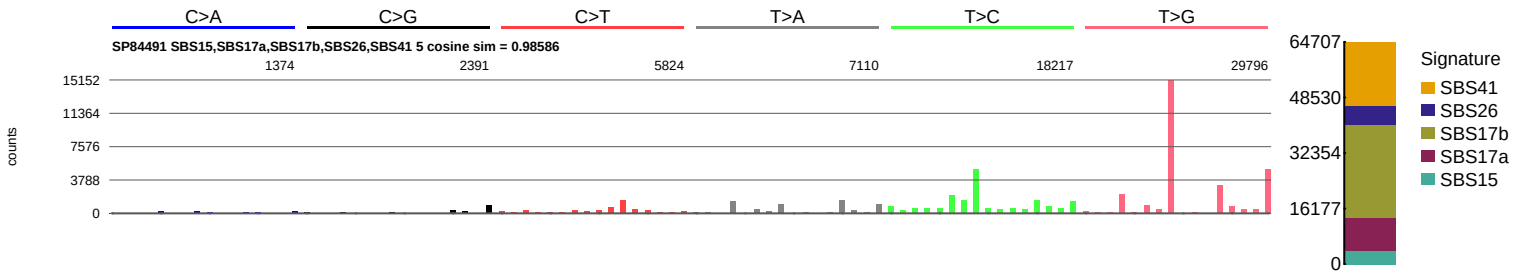

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98603
5947
3753
8325
4653
13436
28593
15060
11295
7530
3765
0
counts
64707
48530
32354
16177
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5

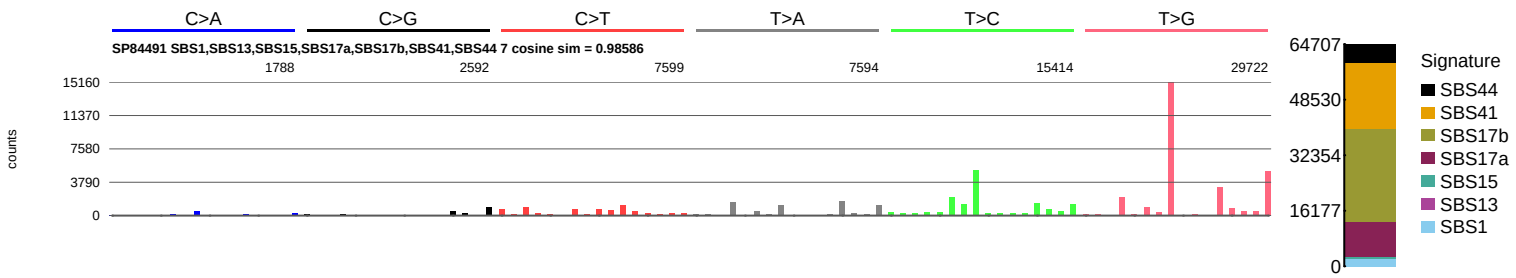

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS26,SBS41 5 cosine sim = 0.98595
923
2478
5549
7279
18546
29934
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS1

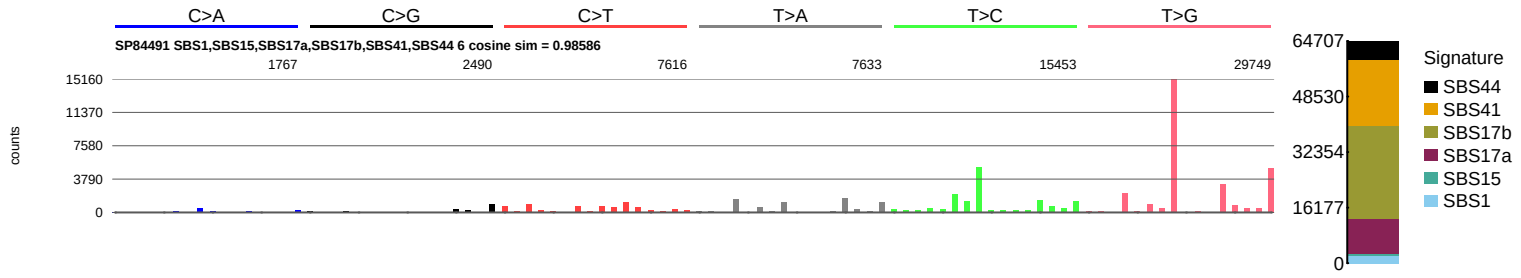

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS41 6 cosine sim = 0.98594
934
2477
5549
7278
18549
29920
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS1

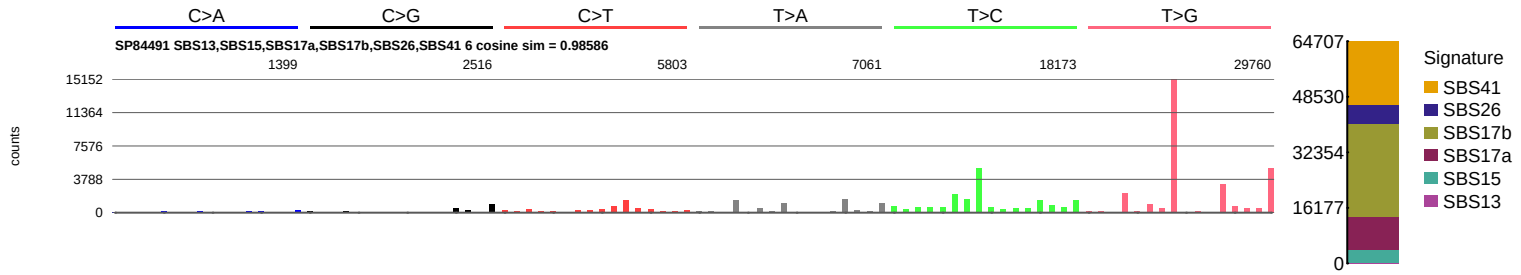

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98594
953
2624
5527
7220
18493
29891
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS1

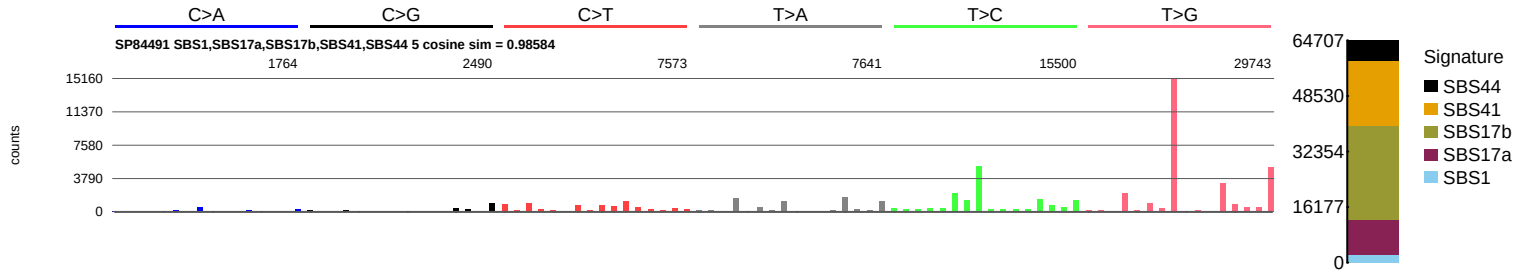

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98594
963
2623
5525
7221
18497
29879
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1

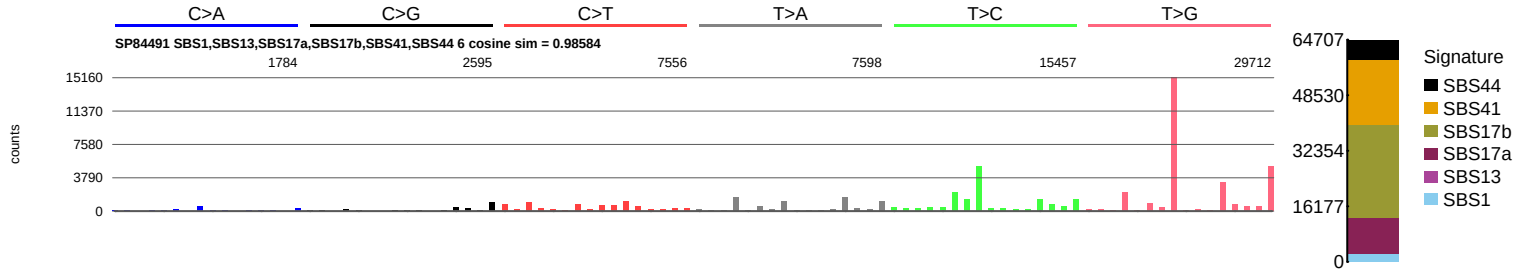

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS26,SBS41 5 cosine sim = 0.98586
1374
2391
5824
7110
18217
29796
15152
11364
7576
3788
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15

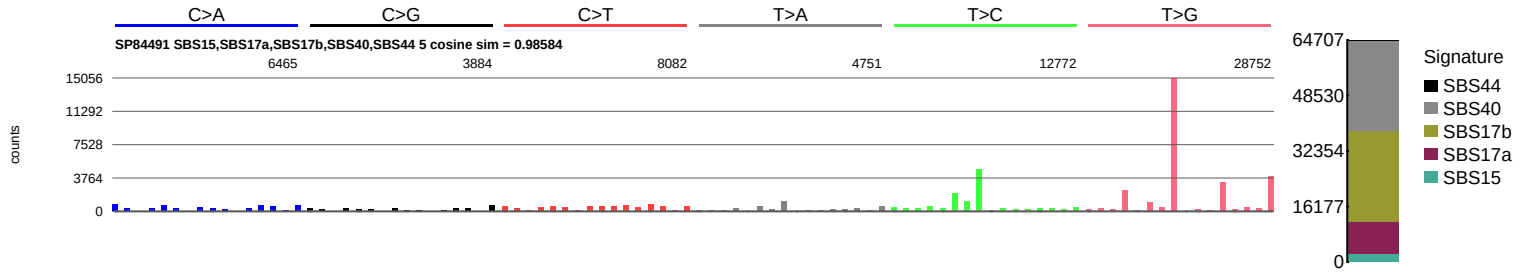

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98586
1788
2592
7599
7594
15414
29722
15160
11370
7580
3790
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS1

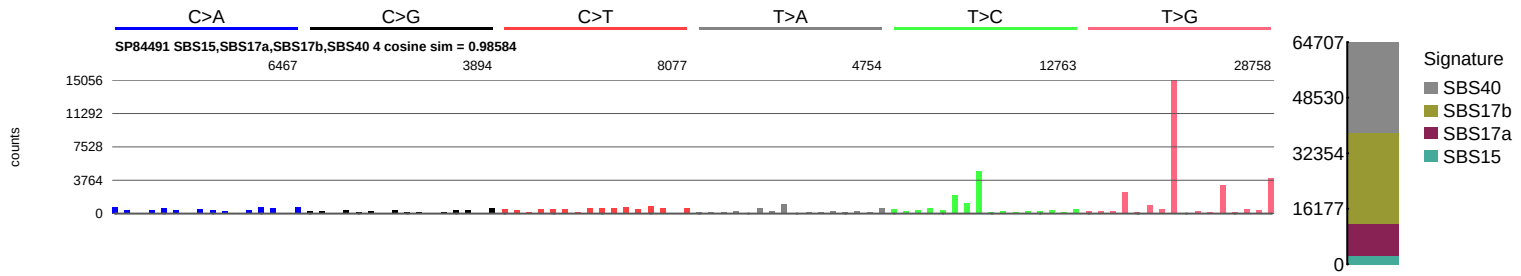

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98586
1767
2490
7616
7633
15453
29749
15160
11370
7580
3790
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS1

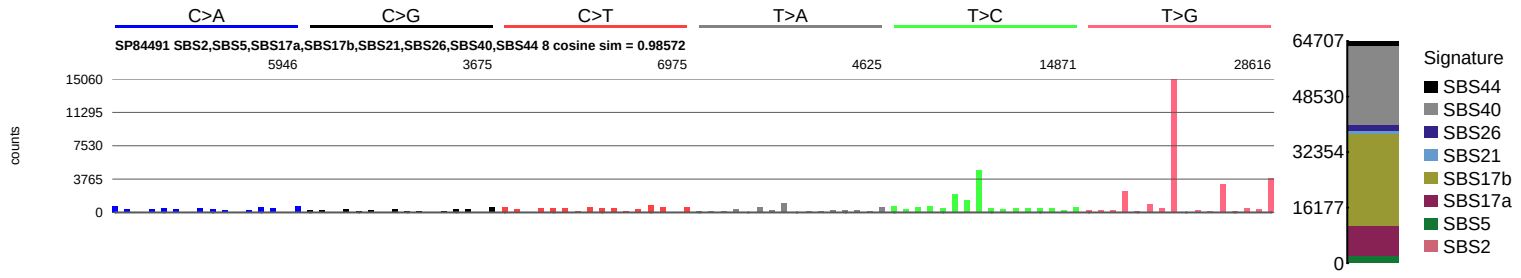

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98586
1399
2516
5803
7061
18173
29760
15152
11364
7576
3788
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13

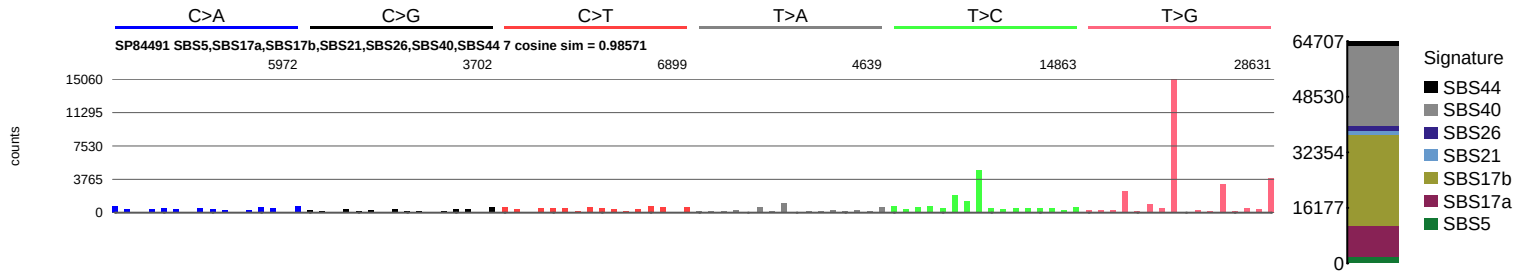

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.98584
1764
2490
7573
7641
15500
29743
15160
11370
7580
3790
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS1

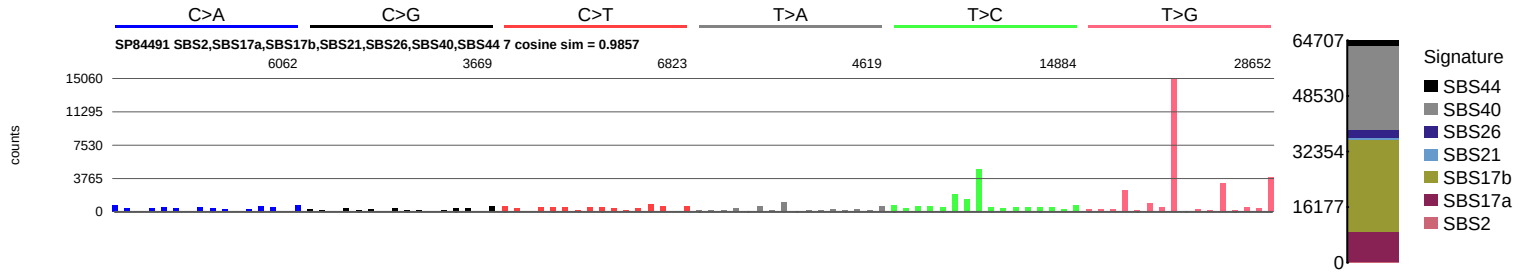

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98584
1784
2595
7556
7598
15457
29712
15160
11370
7580
3790
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS1

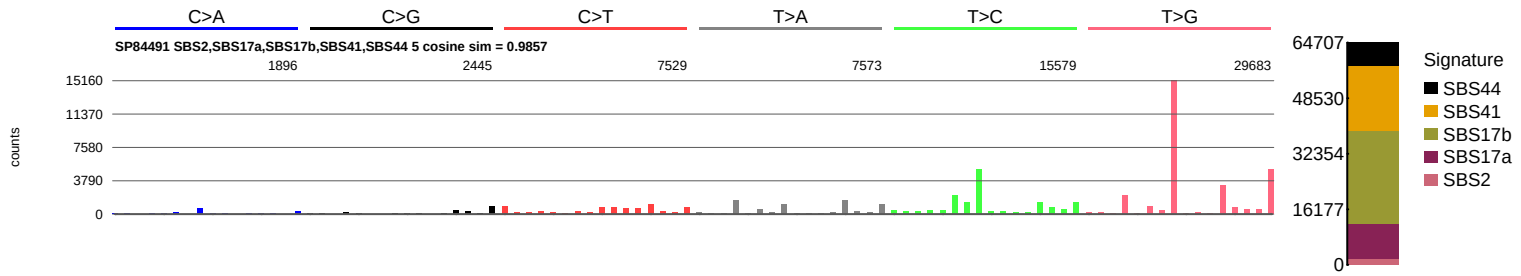

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98584
6465
3884
8082
4751
12772
28752
15056
11292
7528
3764
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15

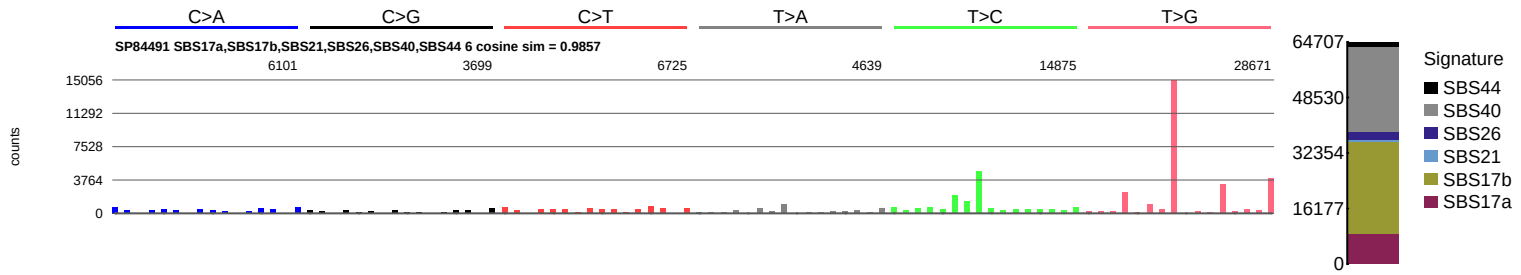

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98584
6467
3894
8077
4754
12763
28758
15056
11292
7528
3764
0
counts
64707
48530
32354
16177
0
Signature
SBS40
SBS17b
SBS17a
SBS15

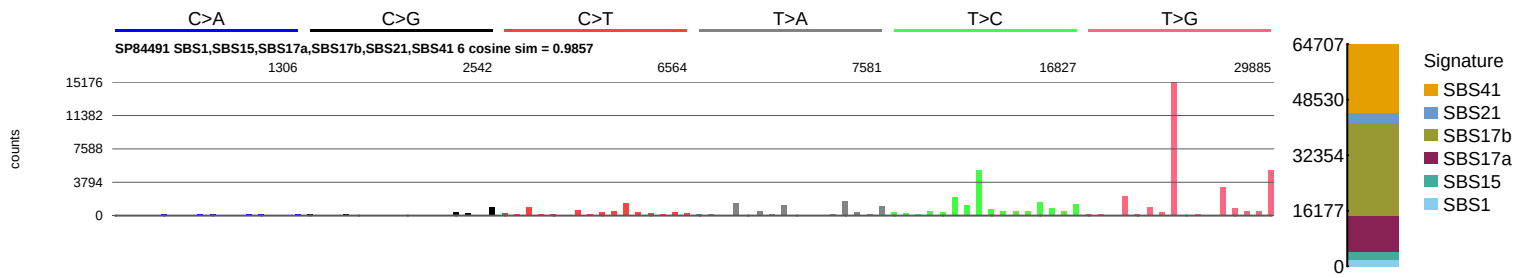

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98572
5946
3675
6975
4625
14871
28616
counts
15060
11295
7530
3765
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS5
SBS2

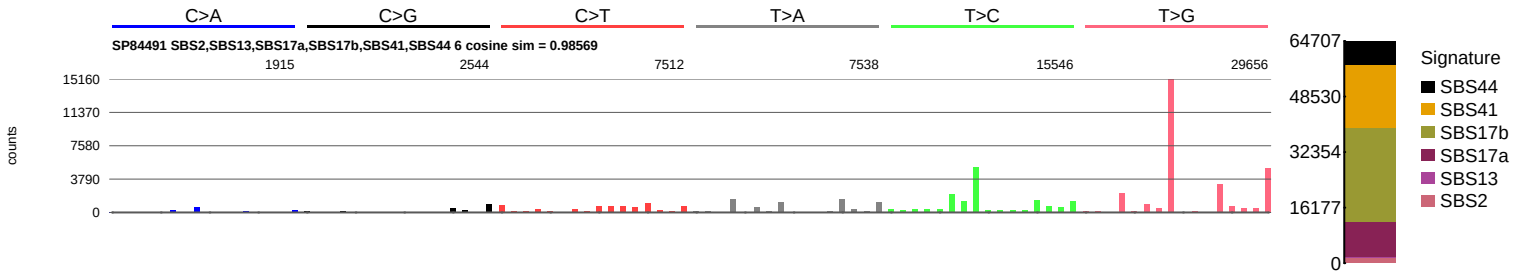

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98571
5972
3702
6899
4639
14863
28631
counts
15060
11295
7530
3765
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS5

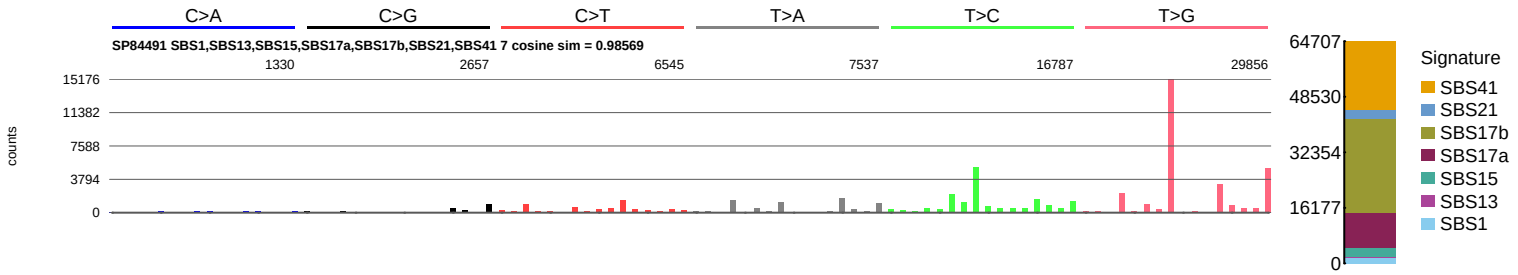

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.9857
6062
3669
6823
4619
14884
28652
counts
15060
11295
7530
3765
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2

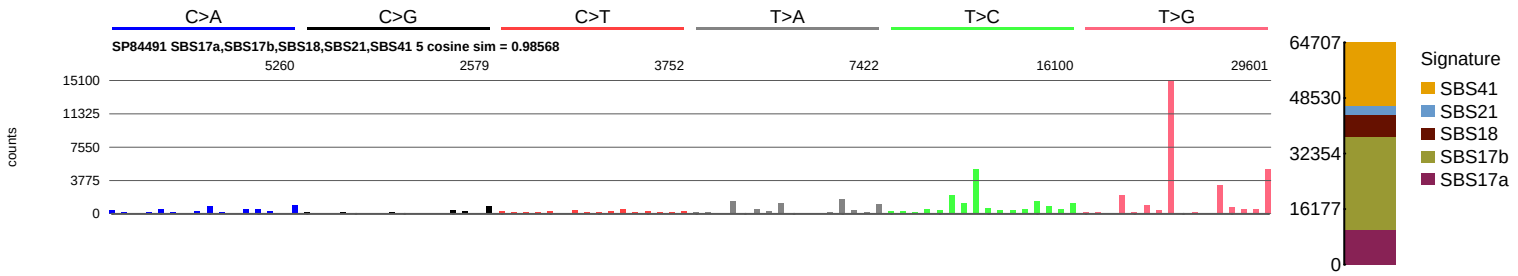

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.9857
1896
2445
7529
7573
15579
29683
counts
15160
11370
7580
3790
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS2

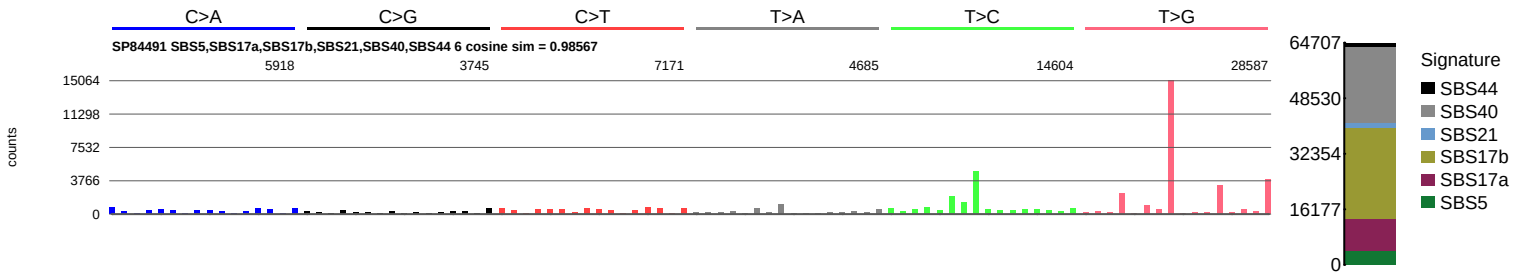

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 6 cosine sim = 0.9857
6101
3699
6725
4639
14875
28671
counts
15056
11292
7528
3764
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a

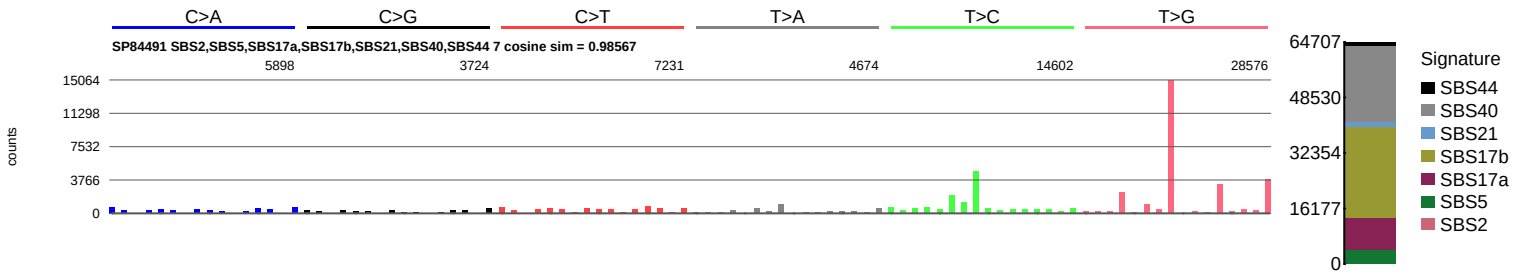

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.9857
1306
2542
6564
7581
16827
29885
counts
15176
11382
7588
3794
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS1

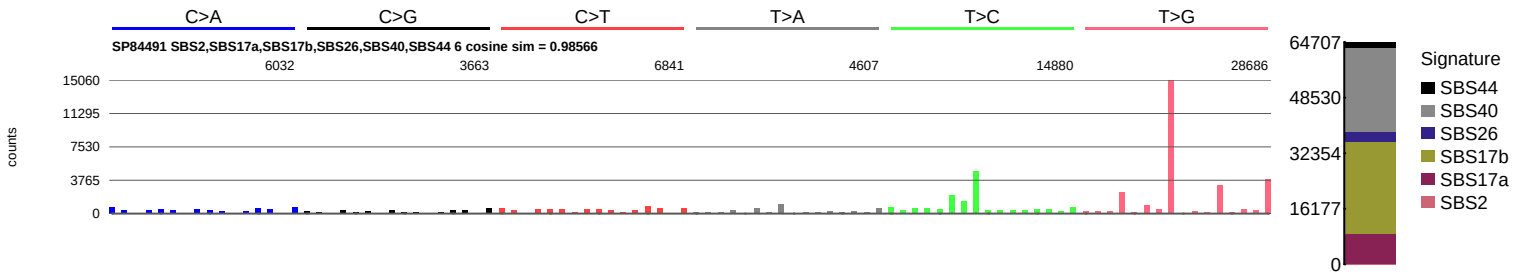

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98569
1915
2544
7512
7538
15546
29656
15160
11370
7580
3790
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS2

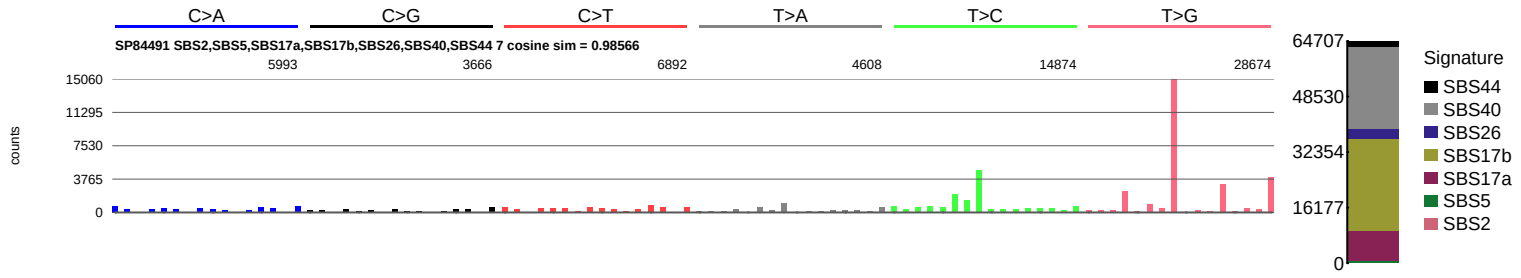

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98569
1330
2657
6545
7537
16787
29856
15176
11382
7588
3794
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS18,SBS21,SBS41 5 cosine sim = 0.98568
5260
2579
3752
7422
16100
29601
15100
11325
7550
3775
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98567
5918
3745
7171
4685
14604
28587
15064
11298
7532
3766
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98567
5898
3724
7231
4674
14602
28576
15064
11298
7532
3766
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98566
6032
3663
6841
4607
14880
28686
15060
11295
7530
3765
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS2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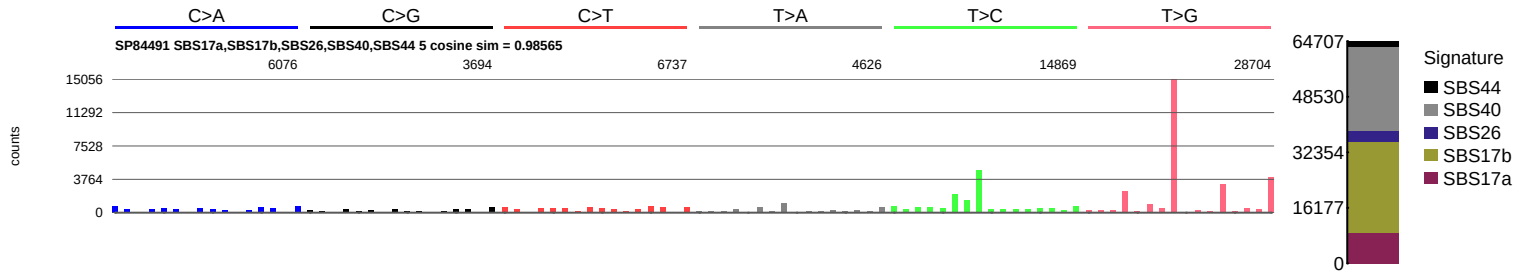
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98566
5993
3666
6892
4608
14874
28674
counts
15060
11295
7530
3765
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2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C>G
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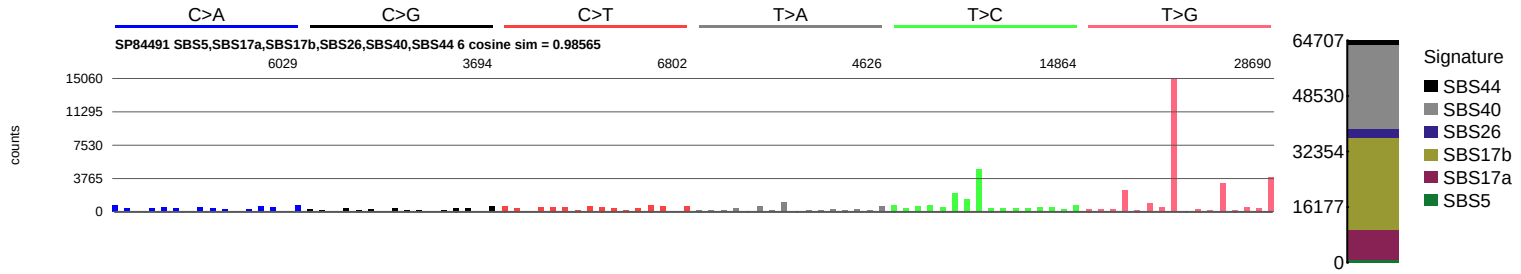
T>G
SP84491 SBS17a,SBS17b,SBS26,SBS40,SBS44 5 cosine sim = 0.98565
6076
3694
6737
4626
14869
28704
counts
15056
11292
7528
3764
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a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C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98565
6029
3694
6802
4626
14864
28690
counts
15060
11295
7530
3765
0
64707
48530
32354
16177
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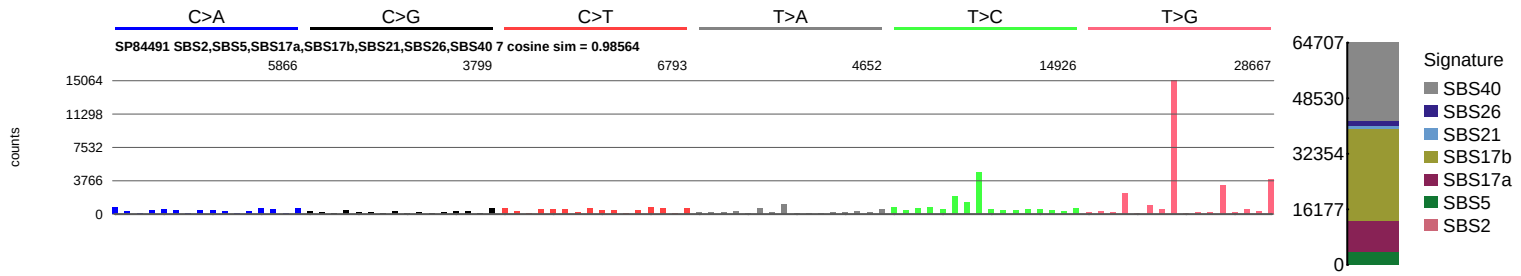
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS5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C>G
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SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98564
5866
3799
6793
4652
14926
28667
counts
15064
11298
7532
3766
0
64707
48530
32354
16177
0
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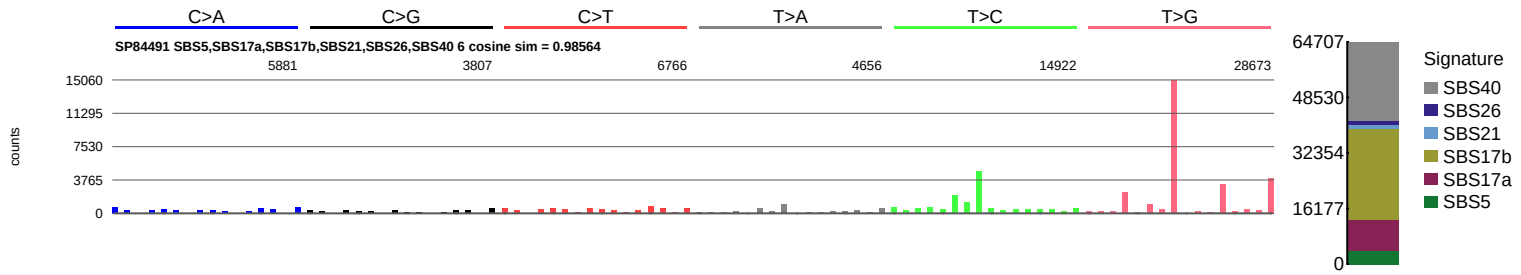
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS5
SBS2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C>G
C>T
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SP84491 SBS5,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.98564
5881
3807
6766
4656
14922
28673
counts
15060
11295
7530
3765
0
64707
48530
32354
16177
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SBS40
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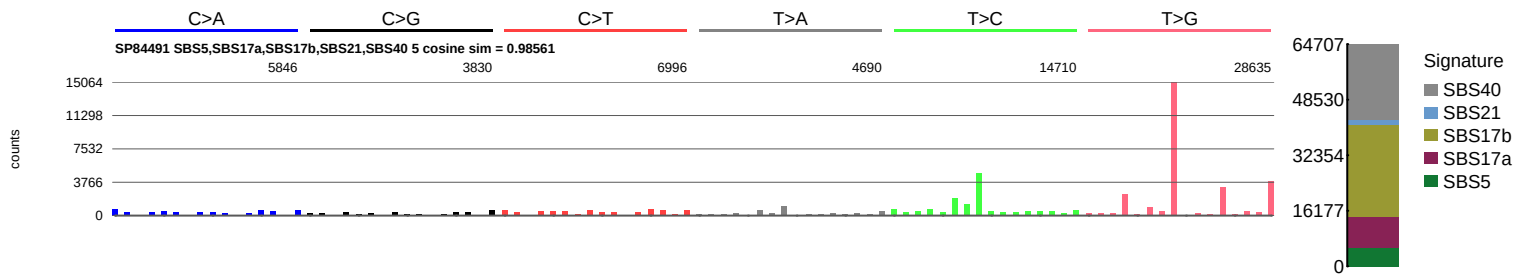
SBS21
SBS17b
SBS17a
SBS5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C>G
C>T
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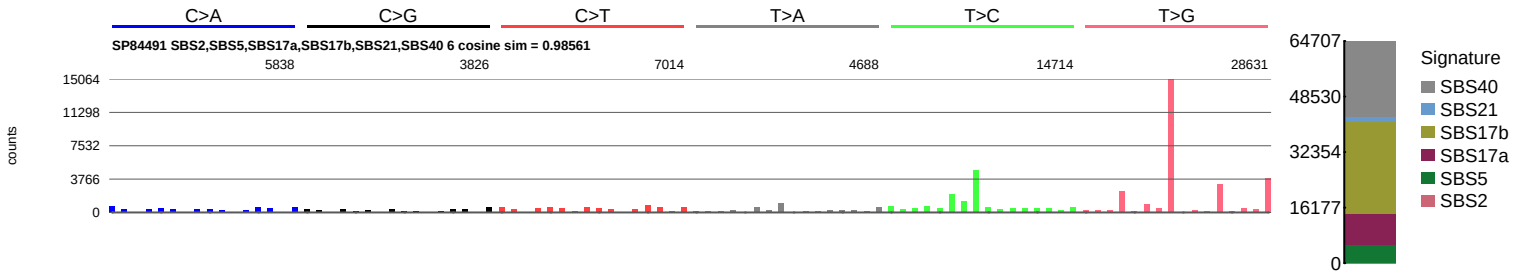
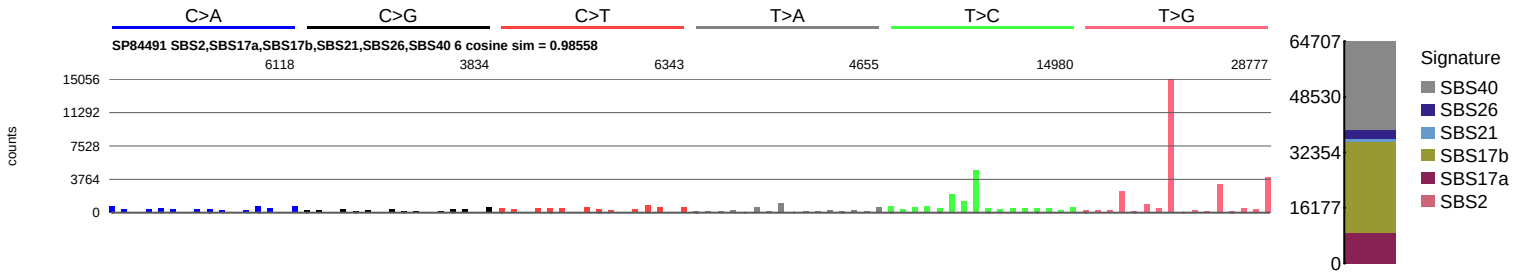
SP84491 SBS5,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98561
5846
3830
6996
4690
14710
28635
counts
15064
11298
7532
3766
0
64707
48530
32354
16177
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98561
5838
3826
7014
4688
14714
28631
counts
15064
11298
7532
3766
0
64707
48530
32354
16177
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.98558
6118
3834
6343
4655
14980
28777
counts
15056
11292
7528
3764
0
64707
48530
32354
16177
0
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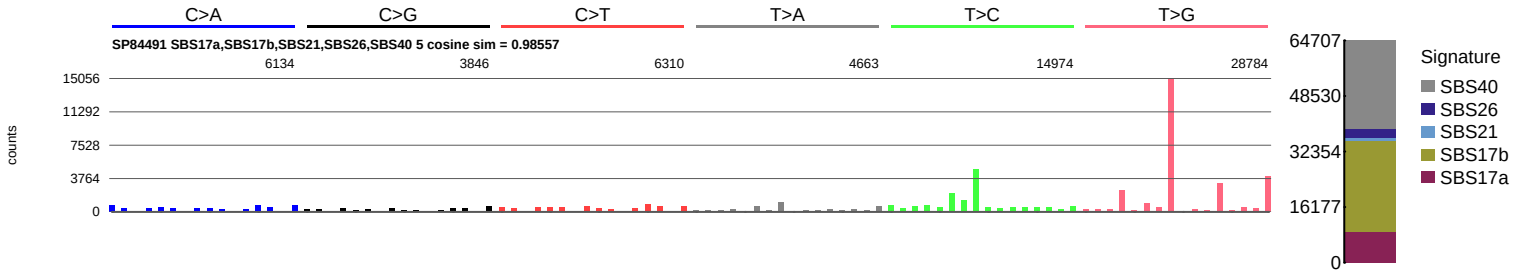
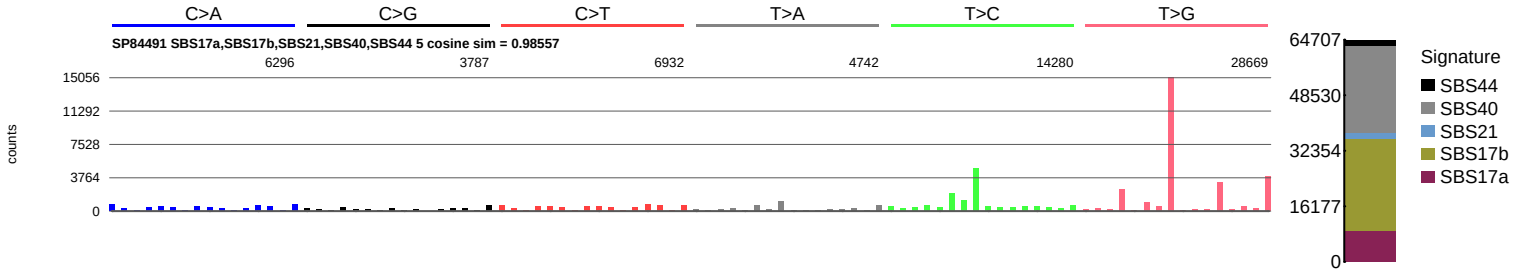
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS26,SBS40 5 cosine sim = 0.98557
6134
3846
6310
4663
14974
28784
counts
15056
11292
7528
3764
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a


C>A
C>G
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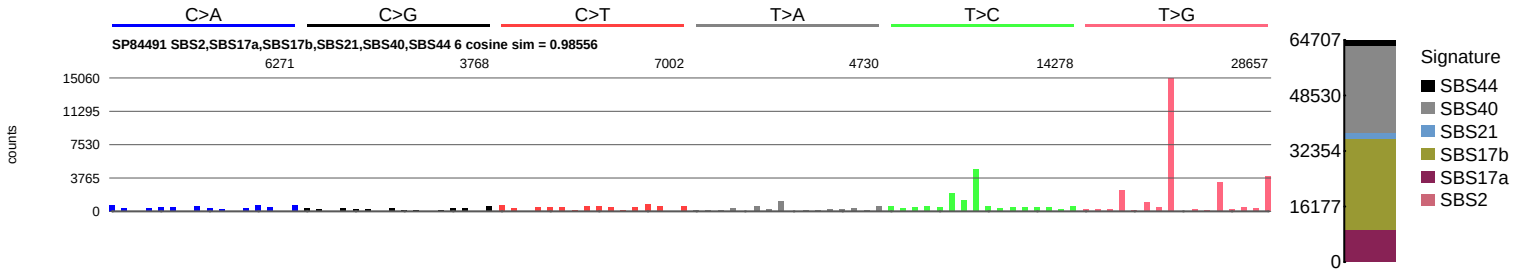
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS40,SBS44 5 cosine sim = 0.98557
6296
3787
6932
4742
14280
28669
counts
15056
11292
7528
3764
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a


C>A
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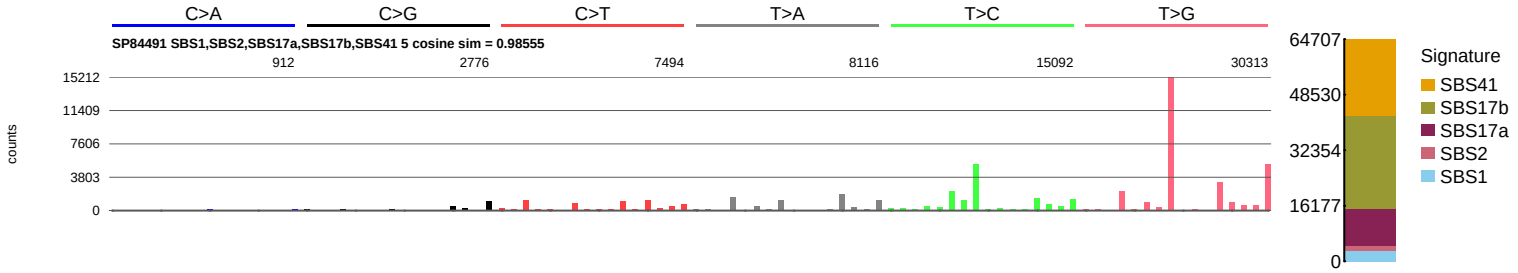
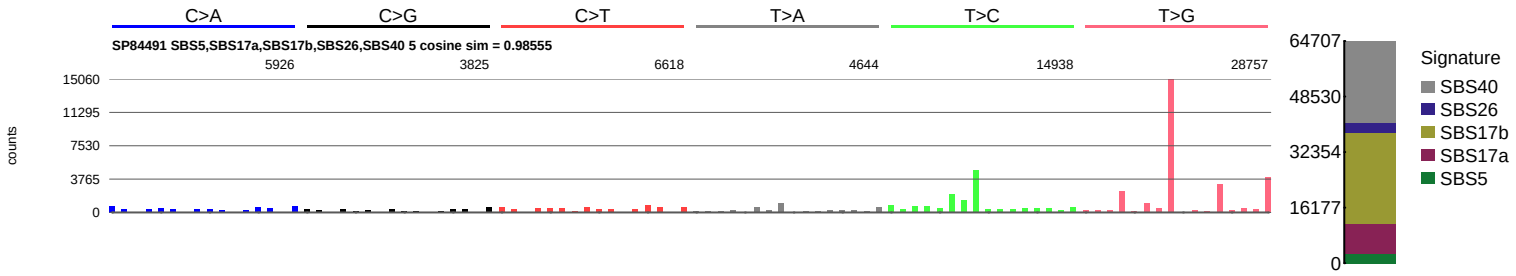
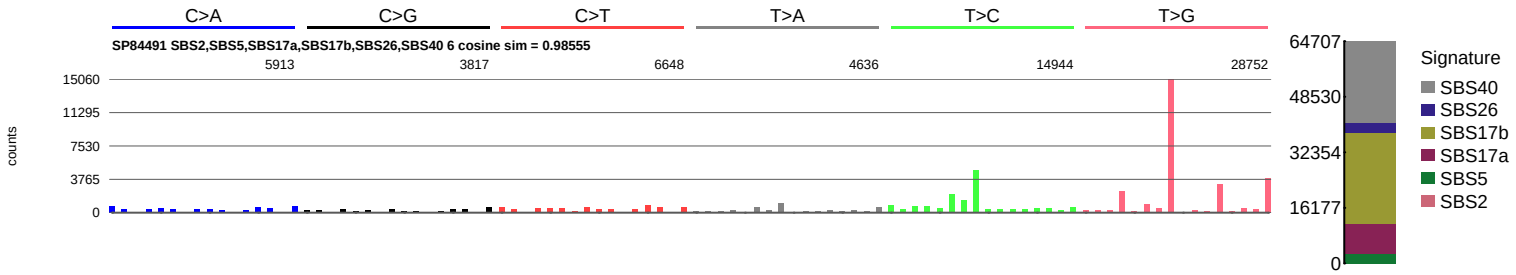
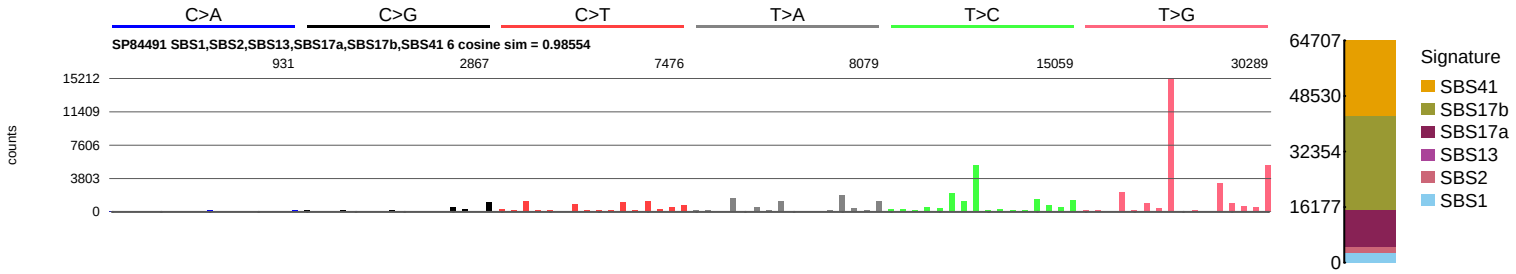
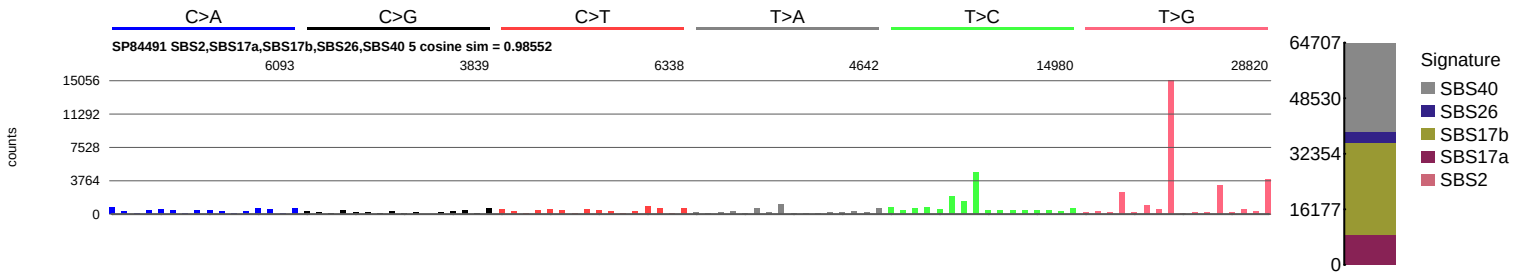
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98556
6271
3768
7002
4730
14278
28657
counts
15060
11295
7530
3765
0
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98555
912
2776
7494
8116
15092
30313
counts
15212
11409
7606
3803
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98555
5926
3825
6618
4644
14938
28757
counts
15060
11295
7530
3765
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98555
5913
3817
6648
4636
14944
28752
counts
15060
11295
7530
3765
0
64707
48530
32354
16177
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98554
931
2867
7476
8079
15059
30289
counts
15212
11409
7606
3803
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98552
6093
3839
6338
4642
14980
28820
counts
15056
11292
7528
3764
0
64707
48530
32354
16177
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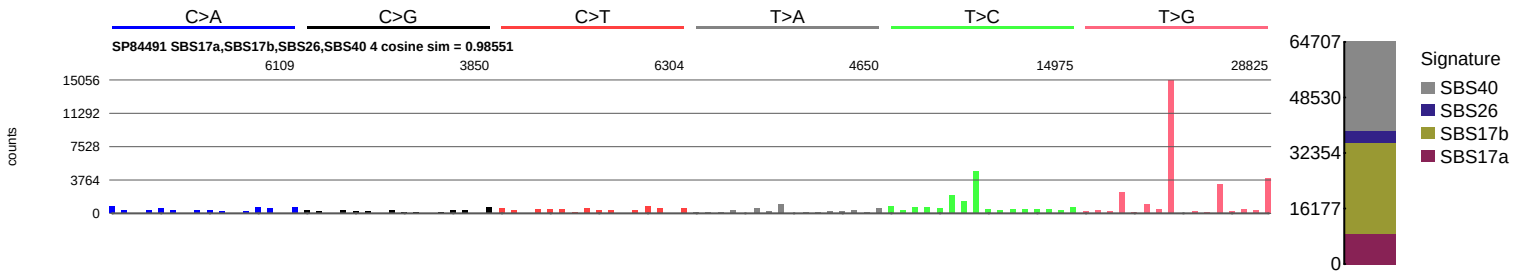
SBS40
SBS26
SBS17b
SBS17a
SBS2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T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS26,SBS40 4 cosine sim = 0.98551
6109
3850
6304
4650
14975
28825
counts
15056
11292
7528
3764
0
64707
48530
32354
16177
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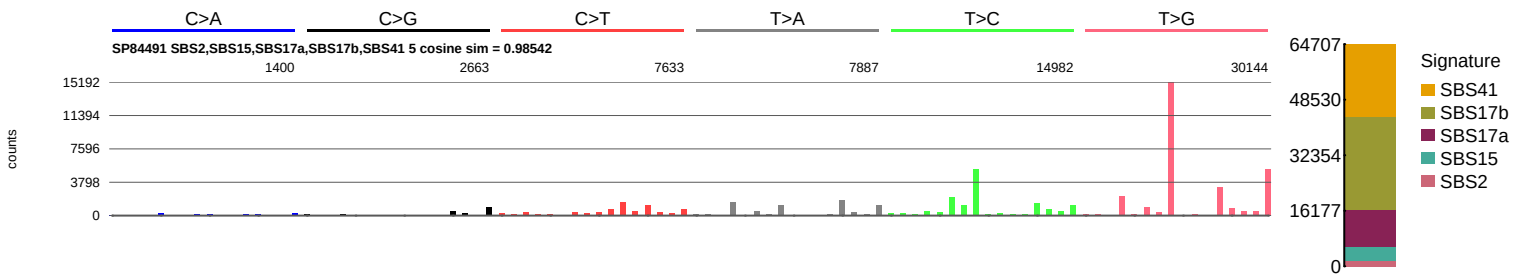
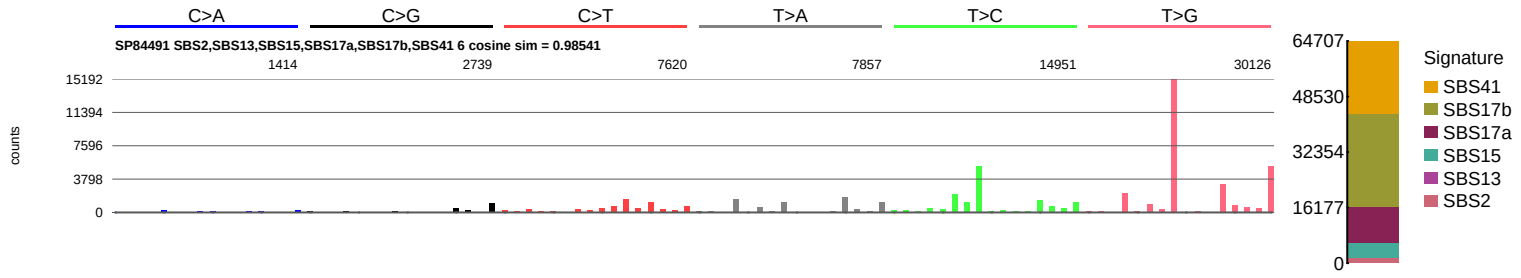
Signature
SBS40
SBS26
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98542
1400
2663
7633
7887
14982
30144
counts
15192
11394
7596
3798
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98541
1414
2739
7620
7857
14951
30126
15192
11394
7596
3798
0
counts
64707
48530
32354
16177
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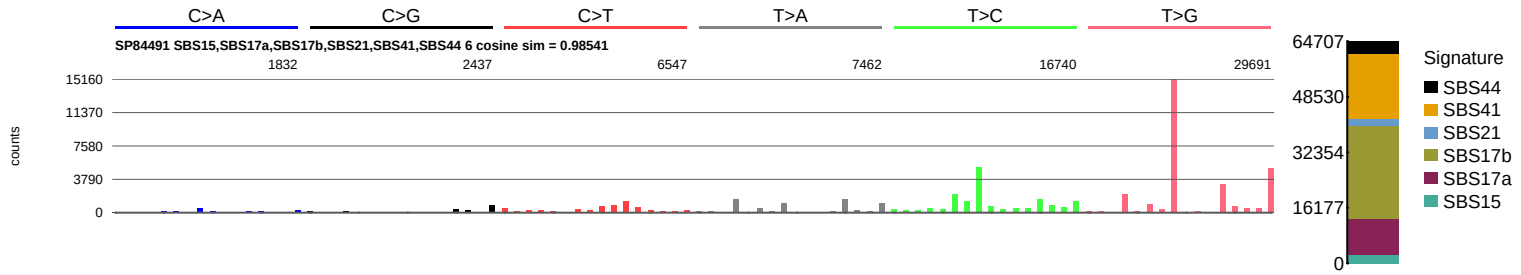
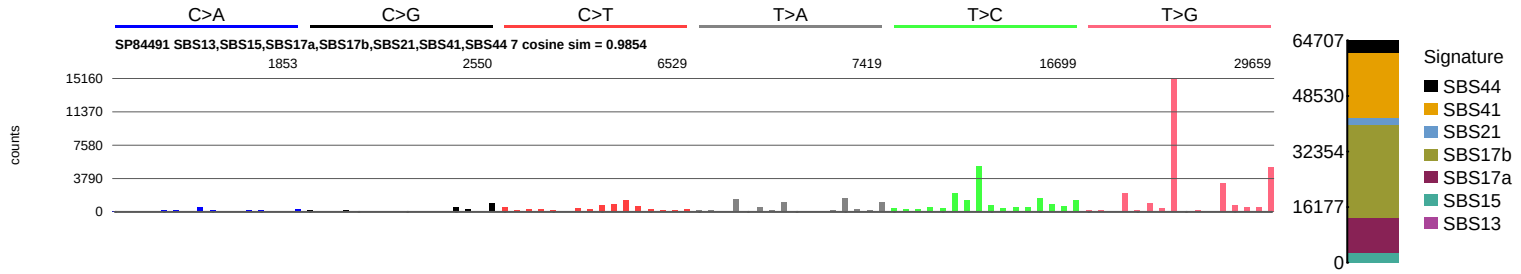
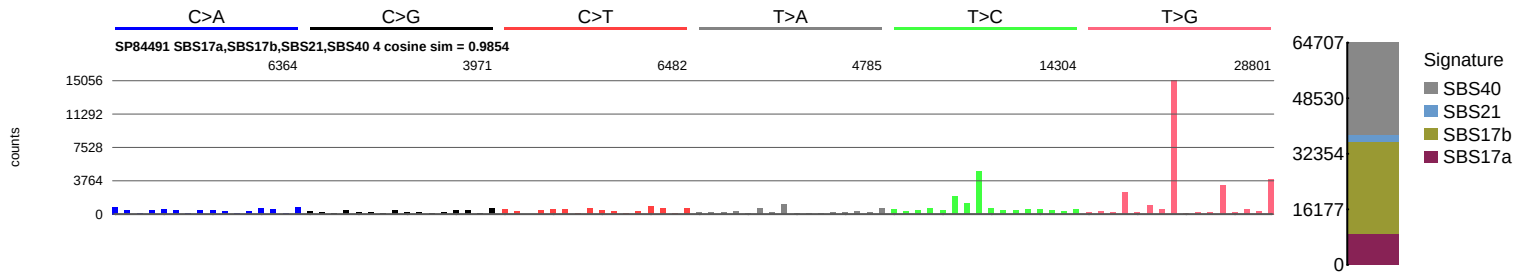
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.98541
1832
2437
6547
7462
16740
29691
15160
11370
7580
3790
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.9854
1853
2550
6529
7419
16699
29659
15160
11370
7580
3790
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS40 4 cosine sim = 0.9854
6364
3971
6482
4785
14304
28801
15056
11292
7528
3764
0
counts
64707
48530
32354
16177
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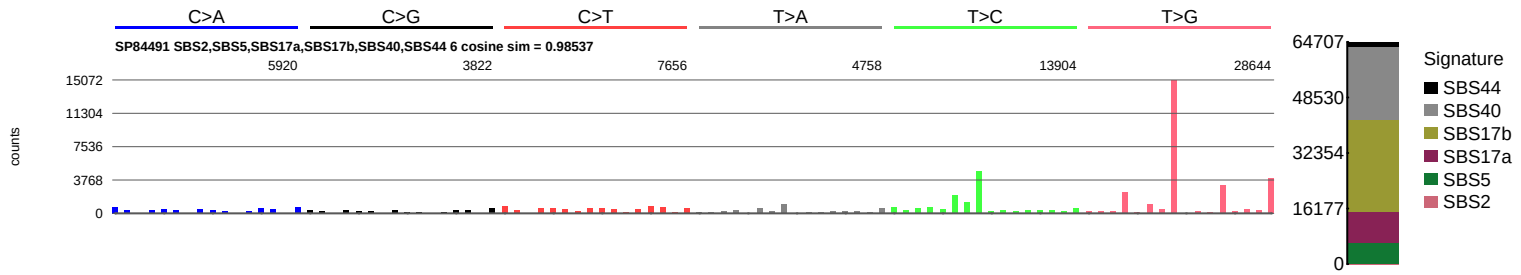
Signature
SBS40
SBS21
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98537
5920
3822
7656
4758
13904
28644
15072
11304
7536
3768
0
counts
64707
48530
32354
16177
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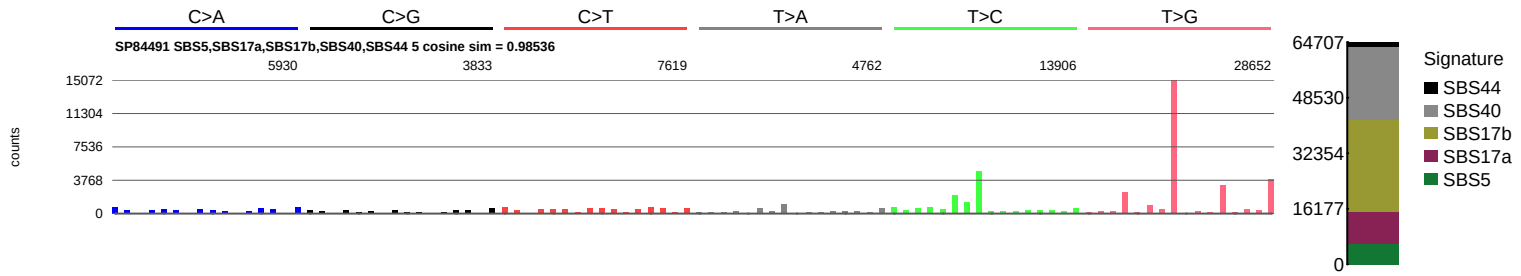
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98536
5930
3833
7619
4762
13906
28652
15072
11304
7536
3768
0
counts
64707
48530
32354
16177
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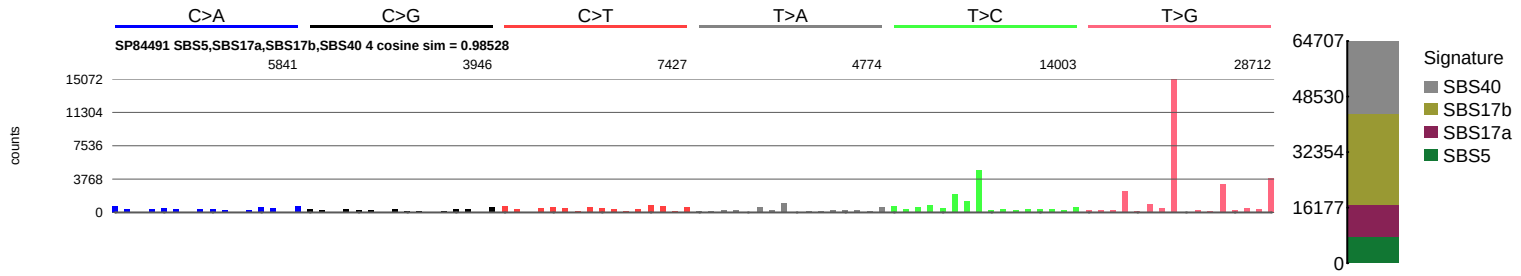
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5


C>A
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C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98528
5841
3946
7427
4774
14003
28712
15072
11304
7536
3768
0
counts
64707
48530
32354
16177
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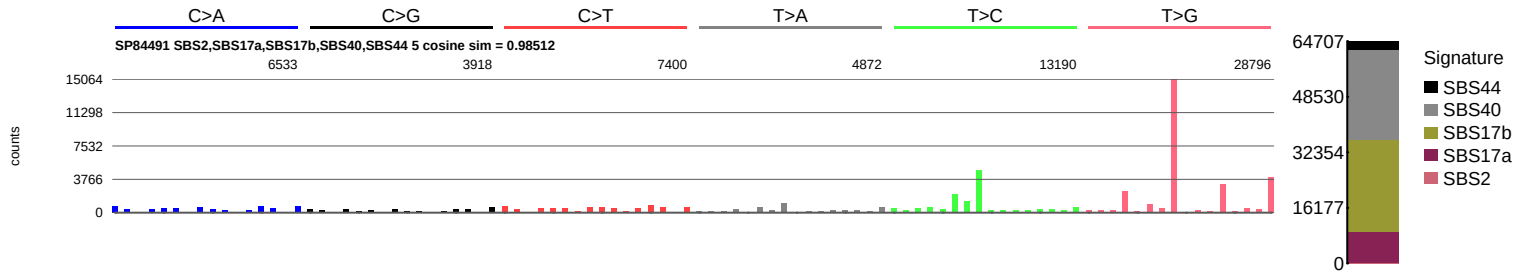
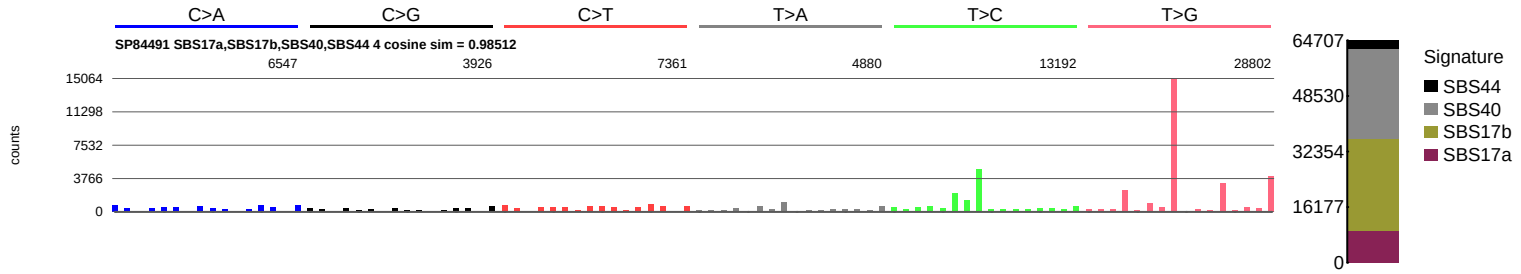
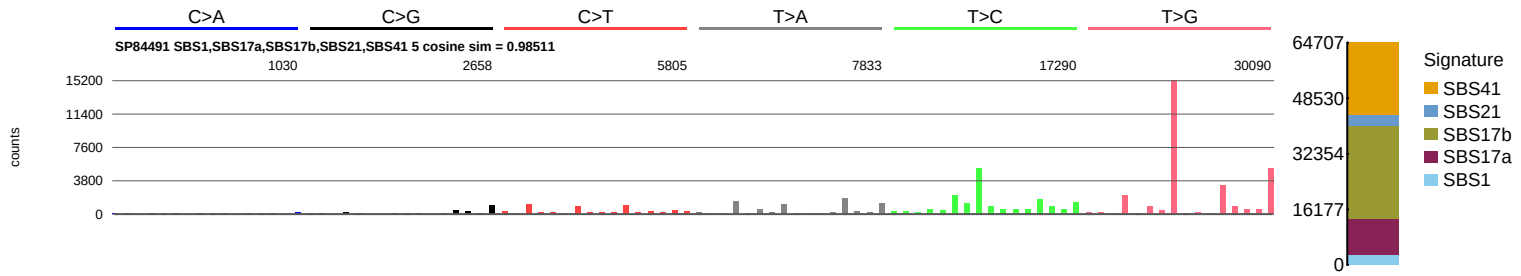
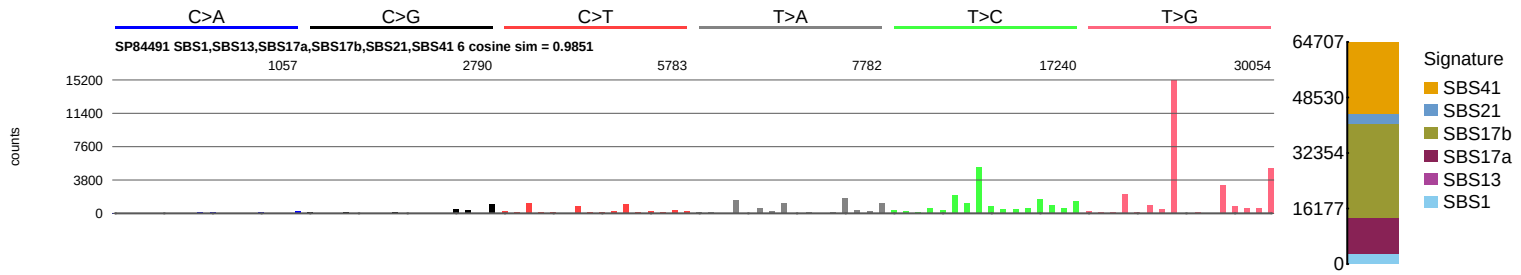
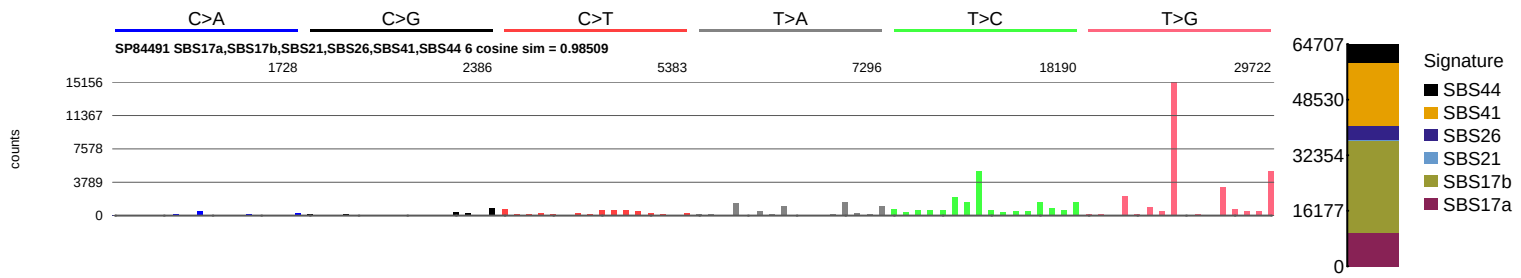
Signature
SBS40
SBS17b
SBS17a
SBS5

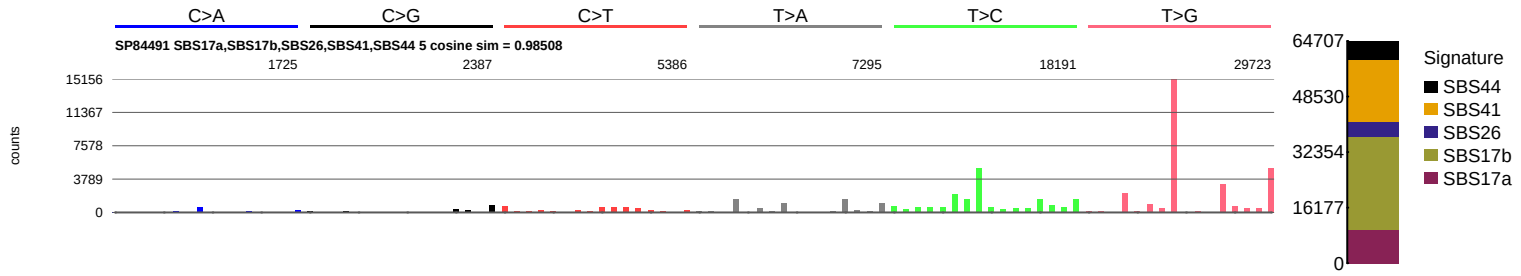

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98512
6533
3918
7400
4872
13190
28796
15064
11298
7532
3766
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS2

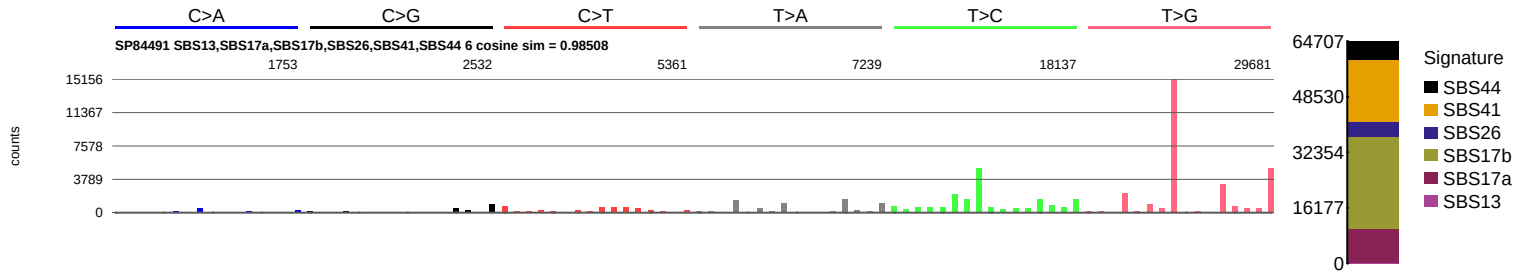

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS40,SBS44 4 cosine sim = 0.98512
6547
3926
7361
4880
13192
28802
15064
11298
7532
3766
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS40
SBS17b
SBS17a

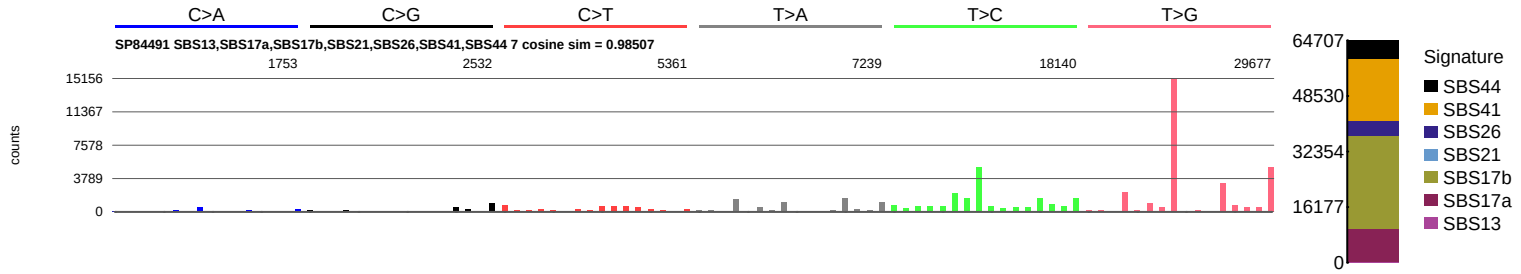

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98511
1030
2658
5805
7833
17290
30090
15200
11400
7600
3800
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS1

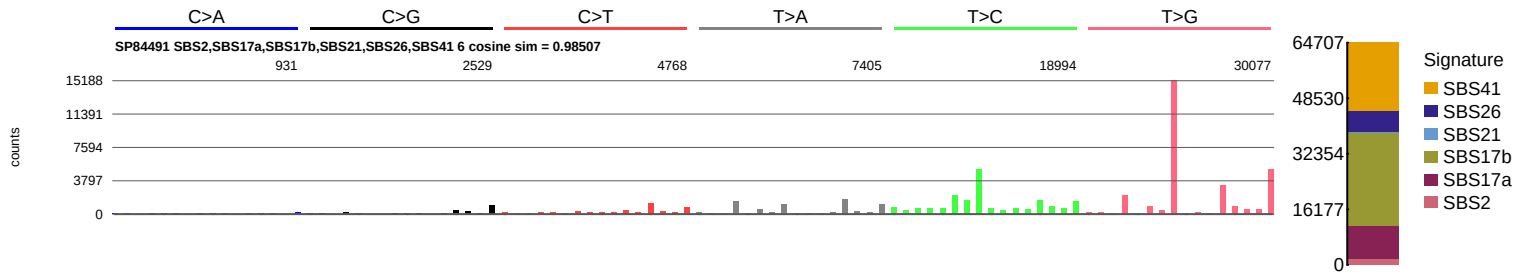

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.9851
1057
2790
5783
7782
17240
30054
15200
11400
7600
3800
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS1

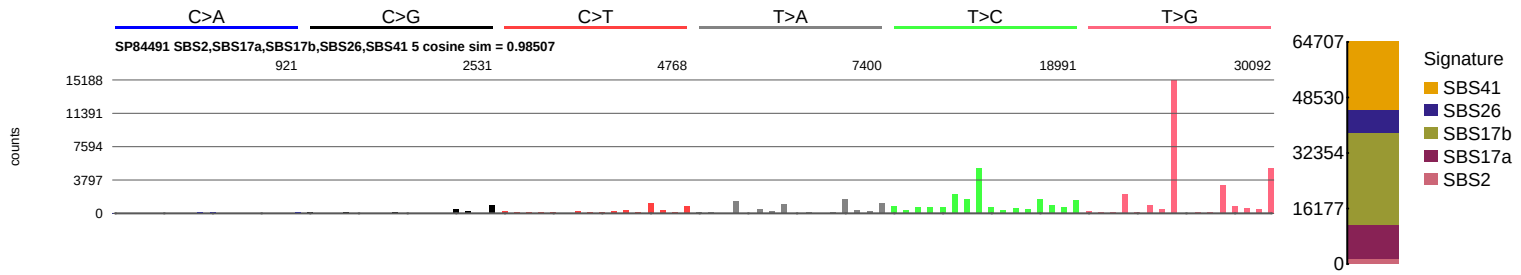

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 6 cosine sim = 0.98509
1728
2386
5383
7296
18190
29722
15156
11367
7578
3789
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a

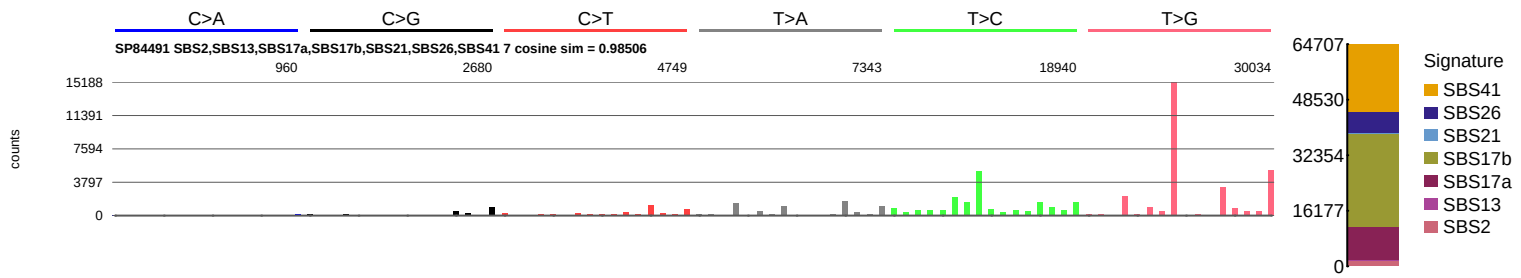

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS26,SBS41,SBS44 5 cosine sim = 0.98508
1725
2387
5386
7295
18191
29723
15156
11367
7578
3789
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a

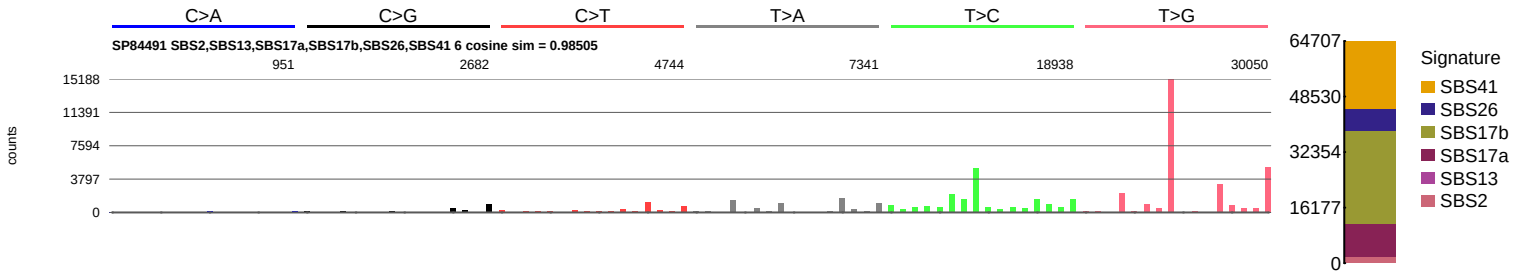

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98508
1753
2532
5361
7239
18137
29681
15156
11367
7578
3789
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13

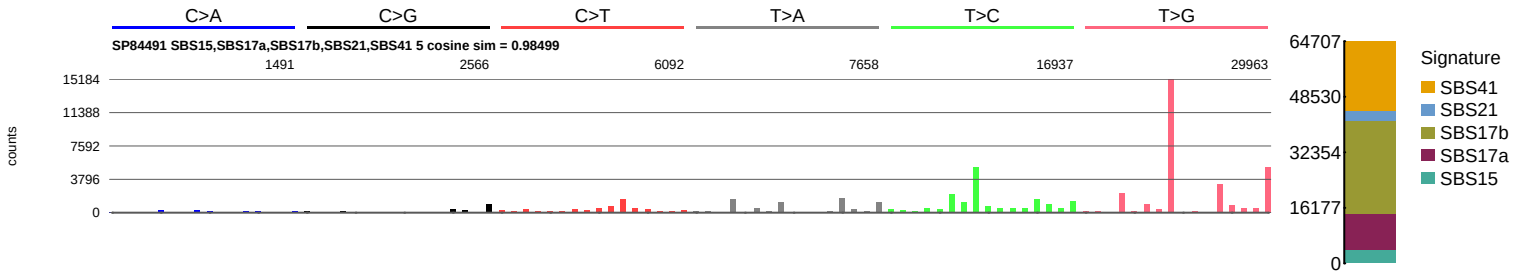

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 7 cosine sim = 0.98507
1753
2532
5361
7239
18140
29677
15156
11367
7578
3789
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13

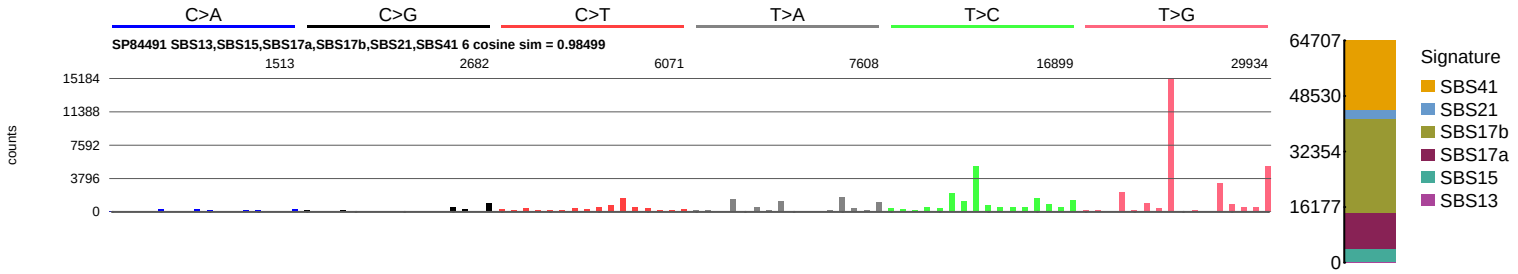

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS41 6 cosine sim = 0.98507
931
2529
4768
7405
18994
30077
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS2

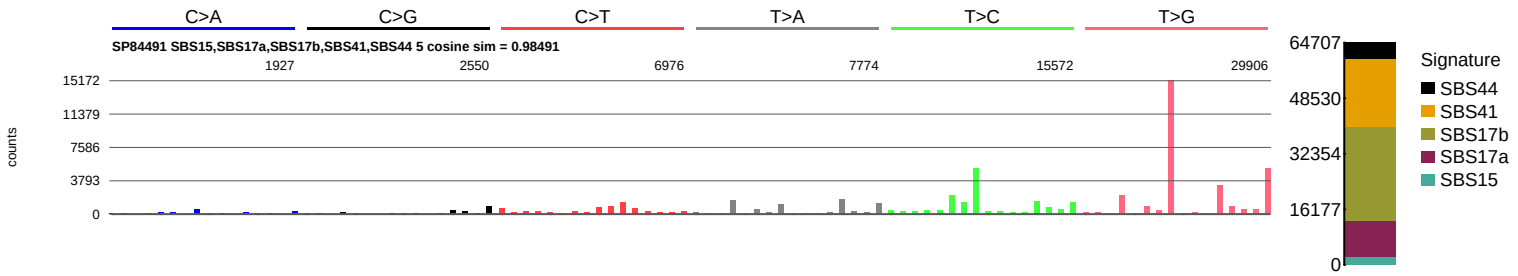

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS26,SBS41 5 cosine sim = 0.98507
921
2531
4768
7400
18991
30092
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS2

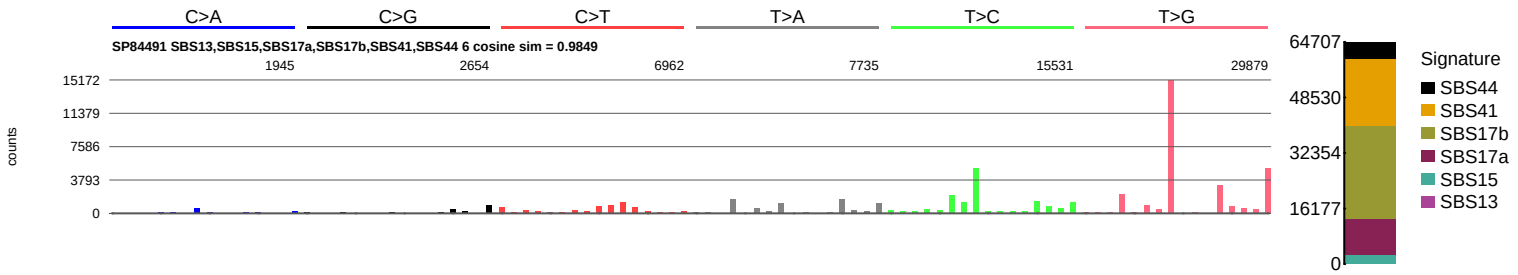

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98506
960
2680
4749
7343
18940
30034
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2

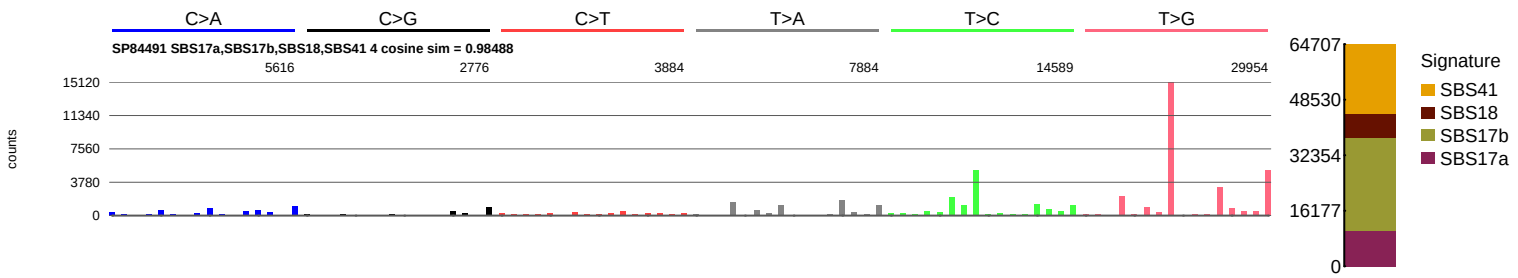

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98505
951
2682
4744
7341
18938
30050
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2

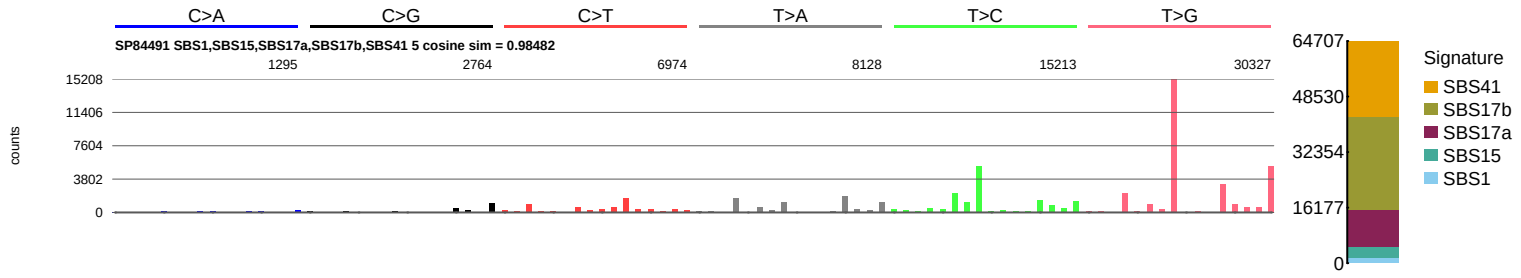

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98499
1491
2566
6092
7658
16937
29963
15184
11388
7592
3796
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15

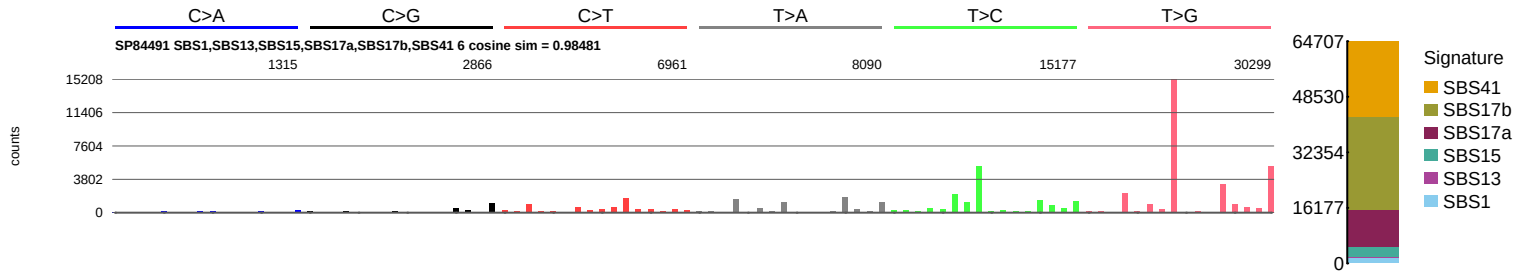

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98499
1513
2682
6071
7608
16899
29934
15184
11388
7592
3796
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13

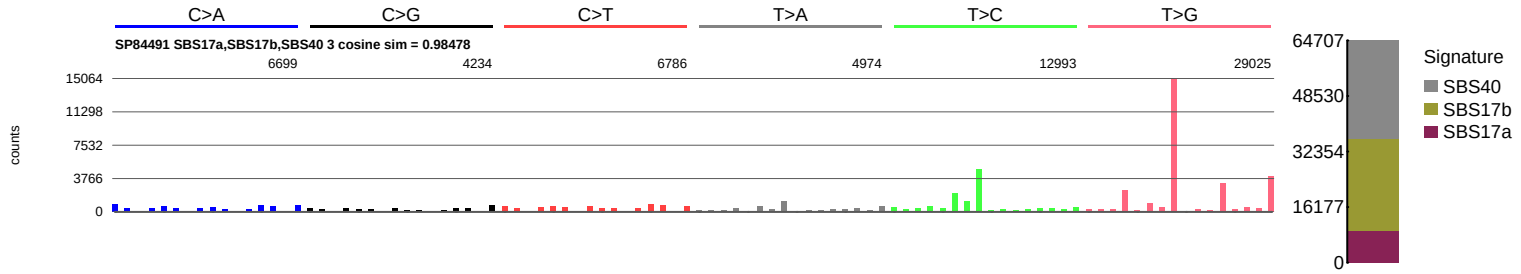

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.98491
1927
2550
6976
7774
15572
29906
15172
11379
7586
3793
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15

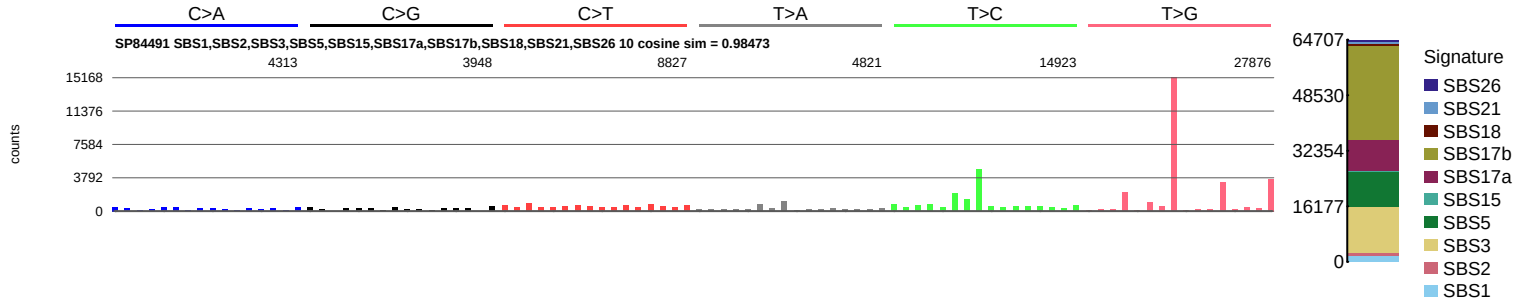

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.9849
1945
2654
6962
7735
15531
29879
15172
11379
7586
3793
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13

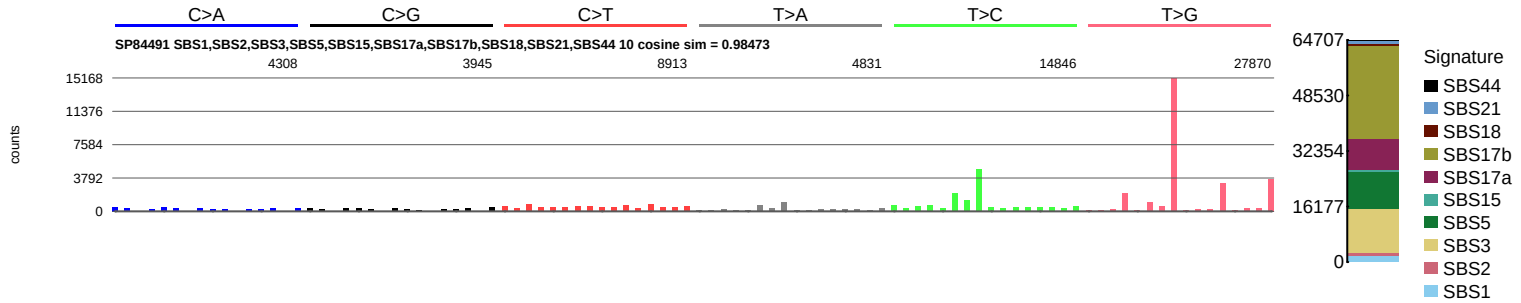

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS18,SBS41 4 cosine sim = 0.98488
5616
2776
3884
7884
14589
29954
15120
11340
7560
3780
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS18
SBS17b
SBS17a

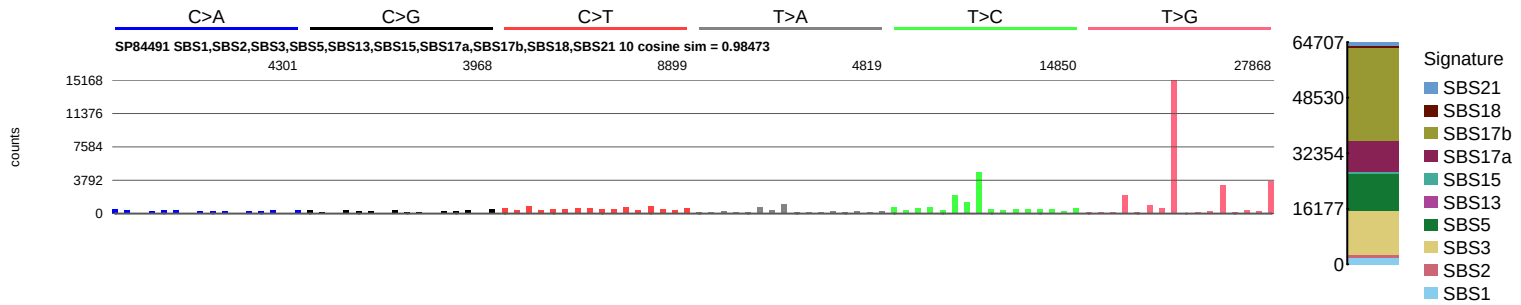

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98482
1295
2764
6974
8128
15213
30327
counts
15208
11406
7604
3802
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS1

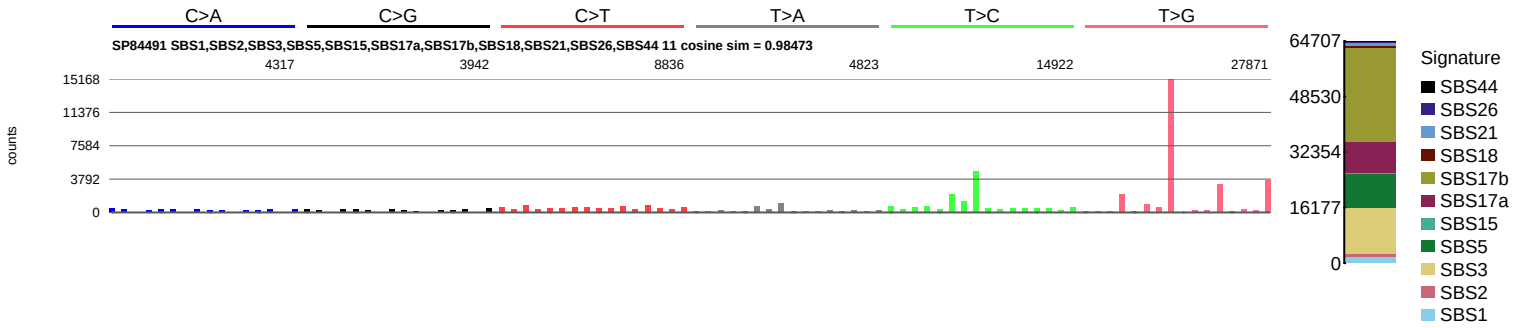

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98481
1315
2866
6961
8090
15177
30299
counts
15208
11406
7604
3802
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS1

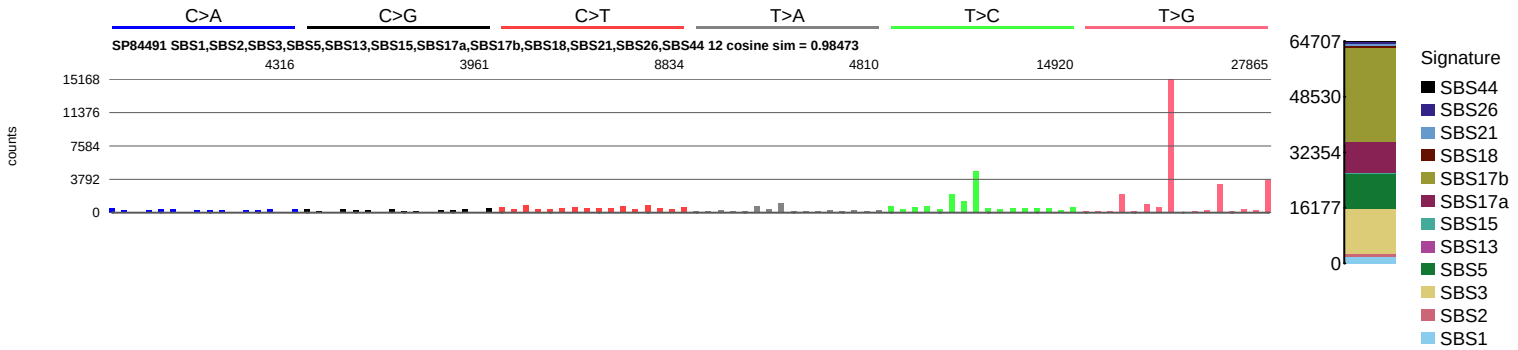

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS40 3 cosine sim = 0.98478
6699
4234
6786
4974
12993
29025
counts
15064
11298
7532
3766
0
64707
48530
32354
16177
0
Signature
SBS40
SBS17b
SBS17a

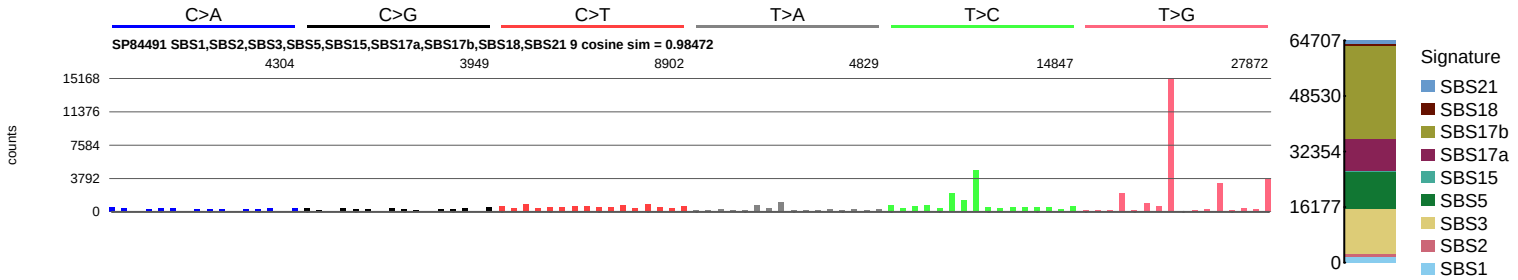

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98473
4313
3948
8827
4821
14923
27876
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

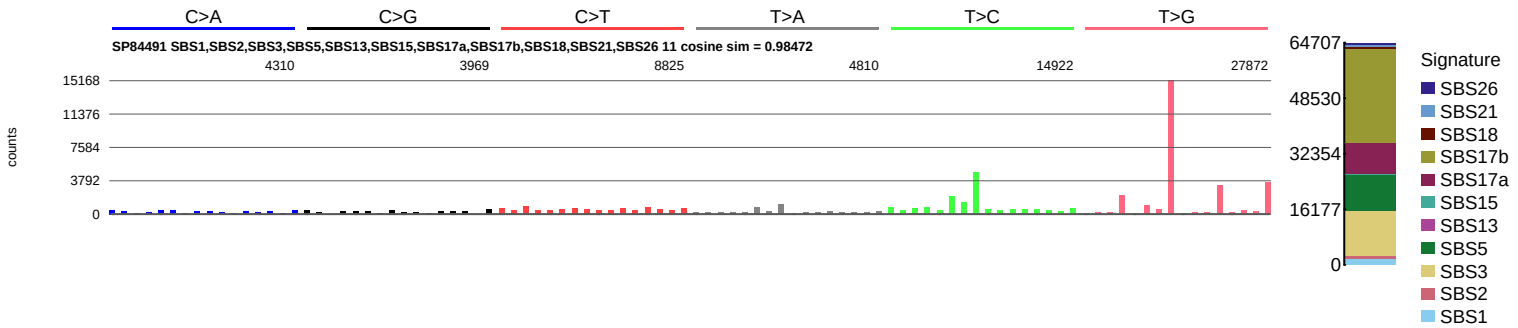

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98473
4308
3945
8913
4831
14846
27870
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98473
4301
3968
8899
4819
14850
27868
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.98473
4317
3942
8836
4823
14922
27871
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 12 cosine sim = 0.98473
4316
3961
8834
4810
14920
27865
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98472
4304
3949
8902
4829
14847
27872
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 11 cosine sim = 0.98472
4310
3969
8825
4810
14922
27872
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
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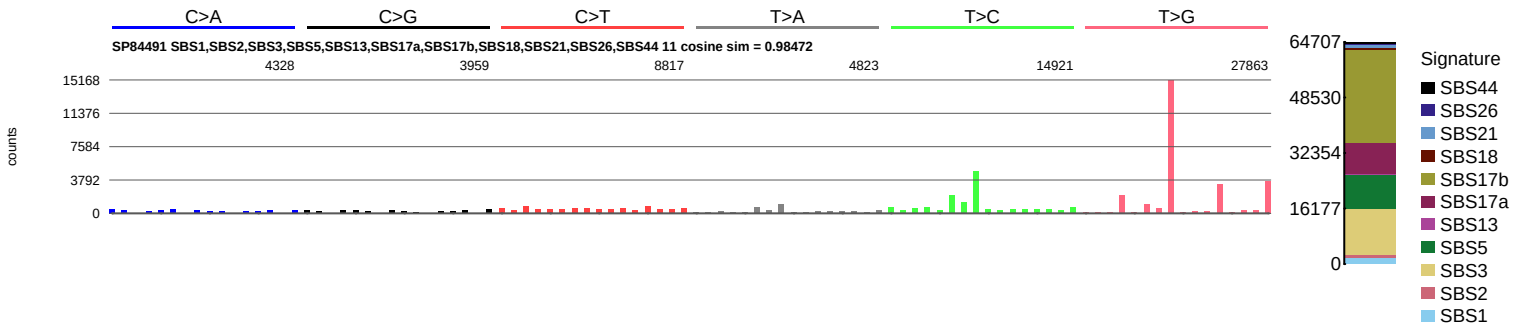
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.98472
4328
3959
8817
4823
14921
27863
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
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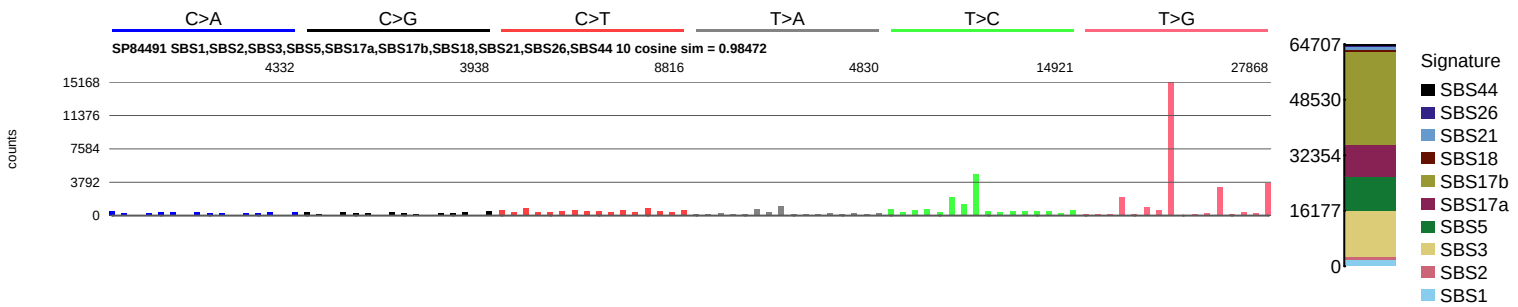
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98472
4332
3938
8816
4830
14921
27868
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
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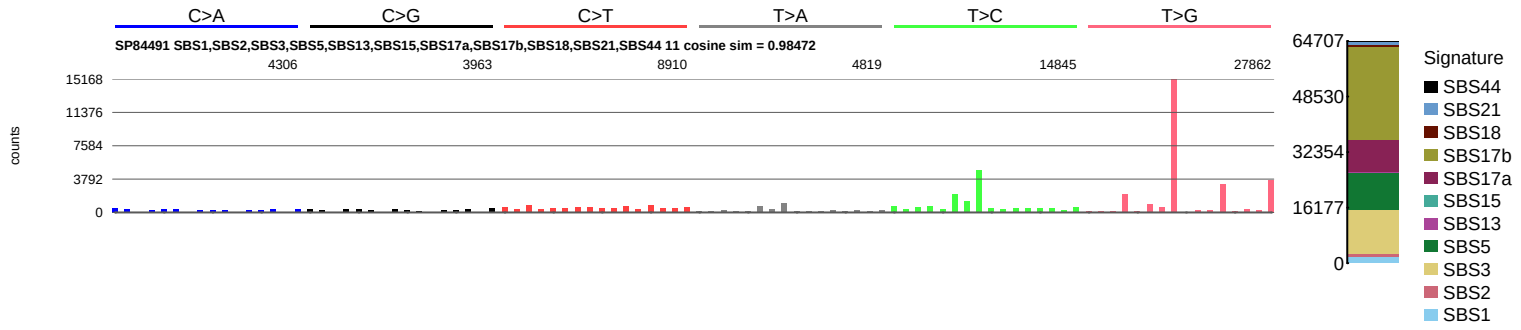
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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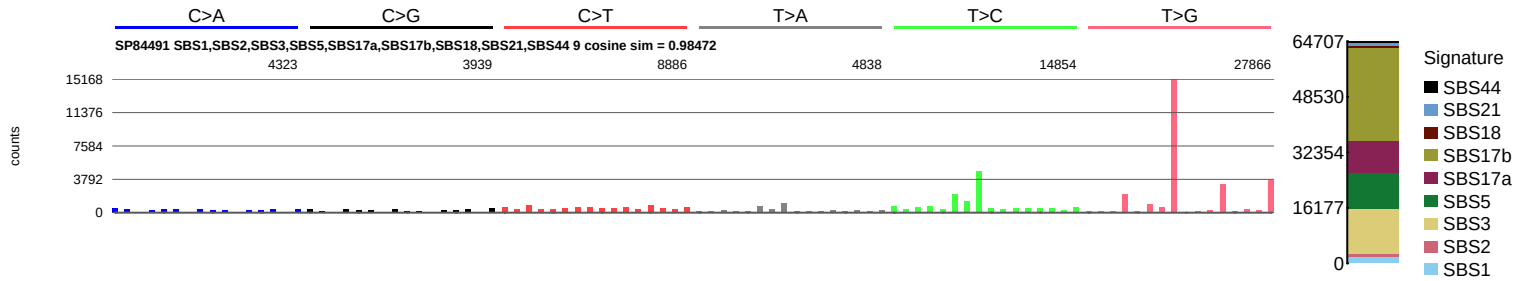
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98472
4306
3963
8910
4819
14845
27862
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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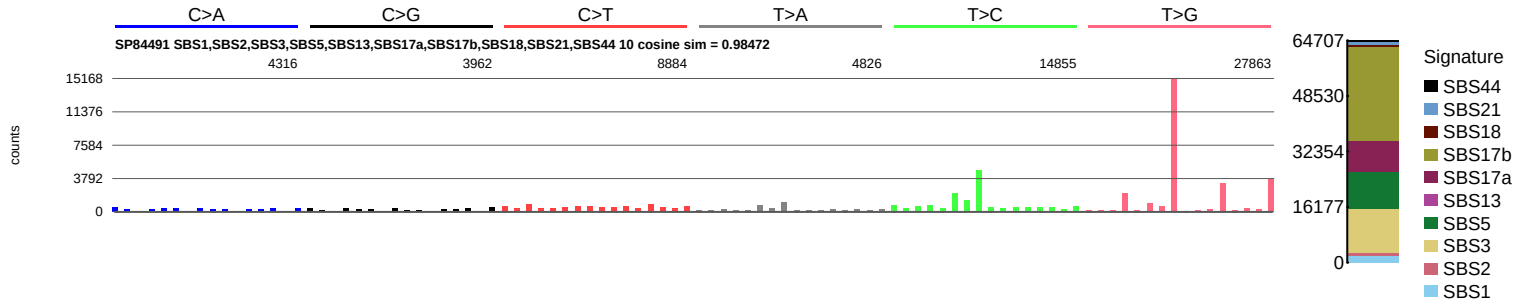
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98472
4323
3939
8886
4838
14854
27866
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98472
4316
3962
8884
4826
14855
27863
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
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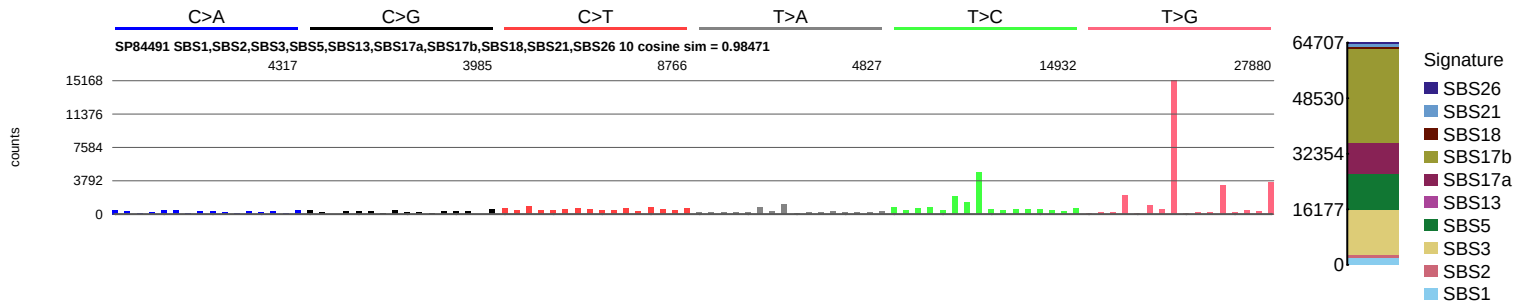
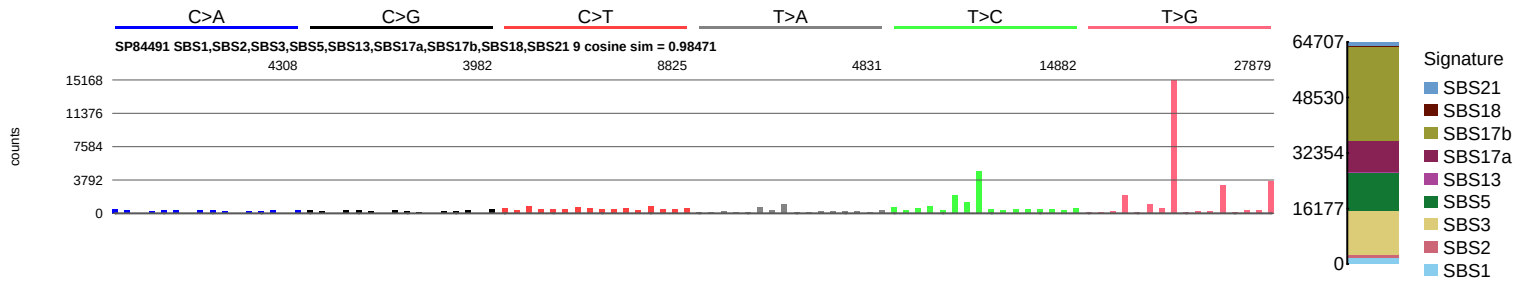
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98471
4317
3985
8766
4827
14932
27880
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98471
4308
3982
8825
4831
14882
27879
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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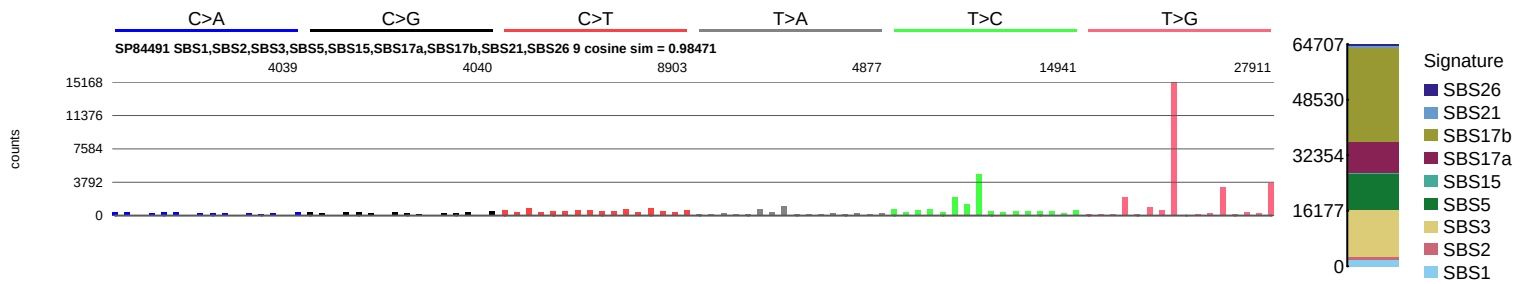
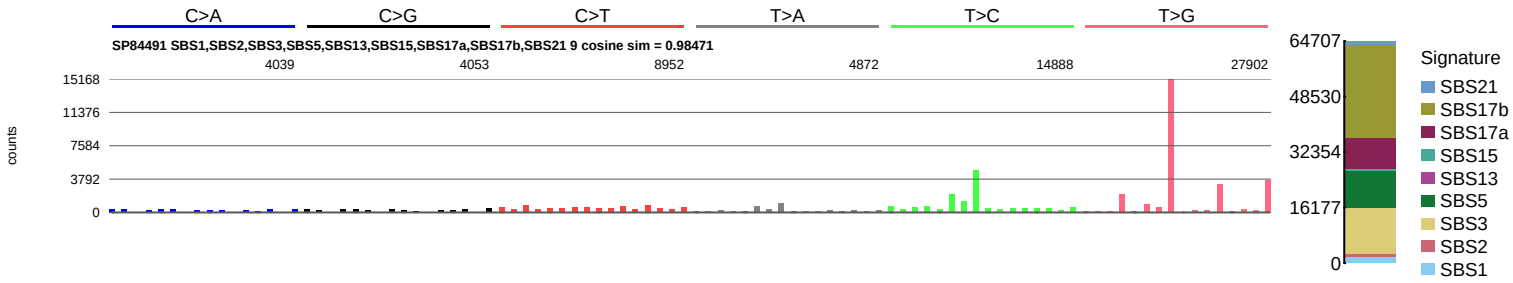
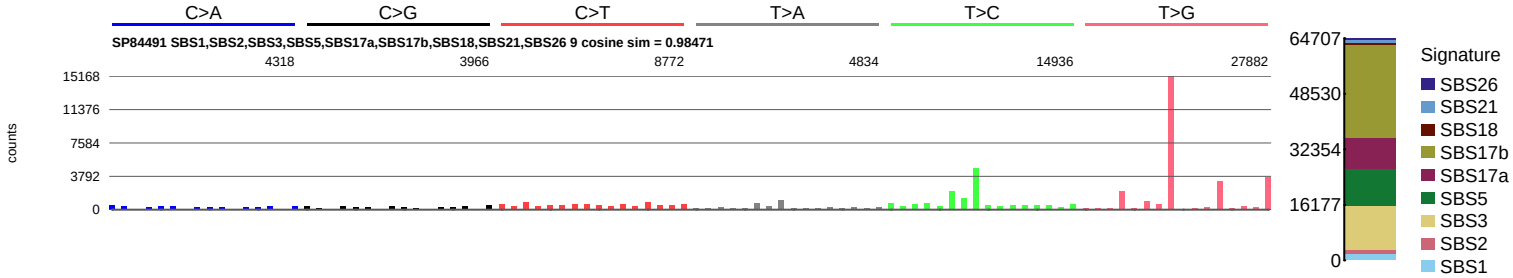
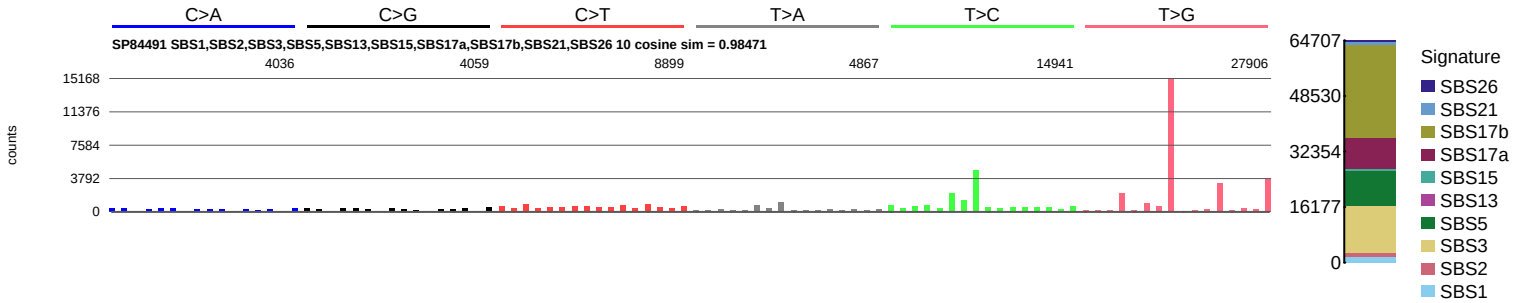
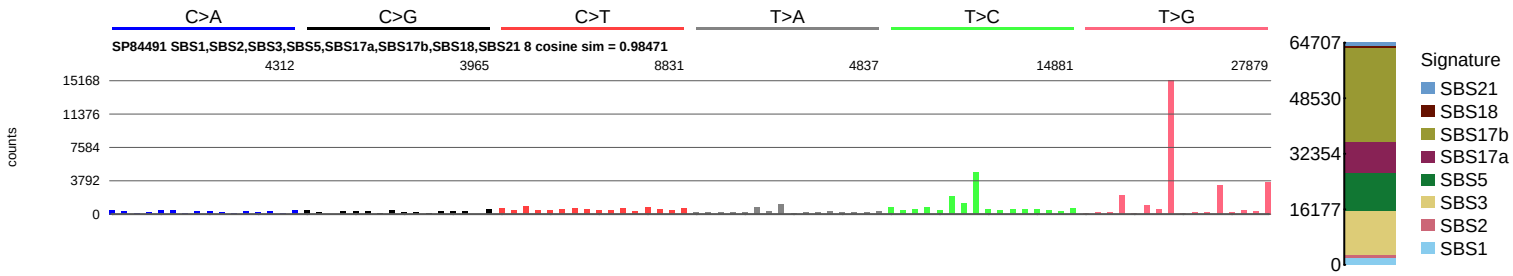
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

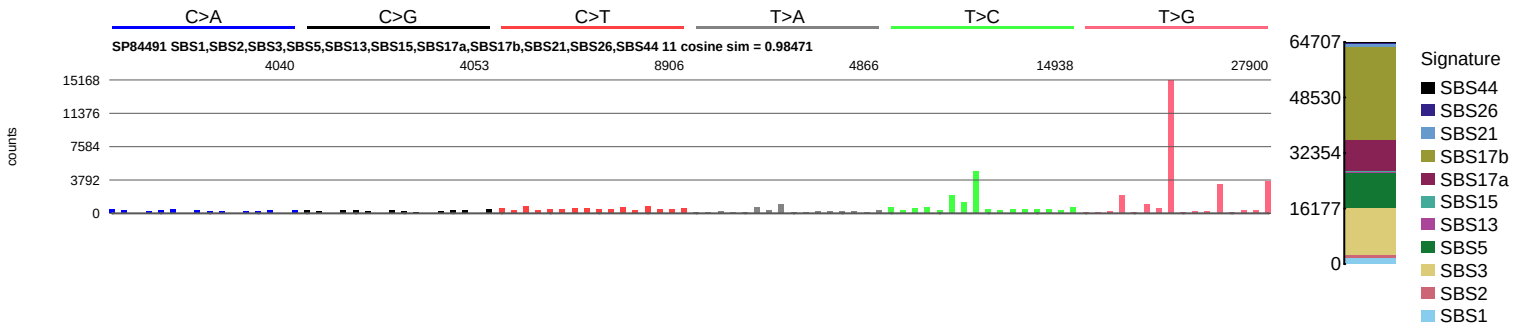

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26 9 cosine sim = 0.98471
4039
4040
8903
4877
14941
27911
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

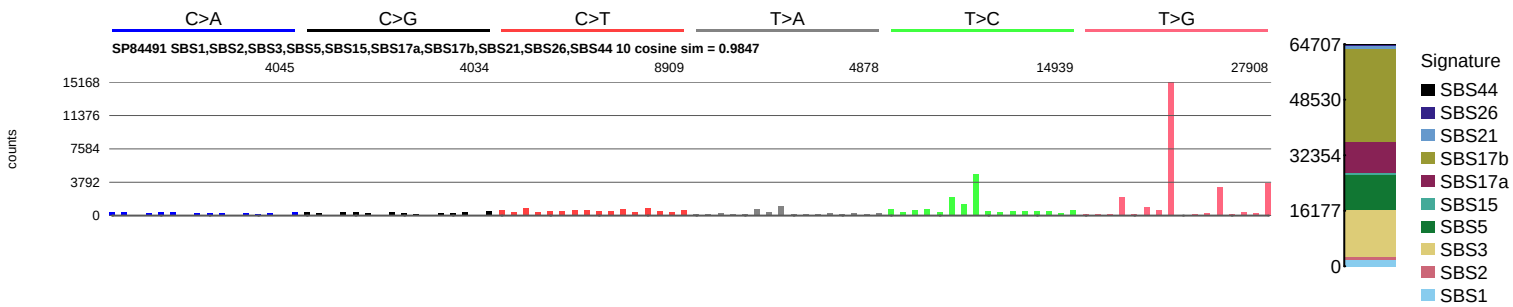

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 9 cosine sim = 0.98471
4039
4053
8952
4872
14888
27902
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

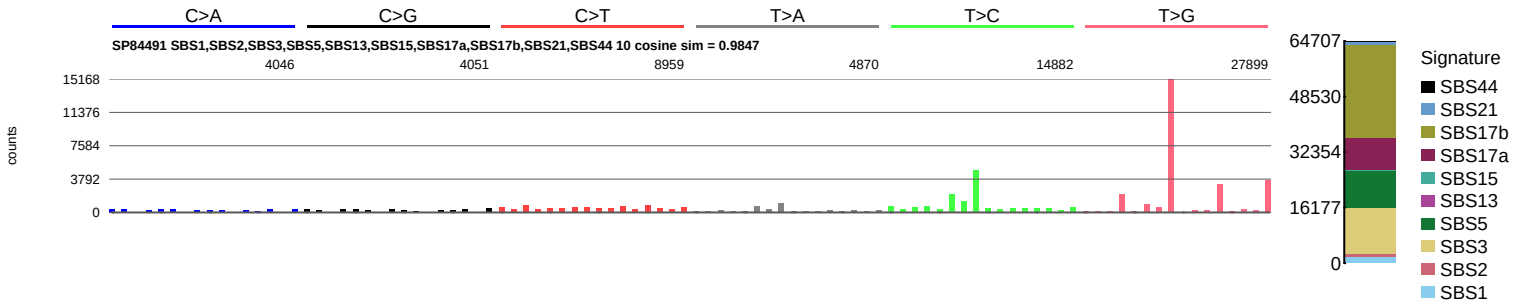

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98471
4318
3966
8772
4834
14936
27882
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26 10 cosine sim = 0.98471
4036
4059
8899
4867
14941
27906
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98471
4312
3965
8831
4837
14881
27879
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 11 cosine sim = 0.98471
4040
4053
8906
4866
14938
27900
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 10 cosine sim = 0.9847
4045
4034
8909
4878
14939
27908
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 10 cosine sim = 0.9847
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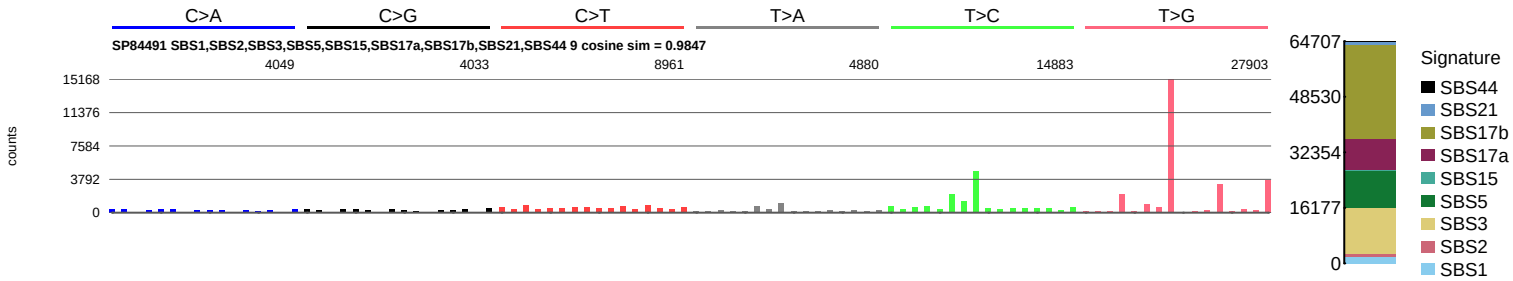
4051
8959
4870
14882
27899
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.9847
4049
4033
8961
4880
14883
27903
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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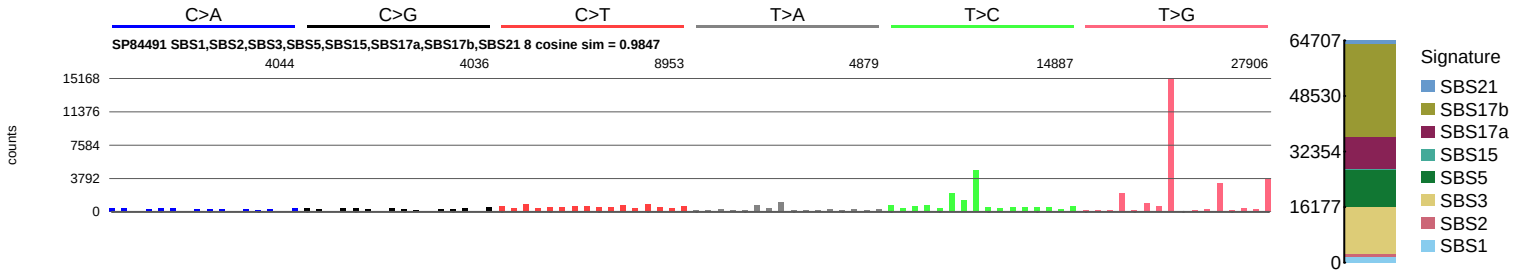
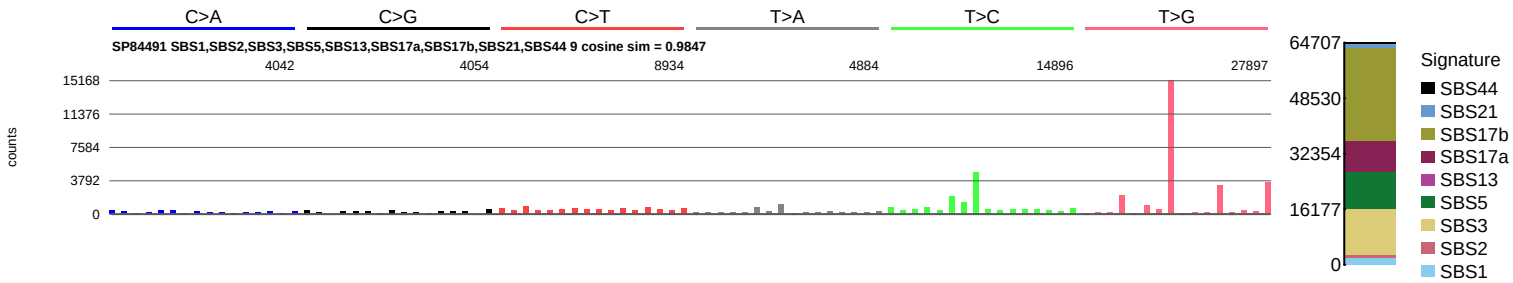
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.9847
4044
4036
8953
4879
14887
27906
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.9847
4042
4054
8934
4884
14896
27897
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
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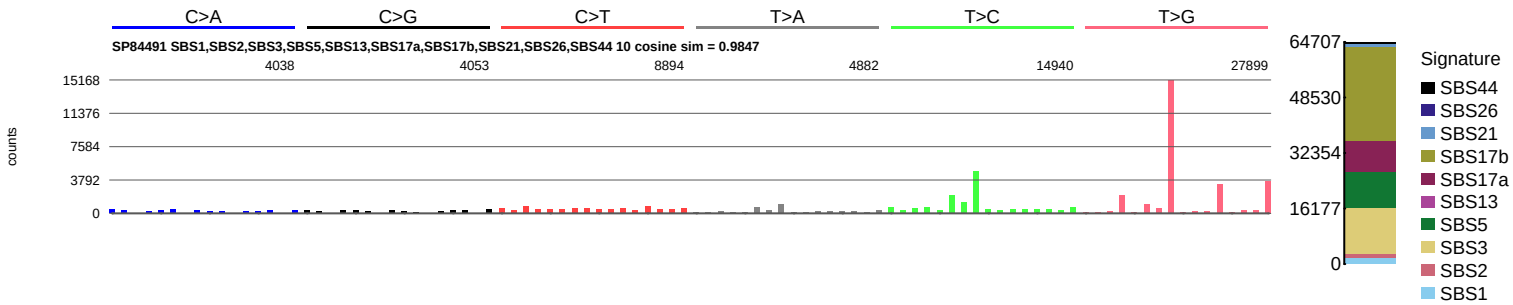
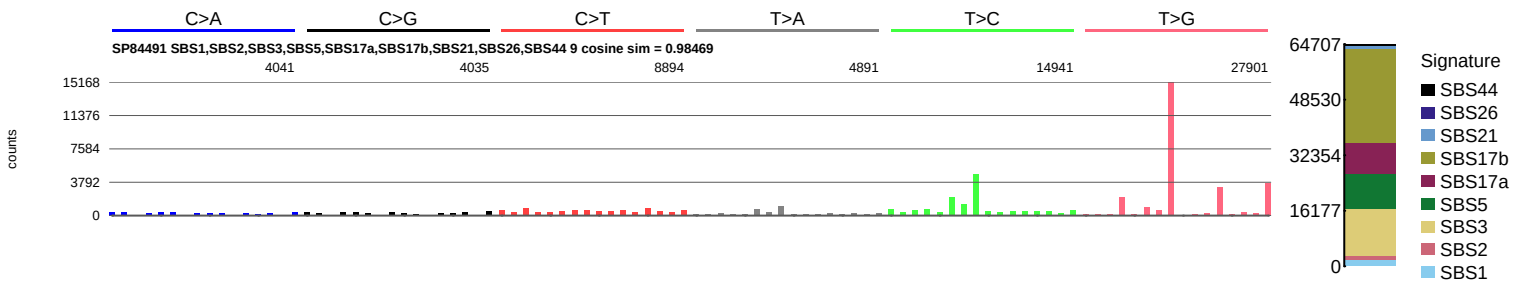
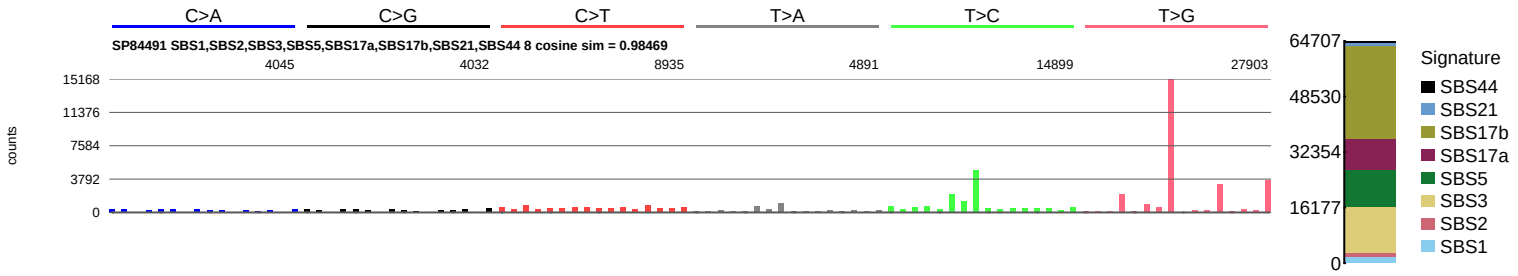
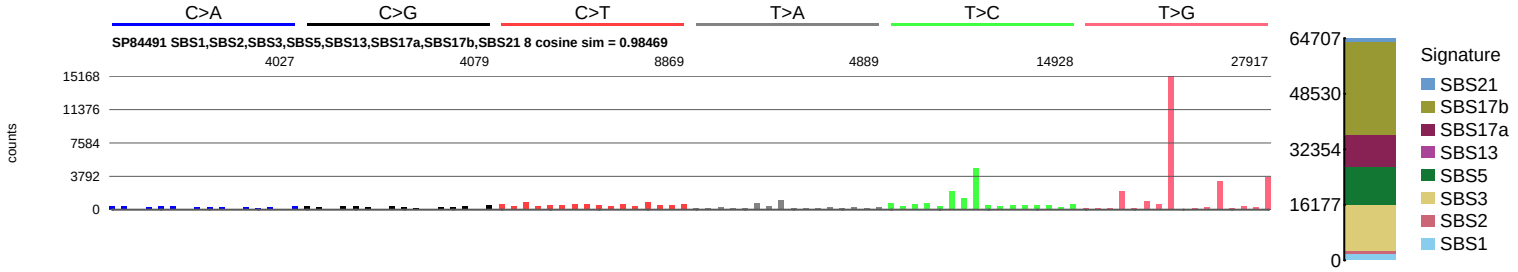
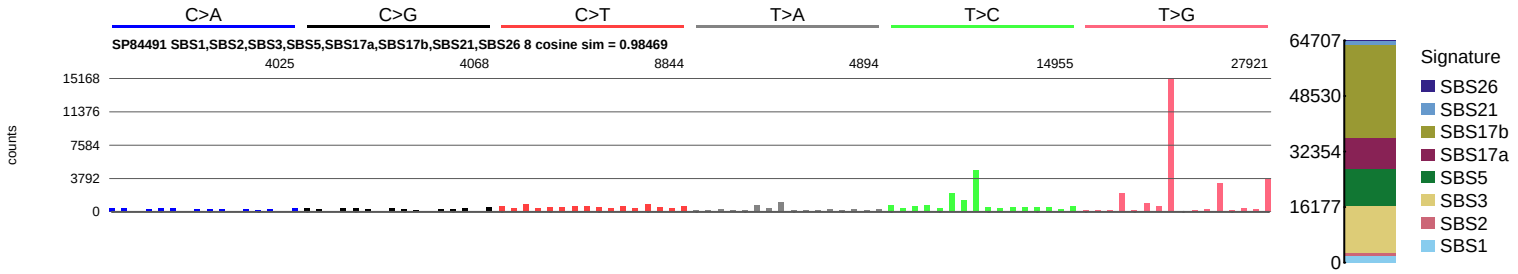
SBS3
SBS2
SBS1

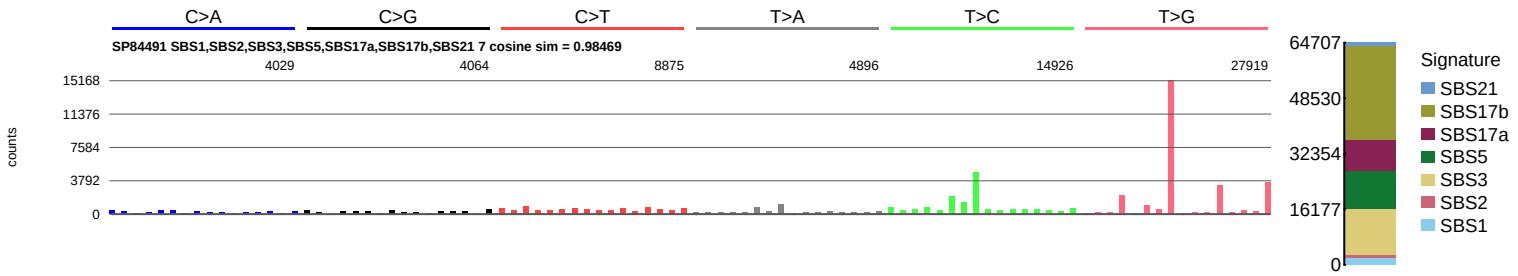

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS44 10 cosine sim = 0.9847
4038
4053
8894
4882
14940
27899
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.98469
4041
4035
8894
4891
14941
27901
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98469
4045
4032
8935
4891
14899
27903
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98469
4027
4079
8869
4889
14928
27917
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98469
4025
4068
8844
4894
14955
27921
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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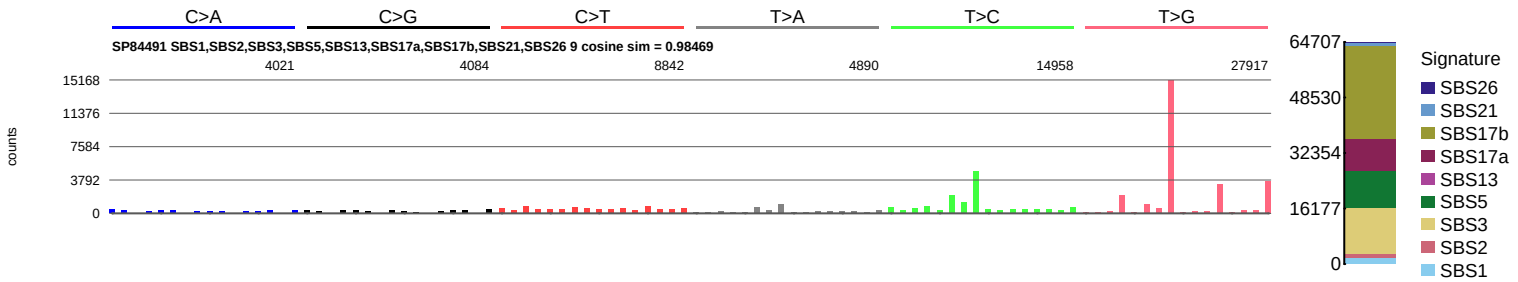
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98469
4029
4064
8875
4896
14926
27919
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS26 9 cosine sim = 0.98469
4021
4084
8842
4890
14958
27917
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
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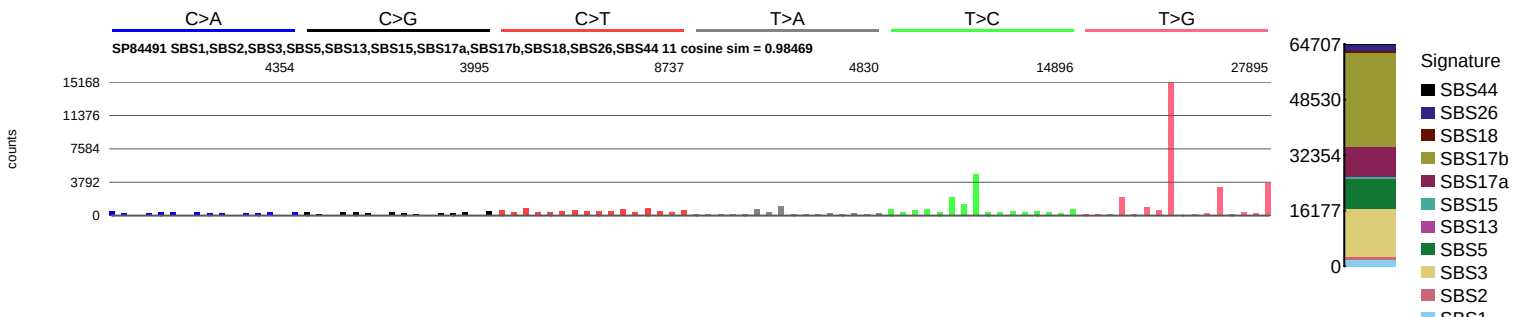
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 11 cosine sim = 0.98469
4354
3995
8737
4830
14896
27895
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
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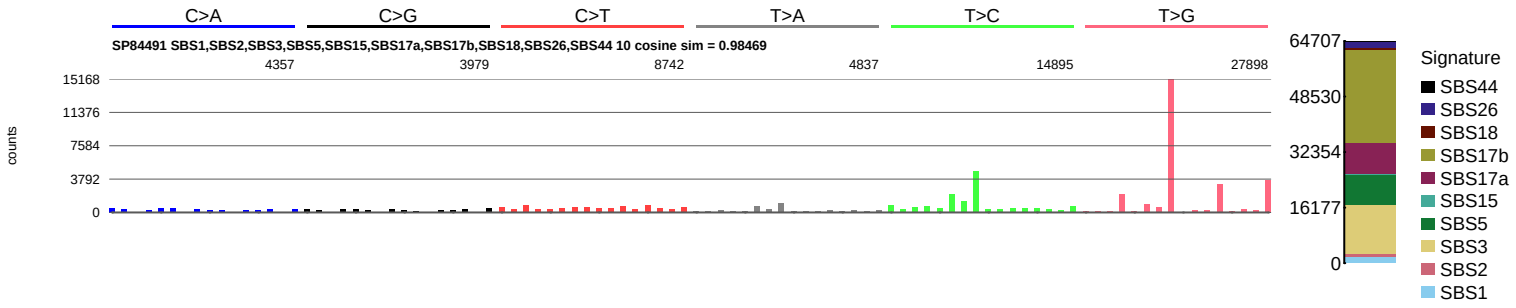
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98469
4357
3979
8742
4837
14895
27898
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
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SBS44
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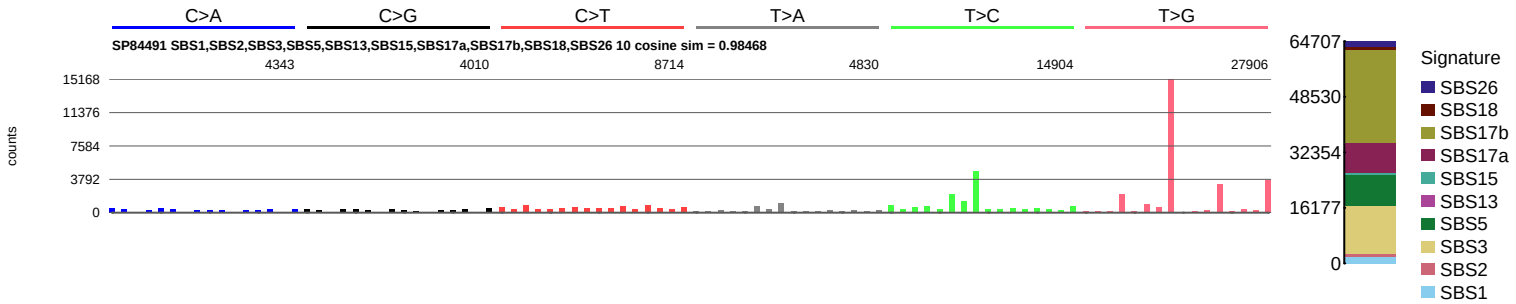
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 10 cosine sim = 0.98468
4343
4010
8714
4830
14904
27906
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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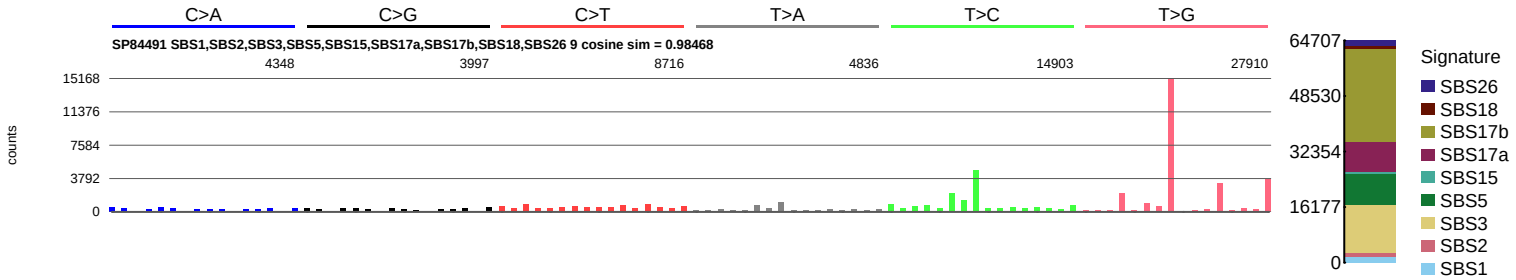
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98468
4348
3997
8716
4836
14903
27910
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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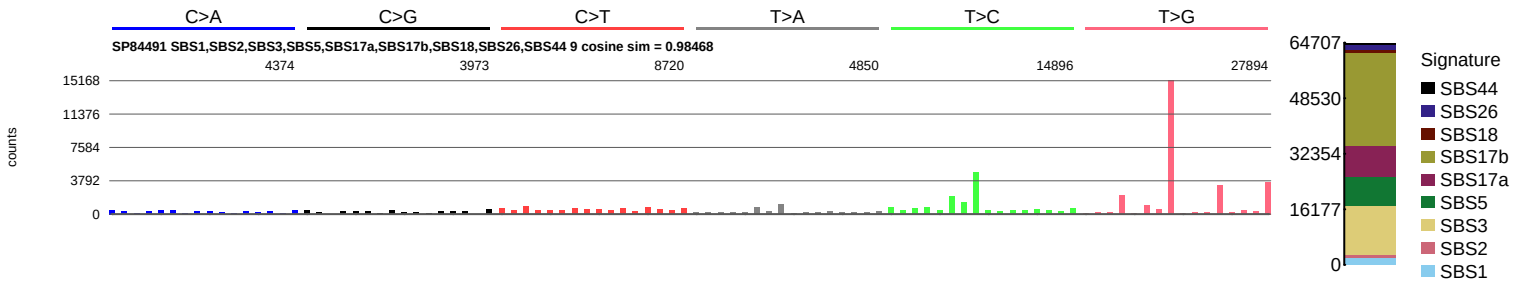
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98468
4374
3973
8720
4850
14896
27894
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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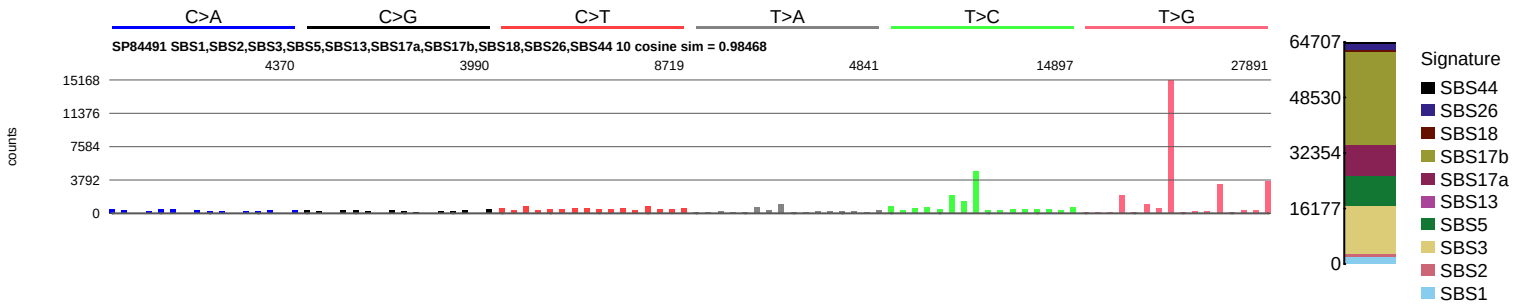
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98468
4370
3990
8719
4841
14897
27891
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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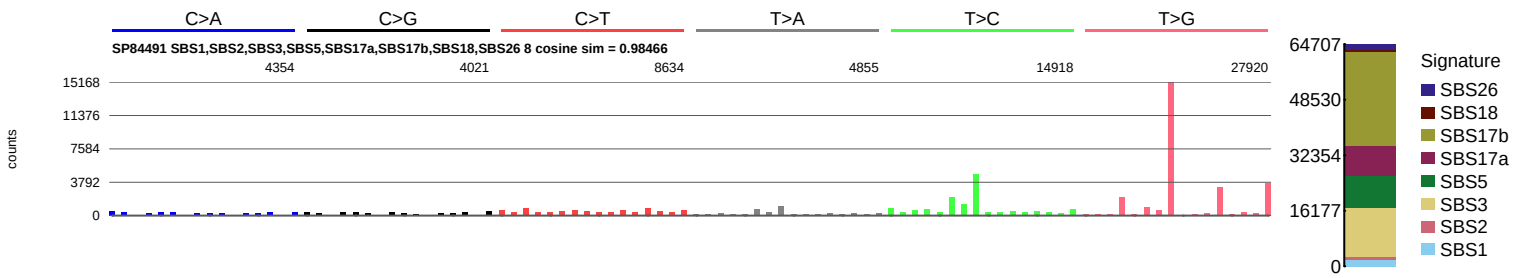
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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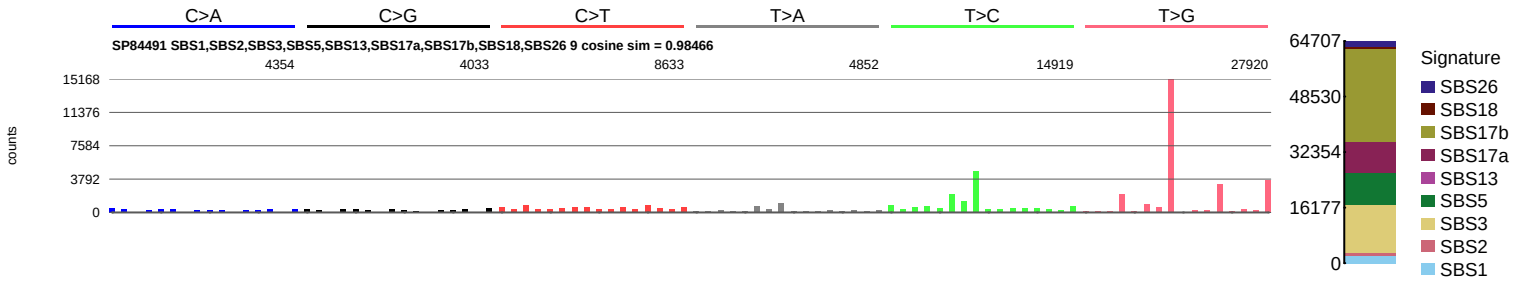
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98466
4354
4021
8634
4855
14918
27920
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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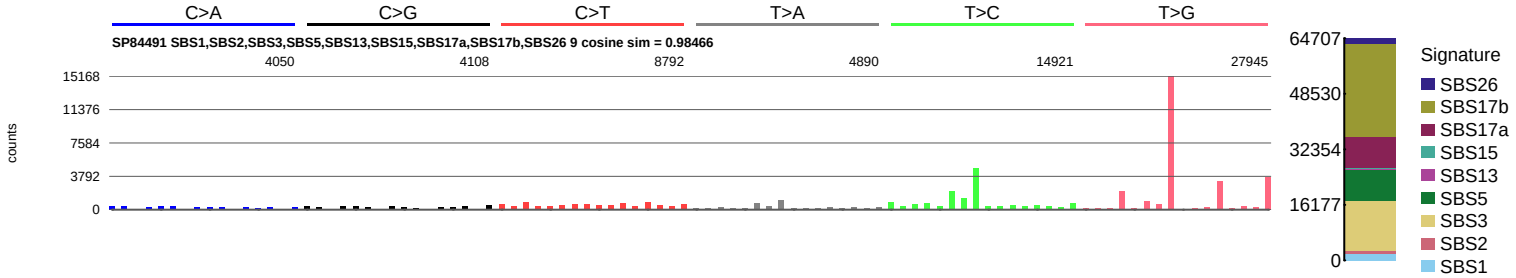
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98466
4354
4033
8633
4852
14919
27920
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 9 cosine sim = 0.98466
4050
4108
8792
4890
14921
27945
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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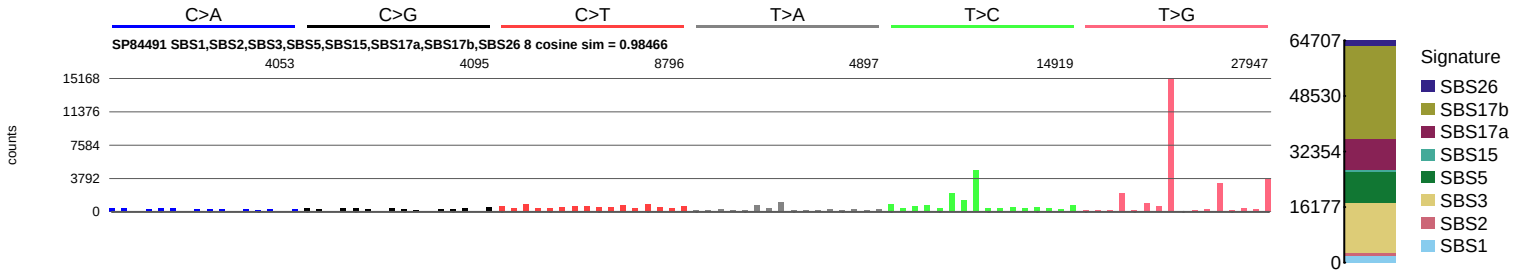
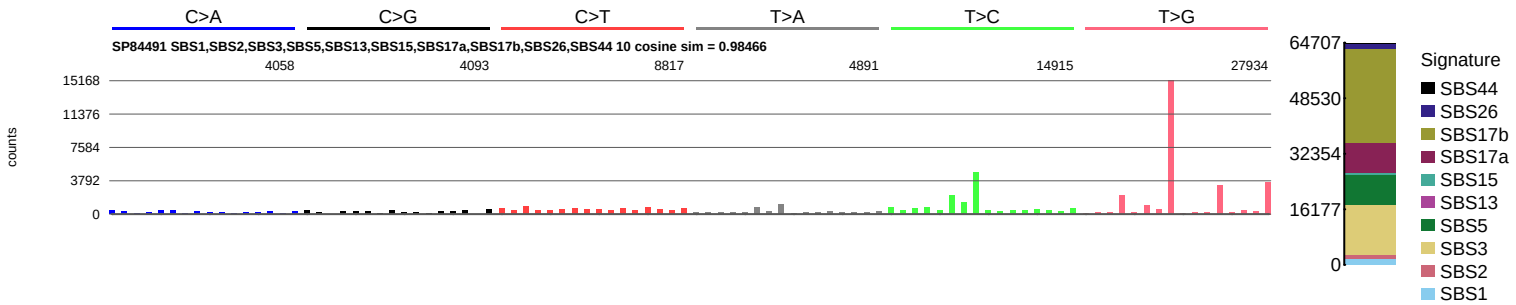
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98466
4053
4095
8796
4897
14919
27947
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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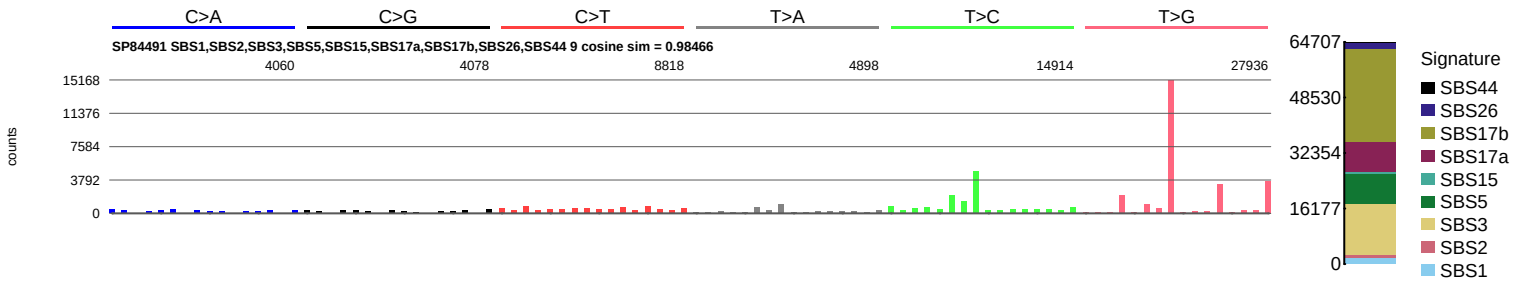
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 10 cosine sim = 0.98466
4058
4093
8817
4891
14915
27934
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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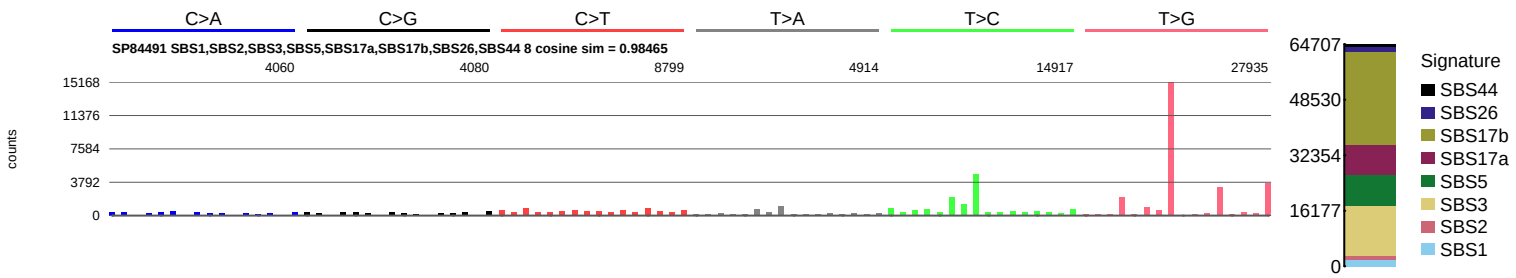
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98466
4060
4078
8818
4898
14914
27936
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98465
4060
4080
8799
4914
14917
27935
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
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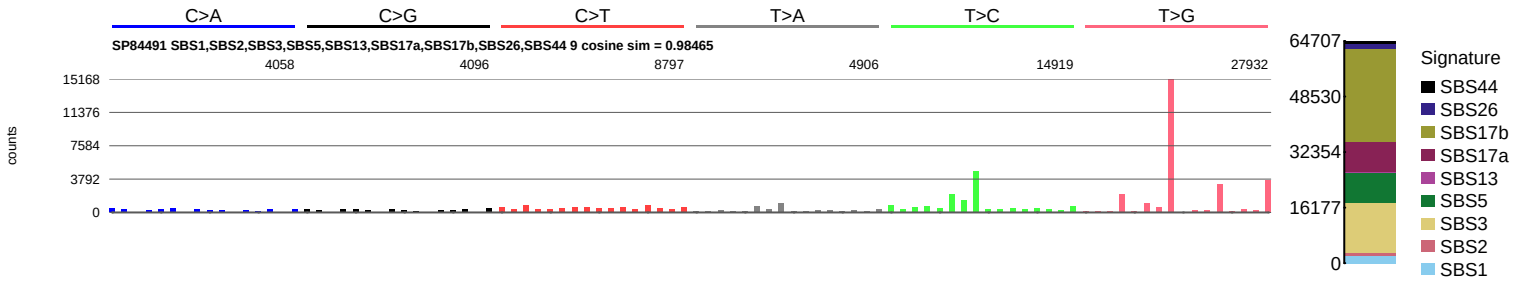
SBS44
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98465
4058
4096
8797
4906
14919
27932
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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Signature
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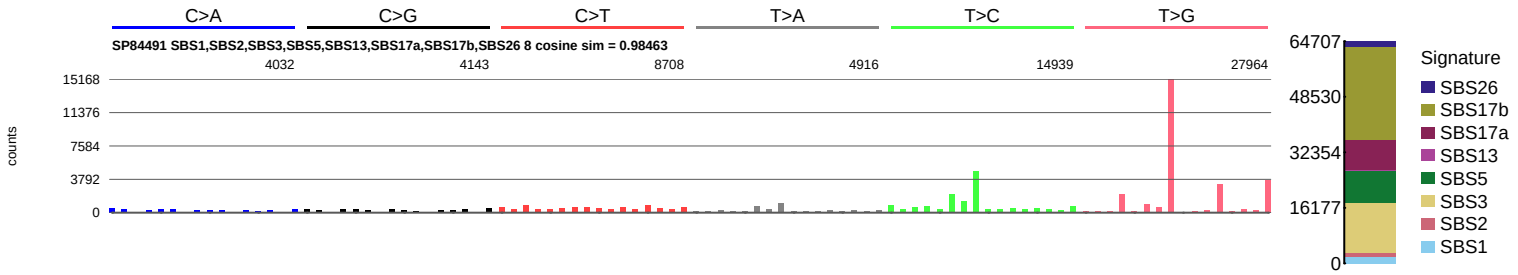
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98463
4032
4143
8708
4916
14939
27964
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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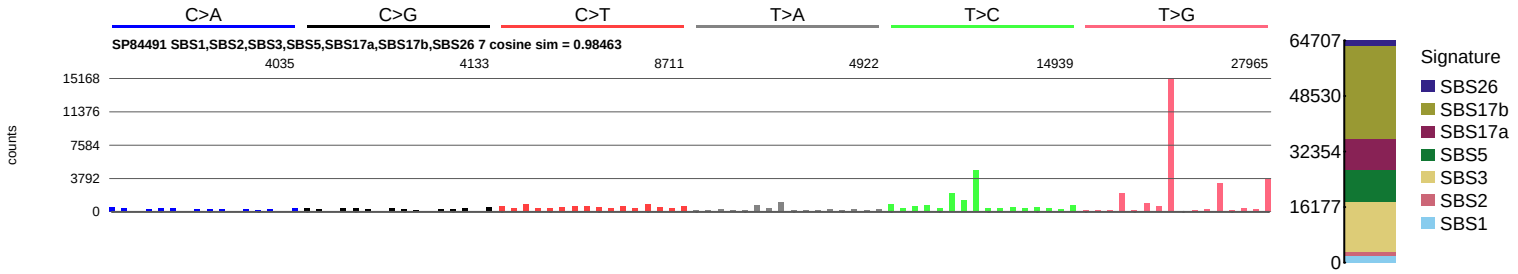
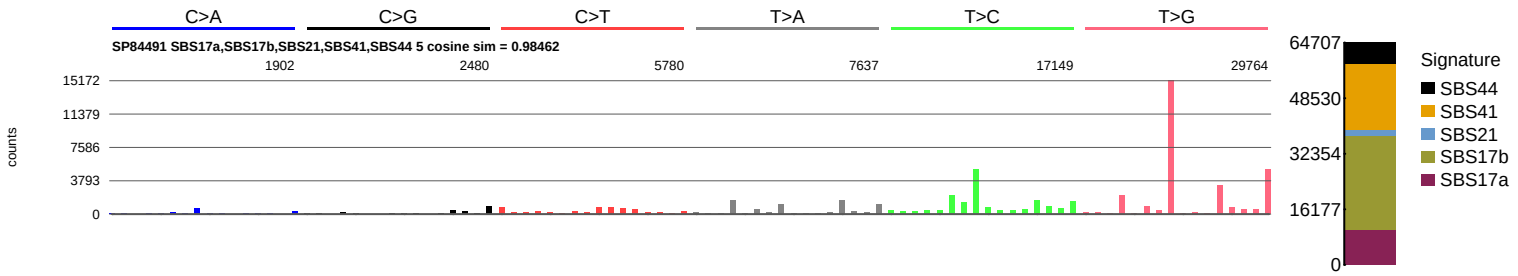
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98463
4035
4133
8711
4922
14939
27965
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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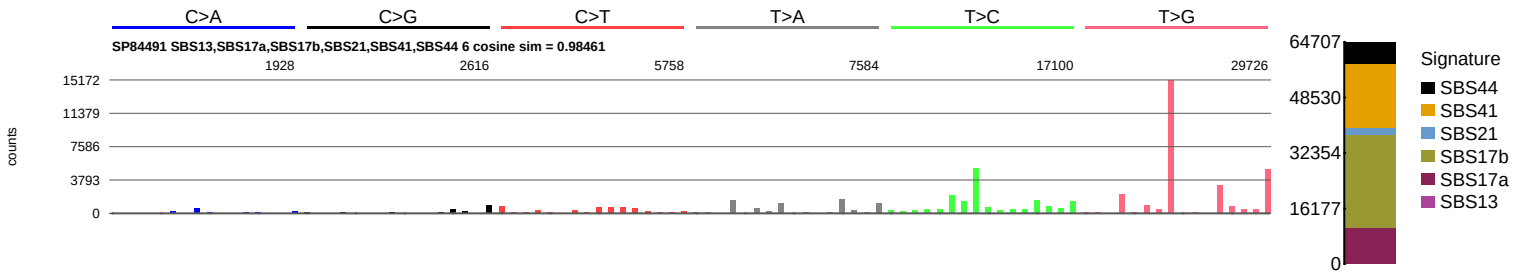
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS41,SBS44 5 cosine sim = 0.98462
1902
2480
5780
7637
17149
29764
counts
15172
11379
7586
3793
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a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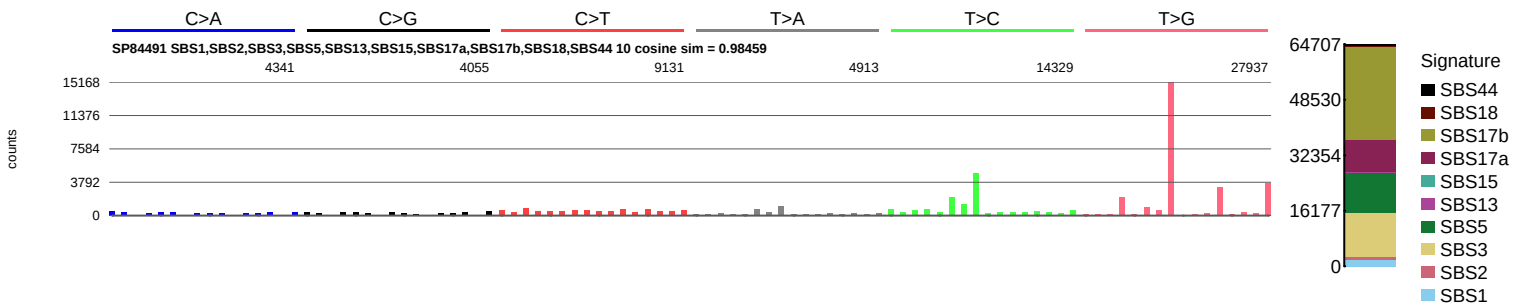
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.98461
1928
2616
5758
7584
17100
29726
counts
15172
11379
7586
3793
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98459
4341
4055
9131
4913
14329
27937
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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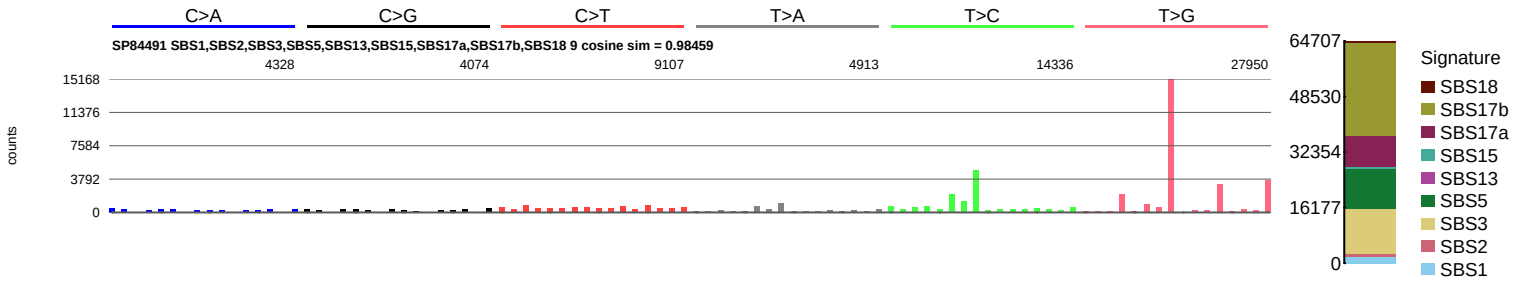
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98459
4328
4074
9107
4913
14336
27950
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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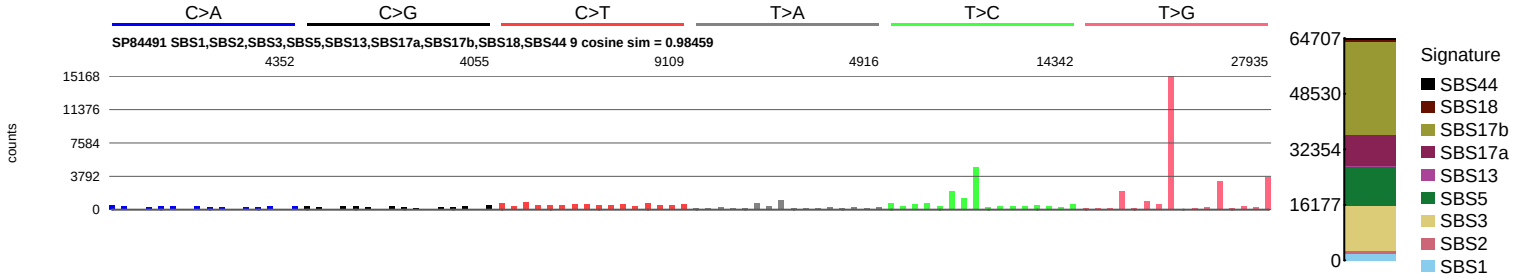
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98459
4352
4055
9109
4916
14342
27935
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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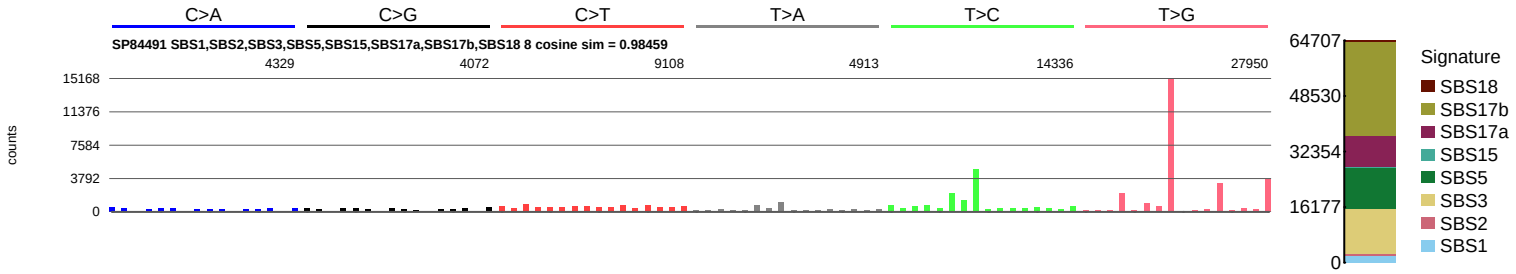
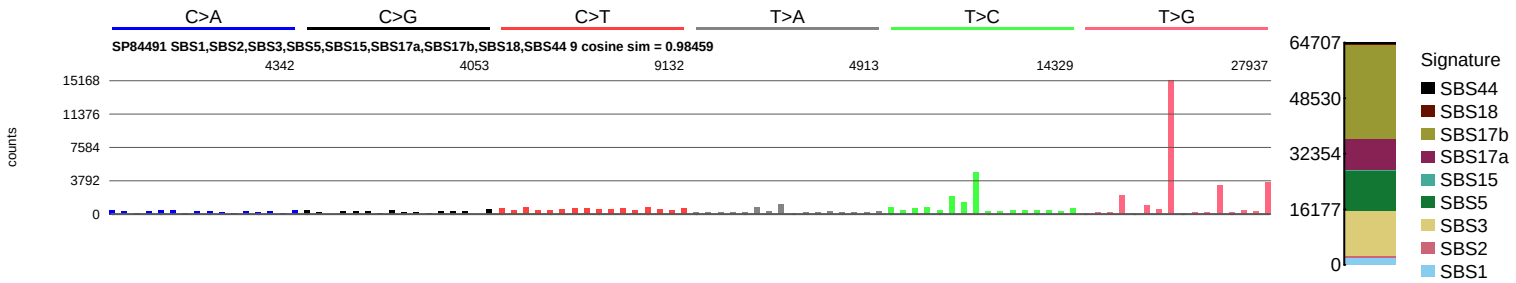
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98459
4329
4072
9108
4913
14336
27950
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98459
4342
4053
9132
4913
14329
27937
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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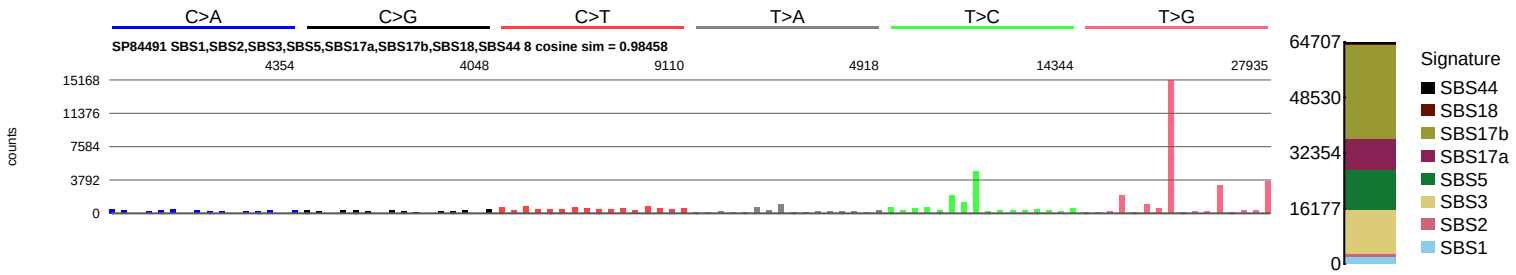
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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4354
4048
9110
4918
14344
27935
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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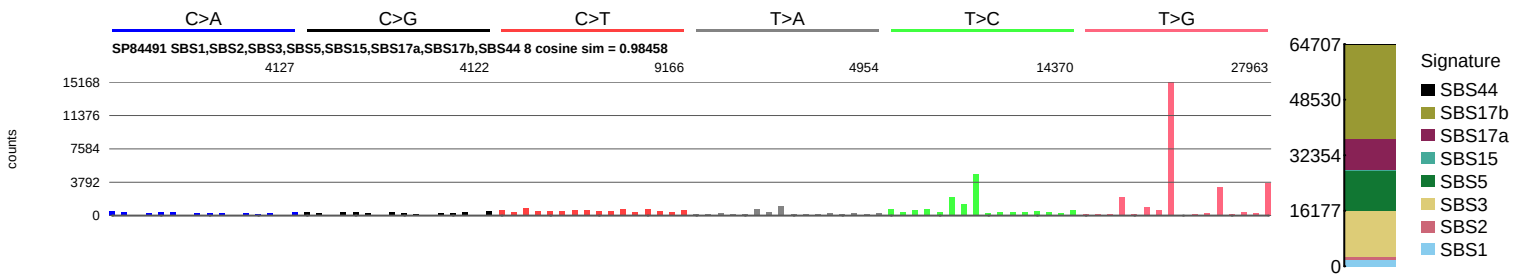
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98458
4127
4122
9166
4954
14370
27963
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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SBS44
SBS17b
SBS17a
SBS15
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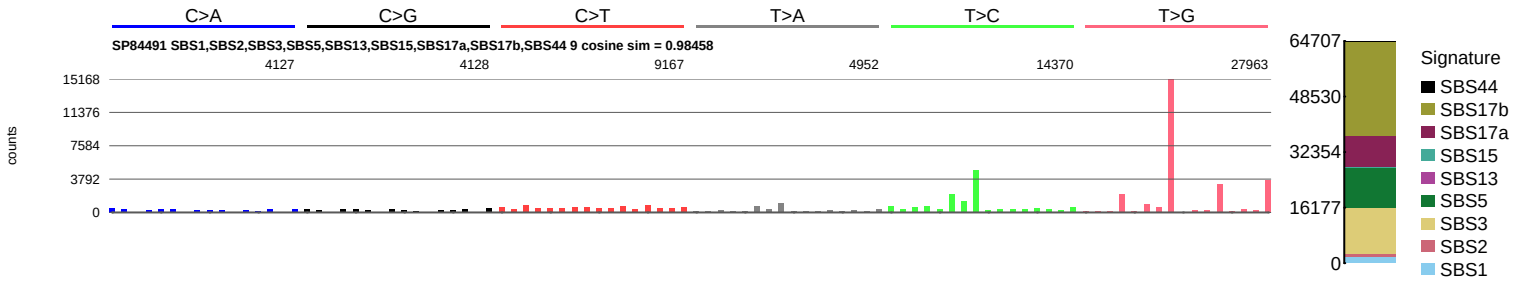
SBS2
SBS1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SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS44 9 cosine sim = 0.98458
4127
4128
9167
4952
14370
27963
counts
15168
11376
7584
3792
0
64707
48530
32354
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SBS44
SBS17b
SBS17a
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SBS13
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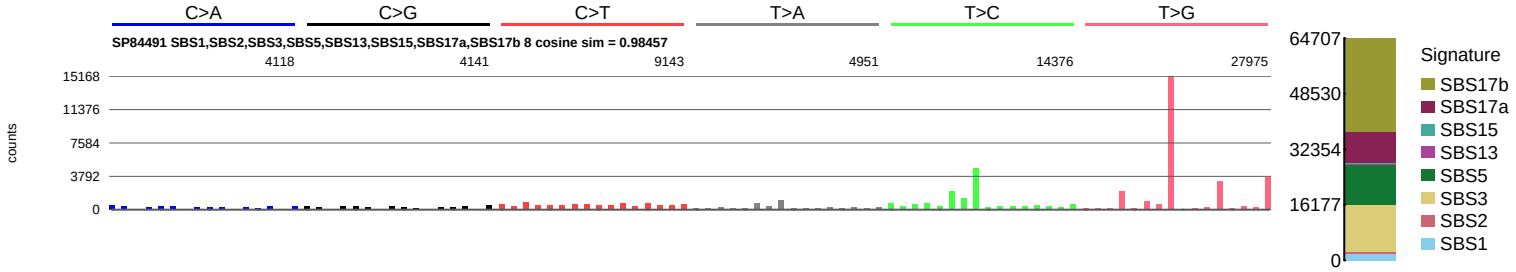
SBS3
SBS2
SBS1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C>G
C>T
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SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 8 cosine sim = 0.98457
4118
4141
9143
4951
14376
27975
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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SBS17b
SBS17a
SBS15
SBS13
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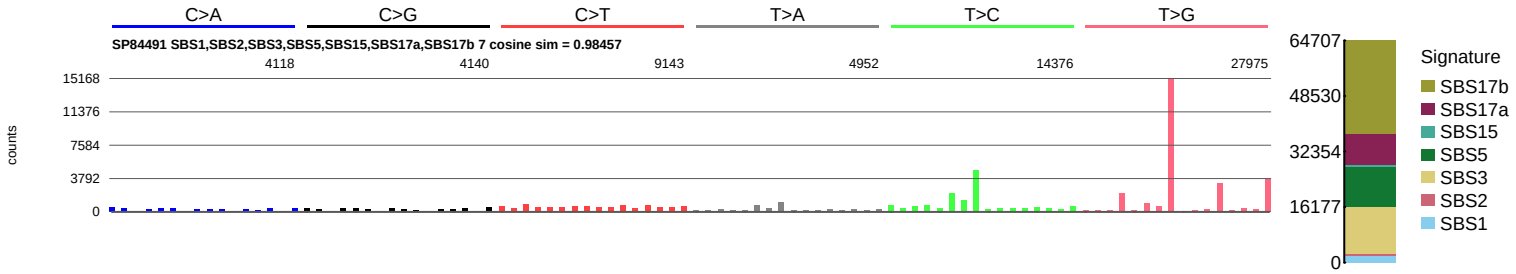
SBS3
SBS2
SBS1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C>T
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SP84491 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98457
4118
4140
9143
4952
14376
27975
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
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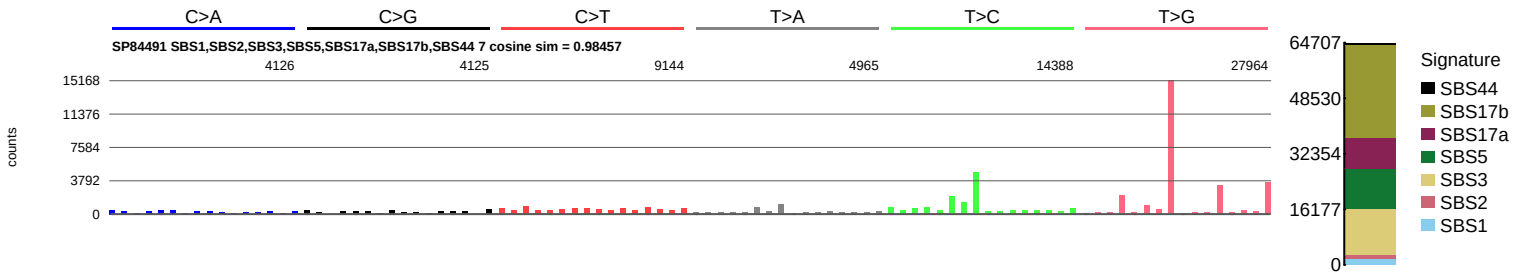
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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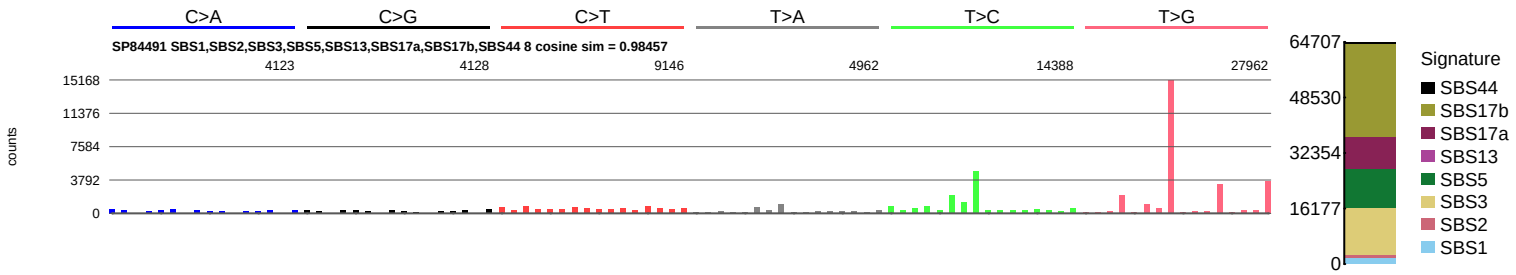
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98457
4126
4125
9144
4965
14388
27964
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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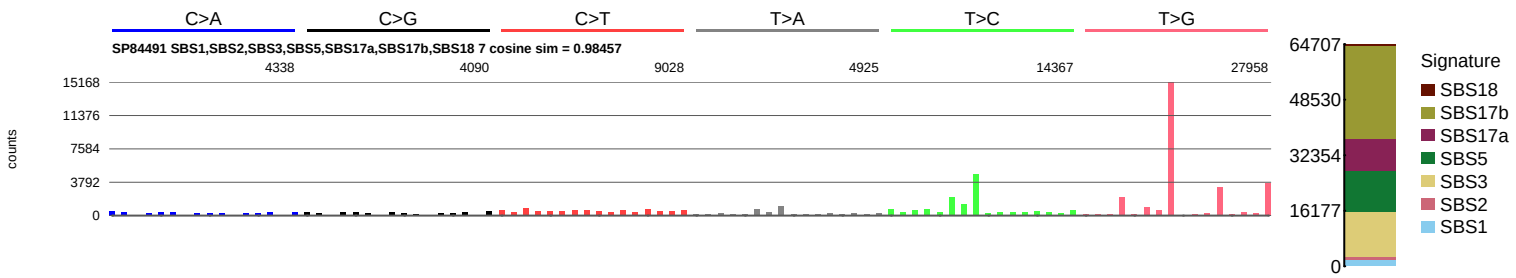
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98457
4123
4128
9146
4962
14388
27962
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98457
4338
4090
9028
4925
14367
27958
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
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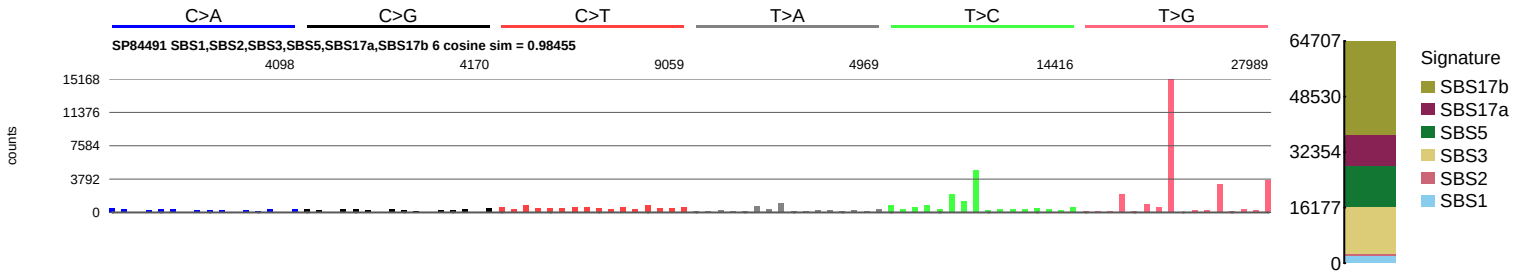
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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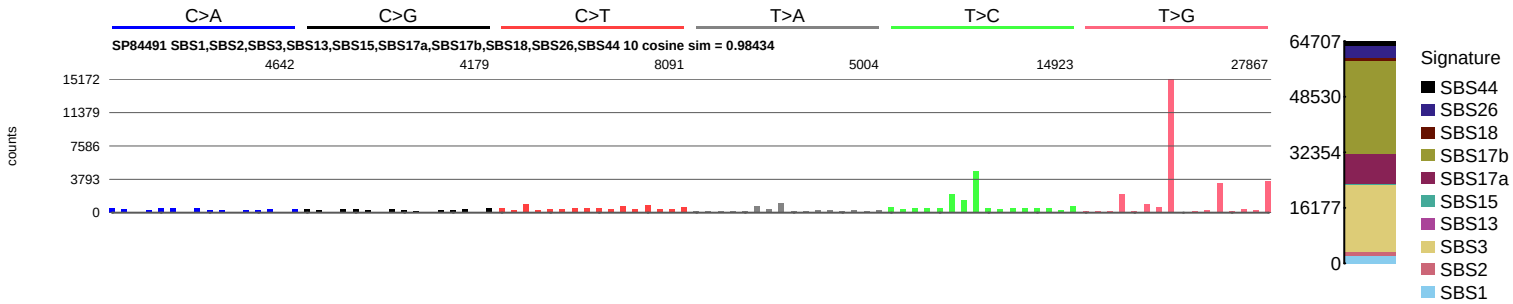
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b 6 cosine sim = 0.98455
4098
4170
9059
4969
14416
27989
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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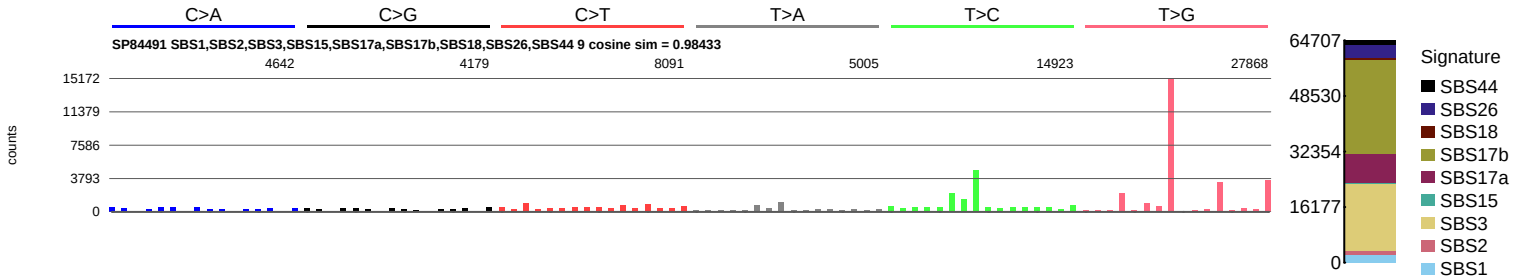
T>G
SP84491 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98434
4642
4179
8091
5004
14923
27867
15172
11379
7586
3793
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98433
4642
4179
8091
5005
14923
27868
15172
11379
7586
3793
0
counts
64707
48530
32354
16177
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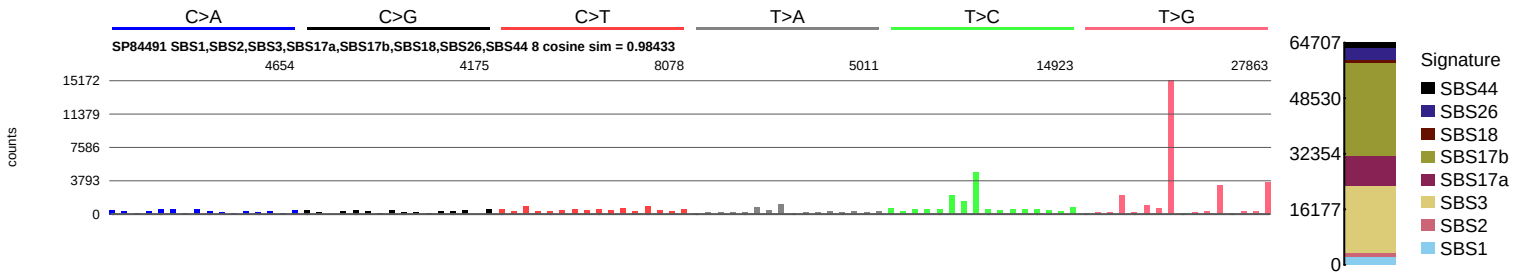
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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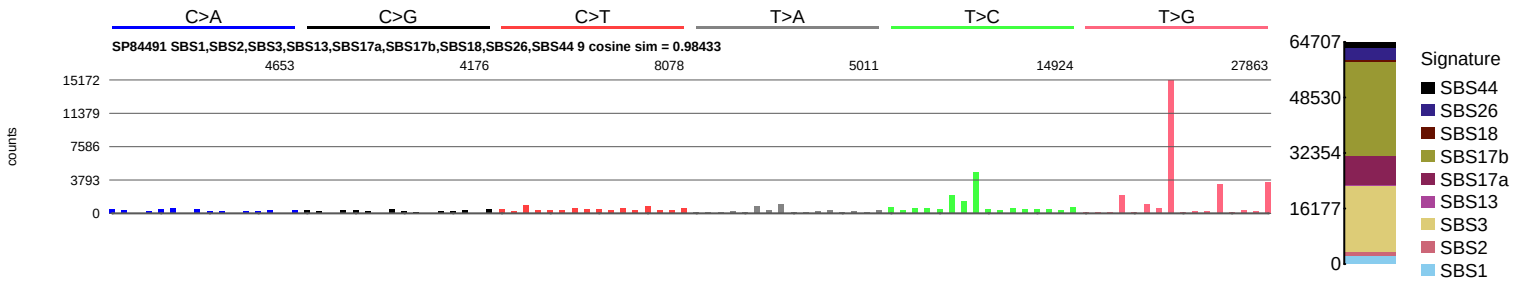
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98433
4654
4175
8078
5011
14923
27863
15172
11379
7586
3793
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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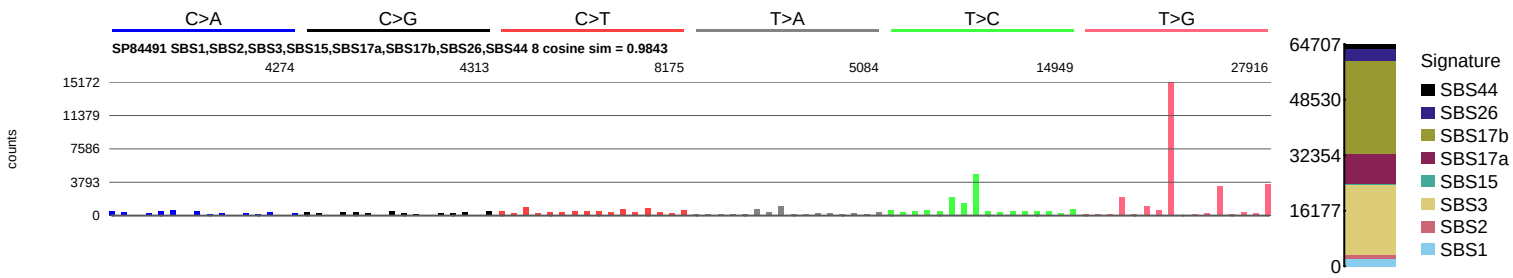
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98433
4653
4176
8078
5011
14924
27863
15172
11379
7586
3793
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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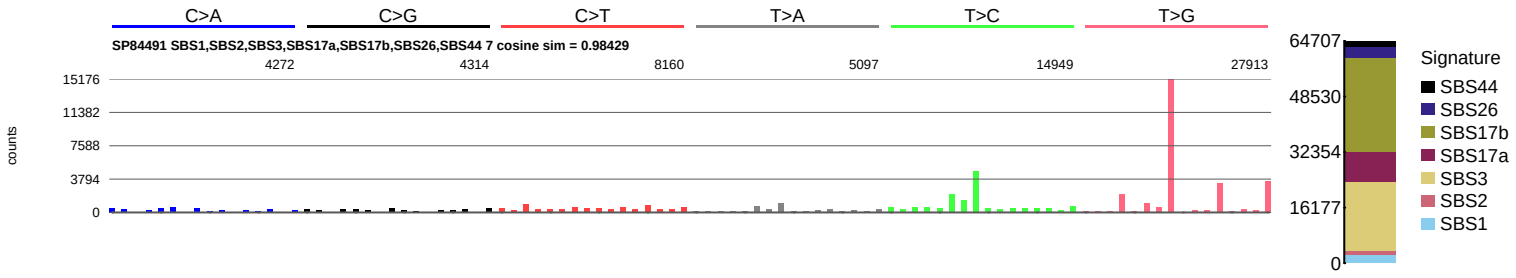
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.9843
4274
4313
8175
5084
14949
27916
15172
11379
7586
3793
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98429
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4314
8160
5097
14949
27913
15176
11382
7588
3794
0
counts
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48530
32354
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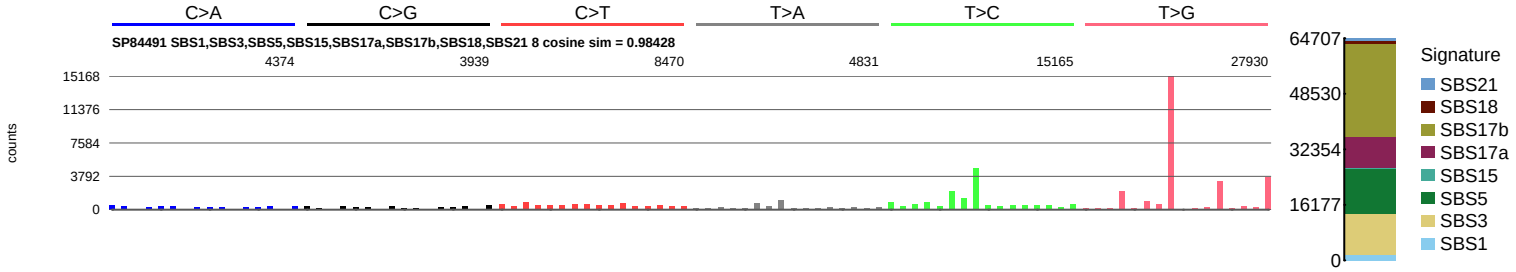
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
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T>G
SP84491 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98428
4374
3939
8470
4831
15165
27930
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
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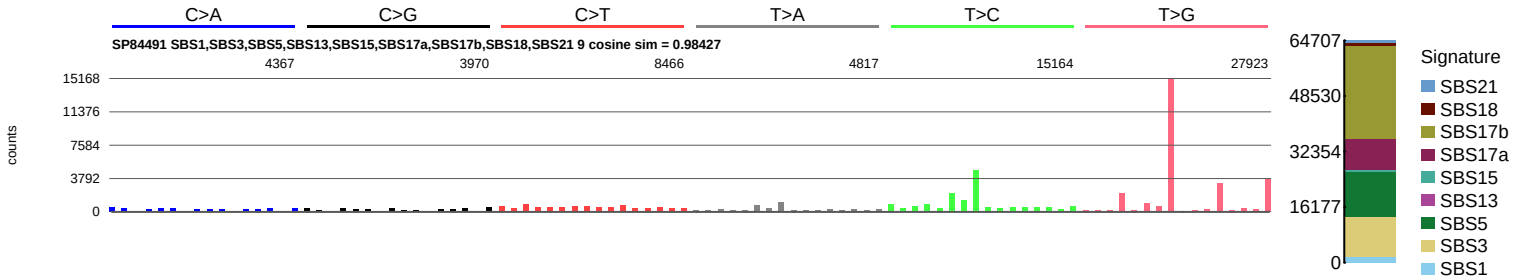
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
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T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98427
4367
3970
8466
4817
15164
27923
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
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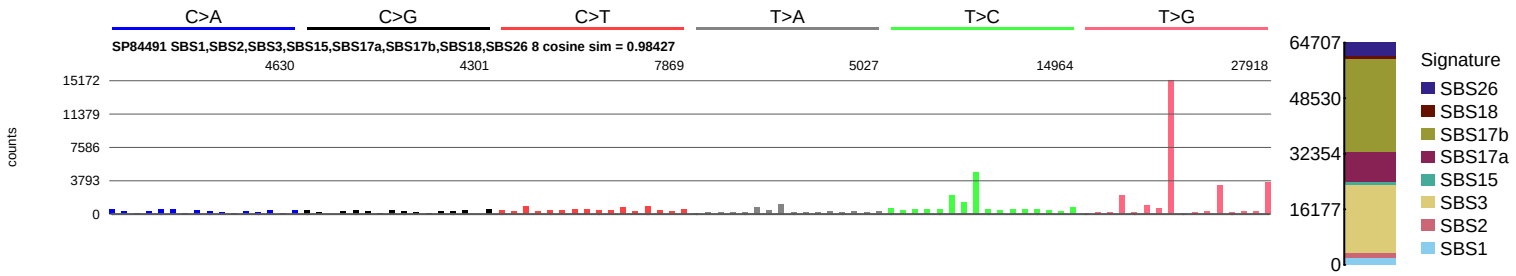
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98427
4630
4301
7869
5027
14964
27918
15172
11379
7586
3793
0
counts
64707
48530
32354
16177
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Signature
SBS26
SBS18
SBS17b
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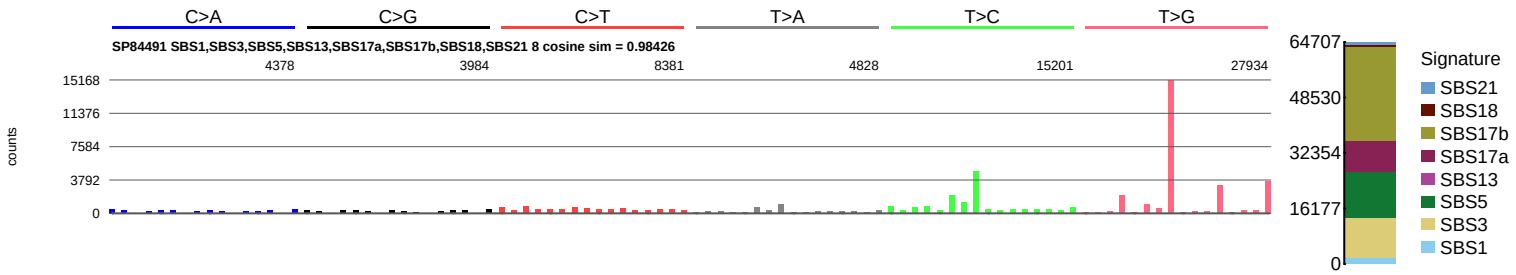
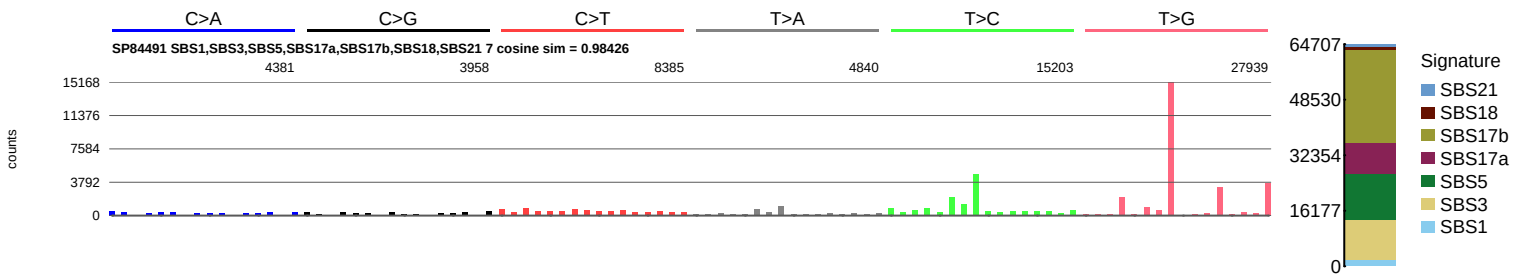
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98426
4378
3984
8381
4828
15201
27934
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98426
4381
3958
8385
4840
15203
27939
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
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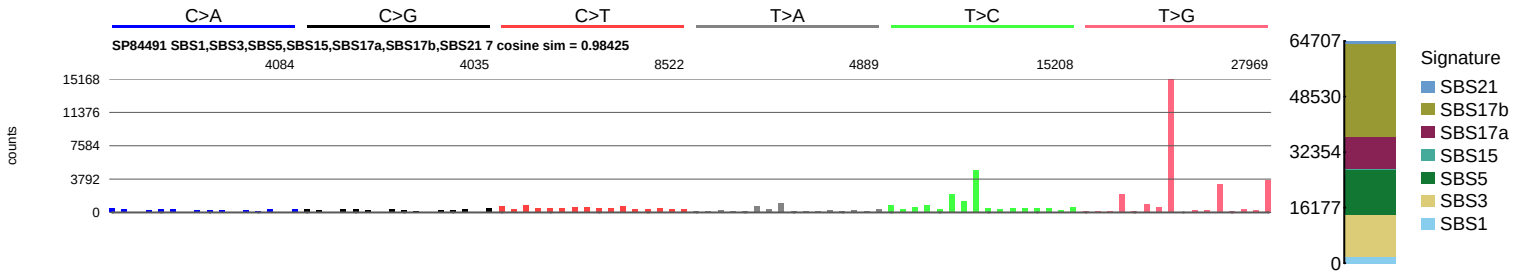
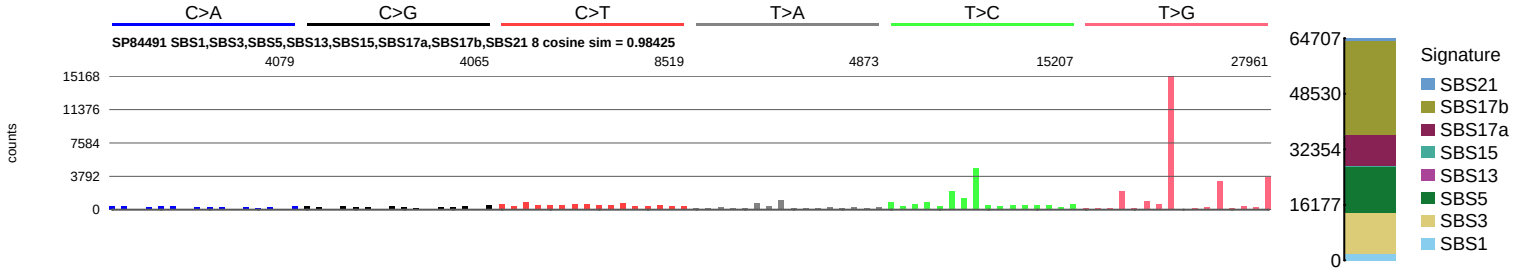
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98425
4084
4035
8522
4889
15208
27969
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98425
4079
4065
8519
4873
15207
27961
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
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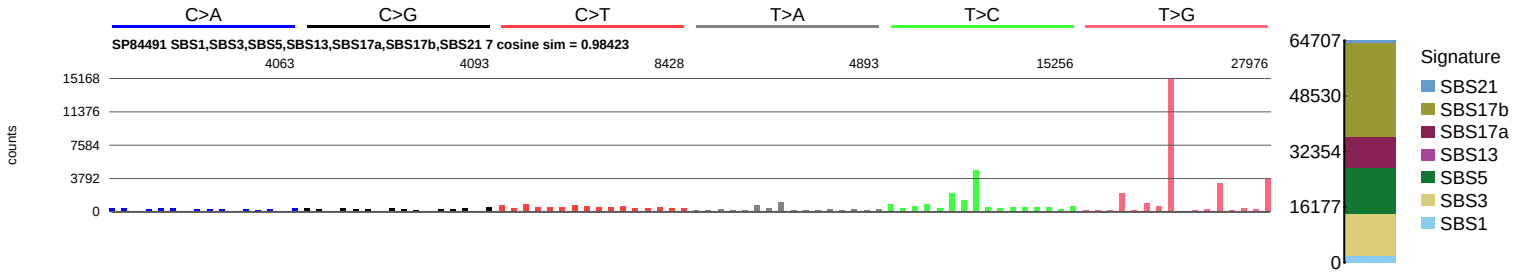
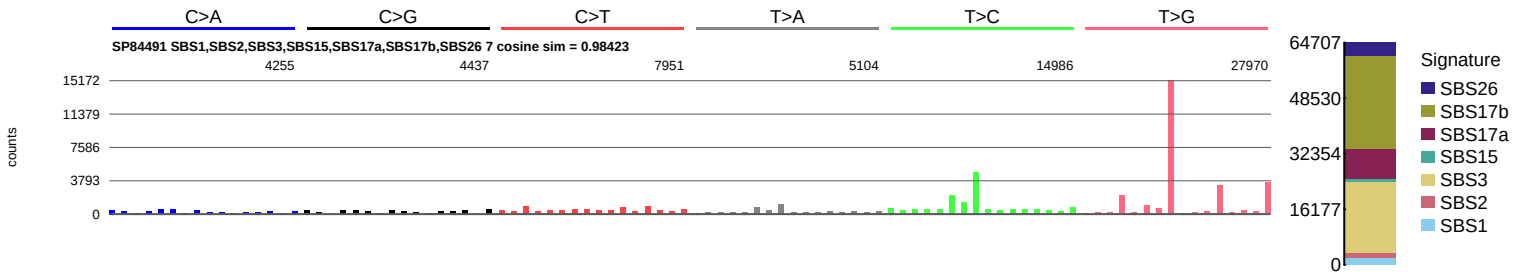
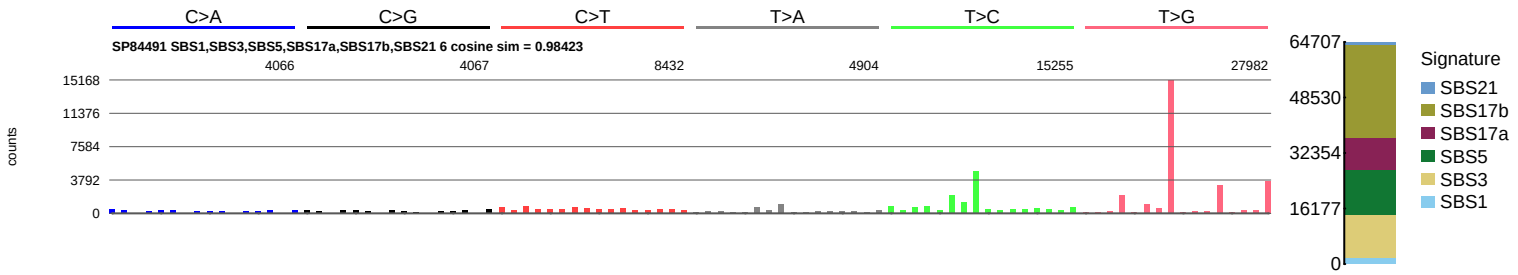
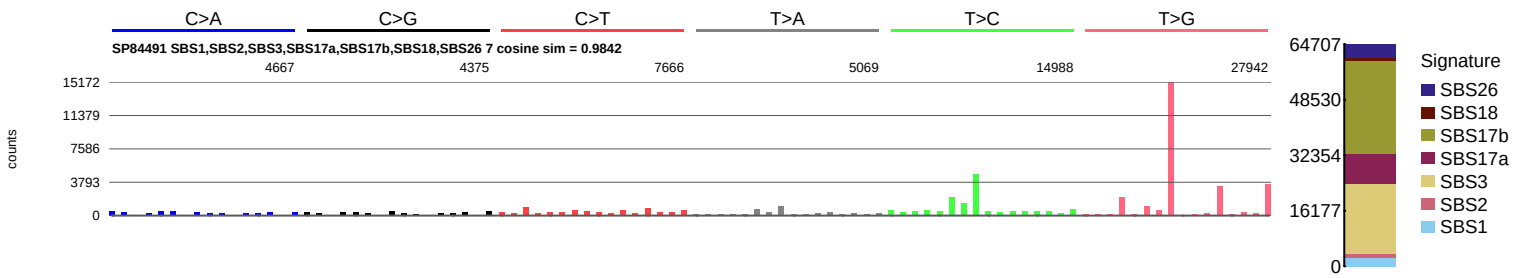
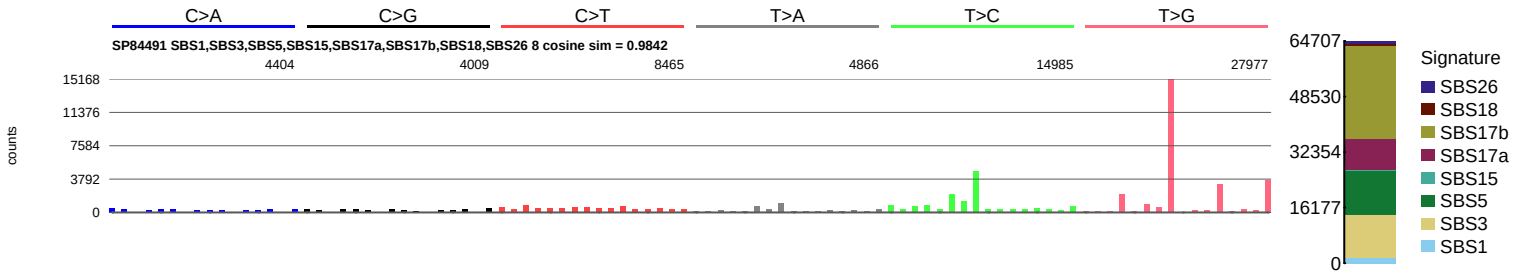
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

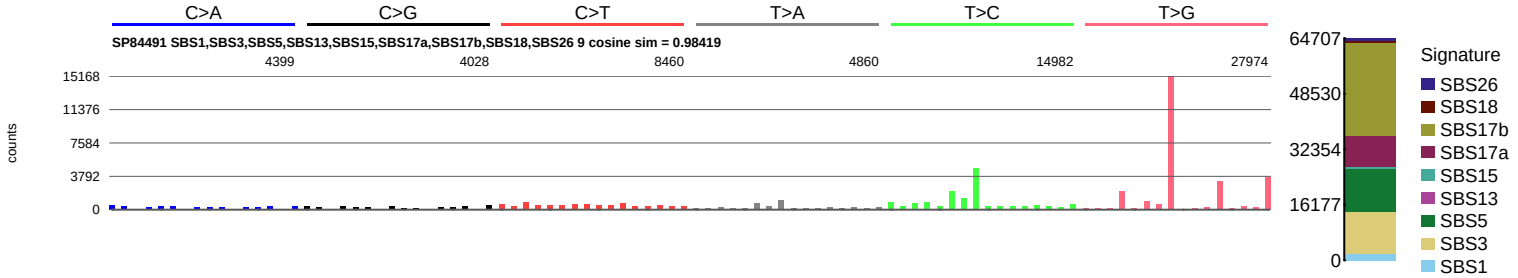

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98423
4063
4093
8428
4893
15256
27976
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

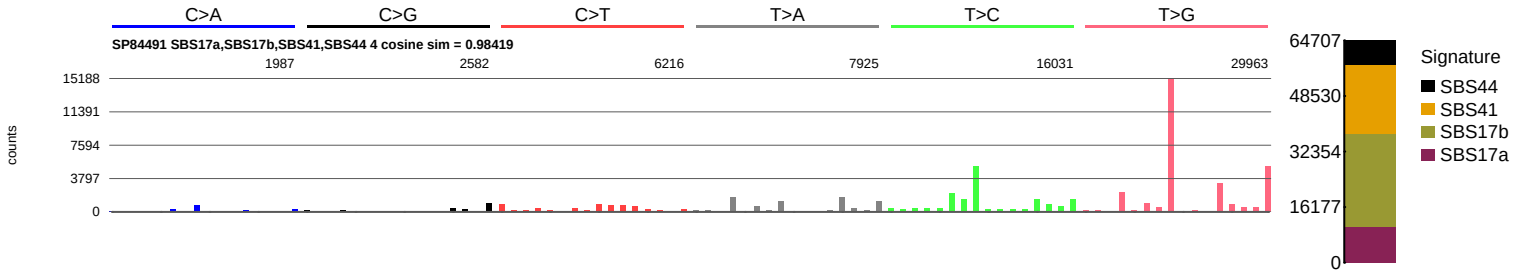

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98423
4255
4437
7951
5104
14986
27970
counts
15172
11379
7586
3793
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

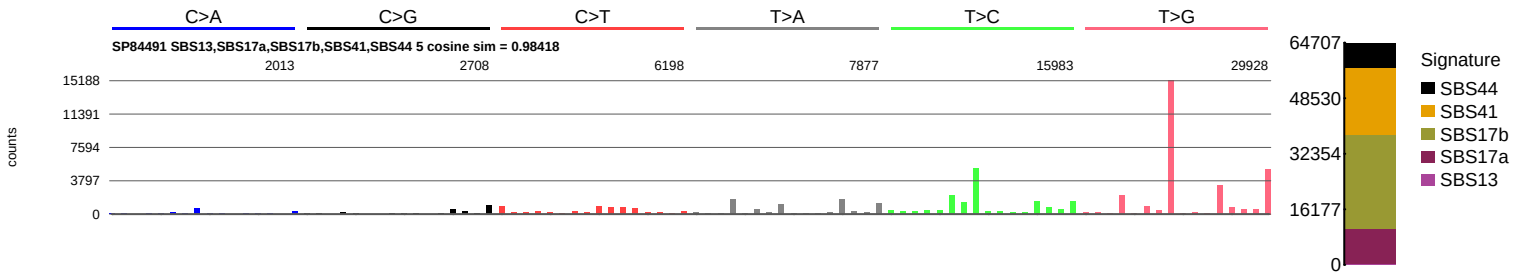

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98423
4066
4067
8432
4904
15255
27982
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

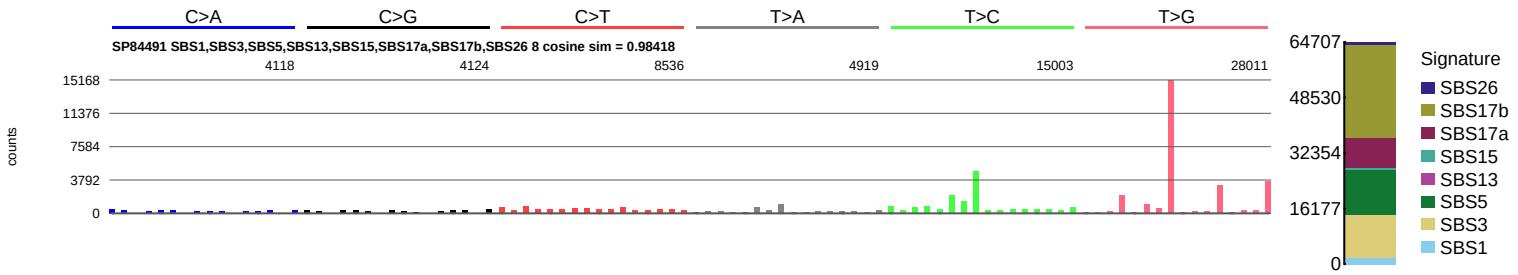

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.9842
4667
4375
7666
5069
14988
27942
counts
15172
11379
7586
3793
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

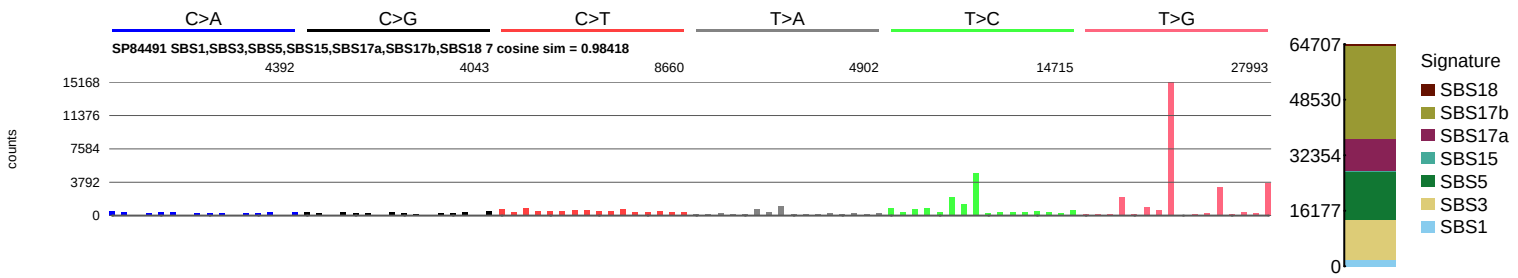

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.9842
4404
4009
8465
4866
14985
27977
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

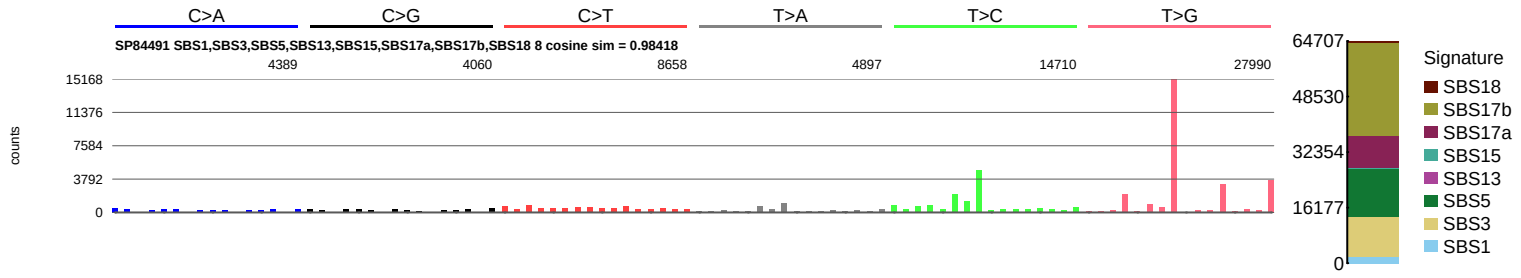

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98419
4399
4028
8460
4860
14982
27974
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

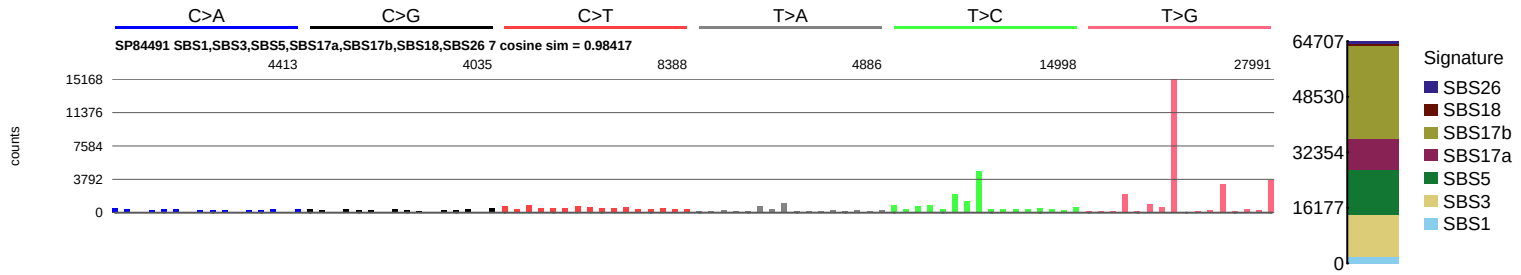

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS41,SBS44 4 cosine sim = 0.98419
1987
2582
6216
7925
16031
29963
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a

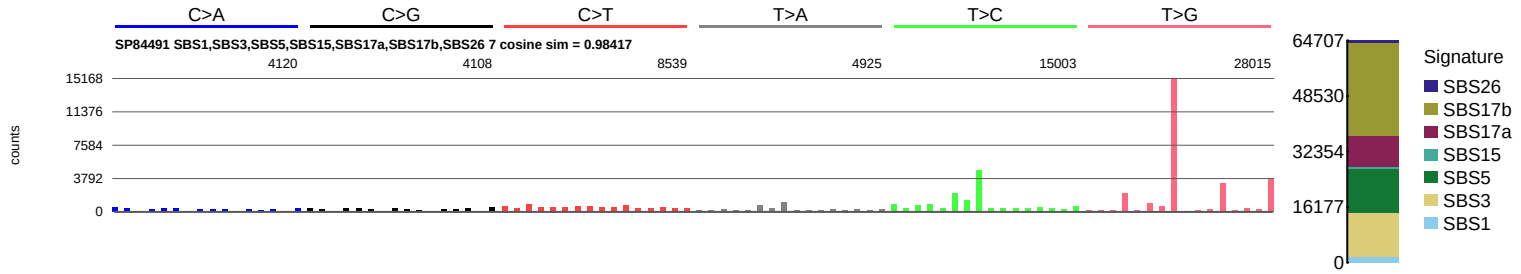

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.98418
2013
2708
6198
7877
15983
29928
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13

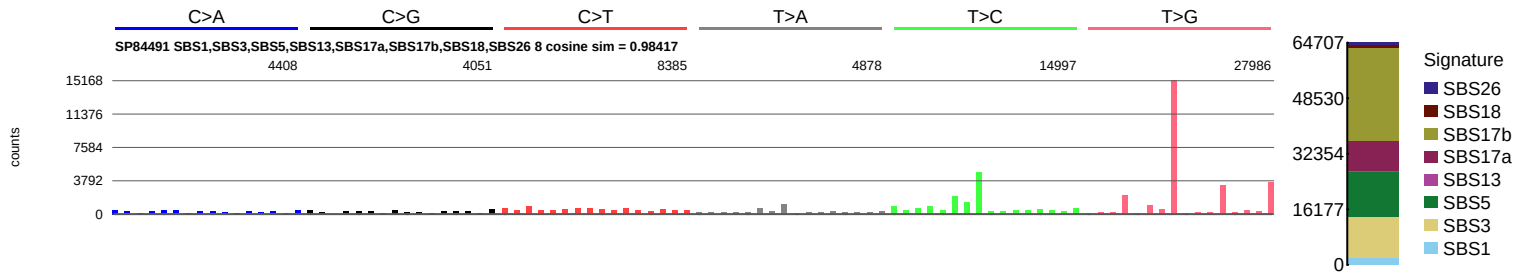

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98418
4118
4124
8536
4919
15003
28011
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

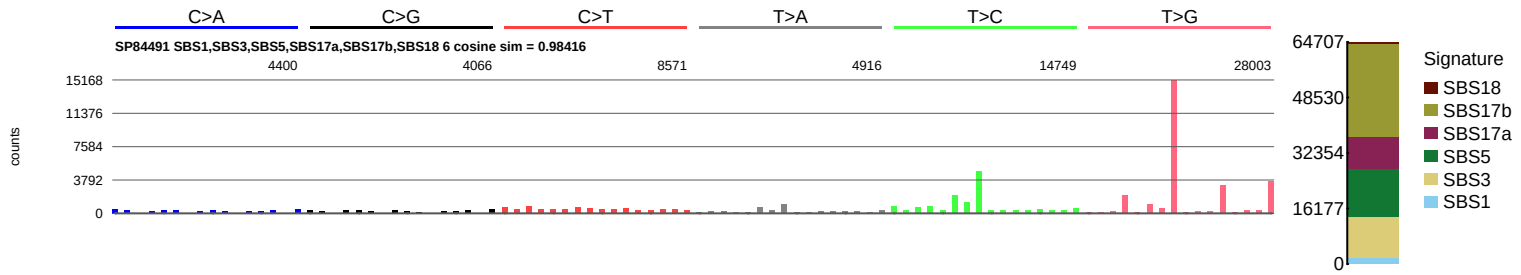

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98418
4392
4043
8660
4902
14715
27993
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

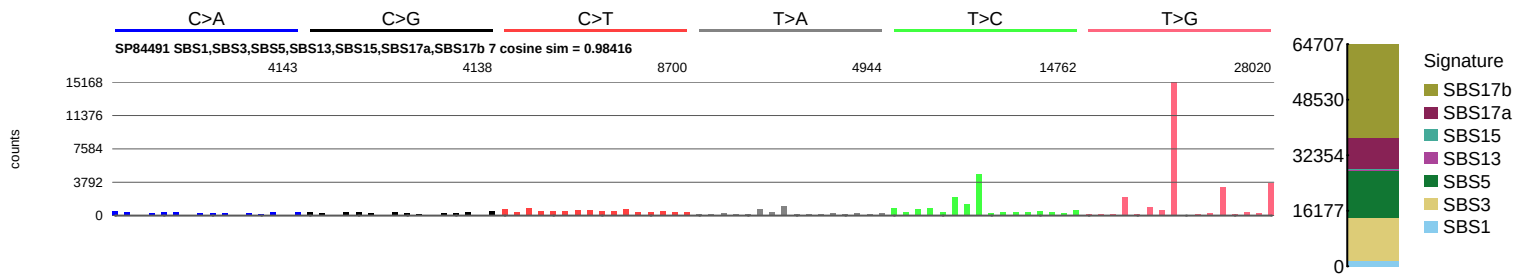

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98418
4389
4060
8658
4897
14710
27990
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

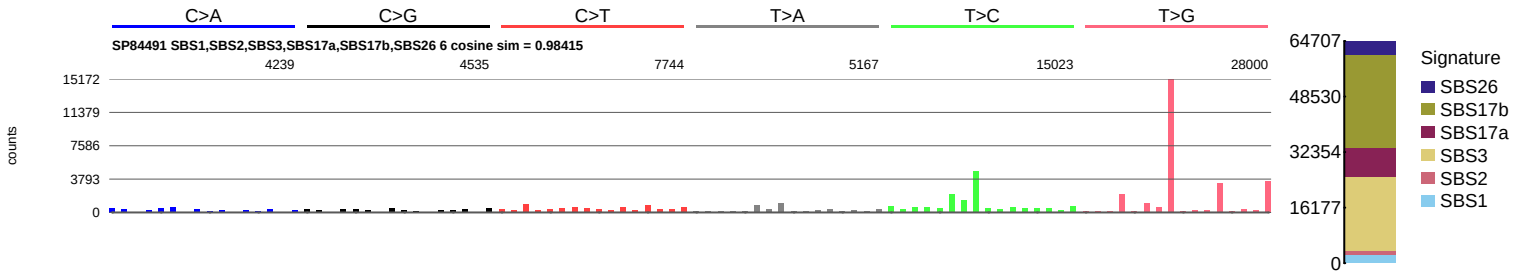

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98417
4413
4035
8388
4886
14998
27991
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

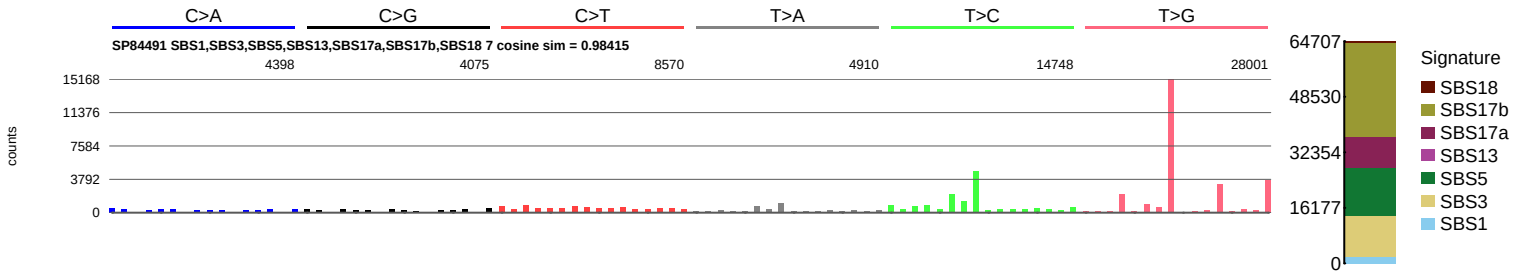

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98417
4120
4108
8539
4925
15003
28015
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

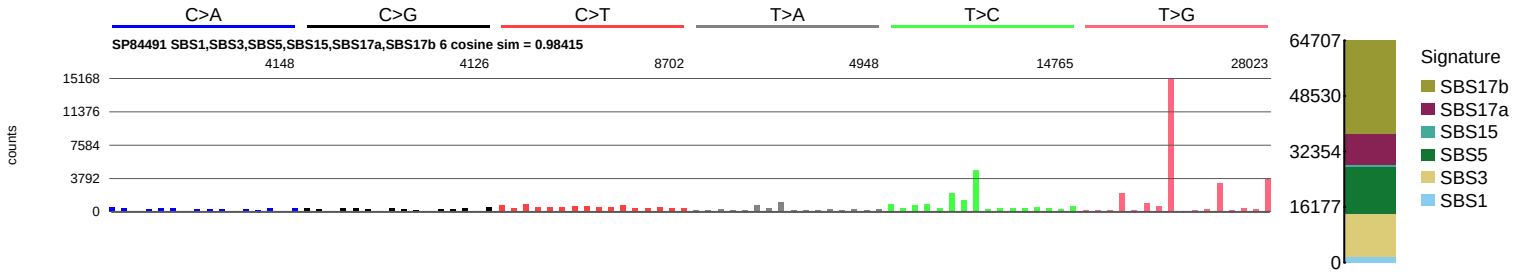

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98417
4408
4051
8385
4878
14997
27986
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

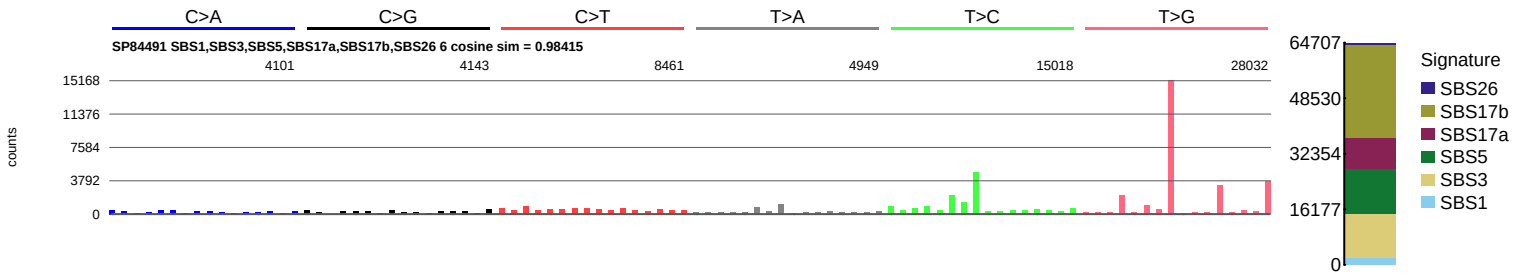

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98416
4400
4066
8571
4916
14749
28003
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

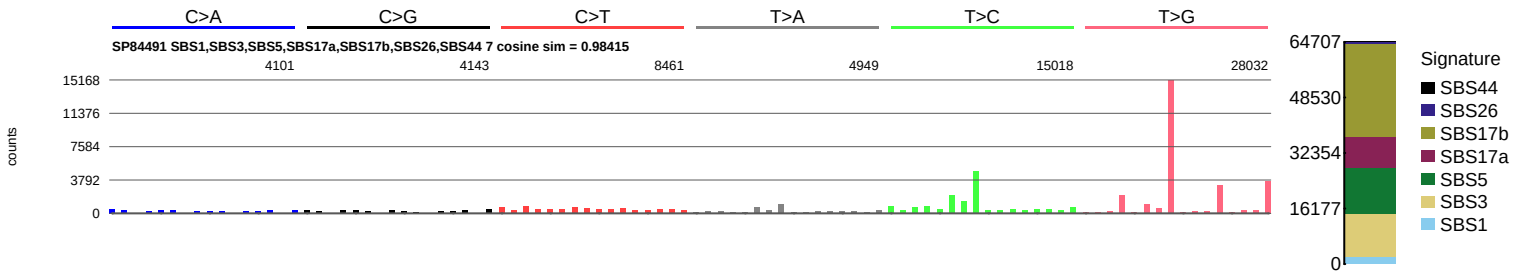

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98416
4143
4138
8700
4944
14762
28020
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

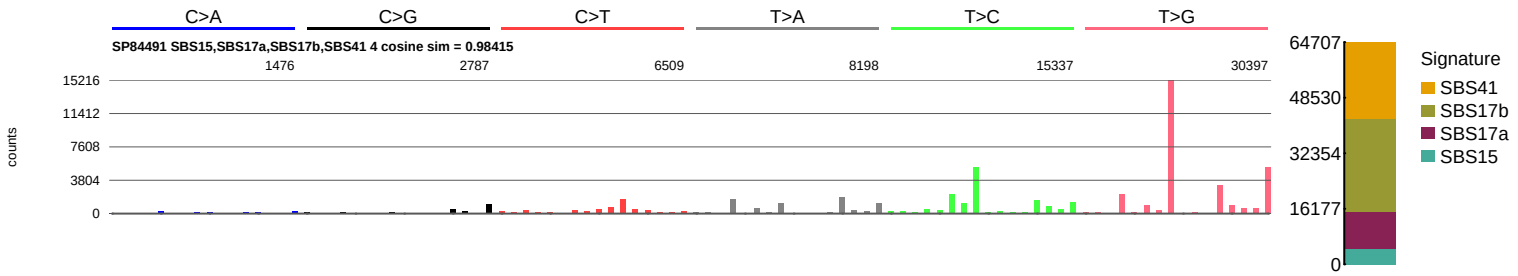

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98415
4239
4535
7744
5167
15023
28000
15172
11379
7586
3793
0
counts
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

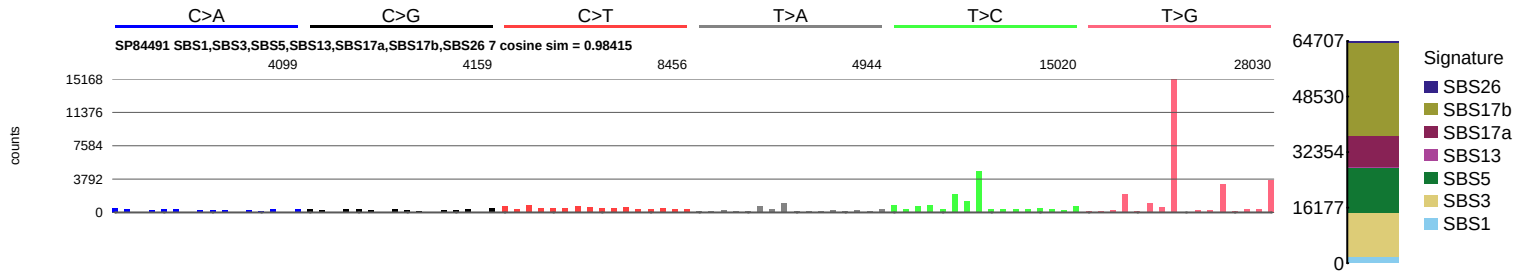

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98415
4398
4075
8570
4910
14748
28001
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

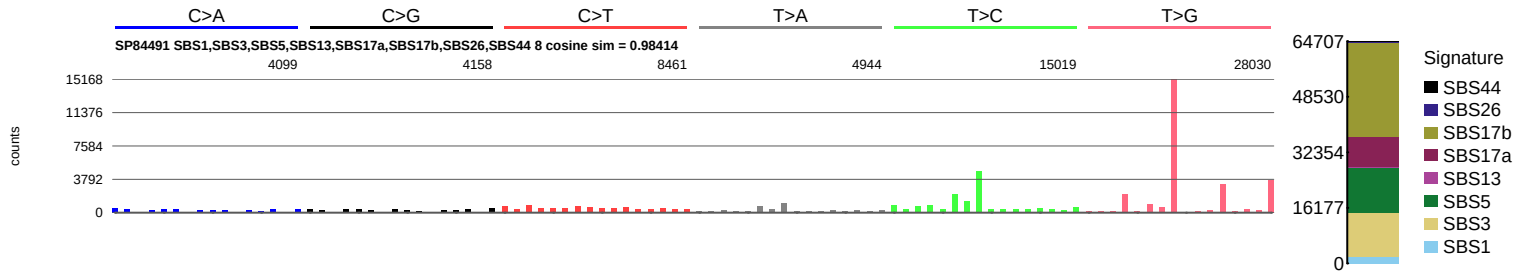

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98415
4148
4126
8702
4948
14765
28023
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

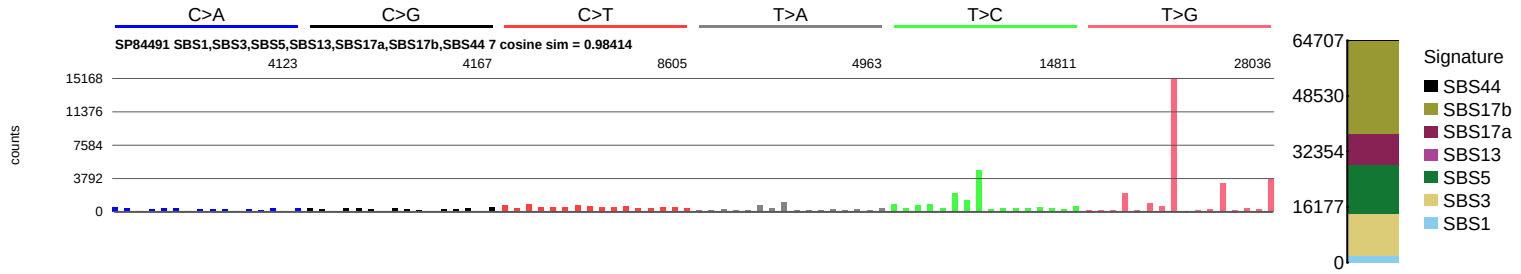

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98415
4101
4143
8461
4949
15018
28032
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS1

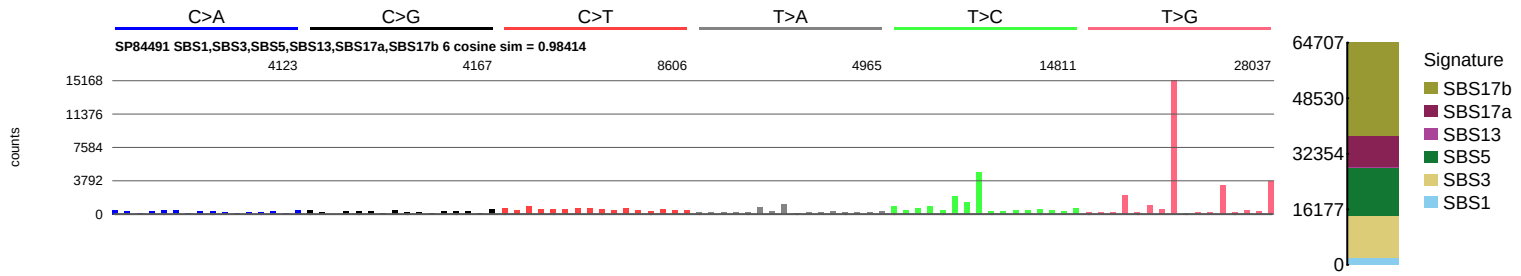

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98415
4101
4143
8461
4949
15018
28032
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS1

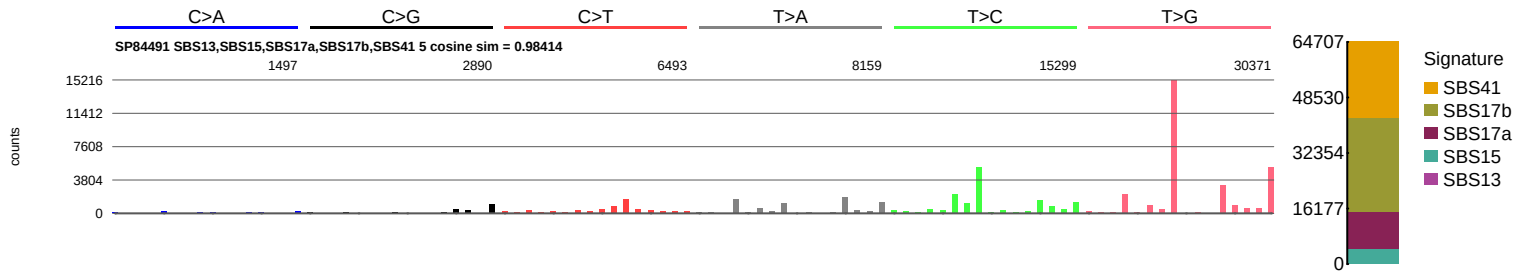

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS15,SBS17a,SBS17b,SBS41 4 cosine sim = 0.98415
1476
2787
6509
8198
15337
30397
15216
11412
7608
3804
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15

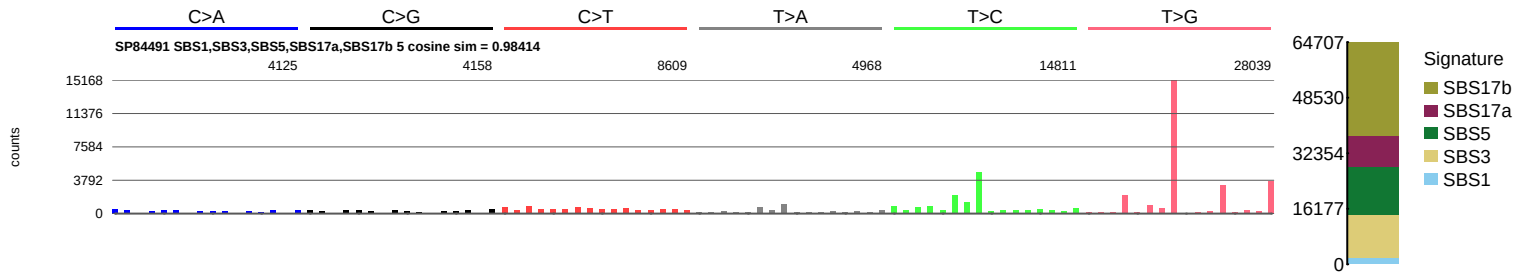

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98415
4099
4159
8456
4944
15020
28030
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

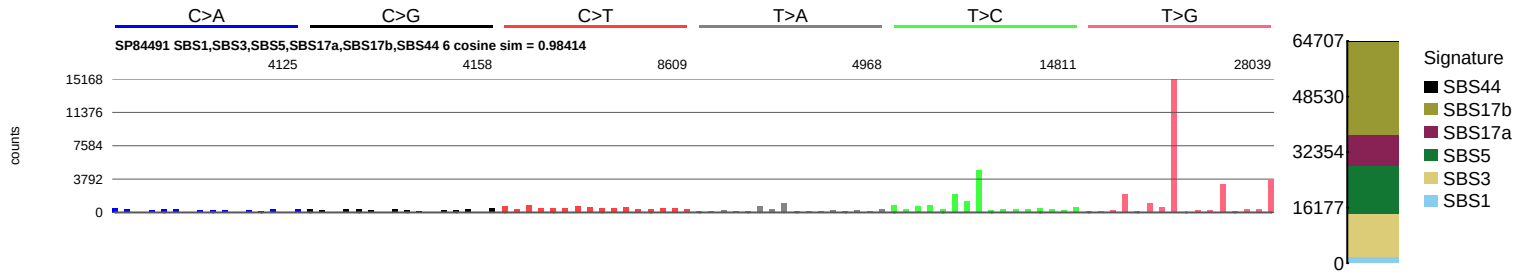

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98414
4099
4158
8461
4944
15019
28030
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98414
4123
4167
8605
4963
14811
28036
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98414
4123
4167
8606
4965
14811
28037
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98414
1497
2890
6493
8159
15299
30371
15216
11412
7608
3804
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b 5 cosine sim = 0.98414
4125
4158
8609
4968
14811
28039
15168
11376
7584
3792
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS5
SBS3
SBS1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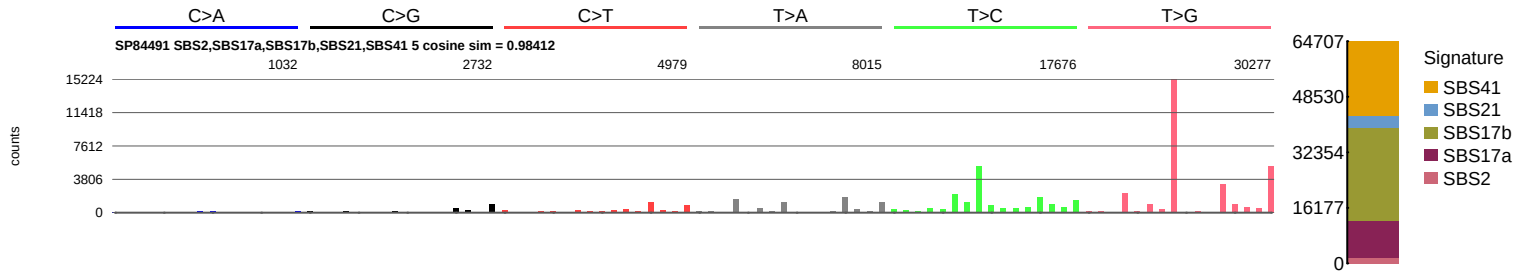
SP84491 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98414
4125
4158
8609
4968
14811
28039
counts
15168
11376
7584
3792
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
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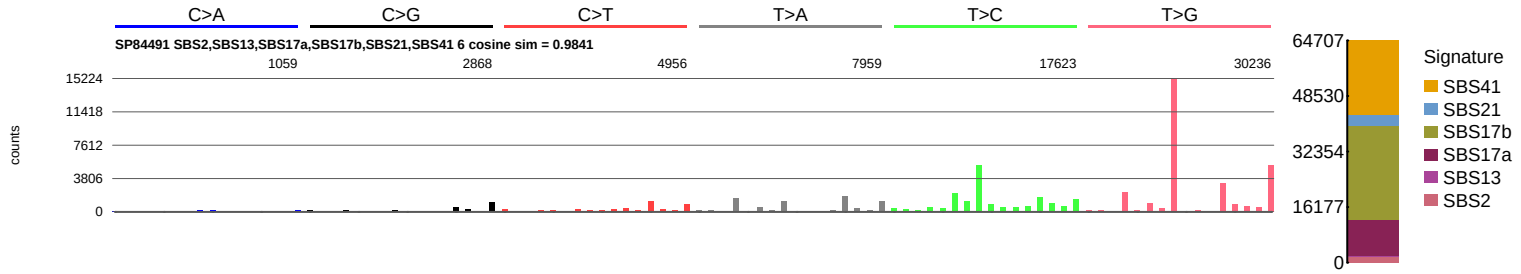
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98412
1032
2732
4979
8015
17676
30277
counts
15224
11418
7612
3806
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS2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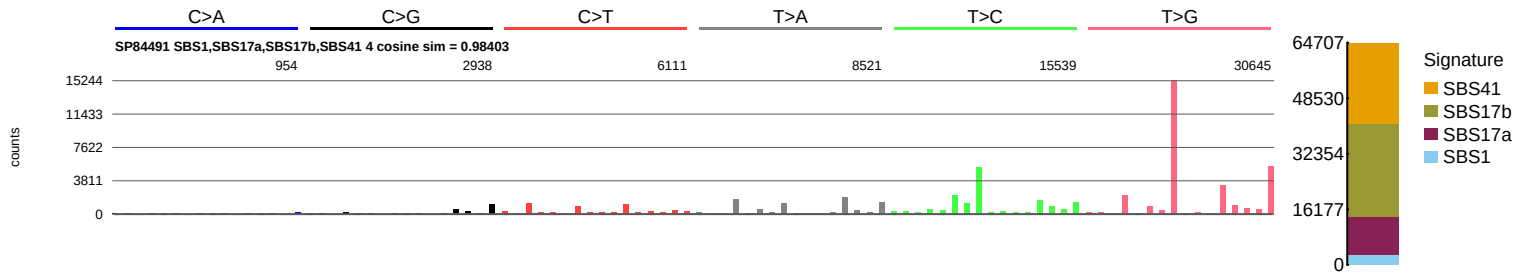
SP84491 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.9841
1059
2868
4956
7959
17623
30236
counts
15224
11418
7612
3806
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS2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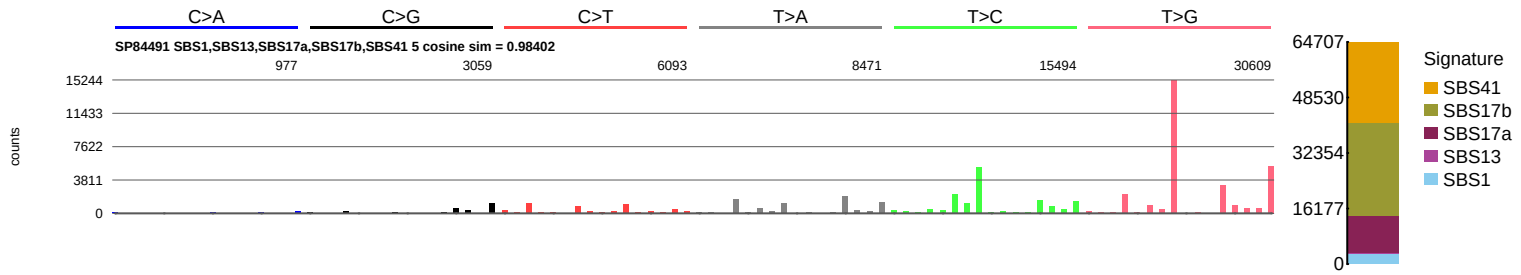
SP84491 SBS1,SBS17a,SBS17b,SBS41 4 cosine sim = 0.98403
954
2938
6111
8521
15539
30645
counts
15244
11433
7622
3811
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
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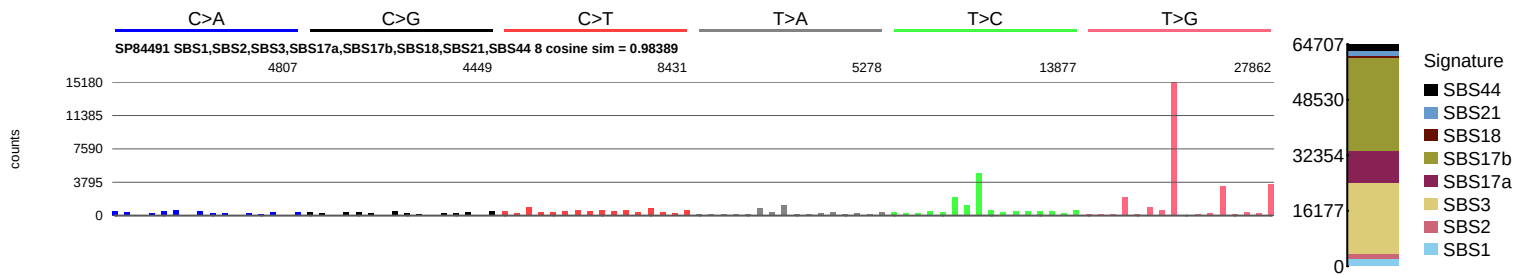
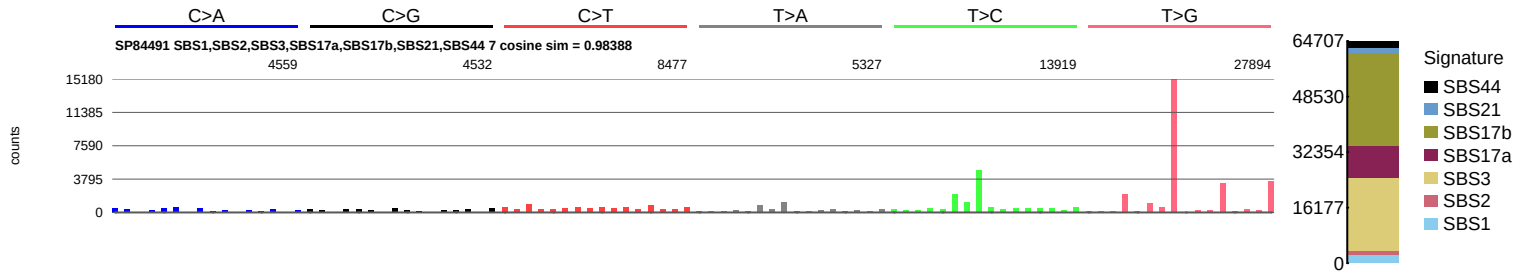
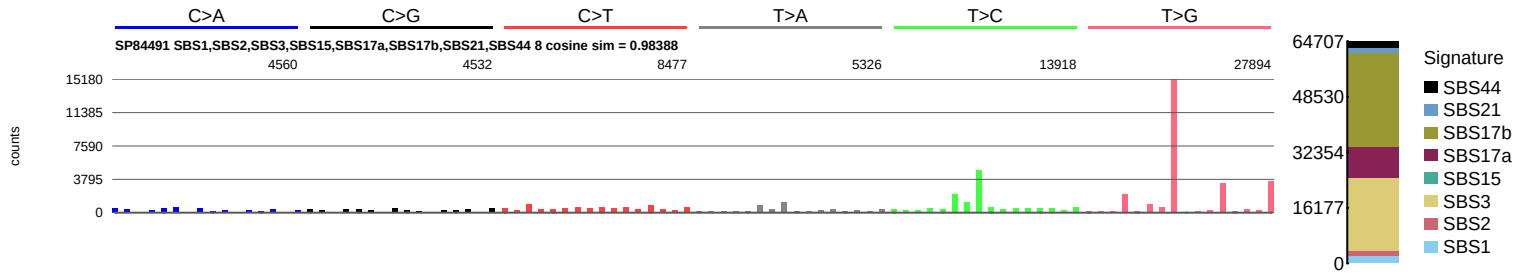
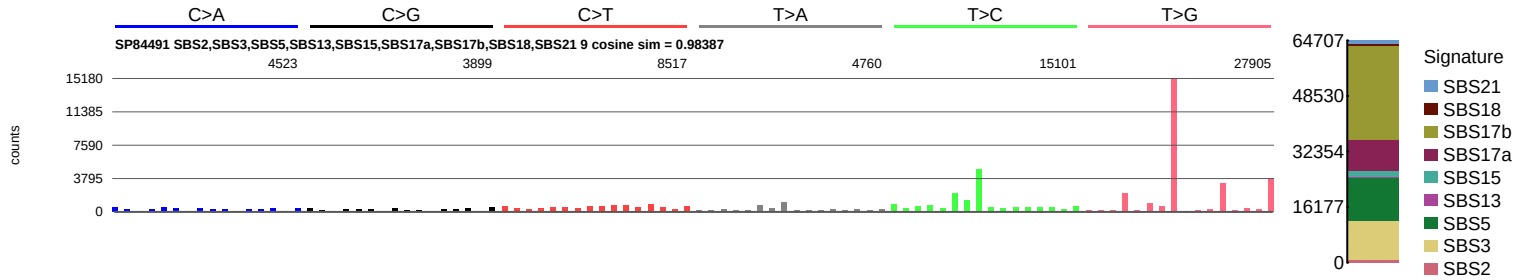
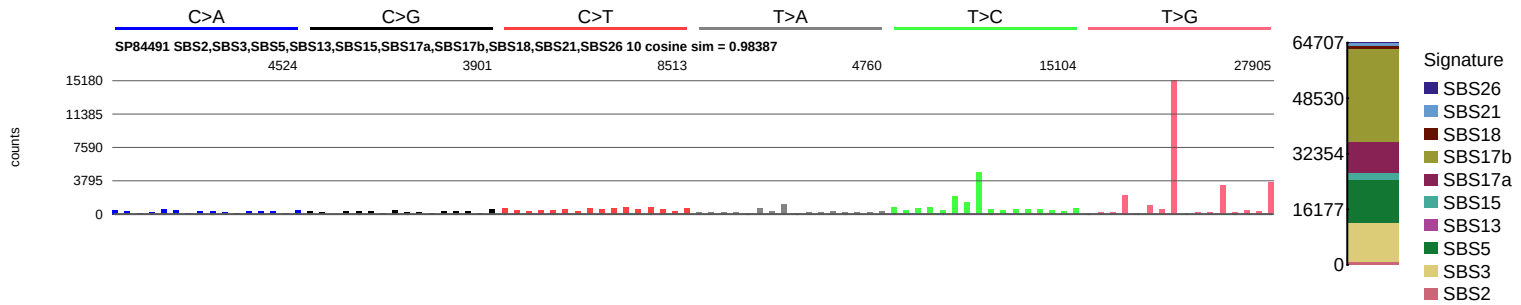
T>G
SP84491 SBS1,SBS13,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98402
977
3059
6093
8471
15494
30609
counts
15244
11433
7622
3811
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS1

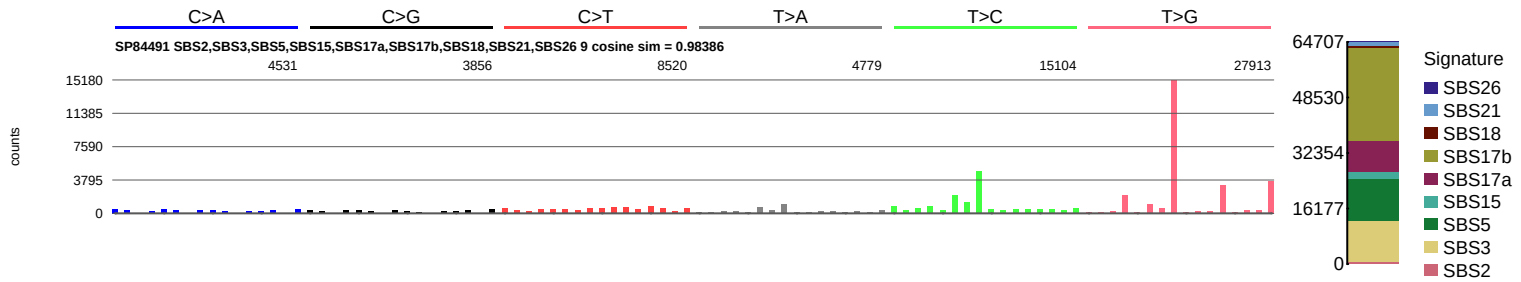

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98389
4807
4449
8431
5278
13877
27862
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

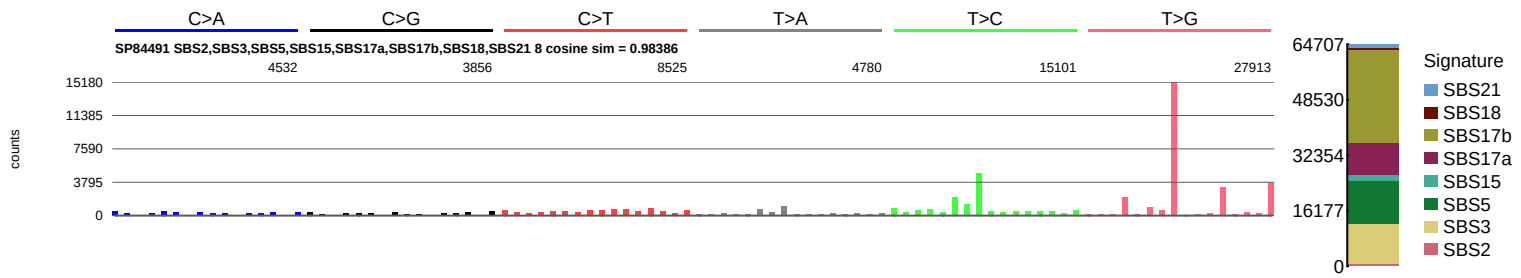

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98388
4559
4532
8477
5327
13919
27894
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

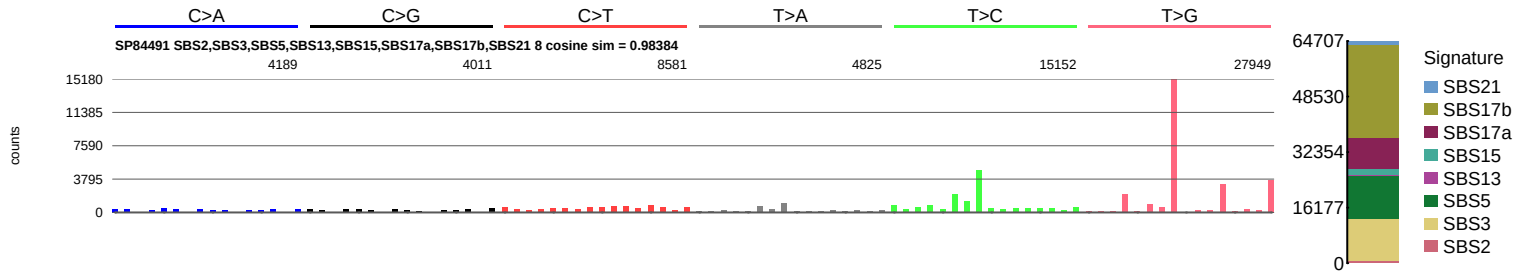

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98388
4560
4532
8477
5326
13918
27894
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

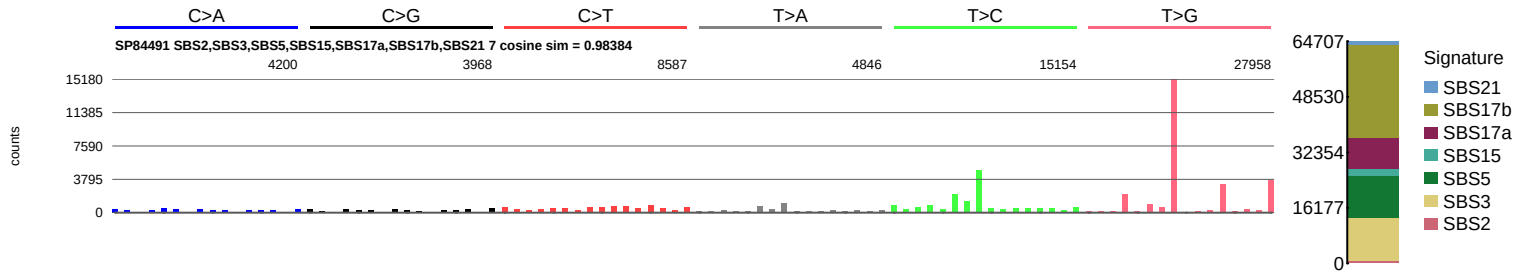

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98387
4523
3899
8517
4760
15101
27905
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

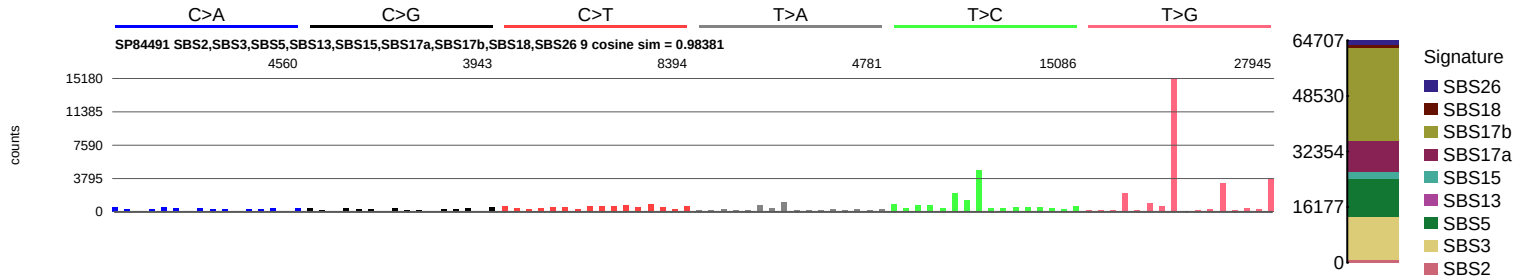

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98387
4524
3901
8513
4760
15104
27905
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

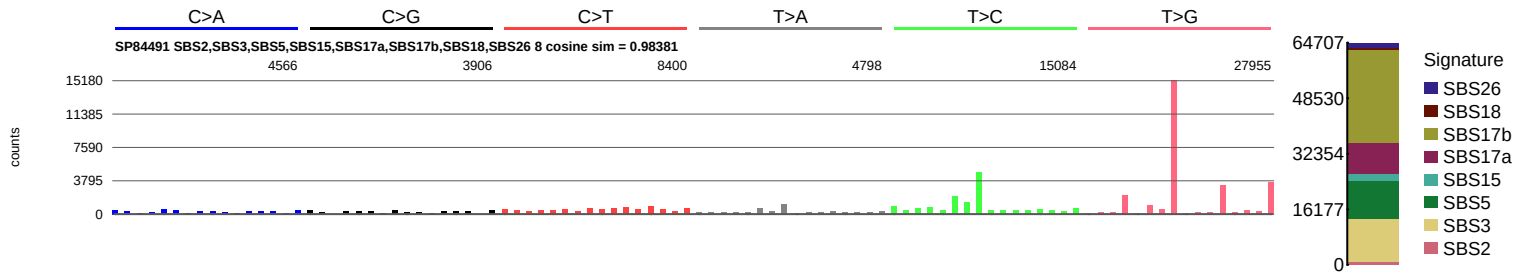

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98386
4531
3856
8520
4779
15104
27913
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

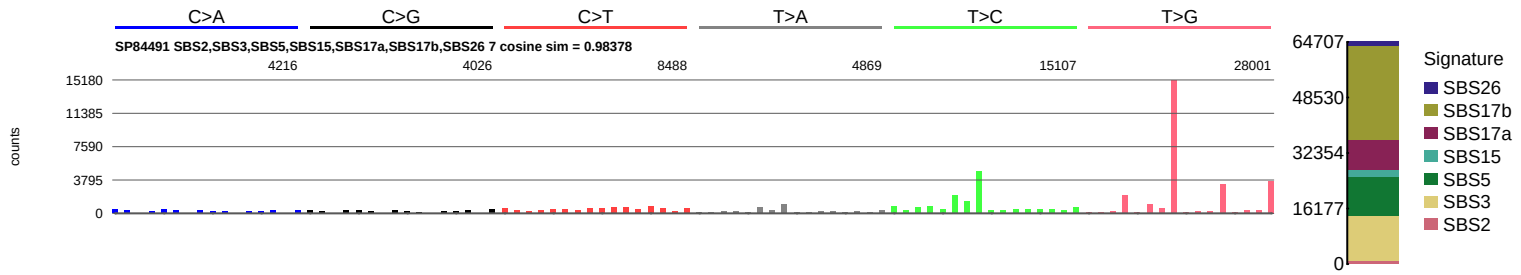

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98386
4532
3856
8525
4780
15101
27913
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

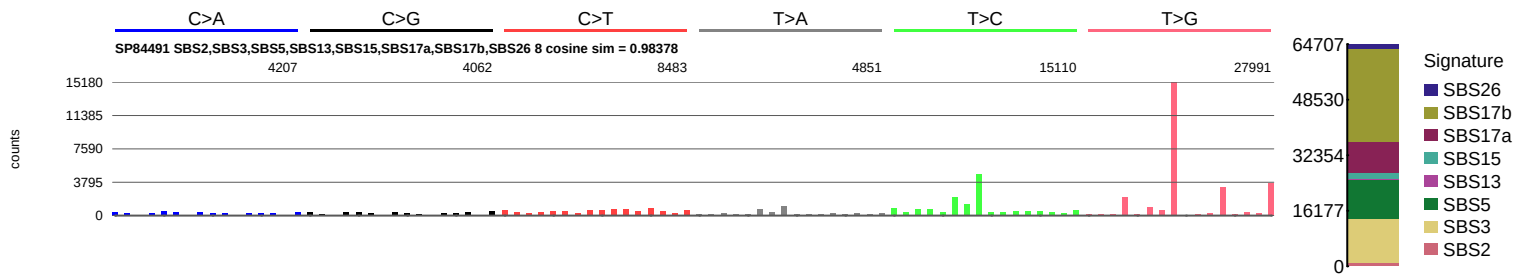

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98384
4189
4011
8581
4825
15152
27949
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98384
4200
3968
8587
4846
15154
27958
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98381
4560
3943
8394
4781
15086
27945
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98381
4566
3906
8400
4798
15084
27955
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98378
4216
4026
8488
4869
15107
28001
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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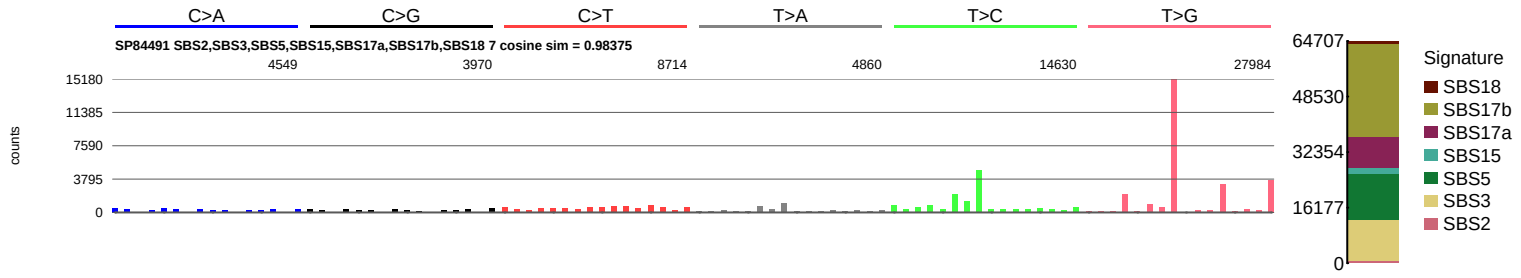
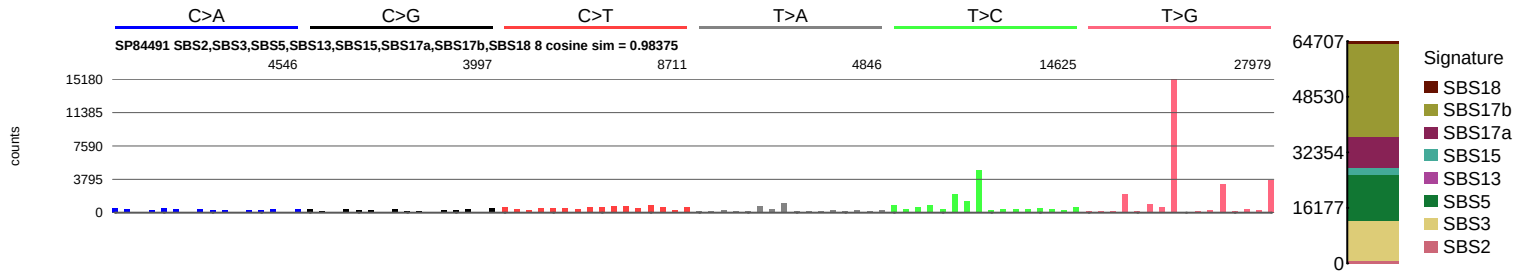
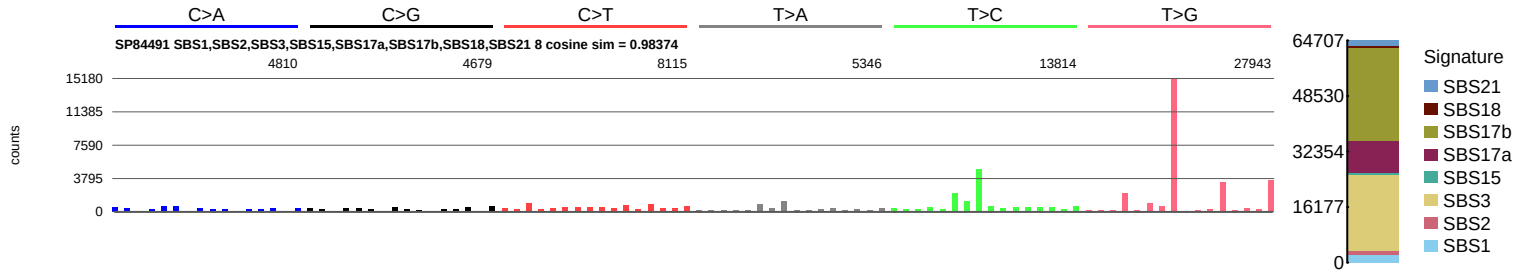
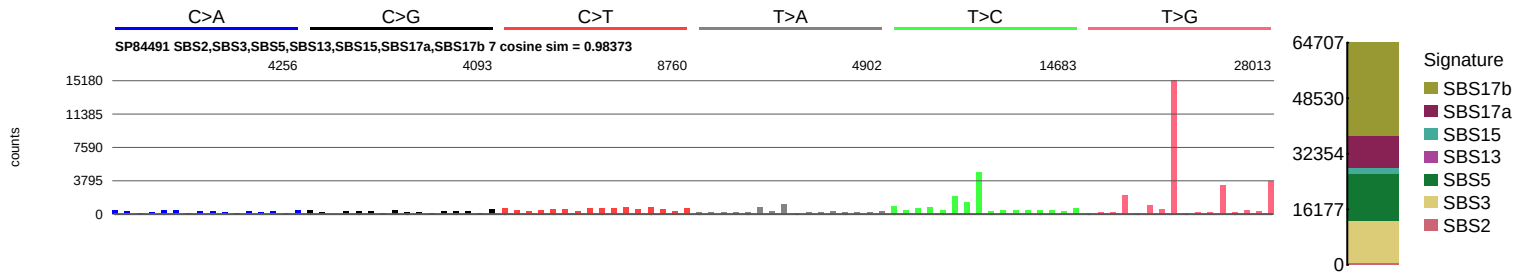
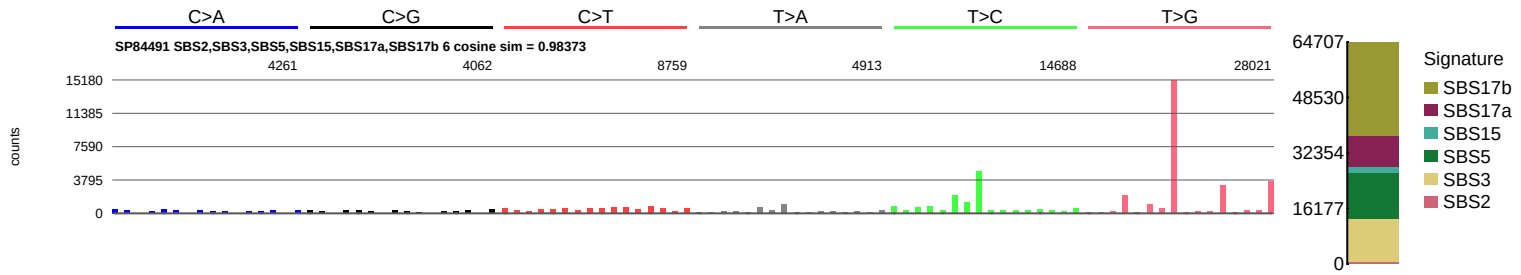
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98378
4207
4062
8483
4851
15110
27991
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

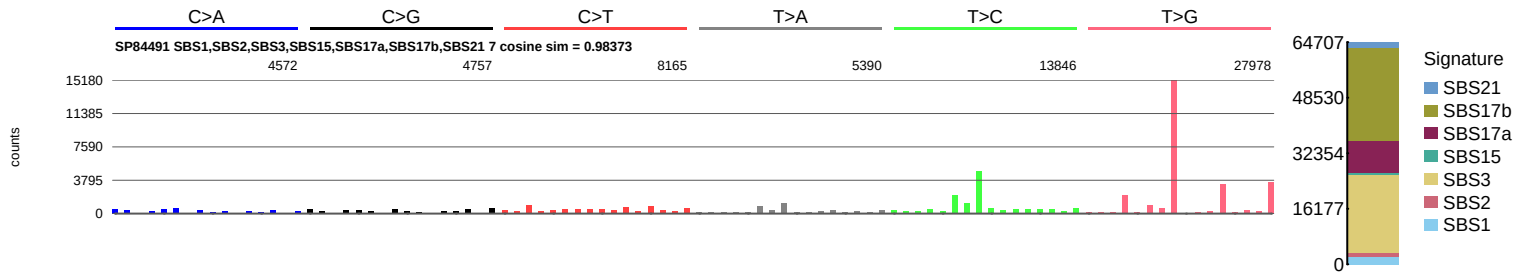

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98375
4549
3970
8714
4860
14630
27984
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

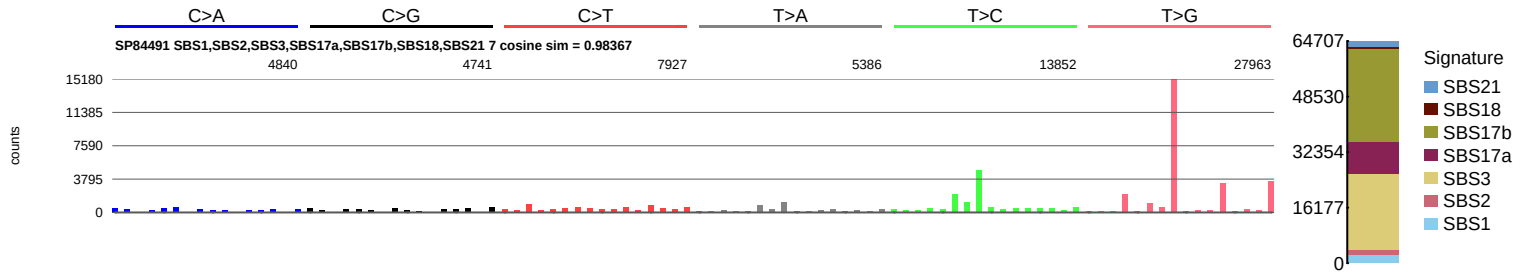

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98375
4546
3997
8711
4846
14625
27979
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98374
4810
4679
8115
5346
13814
27943
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98373
4256
4093
8760
4902
14683
28013
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98373
4261
4062
8759
4913
14688
28021
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98373
4572
4757
8165
5390
13846
27978
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98367
4840
4741
7927
5386
13852
27963
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
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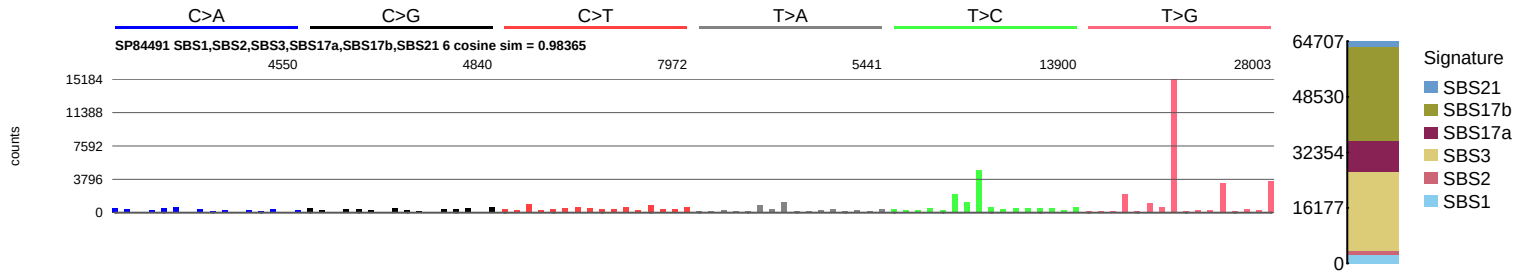
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
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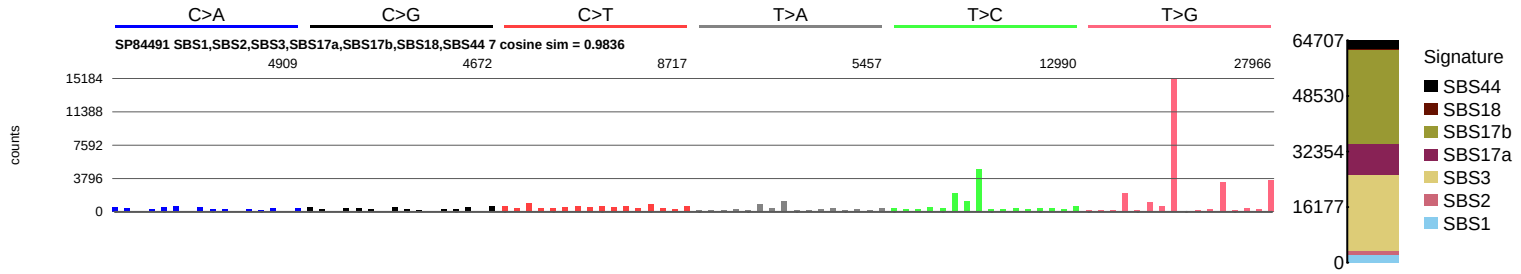
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98365
4550
4840
7972
5441
13900
28003
counts
15184
11388
7592
3796
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.9836
4909
4672
8717
5457
12990
27966
counts
15184
11388
7592
3796
0
64707
48530
32354
16177
0
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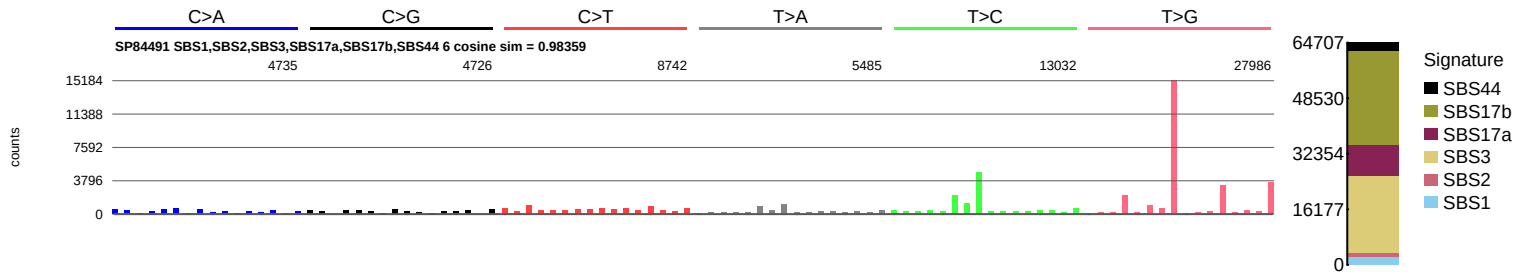
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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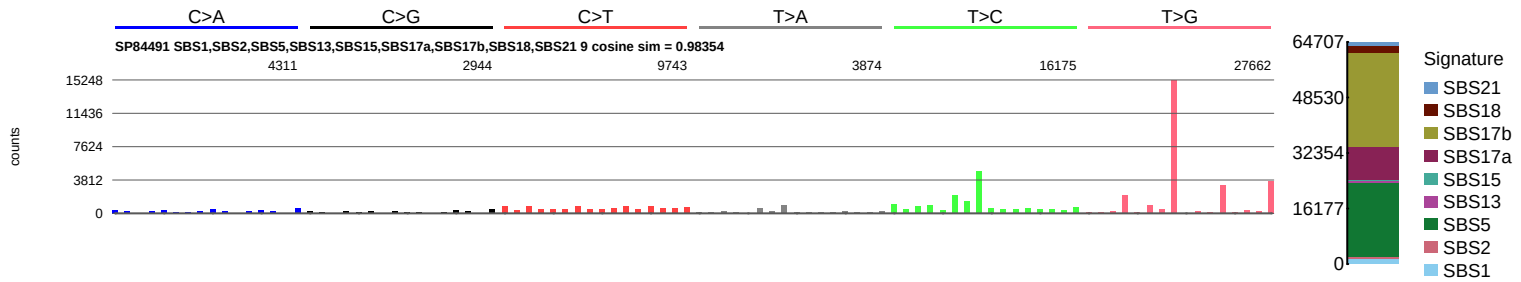
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98359
4735
4726
8742
5485
13032
27986
counts
15184
11388
7592
3796
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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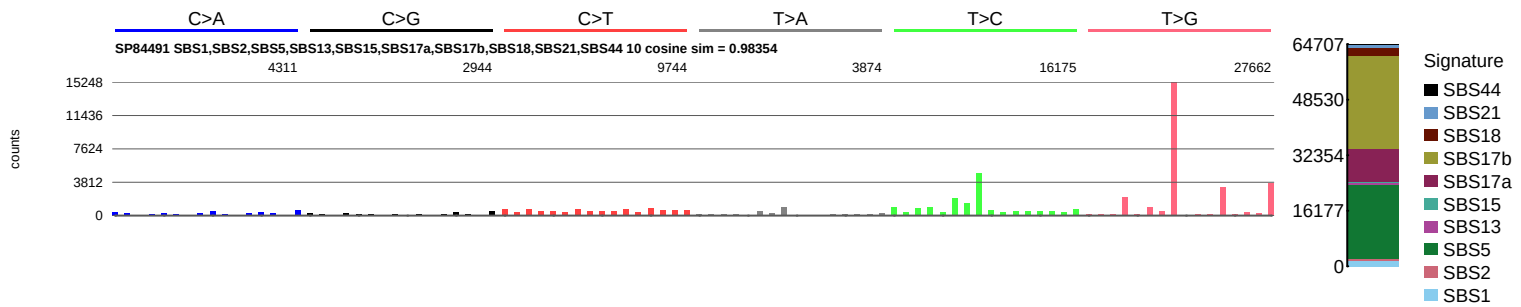
SP84491 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98354
4311
2944
9743
3874
16175
27662
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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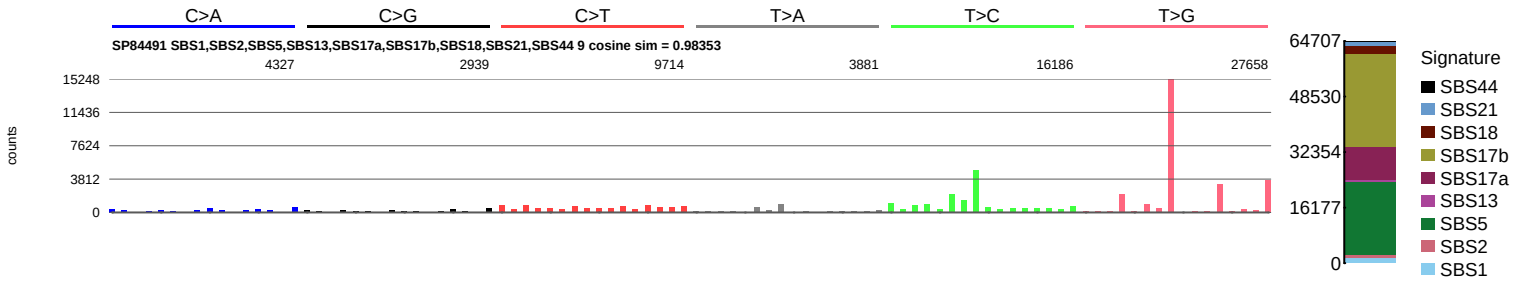
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98354
4311
2944
9744
3874
16175
27662
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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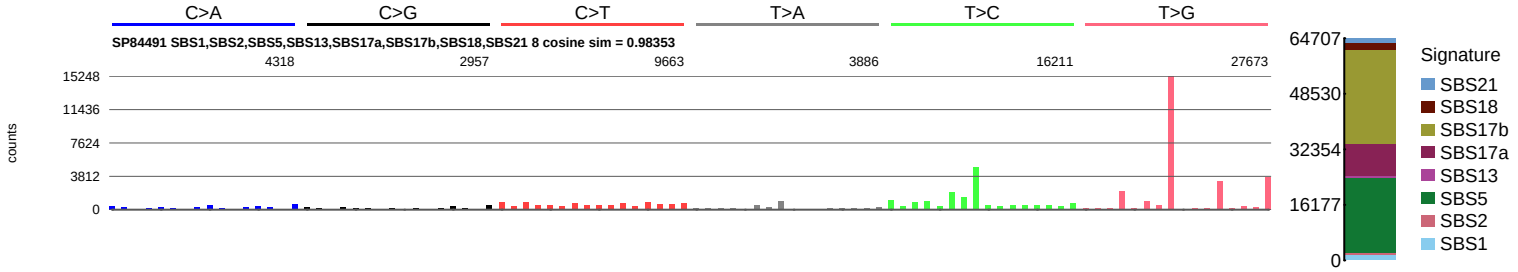
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98353
4327
2939
9714
3881
16186
27658
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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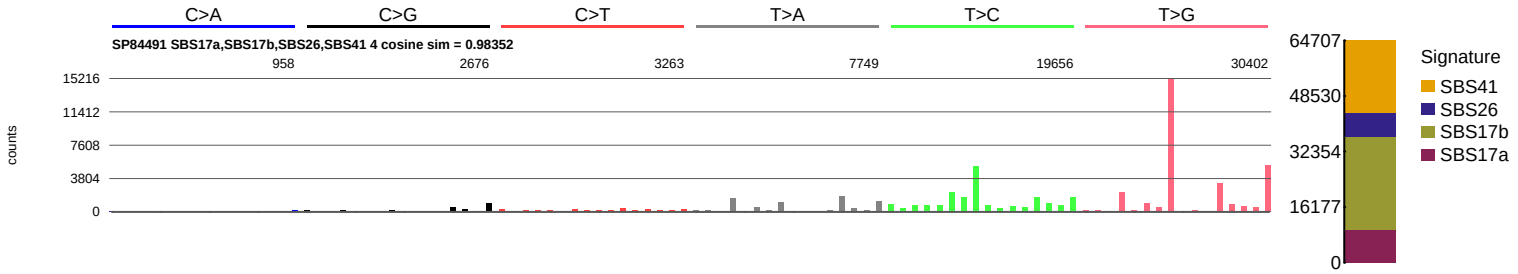
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98353
4318
2957
9663
3886
16211
27673
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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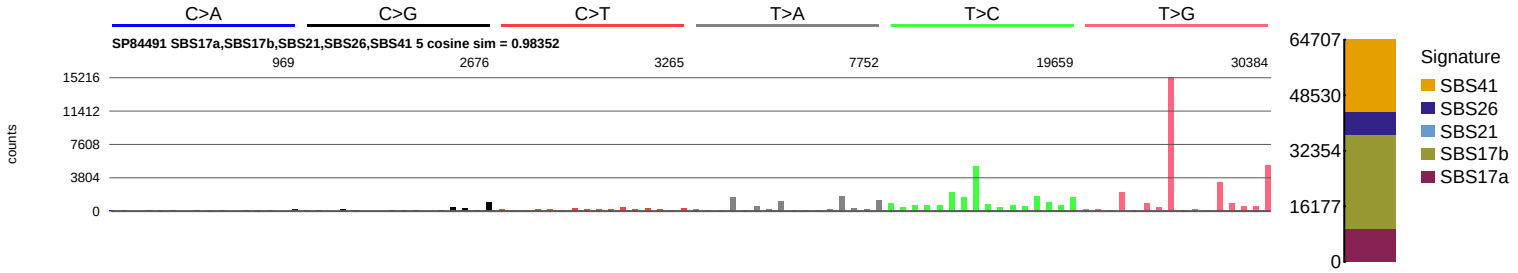
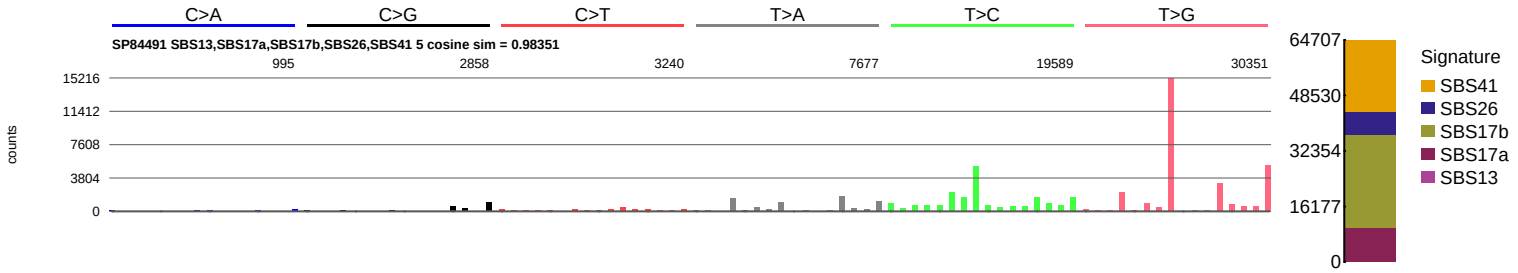
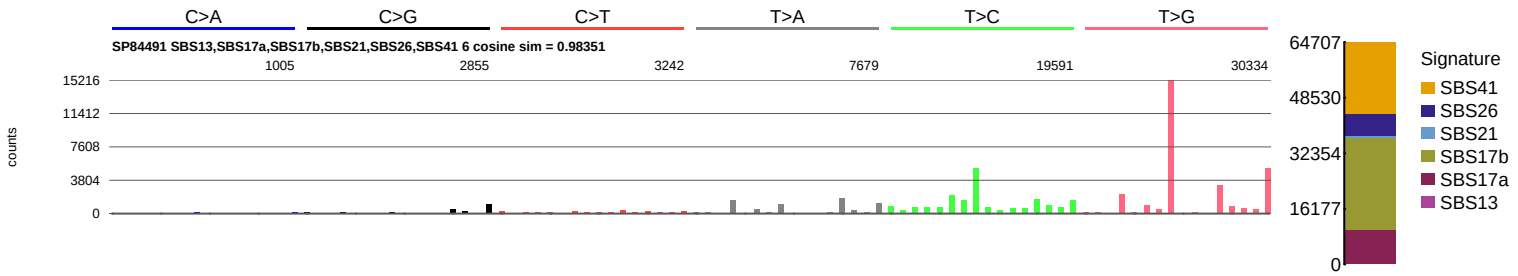
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS26,SBS41 4 cosine sim = 0.98352
958
2676
3263
7749
19656
30402
counts
15216
11412
7608
3804
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS26,SBS41 5 cosine sim = 0.98352
969
2676
3265
7752
19659
30384
counts
15216
11412
7608
3804
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS17a,SBS17b,SBS26,SBS41 5 cosine sim = 0.98351
995
2858
3240
7677
19589
30351
counts
15216
11412
7608
3804
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13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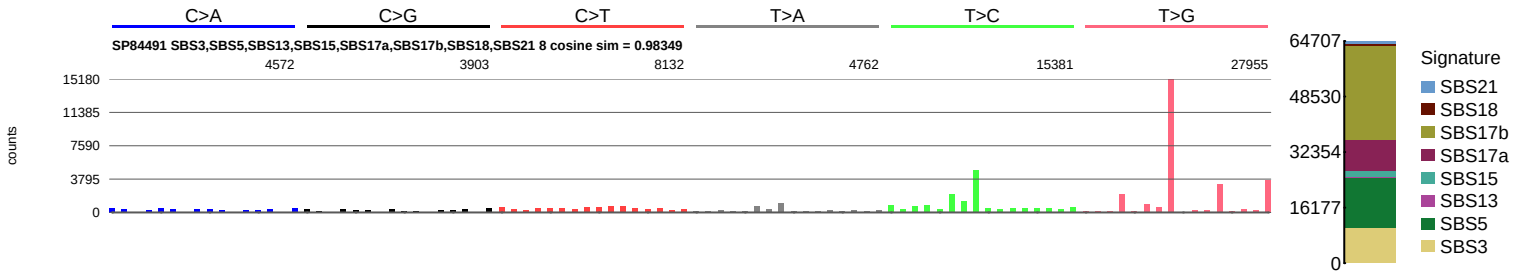
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 6 cosine sim = 0.98351
1005
2855
3242
7679
19591
30334
counts
15216
11412
7608
3804
0
64707
48530
32354
16177
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13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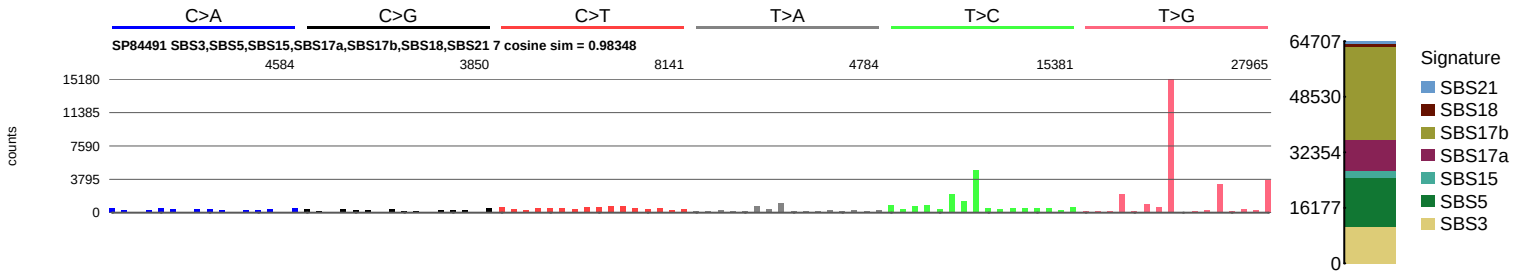
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98349
4572
3903
8132
4762
15381
27955
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3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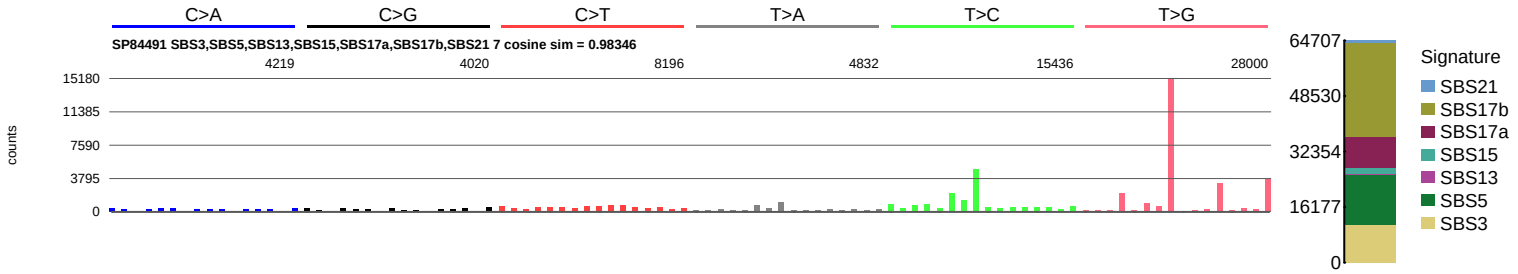
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98348
4584
3850
8141
4784
15381
27965
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3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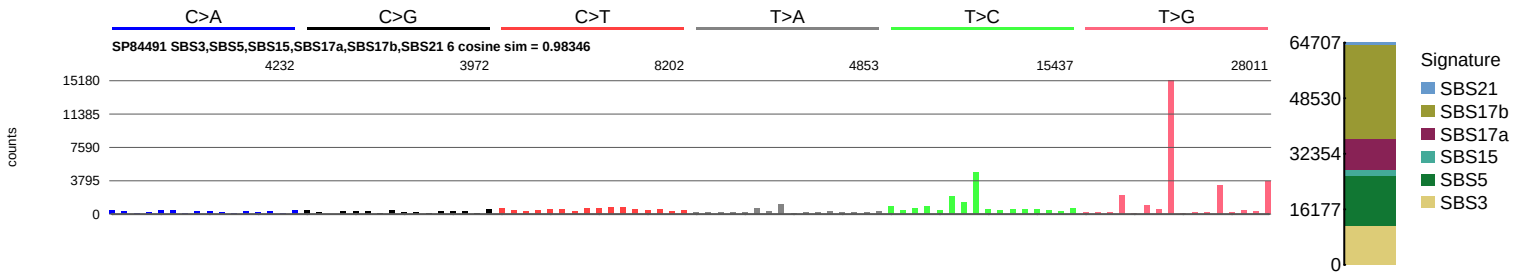
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98346
4219
4020
8196
4832
15436
28000
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3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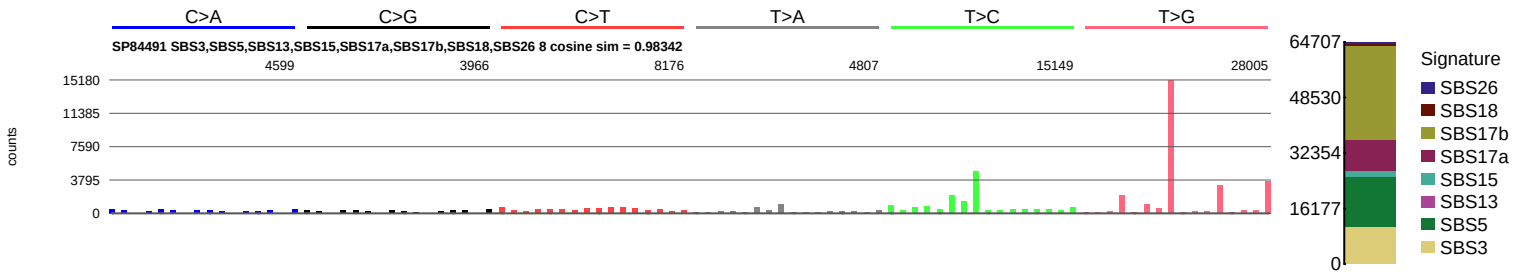
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98346
4232
3972
8202
4853
15437
28011
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3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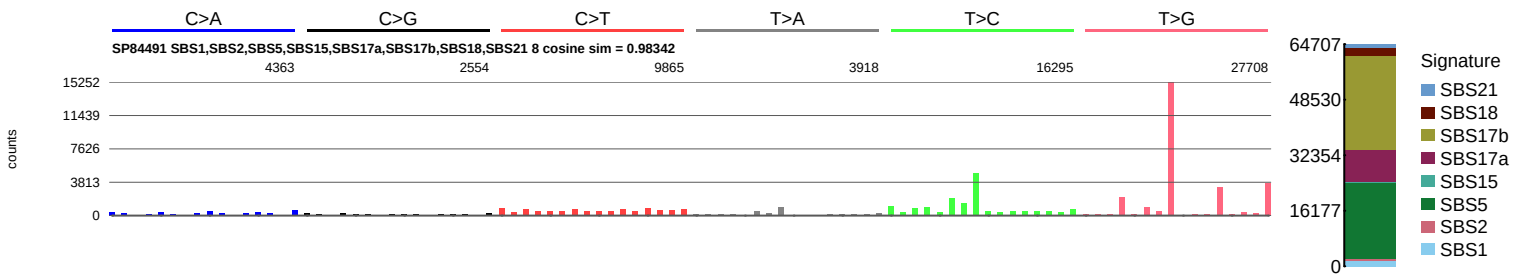
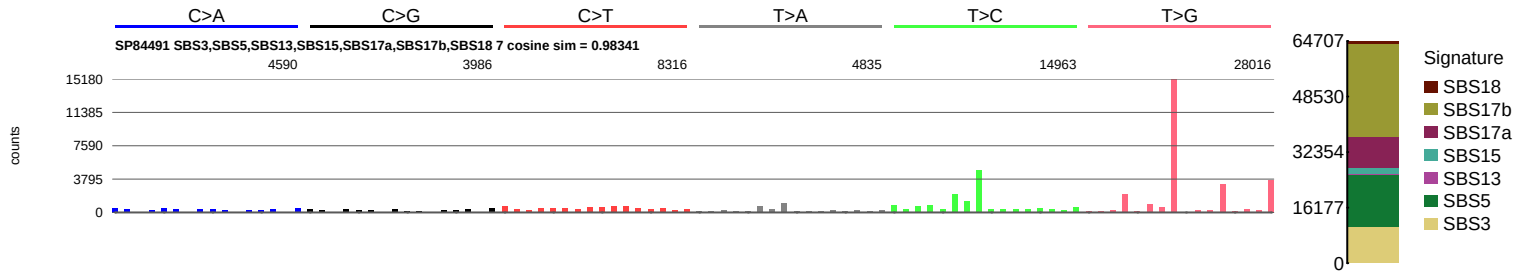
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98342
4599
3966
8176
4807
15149
28005
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98342
4363
2554
9865
3918
16295
27708
counts
15252
11439
7626
3813
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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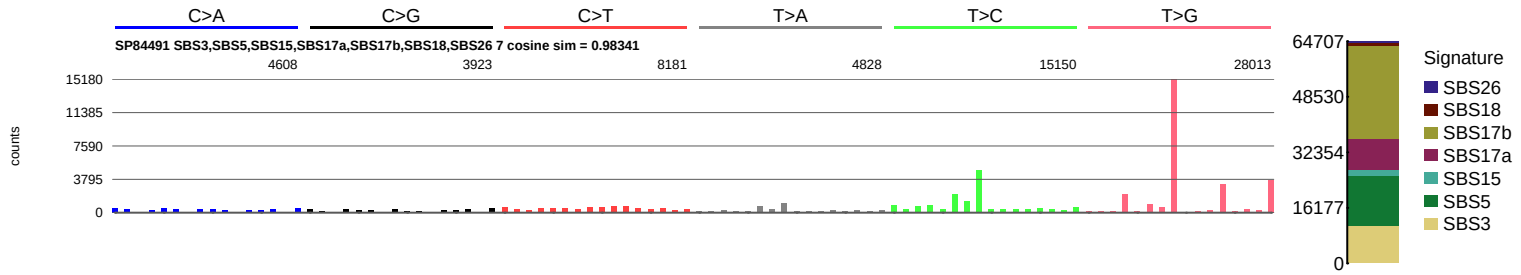
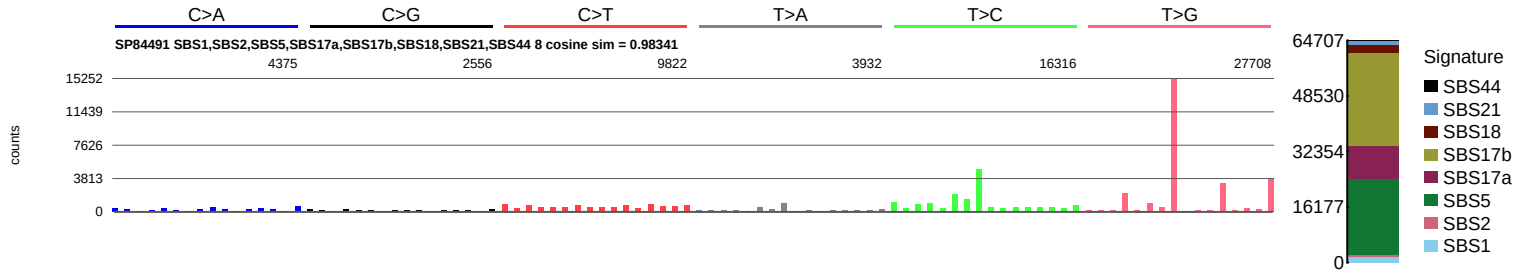
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98341
4590
3986
8316
4835
14963
28016
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98341
4608
3923
8181
4828
15150
28013
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3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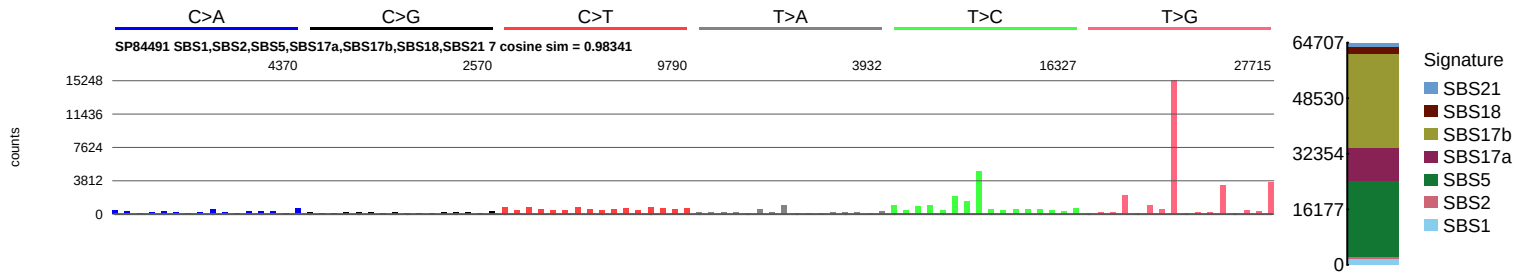
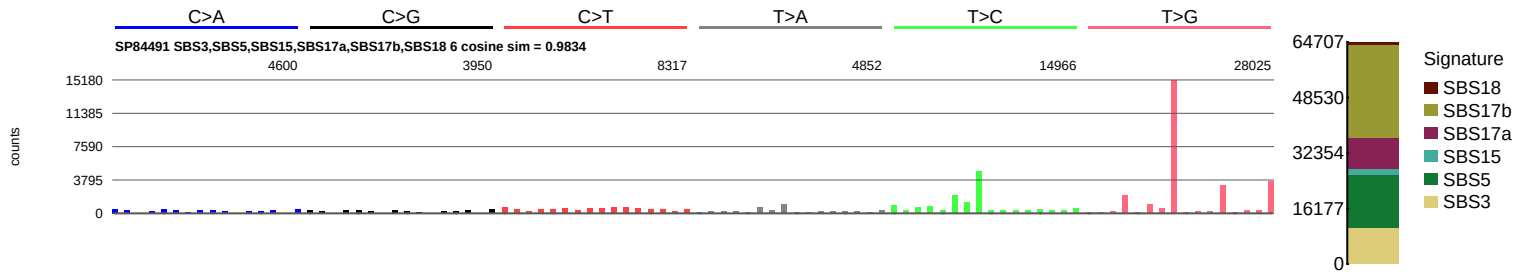
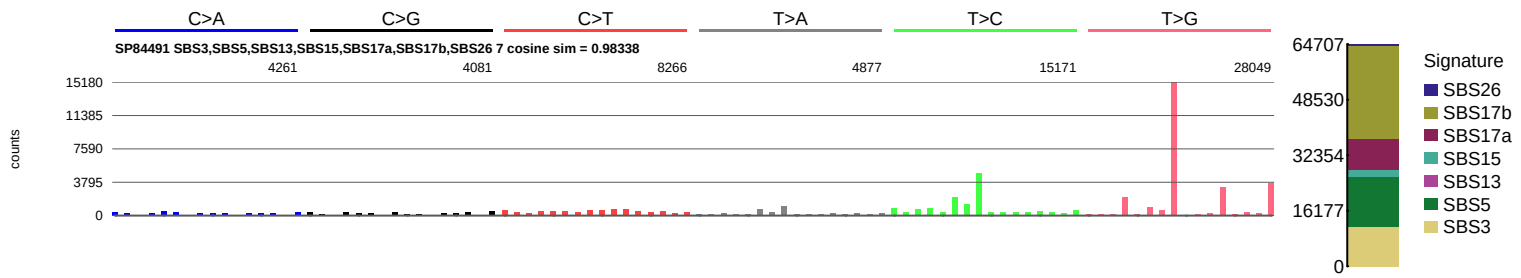
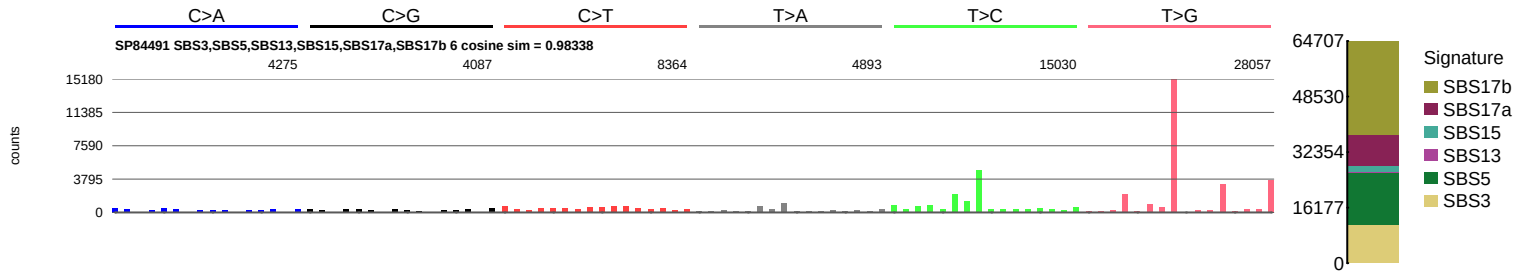
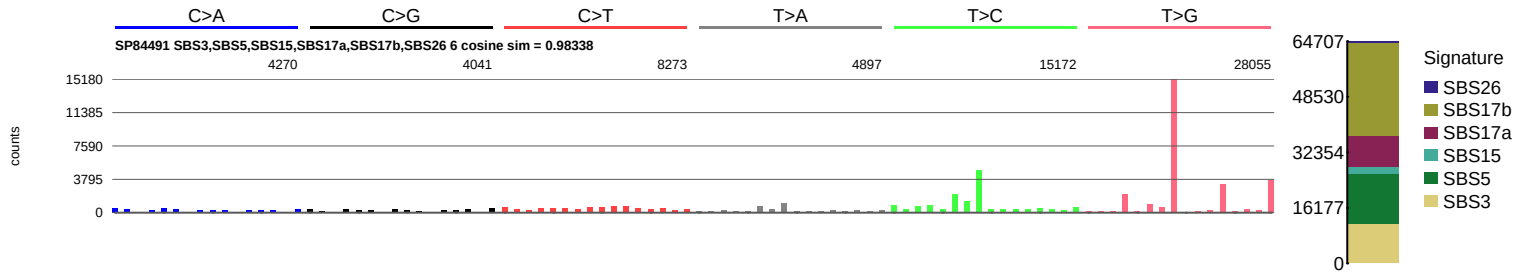
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98341
4375
2556
9822
3932
16316
27708
counts
15252
11439
7626
3813
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98341
4370
2570
9790
3932
16327
27715
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.9834
4600
3950
8317
4852
14966
28025
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98338
4261
4081
8266
4877
15171
28049
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98338
4275
4087
8364
4893
15030
28057
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
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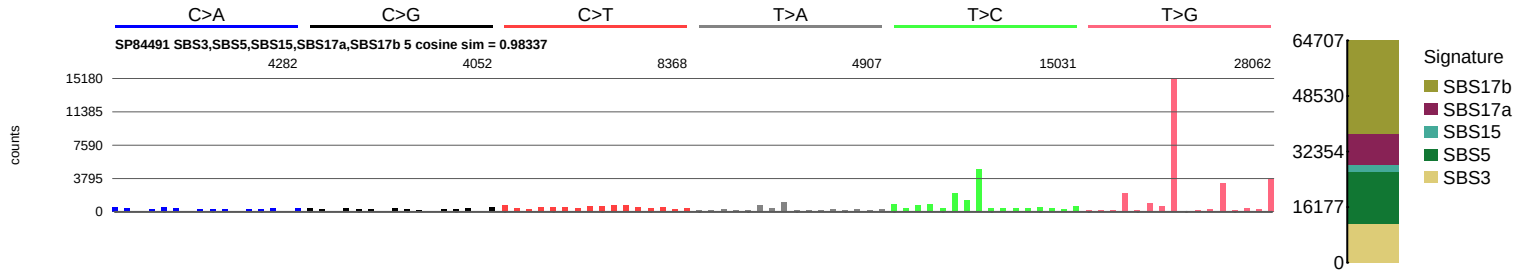
T>C
T>G
SP84491 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98338
4270
4041
8273
4897
15172
28055
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3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C>G
C>T
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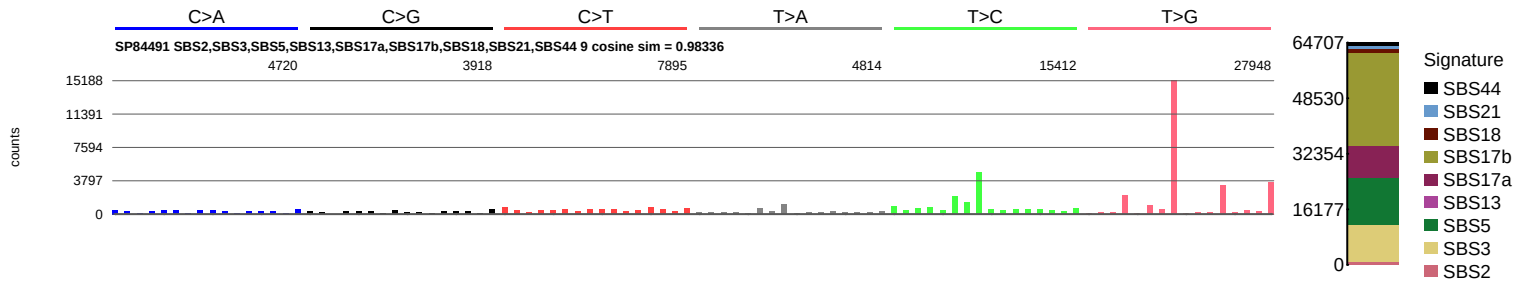
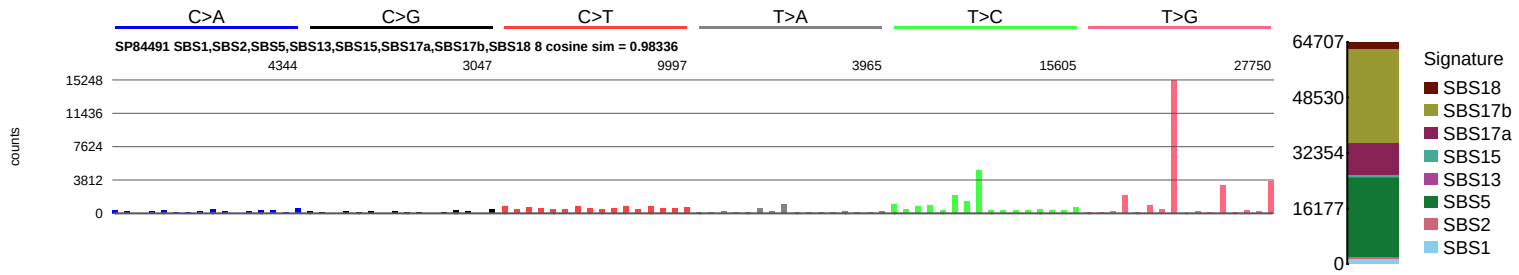
T>G
SP84491 SBS3,SBS5,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98337
4282
4052
8368
4907
15031
28062
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98336
4720
3918
7895
4814
15412
27948
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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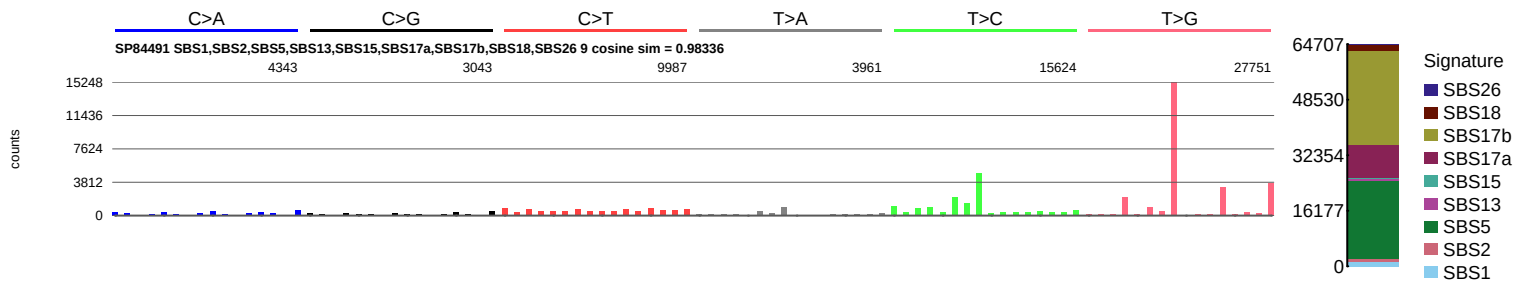
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98336
4344
3047
9997
3965
15605
27750
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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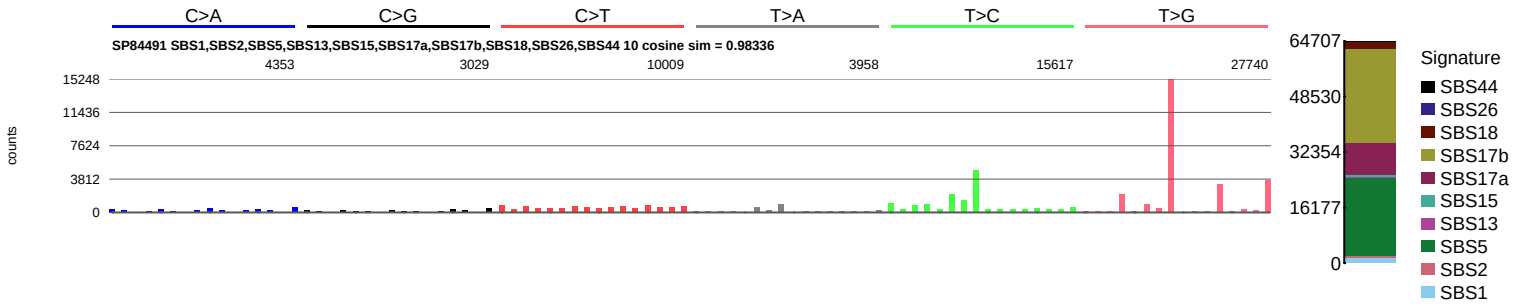
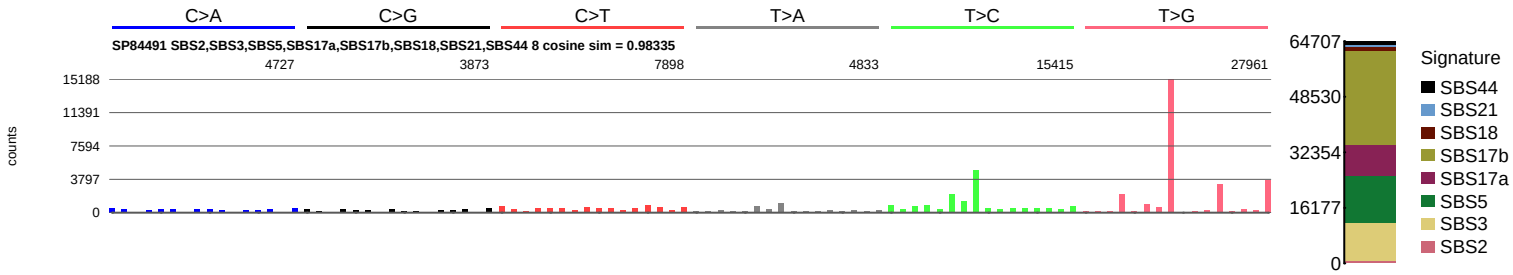
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98336
4343
3043
9987
3961
15624
27751
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98336
4353
3029
10009
3958
15617
27740
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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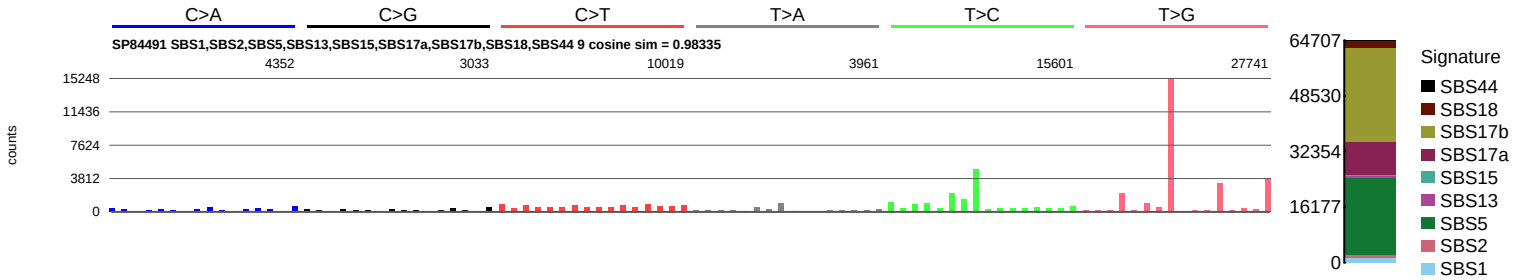
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98335
4727
3873
7898
4833
15415
27961
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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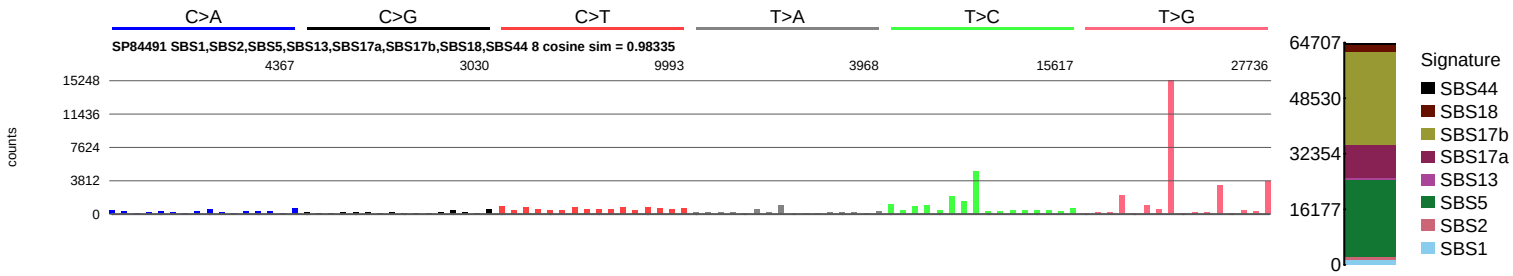
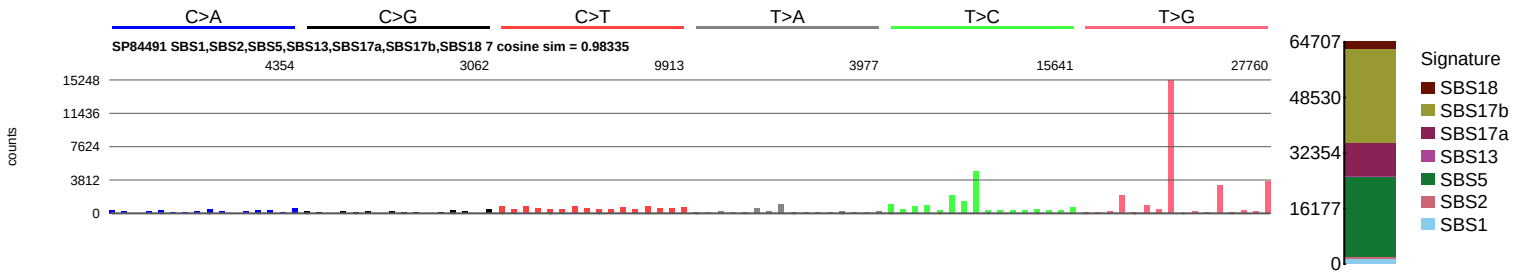
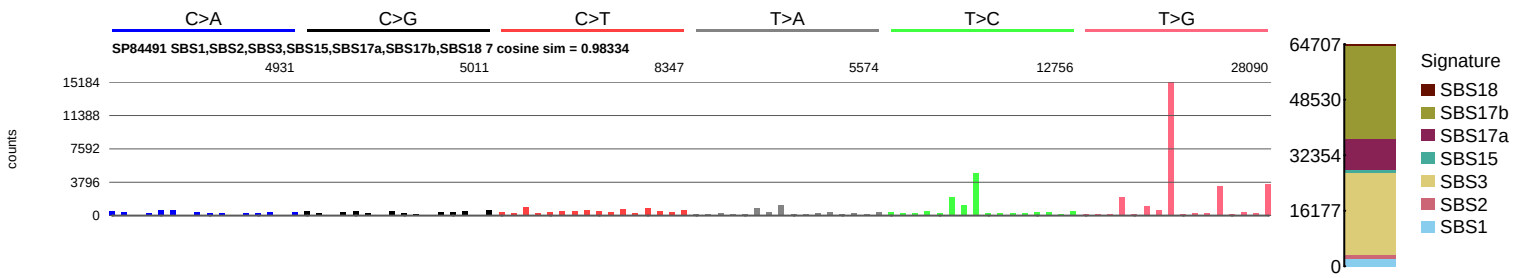
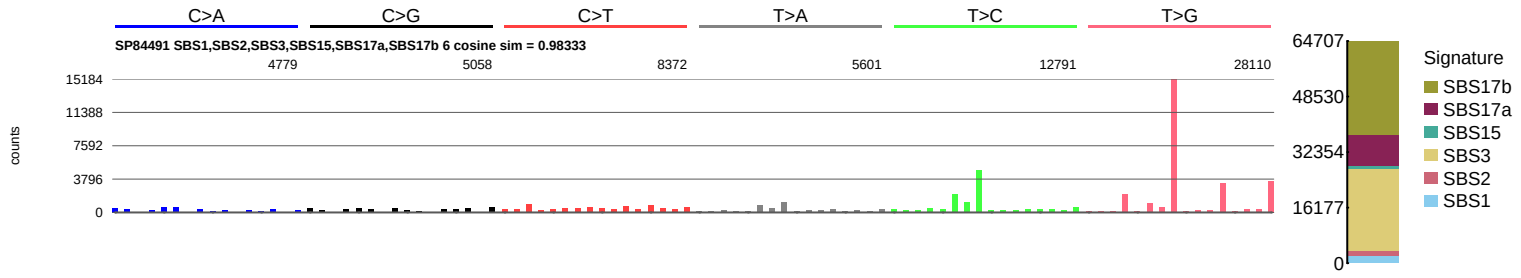
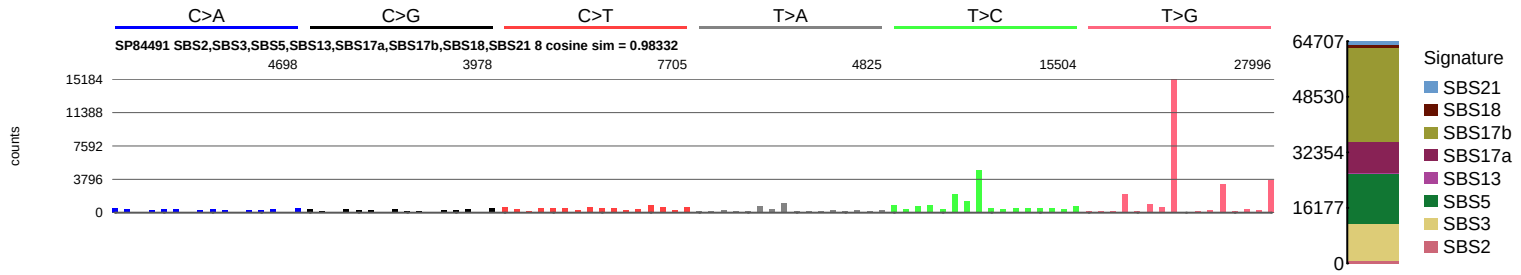
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98335
4352
3033
10019
3961
15601
27741
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

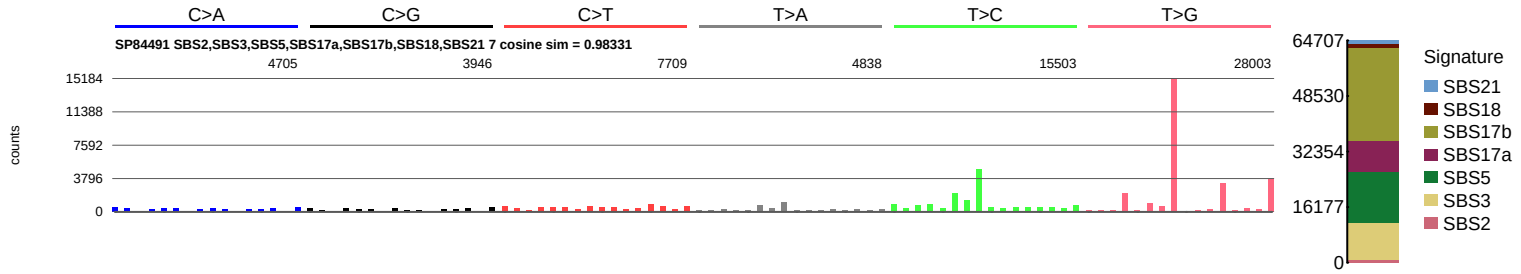

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98335
4367
3030
9993
3968
15617
27736
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

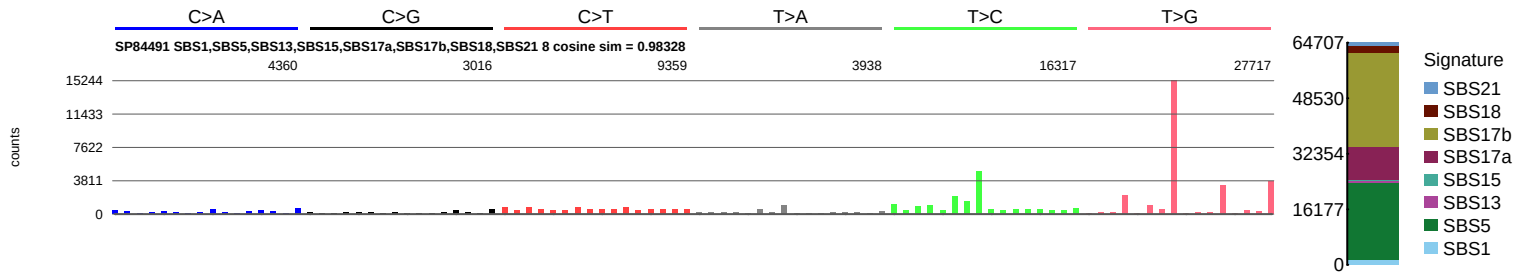

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98335
4354
3062
9913
3977
15641
27760
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

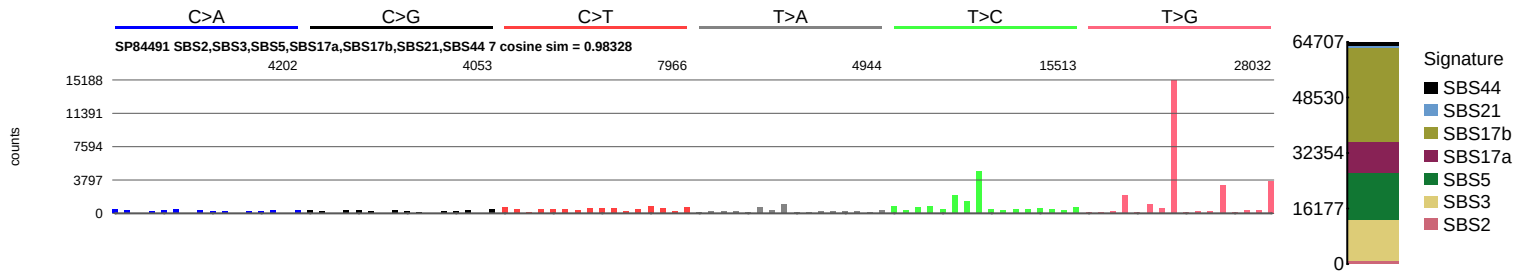

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98334
4931
5011
8347
5574
12756
28090
counts
15184
11388
7592
3796
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

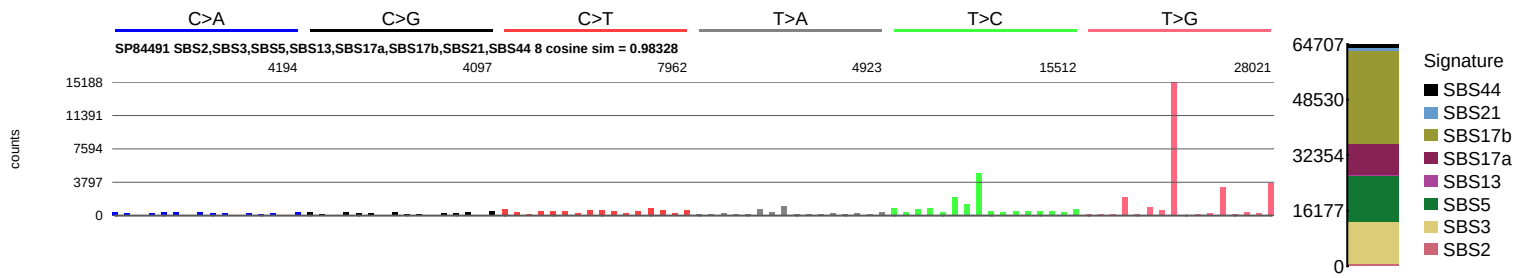

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98333
4779
5058
8372
5601
12791
28110
counts
15184
11388
7592
3796
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

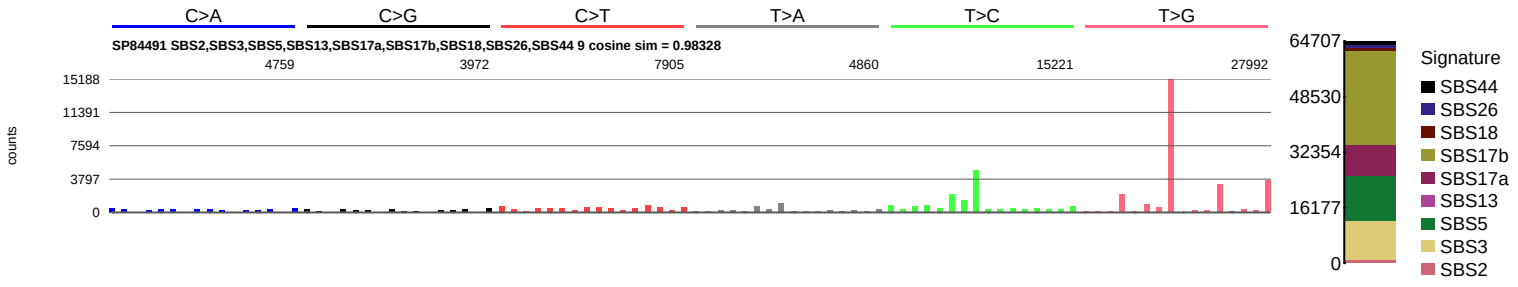

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98332
4698
3978
7705
4825
15504
27996
counts
15184
11388
7592
3796
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

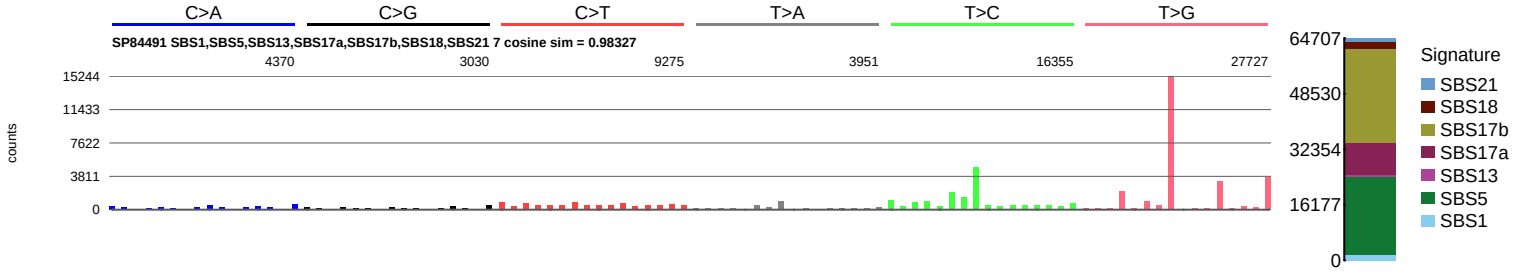

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98331
4705
3946
7709
4838
15503
28003
counts
15184
11388
7592
3796
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

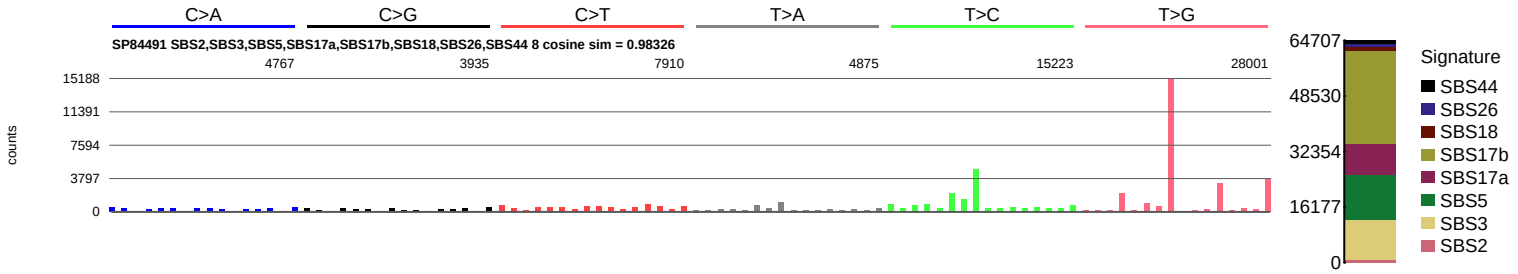

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98328
4360
3016
9359
3938
16317
27717
counts
15244
11433
7622
3811
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

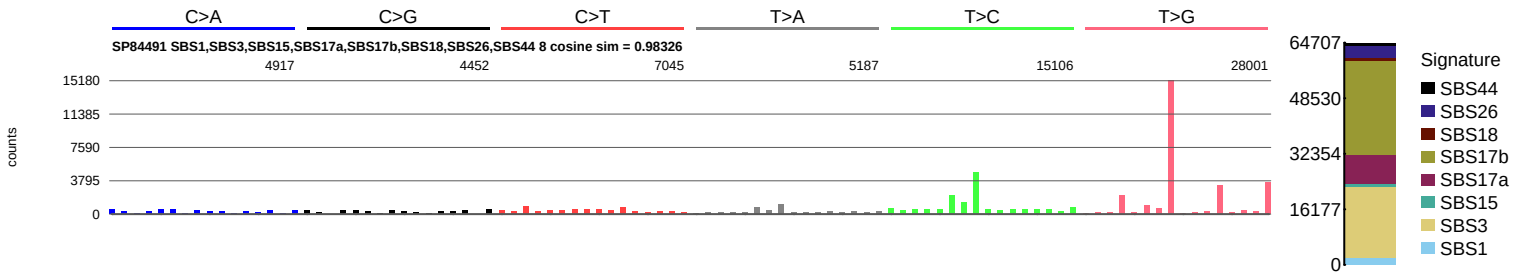

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98328
4202
4053
7966
4944
15513
28032
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2

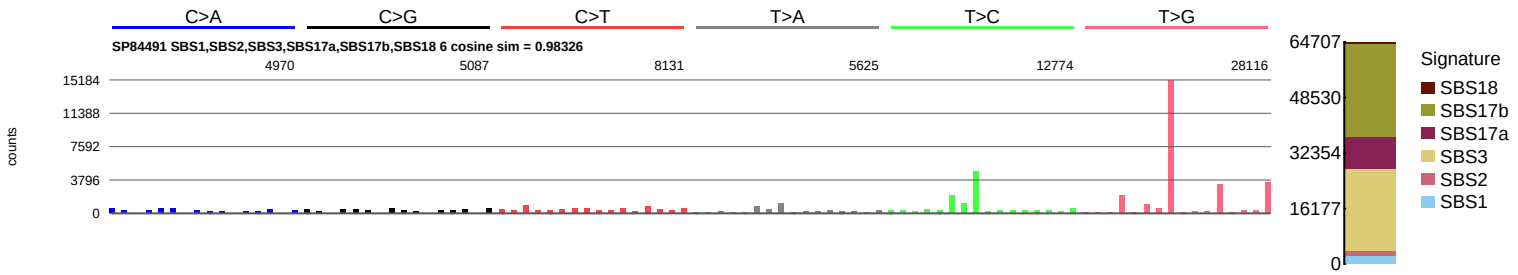

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98328
4194
4097
7962
4923
15512
28021
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

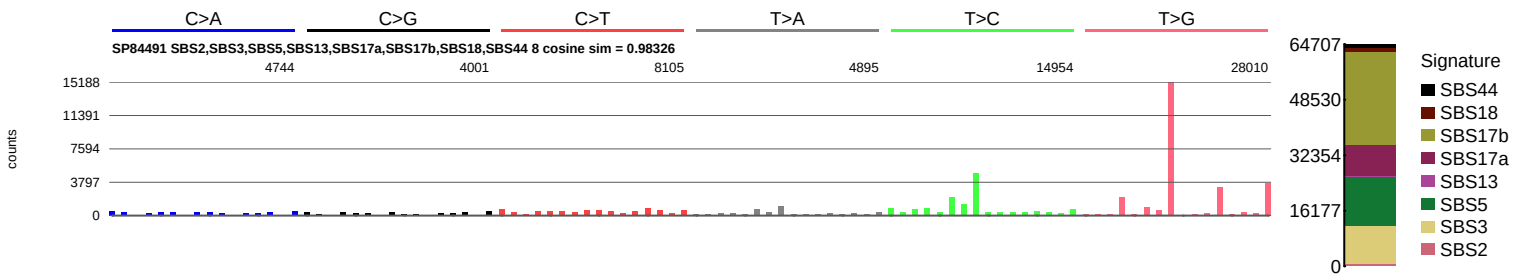

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98328
4759
3972
7905
4860
15221
27992
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

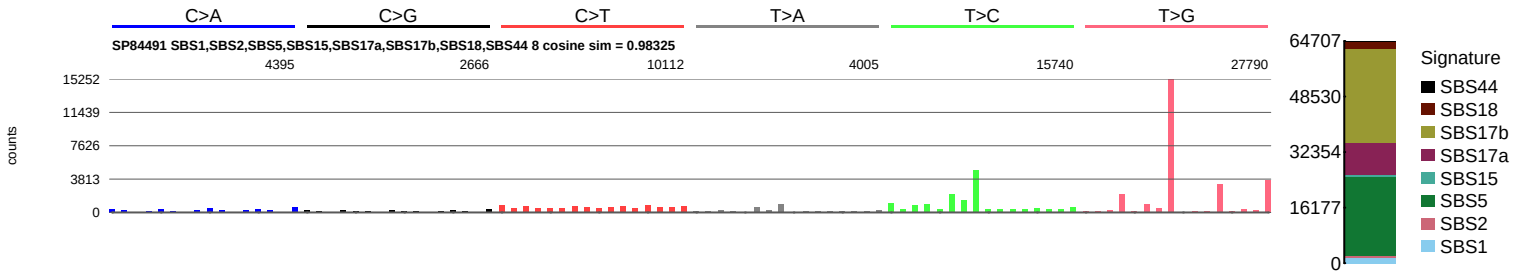

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98327
4370
3030
9275
3951
16355
27727
counts
15244
11433
7622
3811
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

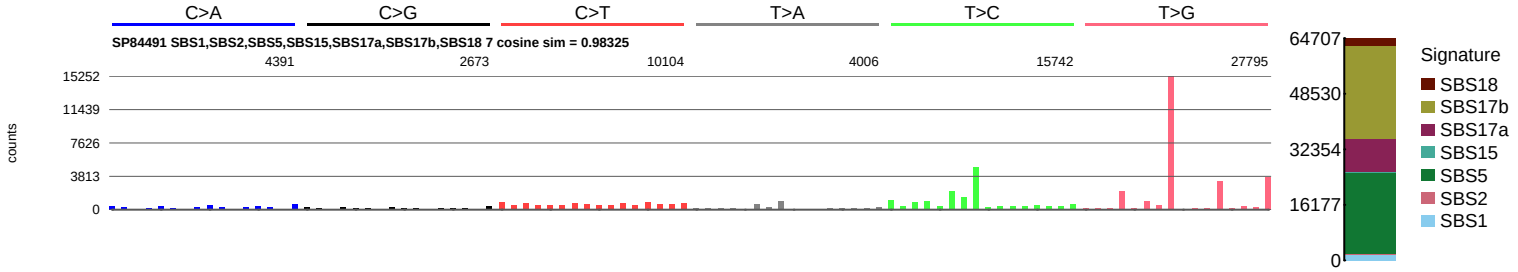

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98326
4767
3935
7910
4875
15223
28001
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98326
4917
4452
7045
5187
15106
28001
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98326
4970
5087
8131
5625
12774
28116
counts
15184
11388
7592
3796
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98326
4744
4001
8105
4895
14954
28010
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98325
4395
2666
10112
4005
15740
27790
counts
15252
11439
7626
3813
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98325
4391
2673
10104
4006
15742
27795
counts
15252
11439
7626
3813
0
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48530
32354
16177
0
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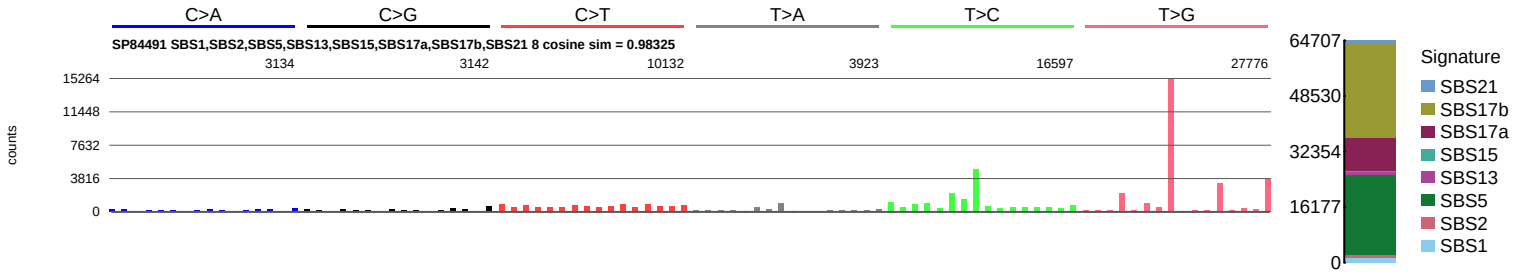
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98325
3134
3142
10132
3923
16597
27776
counts
15264
11448
7632
3816
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
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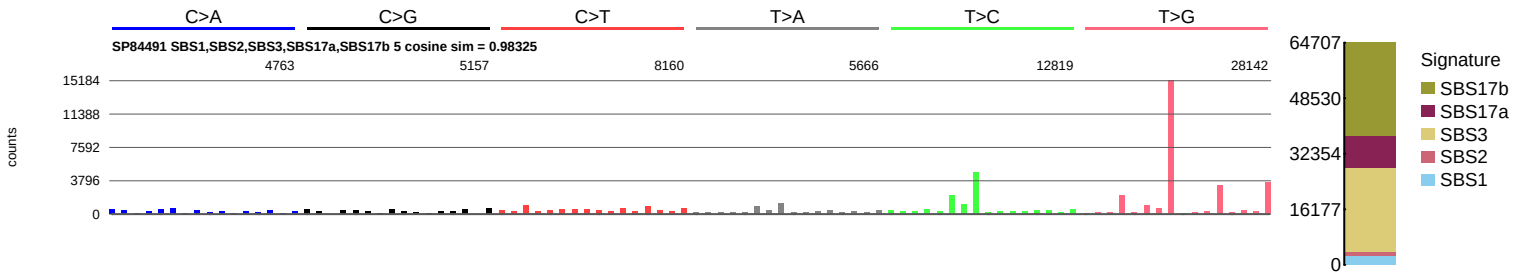
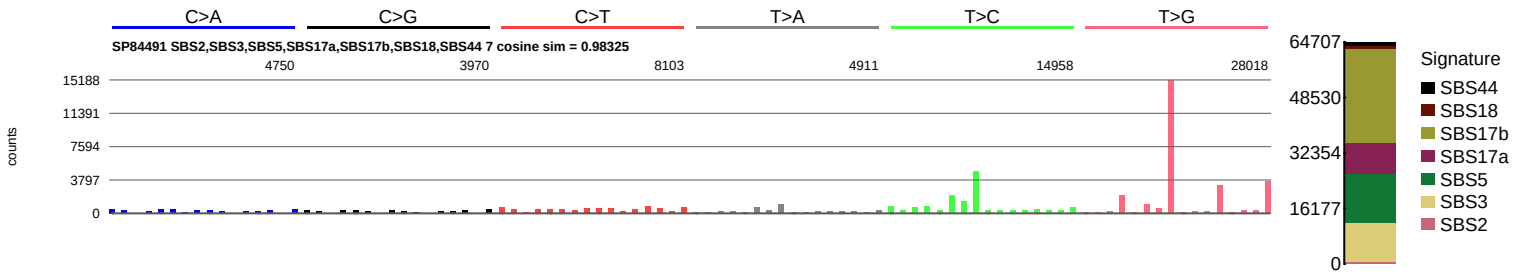
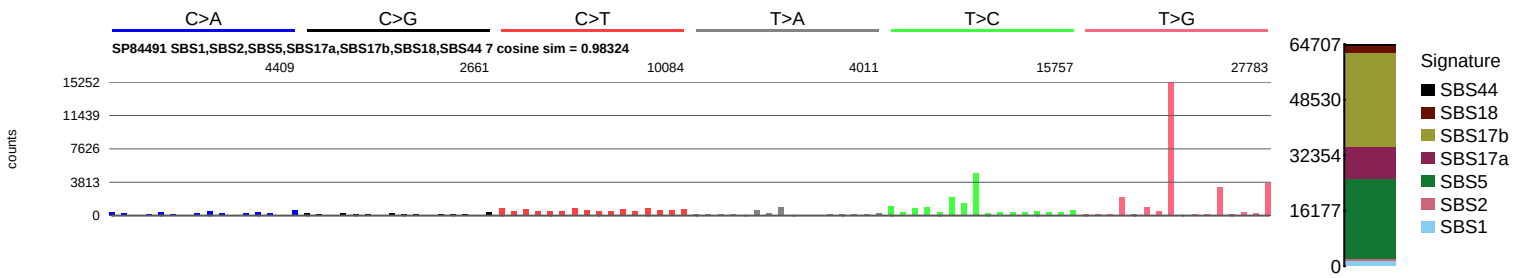
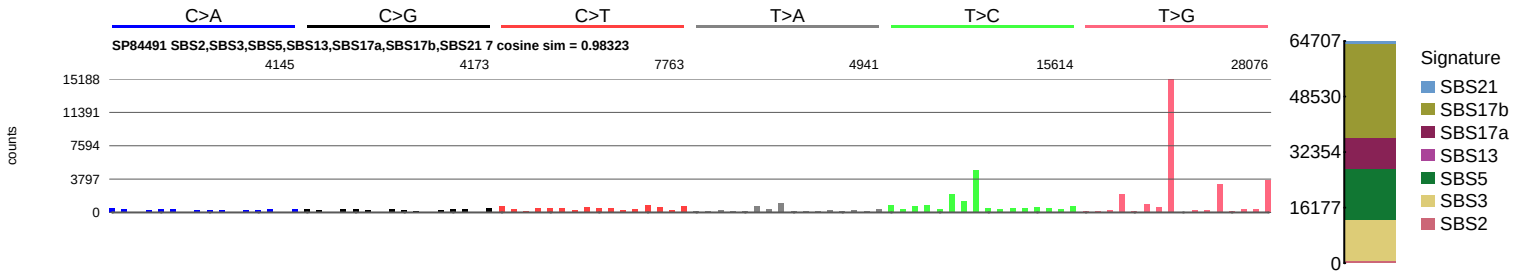
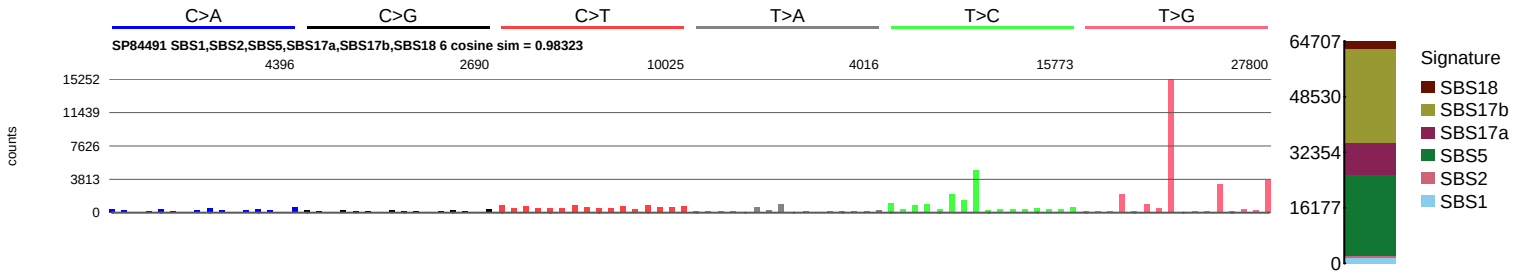
SBS5
SBS2
SBS1

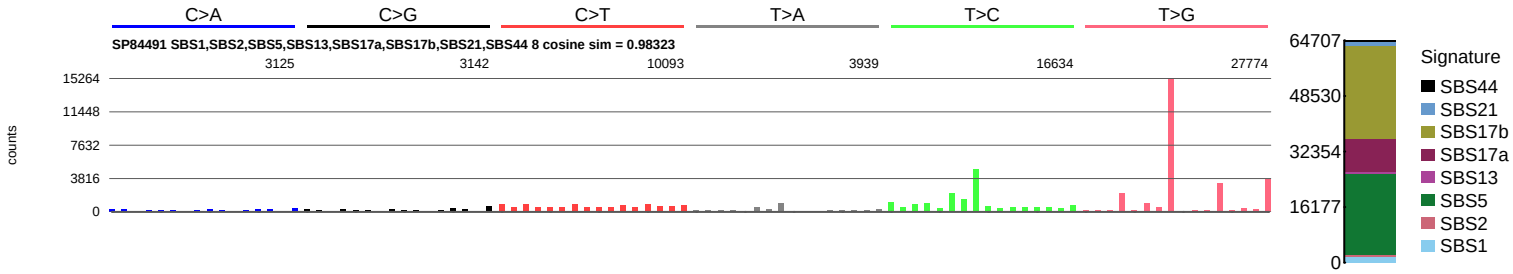

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS3,SBS17a,SBS17b 5 cosine sim = 0.98325
4763
5157
8160
5666
12819
28142
counts
15184
11388
7592
3796
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS3
SBS2
SBS1

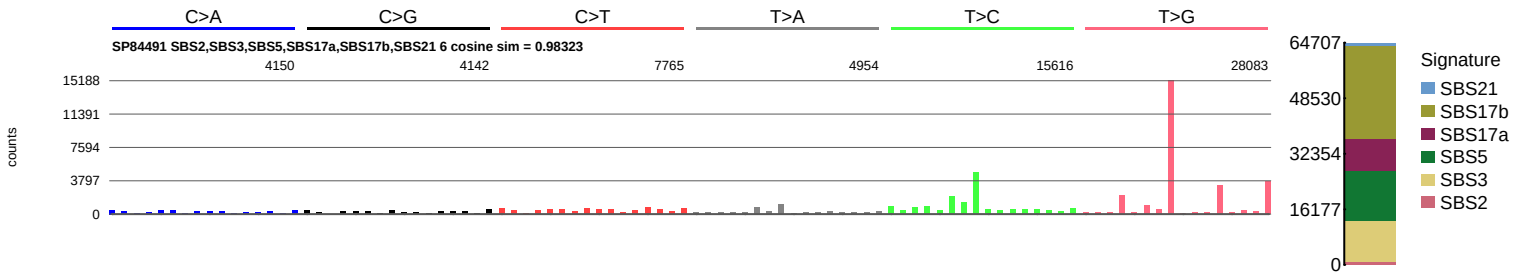

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98325
4750
3970
8103
4911
14958
28018
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

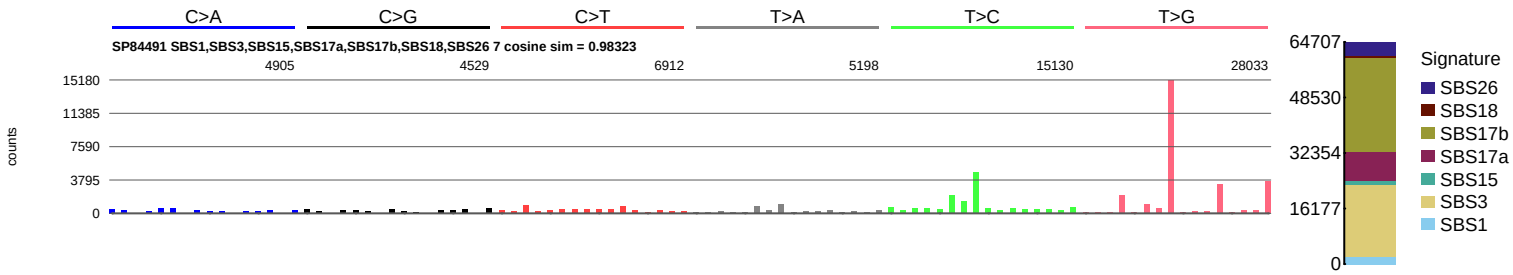

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98324
4409
2661
10084
4011
15757
27783
counts
15252
11439
7626
3813
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

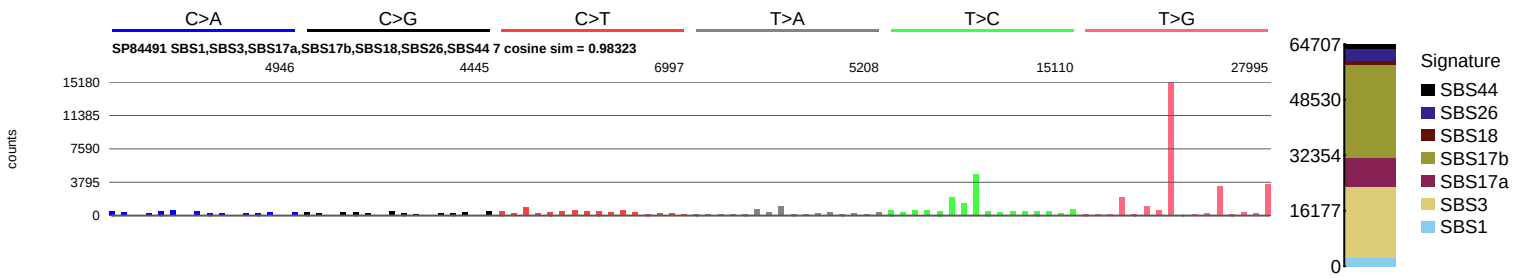

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98323
4145
4173
7763
4941
15614
28076
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

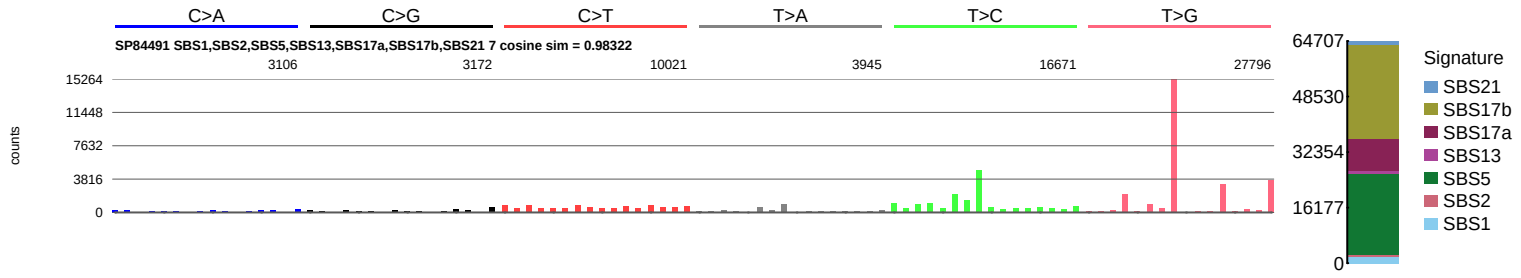

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98323
4396
2690
10025
4016
15773
27800
counts
15252
11439
7626
3813
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

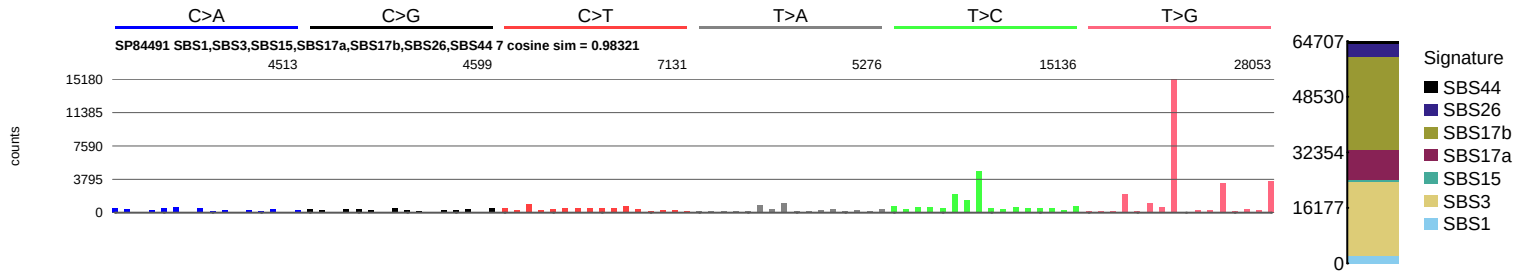

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98323
3125
3142
10093
3939
16634
27774
counts
15264
11448
7632
3816
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

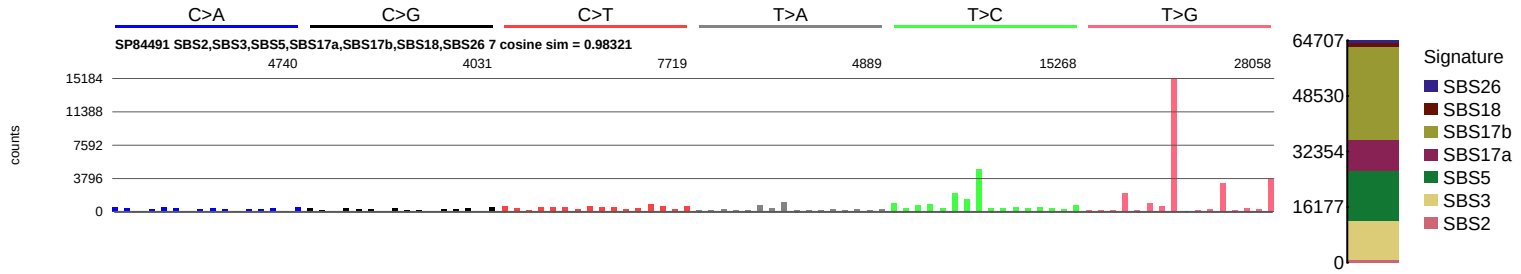

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98323
4150
4142
7765
4954
15616
28083
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2

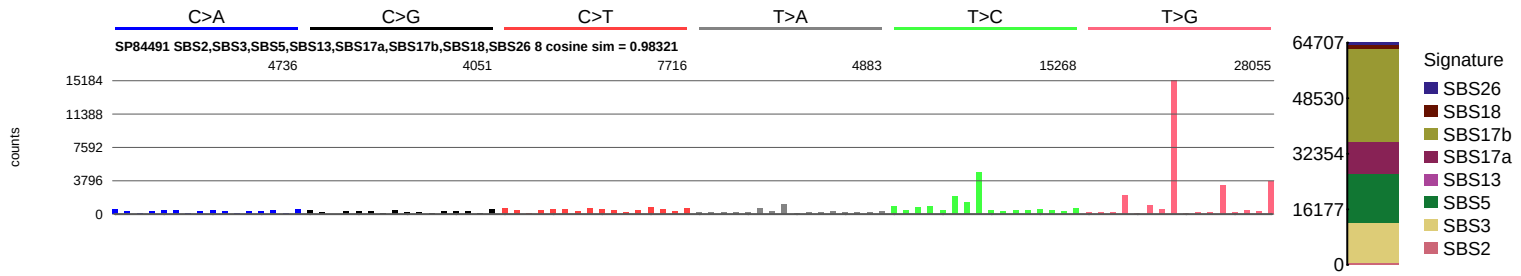

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98323
4905
4529
6912
5198
15130
28033
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

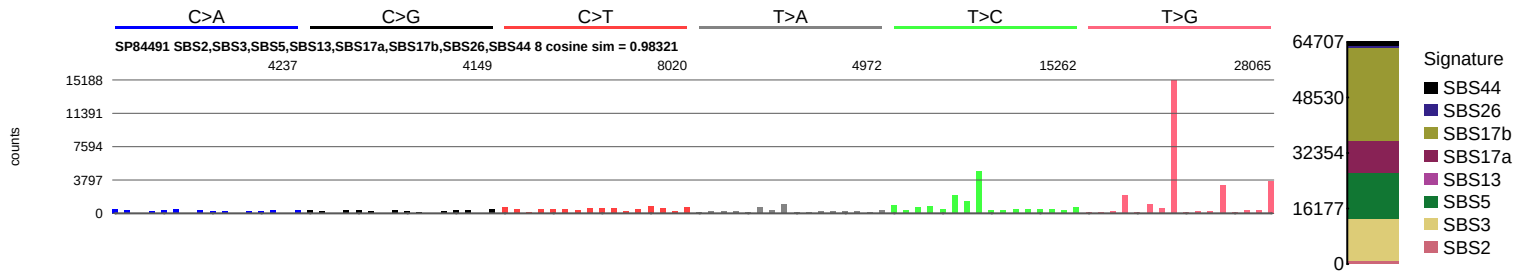

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98323
4946
4445
6997
5208
15110
27995
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

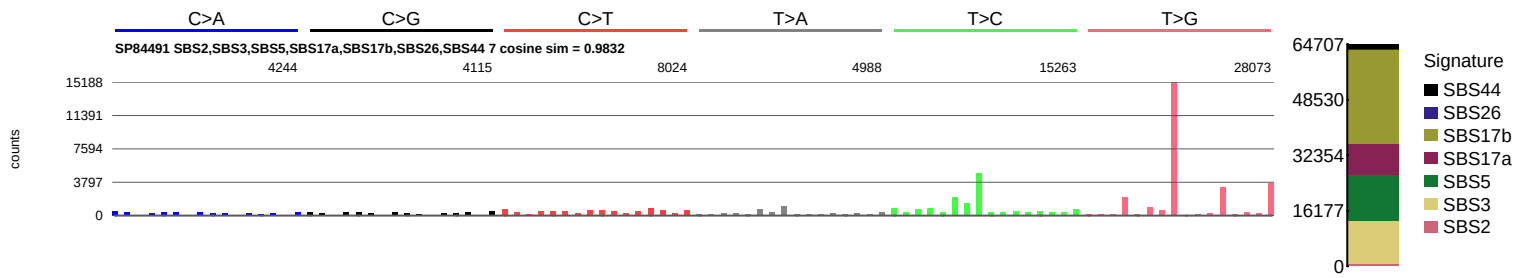

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98322
3106
3172
10021
3945
16671
27796
15264
11448
7632
3816
0
counts
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98321
4513
4599
7131
5276
15136
28053
15180
11385
7590
3795
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98321
4740
4031
7719
4889
15268
28058
15184
11388
7592
3796
0
counts
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98321
4736
4051
7716
4883
15268
28055
15184
11388
7592
3796
0
counts
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98321
4237
4149
8020
4972
15262
28065
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.9832
4244
4115
8024
4988
15263
28073
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
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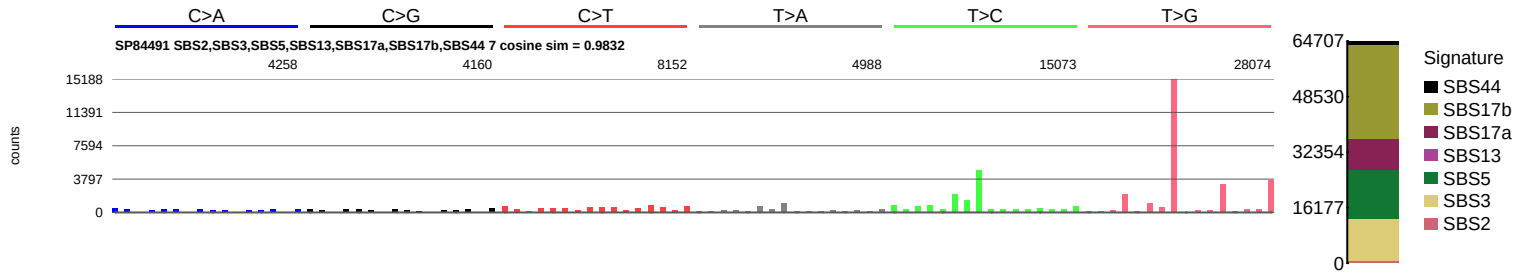
SBS44
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.9832
4258
4160
8152
4988
15073
28074
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
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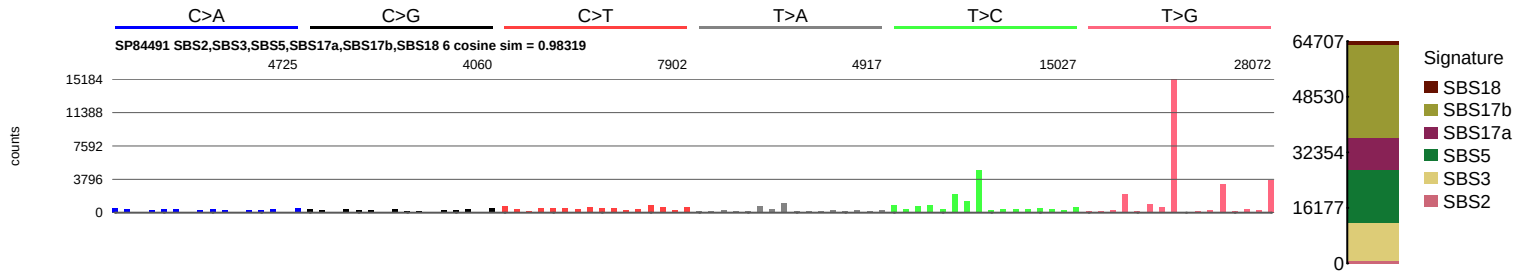
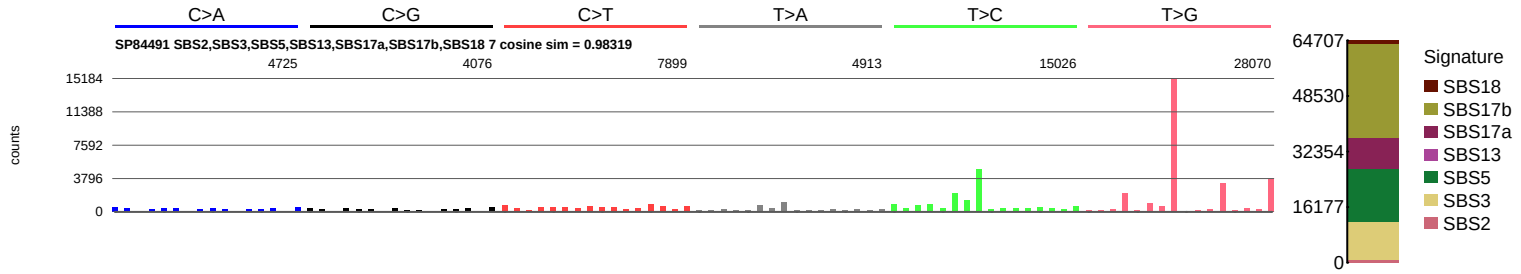
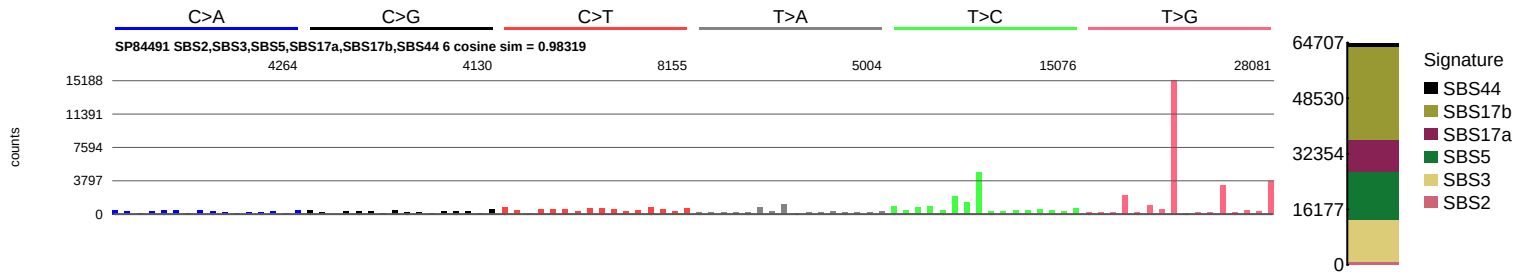
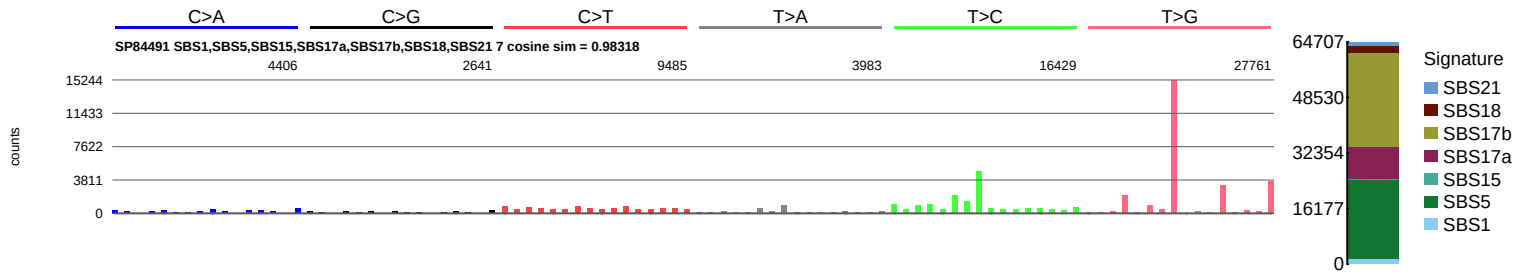
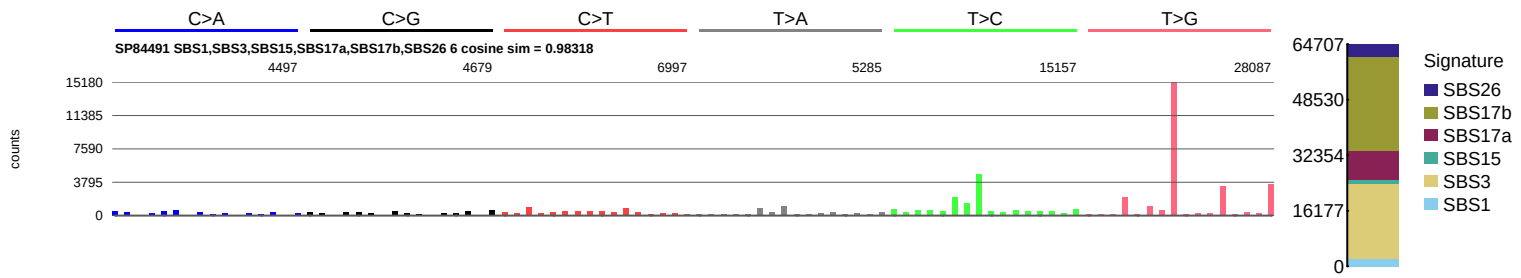
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

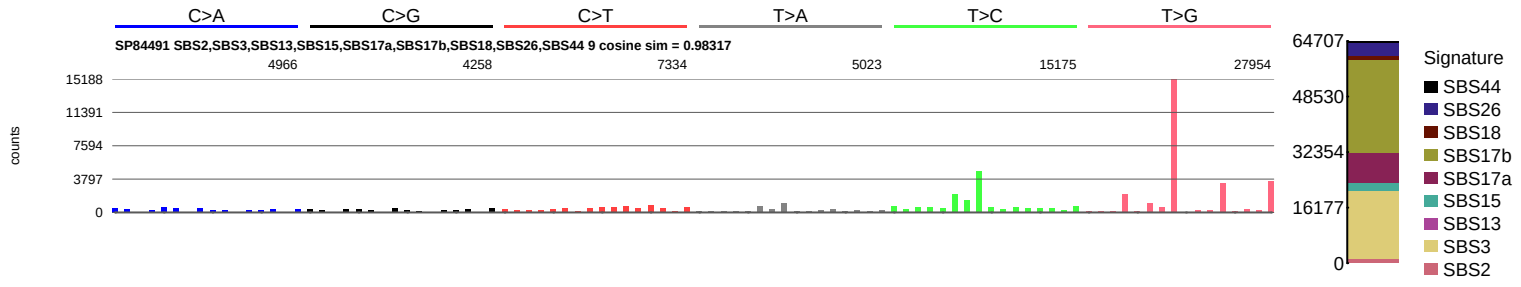

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98319
4725
4060
7902
4917
15027
28072
15184
11388
7592
3796
0
counts
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98319
4725
4076
7899
4913
15026
28070
15184
11388
7592
3796
0
counts
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98319
4264
4130
8155
5004
15076
28081
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98318
4406
2641
9485
3983
16429
27761
15244
11433
7622
3811
0
counts
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98318
4497
4679
6997
5285
15157
28087
15180
11385
7590
3795
0
counts
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98317
4966
4258
7334
5023
15175
27954
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
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Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
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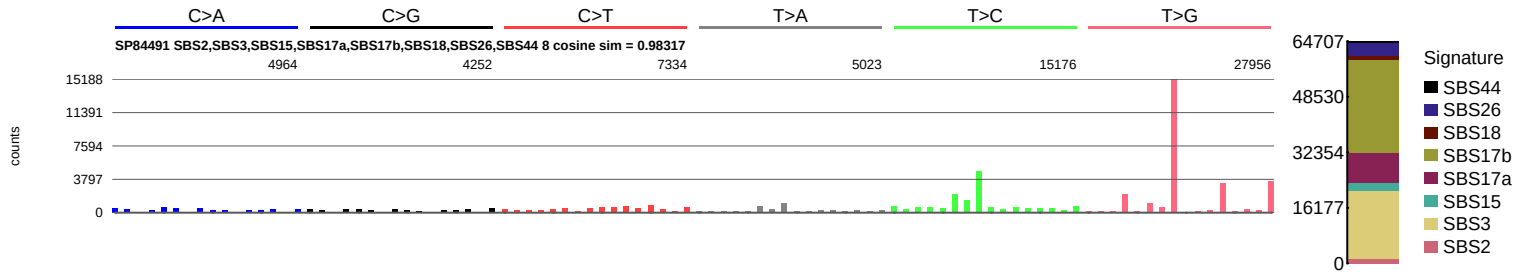
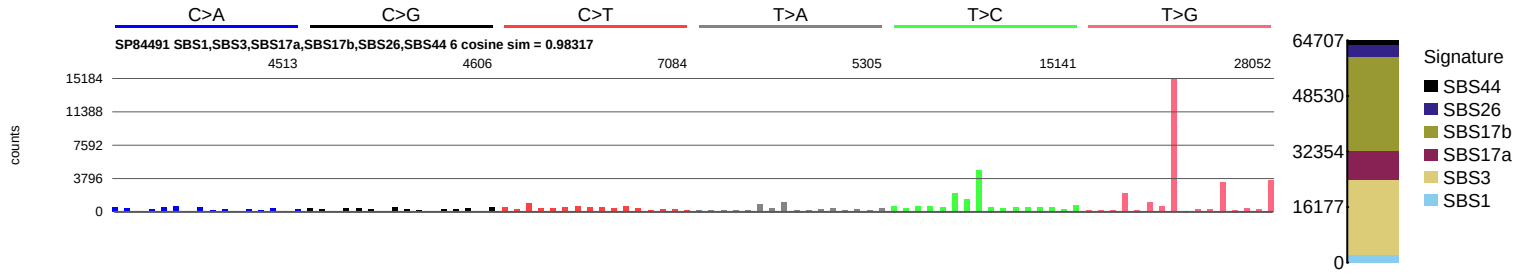
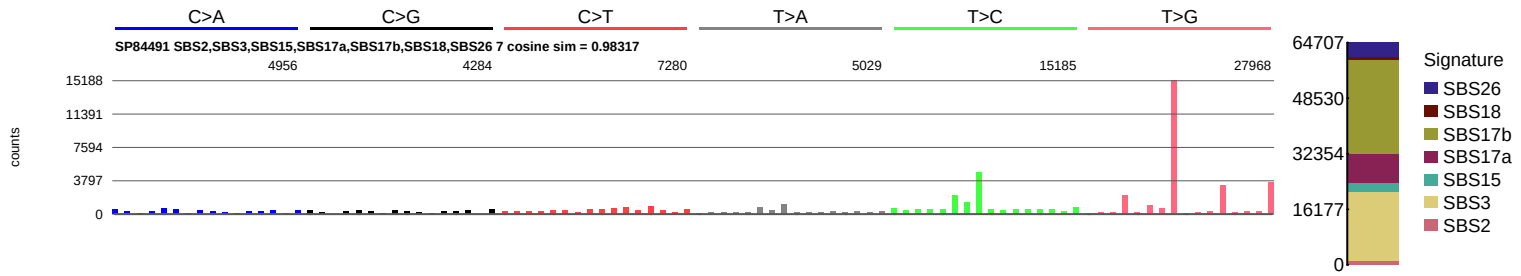
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98317
4964
4252
7334
5023
15176
27956
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98317
4513
4606
7084
5305
15141
28052
counts
15184
11388
7592
3796
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS1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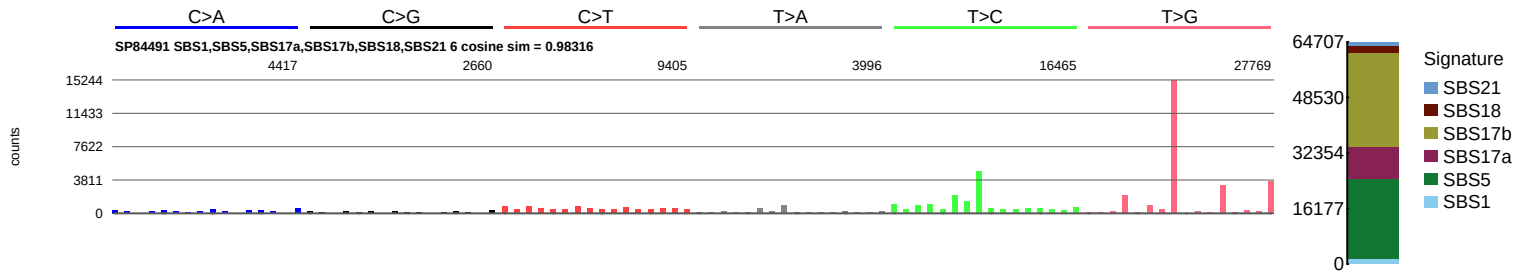
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98317
4956
4284
7280
5029
15185
27968
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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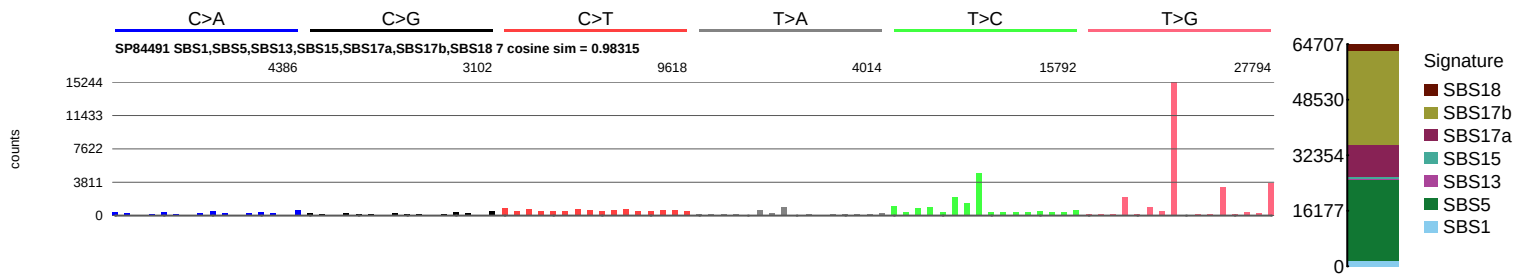
T>G
SP84491 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98316
4417
2660
9405
3996
16465
27769
counts
15244
11433
7622
3811
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>T
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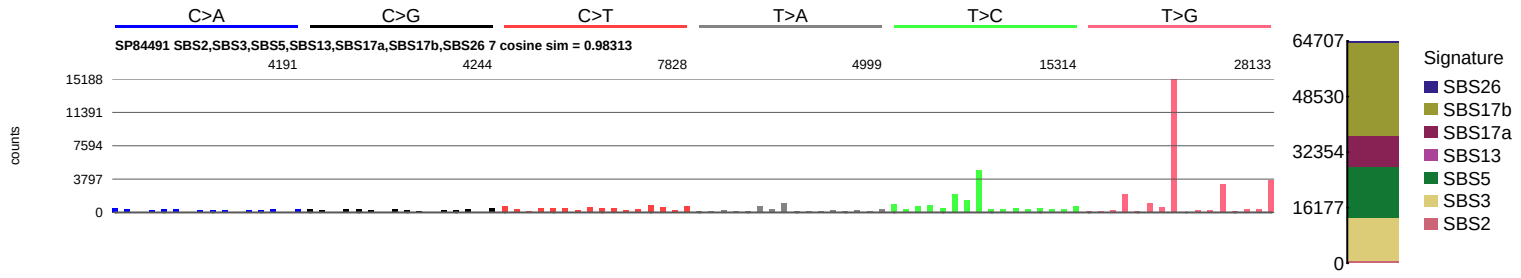
SP84491 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98315
4386
3102
9618
4014
15792
27794
counts
15244
11433
7622
3811
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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C>G
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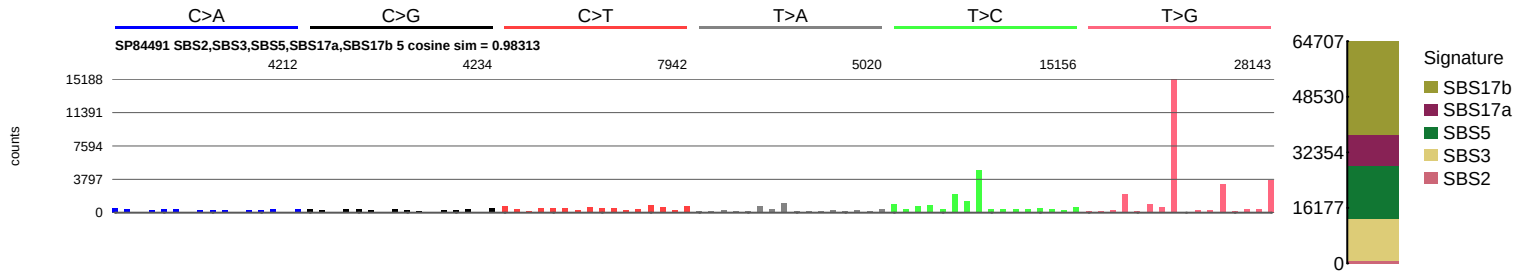
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98313
4191
4244
7828
4999
15314
28133
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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C>G
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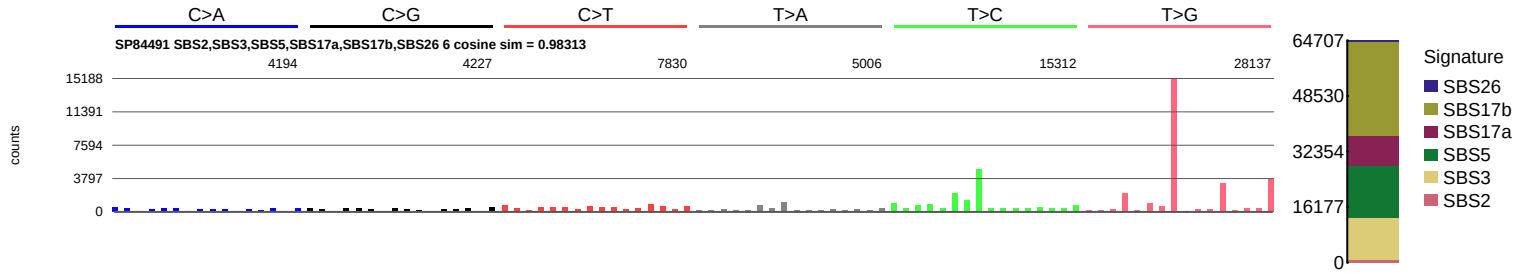
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b 5 cosine sim = 0.98313
4212
4234
7942
5020
15156
28143
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS5
SBS3
SBS2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C>G
C>T
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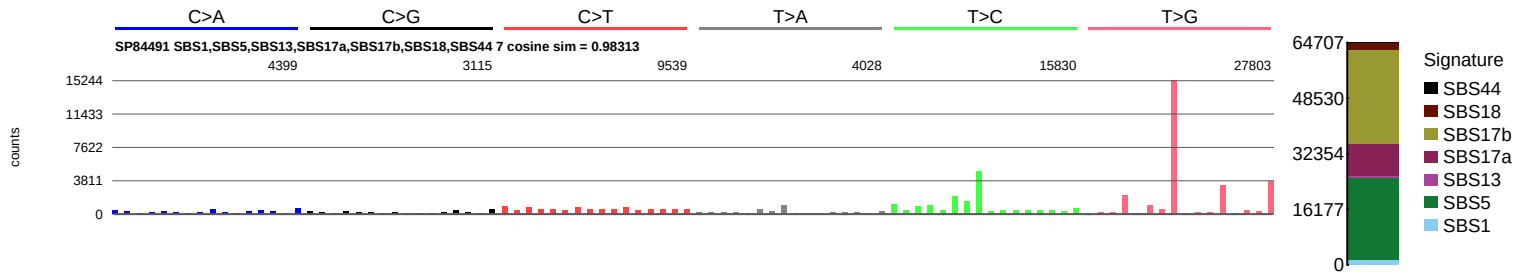
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98313
4194
4227
7830
5006
15312
28137
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2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C>G
C>T
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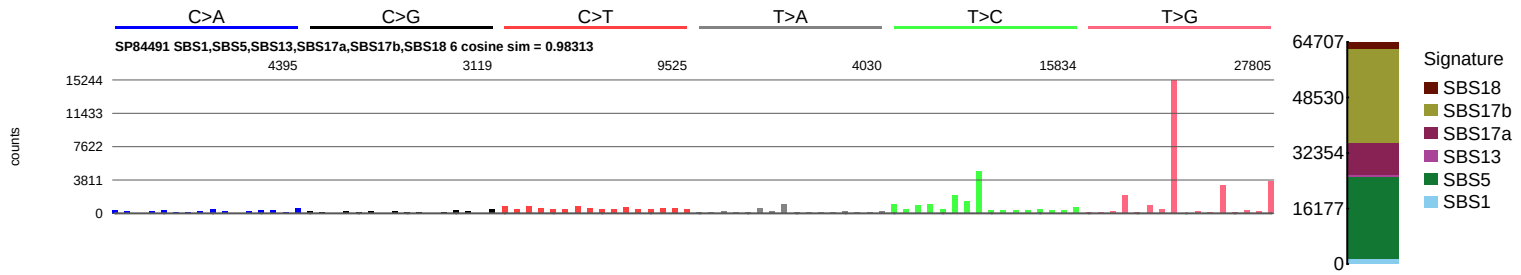
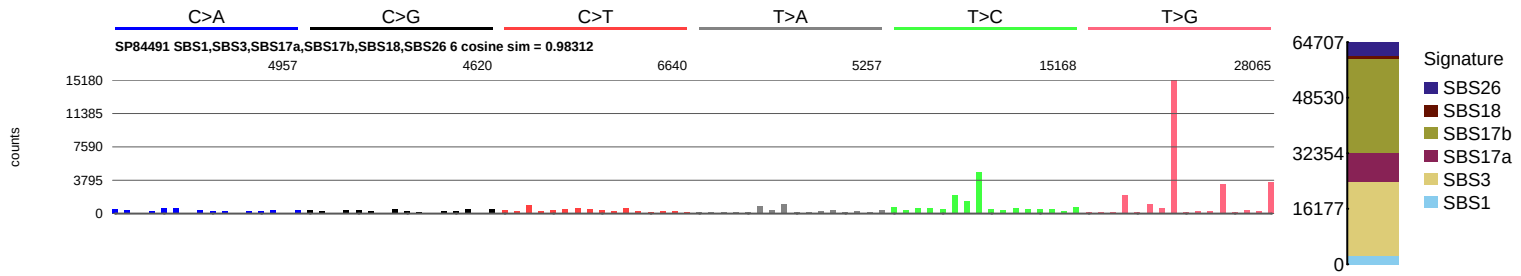
T>C
T>G
SP84491 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98313
4399
3115
9539
4028
15830
27803
counts
15244
11433
7622
3811
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98313
4395
3119
9525
4030
15834
27805
counts
15244
11433
7622
3811
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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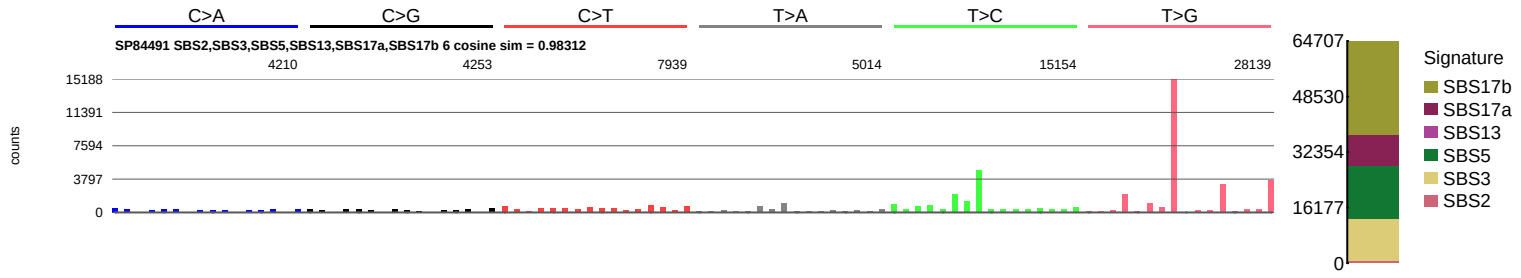
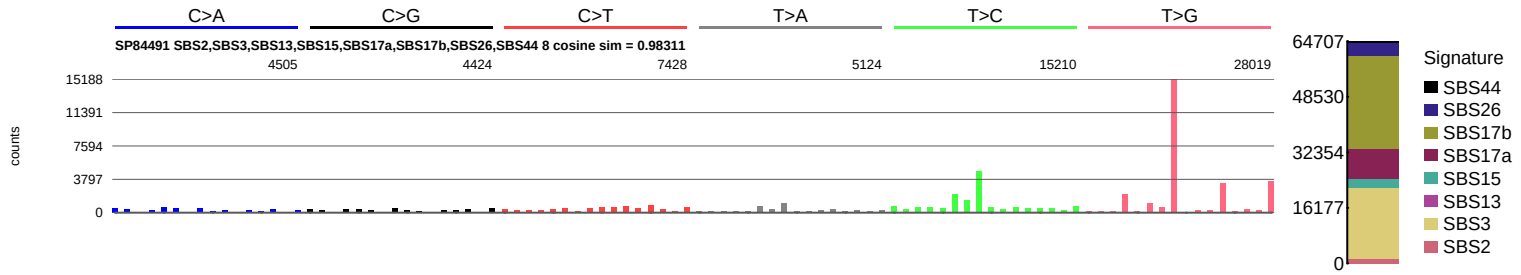
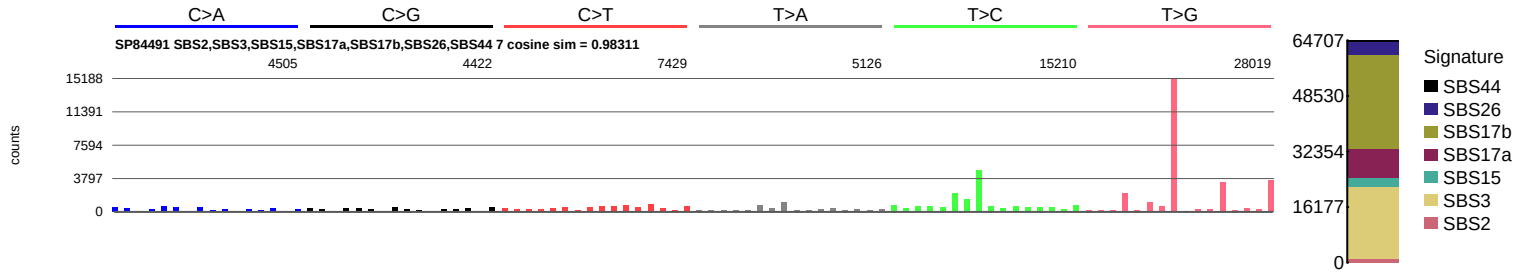
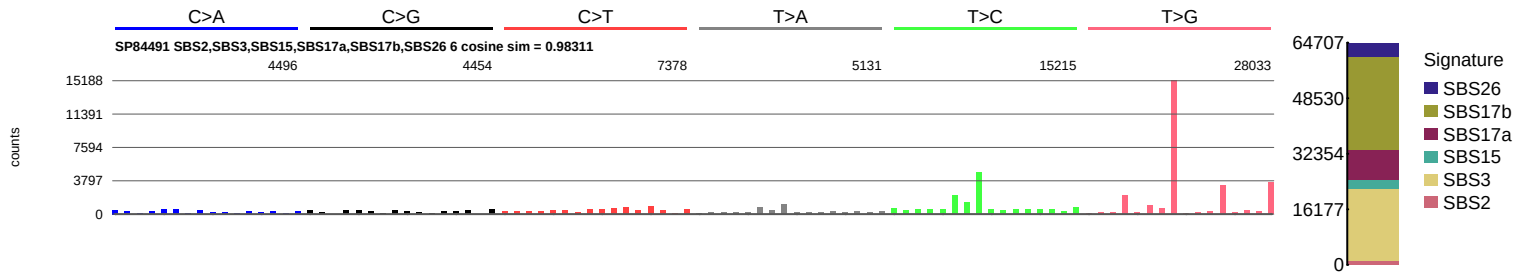
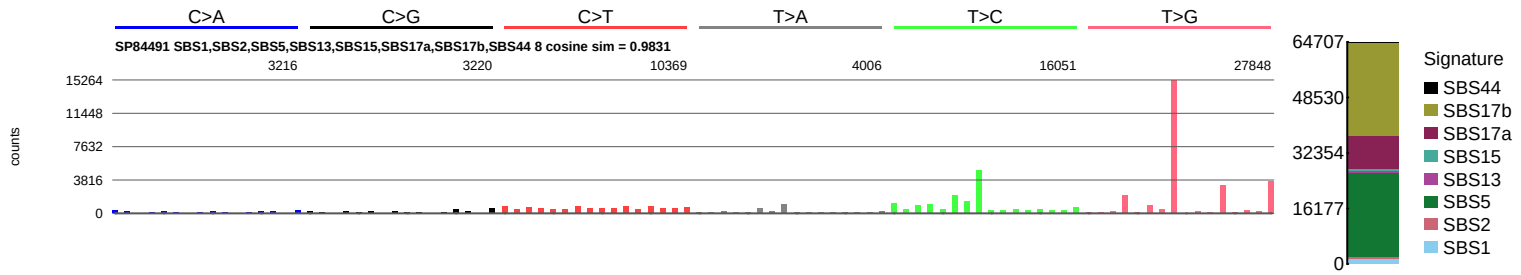
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98312
4957
4620
6640
5257
15168
28065
counts
15180
11385
7590
3795
0
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98312
4210
4253
7939
5014
15154
28139
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98311
4505
4424
7428
5124
15210
28019
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98311
4505
4422
7429
5126
15210
28019
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98311
4496
4454
7378
5131
15215
28033
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2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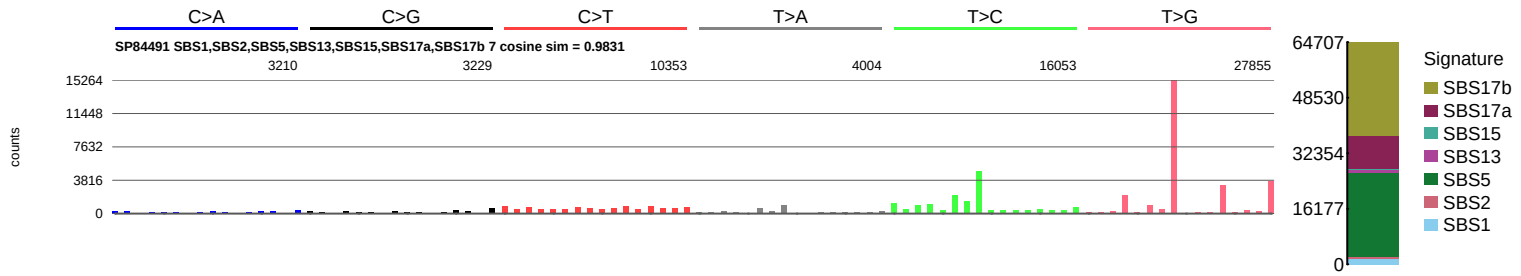
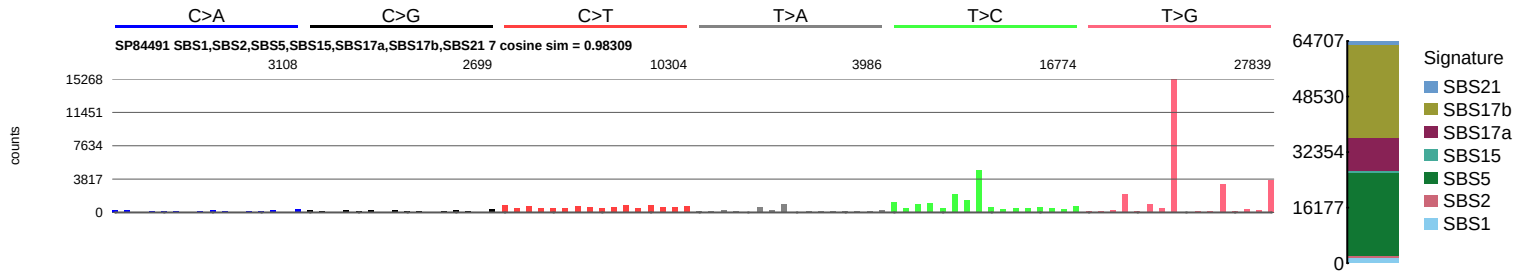
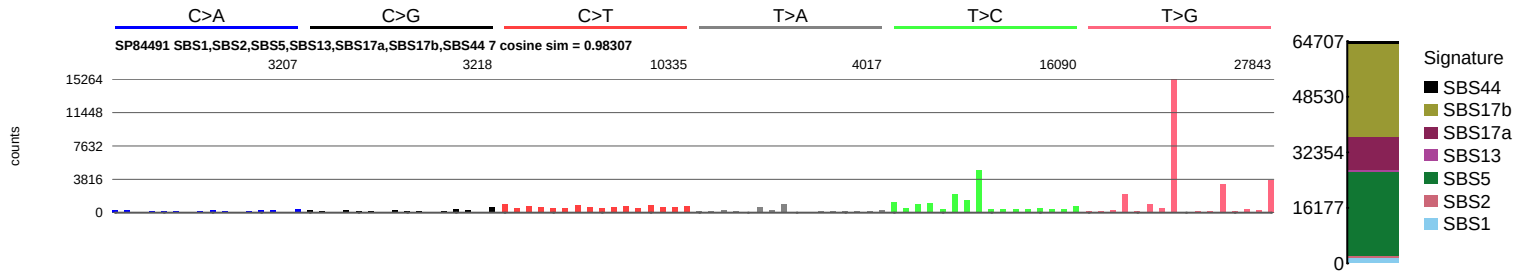
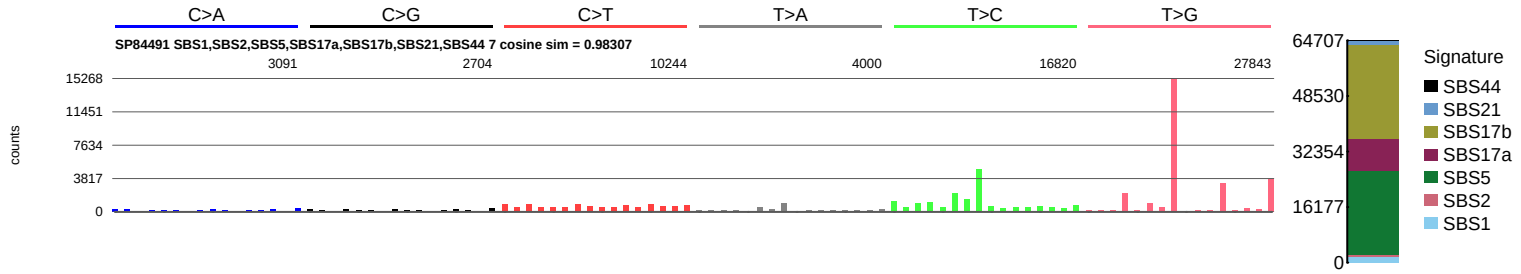
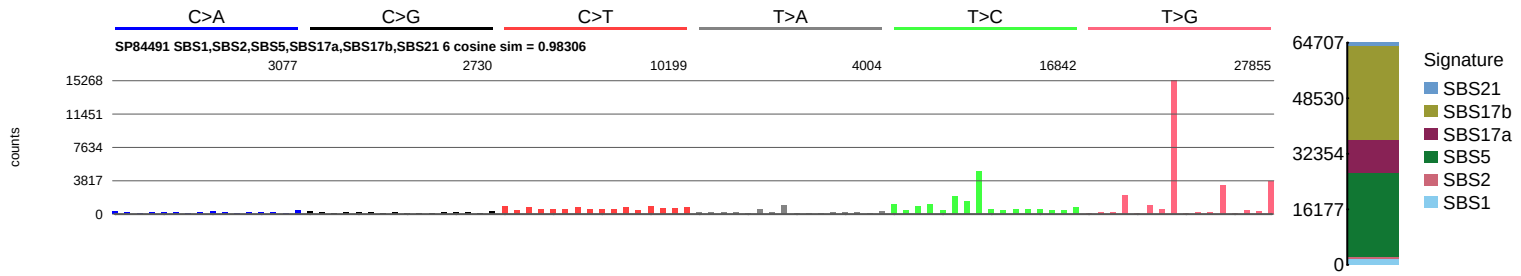
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS44 8 cosine sim = 0.9831
3216
3220
10369
4006
16051
27848
counts
15264
11448
7632
3816
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

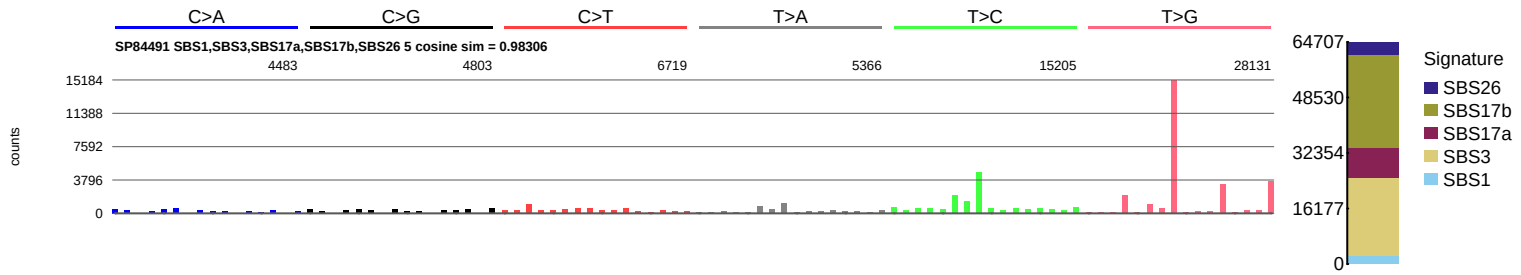

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.9831
3210
3229
10353
4004
16053
27855
counts
15264
11448
7632
3816
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

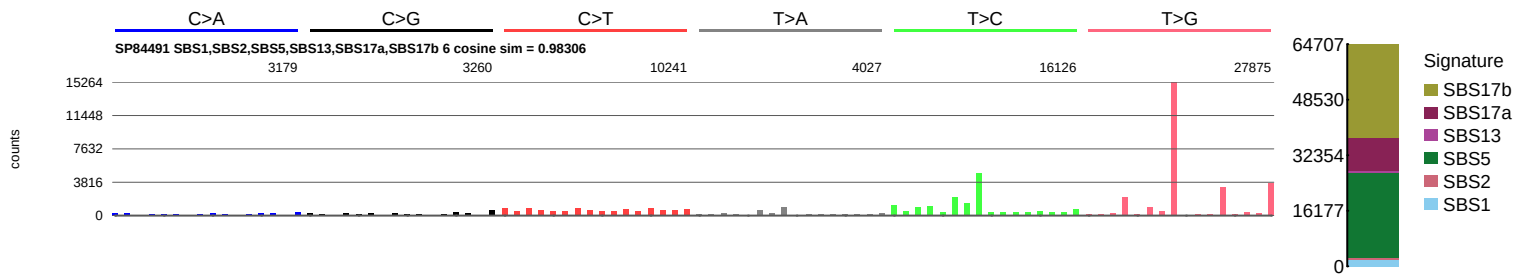

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98309
3108
2699
10304
3986
16774
27839
counts
15268
11451
7634
3817
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

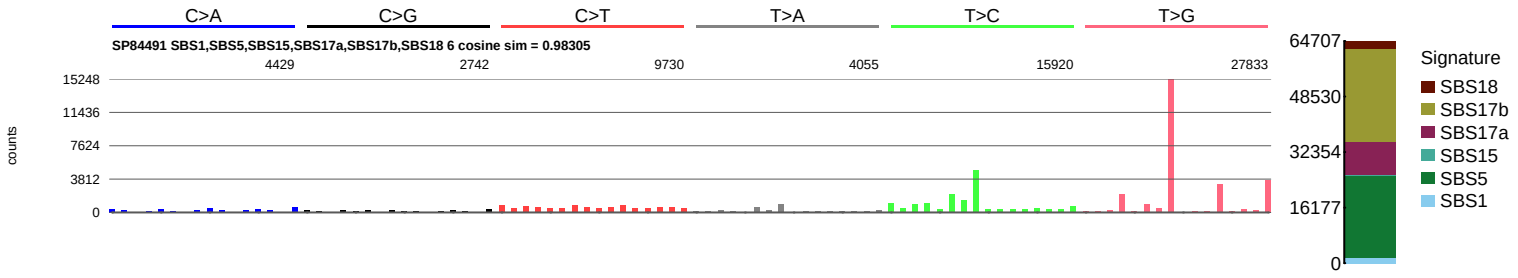

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98307
3207
3218
10335
4017
16090
27843
counts
15264
11448
7632
3816
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

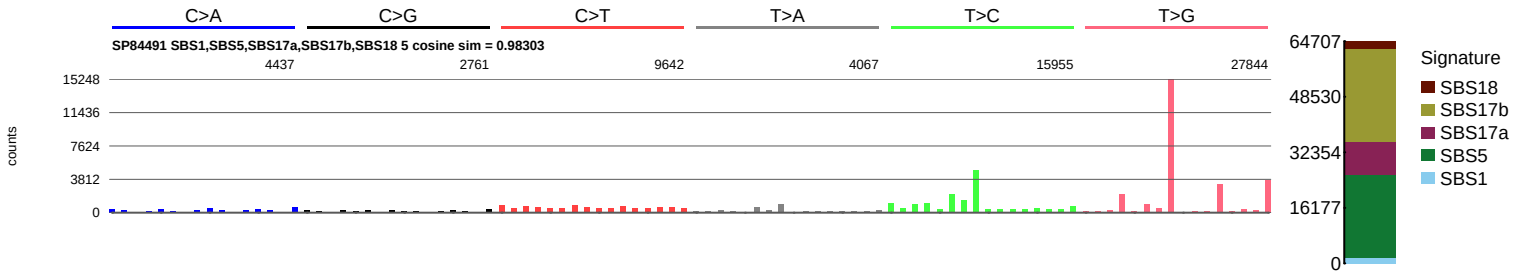

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98307
3091
2704
10244
4000
16820
27843
counts
15268
11451
7634
3817
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

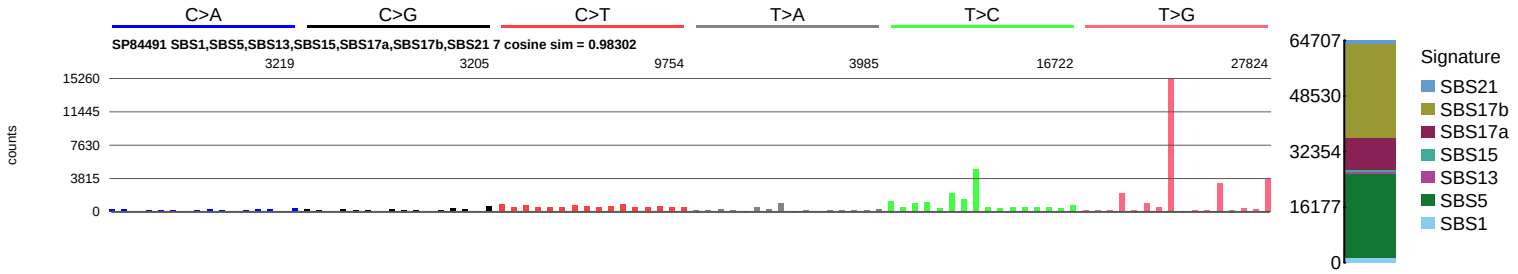

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98306
3077
2730
10199
4004
16842
27855
counts
15268
11451
7634
3817
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

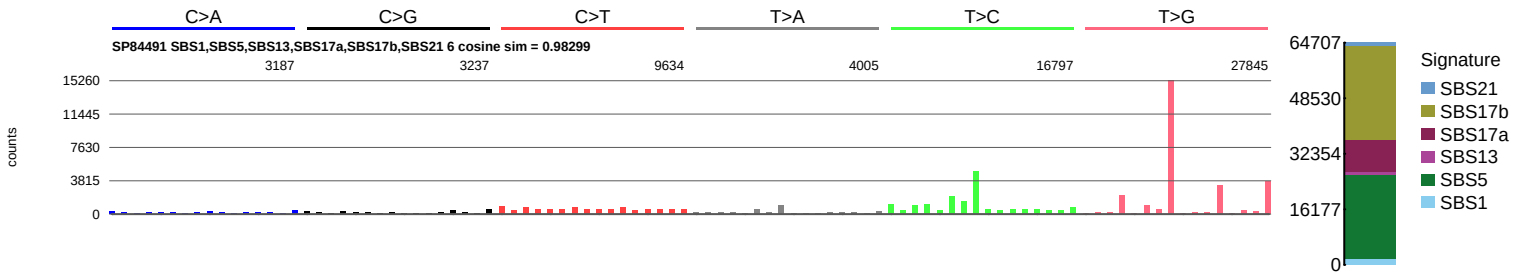

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS26 5 cosine sim = 0.98306
4483
4803
6719
5366
15205
28131
counts
15184
11388
7592
3796
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS1

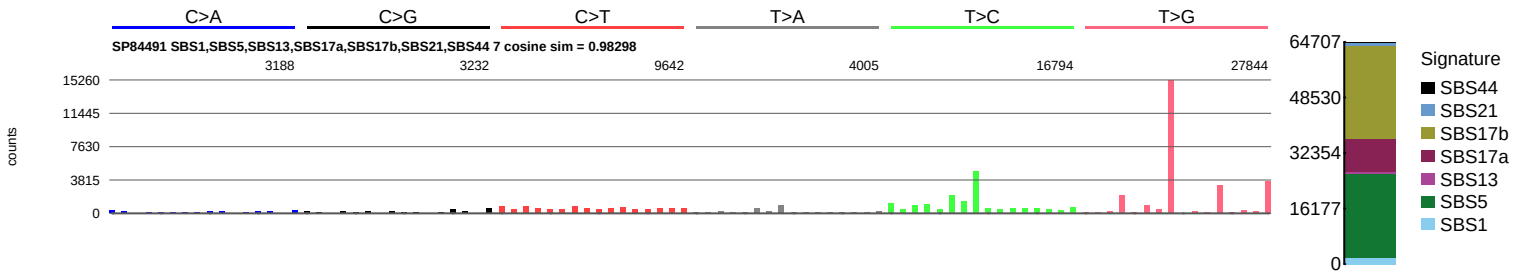

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98306
3179
3260
10241
4027
16126
27875
counts
15264
11448
7632
3816
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

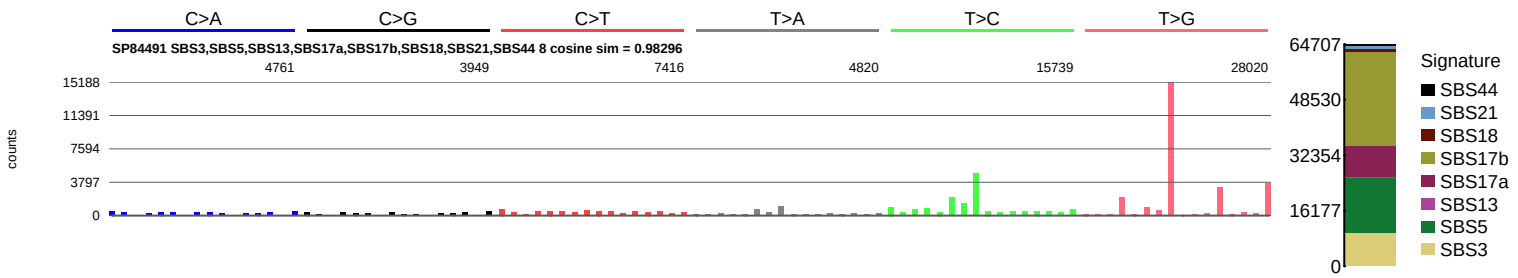

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98305
4429
2742
9730
4055
15920
27833
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

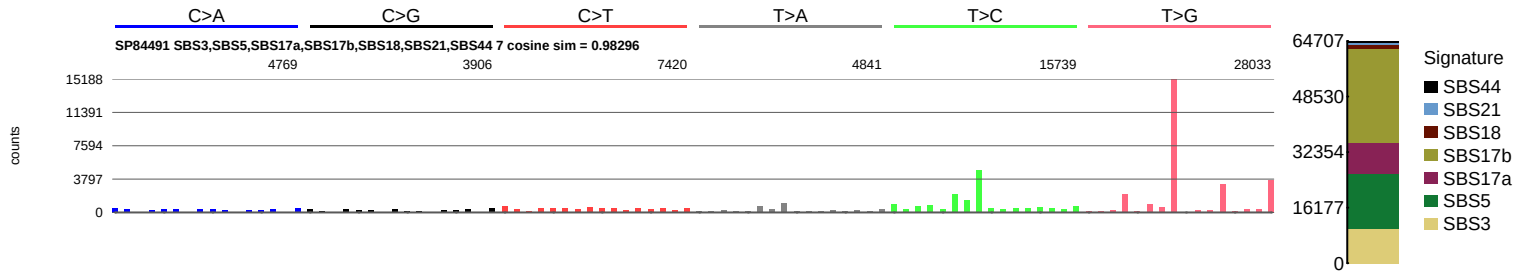

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98303
4437
2761
9642
4067
15955
27844
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS1

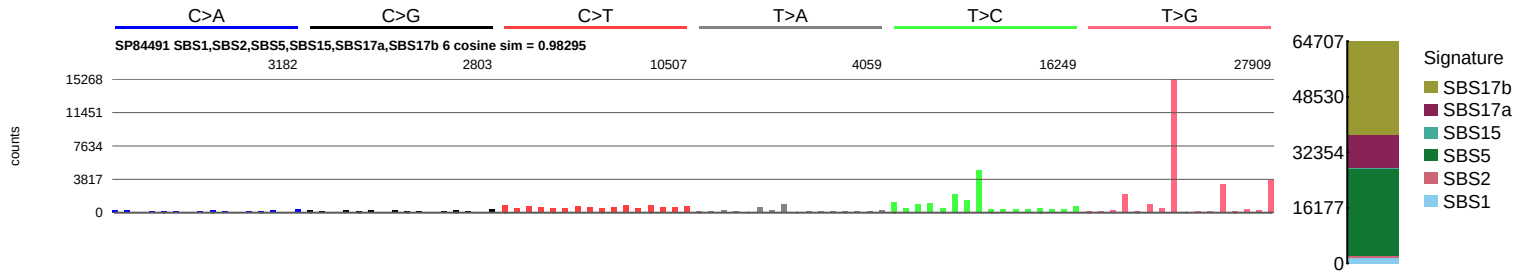

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98302
3219
3205
9754
3985
16722
27824
counts
15260
11445
7630
3815
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

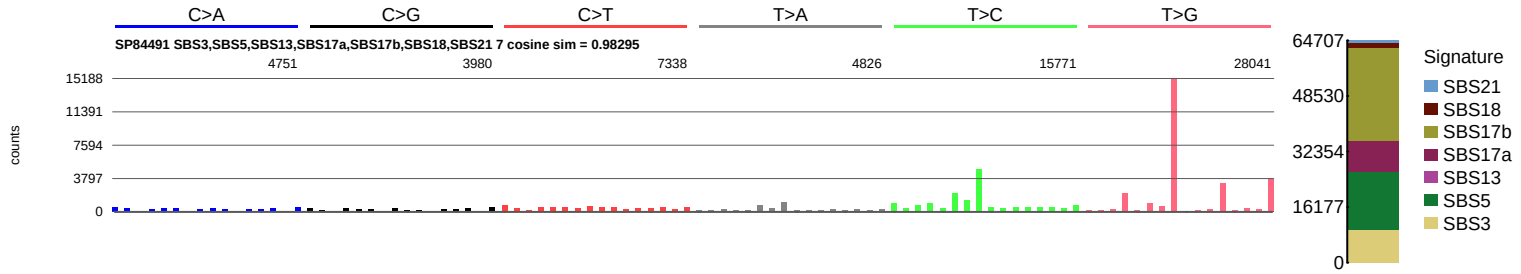

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98299
3187
3237
9634
4005
16797
27845
counts
15260
11445
7630
3815
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

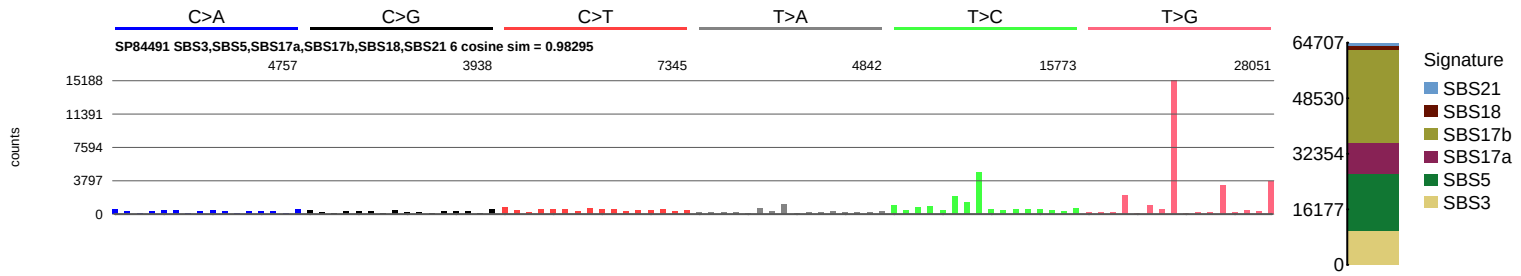

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98298
3188
3232
9642
4005
16794
27844
counts
15260
11445
7630
3815
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

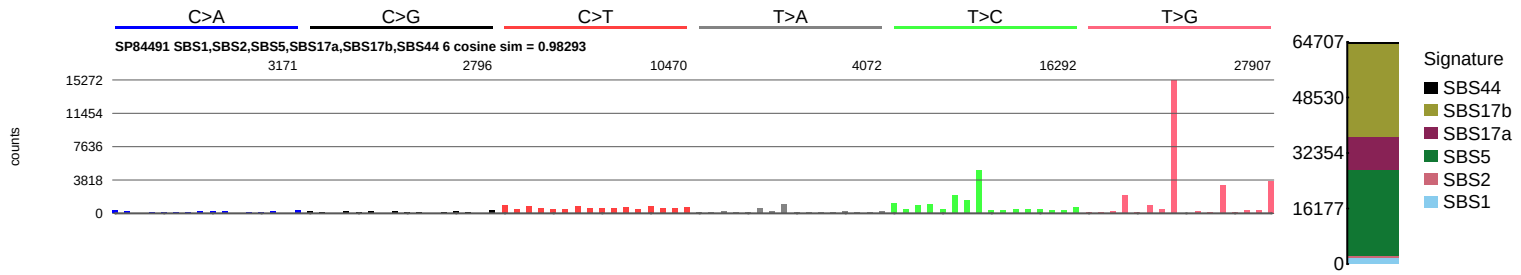

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98296
4761
3949
7416
4820
15739
28020
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3

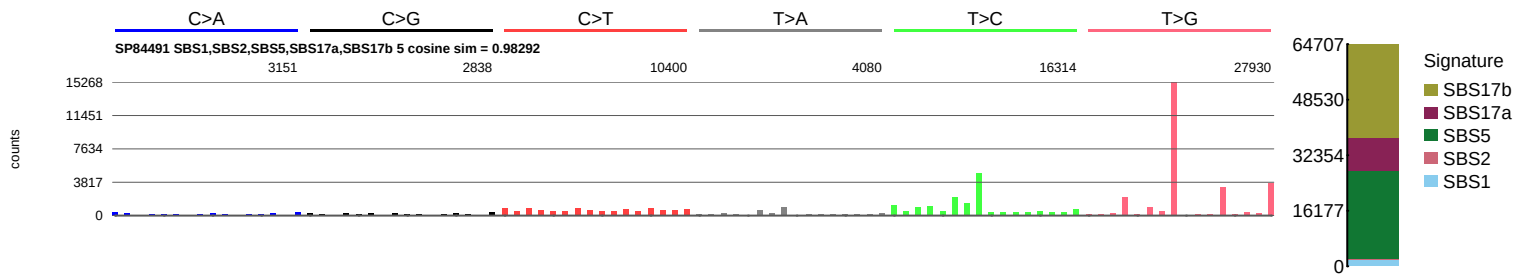

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98296
4769
3906
7420
4841
15739
28033
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3

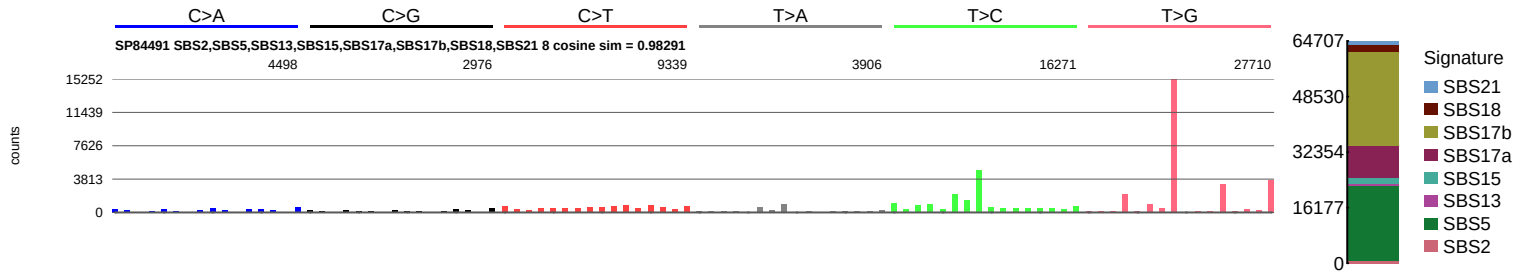

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98295
3182
2803
10507
4059
16249
27909
counts
15268
11451
7634
3817
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

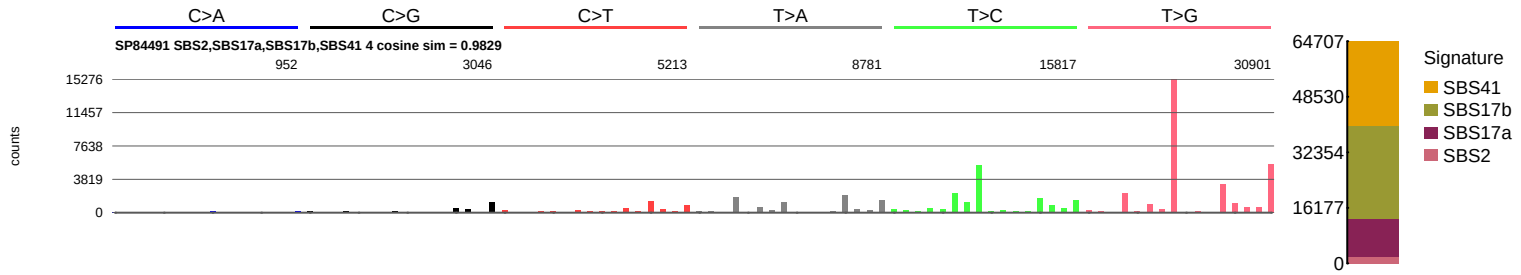

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98295
4751
3980
7338
4826
15771
28041
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3

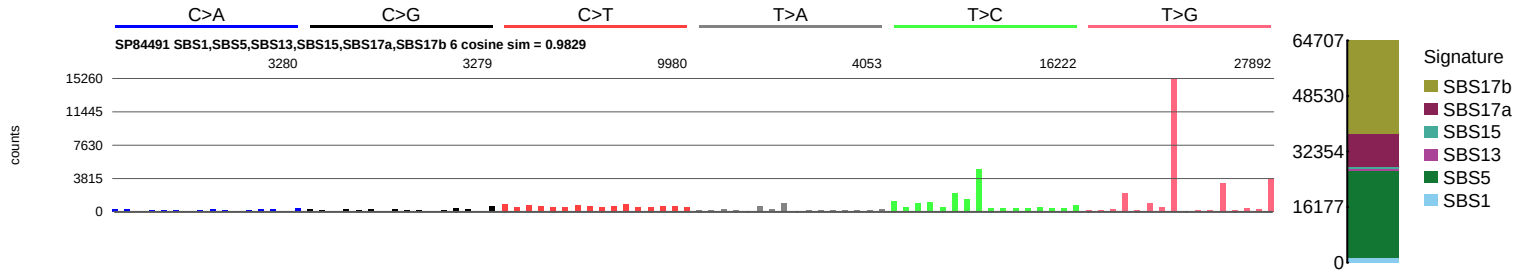

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98295
4757
3938
7345
4842
15773
28051
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3

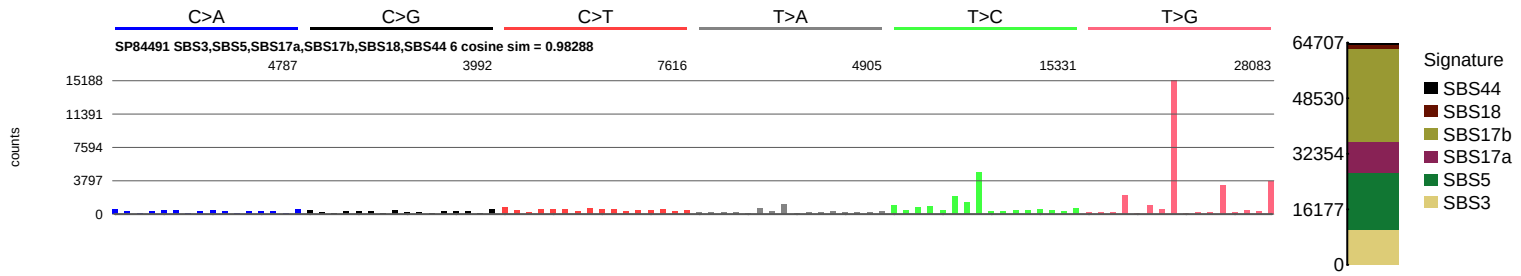

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98293
3171
2796
10470
4072
16292
27907
counts
15272
11454
7636
3818
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS2
SBS1

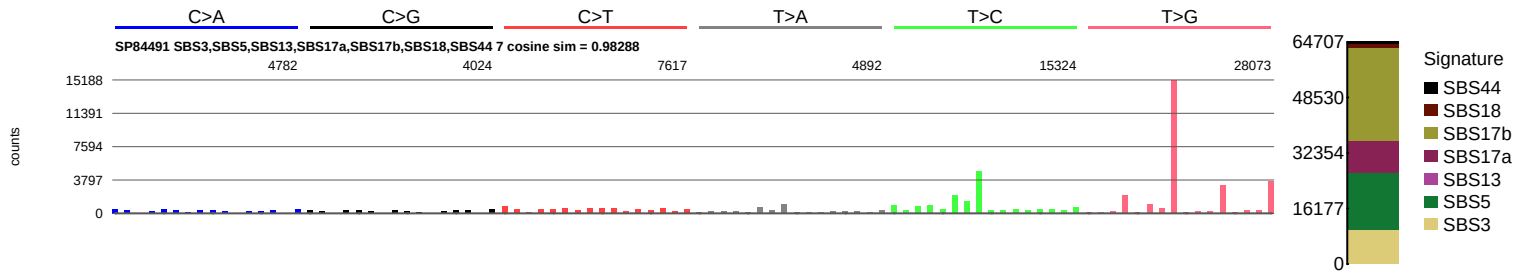

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS2,SBS5,SBS17a,SBS17b 5 cosine sim = 0.98292
3151
2838
10400
4080
16314
27930
counts
15268
11451
7634
3817
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS5
SBS2
SBS1

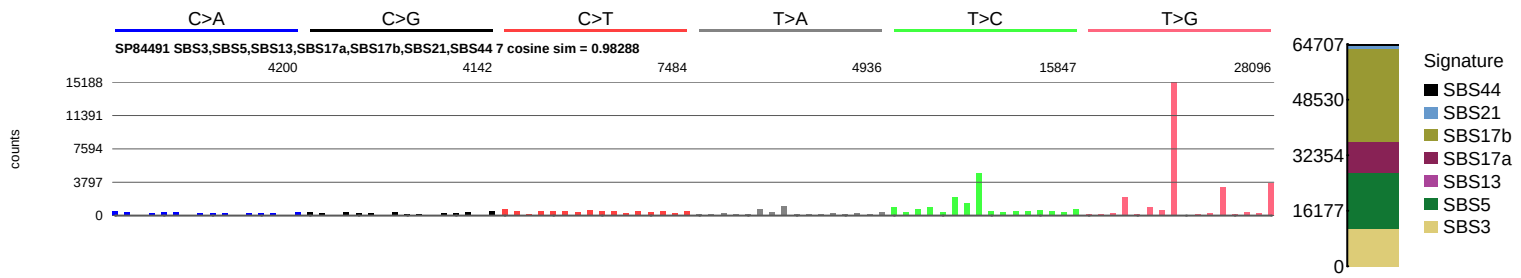

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98291
4498
2976
9339
3906
16271
27710
15252
11439
7626
3813
0
counts
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

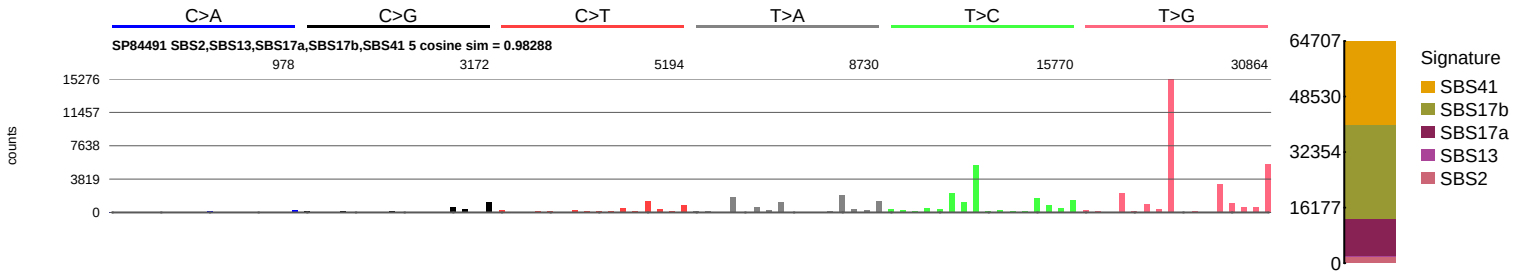

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS17a,SBS17b,SBS41 4 cosine sim = 0.9829
952
3046
5213
8781
15817
30901
15276
11457
7638
3819
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS2

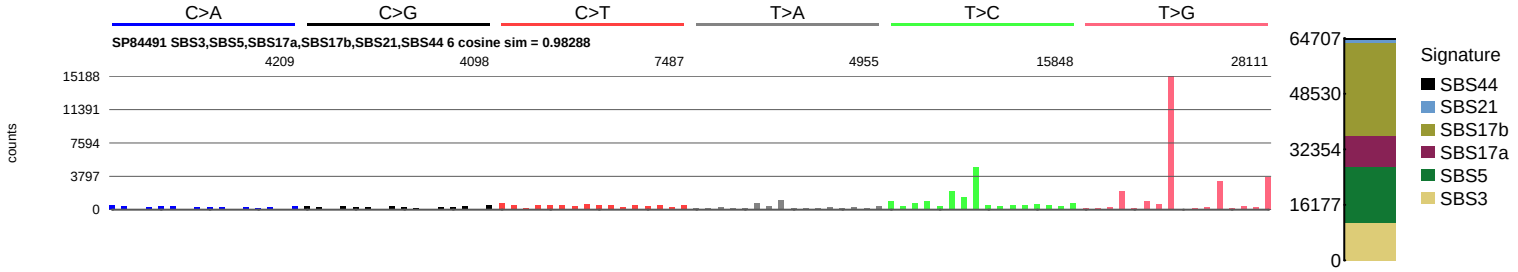

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.9829
3280
3279
9980
4053
16222
27892
15260
11445
7630
3815
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

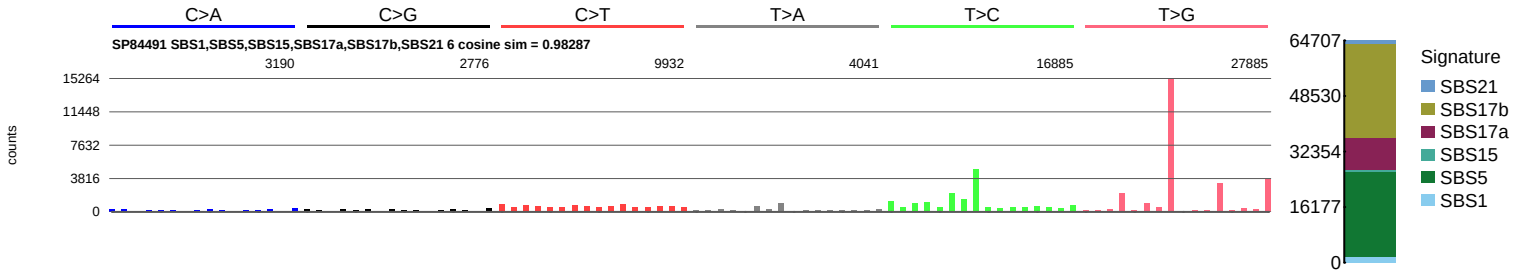

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98288
4787
3992
7616
4905
15331
28083
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3

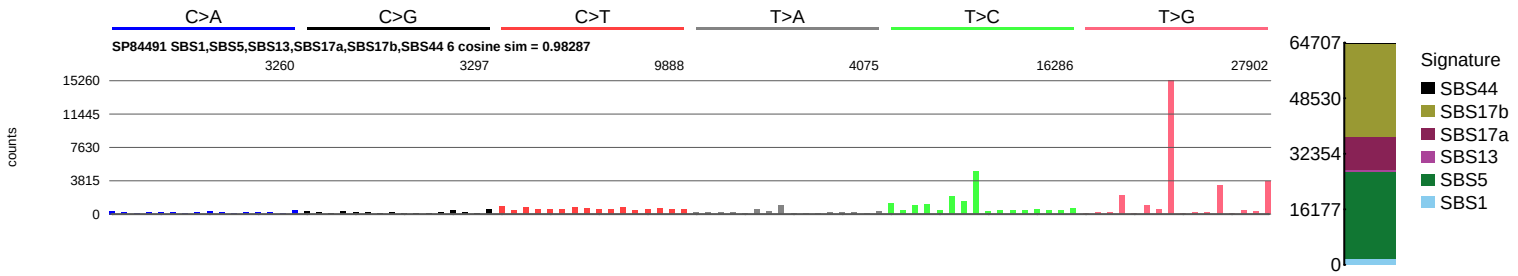

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98288
4782
4024
7617
4892
15324
28073
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3

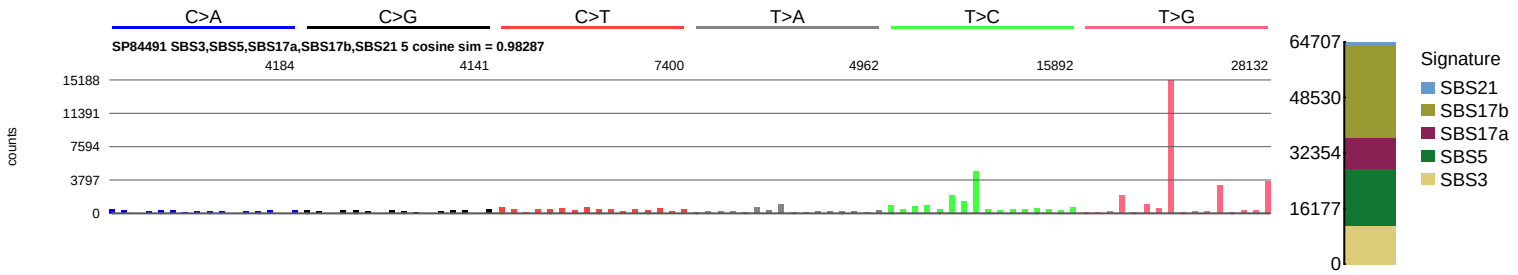

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98288
4200
4142
7484
4936
15847
28096
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3

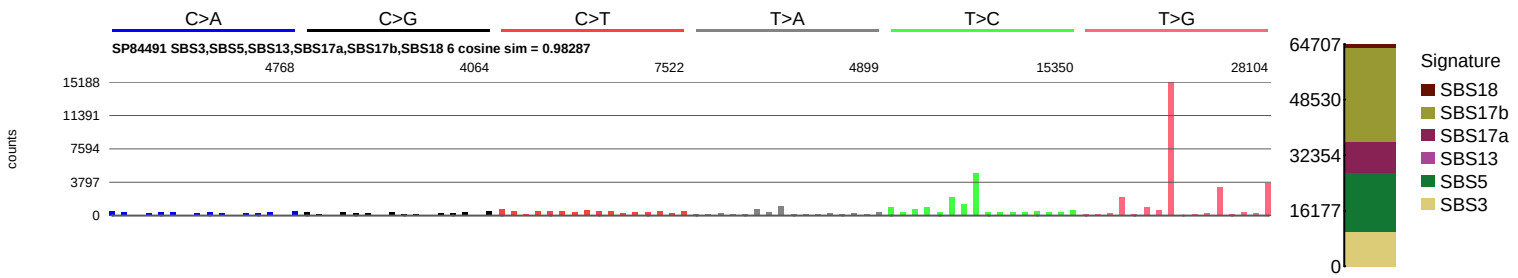

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS13,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98288
978
3172
5194
8730
15770
30864
counts
15276
11457
7638
3819
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS2

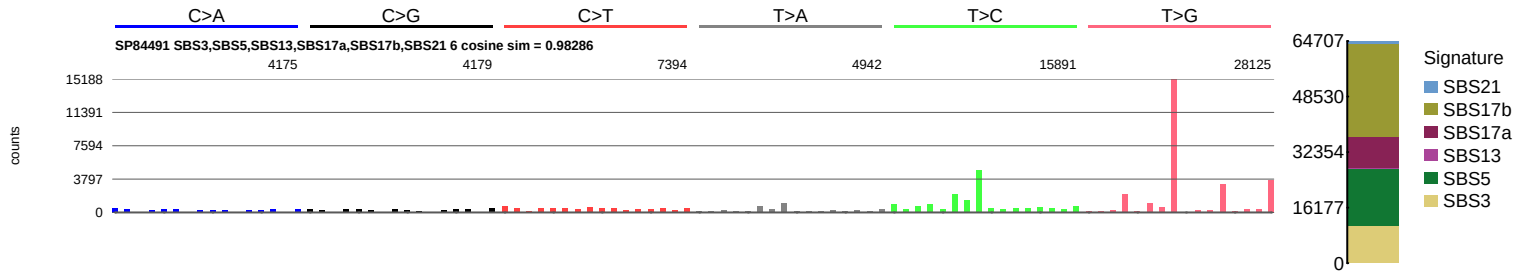

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98288
4209
4098
7487
4955
15848
28111
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS3

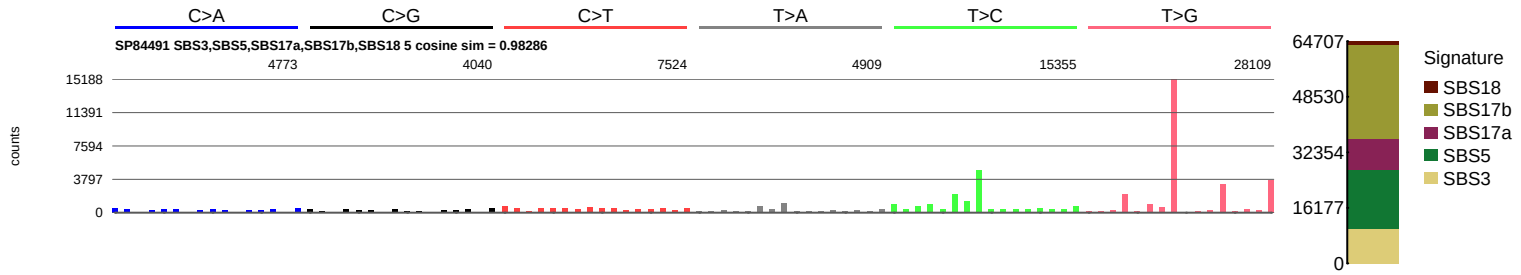

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98287
3190
2776
9932
4041
16885
27885
counts
15264
11448
7632
3816
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

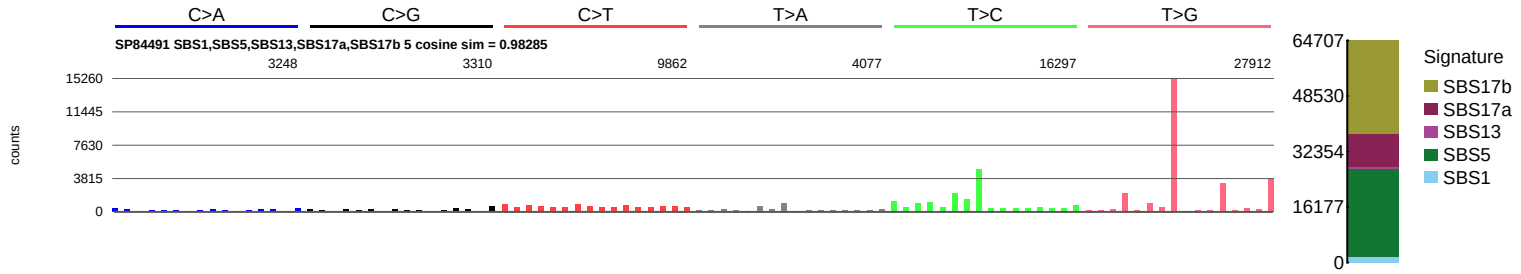

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98287
3260
3297
9888
4075
16286
27902
counts
15260
11445
7630
3815
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS1

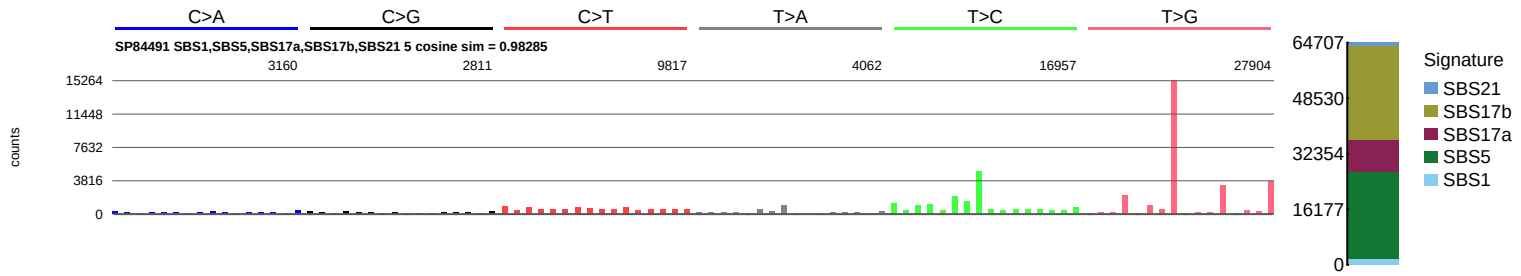

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98287
4184
4141
7400
4962
15892
28132
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3

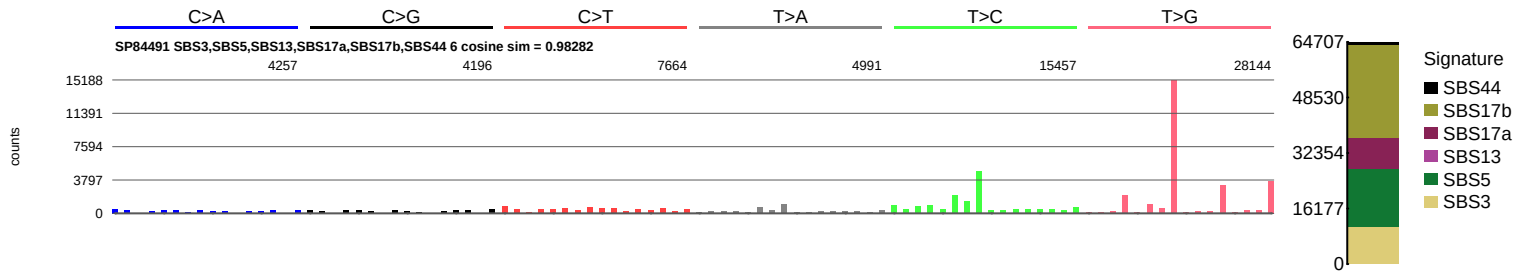

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98287
4768
4064
7522
4899
15350
28104
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3

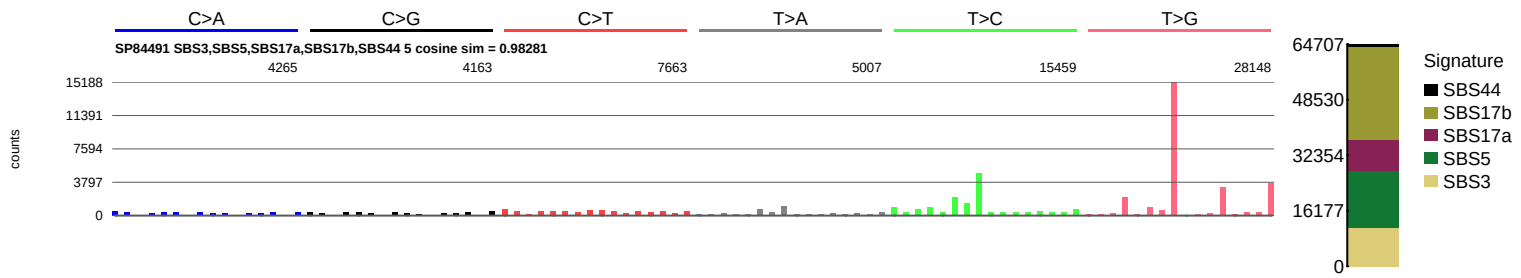

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98286
4175
4179
7394
4942
15891
28125
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3

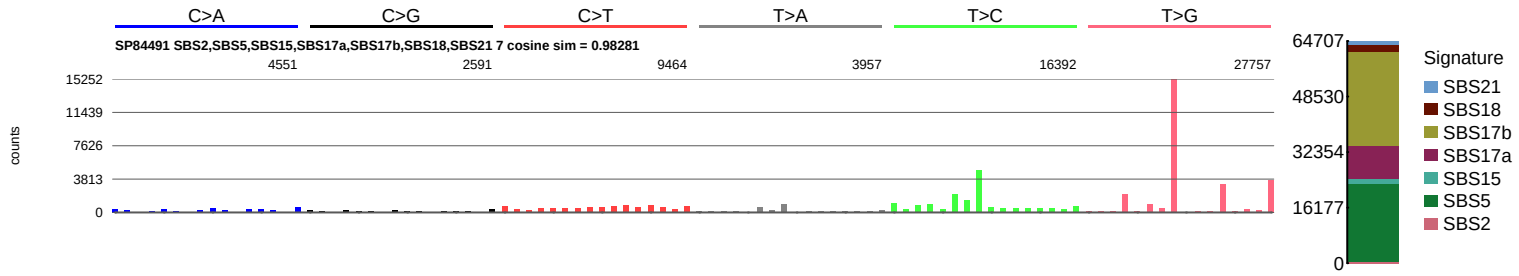

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98286
4773
4040
7524
4909
15355
28109
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3

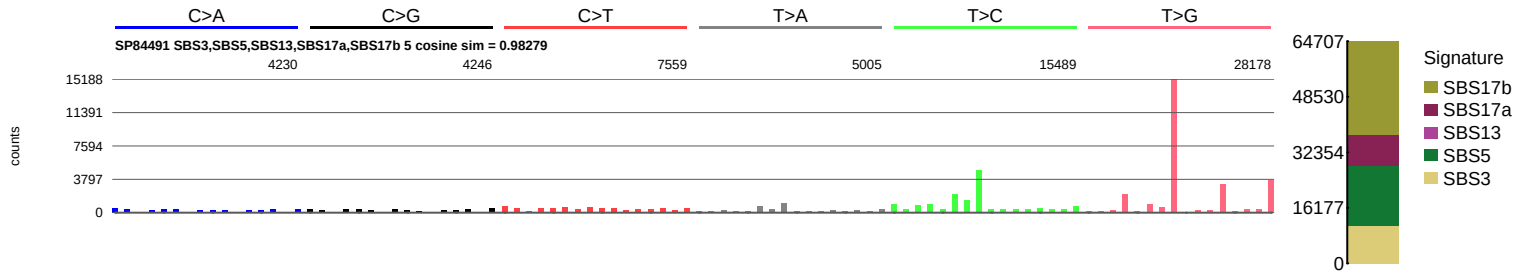

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS13,SBS17a,SBS17b 5 cosine sim = 0.98285
3248
3310
9862
4077
16297
27912
counts
15260
11445
7630
3815
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS1

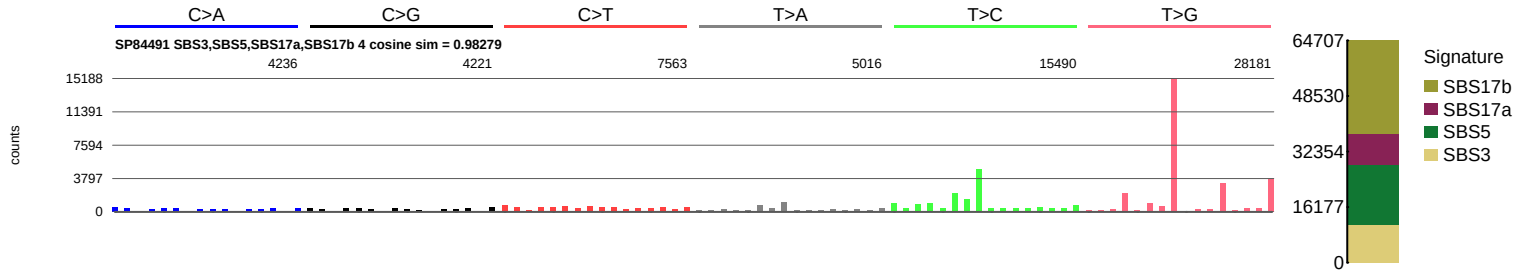

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98285
3160
2811
9817
4062
16957
27904
counts
15264
11448
7632
3816
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS1

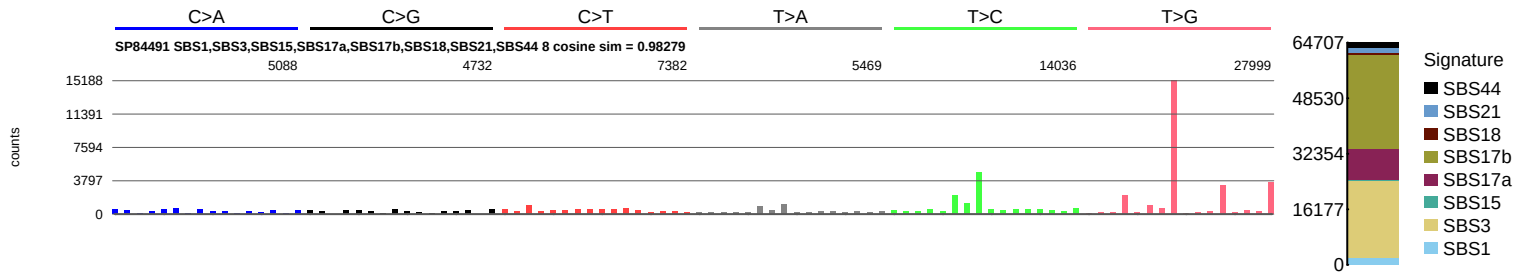

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98282
4257
4196
7664
4991
15457
28144
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3

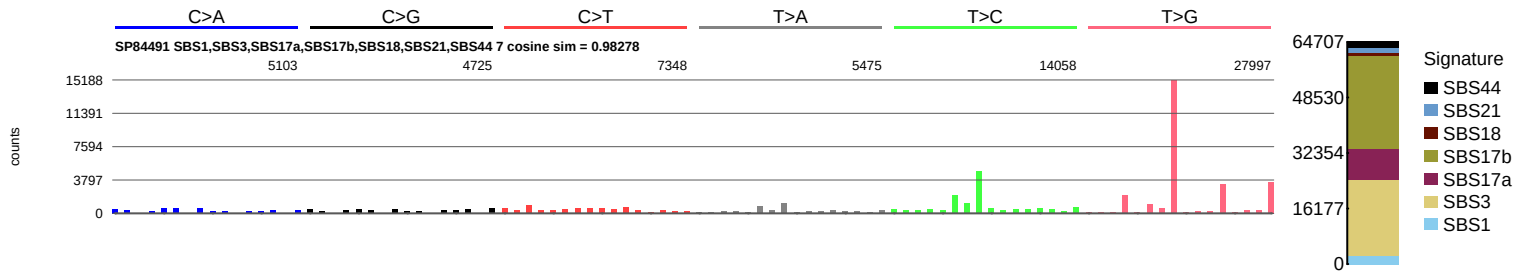

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98281
4265
4163
7663
5007
15459
28148
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3

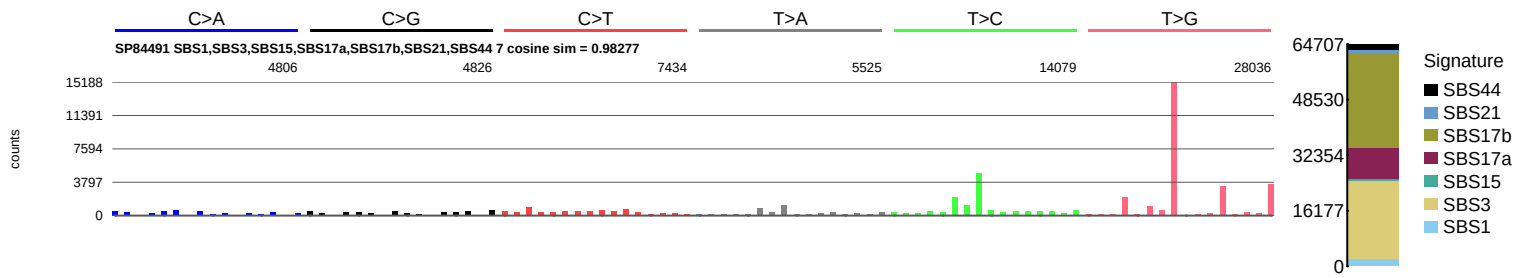

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98281
4551
2591
9464
3957
16392
27757
15252
11439
7626
3813
0
counts
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

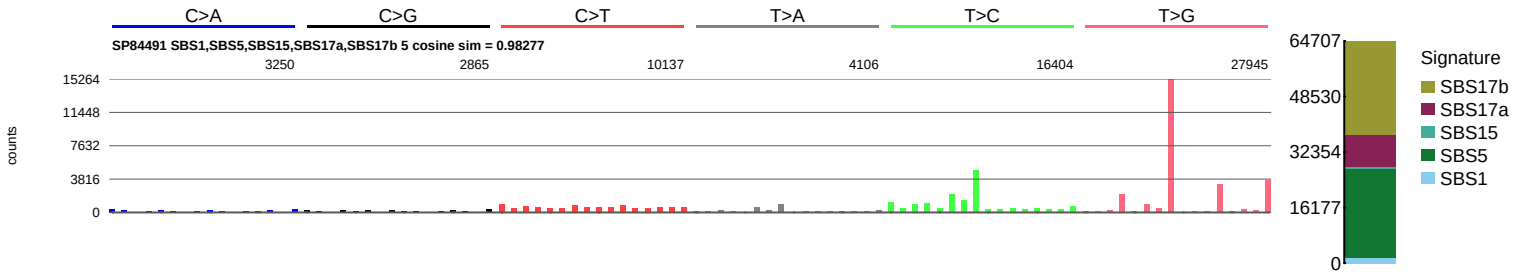

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS13,SBS17a,SBS17b 5 cosine sim = 0.98279
4230
4246
7559
5005
15489
28178
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3

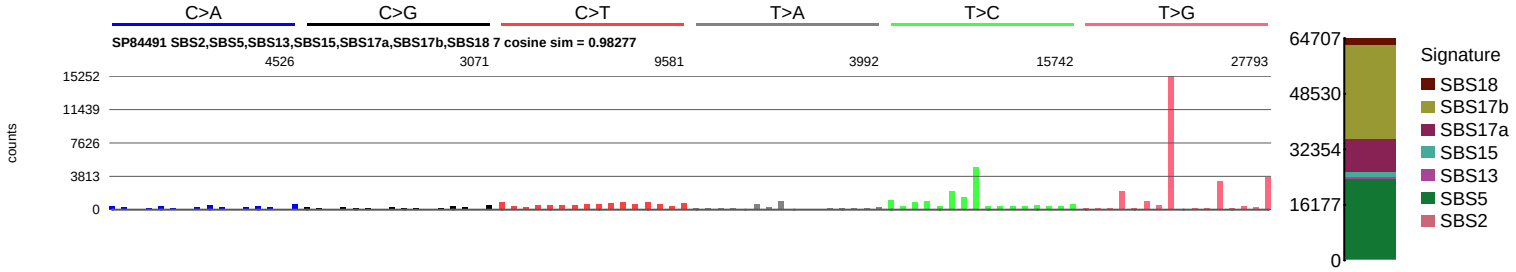

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS5,SBS17a,SBS17b 4 cosine sim = 0.98279
4236
4221
7563
5016
15490
28181
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS5
SBS3

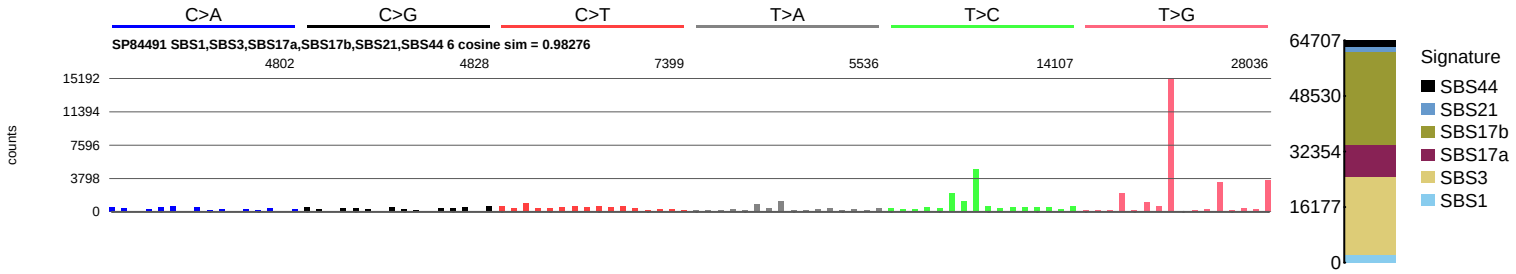

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98279
5088
4732
7382
5469
14036
27999
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

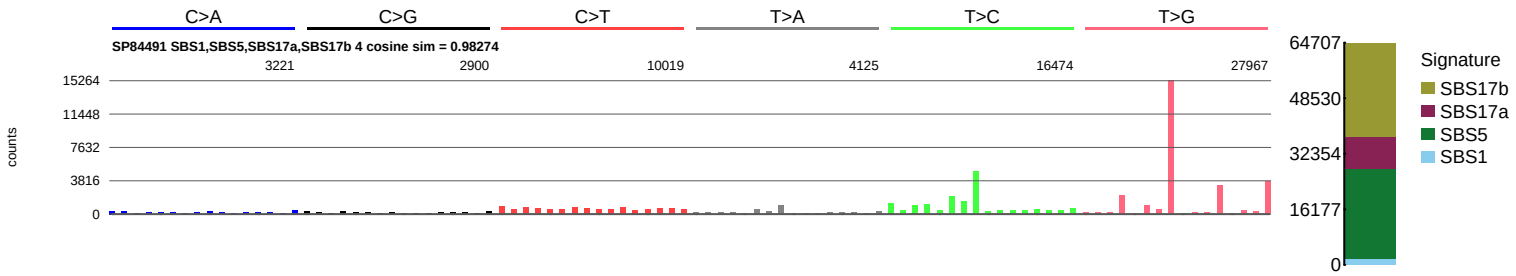

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98278
5103
4725
7348
5475
14058
27997
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98277
4806
4826
7434
5525
14079
28036
15188
11391
7594
3797
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98277
3250
2865
10137
4106
16404
27945
15264
11448
7632
3816
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98277
4526
3071
9581
3992
15742
27793
15252
11439
7626
3813
0
counts
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98276
4802
4828
7399
5536
14107
28036
15192
11394
7596
3798
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS1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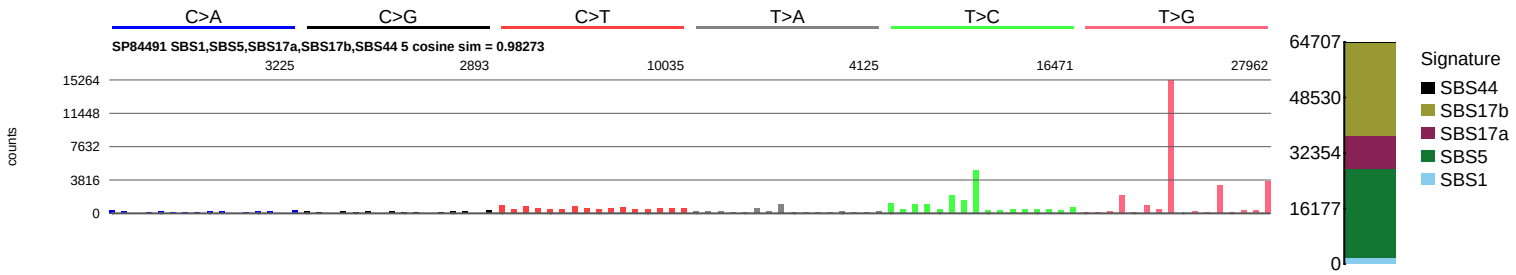
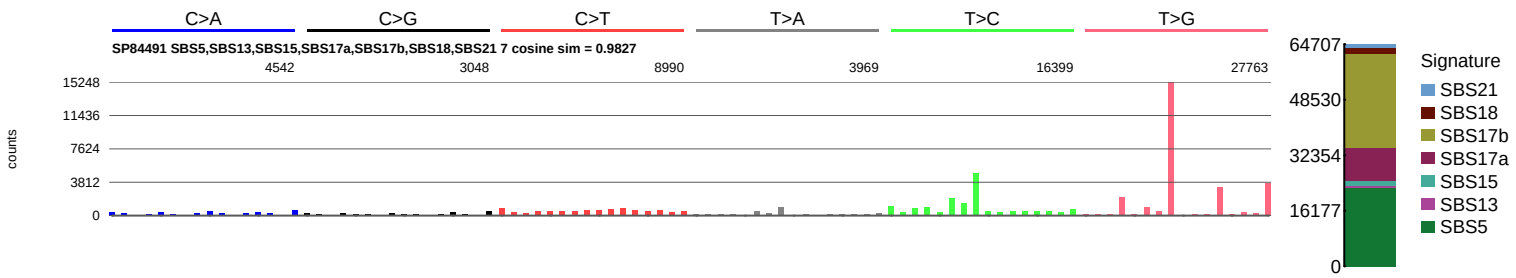
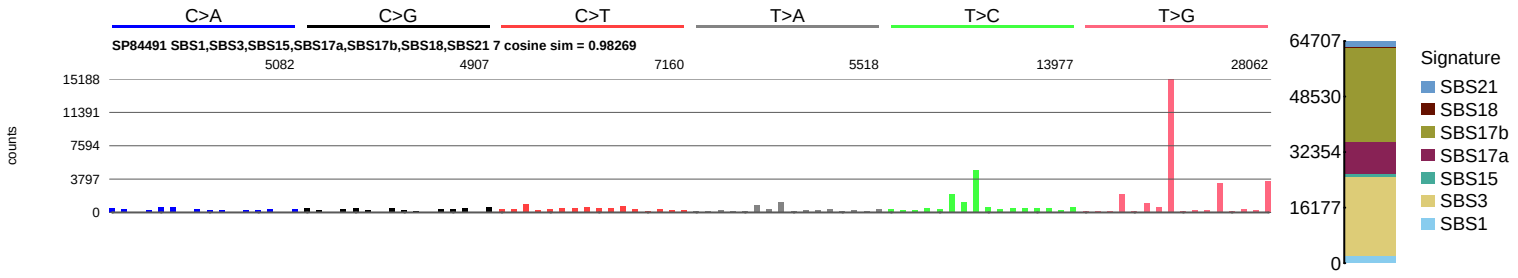
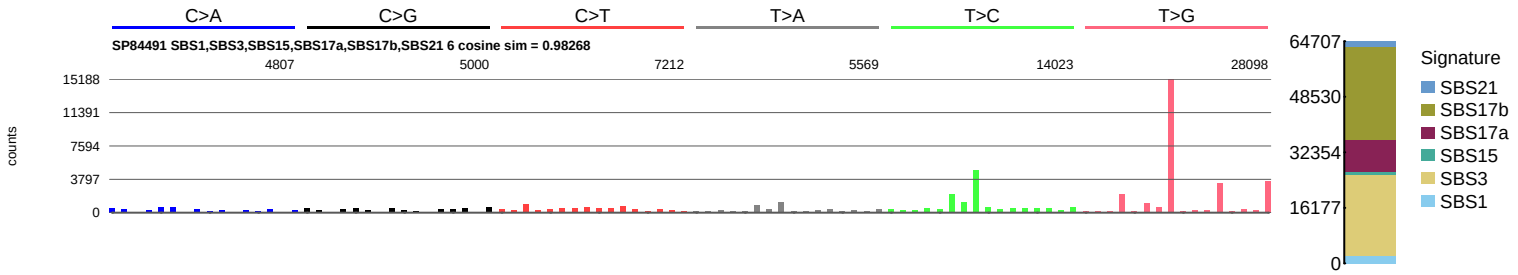
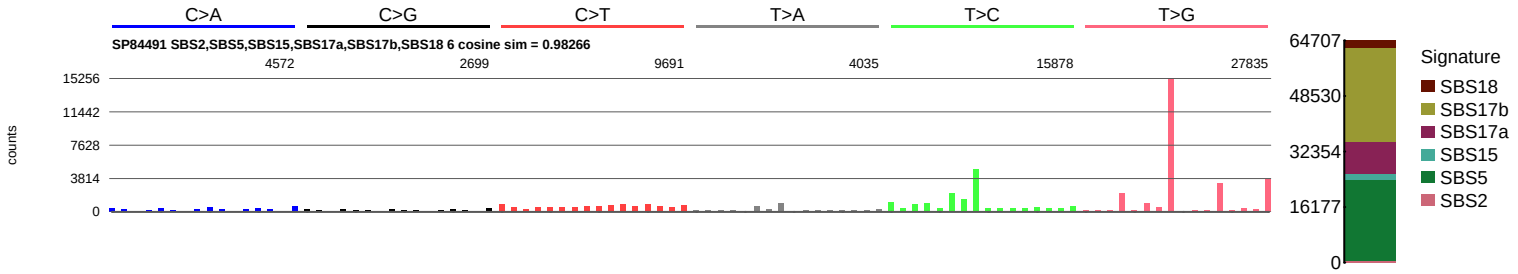
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS17a,SBS17b 4 cosine sim = 0.98274
3221
2900
10019
4125
16474
27967
15264
11448
7632
3816
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS5
SBS1

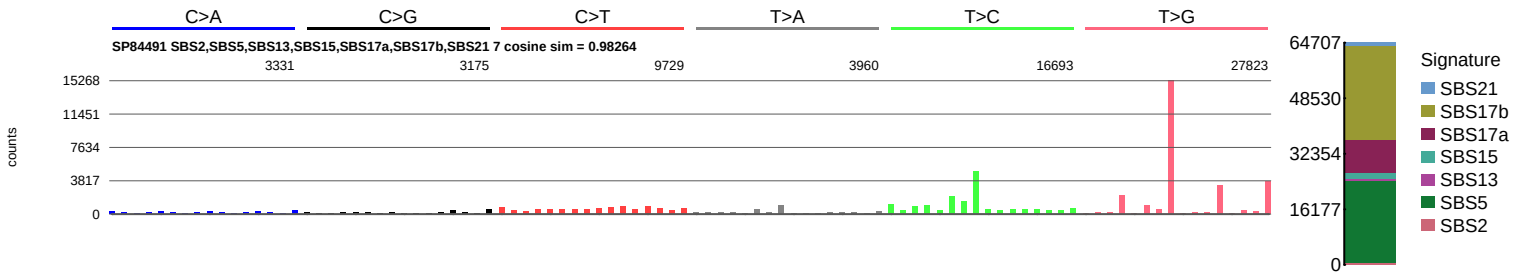

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS5,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98273
3225
2893
10035
4125
16471
27962
15264
11448
7632
3816
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS1

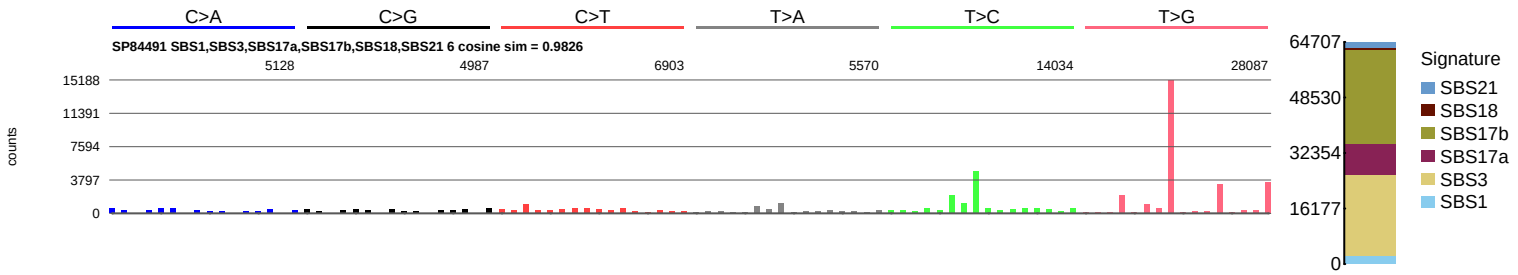

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.9827
4542
3048
8990
3969
16399
27763
15248
11436
7624
3812
0
counts
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

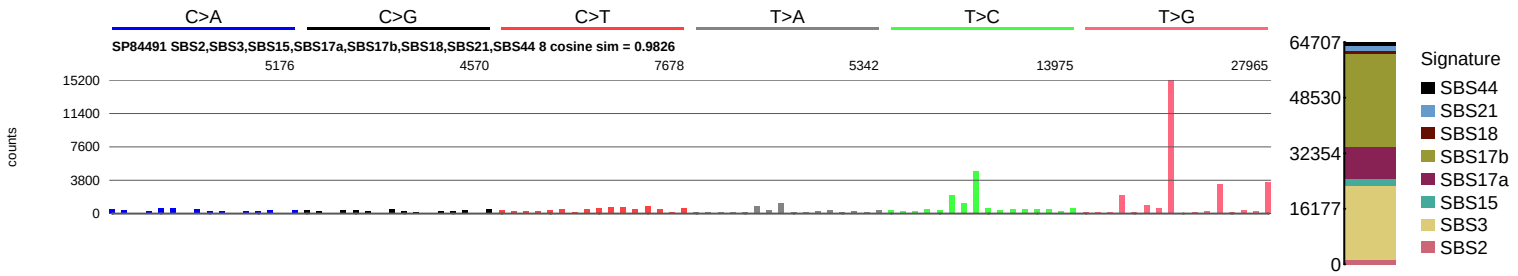

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98269
5082
4907
7160
5518
13977
28062
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

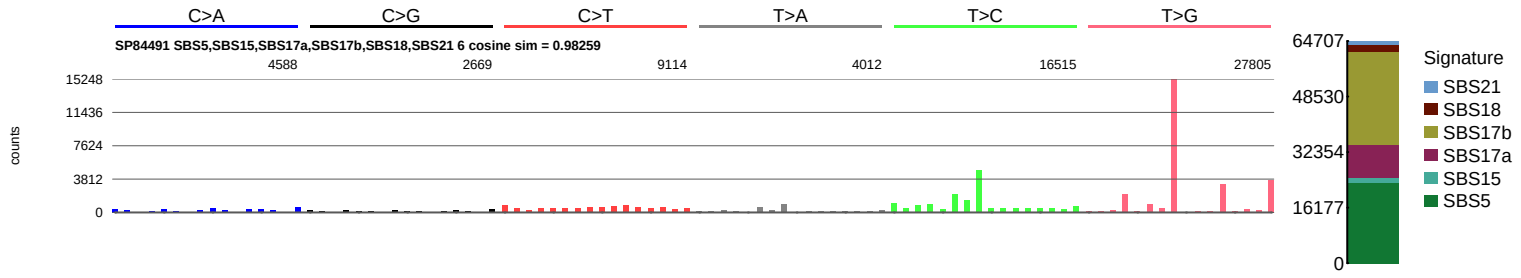

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98268
4807
5000
7212
5569
14023
28098
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

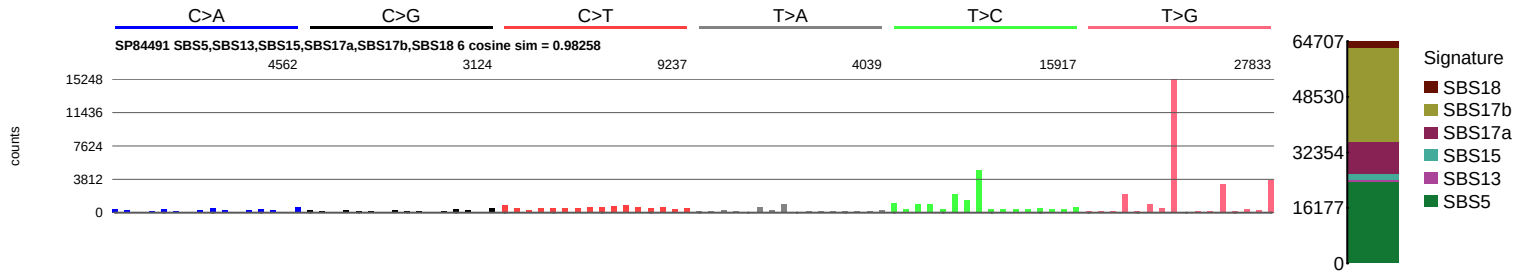

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98266
4572
2699
9691
4035
15878
27835
counts
15256
11442
7628
3814
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

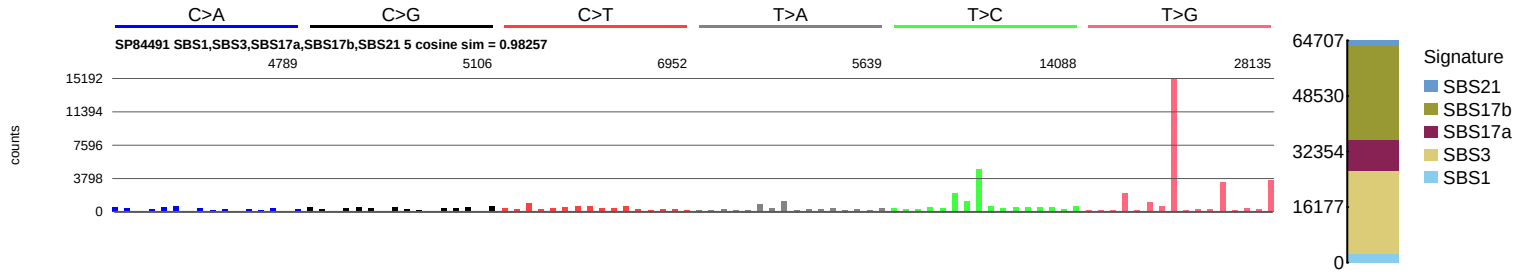

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98264
3331
3175
9729
3960
16693
27823
counts
15268
11451
7634
3817
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

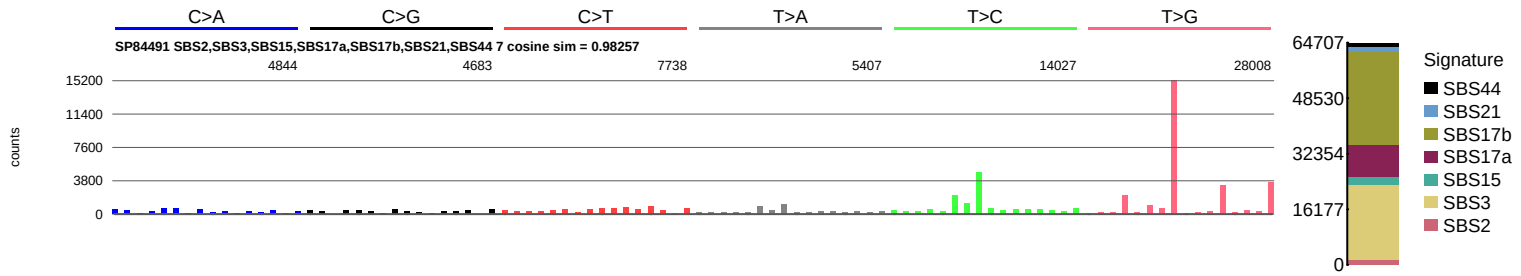

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.9826
5128
4987
6903
5570
14034
28087
counts
15188
11391
7594
3797
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

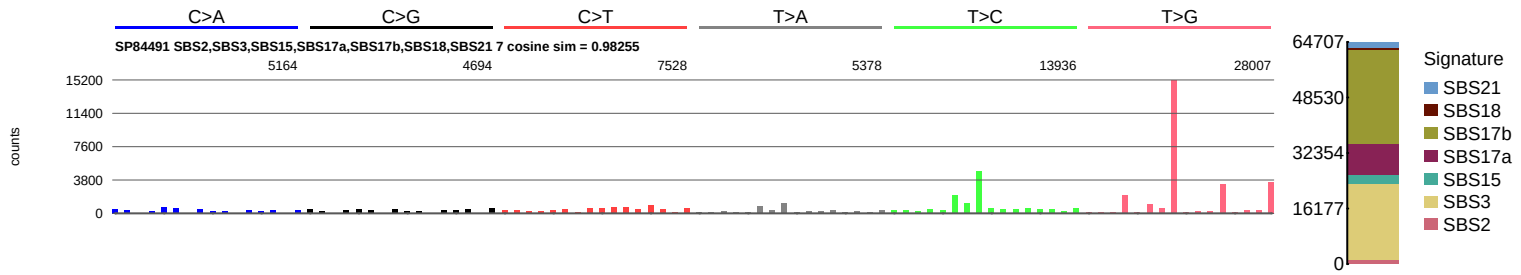

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.9826
5176
4570
7678
5342
13975
27965
counts
15200
11400
7600
3800
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

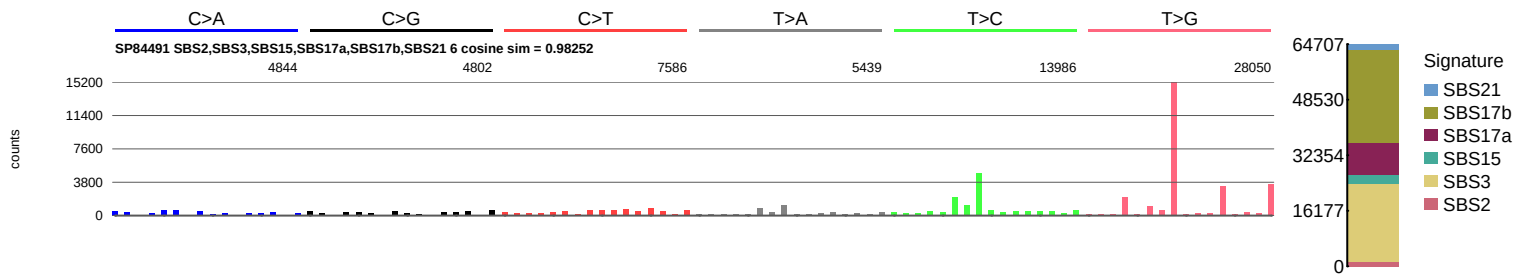

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98259
4588
2669
9114
4012
16515
27805
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5

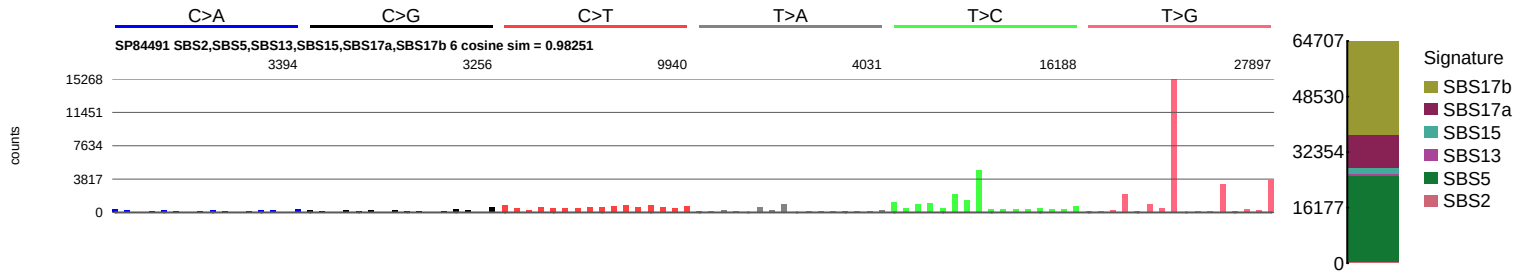

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98258
4562
3124
9237
4039
15917
27833
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98257
4789
5106
6952
5639
14088
28135
counts
15192
11394
7596
3798
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98257
4844
4683
7738
5407
14027
28008
counts
15200
11400
7600
3800
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98255
5164
4694
7528
5378
13936
28007
counts
15200
11400
7600
3800
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98252
4844
4802
7586
5439
13986
28050
counts
15200
11400
7600
3800
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98251
3394
3256
9940
4031
16188
27897
15268
11451
7634
3817
0
counts
64707
48530
32354
16177
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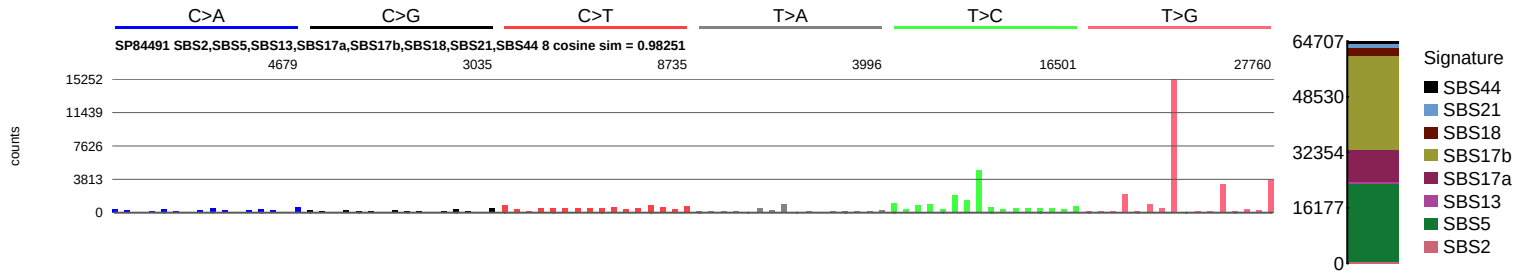
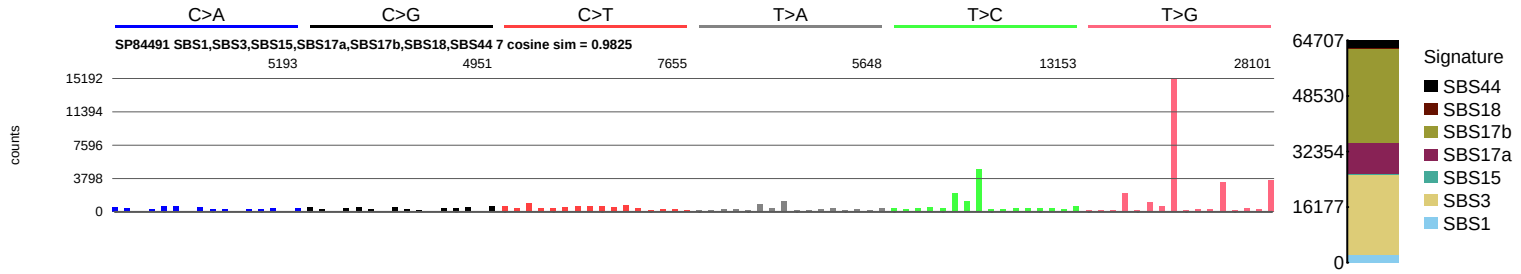
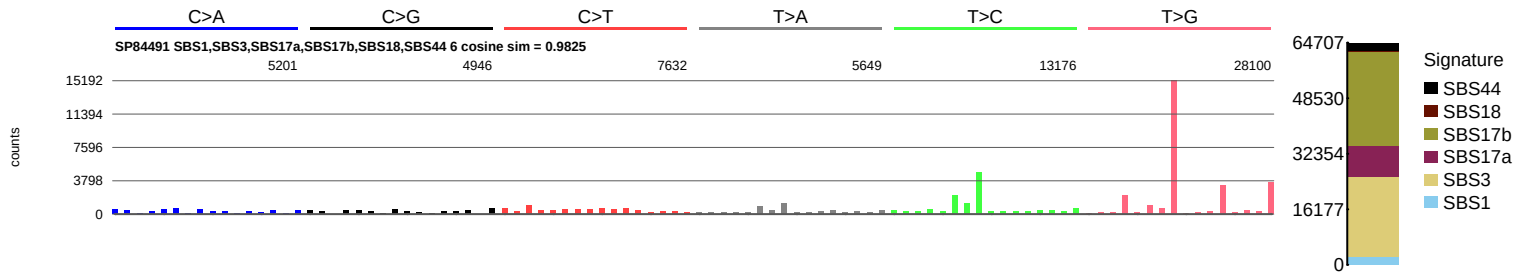
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98251
4679
3035
8735
3996
16501
27760
15252
11439
7626
3813
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.9825
5193
4951
7655
5648
13153
28101
15192
11394
7596
3798
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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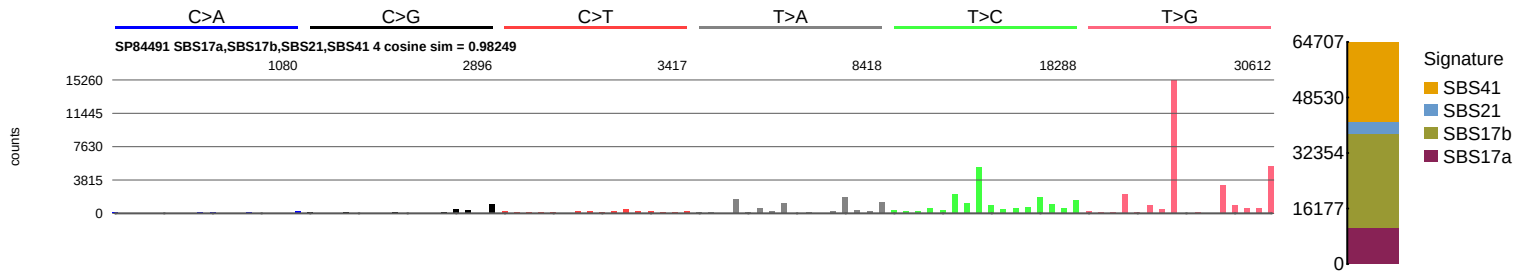
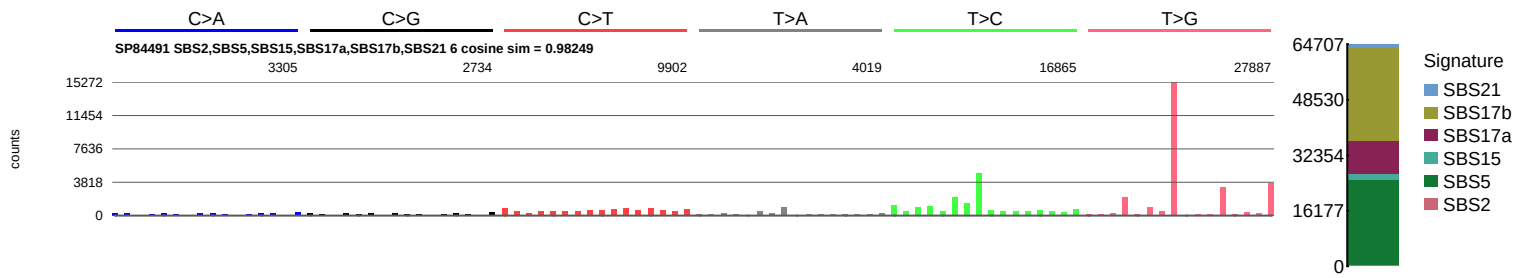
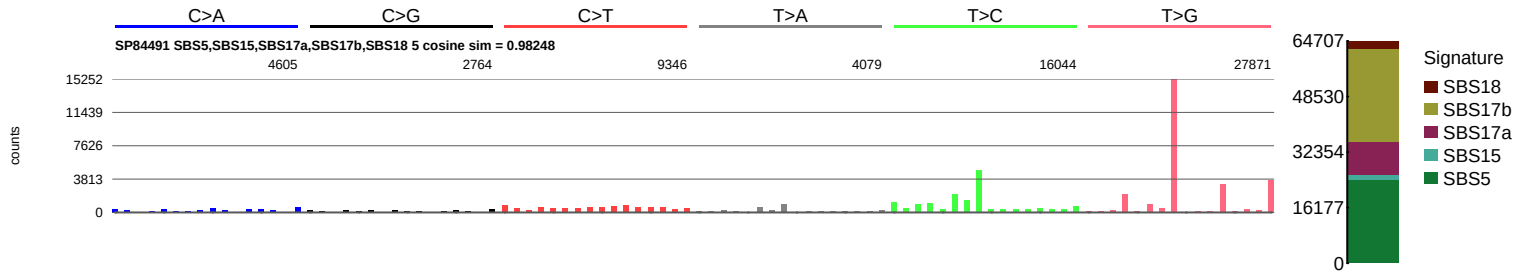
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.9825
5201
4946
7632
5649
13176
28100
15192
11394
7596
3798
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS21,SBS41 4 cosine sim = 0.98249
1080
2896
3417
8418
18288
30612
15260
11445
7630
3815
0
counts
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98249
3305
2734
9902
4019
16865
27887
15272
11454
7636
3818
0
counts
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2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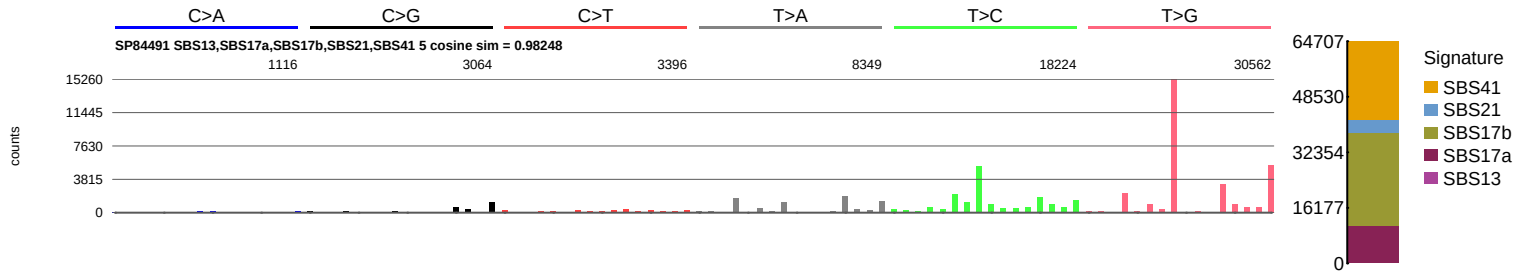
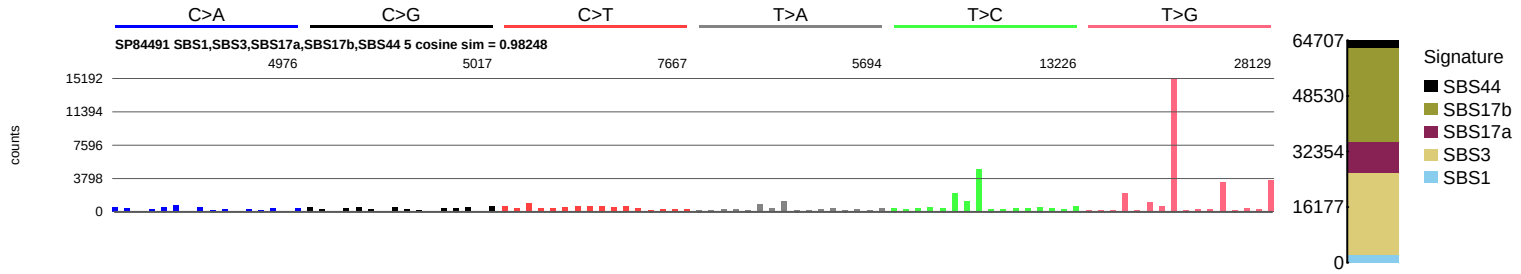
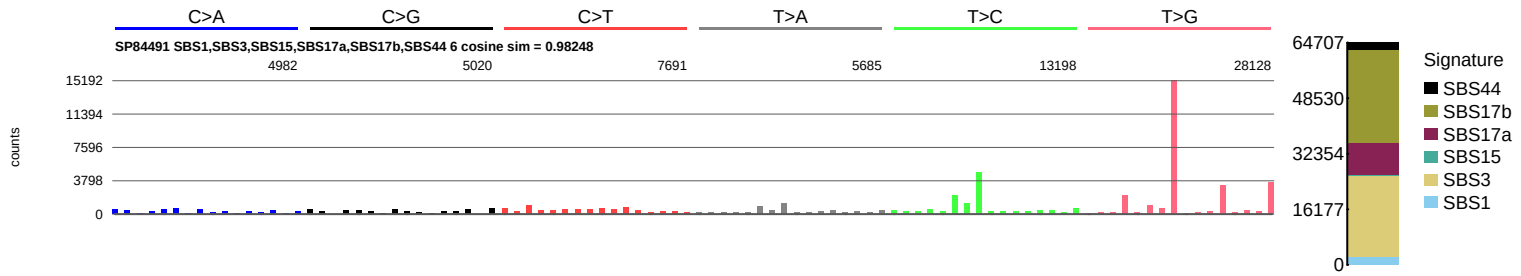
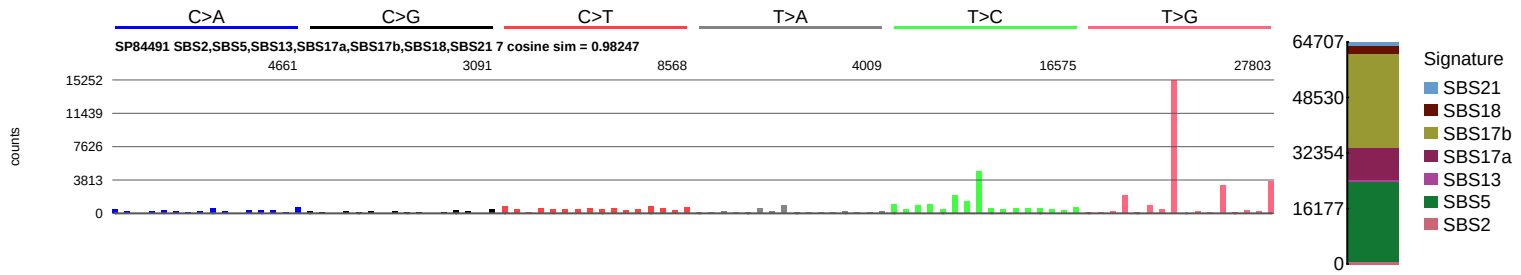
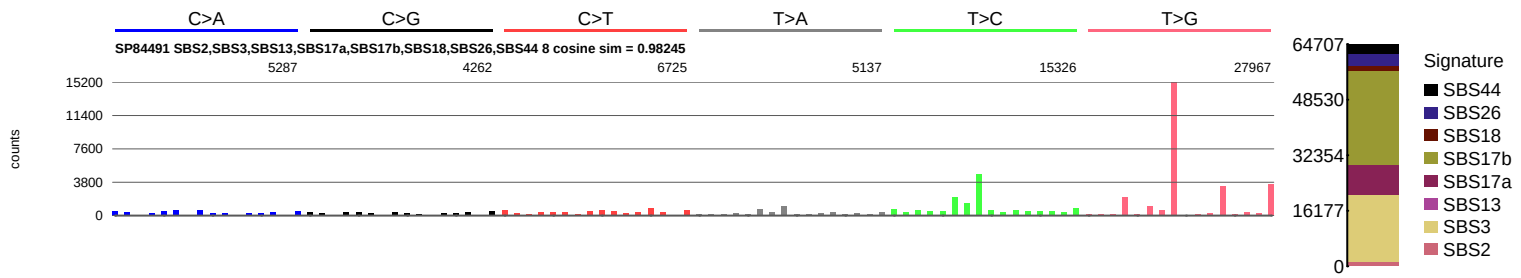
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS15,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98248
4605
2764
9346
4079
16044
27871
counts
15252
11439
7626
3813
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5

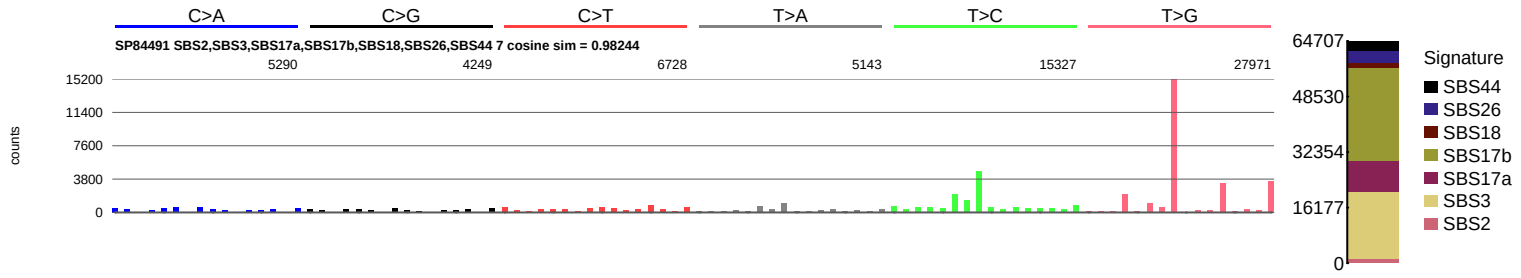

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98248
1116
3064
3396
8349
18224
30562
counts
15260
11445
7630
3815
0
64707
48530
32354
16177
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13

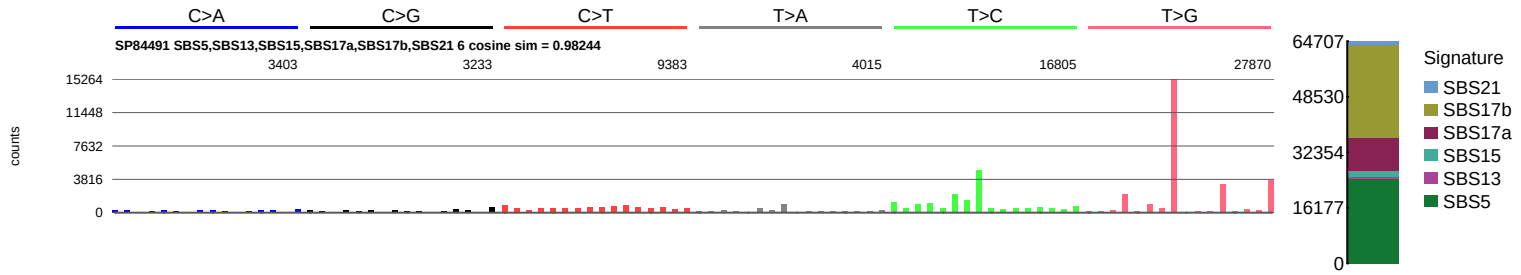

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98248
4976
5017
7667
5694
13226
28129
counts
15192
11394
7596
3798
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS3
SBS1

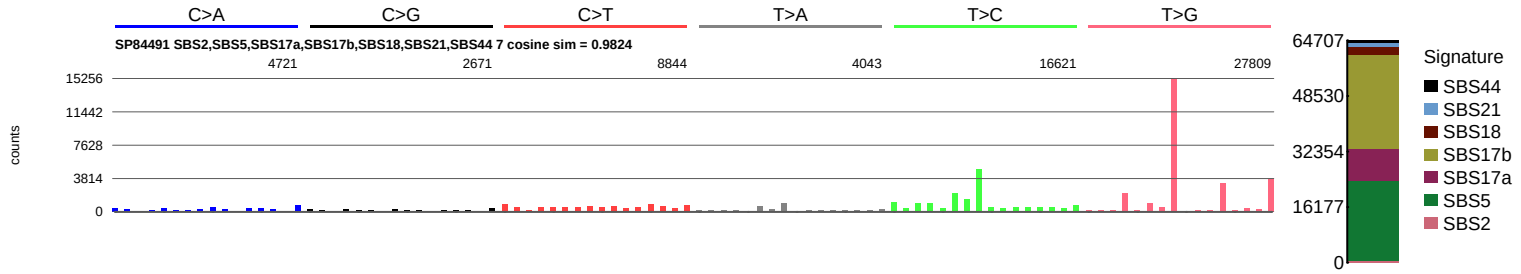

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98248
4982
5020
7691
5685
13198
28128
counts
15192
11394
7596
3798
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS3
SBS1

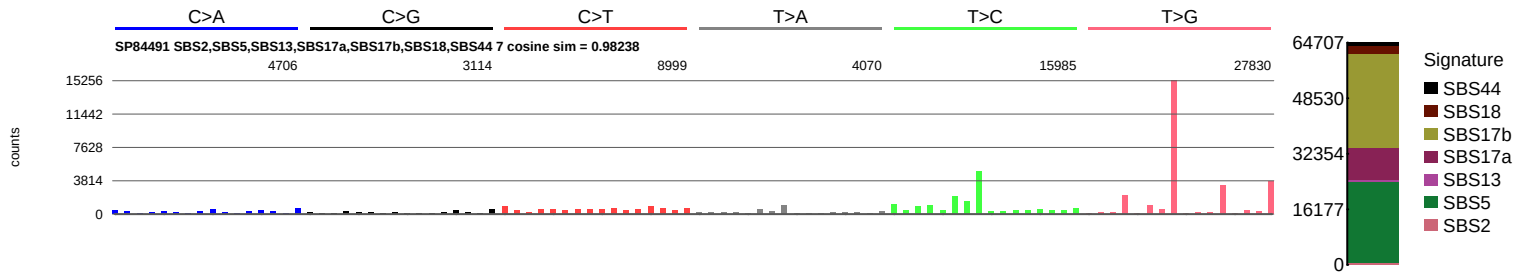

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98247
4661
3091
8568
4009
16575
27803
counts
15252
11439
7626
3813
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

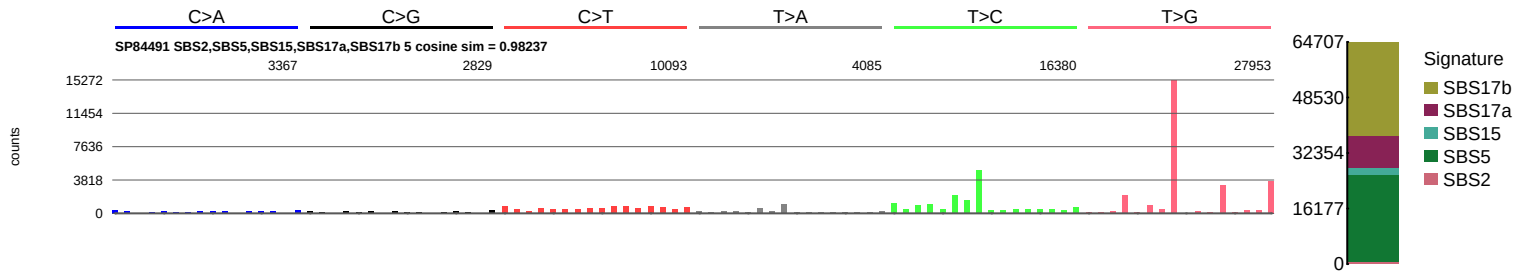

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98245
5287
4262
6725
5137
15326
27967
counts
15200
11400
7600
3800
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

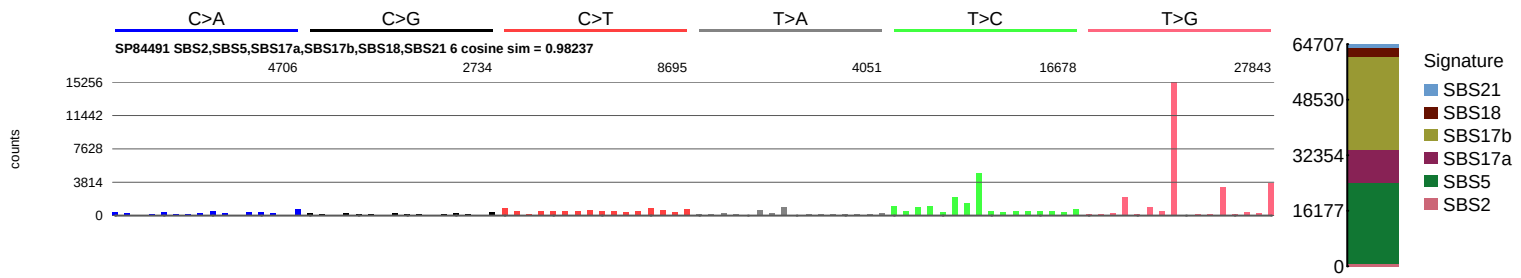

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98244
5290
4249
6728
5143
15327
27971
counts
15200
11400
7600
3800
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2

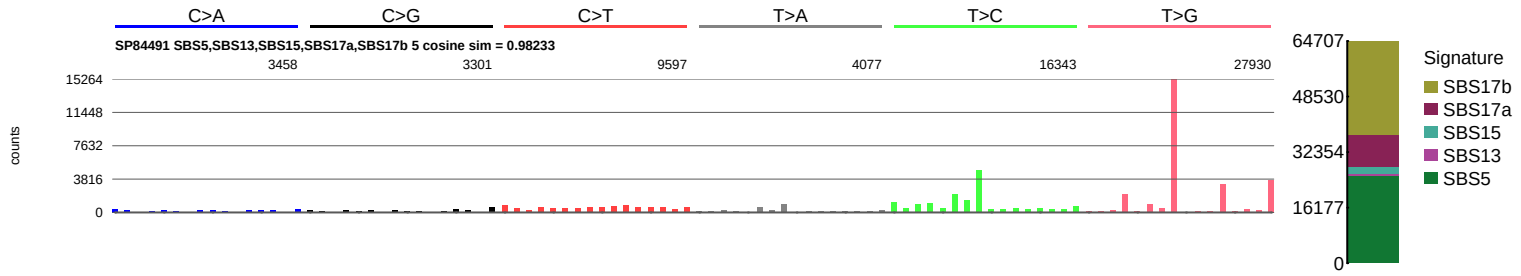

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98244
3403
3233
9383
4015
16805
27870
counts
15264
11448
7632
3816
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5

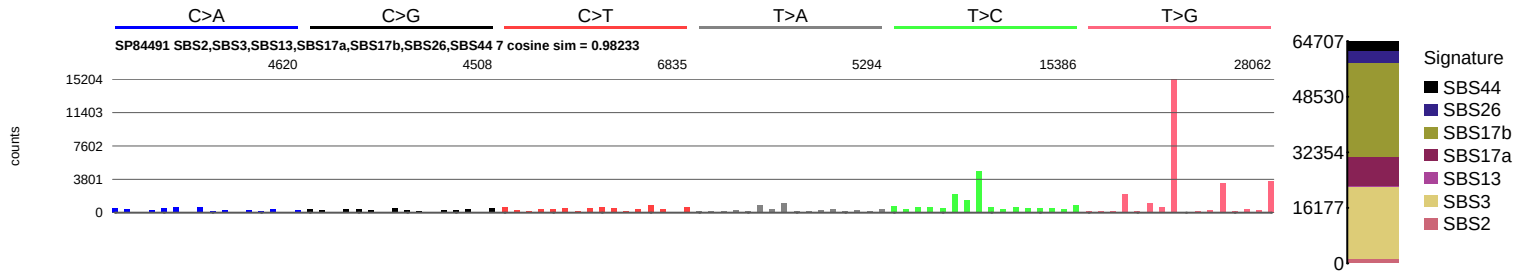

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.9824
4721
2671
8844
4043
16621
27809
counts
15256
11442
7628
3814
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

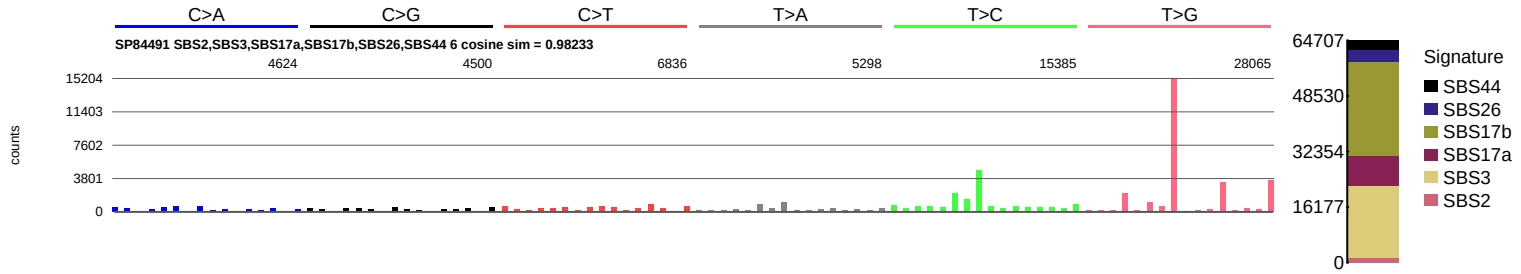

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98238
4706
3114
8999
4070
15985
27830
counts
15256
11442
7628
3814
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

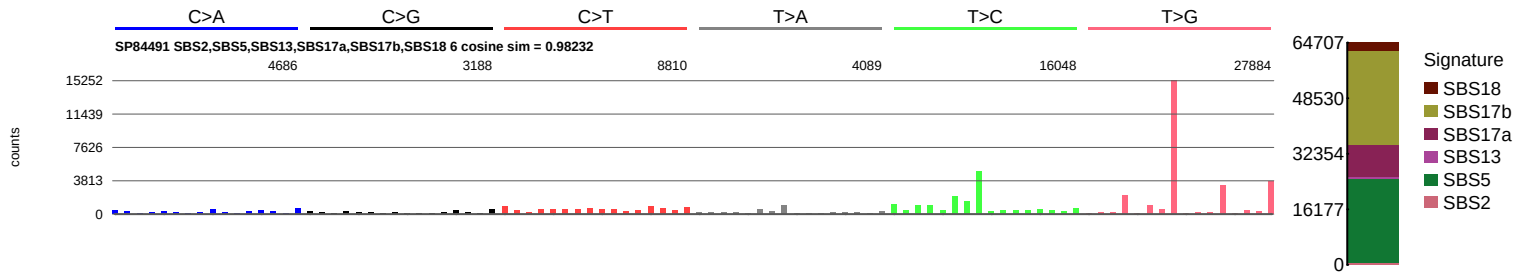

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98237
3367
2829
10093
4085
16380
27953
counts
15272
11454
7636
3818
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS2

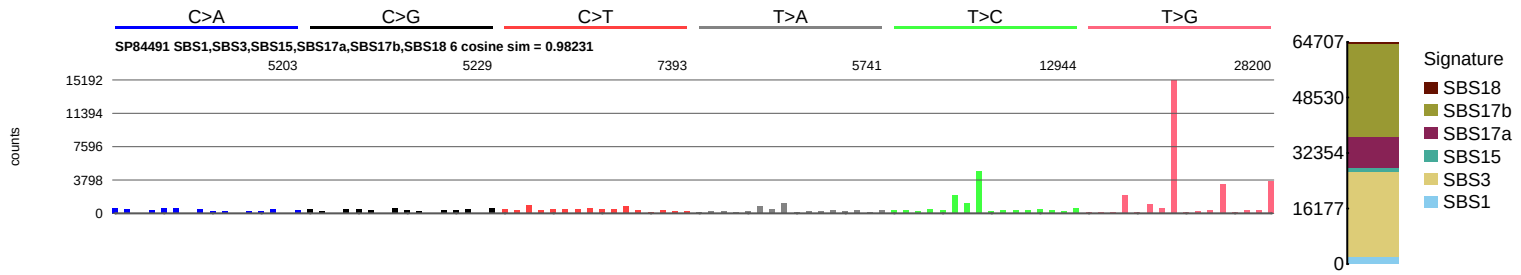

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98237
4706
2734
8695
4051
16678
27843
counts
15256
11442
7628
3814
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

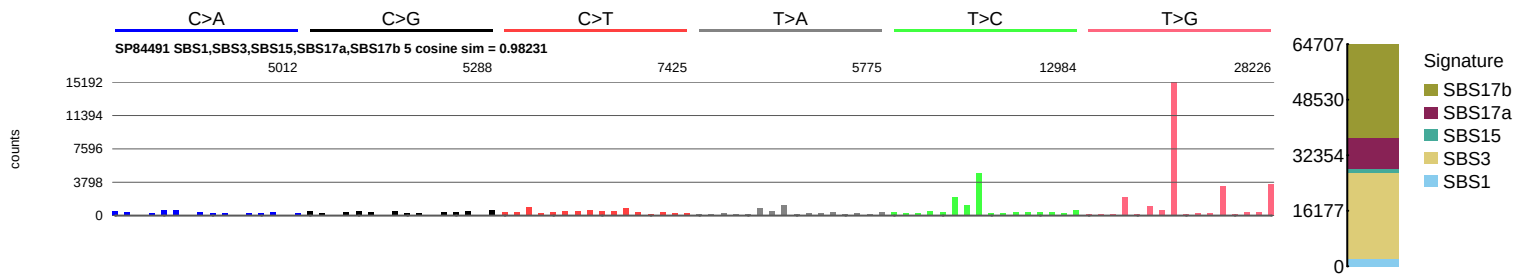

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS13,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98233
3458
3301
9597
4077
16343
27930
15264
11448
7632
3816
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5

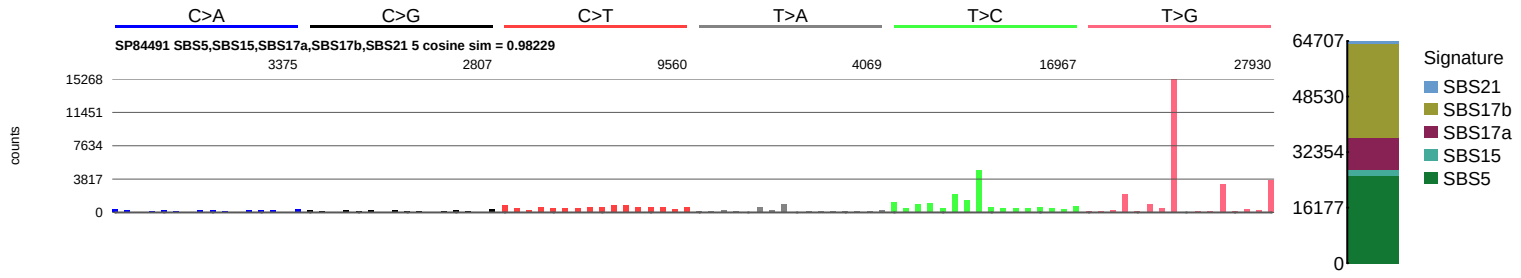

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98233
4620
4508
6835
5294
15386
28062
15204
11403
7602
3801
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2

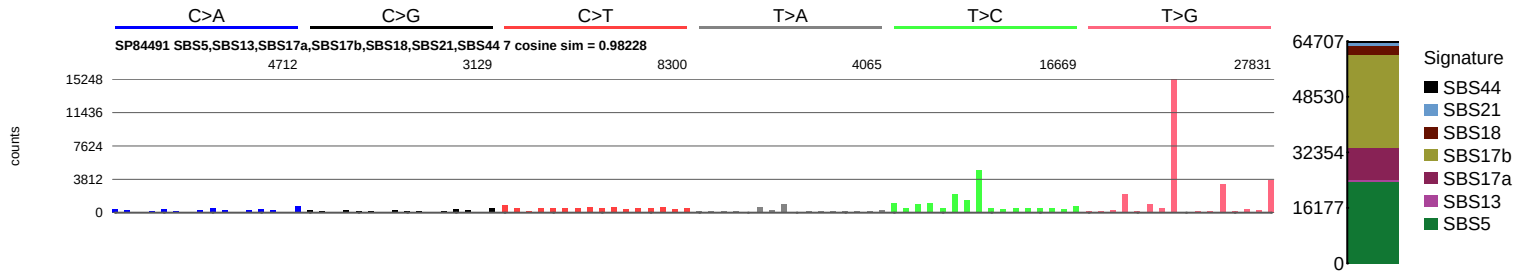

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98233
4624
4500
6836
5298
15385
28065
15204
11403
7602
3801
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS2

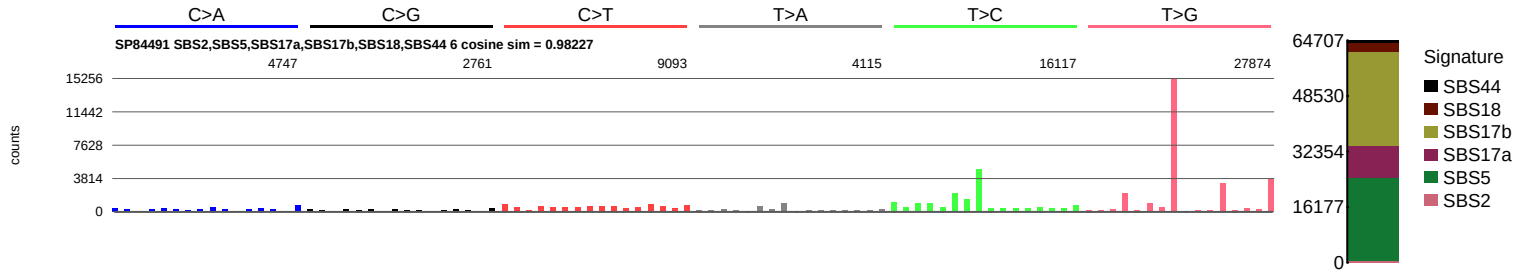

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98232
4686
3188
8810
4089
16048
27884
15252
11439
7626
3813
0
counts
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

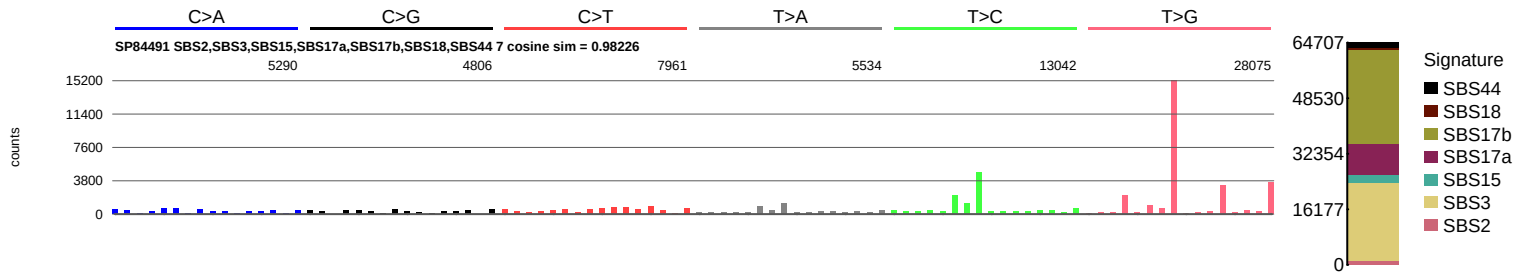

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98231
5203
5229
7393
5741
12944
28200
15192
11394
7596
3798
0
counts
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

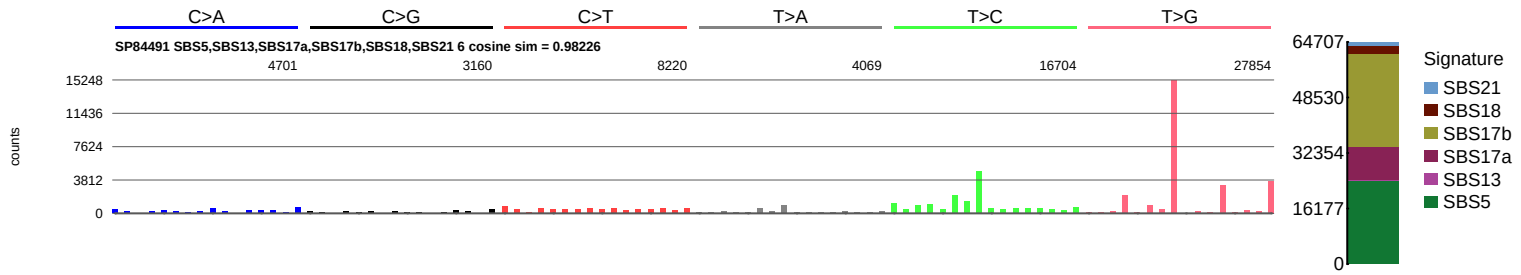

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS1,SBS3,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98231
5012
5288
7425
5775
12984
28226
15192
11394
7596
3798
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS15,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98229
3375
2807
9560
4069
16967
27930
counts
15268
11451
7634
3817
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98228
4712
3129
8300
4065
16669
27831
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98227
4747
2761
9093
4115
16117
27874
counts
15256
11442
7628
3814
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98226
5290
4806
7961
5534
13042
28075
counts
15200
11400
7600
3800
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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SP84491 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98226
4701
3160
8220
4069
16704
27854
counts
15248
11436
7624
3812
0
64707
48530
32354
16177
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SBS21
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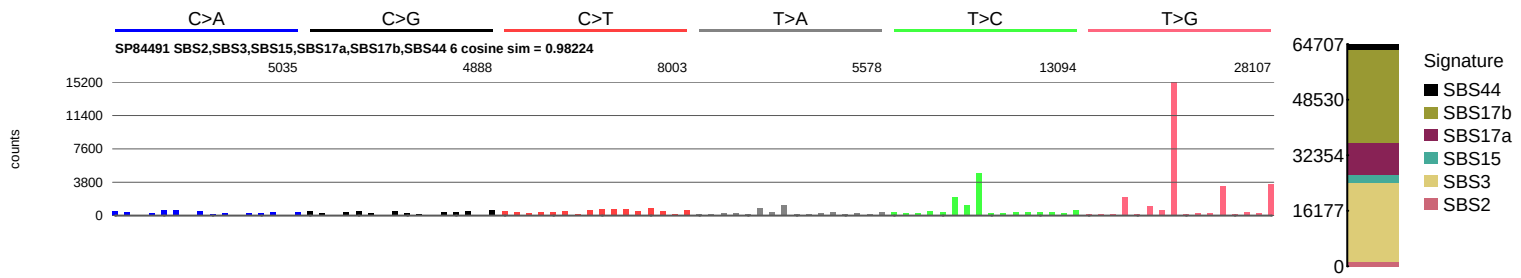
SBS17b
SBS17a
SBS13
SBS5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SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98224
5035
4888
8003
5578
13094
28107
counts
15200
11400
7600
3800
0
64707
48530
32354
16177
0
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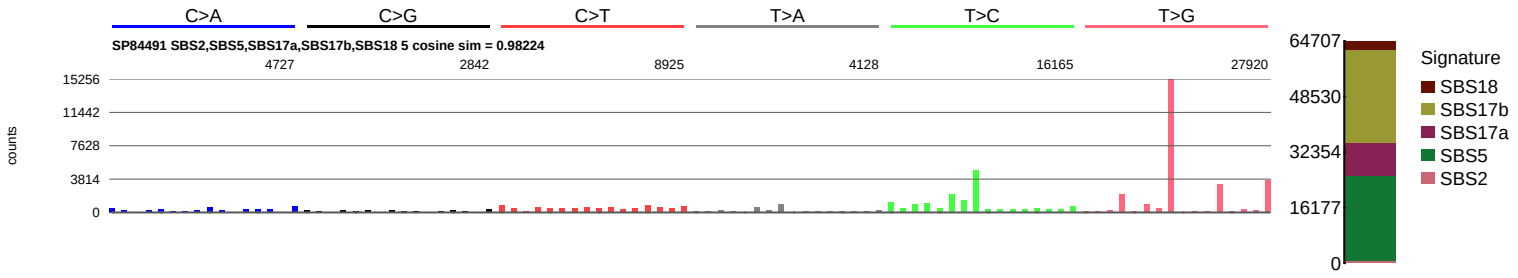
SBS44
SBS17b
SBS17a
SBS15
SBS3
SBS2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SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98224
4727
2842
8925
4128
16165
27920
15256
11442
7628
3814
0
counts
64707
48530
32354
16177
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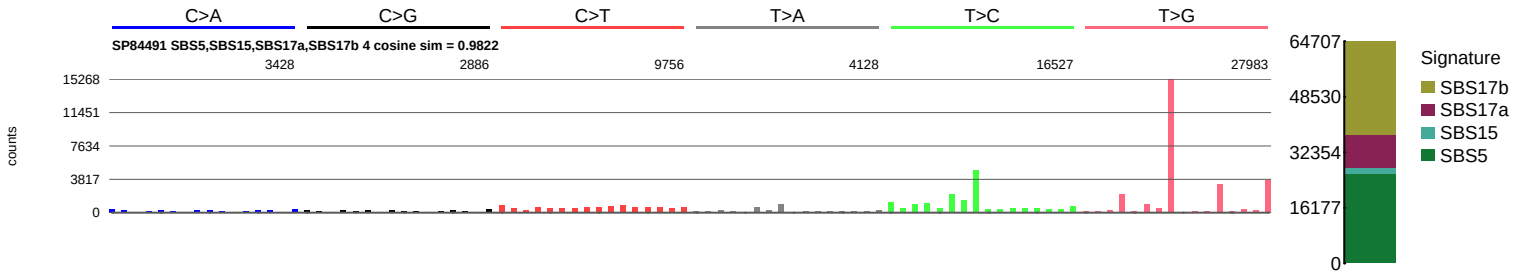
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2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SP84491 SBS5,SBS15,SBS17a,SBS17b 4 cosine sim = 0.9822
3428
2886
9756
4128
16527
27983
15268
11451
7634
3817
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counts
64707
48530
32354
16177
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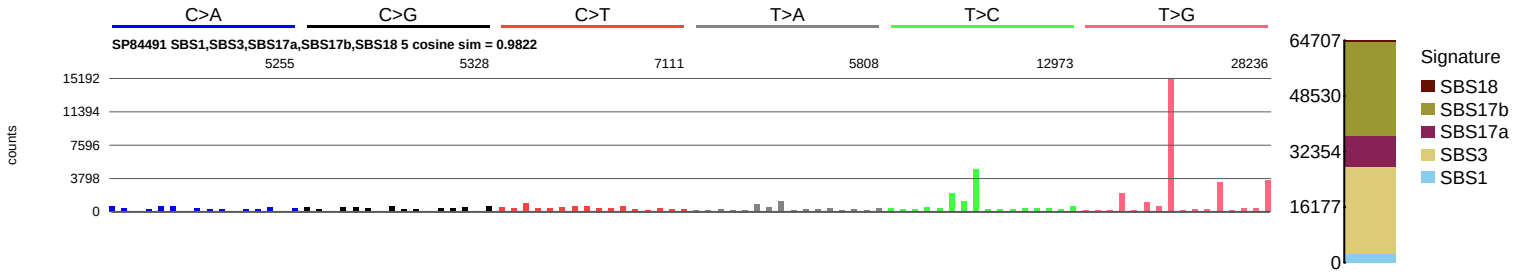
Signature
SBS17b
SBS17a
SBS15
SBS5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SP84491 SBS1,SBS3,SBS17a,SBS17b,SBS18 5 cosine sim = 0.9822
5255
5328
7111
5808
12973
28236
15192
11394
7596
3798
0
counts
64707
48530
32354
16177
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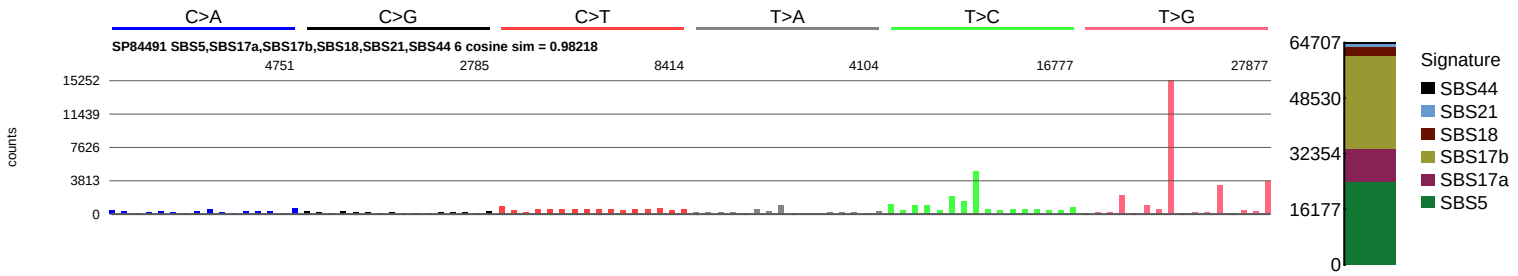
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>T
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T>G
SP84491 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 6 cosine sim = 0.98218
4751
2785
8414
4104
16777
27877
15252
11439
7626
3813
0
counts
64707
48530
32354
16177
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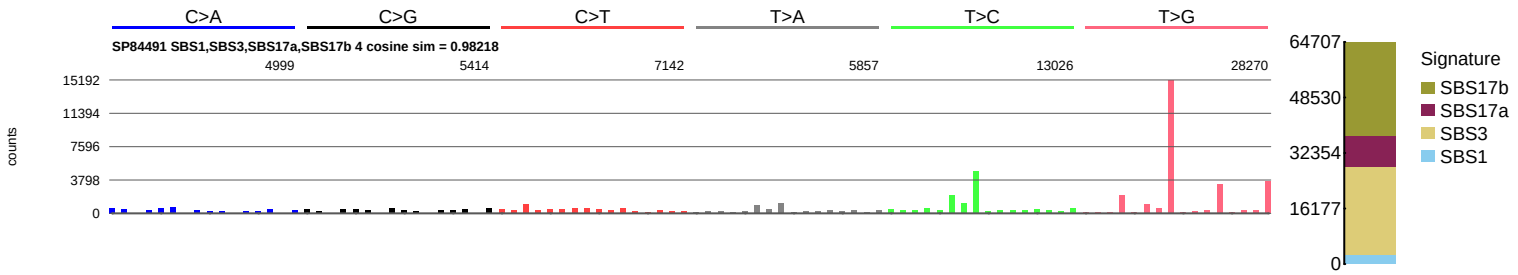
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5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T>C
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SP84491 SBS1,SBS3,SBS17a,SBS17b 4 cosine sim = 0.98218
4999
5414
7142
5857
13026
28270
15192
11394
7596
3798
0
counts
64707
48530
32354
16177
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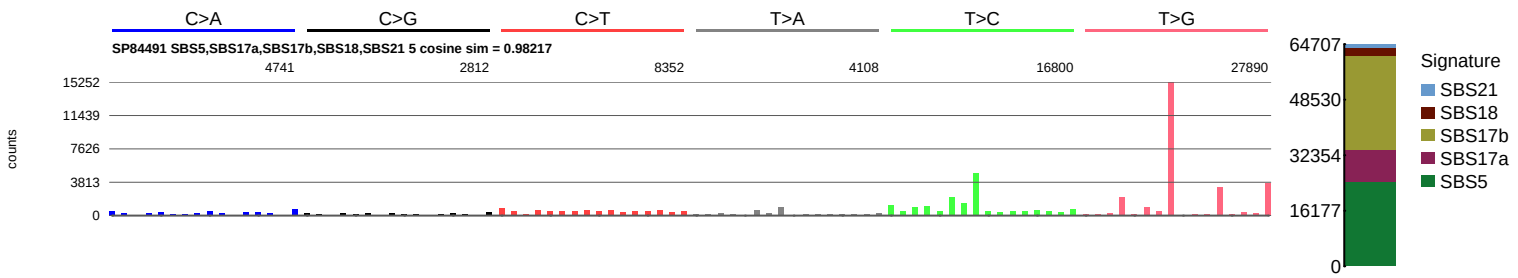
Signature
SBS17b
SBS17a
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS18,SBS21 5 cosine sim = 0.98217
4741
2812
8352
4108
16800
27890
15252
11439
7626
3813
0
counts
64707
48530
32354
16177
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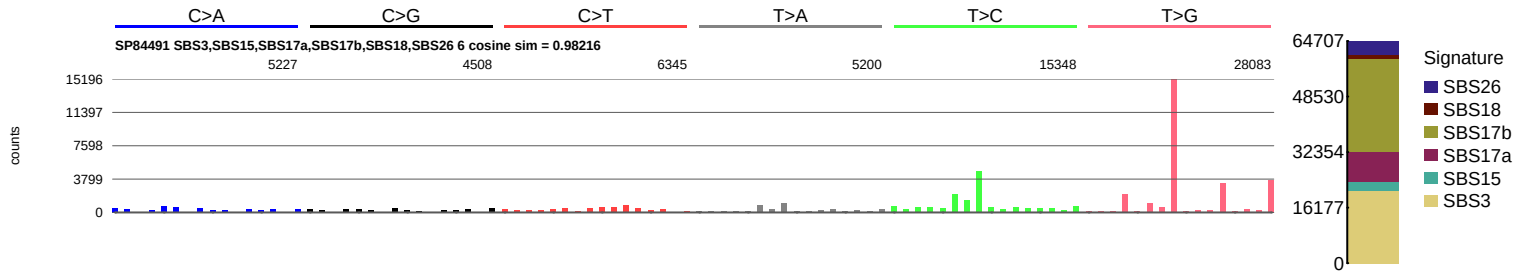
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5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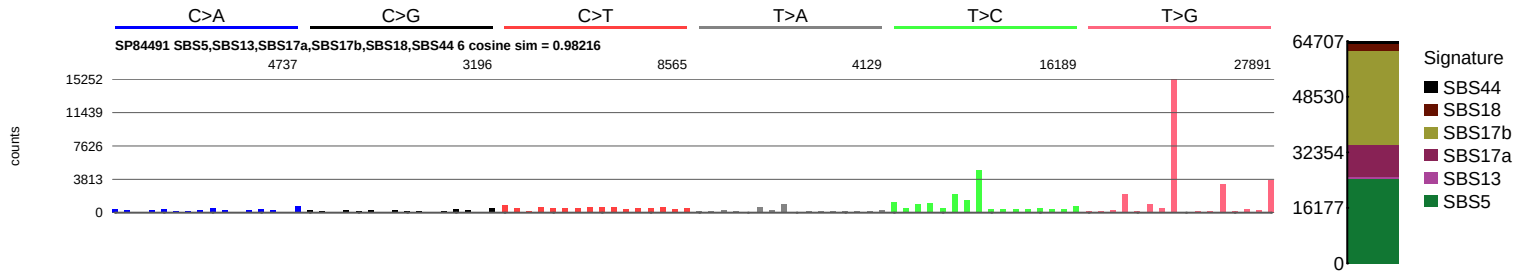
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98216
5227
4508
6345
5200
15348
28083
15196
11397
7598
3799
0
counts
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98216
4737
3196
8565
4129
16189
27891
15252
11439
7626
3813
0
counts
64707
48530
32354
16177
0
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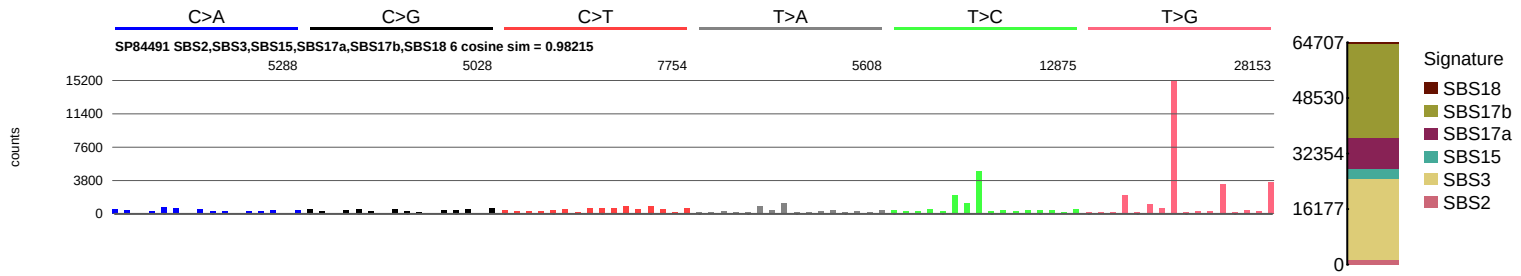
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5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C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98215
5288
5028
7754
5608
12875
28153
15200
11400
7600
3800
0
counts
64707
48530
32354
16177
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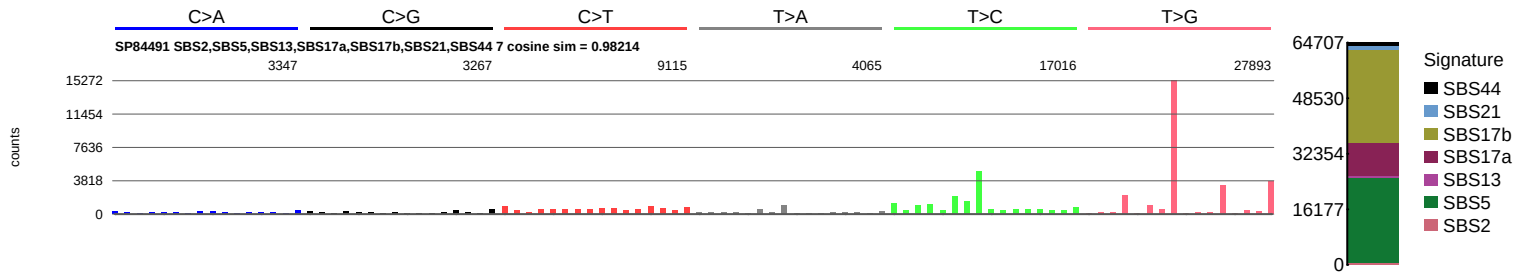
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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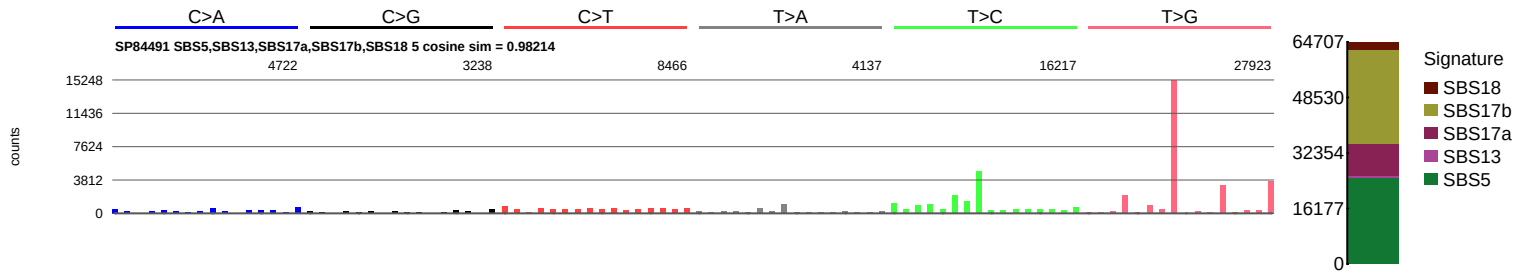
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98214
3347
3267
9115
4065
17016
27893
15272
11454
7636
3818
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS13,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98214
4722
3238
8466
4137
16217
27923
15248
11436
7624
3812
0
counts
64707
48530
32354
16177
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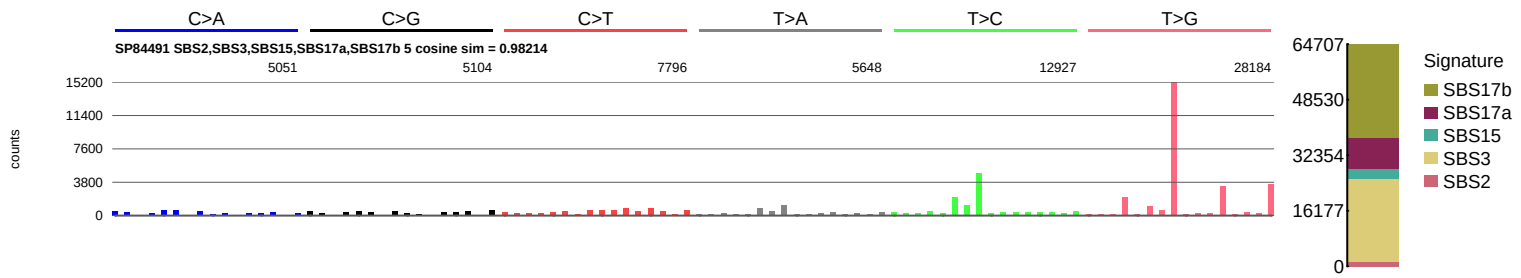
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5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C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98214
5051
5104
7796
5648
12927
28184
15200
11400
7600
3800
0
counts
64707
48530
32354
16177
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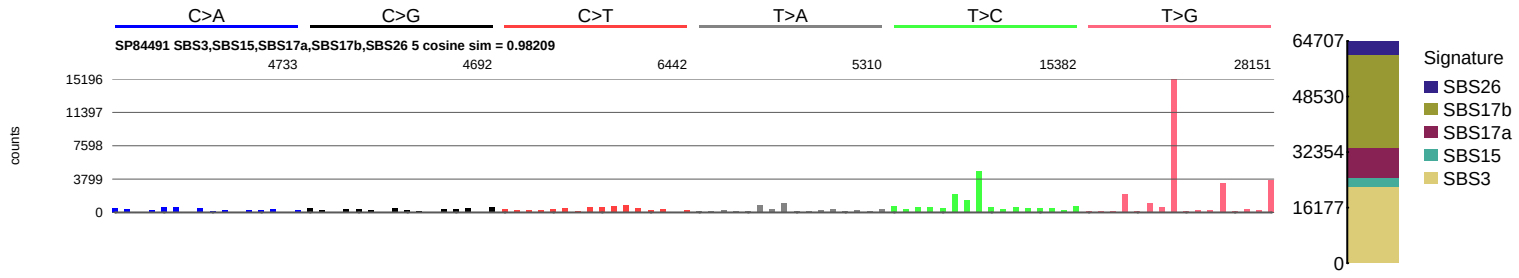
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS2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C>G
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T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS26 5 cosine sim = 0.98209
4733
4692
6442
5310
15382
28151
counts
15196
11397
7598
3799
0
64707
48530
32354
16177
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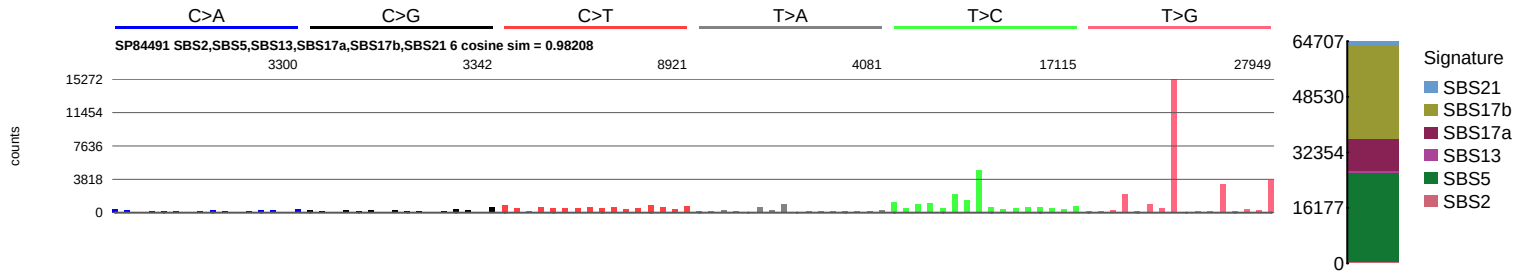
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3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C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98208
3300
3342
8921
4081
17115
27949
counts
15272
11454
7636
3818
0
64707
48530
32354
16177
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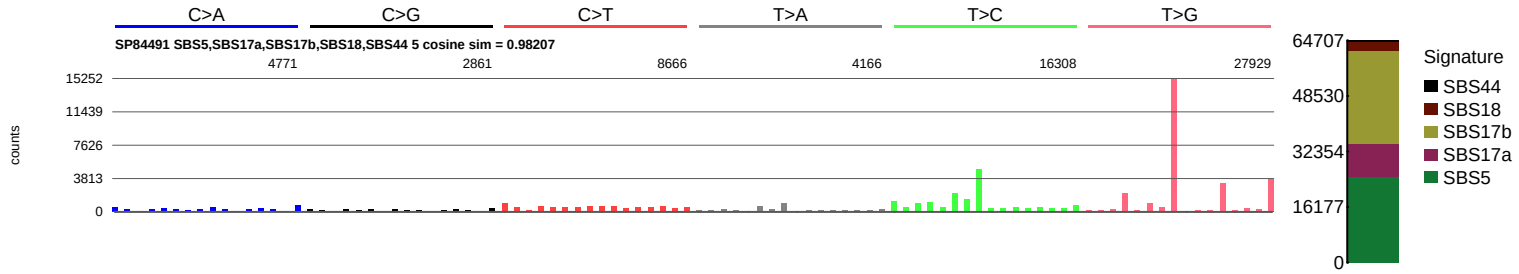
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2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C>G
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T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS18,SBS44 5 cosine sim = 0.98207
4771
2861
8666
4166
16308
27929
counts
15252
11439
7626
3813
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64707
48530
32354
16177
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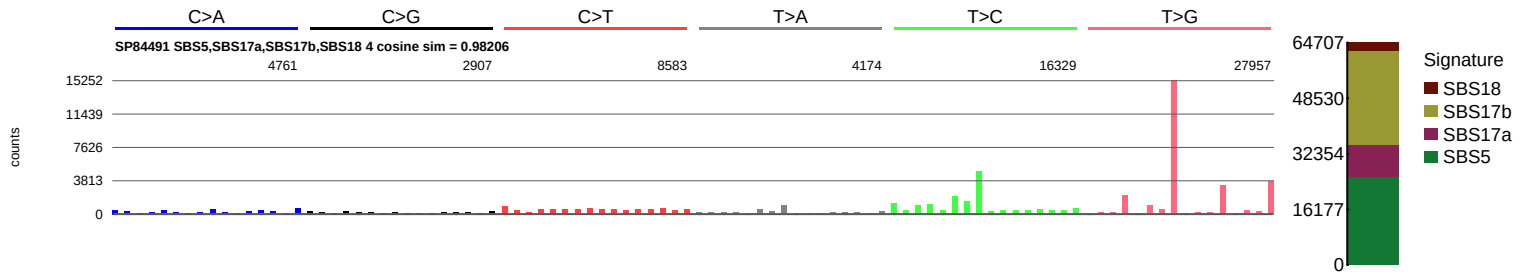
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5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T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS18 4 cosine sim = 0.98206
4761
2907
8583
4174
16329
27957
counts
15252
11439
7626
3813
0
64707
48530
32354
16177
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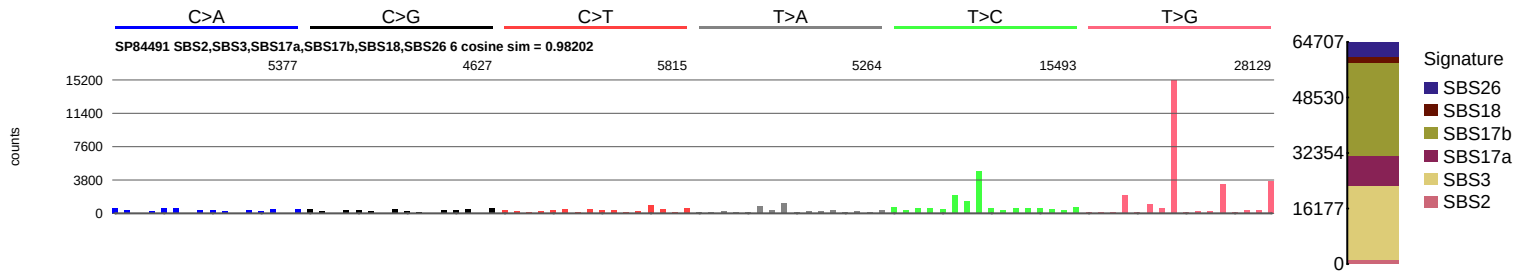
Signature
SBS18
SBS17b
SBS17a
SBS5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T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98202
5377
4627
5815
5264
15493
28129
counts
15200
11400
7600
3800
0
64707
48530
32354
16177
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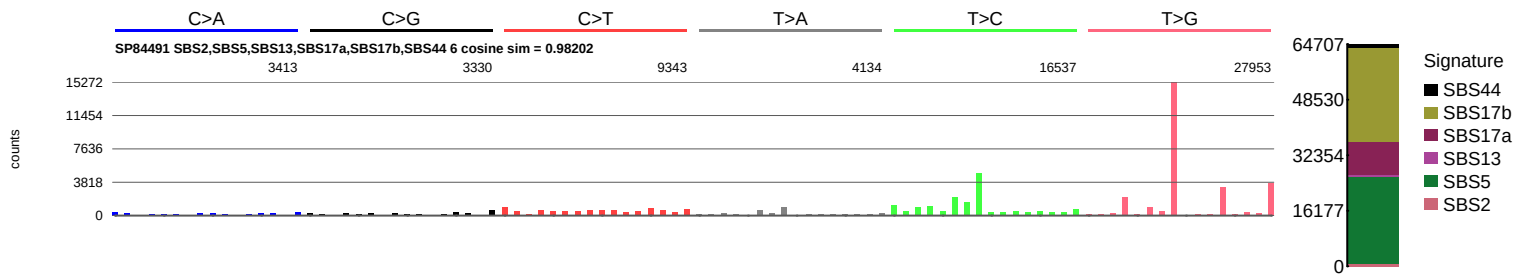
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98202
3413
3330
9343
4134
16537
27953
counts
15272
11454
7636
3818
0
64707
48530
32354
16177
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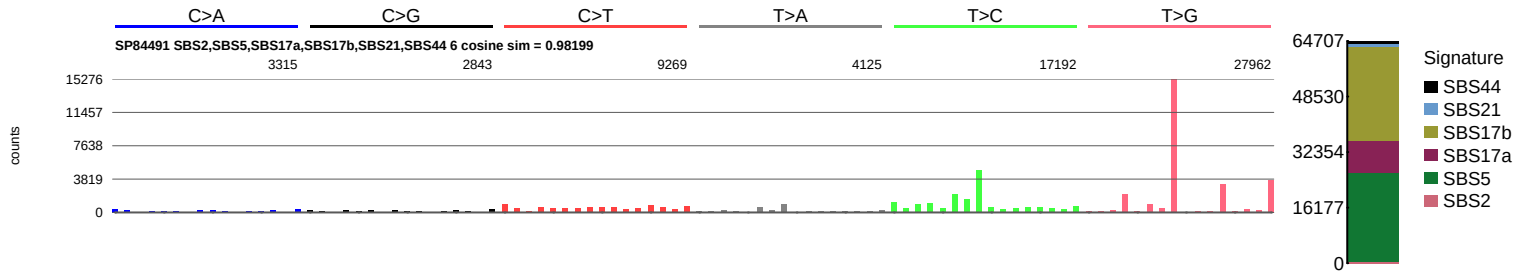
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS2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T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98199
3315
2843
9269
4125
17192
27962
counts
15276
11457
7638
3819
0
64707
48530
32354
16177
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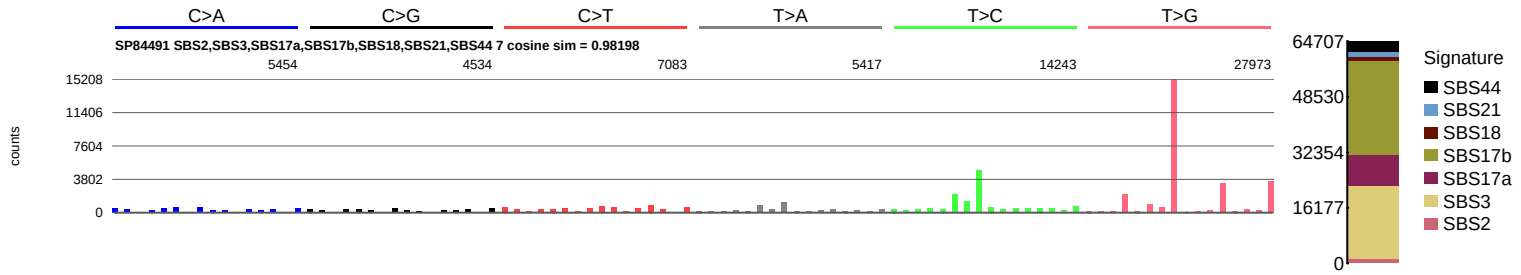
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98198
5454
4534
7083
5417
14243
27973
counts
15208
11406
7604
3802
0
64707
48530
32354
16177
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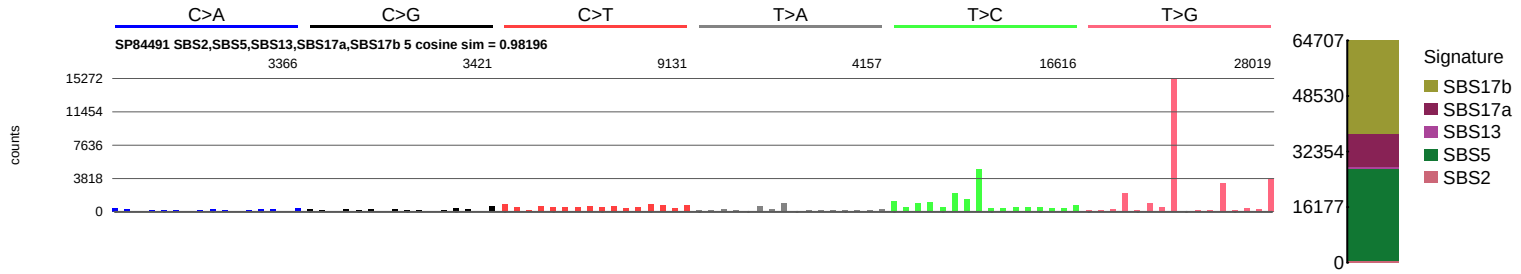
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS13,SBS17a,SBS17b 5 cosine sim = 0.98196
3366
3421
9131
4157
16616
28019
counts
15272
11454
7636
3818
0
64707
48530
32354
16177
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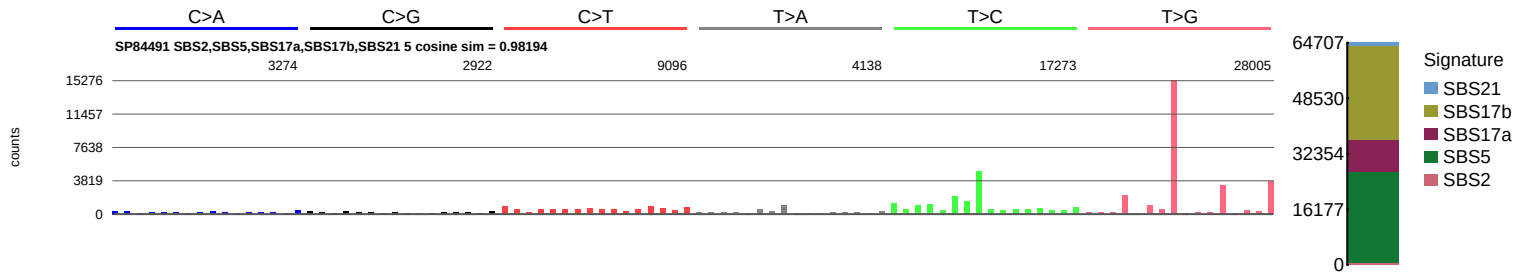
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98194
3274
2922
9096
4138
17273
28005
counts
15276
11457
7638
3819
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64707
48530
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16177
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SBS17b
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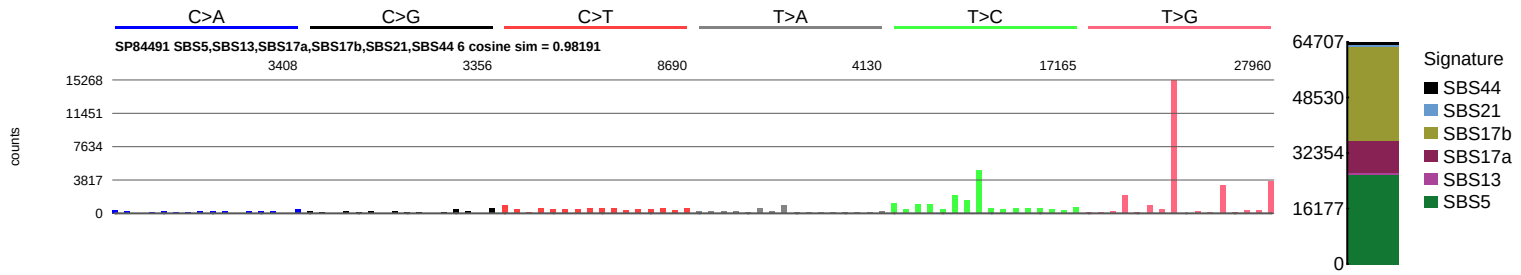
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98191
3408
3356
8690
4130
17165
27960
counts
15268
11451
7634
3817
0
64707
48530
32354
16177
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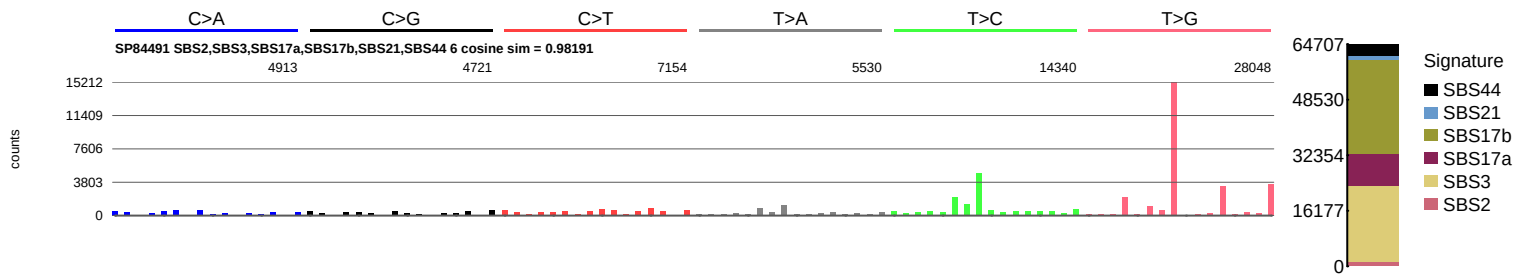
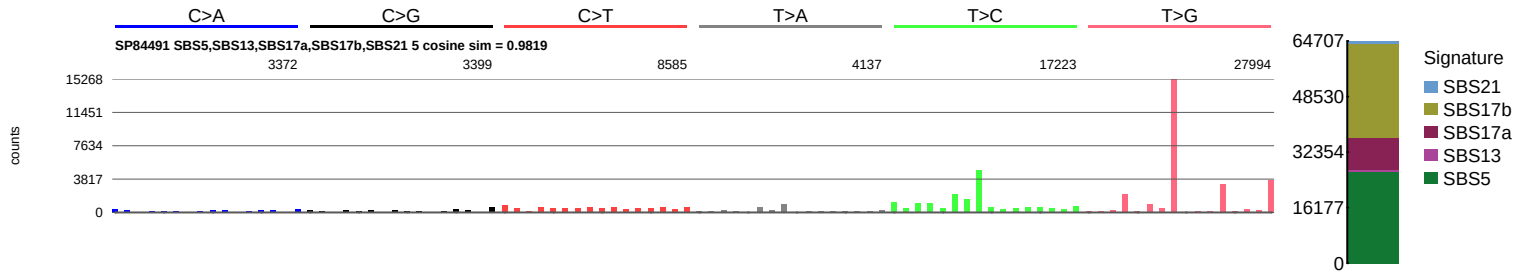
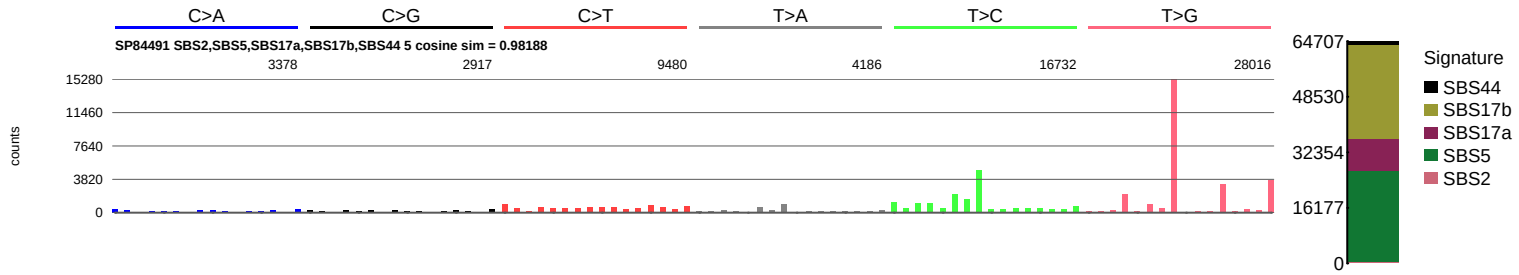
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98191
4913
4721
7154
5530
14340
28048
counts
15212
11409
7606
3803
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS13,SBS17a,SBS17b,SBS21 5 cosine sim = 0.9819
3372
3399
8585
4137
17223
27994
15268
11451
7634
3817
0
counts
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5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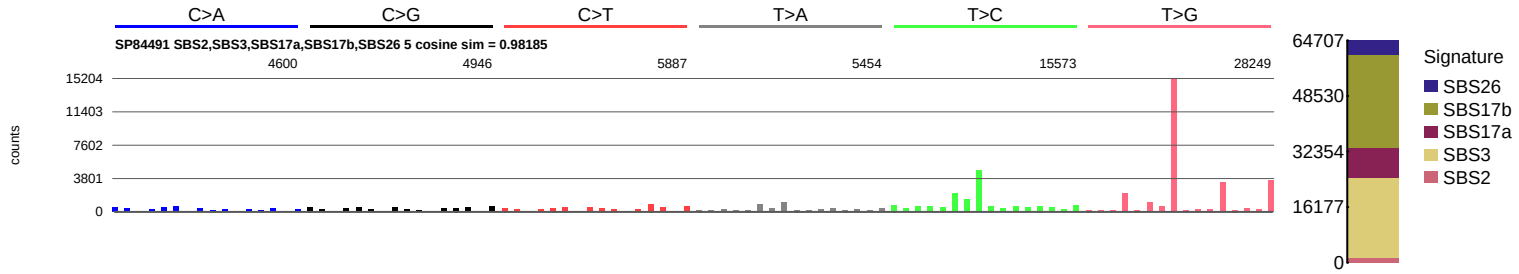
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98188
3378
2917
9480
4186
16732
28016
15280
11460
7640
3820
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS2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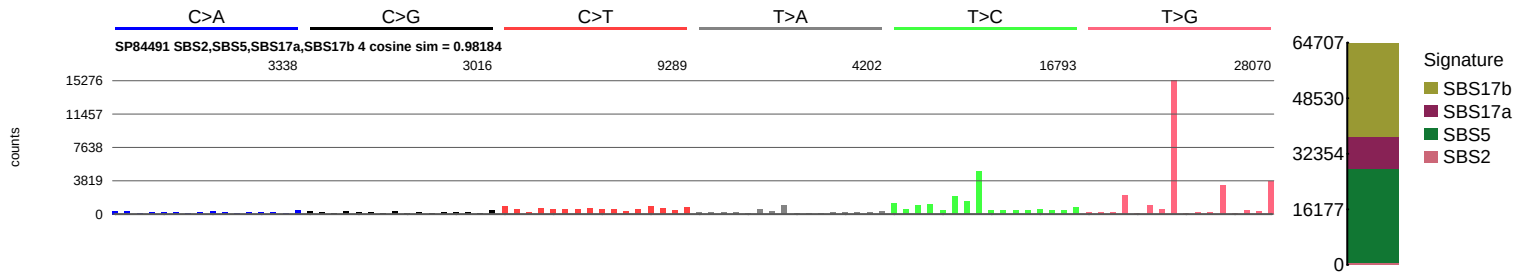
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS26 5 cosine sim = 0.98185
4600
4946
5887
5454
15573
28249
15204
11403
7602
3801
0
counts
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS2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C>G
C>T
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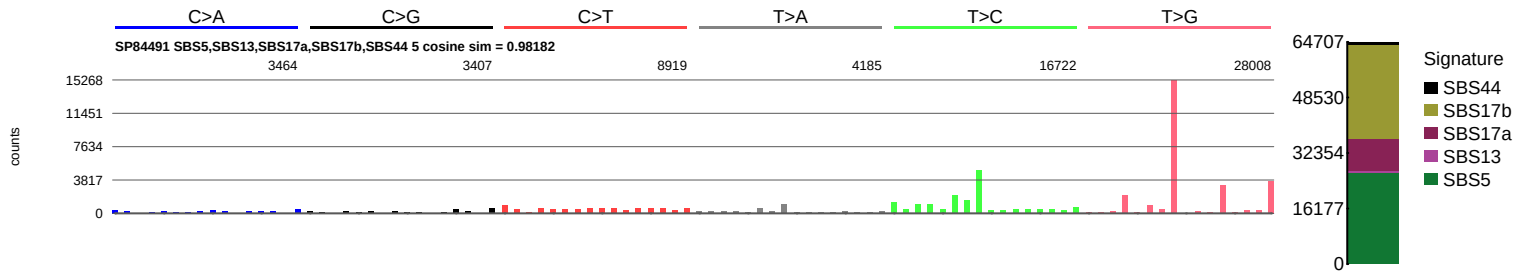
T>C
T>G
SP84491 SBS2,SBS5,SBS17a,SBS17b 4 cosine sim = 0.98184
3338
3016
9289
4202
16793
28070
15276
11457
7638
3819
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
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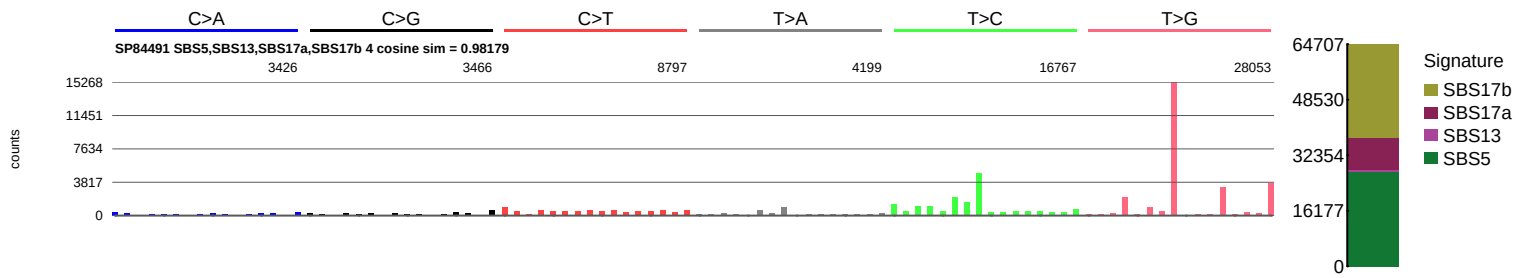
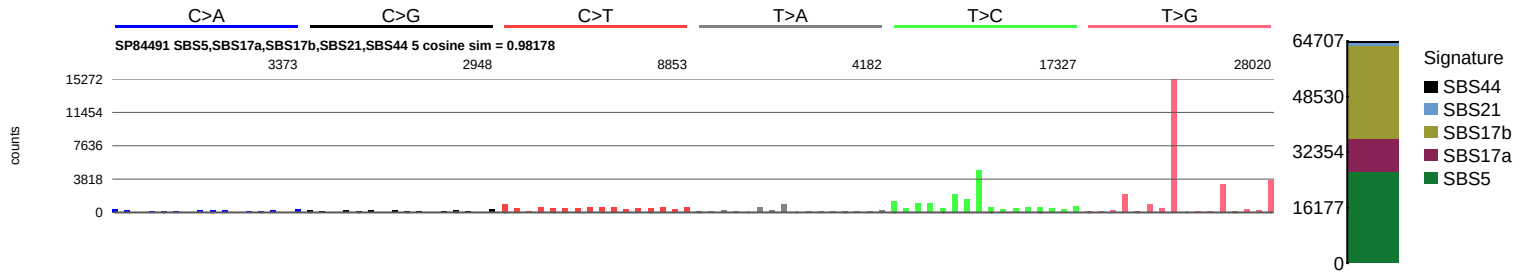
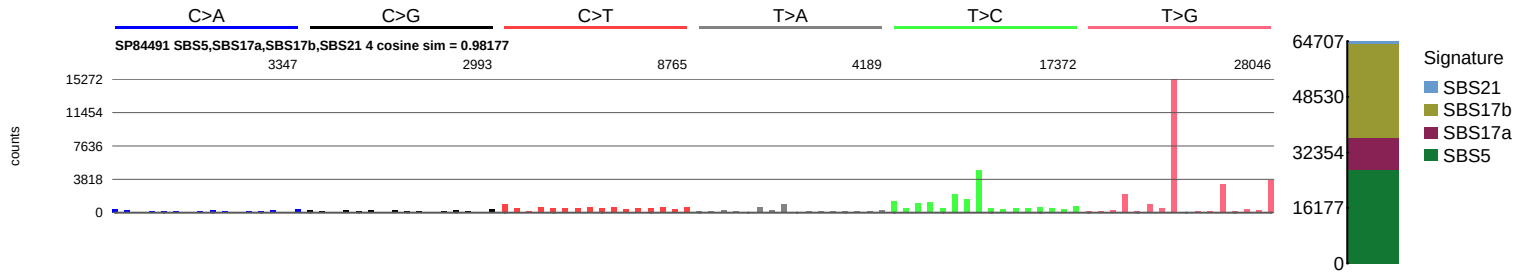
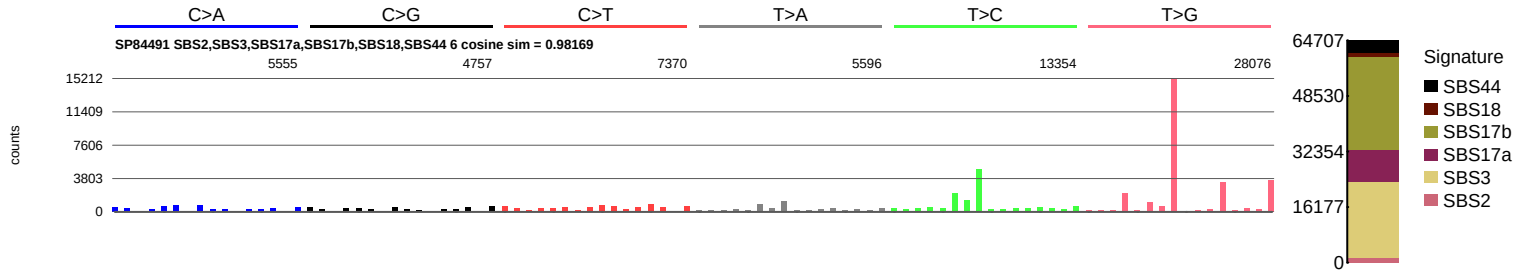
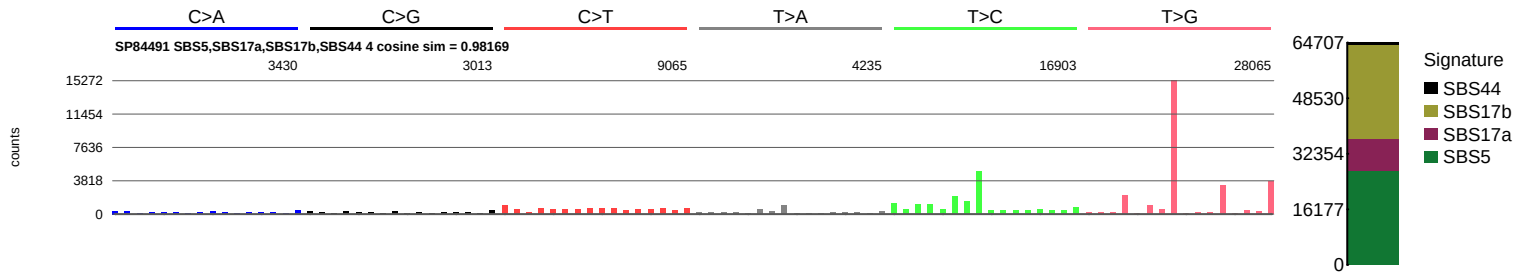
T>C
T>G
SP84491 SBS5,SBS13,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98182
3464
3407
8919
4185
16722
28008
15268
11451
7634
3817
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5

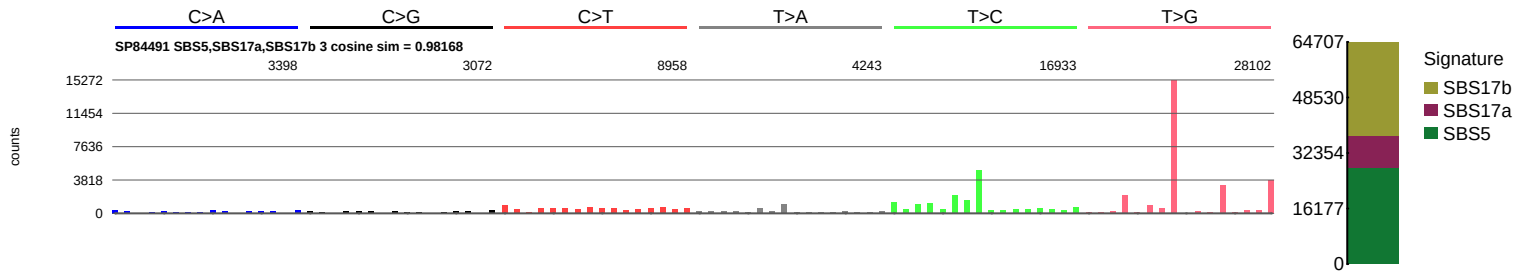

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS13,SBS17a,SBS17b 4 cosine sim = 0.98179
3426
3466
8797
4199
16767
28053
15268
11451
7634
3817
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS21,SBS44 5 cosine sim = 0.98178
3373
2948
8853
4182
17327
28020
counts
15272
11454
7636
3818
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS21 4 cosine sim = 0.98177
3347
2993
8765
4189
17372
28046
counts
15272
11454
7636
3818
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98169
5555
4757
7370
5596
13354
28076
counts
15212
11409
7606
3803
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b,SBS44 4 cosine sim = 0.98169
3430
3013
9065
4235
16903
28065
counts
15272
11454
7636
3818
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS5


C>A
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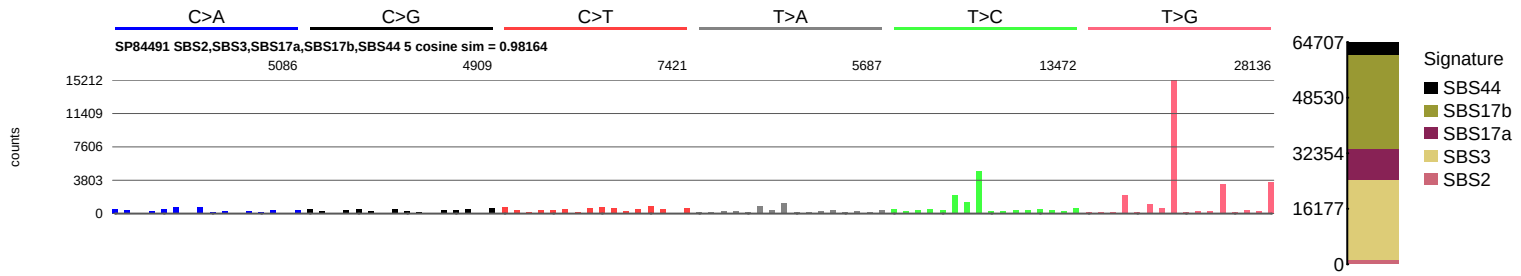
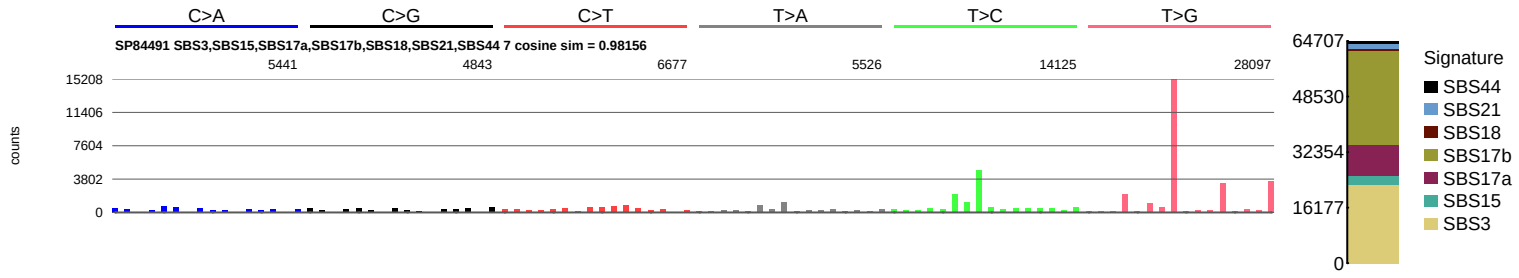
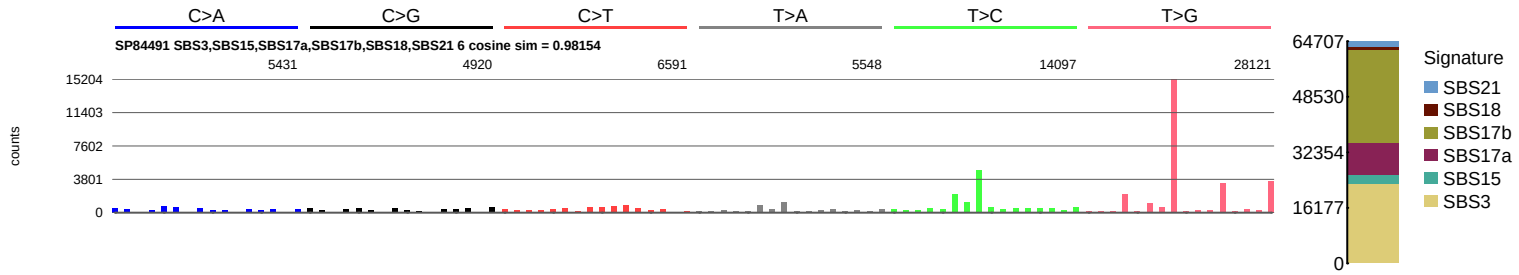
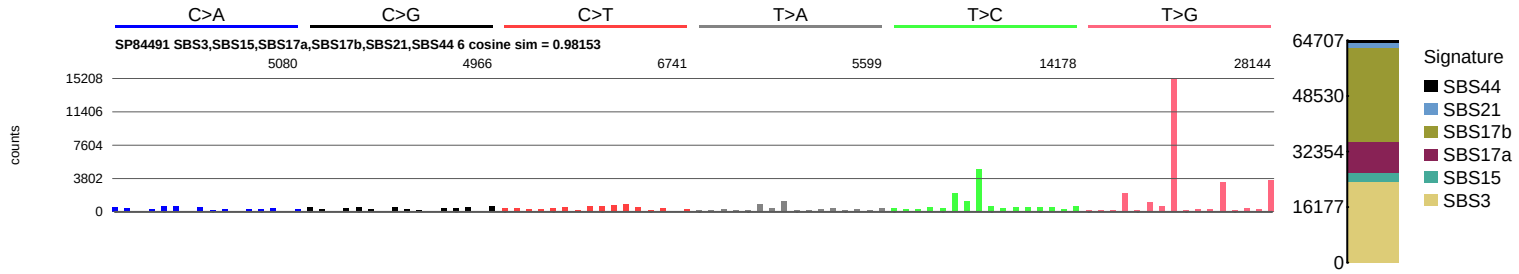
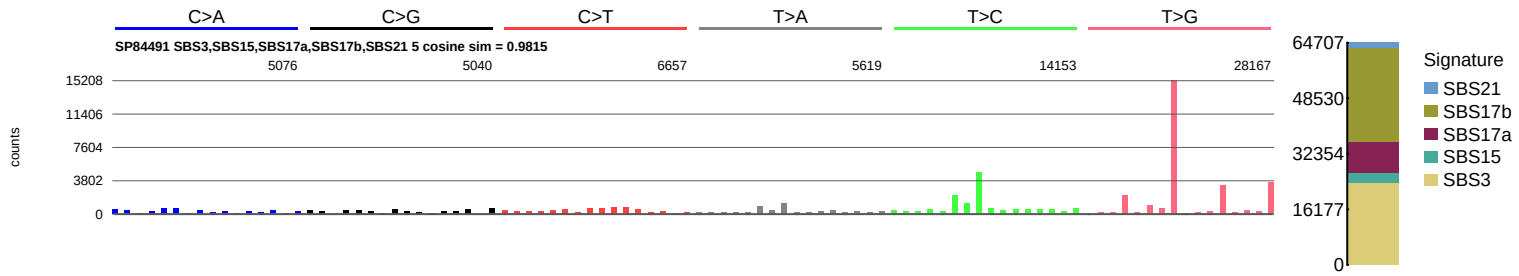
C>T
T>A
T>C
T>G
SP84491 SBS5,SBS17a,SBS17b 3 cosine sim = 0.98168
3398
3072
8958
4243
16933
28102
counts
15272
11454
7636
3818
0
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS5

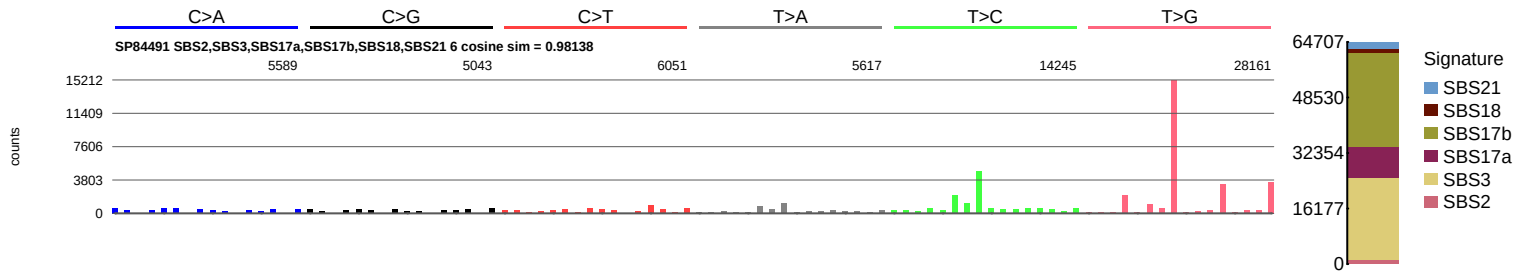

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98164
5086
4909
7421
5687
13472
28136
counts
15212
11409
7606
3803
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS3
SBS2

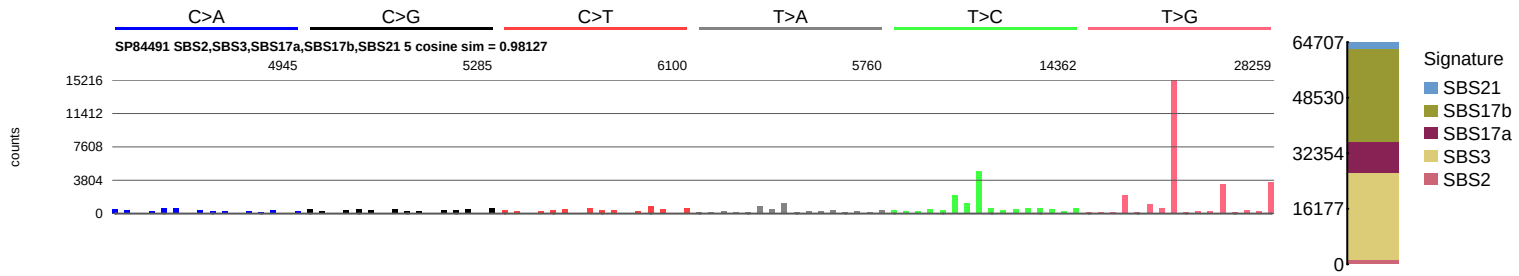

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98156
5441
4843
6677
5526
14125
28097
counts
15208
11406
7604
3802
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

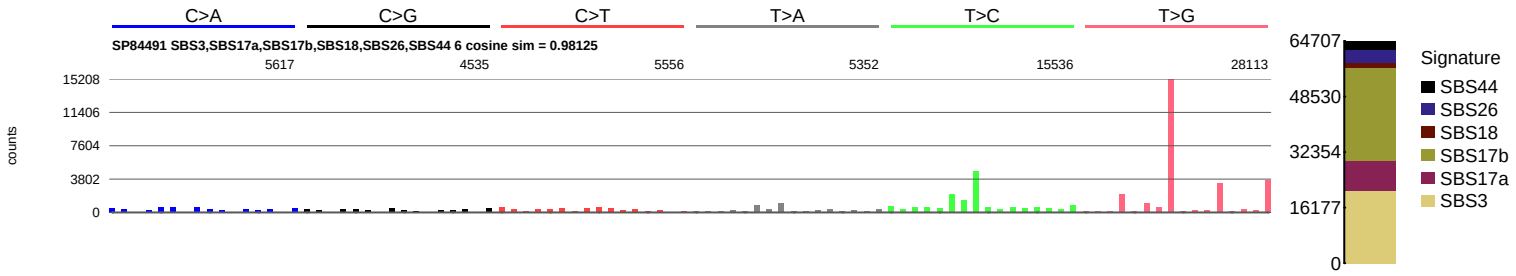

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98154
5431
4920
6591
5548
14097
28121
counts
15204
11403
7602
3801
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

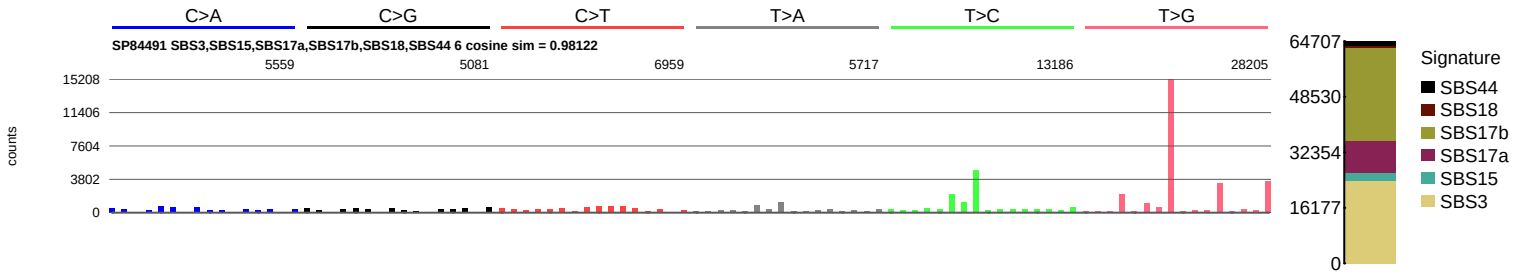

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98153
5080
4966
6741
5599
14178
28144
counts
15208
11406
7604
3802
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS21 5 cosine sim = 0.9815
5076
5040
6657
5619
14153
28167
counts
15208
11406
7604
3802
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98138
5589
5043
6051
5617
14245
28161
counts
15212
11409
7606
3803
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98127
4945
5285
6100
5760
14362
28259
counts
15216
11412
7608
3804
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 6 cosine sim = 0.98125
5617
4535
5556
5352
15536
28113
counts
15208
11406
7604
3802
0
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98122
5559
5081
6959
5717
13186
28205
counts
15208
11406
7604
3802
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64707
48530
32354
16177
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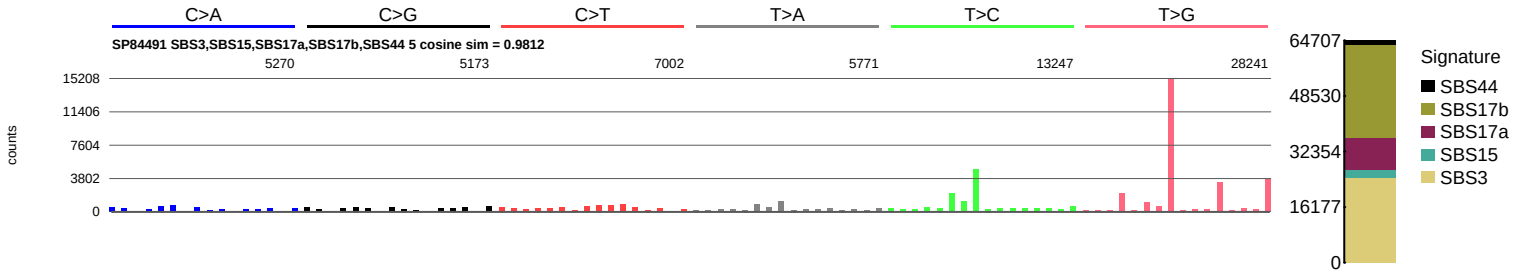
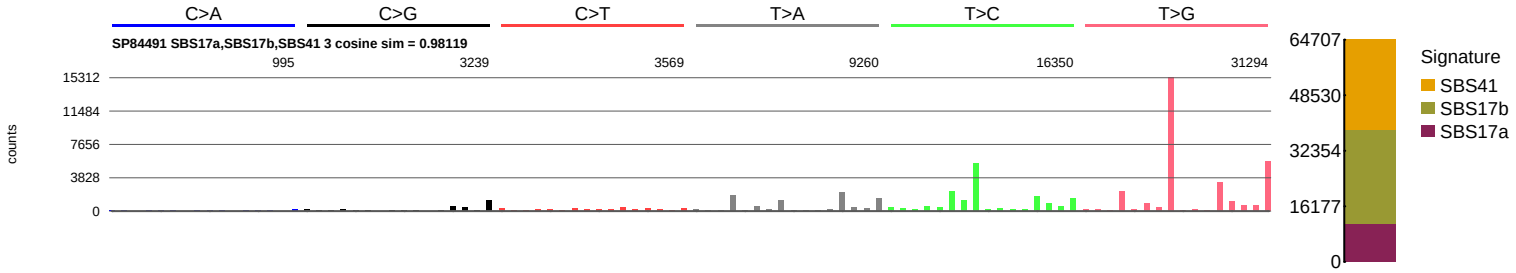
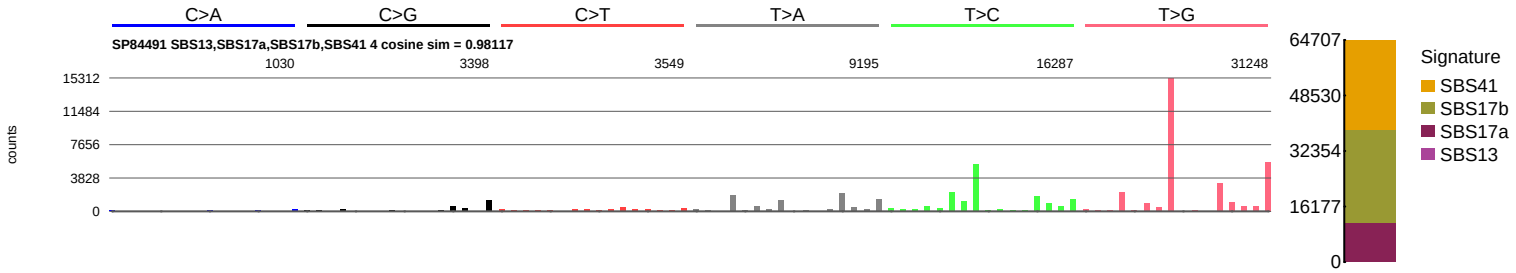
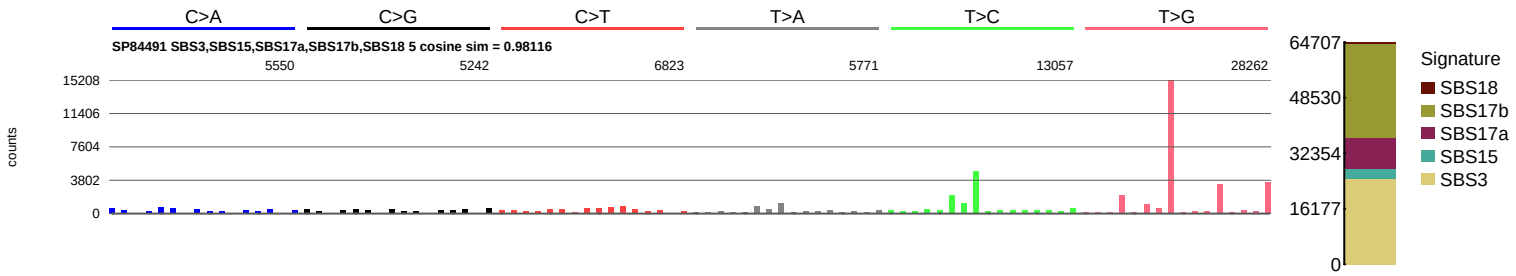
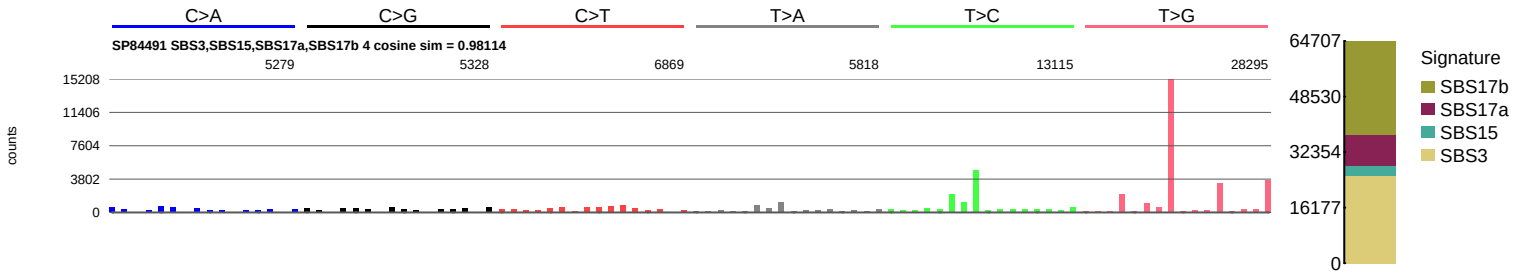
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3

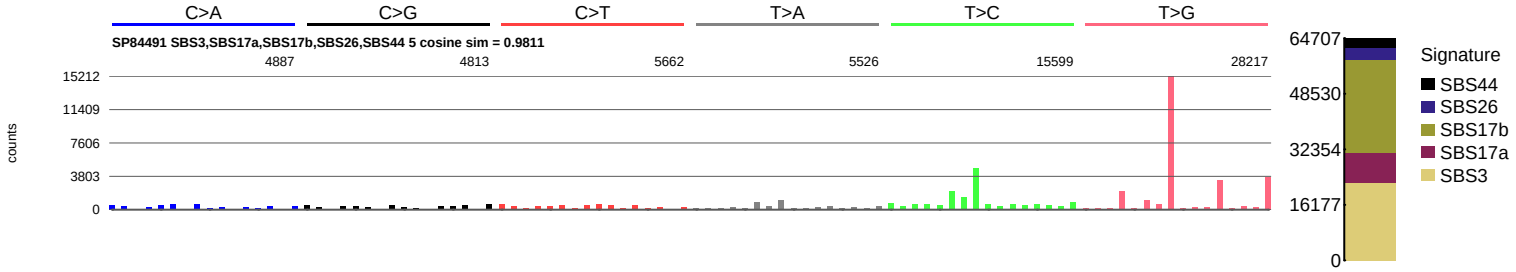

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS44 5 cosine sim = 0.9812
5270
5173
7002
5771
13247
28241
counts
15208
11406
7604
3802
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS3

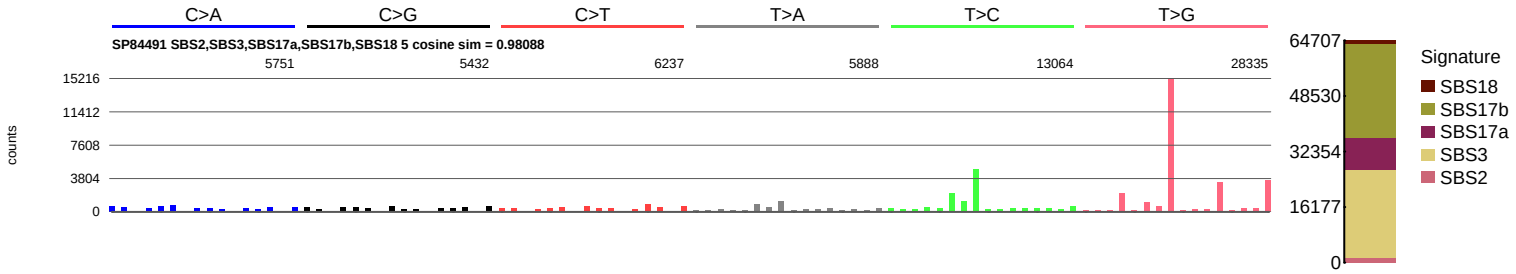

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS17a,SBS17b,SBS41 3 cosine sim = 0.98119
995
3239
3569
9260
16350
31294
counts
15312
11484
7656
3828
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a

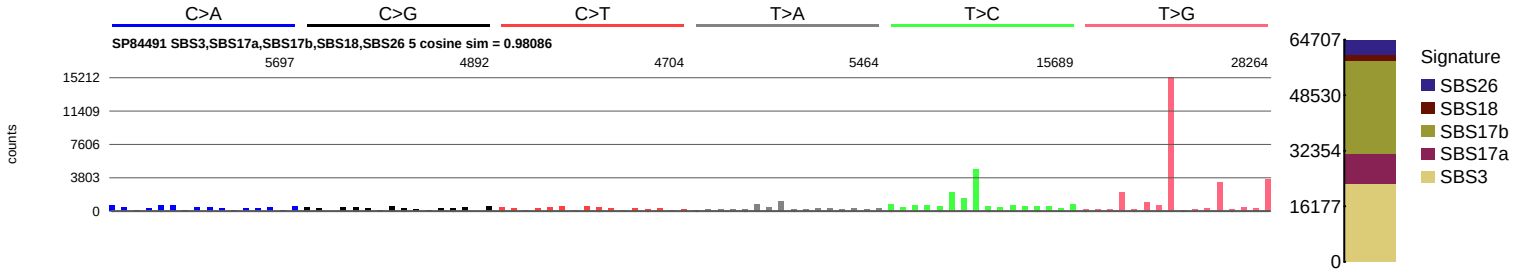

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS13,SBS17a,SBS17b,SBS41 4 cosine sim = 0.98117
1030
3398
3549
9195
16287
31248
counts
15312
11484
7656
3828
0
64707
48530
32354
16177
0
Signature
SBS41
SBS17b
SBS17a
SBS13

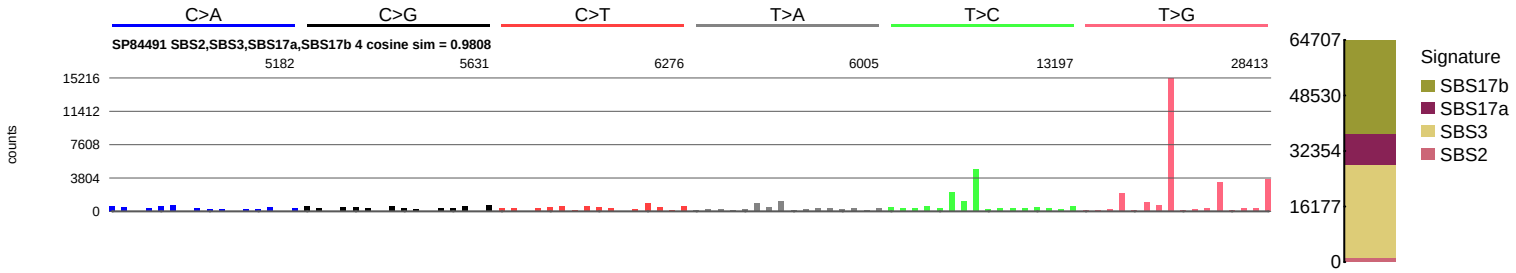

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98116
5550
5242
6823
5771
13057
28262
counts
15208
11406
7604
3802
0
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS15,SBS17a,SBS17b 4 cosine sim = 0.98114
5279
5328
6869
5818
13115
28295
15208
11406
7604
3802
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS26,SBS44 5 cosine sim = 0.9811
4887
4813
5662
5526
15599
28217
15212
11409
7606
3803
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98088
5751
5432
6237
5888
13064
28335
15216
11412
7608
3804
0
counts
64707
48530
32354
16177
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS18,SBS26 5 cosine sim = 0.98086
5697
4892
4704
5464
15689
28264
15212
11409
7606
3803
0
counts
64707
48530
32354
16177
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3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C>G
C>T
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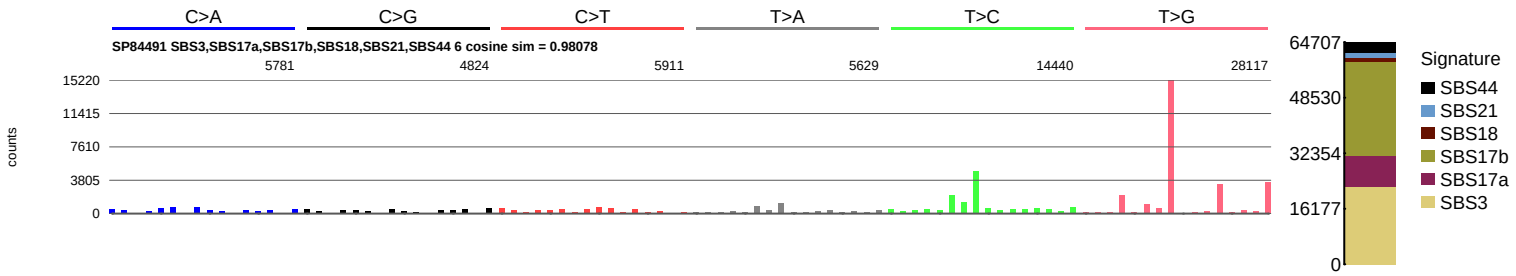
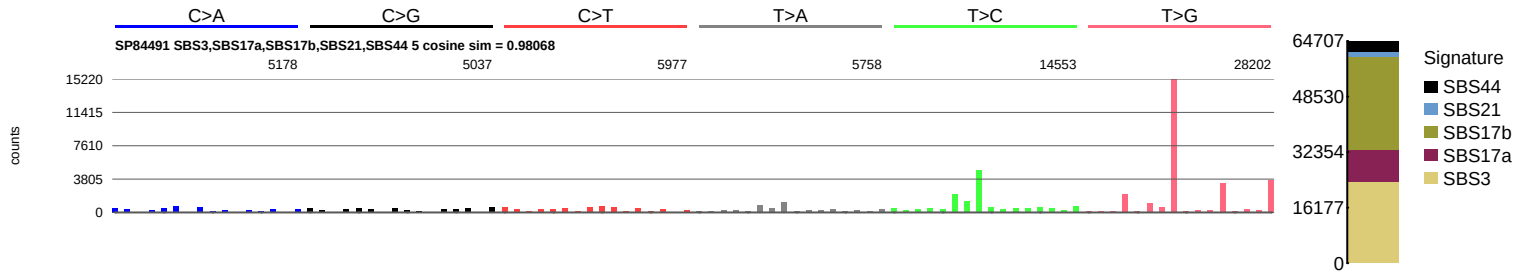
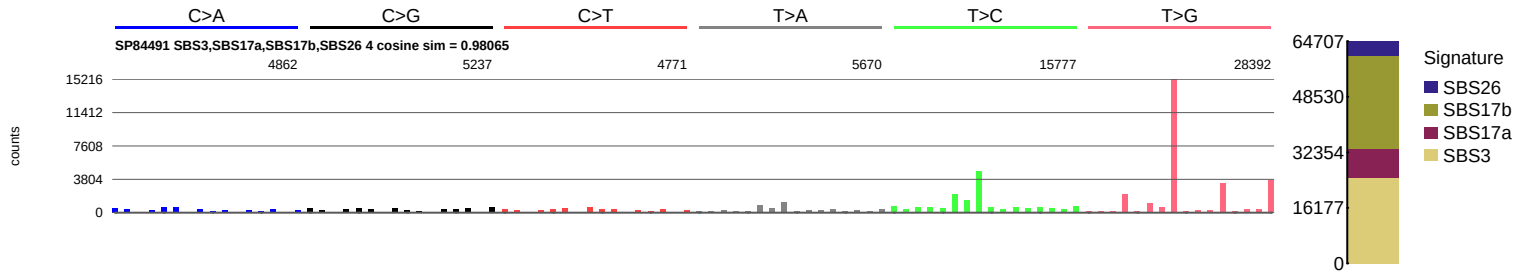
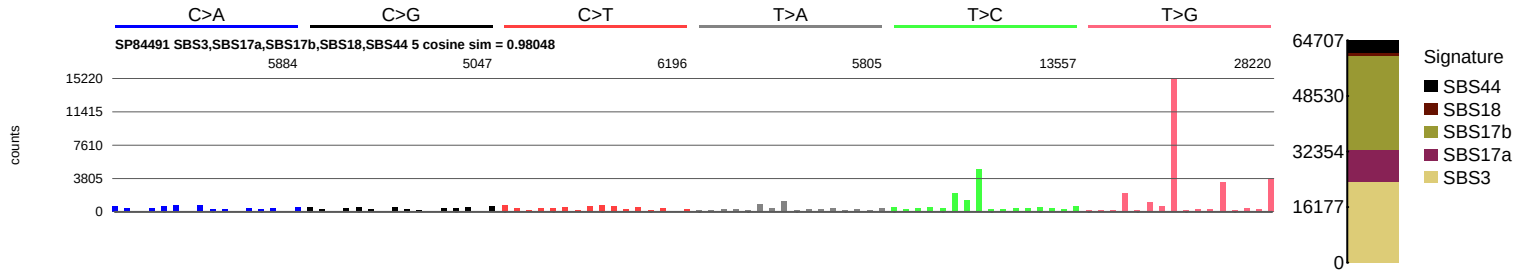
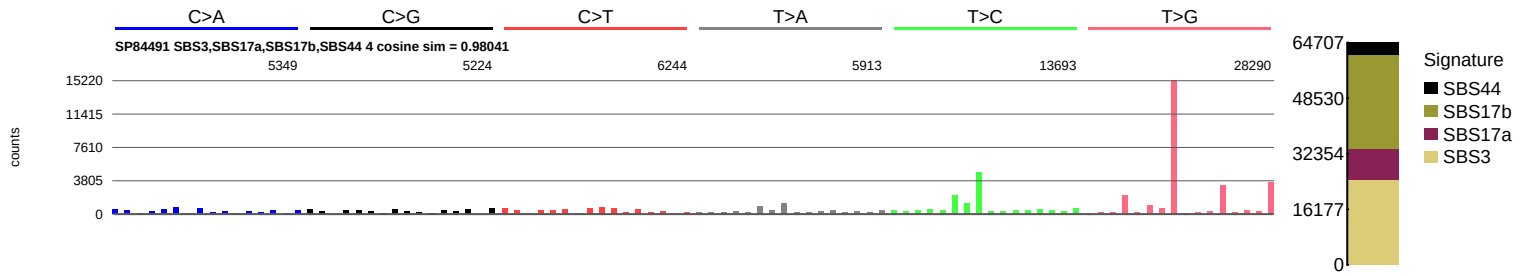
T>C
T>G
SP84491 SBS2,SBS3,SBS17a,SBS17b 4 cosine sim = 0.9808
5182
5631
6276
6005
13197
28413
15216
11412
7608
3804
0
counts
64707
48530
32354
16177
0
Signature
SBS17b
SBS17a
SBS3
SBS2

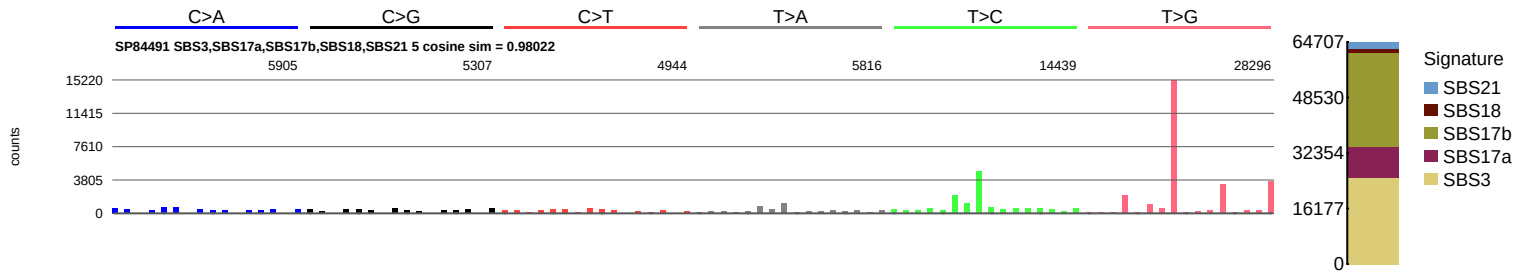

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 6 cosine sim = 0.98078
5781
4824
5911
5629
14440
28117
15220
11415
7610
3805
0
counts
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3

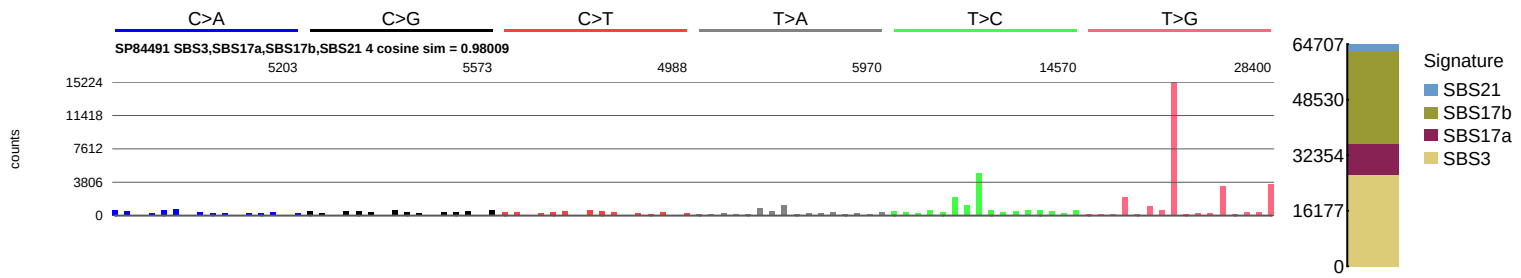

C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS21,SBS44 5 cosine sim = 0.98068
5178
5037
5977
5758
14553
28202
counts
15220
11415
7610
3805
0
64707
48530
32354
16177
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS26 4 cosine sim = 0.98065
4862
5237
4771
5670
15777
28392
counts
15216
11412
7608
3804
0
64707
48530
32354
16177
0
Signature
SBS26
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS18,SBS44 5 cosine sim = 0.98048
5884
5047
6196
5805
13557
28220
counts
15220
11415
7610
3805
0
64707
48530
32354
16177
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS44 4 cosine sim = 0.98041
5349
5224
6244
5913
13693
28290
counts
15220
11415
7610
3805
0
64707
48530
32354
16177
0
Signature
SBS44
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS18,SBS21 5 cosine sim = 0.98022
5905
5307
4944
5816
14439
28296
counts
15220
11415
7610
3805
0
64707
48530
32354
16177
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84491 SBS3,SBS17a,SBS17b,SBS21 4 cosine sim = 0.98009
5203
5573
4988
5970
14570
28400
counts
15224
11418
7612
3806
0
64707
48530
32354
16177
0
Signature
SBS21
SBS17b
SBS17a
SBS3